Machine-Native Reasoning

Representation, Learning, and Computational Organization

VOLUME II
Mathematical and Architectural Foundations

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Collection edition 1.0.0 / 9 October 2026
Komorebi P.

Mathematical and Architectural Foundations

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SR / Sufficient Resolution 269
Standardized publication · 47 source pages

PRIQ / Hypergraph Lumpability and Bayesian Refinement 317
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SRWT / Stratified Relational Workspace Theory 370
209-page monograph / repository 1.0.0 · 209 source pages

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PART II

Mathematical and Architectural Foundations

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SU / SOURCE READING CARD

Stratified Universality for Resolved Representations

Mathematical base 2.0.0 / standardized publication · 262 original pages

READ THIS FOR

Typed representation transport, semantic quotients, carrier changes and profiled descent.

KEEP THIS BOUNDARY

Relative to declared grammars, profiles and bridges; not equivalence of all mathematical carriers.

CONNECTED SOURCES

SR LSM SRWT

Original text begins on the next page.

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OPERATION KOMOREBI

REALIZATION CALCULUS



RESEARCH LIBRARY

Stratified Universality for Resolved Representations

Profiled lane transport, semantic quotients, carrier change, and certified descent

Traditional paper

Komorebi P.

Standardized publication edition / 8 October 2026

REALIZATION CALCULUS

Reading scope

Mathematical base: version 2.0.0 (23 September 2026). This edition retains the finite atlas and finite profiled and holonomy-twisted lane theorems, with the local corrections recorded in the publication note. The mathematical base adds a formal calculus of task-relative semantic quotients, exact bridge loss, carrier factorization, observer base change, admissibility-relative obstruction, and enrichment lifting. Inherited claims retain their original proof boundaries. New claims are separated in Part IV and in the expanded theorem concordance.

Universality is asserted only relative to declared grammars, strata, bridges, preservation profiles, normalization targets, and descent doctrines.

Abstract

Komorebi Math is formulated as a relative classifying and transport theory of resolved representations. A resolved theory records typed operations, packet decompositions, refinement laws, residuals, certificates, and only those analytic, tensor, metric, or arithmetic structures included in a declared preservation profile. Its realizations are organized into normalized strata. Reversible changes of presentation form representation groupoids, while quotients, completions, reductions, and other information-losing constructions remain in a directed category with explicit loss passports.

The first theorem layer establishes the additive–Karoubi resolved envelope, finite profiled atlas assembly, and componentwise invariant descent. The finite profiled Universal Lane Transport Theorem identifies every morphism between complete finite split packet systems with its full lane matrix. Composition becomes block contraction, refinement is functorial, and selectors split retained transport from a named residual. Under finite address holonomy, the same calculus descends as sections of a matrix local system. Monodromy orbits give the finest ordinary global lane coarsening, and pair orbits carry the compatible selector algebra.

The second theorem layer formalizes task-relative semantics. For a declared continuation grammar and observation family, the behavioral quotient is the coarsest exact representation preserving every future observation. In module-valued slices, task combination, common coarsening, relative residuals, bridge loss, essential-image failure, and task weakening fit into canonical exact sequences. Carrier replacement is possible exactly on a kernel-factorization locus. Set-valued residual equality and linear residual span are separated by an explicit relation module. Convex value packets, provenance states, additive route packets, and weighted-linear states therefore remain distinct unless certified comparison maps are supplied.

Finally, obstruction is indexed by the admissible realization class. Enlarging that class can reduce an optimized defect, while zero infimum may still fail to be attained. Rich structure descends under forgetting, but reconstructing it from a bare bridge is a typed lifting problem. The resulting theory is finite, profiled, task-relative, and auditable. It does not assert one global coordinate system, one observer-independent minimum, equivalence of all carriers, or effectivity of an unrestricted heterogeneous stack.

Keywords: resolved representation, stratified universality, lane transport, semantic quotient, behavioral core, carrier change, observer base change, exact loss, address holonomy, admissibility-relative obstruction, certified descent.

Central claim. Komorebi Math is a stratified classifying and transport theory of resolved representations. Within a declared profile, it separates exact representation change from directional information loss. Within a declared semantic slice, it constructs the coarsest representation preserving all future

task observations. Changes of carrier, observer, geometry, completion, or enrichment are legal only through certified comparison or lifting data.

Claim boundary. The paper does not assert that every mathematical object has a finite Komorebi presentation, that all strata or carriers are equivalent, that every bridge is reversible, that arbitrary presentations admit one observer-independent minimum, or that the unrestricted heterogeneous atlas is an effective stack. Internal proof status, exact computation, classical import, historical novelty, and independent peer validation remain distinct classifications.

Publication edition and source availability. This publication edition preserves the theorem labels, concordance component identities, and claim boundaries of the 23 September 2026 version 2.0.0, with five local amendments recorded explicitly: orbit supports are distinguished from nonzero selector action; observer updates have finite branching, finite raw-history aggregation, and declared empty-fiber conventions; the compactness proof of escape uses an attained positive minimum; the coinduction and finite-certification proofs make their fixed-point implications explicit; and metric statements declare positive dimension, determinant normalization, and inverse edge traversal. The original sources and inherited records remain included unchanged. Its companion visual guide and Lean guide are reading and evidence guides to that mathematical base.

The complete release includes the Markdown (`.md`) files, LaTeX sources, figure generators, data, exact validators, and inherited validation reports. Inherited reports retain their original dates and scope. Written formal mathematics, exact finite computation, publication checks, and Lean certification are separate forms of evidence. No Lean source files or Lean build project are present in the supplied version 2.0.0 source tree. Separate related Lean developments, where identified, require their own source and theorem correspondence. This publication edition does not claim a fresh Lean build or a new independent proof audit of the complete manuscript.

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1 Introduction

1.1 The assembly problem

The source manuscripts already contain typed quiver and module semantics, seed presentations, exact packet coordinates, refinement and residual calculus, nonlinear packet composition, finite holonomy, relative obstruction objects, operator-algebraic recognition, Galois descent, and several exact sequence and arithmetic realizations [1, 2, 3, 6, 7]. The central problem is therefore no longer to reconstruct these foundations independently. It is to determine when they assemble into one finite, typed, coherent classifying object.

The proposed architecture has six layers:

$$\boxed{\begin{array}{c}\text{resolved syntax} \longrightarrow \text{normalized strata} \longrightarrow \text{typed overlaps} \longrightarrow \text{coherence} \\ \longrightarrow \text{effective descent} \longrightarrow \text{compatible complete invariants.}\end{array}}$$

Each arrow carries a proof obligation. A semantic analogy is not a bridge. A bridge is not reversible merely because formulas exist in both directions. Local completeness is not global completeness until the local isomorphisms themselves satisfy descent.

1.2 Main theorem and classification domain

Three theorem levels are kept distinct. First, the finite profiled Universal Lane Transport Theorem is an internal theorem about complete finite split packet systems in an additive idempotent-complete category. Its holonomy-twisted extension retains exact analysis, synthesis, composition, refinement, selectors, and residuals when chart loops permute lane addresses. The global coordinate object is then a matrix local system, and ordinary lane projectors descend at the monodromy-orbit resolution.

Second, the finite atlas assembly theorem is conditional in general. Given profile-preserving normalized strata, typed reversible overlaps, coherent comparison cells, effective descent, and compatible native invariants, the descended invariant reflects isomorphism. The manuscript verifies these hypotheses on a finite certified atlas whose reversible core has five connected semantic components.

Third, the semantic quotient theorem works inside a fixed doctrine and declared observation category. It constructs the maximal continuation-stable invisible core and the associated task-relative quotient. In module-valued slices, semantic combination and bridge transport admit canonical exact sequences. Carrier and observer changes are separate factorization problems, while changes of admissible geometry or enrichment are typed lifting problems.

The statements are strongest precisely because their targets are explicit. Lane transport concerns declared finite resolutions. Atlas classification concerns the declared normalized finite atlas. Behavioral minimality concerns one semantic slice. None of these statements classifies arbitrary presentations, identifies all carrier algebras, or promotes unverified doctrine changes to equivalences.

1.3 Contributions

The paper makes fourteen consolidated contributions.

1. It provides a source-controlled theorem concordance and an explicit profile-indexed additive–Karoubi resolved envelope.

2. It separates representation groupoids from the directed transformation category and assigns every certified bridge a preservation and information-loss passport.
3. It proves a finite profiled atlas assembly theorem and distinguishes holonomy, relative cokernel, Brauer, and higher-coherence defects.
4. It establishes normalized effective-descent atlases and complete native invariants on the five-component certified semantic union.
5. It proves exact bridge comparison and higher coherence, including the nonlinear–algebraic–holonomic Fuss–Catalan triangle.
6. It proves the finite profiled Universal Lane Transport Theorem with exact block transport, refinement, selectors, residuals, and separately typed fold and kernel clauses.
7. It removes the trivial-address-holonomy restriction from transport by proving matrix-local-system descent, orbit coarsening, pair-orbit selectors, and orbital path composition.
8. It defines the task-relative behavioral quotient as the coarsest exact representation preserving all declared future observations.
9. It identifies the behavioral kernel as the maximal continuation-stable invisible core and gives a greatest-fixed-point computation with finite-dimensional termination.
10. It proves the semantic diamond, relative semantic residual, task-change exact sequence, bridge-loss exact sequence, and extension law for composite loss in module-valued slices.
11. It proves the carrier factorization gate and the universal joint completion preserving two incomparable carrier semantics.
12. It separates Set-valued continuation minimization from linear residual minimization by an exact relation-module sequence and an explicit polyphase family with infinite discrete state and finite Hankel rank.
13. It proves carrier-indexed observer aggregation and observer-conditioned convex packet transport, while isolating cancellation, idempotence, singular-resurrection, and provenance boundaries.
14. It formalizes admissibility monotonicity, completion/gap/escape stratification, a sharp convex-versus-quadratic example, and the forgetful-descent versus enrichment-lifting principle.

1.4 Proof status and organization

Classical inputs, internally proved theorems, exact finite verification, conditional statements, framework definitions, and conjectures are separately marked. The theorem concordance is source control for this distinction. Part I gives the common resolved foundation and finite assembly theorem. Part II develops the certified native strata. Part III gives global synthesis and finite profiled and holonomy-twisted lane transport. Part IV develops task-relative semantic quotients, carrier and observer change, exact loss, admissibility-relative obstruction, and doctrine lifting. The final sections state scope and conclusion. Detailed development, claim-boundary, and validation records are retained in the appendices.

2 Main theorem suite and proof architecture

Let

$$\mathfrak{A}_{\mathrm{cert}} := \mathfrak{A}_{13}$$

denote the finite certified atlas constructed in Section 19. Its reversible core is the disjoint union of five connected semantic components, while its directional satellites retain explicit loss or obstruction passports.

Theorem 2.1 (Stratified resolved representation, descent, and complete invariant theorem)**.** *Let $\mathbb{K}$ be the resolved theory used in this manuscript and let $\mathfrak{A}_{\mathrm{cert}}$ be the declared finite atlas. Then the following statements hold.*

1. *Every chart is a profile-preserving realization of the common resolved syntax, and every reversible edge is an equivalence on its named overlap category.*
2. *The descent category exists as the finite pseudolimit of the atlas and decomposes as the coproduct of its five connected semantic components.*
3. *Every proof whose constructor support lies in the common verified preservation profile transports to compatible local theorems and hence to the corresponding global resolved object.*
4. *The semantic normalizers induce the comparison 2-cells on all nonempty triple overlaps and force the quadruple-overlap coherence equation.*
5. *The compatible native invariant family descends to a tagged global invariant*

$$\mathcal{J}_{\mathrm{cert}} : \mathrm{Desc}(\mathfrak{A}_{\mathrm{cert}}) \longrightarrow \mathrm{Desc}(\mathcal{V}_{\mathrm{cert}}).$$

6. *The descended invariant is complete on the declared normalized global category:*

$$\mathcal{J}_{\mathrm{cert}}(X) \cong \mathcal{J}_{\mathrm{cert}}(Y) \quad \Longleftrightarrow \quad X \cong Y.$$

7. *Every noninvertible construction remains outside the reversible core and carries a typed record of its preserved structure, forgotten information, named residual, or obstruction class.*

Proof. The common resolved syntax and preservation-profile transport are established in Section 3. The finite atlas and its pseudolimit are supplied by theorem 6.3. The normalizer comparison cells and their higher coherence are theorem 19.2. Descent of compatible invariant families is theorem 19.4. Completeness is theorem 19.11; its hypotheses are verified component by component in Section 19. The bridge passports and obstruction taxonomy in Section 4 keep every directional construction outside the maximal subgroupoid. Combining these results proves all seven assertions. □

Corollary 2.2 (Relative universality)**.** *The certified atlas is universal for profile-preserving models of the declared resolved grammar on its five normalized semantic components. This universality is relative to the chosen grammar, strata, bridges, preservation profiles, normalization targets, and descent doctrine.*

Theorem 2.3 (Relative semantic quotient and typed contract-change theorem)**.** *Fix a declared semantic slice in a stratum of the resolved theory. Assume the required effective quotients exist. Then:*

1. *the behavioral quotient is the coarsest exact representation preserving all declared future observations;*
2. *its kernel or congruence is the maximal continuation-stable invisible core;*
3. *strengthening the task induces a canonical map from the stronger quotient to the weaker quotient;*
4. *in module-valued slices, joint semantics, shared semantics, relative residuals, bridge loss, essential-image failure, and task weakening fit into canonical exact sequences;*

5. *one carrier semantics replaces another exactly on the corresponding kernel-factorization locus;*
6. *observer base change may introduce an independent relation object and therefore does not follow from carrier compatibility;*
7. *enlarging an admissible realization class can only decrease optimized defect, but zero infimum need not be attained;*
8. *lifting a bare bridge to richer metric, tensor, topological, projective, or completed structure requires a separate enhancement certificate.*

Proof. The universal quotient and maximal-core statements are theorems 22.4 and 22.6. Task change and the module exact sequences are theorems 22.16, 22.20, 23.5 and 23.7. Carrier and observer change are theorems 23.1 and 23.13. Admissibility monotonicity is theorem 24.2, and the enrichment statement is proposition 24.9. Each assertion retains its printed doctrine and existence hypotheses. □

Corollary 2.4 (No absolute minimal representation)**.** *There is no observer-independent or doctrine-independent minimum asserted by this manuscript. Minimality is relative to the carrier, continuation grammar, observation category, query family, and admissible enrichment class.*

Boundary 2.5 (Publication boundary). Theorem 2.1 does not imply a global canonical normal form for arbitrary mathematical objects, equivalence of all broad strata, decidability of arbitrary presentation equivalence, or effectivity of an unrestricted heterogeneous stack. It also does not identify functional-Mahler equations with Newton-Mahler interpolation, forget branch data in the algebraic-to-holonomic direction, or infer Tannakian symmetry from a merely faithful additive realization.

Part I

Common foundations and finite assembly

3 Resolved theories, packet coordinates, and preservation profiles

This section expands the common formal core used by every later atlas construction. The purpose is not to force every domain into one category. It is to separate four layers that are often conflated:

$$\boxed{\text{typed syntax} \longrightarrow \text{resolved coordinates} \longrightarrow \text{profile-preserving realization} \longrightarrow \text{domain semantics.}}$$

The first two layers are algebraic. The third records which algebraic or universal constructions survive a realization. The fourth supplies domain-specific hypotheses such as completeness, positivity, convergence, integrality, measurability, or analytic continuation. A cross-domain statement is a theorem only when all four layers are present.

3.1 Typed resolved signatures

Definition 3.1 (Finite resolved signature)**.** Fix a commutative coefficient ring $\Bbbk$. A finite resolved signature is a tuple

$$\mathbb{K}_{\mathrm{fin}} = (Q, I, \mathsf{P}, \mathsf{Ref}, \mathsf{Comp}, \mathsf{Cert})$$

with the following data.

Table 1: The publication theorem spine. Section numbers are resolved by the final build; the logical order, rather than the numbering, is authoritative.

Stage	Principal result	Exact role in the proof	Location
Resolved source	Additive-Karoubi envelope	Literal finite packet objects and profile-indexed realization source	theorem 3.36
Typed atlas	Bridge passports and obstruction separation	Reversible overlaps are separated from directional loss	Section 4
Finite assembly	Profiled atlas assembly	Bilimit, theorem transport, groupoid core, and global directional functors	theorem 6.3
Native normalization	Certified stratum atlases	Complete local coordinates on declared semantic slices	Part II
Higher coherence	Normalizer-induced coherence	Unique semantic normalizers force all certified overlap cycles to commute	theorem 19.2
Invariant descent	Certified invariant-descent theorem	Coherent local invariants descend through effective object and morphism gluing	theorem 19.4
Global completeness	Tagged componentwise invariant	The descended invariant reflects isomorphism on the five-component union	theorems 2.1 and 19.11
Lane transport	Finite profiled Universal Lane Transport Theorem	Exact block transport, refinement, selection, residuals, conditional fold loss, and certified descent	theorems 20.3 and 20.24
Twisted transport	Holonomy-Twisted Universal Lane Transport Theorem	Matrix-local-system synthesis, base-fiber invariant normal form, orbit coarsening, and orbital path algebra	theorems 21.3, 21.6 and 21.8 and corollary 21.4
Task semantics	Relative semantic quotient theorem	Behavioral cores, task quotients, semantic diamonds, and exact task loss	theorems 2.3, 22.16 and 22.20
Carrier and observer change	Factorization and base-change theorems	Joint semantic completion, exact bridge loss, relation modules, and carrier-indexed aggregation	theorems 23.1, 23.5, 23.9 and 23.13
Admissibility and lifting	Class-relative defect and enhancement gates	Completion/gap/escape, geometry-class separation, and typed enrichment lifting	theorems 24.2 and 24.7 and proposition 24.9

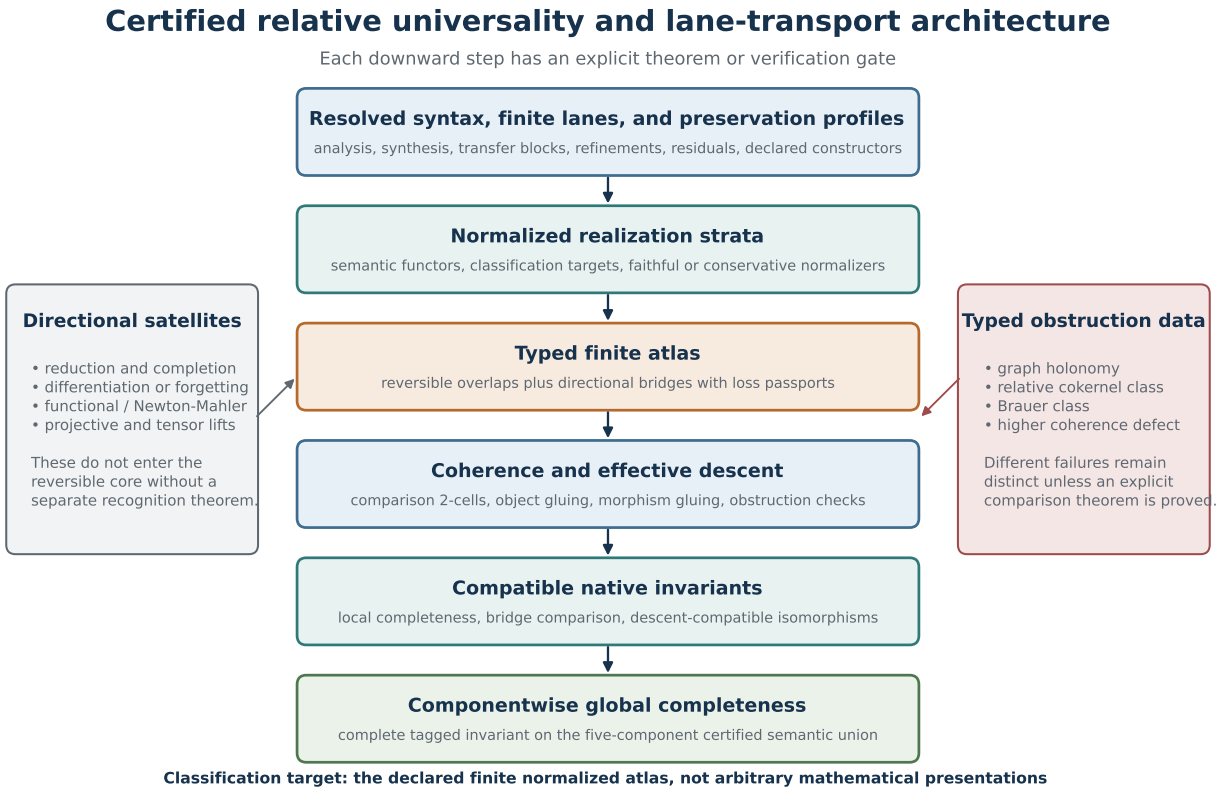


Figure 1: Proof architecture of the journal-facing theorem. The vertical spine contains the reversible construction and descent argument. Directional satellites and obstruction data remain typed external inputs rather than silent equivalences.

(i) $Q = (Q_0, Q_1, s, t)$ is a finite quiver. Its vertices are sorts and its arrows are typed primitive operations.

(ii) $I \triangleleft \Bbbk Q$ is a finitely generated two-sided ideal generated by $\Bbbk$-linear combinations of paths with common source and common target.

(iii) P is a finite list of packet declarations. A declaration on a sort X consists of endomorphism words $p_{X,r,i}$ and the relations

$$p_{X,r,i}p_{X,r,j} = \delta_{ij}p_{X,r,i}, \qquad \sum_i p_{X,r,i} = 1_X.$$

(iv) Ref is a finite list of refinement declarations. For $s \succeq r$, an address map

$$\rho_{sr} : I_{X,s} \twoheadrightarrow I_{X,r}$$

is accompanied by the equations

$$p_{X,r,i} = \sum_{\rho_{sr}(j)=i} p_{X,s,j}.$$

(v) Comp is a list of optional completion tags. A tag names a doctrine to be added later, such as Banach completion, a C^*-involution, a multiplier completion, or a monoidal structure. A tag is not itself an existence theorem.

(vi) Cert is a list of named residual and certificate types. Each certificate type specifies its source object, target object, and the equations or inequalities it is intended to record.

The finite signature is the presentation-level object. It records legal words and declared equations. It does not identify a preferred realization, a preferred coordinate basis, or a preferred semantic domain.

Definition 3.2 (Realization of a finite resolved signature)**.** Let $\mathcal{C}$ be a $\Bbbk$-linear category. A realization of $\mathbb{K}_{\mathrm{fin}}$ in $\mathcal{C}$ assigns an object M_X to every vertex $X \in Q_0$ and a morphism

$$M_a : M_{s(a)} \longrightarrow M_{t(a)}$$

to every arrow $a \in Q_1$, such that all relations in I, all packet equations, and all declared refinement equations hold. Completion and certificate tags are interpreted only when the realization profile explicitly includes them.

Theorem 3.3 (Typed realization and module equivalence, imported)**.** *Let*

$$\Lambda_{\mathbb{K}} := \Bbbk Q / I.$$

In $\Bbbk$-modules, the category of typed realizations of the linear grammar (Q, I) is equivalent to the category of unital left $\Lambda_{\mathbb{K}}$-modules. Under the equivalence, the vertex idempotent e_X acts as the projector onto the component of sort X, and each arrow acts in its declared source-target block.

Proof. A typed realization defines the action of a path by composition and then extends $\Bbbk$-linearly. Every element of I acts by zero, so the action factors through $\Lambda_{\mathbb{K}}$. Conversely, for a unital module M, the pairwise orthogonal vertex idempotents give

$$M = \bigoplus_{X \in Q_0} e_X M,$$

and an arrow $a : X \to Y$ maps $e_X M$ into $e_Y M$. These constructions are inverse up to their canonical natural isomorphisms. This is the typed-module theorem of Paper I [1]. □

Remark 3.4 (Presentation is not classification). The theorem fixes the semantics of one declared grammar. It does not say that the grammar is unique, minimal, or recoverable from a finite list of observations. Distinct quivers with relations can present equivalent module categories, and one realization can annihilate additional syntax through the kernel of its action.

Example 3.5 (Two-sort transfer grammar)**.** Take two sorts X, Y, one arrow $a : X \to Y$, and no nontrivial relation. A realization is a single typed map

$$M_X \xrightarrow{M_a} M_Y.$$

After collapsing to $M_X \oplus M_Y$, the vertex idempotents are coordinate packets and M_a is one off-diagonal transfer block. The smallest typed grammar therefore already contains the distinction between object packets and transfer channels.

3.2 Packet objects, analysis, and exact transfer coordinates

Definition 3.6 (Finite packet system)**.** Let X be an object of a preadditive category. A finite packet system on X is a family

$$\mathcal{P} = (p_i)_{i \in I} \subseteq \mathrm{End}(X)$$

satisfying

$$p_i^2 = p_i, \qquad p_i p_j = 0 \quad (i \neq j), \qquad \sum_{i \in I} p_i = 1_X.$$

No self-adjointness or metric orthogonality is assumed unless stated separately.

Assume now that $\mathcal{C}$ is additive and idempotent complete. Choose a splitting

$$X_i \xrightarrow{\iota_i} X \xrightarrow{\pi_i} X_i, \qquad \iota_i \pi_i = p_i, \qquad \pi_i \iota_i = 1_{X_i}.$$

The packet equations imply

$$\pi_i \iota_j = 0 \quad (i \neq j).$$

Definition 3.7 (Packet analysis and synthesis)**.** The packet analysis and synthesis maps are

$$\mathcal{A}_{\mathcal{P}} := (\pi_i)_{i \in I} : X \longrightarrow \bigoplus_{i \in I} X_i,$$

$$\mathcal{S}_{\mathcal{P}} := (\iota_i)_{i \in I} : \bigoplus_{i \in I} X_i \longrightarrow X.$$

Theorem 3.8 (Exact packet-coordinate isomorphism, imported)**.** *For every finite packet system in an additive idempotent-complete category,*

$$\boxed{\mathcal{S}_{\mathcal{P}} \mathcal{A}_{\mathcal{P}} = 1_X, \qquad \mathcal{A}_{\mathcal{P}} \mathcal{S}_{\mathcal{P}} = 1_{\oplus_i X_i}.}$$

Consequently,

$$X \cong \bigoplus_{i \in I} X_i.$$

Proof. The first identity is

$$\mathcal{S}_{\mathcal{P}} \mathcal{A}_{\mathcal{P}} = \sum_i \iota_i \pi_i = \sum_i p_i = 1_X.$$

The (i, j)-entry of the second composite is $\pi_i \iota_j$, which equals 1_{X_i} for $i = j$ and zero otherwise. This is the exact packet-coordinate theorem of Paper I [1]. □

Corollary 3.9 (Packetwise equality)**.** *For parallel morphisms $f, g : Z \to X$,*

$$f = g \quad \Longleftrightarrow \quad \pi_i f = \pi_i g \text{ for every } i.$$

For parallel morphisms $f, g : X \to Z$,

$$f = g \quad \Longleftrightarrow \quad f\iota_i = g\iota_i \text{ for every } i.$$

Proof. One direction is immediate. In the other direction, insert the reconstruction identity on the appropriate side and sum the packetwise equalities. □

Corollary 3.10 (Universal mapping property of packet coordinates)**.** *For every object Z, restriction to the packet objects gives natural bijections*

$$\mathrm{Hom}(X, Z) \cong \prod_i \mathrm{Hom}(X_i, Z), \qquad \mathrm{Hom}(Z, X) \cong \prod_i \mathrm{Hom}(Z, X_i),$$

where the products are finite. Packetwise maps therefore assemble uniquely into a global map.

Definition 3.11 (Mixed transfer blocks)**.** Let $\mathcal{P} = (p_i)$ be a packet system on X, let $\mathcal{Q} = (q_a)$ be a packet system on Y, and choose splittings (ι_i, π_i) and $(\jmath_a, \varpi_a)$. For $f : X \to Y$, define the packet-to-packet map

$$f_{ai} := \varpi_a f \iota_i : X_i \longrightarrow Y_a.$$

Its ambient corner is

$$q_a f p_i = \jmath_a f_{ai} \pi_i.$$

Theorem 3.12 (Exact transfer reconstruction and block composition, imported)**.** *Every morphism $f : X \to Y$ has the exact expansion*

$$\boxed{f = \sum_{a,i} q_a f p_i = \sum_{a,i} \jmath_a f_{ai} \pi_i.}$$

If $g : Y \to Z$ is resolved by a packet system $\mathcal{R} = (r_c)$, then

$$\boxed{(gf)_{ci} = \sum_a g_{ca} f_{ai}.}$$

Proof. Insert the finite packet identities on both sides of f:

$$f = \left(\sum_a q_a \right) f \left(\sum_i p_i \right).$$

For composition, insert $1_Y = \sum_a q_a$ between g and f. This is the transfer-block reconstruction and multiplication law of Paper I [1]. □

Corollary 3.13 (Path expansion)**.** *For a composable string*

$$X_0 \xrightarrow{f_1} X_1 \xrightarrow{f_2} \cdots \xrightarrow{f_m} X_m$$

with packet systems on every X_j, each source-target block of $f_m \cdots f_1$ is the sum over all compatible intermediate packet paths of the corresponding block products.

Example 3.14 (Exact oblique packets)**.** Over any field, set

$$p_1 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \qquad p_2 = \begin{pmatrix} 0 & -1 \\ 0 & 1 \end{pmatrix}.$$

Then $p_1^2 = p_1$, $p_2^2 = p_2$, $p_1 p_2 = p_2 p_1 = 0$, and $p_1 + p_2 = I$. The decomposition of $\mathbb{k}^2$ is exact, although the packets are not orthogonal for the standard inner product. This separates the algebraic reconstruction theorem from later metric statements about energy or best approximation.

3.3 Refinement, selected support, and named residuals

Definition 3.15 (Finite refinement)**.** A packet system $\mathcal{P}^s = (p_j^s)_{j\in I_s}$ refines $\mathcal{P}^r = (p_i^r)_{i\in I_r}$ when there is a surjection

$$\rho_{sr} : I_s \twoheadrightarrow I_r$$

such that

$$p_i^r = \sum_{\rho_{sr}(j)=i} p_j^s.$$

The map ρ_{sr} records the address of each fine packet inside its coarse ancestor.

Proposition 3.16 (Aggregation and transitivity)**.** *A finite aggregation of pairwise annihilating packets is again a packet system. If $t \succeq s \succeq r$, then refinement is transitive with address map*

$$\rho_{tr} = \rho_{sr} \circ \rho_{ts}.$$

Proof. For an aggregated packet, idempotence follows because all cross terms vanish. Distinct aggregates annihilate each other, and the total sum remains the identity. Substituting the t-level expansion into the s-level expansion gives the composite address law. □

Theorem 3.17 (Exact refinement law, imported)**.** *Let $p_i^r = \sum_{\rho_{sr}(j)=i} p_j^s$ and $q_a^u = \sum_{\sigma_{vu}(b)=a} q_b^v$. For every morphism $f : X \to Y$,*

$$\boxed{q_a^u f p_i^r = \sum_{\sigma_{vu}(b)=a} \sum_{\rho_{sr}(j)=i} q_b^v f p_j^s.}$$

Coarsening the fine block product of two composable maps gives the coarse block product.

Proof. Insert the two refinement identities on the left and right of f. For composition, expand both maps at the fine level, multiply, and sum over descendant intermediate packets. The fine intermediate projectors sum to the coarse intermediate projector. This is the exact refinement law of Resolution Theory [2]. □

Definition 3.18 (Support compression and residual)**.** For packet systems $\mathcal{P}$ on X and $\mathcal{Q}$ on Y, let $E \subseteq I_{\mathcal{Q}} \times I_{\mathcal{P}}$. Define

$$C_E(f) := \sum_{(a,i)\in E} q_a f p_i, \qquad R_E(f) := f - C_E(f).$$

The first term is the retained transfer support and the second is its named residual.

Proposition 3.19 (Boolean algebra of block selectors)**.** *As endomorphisms of the $\Bbbk$-module $\mathrm{Hom}(X, Y)$, the selectors satisfy*

$$C_E^2 = C_E, \qquad C_E C_F = C_{E\cap F}, \qquad R_E = I - C_E.$$

In particular,

$$C_E R_E = R_E C_E = 0, \qquad f = C_E(f) + R_E(f).$$

Proof. Applying C_E to a block $q_b f p_j$ retains it exactly when $(b, j) \in E$. Pairwise annihilation removes every mismatched source or target index. The formulas follow block by block. □

Definition 3.20 (Relational support product)**.** For $E \subseteq I_Y \times I_X$ and $F \subseteq I_Z \times I_Y$, define

$$F \circ E := \{(c, i) : \text{there exists } a \in I_Y \text{ with } (c, a) \in F \text{ and } (a, i) \in E\}.$$

Proposition 3.21 (Support propagation under composition)**.** *If $f = C_E(f)$ and $g = C_F(g)$, then*

$$gf = C_{F\circ E}(gf).$$

Thus exact zero patterns propagate by relational composition.

Proof. Every block of gf is a sum $\sum_a g_{ca} f_{ai}$. A nonzero summand can occur only along an intermediate address a belonging to the relational product. □

Proposition 3.22 (Exact residual product identity)**.** *Let*

$$f_E = C_E(f), \quad r_f = R_E(f), \qquad g_F = C_F(g), \quad r_g = R_F(g).$$

Then

$$\boxed{gf - g_F f_E = r_g f_E + g_F r_f + r_g r_f.}$$

If the morphisms lie in a normed operator algebra with a submultiplicative norm, then

$$\|gf - g_F f_E\| \le \|r_g\|\,\|f_E\| + \|g_F\|\,\|r_f\| + \|r_g\|\,\|r_f\|.$$

Proof. Substitute $g = g_F + r_g$ and $f = f_E + r_f$ and expand. The norm estimate is immediate from the triangle inequality and submultiplicativity. □

The identity is a basic certificate-propagation law. It explains exactly which residual terms enter after composing two compressed transfers. No asymptotic claim is required.

Theorem 3.23 (Hilbert–Schmidt optimality, imported)**.** *Assume X, Y are Hilbert spaces, the packet idempotents are orthogonal projectors, and $f : X \to Y$ is Hilbert–Schmidt. Then C_E is the Hilbert–Schmidt orthogonal projection onto the selected block subspace and*

$$\|R_E(f)\|_{\mathrm{HS}}^2 = \sum_{(a,i)\notin E} \|q_a f p_i\|_{\mathrm{HS}}^2.$$

Consequently, $C_E(f)$ is the unique best Hilbert–Schmidt approximation to f among operators supported on E.

Boundary 3.24 (Algebraic exactness is not metric optimality)*.* For arbitrary idempotent packets, reconstruction and residual decomposition remain exact. Orthogonal energy identities and best-approximation statements require a Hilbert structure, orthogonal projectors, and the stated operator class. Those hypotheses are part of the theorem rather than consequences of the packet notation.

3.4 Preservation profiles and proof-support certificates

A realization can preserve one part of the resolved calculus and forget another. The preservation profile is therefore a mathematical interface, not informal metadata.

Definition 3.25 (Constructor universe)**.** Fix a declared universe $\mathfrak{C}$ of constructors and universal properties. Typical entries are

$$0,\ 1,\ \text{finite sums},\ \text{composition},\ \text{scalar action},\ \text{biproducts},$$
$$\text{kernels},\ \text{cokernels},\ \text{quotients},\ \text{tensor products},$$
$$\text{packet idempotents},\ \text{transfer blocks},\ \text{refinements},\ \text{named residuals},$$
$$\text{finite limits},\ \text{finite colimits},\ \text{topological completions},\ \text{adjoints}.$$

The list may be enlarged, but every new entry must specify the exact identity or universal property to be preserved.

Definition 3.26 (Verified preservation profile)**.** A verified preservation profile Π_F for a functor $F : \mathcal{C} \to \mathcal{D}$ consists of a subset of $\mathfrak{C}$ together with a proof or declared comparison isomorphism showing that F preserves each listed constructor. We write $c \in \Pi_F$ only after the corresponding preservation claim has been verified.

Definition 3.27 (Proof-support certificate)**.** A proof-support certificate for a finite proof $\mathcal{P}$ is a finite subset

$$\mathrm{csupp}(\mathcal{P}) \subseteq \mathfrak{C}$$

containing every constructor and universal property used in the proof. The certificate need not be minimal. It must be sufficient and auditable.

Proposition 3.28 (Common profile and composition)**.** *Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{E}$. Every constructor preserved by both functors, with compatible chosen comparison data, is preserved by GF. For a finite diagram of realizations and bridges, the intersection*

$$\Pi_{\mathrm{common}} := \bigcap_F \Pi_F$$

is therefore a valid common profile for every proof transported through the whole diagram.

Proof. Preservation equations and comparison isomorphisms compose. Universal properties remain universal after applying the second functor whenever both preservation witnesses are present. Intersecting the verified lists retains only constructors supported at every stage. □

Theorem 3.29 (Equational realization principle, imported)**.** *Let $F : \mathcal{C} \to \mathcal{D}$ be additive between preadditive categories. Every equality of morphism expressions built from declared generators by finite sums, additive inverses, identities, zero maps, and composition is preserved by F. Scalar multiplication is preserved when the categories are enriched over the same coefficient ring and F is linear. If a proof additionally invokes a chosen biproduct, kernel, cokernel, quotient, tensor product, completion, or another universal construction, the proof transports only when that constructor belongs to Π_F.*

Proof. Functoriality preserves identities and composition. Additivity preserves zero, sums, and additive inverses. Linearity preserves the declared scalar action. Structural induction on the morphism expression transports every equational step. When a step is characterized by a universal property, the verified profile witness identifies the image construction with the corresponding target construction. This is the equational realization principle of Paper I [1]. □

Proposition 3.30 (Profile-safe theorem transport)**.** *Let $\mathcal{P}$ prove a statement T in the resolved syntax, and let F be a realization or bridge. Suppose*

$$\mathrm{csupp}(\mathcal{P}) \subseteq \Pi_F.$$

After every additional domain hypothesis used by T has been verified in the target, $F(T)$ is a valid target theorem. If the inclusion fails, the source proof does not by itself establish the target statement.

Proof. Apply theorem 3.29 to every equational and universal-construction step in the proof certificate. The remaining assumptions are precisely the domain-specific hypotheses, which are checked separately. □

Definition 3.31 (Semantic realization certificate)**.** A semantic realization certificate for transporting a theorem consists of four fields:

(C1) an explicit functor or intertwining construction;

(C2) its verified preservation profile;

(C3) a proof-support certificate for the source theorem together with the inclusion into the profile;

(C4) verification of every target-specific hypothesis and a restatement of the conclusion in native target objects.

Table 2: Typical preservation gates. The right column names the extra obligation rather than asserting it automatically.

Proof step	Required profile entry	Additional target obligation
Finite block identity	finite sums, composition, packet idempotents	verify that the declared packet equations hold
Kernel or cokernel argument	the corresponding universal property	check the target category has the selected construction
Quotient descent	quotient preservation	identify the target quotient and, topologically, its quotient topology
Infinite reconstruction	declared convergence/completion doctrine	specify norm, strong, weak, or unconditional convergence
Tensor reconstruction	tensor preservation and coherence	verify associativity, unit, and any symmetry or dagger structure used
Positivity or best approximation	adjoint, positivity, Hilbert structure	verify orthogonality and the relevant operator ideal
Arithmetic congruence	integral or local realization data	verify valuation, denominator, and reduction hypotheses

Boundary 3.32 (Terminology is not semantic transport). Calling a frequency band, residue class, ramification sector, or latent channel a "packet" is only terminological reuse until the packet idempotents, reconstruction law, transfer blocks, and required preservation maps are supplied. Failure of a profile gate is an explicit open obligation, not permission to infer the desired theorem by analogy.

3.5 The finite additive-Karoubi resolved envelope

The full profile-indexed envelope remains a multi-doctrine problem. The finite linear, additive, and split-idempotent core can be constructed exactly.

Definition 3.33 (Linear grammar category)**.** For a finite linear resolved presentation $\mathbb{K}_{\mathrm{lin}} = (Q, I, \mathsf{P}, \mathsf{Ref})$, let

$$\mathcal{A}_{\mathbb{K}} := \Bbbk Q / I.$$

This is the $\Bbbk$-linear category whose objects are the vertices of Q and whose morphisms are $\Bbbk$-linear combinations of typed paths modulo the declared relations.

Definition 3.34 (Finite additive hull)**.** The category $\mathrm{Add}(\mathcal{A}_{\mathbb{K}})$ has finite tuples

$$A = (A_1, \dots, A_m)$$

as objects. A morphism $A \to B = (B_1, \dots, B_n)$ is an $n \times m$ matrix with entries in $\mathcal{A}_{\mathbb{K}}(A_j, B_i)$. Composition is matrix multiplication using composition and addition in $\mathcal{A}_{\mathbb{K}}$.

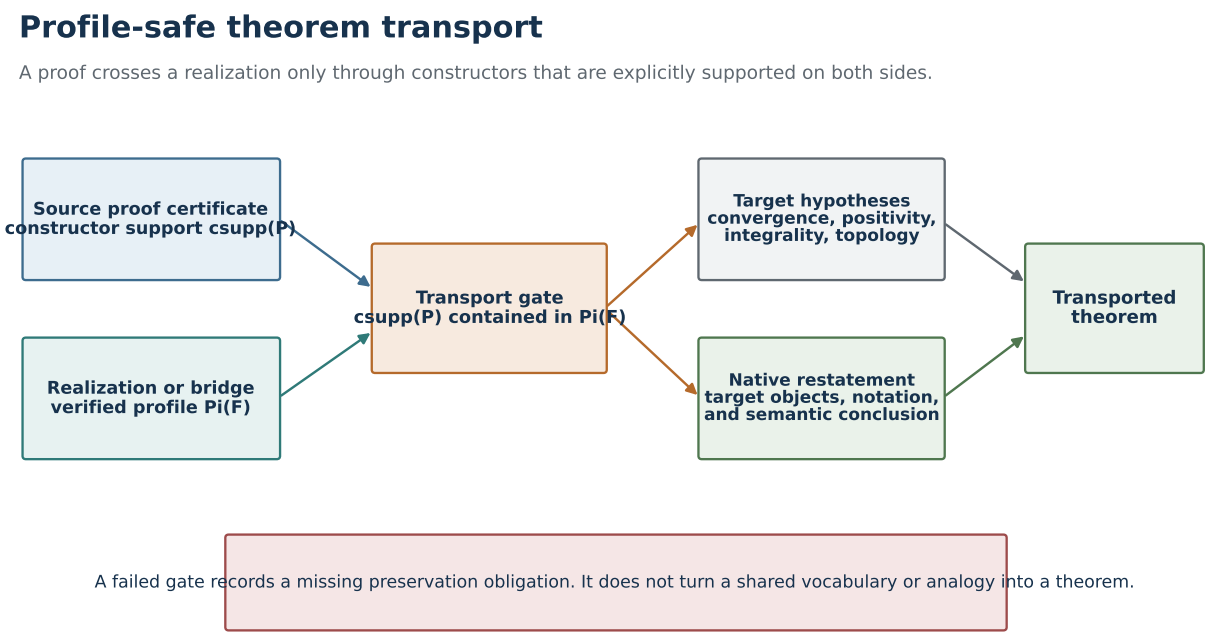


Figure 2: The theorem-transport gate. A shared vocabulary is not enough: the constructors used by the proof must lie inside the verified profile, and target-specific hypotheses must still be checked.

Definition 3.35 (Karoubi envelope)**.** The Karoubi envelope $\mathrm{Kar}(\mathrm{Add}(\mathcal{A}_{\mathbb{K}}))$ has pairs (A,e), where $e : A \to A$ is idempotent, as objects. A morphism

$$f : (A,e) \longrightarrow (B,d)$$

is a morphism $f : A \to B$ satisfying

$$f = dfe.$$

The identity on (A,e) is e.

Theorem 3.36 (Finite additive-Karoubi resolved envelope)**.** *The category*

$$\boxed{\mathrm{Syn}_{\mathrm{ak}}(\mathbb{K}_{\mathrm{lin}}) := \mathrm{Kar}\big(\mathrm{Add}(\Bbbk Q/I)\big)}$$

is small, additive, idempotent complete, and $\Bbbk$-linear. For every additive idempotent-complete $\Bbbk$-linear category $\mathcal{C}$, restriction to the generating quiver induces an equivalence

$$\mathrm{Fun}^{\oplus}_{\Bbbk}\big(\mathrm{Syn}_{\mathrm{ak}}(\mathbb{K}_{\mathrm{lin}}),\mathcal{C}\big) \simeq \mathrm{Mod}_{\mathrm{ak}}(\mathbb{K}_{\mathrm{lin}},\mathcal{C}),$$

where the right side is the category of typed realizations satisfying all declared relations and carrying every declared packet idempotent to a split idempotent.

Proof. There are four universal stages.

Free typed stage. The free $\Bbbk$-linear category on Q is universal for assigning objects to vertices and morphisms to arrows.

Relation stage. Quotienting by I imposes exactly the declared linear, packet, and refinement equations. A functor out of $\Bbbk Q/I$ is therefore the same as a typed realization satisfying those equations.

Additive stage. Every $\Bbbk$-linear functor from $\Bbbk Q/I$ into an additive category extends, uniquely up to canonical natural isomorphism, to an additive functor from the finite additive hull by sending a tuple to the corresponding biproduct and a matrix to its induced biproduct map.

Idempotent-splitting stage. Every additive functor into an idempotent-complete category extends, again uniquely up to canonical natural isomorphism, across the Karoubi envelope by sending (A, e) to the chosen image of the split idempotent $F(e)$. Morphisms satisfying $f = dfe$ descend to the corresponding split summands.

Composing these universal properties yields the displayed equivalence. Smallness is preserved because only finite tuples and idempotents in hom sets of a small category are adjoined. □

Proposition 3.37 (Fully faithful grammar embedding)**.** *The canonical composite*

$$\mathcal{A}_{\mathbb{K}} \longrightarrow \mathrm{Add}(\mathcal{A}_{\mathbb{K}}) \longrightarrow \mathrm{Kar}(\mathrm{Add}(\mathcal{A}_{\mathbb{K}}))$$

is fully faithful. Thus the envelope adds finite biproducts and split summands without identifying two previously distinct grammar morphisms.

Proof. The first functor sends an object to the one-term tuple and preserves its hom set exactly. The second sends A to $(A, 1_A)$; its morphism condition is $f = 1_B f 1_A$, which imposes no additional relation. □

Corollary 3.38 (Declared packets become literal objects)**.** *Every declared idempotent $p : X \to X$ defines an object (X, p) of $\mathrm{Syn}_{\mathrm{ak}}(\mathbb{K}_{\mathrm{lin}})$. A complete packet family (p_i) gives the canonical isomorphism*

$$(X, 1_X) \cong \bigoplus_i (X, p_i).$$

Corollary 3.39 (Linear resolved strata as model categories)**.** *Every additive split-idempotent realization stratum of $\mathbb{K}_{\mathrm{lin}}$ is a category of profile-preserving functors out of $\mathrm{Syn}_{\mathrm{ak}}(\mathbb{K}_{\mathrm{lin}})$.*

Example 3.40 (One operator and primary packets)**.** Take one sort, one endomorphism T, and one relation $\mu(T) = 0$. The linear grammar algebra is $\Bbbk[x]/(\mu)$. When

$$\mu = \mu_1 \cdots \mu_r$$

is a product of pairwise comaximal factors, the Chinese remainder idempotents define packet objects for the primary summands. The envelope splits those idempotents formally. Over a realization where the same factorization remains valid, the packet objects become the corresponding invariant primary sectors.

Remark 3.41 (Universal does not mean presentation-minimal)*.* The envelope is universal for realizations of the declared presentation. It does not provide a canonical smallest quiver, decide equivalence of arbitrary presentations, or identify a globally minimal packet system. Those are recognition or minimization problems inside particular strata.

3.6 Profile-indexed extensions and the exact boundary of Foundation layer

Definition 3.42 (Profile-indexed resolved envelope)**.** For a preservation profile Π, a profile-indexed resolved envelope $\mathrm{Syn}_{\Pi}(\mathbb{K})$, when it exists, is a universal source satisfying

$$\mathrm{Mod}_{\Pi}(\mathbb{K}, \mathcal{C}) \simeq \mathrm{Fun}_{\Pi}(\mathrm{Syn}_{\Pi}(\mathbb{K}), \mathcal{C})$$

for every target category $\mathcal{C}$ carrying the structures named by Π. The equivalence must preserve the chosen comparison data, not only the underlying functors.

Table 3: Profile extensions beyond the proved additive-Karoubi core.

Doctrine	Additional data required	Boundary
Nonlinear / operadic	colored finite-arity operations, symmetric actions, substitution, coherence	available locally in the nonlinear packet calculus, not freely present in every stratum
Banach / topological	topology, boundedness, convergence mode, completion and quotient-topology rules	algebraic packet sums do not choose a topology
C^* / W^*	involution, norm, positivity, closure, multiplier or weak completion	faithful corner realization does not automatically preserve an independent tensor product
Countable packets	strong, norm, weak, or unconditional convergence with boundedness controls	finite identities do not determine the infinite convergence doctrine
Monoidal / Tannakian	tensor product, unit, coherence, exact faithful fiber functor when reconstruction is claimed	ordinary faithful representation is weaker than monoidal reconstruction
Arithmetic / local	integral models, reductions, valuations, completeness, residue and lift data	denominator and reduction hypotheses remain domain-specific

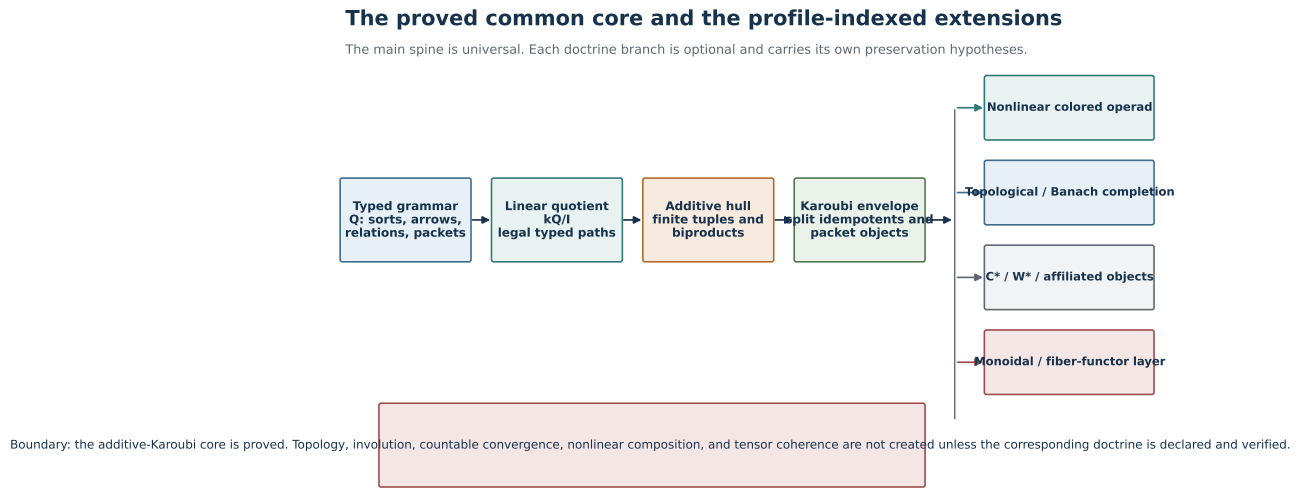


Figure 3: The proved common envelope and the optional profile branches. The branches should be added only for the strata that actually preserve them.

Boundary 3.43 (No single overpowered envelope). The existence of $\mathrm{Syn}_{\mathrm{ak}}$ does not prove the existence of one universal category simultaneously free, complete, C^*, W^*, nonlinear-operadic, monoidal, and arithmetically integral in a way preserved by every target. The mathematically safer target is a family of profile-indexed envelopes or a doctrine-valued construction with explicit forgetful and comparison maps.

Section result. The foundational language is now explicit at four levels: typed grammar semantics, exact finite packet and residual calculus, proof-support-controlled theorem transport, and the universal additive-Karoubi envelope. The nonlinear, analytic, operator-algebraic, monoidal, and arithmetic envelopes remain separate profile obligations rather than hidden assumptions.

4 Strata, normalized charts, typed bridges, and obstruction objects

The common envelope developed in Section 3 is deliberately weaker than most concrete mathematical domains. The next layer records how that envelope is realized, normalized, compared, and sometimes only partially remembered. The central organizing rule is that a chart is not specified by a list of examples. It is specified by a category of models, a preservation profile, a semantic functor, a normalized classification target, and explicit overlap data.

4.1 Stratum passports and normalized model categories

Definition 4.1 (Profiled stratum passport)**.** A profiled Komorebi stratum is a record

$$\mathbf{S}_\alpha = (\mathcal{C}_\alpha, \Pi_\alpha, \mathcal{M}_\alpha, \mathcal{N}_\alpha, \mathrm{Sem}_\alpha, \mathcal{I}_\alpha),$$

with the following fields.

(i) $\mathcal{C}_\alpha$ is the ambient semantic category in which the realized objects live.

(ii) Π_α is the verified preservation profile of the realization.

(iii) $\mathcal{M}_\alpha$ is a full replete category of admissible models of the declared resolved syntax.

(iv) $\mathcal{N}_\alpha \subseteq \mathcal{M}_\alpha$ is a full replete normalized subcategory on which the intended inverse or classification theorem is stated.

(v)
$$\mathrm{Sem}_\alpha : \mathcal{M}_\alpha \longrightarrow \mathcal{O}_\alpha$$
is the native semantics functor.

(vi)
$$\mathcal{I}_\alpha : \mathrm{Core}(\mathcal{N}_\alpha) \longrightarrow \mathcal{D}_\alpha$$
is a declared invariant functor, when one has been constructed. Its codomain $\mathcal{D}_\alpha$ and classification target are part of the record.

The original four-field stratum $(\mathcal{C}_\alpha, \Pi_\alpha, \mathcal{M}_\alpha, \mathrm{Sem}_\alpha)$ remains the underlying realization datum. The normalized and invariant fields are additional assembly data rather than automatic consequences.

Definition 4.2 (Normalization witness)**.** A normalization witness on a full subcategory $\mathcal{U}_\alpha \subseteq \mathcal{M}_\alpha$ consists of a functor

$$N_\alpha : \mathcal{U}_\alpha \longrightarrow \mathcal{N}_\alpha$$

and a natural isomorphism

$$\nu_\alpha : \mathrm{Sem}_\alpha N_\alpha \Longrightarrow \mathrm{Sem}_\alpha |_{\mathcal{U}_\alpha}.$$

It is *classification-normalizing* when every object of $\mathcal{U}_\alpha$ is semantically isomorphic to its image, and two normalized objects with isomorphic semantics are isomorphic in $\mathcal{N}_\alpha$. A normalization may be partial, domain-restricted, or defined only after auxiliary choices. Those restrictions belong to the passport.

Definition 4.3 (Complete invariant on a declared target)**.** An invariant $\mathcal{I}_\alpha$ is complete on $\mathcal{N}_\alpha$ when

$$\mathcal{I}_\alpha(X) \cong \mathcal{I}_\alpha(Y) \quad \Longleftrightarrow \quad X \cong Y$$

for all $X, Y \in \mathcal{N}_\alpha$. This statement classifies the objects of $\mathcal{N}_\alpha$ up to the isomorphisms allowed there. It does not classify arbitrary presentations outside that subcategory.

Proposition 4.4 (Semantic conservativity of a normalized chart)**.** *If N_α is classification-normalizing, then*

$$\mathrm{Sem}_\alpha |_{\mathcal{N}_\alpha}$$

reflects isomorphisms. Consequently, a complete invariant on $\mathcal{N}_\alpha$ can be transported through a reversible overlap only after the overlap is shown to preserve the normalization and reflect the relevant isomorphisms.

Proof. Let $X, Y \in \mathcal{N}_\alpha$ and suppose their semantic objects are isomorphic. The second clause of definition 4.2 gives $X \cong Y$. The final statement follows because completeness is an assertion about isomorphism classes, not merely about equality of numerical outputs. □

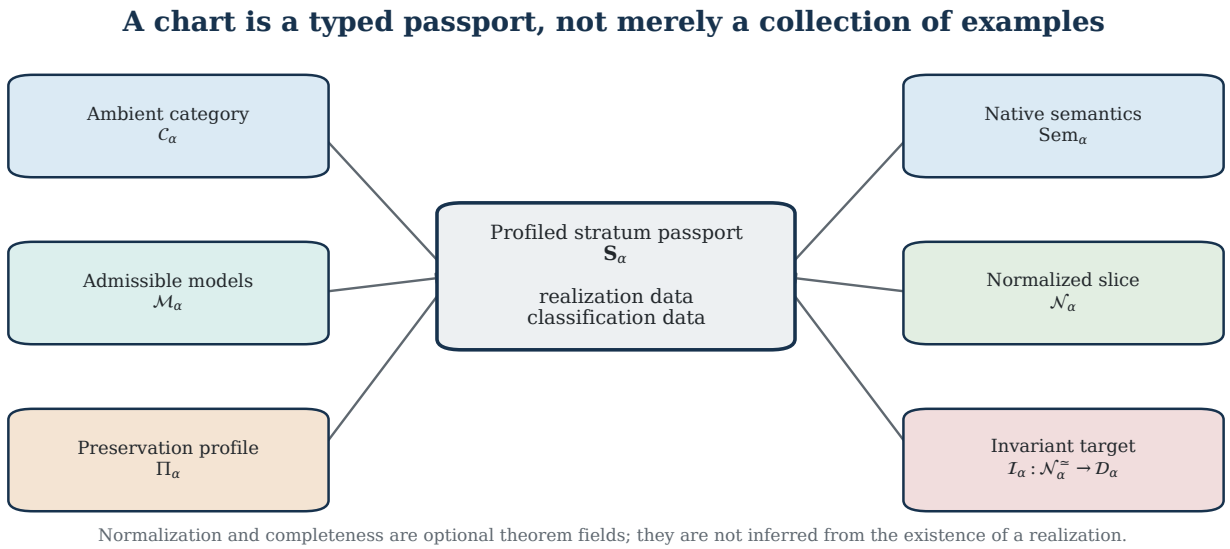


Figure 4: A stratum passport separates the ambient category, admissible models, preservation profile, semantics, normalization, and invariant target. Omitting any one of these fields makes cross-stratum classification claims ambiguous.

The present program uses the following principal strata. The normalized fields are intentionally narrower than the ambient strata. Several rows have exact slices but no canonical normalization for the whole broad class [7, 6, 8, 9].

Table 4: Current stratum passports at the level supported by the audited Library. A dash means that no whole-stratum normalization theorem is asserted.

Stratum	Native objects	Native morphisms	Normalized or exact slice	Exact coordinate or invariant already available	Remaining boundary
Finite linear and C-finite	finite recurrences, rational OGFs, finite linear states	intertwiners, recurrence changes, similarity	minimal recurrence, reduced OGF, reachable-observable state	Hankel rank, reduced denominator degree, minimal state dimension, torsion–Fourier subsequence profile	no classification of arbitrary nonminimal or mixed-grammar presentations
Holonomic	P-recursive sequences, D-finite functions, differential modules	coefficient/differential intertwiners	fixed operator convention and selected local data	effective P-recursive–D-finite translation; finite annihilator modules	minimal multisection operators, singularity orbits, monodromy aliasing, and cancellation signatures
Digital	DFAO, finite digit kernels, regular linear representations	state maps and radix-compatible intertwiners	minimal DFAO for fixed radix and reading direction	unique minimal DFAO up to state renaming; exact finite k-kernel compilation	no all-moduli packet profile or general linear-state minimality for k-regular systems
Algebraic	selected algebraic branches and packet polynomial systems	branch-preserving polynomial maps	a declared branch with a fixed defining system	exact packet polynomial realization and synthesis	minimal system dimension, branch-collision data, and effective packet-degree bounds
Nonlinear/tree	packet jets, colored operations, rooted-tree systems	operadic substitutions and smooth intertwiners	marked polynomial fixed points; positive simply generated tree families	packet Faà di Bruno law, diagonal profile compression, SCC cyclic core, arity-gcd support and dominant-orbit signature	no global nonlinear normal form, mixed-direction universal compression, or coordinate-free minimal system dimension
Product and multiplicative	filtered Euler products, prime-power seed systems	exponent and local-factor maps	fixed Euler basis or declared multiplicative class	unique Euler exponents in the fixed basis; exact reconstruction from prime-power seeds	no canonical factorization across arbitrary product grammars and no general additive–multiplicative equivalence
Local, Mahler, and polyphase	Newton–Mahler coordinates, affine branches, functional masks, local states	affine restriction, refinement, Cartier and polyphase maps	fixed local chart and fixed modulus/radix conventions	unique Mahler coordinates, exact affine-polyphase transfer, exact Cartier automata for rational masks	no canonical global local normal form and no unrestricted bridge between functional and Newton–Mahler theories
Galois and Tannakian	semilinear packets, fixed-field charts, tensor representations	equivariant maps, induction/restriction, tensor morphisms	finite separable semisimple descent or neutral Tannakian fiber functor	Hilbert 90 descent, Brauer obstruction, subgroup/fixed-field classification, tensor reconstruction	modular and wild cases, and monoidal reconstruction without a verified fiber functor

Boundary 4.5 (Multiple valid placements). A semantic object may occur in several strata. Catalan numbers, for example, may be algebraic, hypergeometric, P-recursive, and tree-generated. The atlas records these as distinct presentations with certified bridges. It does not force one placement to erase the others.

4.2 Representation groupoids, gauge levels, and the directed transformation category

For every stratum, the invertible part is the maximal subgroupoid

$$\mathfrak{Rep}_\alpha(\mathbb{K}) = \mathrm{Core}(\mathcal{M}_\alpha).$$

This core records changes of presentation that preserve all structure required by the overlap. Quotients, truncations, completions, specializations, differential annihilator maps, and compression maps generally remain outside it.

Definition 4.6 (Finite coordinate action groupoid)**.** Fix a dimension vector $\mathbf{d} = (d_X)_{X\in\mathsf{S}}$. Let

$$G_{\mathbf{d}} = \prod_{X\in\mathsf{S}} GL_{d_X}(\mathbb{k})$$

act on the finite-dimensional representation space $\mathrm{Rep}_{\mathbf{d}}(\mathbb{K})$ by base change. The action groupoid

$$[\mathrm{Rep}_{\mathbf{d}}(\mathbb{K})/G_{\mathbf{d}}]$$

has representations as objects and coordinate changes carrying one representation to another as morphisms.

Proposition 4.7 (Coordinate groupoid realizes isomorphism)**.** *Two objects of* $\mathrm{Rep}_{\mathbf{d}}(\mathbb{K})$ *are isomorphic as typed realizations exactly when they lie in the same* $G_{\mathbf{d}}$*-orbit. The stabilizer of a point is its coordinate automorphism group.*

Proof. A typed isomorphism is a family of invertible linear maps on the sort spaces intertwining every generating operation. This is precisely an element of $G_{\mathbf{d}}$ carrying one coordinate tuple to the other. Elements fixing the tuple are exactly the typed automorphisms. □

Three symmetry levels must remain separate.

Table 5: Three noninterchangeable symmetry notions.

Symmetry level	Acting object	What it changes or preserves	Extra hypothesis
Gauge symmetry	coordinate-change group $G_{\mathbf{d}}$ or a local gauge family	changes presentation coordinates while preserving the represented object	an explicitly typed action on the chart
Intrinsic symmetry	$\mathrm{Aut}_{\mathcal{M}_\alpha}(M)$	automorphisms of one resolved object inside the stratum	none beyond the native category
Reconstructed symmetry	$\mathrm{Aut}^{\otimes}(\omega)$	tensor automorphisms of a fiber functor and the corresponding affine group scheme	faithful exact tensor fiber functor

Definition 4.8 (Typed directed transformation category)**.** The directed transformation category $\mathfrak{Dir}(\mathbb{K})$ has pairs (α, M), with $M \in \mathcal{M}_\alpha$, as objects. A morphism

$$(\alpha, M) \longrightarrow (\beta, N)$$

is an admitted bridge instance equipped with a bridge passport as in definition 4.10. Composition is defined only when the codomain profile of the first bridge satisfies the domain requirements of the second. The union of the representation groupoids embeds as the wide invertible subcategory.

Boundary 4.9 (Invertibility is evidence, not typography). Drawing a two-headed arrow does not make a bridge reversible. A bridge enters the representation groupoid only after an inverse, quasi-inverse, or recognition theorem has been proved on a named overlap category.

4.3 Bridge passports, bridge classes, and information-loss profiles

Definition 4.10 (Bridge passport)**.** A certified bridge from stratum α to stratum β is a record

$$\mathfrak{b} = (\mathcal{U}_\mathfrak{b}, \mathcal{V}_\mathfrak{b}, B_\mathfrak{b}, \tau_\mathfrak{b}, \Pi_\mathfrak{b}^{\mathrm{pres}}, \Pi_\mathfrak{b}^{\mathrm{refl}}, \Lambda_\mathfrak{b}, \nu_\mathfrak{b}, \chi_\mathfrak{b}),$$

where:

(i) $\mathcal{U}_\mathfrak{b} \subseteq \mathcal{M}_\alpha$ and $\mathcal{V}_\mathfrak{b} \subseteq \mathcal{M}_\beta$ are the declared domain and codomain subcategories;

(ii) $B_\mathfrak{b} : \mathcal{U}_\mathfrak{b} \to \mathcal{V}_\mathfrak{b}$ is the functor;

(iii) $\tau_\mathfrak{b}$ is its bridge class;

(iv) $\Pi_\mathfrak{b}^{\mathrm{pres}}$ records preserved constructors and universal properties;

(v) $\Pi_\mathfrak{b}^{\mathrm{refl}}$ records properties or isomorphisms reflected by the bridge;

(vi) $\Lambda_\mathfrak{b}$ is the information-loss profile;

(vii) $\nu_\mathfrak{b}$ is the semantic comparison map;

(viii) $\chi_\mathfrak{b}$ records normalization, residual, and certificate compatibility.

A passport may state that no inverse or no certificate transport is known. Missing data remain missing rather than being inferred from analogy.

The permitted bridge classes follow the work program in the controlling handoff.

Definition 4.11 (Information-loss profile)**.** For a functor $B : \mathcal{U} \to \mathcal{V}$, its information-loss profile contains at least:

$$\begin{aligned} \mathrm{MorKer}_B(X, Y) &= \{(f, g) \in \mathrm{Hom}_\mathcal{U}(X, Y)^2 : Bf = Bg\}, \\ \mathrm{Coll}_B &= \{(X, Y) \in \mathrm{Ob}(\mathcal{U})^2 : BX \cong BY\}, \\ \mathrm{EssIm}(B) &\subseteq \mathcal{V}, \end{aligned}$$

together with the declared constructors, packet labels, residuals, certificates, branch data, topology, or arithmetic structure not preserved by B. The first component records morphism collapse, the second object collapse, and the third which target objects actually arise.

Definition 4.12 (Faithfulness, conservativity, and reversibility)**.** A bridge is:

Table 6: Bridge classes used by the concrete atlas.

Class	Required evidence	Interpretation
Equivalence	quasi-inverse and coherent unit/counit	reversible overlap of presentation groupoids
Fully faithful inclusion	bijections on hom-sets and full image description	no morphism loss inside the image, but possibly a smaller chart
Conservative forgetful functor	reflected isomorphisms	object equivalence remains detectable although structure may be forgotten
Information-losing directional functor	explicit semantic and loss statement	quotient, projection, annihilator, reduction, or compression
Constructor	proof that source data generate a target object	one-way realization, often with nonunique inverse data
Completion	topology and universal completion property	adds limits or closes a domain; reversibility requires a recognition theorem
Partially defined bridge	exact domain predicate	valid only on a named locus, branch, modulus, or regularity class
No certified bridge	none	no arrow is inserted into the trusted atlas

(i) *morphism-faithful* when every $\mathrm{MorKer}_B(X, Y)$ is diagonal;

(ii) *conservative* when Bf invertible implies that f is invertible;

(iii) *classification-faithful* or *essentially injective on objects* when $BX \cong BY$ implies $X \cong Y$ on the declared domain;

(iv) *lossless on its image* when it is fully faithful;

(v) *reversible* when it is an equivalence onto the declared overlap.

These conditions answer different questions and should not be conflated.

Proposition 4.13 (Composition law for bridge loss)**.** *Let*

$$\mathcal{U} \xrightarrow{B} \mathcal{V} \xrightarrow{C} \mathcal{W}$$

be certified bridges.

(i) *Every constructor or universal property preserved by both B and C is preserved by CB.*

(ii)
$$\mathrm{MorKer}_B(X, Y) \subseteq \mathrm{MorKer}_{CB}(X, Y).$$

(iii)
$$\mathrm{Coll}_B \subseteq \mathrm{Coll}_{CB}\,.$$

(iv)
$$\mathrm{EssIm}(CB) \subseteq \mathrm{EssIm}(C).$$

(v) *If both bridges are conservative, then their composite is conservative. If both are classification-faithful, then their composite is classification-faithful. If both are equivalences, then their composite is an equivalence.*

Thus information forgotten at an earlier stage cannot be recovered by merely composing more forgetful bridges, although a later bridge may attach auxiliary data through a separate constructor.

Proof. Functoriality proves the preservation claim. If $Bf = Bg$, then $CBf = CBg$, proving the second inclusion. If $BX \cong BY$, applying C gives $CBX \cong CBY$, proving the third. Every object in the essential image of CB is visibly in the essential image of C. For conservativity, if CBf is invertible then Bf is invertible because C is conservative, and then f is invertible because B is conservative. For classification-faithfulness, $CBX \cong CBY$ implies $BX \cong BY$ and then $X \cong Y$. Composition of equivalences is standard. □

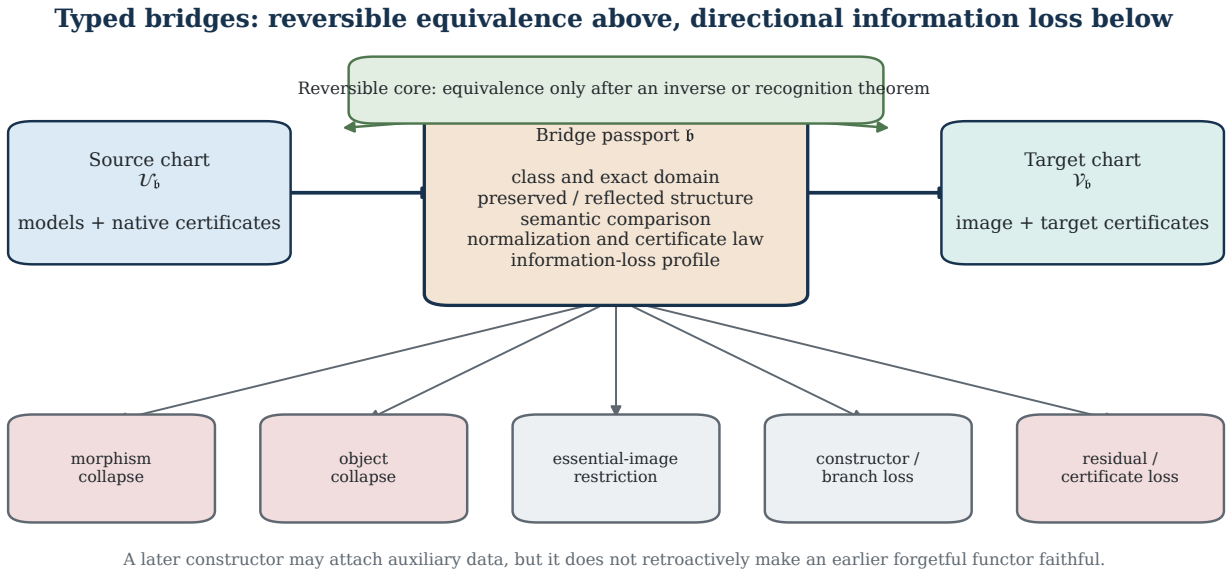


Figure 5: A bridge passport separates reversible equivalence from directional transformation and records the dimensions of information loss. The reversible core is a theorem-level subcategory, not an informal interpretation of the whole bridge.

Three typing cautions are especially important.

1. A selected algebraic branch can produce a D-finite annihilator, but the operator may forget the selected branch and its minimal polynomial. The bridge is directional unless that data are retained.
2. A rational functional-Mahler mask over $\mathbf{F}_p$ produces a Cartier representation and finite automaton under the stated hypotheses, but this does not identify it with a Newton–Mahler interpolation chart [8, 7].
3. A faithful Hilbert-space corner representation realizes packet projectors and transfers, but it does not automatically preserve a tensor product strongly enough for Tannakian reconstruction [2, 6].

4.4 Overlap categories and bridge-law certificates

Definition 4.14 (Typed overlap)**.** A reversible overlap between strata α and β consists of full replete subcategories

$$\mathcal{M}_{\alpha\beta} \subseteq \mathcal{M}_\alpha, \qquad \mathcal{M}_{\beta\alpha} \subseteq \mathcal{M}_\beta,$$

an equivalence

$$\Phi_{\alpha\beta} : \mathcal{M}_{\alpha\beta} \xrightarrow{\sim} \mathcal{M}_{\beta\alpha},$$

and a semantic natural isomorphism

$$\sigma_{\alpha\beta} : J_{\alpha\beta}\,\mathrm{Sem}_\alpha \Longrightarrow \mathrm{Sem}_\beta\,\Phi_{\alpha\beta}$$

for an explicitly declared comparison functor $J_{\alpha\beta}$ between the observable doctrines. An overlap is *normalized* when the equivalence restricts to the selected normalized subcategories.

Definition 4.15 (Bridge-law certificate)**.** A bridge-law certificate for $B : \mathcal{U} \to \mathcal{V}$ records commuting or naturally commuting diagrams for each structure used downstream:

(i) native semantics;

(ii) packet analysis and synthesis;

(iii) refinement and coarsening maps;

(iv) selected-support compression and named residuals;

(v) proof or computation certificates;

(vi) normalization and invariant comparison maps.

For example, packet naturality requires comparison maps under which

$$B(p_{Y,b} f p_{X,a}) \cong p'_{B(Y),b}\,B(f)\,p'_{B(X),a},$$

and refinement naturality requires the image of each coarse packet to equal the declared sum of its refined descendants.

Proposition 4.16 (Bridge-law transport)**.** *Let $\mathcal{P}$ be a proof whose constructor support lies in the preserved part of a bridge-law certificate. Then the image proof is valid in the target chart. If the proof concludes equality of normalized invariants, that equality descends across the bridge only when the normalization and invariant comparison maps are also included in the certificate.*

Proof. The first statement is the equational realization principle of Paper I applied with the bridge-specific preservation profile. The second statement follows because normalization and invariant functors are additional constructors in the proof diagram; they cannot be transported when their comparison cells are absent [1]. □

On triple overlaps, pairwise equivalences are not enough. One needs comparison cells

$$\eta_{\alpha\beta\gamma} : \Phi_{\beta\gamma}\Phi_{\alpha\beta} \Longrightarrow \Phi_{\alpha\gamma}.$$

For four charts, there are two composites from $\Phi_{\gamma\delta}\Phi_{\beta\gamma}\Phi_{\alpha\beta}$ to $\Phi_{\alpha\delta}$. Their ratio is the higher coherence defect.

Definition 4.17 (Quadruple-overlap coherence defect)**.** Write horizontal composition of natural transformations as $*$. Define

$$\Omega_{\alpha\beta\gamma\delta} = \Big[\eta_{\alpha\gamma\delta} \circ (\mathrm{id} * \eta_{\alpha\beta\gamma})\Big] \circ \Big[\eta_{\alpha\beta\delta} \circ (\eta_{\beta\gamma\delta} * \mathrm{id})\Big]^{-1}.$$

It is a natural automorphism of $\Phi_{\alpha\delta}$ on the quadruple overlap. The atlas is coherent there exactly when

$$\Omega_{\alpha\beta\gamma\delta} = \mathrm{id}.$$

Proposition 4.18 (Two elementary coherence mechanisms)**.** *The coherence defect vanishes in either of the following situations.*

(i) *All overlap functors compose strictly and every comparison cell is the identity.*

(ii) *Between each pair of relevant normalized objects there is at most one admissible comparison isomorphism compatible with semantics.*

Proof. The first case is immediate. In the second case, the two sides of the coherence equation are admissible comparison isomorphisms with the same source and target. Uniqueness forces them to agree. This is the mechanism used later in the normalized C-finite three-chart atlas. □

Boundary 4.19 (Prestack versus stack). Typed overlap functors and coherent comparison cells define atlas or prestack data. The word *stack* additionally requires effective descent: compatible local objects and morphisms must arise from global ones in the declared category.

4.5 Reversible cores inside directional bridges

Directional maps can contain a canonical reversible overlap without being globally invertible. Split packet analysis is the basic model.

Theorem 4.20 (Split bridge rectification)**.** *Let $\mathcal{A}$ be an additive idempotent-complete category and suppose*

$$X \xrightarrow{A} Y \xrightarrow{L} X$$

satisfies $LA = \mathrm{id}_X$. Put $E = AL$. Then:

(i) $E^2 = E$;

(ii) $Y \cong \operatorname{im} E \oplus \operatorname{im}(1 - E)$;

(iii) $A : X \to \operatorname{im} E$ *is an isomorphism with inverse induced by L;*

(iv) $L(1 - E) = 0$, *so* $\operatorname{im}(1 - E)$ *is a named residual invisible to the decoder;*

(v) *the directional two-chart system becomes reversible after restricting the packet chart to the consistent core* $\operatorname{im} E$.

In a module or vector-space realization, $\operatorname{im}(1 - E) = \ker L$.

Proof. The identity $E^2 = A(LA)L = AL = E$ proves the first statement. Idempotent completeness splits E, producing the direct sum. The image of A lies in $\operatorname{im} E$, and $LA = \mathrm{id}_X$ while $AL = E$ acts as the identity on $\operatorname{im} E$, so the restricted maps are inverse. Finally,

$$L(1 - E) = L - LAL = L - L = 0.$$

For modules, every $y \in \ker L$ satisfies $(1 - E)y = y$, while every vector in $\operatorname{im}(1 - E)$ is killed by L. □

Corollary 4.21 (Exact meaning of a decoded packet core)**.** *A left inverse proves exact recovery on the consistent packet core. It does not prove that every ambient packet vector is consistent, that the residual is small, or that the packet representation is compressed. Those are separate geometric, analytic, or coding claims.*

The integral Mahler packet theorem is a concrete instance: the analysis matrix has an integer left inverse, the consistency projector is AL, and the residual is the complementary packet component. The same decoder survives every base change, while rank loss of individual residue channels remains visible after reduction [5].

4.6 Finite holonomy and the maximal path-flat core

Finite graph transport gives the sharpest reversible gluing model in the Library. Let $\Gamma = (V, E)$ be a finite connected graph. Attach a finite-dimensional $\Bbbk$-vector space F_v to each vertex and an isomorphism $T_e : F_u \to F_v$ to each oriented edge $e : u \to v$, with $T_{\bar{e}} = T_e^{-1}$. Choose a root o and spanning tree S. If $G_v : F_o \to F_v$ is tree transport, then every non-tree edge $e : u \to v$ has fundamental holonomy

$$H_e = G_v^{-1} T_e G_u \in GL(F_o).$$

Define

$$C_o = \bigcap_{e \notin S} \ker(H_e - I) = \mathrm{Fix}\langle H_e : e \notin S \rangle, \qquad C_v = G_v C_o.$$

Theorem 4.22 (Maximal flat core and global sections)**.** *Under the preceding hypotheses [3, 5]:*

(i) *evaluation at the root identifies the space of global parallel sections with C_o;*

(ii) *$(C_v)_v$ is path-flat, meaning that all path transports with the same endpoints agree on the core;*

(iii) *it is the unique greatest path-flat subfamily;*

(iv) *it is independent of the chosen spanning tree and equals the common fixed space of all closed-path holonomies;*

(v) *for*

$$\Delta_S : F_o \longrightarrow \bigoplus_{e \notin S} F_o, \qquad \Delta_S x = ((H_e - I)x)_{e \notin S},$$

one has

$$C_o = \ker \Delta_S, \qquad \dim C_o = \dim F_o - \operatorname{rank} \Delta_S;$$

(vi) *the full fibers glue to one global frame exactly when every fundamental holonomy is the identity;*

(vii) *a gauge change conjugates the holonomies and transports the core, so $\dim C_o$ is gauge-invariant.*

Proof. For $x \in C_o$, set $x_v = G_v x$. Tree edges satisfy the parallel relation by construction, and a non-tree edge satisfies it because $H_e x = x$. Conversely, any global section is determined along the tree by its root value; every remaining edge forces that value to be fixed by the corresponding H_e. This proves the section classification. Closed paths are words in fundamental cycles, which gives path-flatness and tree independence. Any path-flat family must be fixed by every fundamental cycle and is therefore contained in $(C_v)_v$. The rank formula is rank–nullity. Full flatness and gauge covariance follow from spanning-tree normalization and conjugation of loop products. □

Corollary 4.23 (Global section versus global frame)**.** *Nonidentity holonomy can leave a nonzero maximal core. Therefore the existence of some global compatible vectors is weaker than the existence of one global frame on the full fibers. The former is measured by $\dim C_o$; the latter requires $C_o = F_o$.*

4.7 Relative cokernel obstructions and loss under coarsening

Holonomy addresses compatibility of invertible transports. Correction problems have a different canonical object.

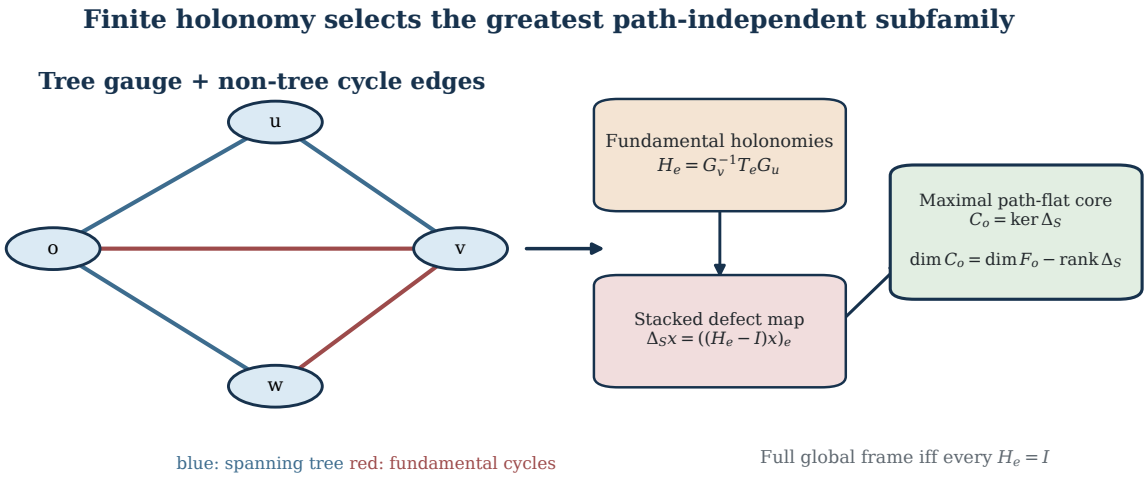


Figure 6: A spanning tree removes local gauge redundancy. Each remaining edge contributes a fundamental holonomy. Their common fixed space is the maximal path-flat core; the stacked defect map Δ_S measures the lost dimension.

Definition 4.24 (Linear correction datum)**.** A linear correction datum is a map

$$D : U \longrightarrow V$$

and a defect $r \in V$. A correction is $u \in U$ satisfying $Du = -r$. The relative obstruction class is

$$\mathrm{Ob}_D(r) = [r] \in \operatorname{coker} D = V / \operatorname{im} D.$$

Theorem 4.25 (Cokernel criterion, universality, and naturality)**.** *For a correction datum (D, r) [3]:*

(i) *a correction exists if and only if* $\mathrm{Ob}_D(r) = 0$*;*

(ii) *every linear obstruction test $\varphi : V \to W$ satisfying $\varphi D = 0$ factors uniquely through $V \to \operatorname{coker} D$;*

(iii) *for every commuting square*

$$\begin{array}{ccc} U & \xrightarrow{D} & V \\ \downarrow a & & \downarrow b \\ U' & \xrightarrow{D'} & V', \end{array} \qquad bD = D'a,$$

there is a unique induced map

$$\overline{b} : \operatorname{coker} D \longrightarrow \operatorname{coker} D'$$

with

$$\overline{b}\ \mathrm{Ob}_D(r) = \mathrm{Ob}_{D'}(br).$$

Proof. The class vanishes exactly when $r \in \operatorname{im} D$, which is equivalent to solvability of $Du = -r$. A map annihilating $\operatorname{im} D$ factors uniquely through the quotient. In the square, the quotient map for D', composed with b, vanishes on $\operatorname{im} D$, so the same universal property produces $\overline{b}$. □

Proposition 4.26 (Packet coordinates on the obstruction quotient)**.** *Suppose (Q_a) is a complete finite family of pairwise annihilating idempotents on V and*

$$Q_a(\operatorname{im} D) \subseteq \operatorname{im} D$$

for every a. Then the Q_a descend to quotient idempotents $\overline{Q}_a$ on coker D*,*

$$\operatorname{coker} D \cong \bigoplus_a \overline{Q}_a(\operatorname{coker} D), \qquad \overline{Q}_a \operatorname{Ob}_D(r) = [Q_a r].$$

If U and V already split into matching packet summands and D is block diagonal, this becomes

$$\operatorname{coker} D \cong \bigoplus_a \operatorname{coker} D_a.$$

Off-diagonal correction blocks can couple packet defects, so the block-diagonal hypothesis cannot be omitted.

Proof. The invariance of $\operatorname{im} D$ makes $[v] \mapsto [Q_a v]$ well defined. Idempotence, annihilation, and reconstruction descend to the quotient. In the block-diagonal case the image of D is the direct sum of the images of the restrictions. □

A coarse representation can erase a nonzero fine obstruction. The strengthened Paper III gives the exact bookkeeping device.

Theorem 4.27 (Relative obstruction exact sequence)**.** *Suppose the diagram*

$$\begin{array}{ccccccccc} 0 & \to & U_0 & \to & U_f & \to & U_c & \to & 0 \\ & & \downarrow D_0 & & \downarrow D_f & & \downarrow D_c & & \\ 0 & \to & V_0 & \to & V_f & \to & V_c & \to & 0 \end{array}$$

commutes and has exact rows. Then there is a natural exact sequence

$$\begin{aligned} 0 \longrightarrow \ker D_0 \longrightarrow \ker D_f \longrightarrow \ker D_c \xrightarrow{\delta} \operatorname{coker} D_0 \\ \longrightarrow \operatorname{coker} D_f \longrightarrow \operatorname{coker} D_c \longrightarrow 0. \end{aligned}$$

If a fine obstruction maps to zero in coker D_c*, then it comes from* coker D_0*, the obstruction sector removed by coarsening. Coarse vanishing therefore does not imply fine vanishing.*

Proof. This is the direct kernel–cokernel sequence proved in the strengthened Paper III. The connecting map lifts an element of $\ker D_c$ to U_f, applies D_f, identifies the result in V_0, and takes its class modulo $\operatorname{im} D_0$. Exactness verifies independence of the lift and identifies each successive kernel with the preceding image [3, 4]. □

Example 4.28 (A fine obstruction erased by summation)**.** Take all correction spaces to be zero,

$$V_f = \Bbbk^2, \qquad V_c = \Bbbk, \qquad p(x, y) = x + y.$$

The fine defect $(1, -1)$ is nonzero in $\operatorname{coker} D_f = V_f$, but its coarse image is zero. It lies entirely in

$$V_0 = \ker p = \operatorname{span}\{(1, -1)\}.$$

This is cancellation by coarsening, not correction of the fine defect.

Definition 4.29 (Mapping cone)**.** For a cochain map $D : C^\bullet \to E^\bullet$, define

$$\mathrm{Cone}(D)^n = E^n \oplus C^{n+1}, \qquad d_{\mathrm{Cone}}(e,c) = (d_E e + Dc, -d_C c).$$

Proposition 4.30 (Cohomological correction criterion)**.** *For a cocycle $e \in E^n$, the class of $(e,0)$ in $H^n(\mathrm{Cone}(D))$ vanishes exactly when $[e]$ lies in the image of $H^n(D)$. When the complexes are concentrated in degree zero,*

$$H^0(\mathrm{Cone}(D)) \cong \operatorname{coker} D.$$

Proof. The class of $(e,0)$ is a boundary exactly when there are $e' \in E^{n-1}$ and $c \in C^n$ with

$$e = d_E e' + Dc, \qquad d_C c = 0.$$

This is equivalent to $[e] = H^n(D)[c]$. Concentration in degree zero reduces the cone quotient to $E^0/\operatorname{im} D$. □

Boundary 4.31 (The correction map is problem data). The cokernel is canonical relative to D. The theory does not yet supply one choice-free rule that constructs the correct D for every mathematical problem and every stratum. Calling $[r] \in \operatorname{coker} D$ canonical must not hide this dependence.

4.8 Semilinear effectivity, projective descent, and the Brauer class

Galois descent gives a second exact local-to-global model. Its linear and projective cases separate effectivity from obstruction sharply.

Theorem 4.32 (Effective vector descent by Hilbert 90)**.** *Let L/K be finite Galois with group G. Suppose*

$$a_{\sigma\tau} = a_\sigma\, \sigma(a_\tau), \qquad a_\sigma \in GL_n(L),$$

defines a semilinear packet cocycle. Then there is $B \in GL_n(L)$ such that

$$a_\sigma = B^{-1}\sigma(B).$$

The twisted fixed packet

$$W = (L^n)^{G,\star}$$

is an n-dimensional K-vector space and

$$W \otimes_K L \xrightarrow{\sim} L^n.$$

Thus every finite linear semilinear atlas of this form is effective [6].

Theorem 4.33 (Brauer obstruction to projective packet descent)**.** *Projective semilinear descent data define a class*

$$\alpha \in H^1(G, PGL_n(L)).$$

The connecting map for

$$1 \longrightarrow L^\times \longrightarrow GL_n(L) \longrightarrow PGL_n(L) \longrightarrow 1$$

produces

$$\delta(\alpha) \in H^2(G, L^\times) = \mathrm{Br}(L/K).$$

The projective atlas lifts to linear semilinear descent data exactly when $\delta(\alpha) = 0$. When the class is nonzero, the descended object is naturally a module over a central simple K-algebra split by L, rather than an ordinary K-vector space [6].

Proof. This is the nonabelian cohomology exact sequence together with nonabelian Hilbert 90. Vanishing of the connecting class gives a $GL_n(L)$-valued cocycle; theorem 4.32 then trivializes it and constructs the descended vector space. A nonzero class is the classical relative Brauer obstruction. □

Corollary 4.34 (Projective compatibility is not linear effectivity)**.** *A projective atlas may satisfy every visible overlap equation in PGL_n and still have no global vector-space realization. Projective cocycle coherence and vanishing Brauer class are distinct conditions.*

The tensor reconstruction problem lies beyond ordinary corner realization.

Theorem 4.35 (Tannakian boundary)**.** *Let $\mathcal{T}$ be a neutral Tannakian category over k and*

$$\omega : \mathcal{T} \longrightarrow \mathrm{Vect}_k$$

a faithful exact tensor fiber functor. Then

$$\mathcal{G} = \mathrm{Aut}^{\otimes}(\omega)$$

is an affine group scheme and

$$\mathcal{T} \simeq \mathrm{Rep}_k(\mathcal{G})$$

as tensor categories. A faithful projector-corner representation without verified exact strong monoidal structure does not cross this boundary [6, 2].

4.9 Higher coherence defects and typed obstruction comparison

The four obstruction families now admit a precise comparison. Each starts from different data and answers a different vanishing question.

Table 7: Typed obstruction comparison.

Failure mode	Required datum	Obstruction object	Vanishing consequence	Nonzero meaning
Finite transport holonomy	connected graph, invertible edge transports, chosen structure group	fundamental cycle holonomies or the conjugacy class of $\pi_1(\Gamma) \to G$	one global frame on the full fibers; on a fixed subspace, path-independent transport	path dependence; possibly a smaller nonzero fixed core
Relative correction failure	correction map $D : U \to V$ and defect r	$[r] \in \operatorname{coker} D$, or a mapping-cone cohomology class	an allowed correction exists	the defect is outside the declared correction image
Projective Galois descent	projective semilinear cocycle over L/K	$\delta(\alpha) \in H^2(G, L^\times) = \mathrm{Br}(L/K)$	lift to linear descent, then effective vector descent by Hilbert 90	descent to a twisted central-simple-algebra object rather than an ordinary vector space
Higher atlas coherence	pairwise equivalences and comparison 2-cells	$\Omega_{\alpha\beta\gamma\delta}$, a natural automorphism on a quadruple overlap	pseudofunctor coherence of the atlas	pairwise bridges use incompatible conventions or incomplete comparison data

Proposition 4.36 (Typed comparison principle for obstruction objects)**.** *An obstruction object transports from one problem to another only after a compatibility morphism for the underlying data has been supplied:*

(i) *commuting squares of correction maps induce maps on cokernels;*

(ii) *group homomorphisms or intertwiners compatible with edge transports induce maps on holonomy data and fixed cores;*

(iii) *morphisms of the relevant central extensions induce maps on cohomological descent obstructions;*

(iv) *modifications of atlas pseudofunctors transport higher coherence defects.*

Absent such typed data, there is no canonical reduction of one obstruction family to another.

Proof. Each item follows by applying the functorial construction defining the obstruction: quotient universality for cokernels, conjugation or intertwining for path products, functoriality of cohomology for Brauer classes, and naturality of horizontal and vertical composition for coherence cells. Since the input categories and codomains differ, no comparison exists without an explicit map between the inputs. □

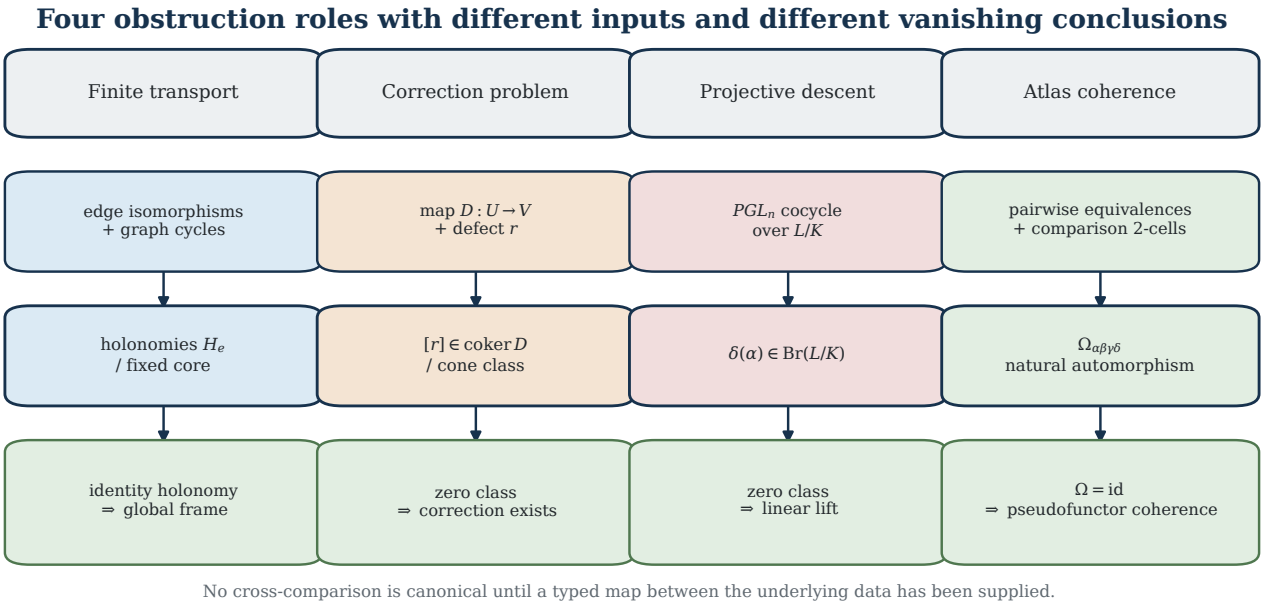


Figure 7: Four obstruction pipelines. The word *obstruction* names a role, not one universal scalar. The required input datum determines both the obstruction object and the exact consequence of vanishing.

Section result. The atlas layer now has explicit stratum passports, normalized subcategories, representation groupoids, a directed transformation category, bridge classes, information-loss profiles, overlap and bridge-law certificates, and a concrete higher-coherence defect. The four obstruction theories are stated with their own data and vanishing criteria. Finite holonomy supplies a maximal path-flat core, split bridges expose reversible consistent cores inside directional maps, relative cokernels separate correction from coarsening loss, and Galois theory separates Hilbert-90 effectivity from Brauer and Tannakian boundaries. The general heterogeneous atlas is still not declared effective until its actual bridge passports and coherence cells are verified.

Table 8: Status distribution in the 75-component theorem concordance.

Classification	Components
proved internally	52
classical input	12
proved conditionally	2
framework only	3
conjectural	6
Total	**75**

Table 9: Concordance coverage by mathematical domain.

Domain	Components
Foundations	17
Groupoids and descent	7
Obstruction theory	4
Galois stratum	4
Linear stratum	3
Holonomic stratum	1
Digital stratum	2
Algebraic stratum	1
Transform stratum	1
Product stratum	1
Multiplicative stratum	1
Local stratum	3
Functional Mahler stratum	3
Assembly	9
Semantic quotient calculus	7
Bridge loss calculus	2
Carrier and observer change	5
Convex and admissibility change	3
Doctrine comparison	1

5 The theorem concordance and edition control

5.1 Audit result

The concordance contains 56 components. Thirty-eight are proved internally, eight are classical theorems imported and reinterpreted, two are proved conditionally, two are definition or framework rows, and six remain conjectural. The new foundations row records the finite profiled Universal Lane Transport Theorem. The conjectural entries remain concentrated in the cross-stratum assembly layer. The local packet, obstruction, Galois, sequence, and Mahler components are substantially more complete than an older open-question ledger suggests.

5.2 Material supersession decisions

Three corrections are mathematically consequential.

Relative obstruction theory. Older Paper III editions leave a broad universal packet-obstruction principle conjectural. The strengthened edition replaces the finite-projector formulation by the relative class $[r] \in \operatorname{coker} D$, with vanishing, naturality, packet decomposition, quotient projectors, relative exact sequences, and mapping-cone formulations proved for declared correction data [3, 4]. The unresolved problem is not the quotient construction itself. It is a choice-free mechanism assigning the appropriate correction datum to arbitrary problems and strata.

Paper I edition policy. The complete symbolic core remains the theorem-ID and proof-status spine. The expanded clarified edition strengthens exposition and hypotheses without replacing the stable record identifiers [1].

Core-Transfer status. Addendum 04 replaces the isolated $m = 6, d = 3$ boundary by the exact composite-modulus rank law, sharp integral packet minimum, universal decoder, and all-degree stabilization [5]. The former example is now a sharp negative control inside a general theorem.

5.3 Existing bridges and exact invariant slices

The strongest source edition already contains the following exact or effective bridges on declared domains:

- constant recurrence $\leftrightarrow$ rational ordinary generating function $\leftrightarrow$ finite linear state;
- P-recursive $\leftrightarrow$ D-finite coefficient translation;
- finite digit realization $\leftrightarrow$ finite k-kernel;
- rational functional-Mahler mask $\rightarrow$ Cartier representation $\rightarrow$ finite automaton;
- finite-mask Cartier transitions $\leftrightarrow$ carry matrices;
- selected algebraic branch $\rightarrow$ packet polynomial system;
- invertible transform $\leftrightarrow$ transported packet resolution;
- Euler-product exponent coordinates $\leftrightarrow$ logarithmic-derivative recurrence data;
- finite modular state $\rightarrow$ local Newton-Mahler interpolation;
- local Mahler coordinates $\leftrightarrow$ affine polyphase branches;
- finite resolved C^*-system $\rightarrow$ faithful Hilbert packet realization.

The persistent C-finite stratum has the strongest complete transfer invariant. Active spectral and torsion-phase data determine the exact minimal recurrence, reduced denominator, Hankel rank, state dimension, rank drops, and periodic packet profile of every arithmetic subsequence [7]. Restricted exact coordinate systems also exist for fixed-radix automata, filtered Euler products, multiplicative prime-power seeds, local Mahler charts, finite semisimple Galois descent, and finite seed lattices. None of these restricted results is a single all-stratum classifier.

6 Finite profiled resolved atlas assembly

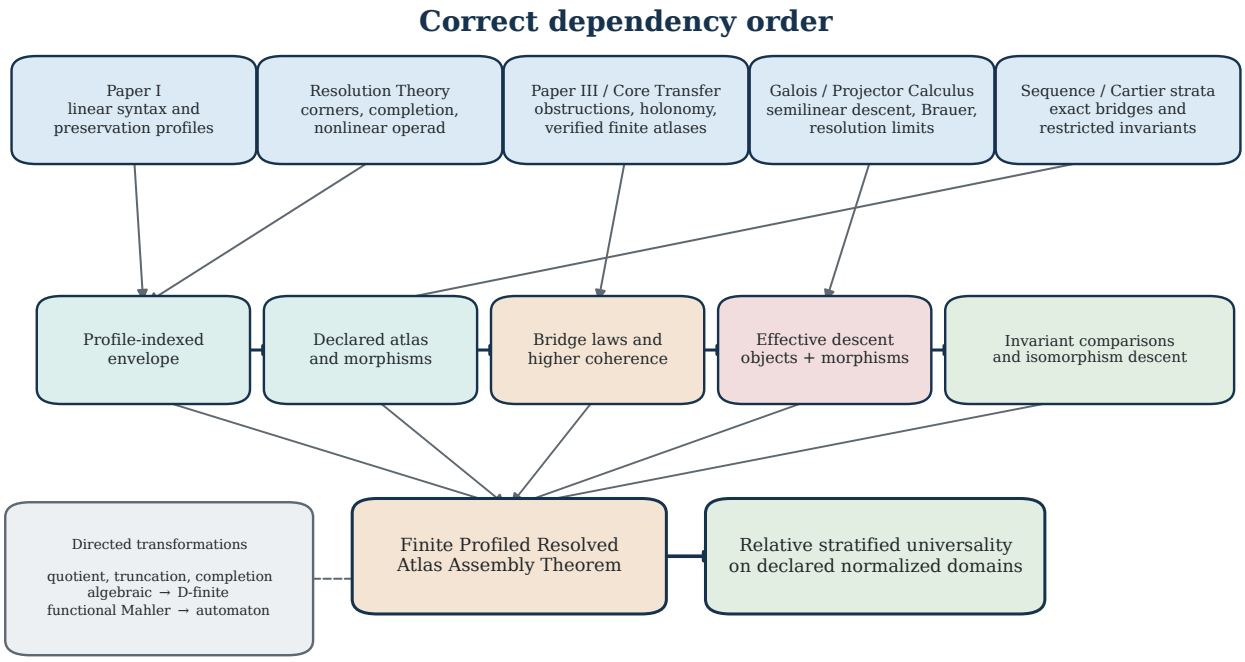


Figure 8: The corrected dependency graph. The theorem concordance and local results feed the assembly theorem, but the heterogeneous conclusion still depends on explicit atlas construction, coherence, effectivity, and isomorphism descent.

6.1 Finite atlases and descent data

Definition 6.1 (Finite profiled resolved atlas)**.** Fix a small resolved theory $\mathbb{K}$. A finite profiled resolved atlas consists of:

(A1) a finite indexing category $\check{\mathcal{A}}$, retained through the overlap degrees needed for the cocycle law;

(A2) a pseudofunctor
$$\mathfrak{R} : \check{\mathcal{A}}^{\mathrm{op}} \to \mathbf{Cat};$$

(A3) a declared preservation profile Π_U on every chart and overlap category $\mathfrak{R}(U)$;

(A4) coherent equivalences on the reversible overlaps and separately typed directional transformations;

(A5) a pseudonatural semantics
$$\mathrm{Sem} : \mathfrak{R} \Rightarrow \mathfrak{O}$$
into a common observable doctrine;

(A6) optional pseudonatural invariant families with explicitly typed comparison cells.

Definition 6.2 (Descent category)**.** The category $\mathrm{Desc}(\mathfrak{R})$ has pseudocones over $\mathfrak{R}$ as objects and modifications as morphisms. Concretely, an object is a compatible family
$$(M_\alpha, \theta_{\alpha\beta})$$
with invertible overlap comparisons satisfying the cocycle law. A morphism is a compatible family of chart morphisms.

6.2 The assembly theorem

Theorem 6.3 (Finite profiled resolved atlas assembly)**.** *Let*
$$\mathfrak{R} : \check{\mathcal{A}}^{\mathrm{op}} \to \mathbf{Cat}$$
be a finite profiled resolved atlas. Assume the chart and overlap categories are small, or locally small in one fixed universe, the reversible transitions and comparison cells satisfy the pseudofunctor coherence equations, and every construction used below belongs to the verified profile of every participating chart and bridge. Then:

(i) $\mathrm{Desc}(\mathfrak{R})$ *exists and has the universal property of the finite pseudolimit of* $\mathfrak{R}$*.*

(ii) *There is a canonical equivalence*
$$\mathrm{Core}\big(\mathrm{Desc}(\mathfrak{R})\big) \simeq \mathrm{Desc}\big(\mathfrak{R}^{\simeq}\big).$$

(iii) *Every pseudonatural transformation* $T : \mathfrak{R} \Rightarrow \mathfrak{S}$ *induces a functor*
$$\mathrm{Desc}(T) : \mathrm{Desc}(\mathfrak{R}) \to \mathrm{Desc}(\mathfrak{S}).$$
If every T_U *is an equivalence, then* $\mathrm{Desc}(T)$ *is an equivalence. A pointwise noninvertible* T *remains directional.*

(iv) *If two global morphism expressions use only constructors in the common verified preservation profile and their equality is proved in the resolved syntax, then, after every domain-specific hypothesis is verified locally, their realizations are equal in* $\mathrm{Desc}(\mathfrak{R})$*.*

(v) *A coherent pseudonatural invariant family* $I : \mathfrak{R} \Rightarrow \mathfrak{V}$ *induces*

$$\mathrm{Desc}(I) : \mathrm{Desc}(\mathfrak{R}) \to \mathrm{Desc}(\mathfrak{V}).$$

(vi) *If every local invariant is complete on its normalized chart and the groupoid of local invariant-compatible isomorphisms has effective descent, then* $\mathrm{Desc}(I)$ *is complete on the corresponding global normalized category.*

If the semantic pseudonatural transformation is effective for objects and morphisms, then $\mathrm{Desc}(\mathrm{Sem})$ *identifies descent data with the declared global semantic objects.*

Proof. Objects and morphisms of $\mathrm{Desc}(\mathfrak{R})$ are finite compatible families. Identities and composition are defined componentwise. The pseudofunctor coherence equations ensure that componentwise composition remains compatible. This explicit construction has the universal property of pseudocones, proving (i).

A descent morphism is invertible exactly when each component is invertible. The componentwise inverses satisfy the overlap equations obtained by inverting the original equations. This proves (ii).

A pseudonatural transformation acts on chart objects, overlap comparisons, and chart morphisms componentwise. Its coherence cells supply the target cocycle law. Pointwise quasi-inverses and their modifications descend, proving (iii).

For (iv), the equational realization principle transports the proof to each chart because every constructor used by the proof is preserved. Bridge naturality makes the local equalities compatible on overlaps. Equality in the descent category is componentwise. The argument creates no missing analytic or arithmetic hypotheses.

Statement (v) is the same pseudonatural-transformation construction applied to the invariant family. For (vi), an isomorphism between descended invariants gives local invariant matches. Local completeness supplies local object isomorphisms. The added isomorphism-effectivity hypothesis is exactly what glues those local isomorphisms to a global one. Without it, a holonomy class, a relative cokernel class, a Brauer class, or a higher coherence defect may obstruct the gluing.

The final assertion is the definition of semantic effectivity applied to the induced descent functor. □

Corollary 6.4 (No automatic promotion of directional bridges)**.** *A directional bridge contributes a global functor only when it is pseudonatural with respect to the chosen atlas. It contributes an equivalence only when the chartwise bridge is an equivalence on the named overlap. No quotient, completion, or compression becomes reversible merely because it appears in the same diagram.*

Corollary 6.5 (Global theorem transport)**.** *A theorem proved from packet reconstruction, refinement, residual calculus, or another preserved constructor descends globally only when every chart and bridge used by the proof carries that constructor in its verified preservation profile.*

Remark 6.6 (Exact status). Theorem 6.3 is a proved conditional theorem. It does not prove that the full proposed Komorebi atlas satisfies the hypotheses. The remaining work is the verification of the actual profiles, overlap categories, naturality cells, higher coherence, and effectivity.

Part II

Certified native strata

7 Verified finite Mahler atlas models

Two exact finite models already realize substantial pieces of the assembly theorem. They remain inside the affine-Mahler and packet-analysis setting, but they verify nontrivial coherence and effectivity mechanisms.

7.1 Affine-Mahler chart groupoid

Fix $d \geq 0$. Over characteristic zero, the affine Mahler coefficient maps $A_{m,s}^{(d)}$ are invertible and satisfy

$$A_{n,t}^{(d)} A_{m,s}^{(d)} = A_{mn,mt+s}^{(d)}.$$

For coprime m, n, the two paths around every declared Chinese-remainder diamond are equal [5]. Hence every finite subdiagram generated by these charts is a coherent finite chart groupoid with trivial holonomy on those diamonds.

Proposition 7.1 (Rational affine-Mahler chart groupoid)**.** *Every finite characteristic-zero diagram generated by the invertible affine Mahler charts and the declared CRT comparison cells defines a strict finite groupoid atlas. Its path transport depends only on the represented affine substitution, and every declared CRT cycle has identity holonomy.*

7.2 Integral split packet descent

Let

$$A = P_{m,S}^{(d)} : \mathbb{Z}^{d+1} \to \mathbb{Z}^{|S|(d+1)}$$

be a stacked affine-Mahler packet analysis map. Whenever an integer left inverse L exists,

$$LA = I, \qquad \Pi = AL, \qquad \Pi^2 = \Pi,$$

and

$$\mathbb{Z}^{|S|(d+1)} = \operatorname{im} A \oplus \ker L.$$

Thus the source chart and the normalized packet core $\operatorname{im} A$ are equivalent, while $\ker L$ is a named residual rather than discarded information.

Addendum 04 proves that the prefix

$$S_* = \{0, 1, \ldots, L_{\mathbb{Z}}(m, d) - 1\}$$

with

$$L_{\mathbb{Z}}(m, d) = \min(d + 1, B(m)), \qquad B(m) = \max\big(\{1\} \cup \{p^{v_p(m)} : p \mid m\}\big),$$

always admits an integer decoder and is cardinality-minimal [5].

Proposition 7.2 (Integral split Mahler packet descent)**.** *The analysis-decoder pair above gives effective object and morphism descent between the source chart and the normalized packet-core chart. The split identity survives every base change. The residual summand* $\ker L$ *is functorial under maps intertwining both* A *and* L*.*

Proof. The identities $LA = I$ and $\Pi = AL$ show that A identifies the source with $\operatorname{im} A$, with inverse $L|_{\operatorname{im} A}$. For any packet vector y,

$$y = ALy + (y - ALy),$$

where $ALy \in \operatorname{im} A$ and $y - ALy \in \ker L$. The intersection is zero because $Ax \in \ker L$ implies $x = LAx = 0$. Tensoring $LA = I$ with any coefficient ring preserves the split identity. Intertwining maps preserve the image and kernel decompositions. □

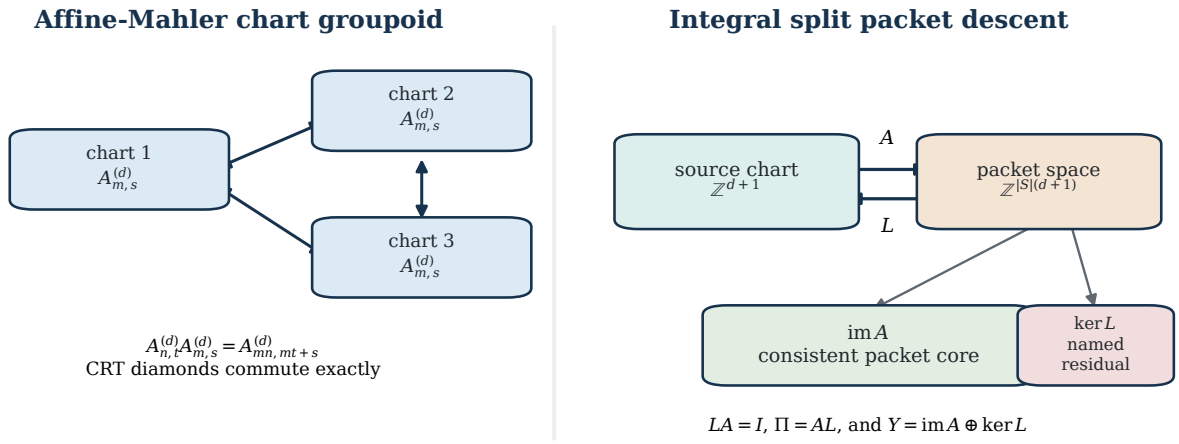


Figure 9: Two verified finite atlas mechanisms. The first is reversible and coherent. The second is split analysis-synthesis with a retained residual.

8 The normalized C-finite three-chart atlas

This section completes the first cross-presentation model in the theorem dependency graph. It does not combine unrelated broad strata, but it does prove a genuine three-chart effective descent theorem with nontrivial morphisms on the state side.

8.1 Persistent C-finite semantics

Fix a field K. A nonzero C-finite sequence $a = (a_n)_{n\geq 0}$ is *persistent* when its minimal characteristic polynomial has nonzero constant term. Equivalently, its active characteristic roots are nonzero, or its minimal state transition is invertible [7].

Let $\mathbf{Seq}_K^{\mathrm{pers}}$ be the discrete groupoid of persistent C-finite sequences over K.

8.2 The recurrence chart

An object of $\mathbf{Rec}_K^{\min}$ is a pair

$$(\chi, \mathbf{a}_0),$$

where

$$\chi(x) = x^d - c_{d-1}x^{d-1} - \cdots - c_1 x - c_0, \qquad c_0 \neq 0,$$

is the monic minimal recurrence polynomial and $\mathbf{a}_0 = (a_0, \ldots, a_{d-1})$ is the initial vector. It generates the sequence

$$a_{n+d} = c_{d-1}a_{n+d-1} + \cdots + c_1 a_{n+1} + c_0 a_n.$$

Because the recurrence is normalized and minimal, two objects representing the same sequence are equal. Thus $\mathbf{Rec}_K^{\min}$ is a discrete groupoid.

8.3 The rational OGF chart

An object of $\mathbf{Rat}_K^{\mathrm{red}}$ is a reduced rational power series

$$A(z) = \frac{P(z)}{Q(z)}, \qquad Q(0) = 1, \qquad \gcd(P, Q) = 1, \qquad \deg P < \deg Q,$$

whose reciprocal characteristic polynomial has nonzero constant term. Again the normalized reduced form is unique, so $\mathbf{Rat}_K^{\mathrm{red}}$ is discrete.

The recurrence-to-OGF bridge is explicit. From

$$\chi(x) = x^d - c_{d-1}x^{d-1} - \cdots - c_0$$

set

$$Q(z) = z^d\chi(z^{-1}) = 1 - c_{d-1}z - \cdots - c_0z^d.$$

The numerator is the degree-$< d$ truncation of $Q(z)\sum_{n\geq 0} a_n z^n$. Conversely, clearing Q recovers the recurrence and the initial vector. Minimality is equivalent to reducedness.

8.4 The minimal state chart

An object of $\mathbf{State}_K^{\mathrm{min}}$ is a triple

$$(C, v, \lambda), \qquad C \in M_d(K), \quad v \in K^d, \quad \lambda \in (K^d)^*,$$

with C invertible, producing

$$a_n = \lambda C^n v,$$

and satisfying reachability and observability. A morphism

$$S : (C, v, \lambda) \longrightarrow (C', v', \lambda')$$

is a similarity satisfying

$$C' = SCS^{-1}, \qquad v' = Sv, \qquad \lambda' = \lambda S^{-1}.$$

The semantic functor sends a state to its coefficient sequence.

The state-to-OGF bridge is the resolvent

$$A(z) = \lambda(I - zC)^{-1}v.$$

The reduced denominator has degree equal to the minimal state dimension, which equals the Hankel rank. Conversely, a reduced rational OGF has a companion realization. Reducedness makes the companion realization reachable, while the companion output is observable. Any two minimal realizations of the same sequence are uniquely similar through their common shift action on the Hankel column space [7].

8.5 Effective descent across the three charts

Theorem 8.1 (Normalized C-finite three-chart atlas)**.** *For every field K, the semantic functors fit into equivalences of groupoids*

$$\mathbf{Rec}_K^{\mathrm{min}} \simeq \mathbf{Rat}_K^{\mathrm{red}} \simeq \mathbf{State}_K^{\mathrm{min}} \simeq \mathbf{Seq}_K^{\mathrm{pers}}.$$

The recurrence-rational bridge is a strict inverse pair. The rational-state bridge is an equivalence with companion quasi-inverse. The comparison isomorphism on the state-side composite is the unique similarity between minimal realizations. Consequently:

(i) *the three charts and their pairwise overlaps form a coherent finite profiled atlas;*

(ii) *the triple-overlap comparison satisfies the cocycle law by uniqueness of minimal similarity;*

(iii) *descent for objects and morphisms is effective;*

(iv) *the descent category is equivalent to* $\mathbf{Seq}_K^{\mathrm{pers}}$.

Proof. The equivalence between normalized minimal recurrences and reduced rational ordinary generating functions follows from the explicit reciprocal-denominator formula and clearing of the denominator. These constructions are mutually inverse.

A minimal state gives a rational resolvent. Conversely, a reduced rational series gives a companion state. Minimal linear dimension equals Hankel rank, and the reduced denominator degree has the same value. Hence the companion state is minimal. Any two minimal states representing the same sequence identify with the forward shift on the Hankel column space and are therefore similar. The similarity is unique because a reachable state is spanned by $C^n v$, so an intertwiner fixing the input vector is determined on a spanning set.

Choose the recurrence-to-state bridge to be the companion realization obtained through the rational chart. Then one triangular composite is strict. On the state side, the composite through the rational and recurrence charts returns the companion realization of the same sequence. The unique minimal similarity supplies the comparison isomorphism. Any two composites around a triple overlap are comparison isomorphisms between the same minimal realizations, so uniqueness forces the cocycle law.

Each semantic functor is an equivalence to the same discrete groupoid of persistent sequences. Compatible local objects and morphisms therefore descend uniquely up to the stated similarity, proving effectivity and the final equivalence. □

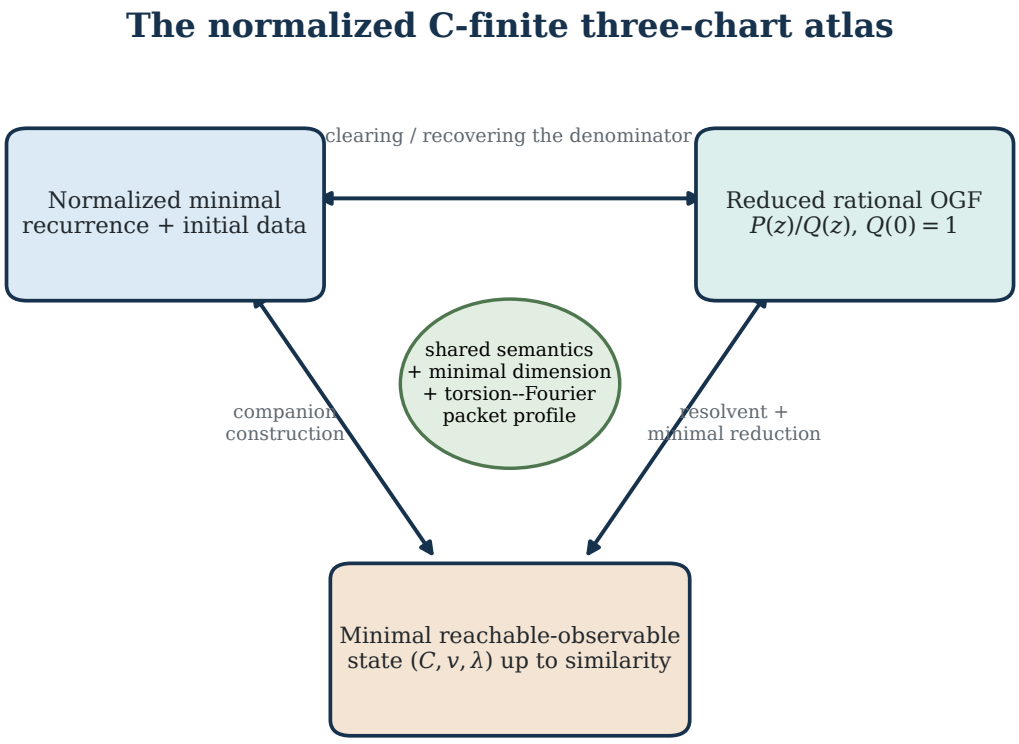


Figure 10: The normalized C-finite atlas. The recurrence and rational charts are canonical normal forms. The state chart has nontrivial presentation coordinates, but minimality makes every semantic identification unique up to similarity.

8.6 Descent of the minimality and torsion-Fourier invariants

For a persistent C-finite sequence, the minimal dimension

$$\rho(a) = \operatorname{linrank}(a) = \operatorname{rank} H(a)$$

is simultaneously the minimal recurrence order, the reduced denominator degree, and the dimension of a minimal reachable-observable state.

Over characteristic zero, write the unique active expansion over an algebraic closure as

$$a_n = \sum_{\lambda \in \Lambda} P_\lambda(n)\lambda^n.$$

For each sampling modulus N and address j, the spectral torsion-Fourier theorem computes the exact minimal recurrence, reduced denominator, Hankel rank, state dimension, rank-drop mechanism, and finite periodic packet signature of a_{Nm+j} [7].

Corollary 8.2 (Cross-presentation torsion-Fourier descent)**.** *The minimal dimension and the complete arithmetic-subsequence packet-complexity profile are independent of whether the input is supplied as a normalized minimal recurrence, a reduced rational OGF, or a minimal state realization. The exact compiler outputs agree under every bridge in theorem 8.1.*

Proof. All three charts represent the same sequence. The compiler theorem accepts any of the three presentation types and computes quantities determined by the active modal decomposition or, equivalently, by exact reachable-observable reduction of the jump state. The bridge functors preserve the coefficient sequence, so the outputs agree. □

The phrase “complete invariant” here has a declared target. The torsion-Fourier signature is complete for the minimal recurrence and complexity profile of every arithmetic subsequence. It is not asserted to classify arbitrary sequence presentations or all sequence phenomena.

8.7 Exact examples

9 A first heterogeneous effective-descent atlas

The previous sections separated the formal assembly theorem from the nonformal work of declaring and verifying an actual atlas. This section discharges that burden for a deliberately restricted but genuinely heterogeneous example. The charts do not merely use different coordinates inside one linear category. They belong to three of the principal strata:

finite linear and C-finite digital and automatic functional-Mahler and Cartier.

The common semantic objects are unit purely periodic sequences over a prime field. The restriction is strong enough to make every bridge explicit, every normalized fiber rigid, every nonempty triple overlap coherent, and descent effective for both objects and morphisms. At the same time it is broad enough to contain repeated-root recurrences in characteristic p, nontrivial minimal automata, rational functional-Mahler masks, and redundant Cartier compiler presentations.

Table 10: Three exact C-finite examples across the normalized atlas.

Sequence	Minimal recurrence	Reduced OGF	Minimal dimension	Packet behavior
Fibonacci F_n	$x^2 - x - 1$	$z/(1 - z - z^2)$	2	The root ratio is not torsion. Every arithmetic subsequence retains rank 2.
$1 + (-1)^n$	$x^2 - 1$	$2/(1 - z^2)$	2	At $N = 2$, the even packet is constant of rank 1 and the odd packet is zero.
$n + 1)$	$(x - 1)^2$	$1/(1 - z)^2$	2	Every packet remains a nonzero affine polynomial in the packet index and retains rank 2.

9.1 The restricted semantic doctrine

Fix a prime p. Let $\mathbf{Per}_p^{\times,\mathrm{per}}$ denote the discrete category whose objects are sequences

$$a = (a_n)_{n\geq 0} \in \mathbb{F}_p^{\mathbb{N}}$$

satisfying

$$a_0 = 1 \qquad \text{and} \qquad a_{n+T} = a_n \quad (n \geq 0)$$

for some $T \geq 1$. The least such positive integer is the *minimal period* $T(a)$. Write

$$w(a) = (a_0, \ldots, a_{T(a)-1})$$

for the primitive period word and

$$W_a(z) = \sum_{r=0}^{T(a)-1} a_r z^r.$$

Then

$$F_a(z) = \sum_{n\geq 0} a_n z^n = \frac{W_a(z)}{1 - z^{T(a)}}. \tag{1}$$

The fraction in Equation (1) need not be reduced. Cancellation is part of the normalized rational chart rather than an error. For example, over $\mathbb{F}_3$, the period word 12 has

$$\frac{1 + 2z}{1 - z^2} = \frac{1}{1 + z}.$$

Boundary 9.1 (Why the unit condition is retained). The functional-Mahler bridge uses the formal quotient

$$R_a(z) = \frac{F_a(z)}{F_a(z^p)}.$$

It is regular at the origin exactly when $F_a(0) \neq 0$. The normalization $a_0 = 1$ removes scalar ambiguity and gives $R_a(0) = 1$. Purely periodic sequences with $a_0 = 0$ remain valid C-finite and automatic objects, but they need not lie in this regular unit-mask overlap. They are not discarded; they lie outside the selected heterogeneous atlas.

Proposition 9.2 (Finite-field persistence equals pure periodicity)**.** *For a sequence over $\mathbb{F}_p$, the following are equivalent.*

(i) *The sequence is C-finite and its minimal recurrence has nonzero constant term.*

(ii) *The sequence has a finite-dimensional state realization*

$$a_n = \lambda C^n v$$

with $C \in GL_d(\mathbb{F}_p)$.

(iii) *The sequence is purely periodic from $n = 0$.*

If $m_a(x)$ is the monic minimal recurrence polynomial and $T = T(a)$, then

$$m_a(x) \mid x^T - 1, \qquad T = \min\{N \geq 1 : m_a(x) \mid x^N - 1\}. \tag{2}$$

Proof. A recurrence with nonzero constant term has an invertible companion matrix, giving (i) $\Rightarrow$ (ii). The orbit of v under an invertible map of the finite set $\mathbb{F}_p^d$ is a cycle containing its initial state, so (ii) $\Rightarrow$ (iii). A sequence of period T is annihilated by $E^T - I$, where E is the forward shift, giving (iii) $\Rightarrow$ (i). The minimal annihilator divides every annihilator, hence $m_a \mid x^T - 1$. Conversely, if $m_a \mid x^N - 1$, then $E^N - I$ annihilates the sequence. Minimality of T proves Equation (2). □

Let

$$F_a(z) = \frac{P_a(z)}{Q_a(z)}$$

be reduced with

$$P_a(0) = Q_a(0) = 1.$$

If $d = \deg m_a$, then

$$Q_a(z) = z^d m_a(z^{-1}),$$

with the evident coefficient normalization. Taking reciprocal polynomials in Equation (2) gives

$$Q_a(z) \mid 1 - z^T, \qquad T = \min\{N \geq 1 : Q_a(z) \mid 1 - z^N\}. \tag{3}$$

Thus pure periodicity is visible both in the linear annihilator and in the reduced rational denominator.

9.2 Four normalized charts

The reversible core of the selected atlas has four charts. Each chart carries its own native objects and morphisms, but every normalized fiber over a fixed sequence will be shown to be contractible.

9.2.1 The minimal recurrence chart

An object of $\mathbf{Rec}_p^{\times,\mathrm{per}}$ is the normalized minimal recurrence polynomial

$$m(x) = x^d + c_1 x^{d-1} + \cdots + c_d, \qquad c_d \neq 0,$$

together with the initial vector $(a_0, \ldots, a_{d-1})$, subject to $a_0 = 1$ and the requirement that the generated sequence is purely periodic. Minimality means that no monic recurrence of lower degree annihilates the sequence. Morphisms are recurrence-presentation isomorphisms preserving the coefficient sequence. Because the minimal annihilator and initial data are unique, every endomorphism is the identity.

9.2.2 The reduced rational chart

An object of $\mathbf{Rat}_p^{\times,\mathrm{per}}$ is a reduced fraction

$$F(z) = \frac{P(z)}{Q(z)} \in \mathbb{F}_p(z), \qquad P(0) = Q(0) = 1,$$

whose coefficient sequence is purely periodic. Equivalently, $Q \mid 1 - z^T$ for some T and the reduced coefficient sequence has no transient prefix. Morphisms preserve the rational function. Reducedness and the constant-term normalization make this chart rigid.

The recurrence-rational bridge is the strict inverse pair already used in the C-finite atlas: multiply the generating series by the reciprocal recurrence denominator, or recover the recurrence by clearing the reduced denominator. On the present finite-field subcategory, the additional condition $Q \mid 1 - z^T$ records pure periodicity.

9.2.3 The canonical digital chart

For $0 \le d < p$, define the Cartier section of a sequence by

$$(\Lambda_d a)_n = a_{pn+d}.$$

The p-kernel is

$$\mathcal{K}_p(a) = \left\{ \Lambda_{d_{\ell-1}} \cdots \Lambda_{d_0} a : \ell \ge 0,\ 0 \le d_i < p \right\}. \tag{4}$$

We use least-significant-digit-first words. Every finite digit word is allowed. Appending high zero digits may change the internal state but never changes the represented coefficient, because the output of a state b is b_0 and $(\Lambda_0 b)_0 = b_0$.

The canonical Moore automaton has state set $\mathcal{K}_p(a)$, start state a, transition

$$\delta_d(b) = \Lambda_d b,$$

and output

$$o(b) = b_0.$$

After reading $d_0, \ldots, d_{\ell-1}$, its state is

$$\Lambda_{d_{\ell-1}} \cdots \Lambda_{d_0} a = \left(n \longmapsto a_{p^\ell n + r}\right), \qquad r = \sum_{j=0}^{\ell-1} d_j p^j.$$

Proposition 9.3 (Finite periodic kernel and exact state model)**.** *Let $a \in \mathbf{Per}_p^{\times,\mathrm{per}}$ have minimal period T and period word w. Every kernel state has the form*

$$b_{u,r}(n) = w_{un+r \bmod T}, \qquad u \in U_p(T), \quad r \in \mathbb{Z}/T\mathbb{Z}, \tag{5}$$

where

$$U_p(T) = \{p^e \bmod T : e \ge 0\}.$$

The digit transition on pair representatives is

$$(u, r) \xrightarrow{d} (pu, r + ud) \pmod T, \tag{6}$$

and the output is w_r. Quotienting pair representatives by equality of the sequences in Equation (5) gives the canonical minimal least-significant-digit-first p-DFAO for a.

Proof. After ℓ digits, the resulting subsequence is $n \mapsto a_{p^\ell n+r}$, which depends only on ($p^\ell \bmod T, r \bmod T$). Applying Λ_d gives

$$\Lambda_d b_{u,r}(n) = b_{u,r}(pn+d) = w_{(pu)n+(r+ud)},$$

proving Equation (6). The pair set is finite, so the kernel is finite. Every kernel state is reachable by construction. If two kernel sequences differ, they differ at some n; reading the base-p digits of that n distinguishes the states by their outputs. Hence the kernel automaton is minimal. Any other reachable minimal automaton producing the same sequence is uniquely isomorphic to it by the usual behavioral quotient. □

The pair model gives a useful finite bound. Write

$$T = p^\nu M, \qquad \gcd(M, p) = 1,$$

and set $\mathrm{ord}_1(p) = 1$.

Lemma 9.4 (Exact multiplier-orbit size)**.** *The set $U_p(T)$ has cardinality*

$$|U_p(T)| = \nu + \mathrm{ord}_M(p). \tag{7}$$

Consequently,

$$|\mathcal{K}_p(a)| \le T\big(\nu + \mathrm{ord}_M(p)\big). \tag{8}$$

Proof. For $0 \le e < \nu$, the residues $p^e \bmod p^\nu$ have distinct p-adic valuations. For $e \ge \nu$, the residue is zero modulo p^ν, while modulo M it runs through one multiplicative cycle of length $\mathrm{ord}_M(p)$. The early and eventual residues cannot coincide. This proves Equation (7). The kernel is a quotient of the pair set $U_p(T) \times \mathbb{Z}/T\mathbb{Z}$, giving Equation (8). □

Let $\mathrm{Aut}_p^{\times,\mathrm{per}}$ be the groupoid of reachable minimal least-significant-digit-first p-DFAOs whose output sequence lies in $\mathbf{Per}_p^{\times,\mathrm{per}}$. Morphisms are automaton isomorphisms preserving the start state, transitions, and outputs. Reachability forces every automorphism to be the identity.

9.2.4 The rational functional-Mahler mask chart

For a unit formal series $F(0) = 1$, define its local functional-Mahler mask by

$$\mathcal{M}_p(F)(z) = \frac{F(z)}{F(z^p)}. \tag{9}$$

Then

$$F(z) = \mathcal{M}_p(F)(z) F(z^p). \tag{10}$$

Iterating Equation (10) gives the coefficientwise product

$$F(z) = \prod_{j \ge 0} \mathcal{M}_p(F)(z^{p^j}). \tag{11}$$

Only finitely many factors contribute to each fixed coefficient.

If $F = P/Q$ is reduced and unit normalized, then

$$R_F(z) = \frac{P(z)Q(z^p)}{Q(z)P(z^p)}. \tag{12}$$

Thus every object of $\mathbf{Rat}_p^{\times,\mathrm{per}}$ has a rational mask. Reduce Equation (12) to

$$R_F(z) = \frac{A(z)}{B(z)}, \qquad A(0) = B(0) = 1, \qquad \gcd(A, B) = 1.$$

Define $\mathbf{Mask}_p^{\times,\mathrm{per}}$ to be the groupoid of such reduced rational masks whose normalized product solution Equation (11) has purely periodic coefficients. Morphisms preserve the reduced mask.

Theorem 9.5 (Unit rational functional-Mahler equivalence)**.** *The assignments*

$$F \longmapsto \frac{F(z)}{F(z^p)} \qquad \textit{and} \qquad R \longmapsto \prod_{j\geq 0} R(z^{p^j})$$

are inverse equivalences

$$\mathbf{Rat}_p^{\times,\mathrm{per}} \simeq \mathbf{Mask}_p^{\times,\mathrm{per}}.$$

The mask is unique for the represented series.

Proof. Starting with F, the finite product through level N telescopes:

$$\prod_{j=0}^{N} \frac{F(z^{p^j})}{F(z^{p^{j+1}})} = \frac{F(z)}{F(z^{p^{N+1}})}.$$

As $N \to \infty$, the denominator converges coefficientwise to $F(0) = 1$, proving that the product recovers F. Conversely, the product solution satisfies $F(z) = R(z)F(z^p)$, so $R = F/F(z^p)$. Reduced normalization makes the rational mask unique. □

Boundary 9.6 (Functional Mahler is not Newton-Mahler). The mask equation in this section is

$$F(z) = R(z)F(z^p).$$

It is a digit-section and automaton construction. It is not the Newton-Mahler interpolation expansion

$$f(x) = \sum_r c_r \binom{x}{r}.$$

Both occur elsewhere in the Library, but this atlas uses only the functional-Mahler meaning.

9.3 Explicit bridge passports

The reversible bridge core is generated by three explicit constructions:

$$\mathbf{Rec}_p^{\times,\mathrm{per}} \leftrightarrow \mathbf{Rat}_p^{\times,\mathrm{per}}, \qquad \mathbf{Rat}_p^{\times,\mathrm{per}} \leftrightarrow \mathrm{Aut}_p^{\times,\mathrm{per}}, \qquad \mathbf{Rat}_p^{\times,\mathrm{per}} \leftrightarrow \mathbf{Mask}_p^{\times,\mathrm{per}}.$$

The remaining reversible bridges are their normalized composites. A separate constructor and reachable-restriction lane connects masks to the digital normal form through the Cartier compiler.

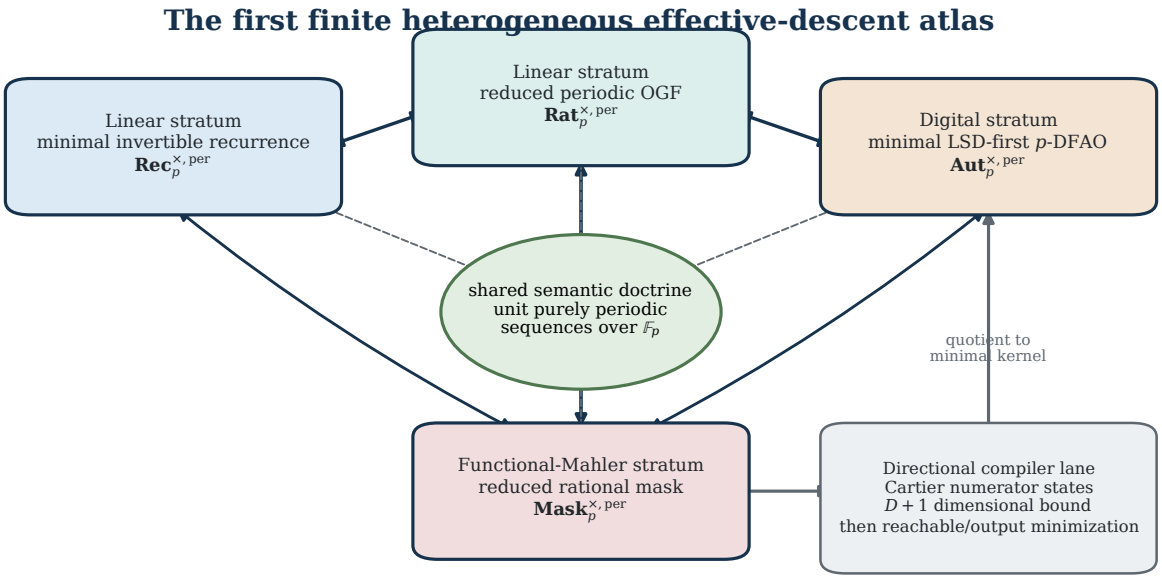


Figure 11: The first heterogeneous atlas proved effective in this manuscript. Four normalized reversible charts span three strata. The raw Cartier compiler is retained as a directional lane because its numerator coordinates need not be minimal.

Table 11: Bridge passports for the selected periodic atlas.

Bridge	Exact class	Domain	Preserved data	Reflected data	Information loss or boundary
Recurrence $\leftrightarrow$ reduced OGF	equivalence	minimal invertible recurrence and reduced purely periodic rational series	coefficients, minimal annihilator, initial data, period, Hankel rank	equality and isomorphism	none after reduced normalization
Reduced OGF $\leftrightarrow$ minimal DFAO	equivalence on the periodic essential image	unit purely periodic series and reachable minimal LSD-first p-DFAOs	coefficient semantics, minimal period, canonical p-kernel, output behavior	equality of represented sequences	a nonminimal residue-pair automaton is quotiented before entering the normalized chart
Reduced OGF $\leftrightarrow$ rational mask	equivalence on the unit-periodic essential image	$F(0) = 1$, periodic rational F, reduced regular mask R	exact series, functional equation, reduced rational mask, minimal period	equality of series and masks	undefined for nonunit series; unrestricted rational masks may have nonperiodic products
Mask $\rightarrow$ Cartier numerator representation	constructor	reduced A/B over $\mathbb{F}_p$	exact coefficient semantics, digit matrices, theorem degree bound	not generally reflective	introduces basis and denominator coordinates and may contain unreachable ambient states
Ambient Cartier compiler $\rightarrow$ minimal DFAO	reachable restriction	full finite numerator-state compiler	every reachable coefficient section and every future digit behavior	equality of the represented p-kernel states	removes only ambient states outside the orbit of the initial numerator
Unrestricted automatic $\dashrightarrow$ rational mask	partial bridge only	automatic sequences whose unit inverse mask is rational and whose product lies in the declared periodic class	coefficient semantics on the domain of definition	no unrestricted reflection theorem	no claim that every automatic sequence has a rational periodic mask

9.3.1 The rational-to-digital construction

For a periodic word w of length T, construct the finite pair automaton from Equation (6), then quotient by equality of the induced subsequences. This gives the canonical minimal DFAO directly. Conversely, a minimal DFAO in the selected chart is promised to produce a purely periodic sequence. Its minimal period can be recovered exactly by enumerating $N = 1, 2, \ldots$, constructing the finite-state transducer for addition by N, and testing equivalence between the original output and its N-shift. Equivalence of finite deterministic output automata is a finite product-graph calculation. The promise of pure periodicity guarantees termination. Once T is known, Equation (1) gives the rational series, which is then reduced.

This inverse is effective but not claimed to be complexity optimal. The point is effectivity, not a sharp automata algorithm.

9.3.2 The rational-mask compiler lane

Let

$$R(z) = \frac{A(z)}{B(z)}, \qquad A(0) = B(0) = 1,$$

with

$$a = \deg A, \qquad b = \deg B, \qquad c = a + (p-2)b,$$

and define

$$D = \max\left\{b, \left\lceil \frac{c}{p-1} \right\rceil\right\}. \tag{13}$$

The rational-mask Cartier theorem gives an invariant numerator space of degree at most D under

$$P \longmapsto \Lambda_d(PAB^{p-2}), \qquad 0 \le d < p.$$

The resulting linear state vector belongs to $\mathbb{F}_p^{D+1}$, so the raw output automaton has at most

$$p^{D+1} \tag{14}$$

states [8]. It computes the coefficient sequence exactly.

Proposition 9.7 (Compiler reachability lands in the canonical digital chart)**.** *For a mask in* $\mathbf{Mask}_p^{\times,\mathrm{per}}$*, the reachable part of the fixed-denominator Cartier compiler is uniquely isomorphic to the canonical p-kernel automaton of proposition 9.3. In particular, the subsequent quotient by equality of future output behavior is trivial.*

Proof. The fixed-denominator intertwining map sends the numerator reached by a digit word w to the Cartier section $\Lambda_w F$. It is injective, as proved in the more general theorems 10.4 and 10.7. Hence distinct reachable numerator states represent distinct section series and are distinguished by some future coefficient. The reachable compiler state set is therefore exactly the canonical p-kernel state set, with matching transitions and outputs. □

9.4 Triple-overlap and quadruple coherence

There are four nonempty triple overlaps:

$$(\mathbf{Rec}, \mathbf{Rat}, \mathrm{Aut}), \qquad (\mathbf{Rec}, \mathbf{Rat}, \mathbf{Mask}), \qquad (\mathbf{Rec}, \mathrm{Aut}, \mathbf{Mask}), \qquad (\mathbf{Rat}, \mathrm{Aut}, \mathbf{Mask}).$$

All four charts meet on the same unit periodic semantic objects.

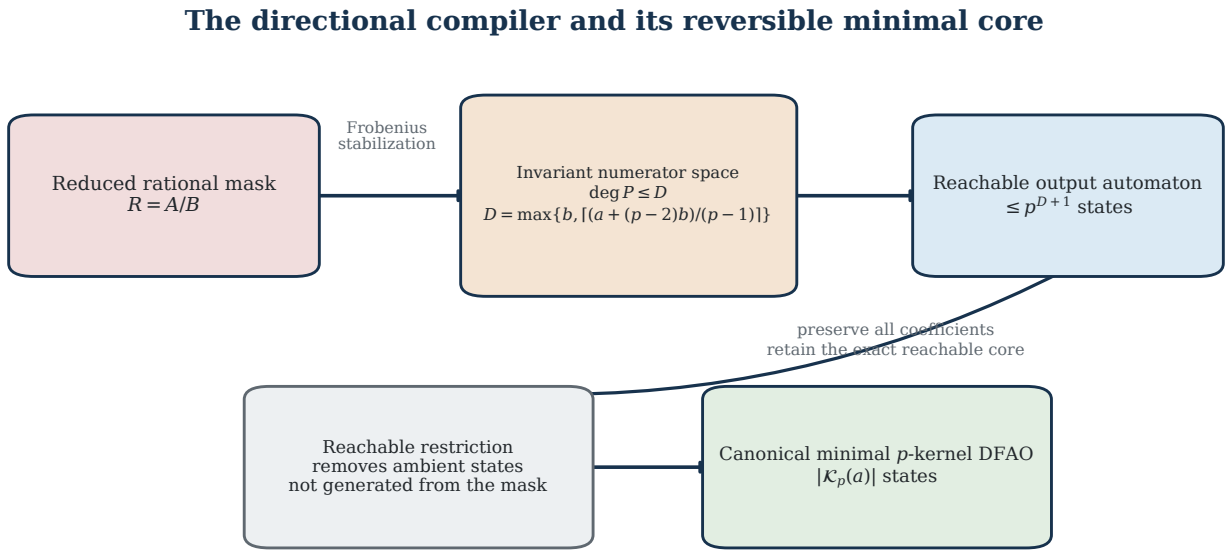


Figure 12: The raw mask compiler is a directional constructor. Reachability removes the ambient states outside the numerator orbit. Injectivity of the fixed-denominator realization then shows that the remaining states are already the unique minimal digital normal form.

Lemma 9.8 (Rigidity of normalized fibers)**.** *Fix $a \in \mathbf{Per}_p^{\times,\mathrm{per}}$. In each of the four normalized charts, the groupoid of presentations of a is contractible.*

Proof. The minimal recurrence with its initial data, the reduced rational function, and the reduced mask are unique, so their fibers are terminal groupoids. A normalized digital presentation is a reachable minimal automaton. Any automorphism fixes the start state and commutes with every transition. Reachability then forces it to fix every state. Hence its automorphism group is trivial, and any two minimal presentations are joined by the unique behavioral isomorphism. □

Theorem 9.9 (Coherence of the periodic heterogeneous atlas)**.** *Choose every pairwise bridge by the explicit constructions above and choose every comparison* 2*-cell to be the unique normalized isomorphism over the common sequence. Then:*

(i) *every one of the four triple-overlap comparison diagrams commutes;*

(ii) *the two composites of comparison cells on the four-chart overlap agree;*

(iii) *the quadruple-overlap coherence defect Ω is the identity.*

Proof. Every comparison in question is an isomorphism between normalized presentations of one fixed semantic sequence. By lemma 9.8, there is exactly one such isomorphism. Any two composites with the same source and target must therefore agree. This proves all triple laws and the quadruple law simultaneously. □

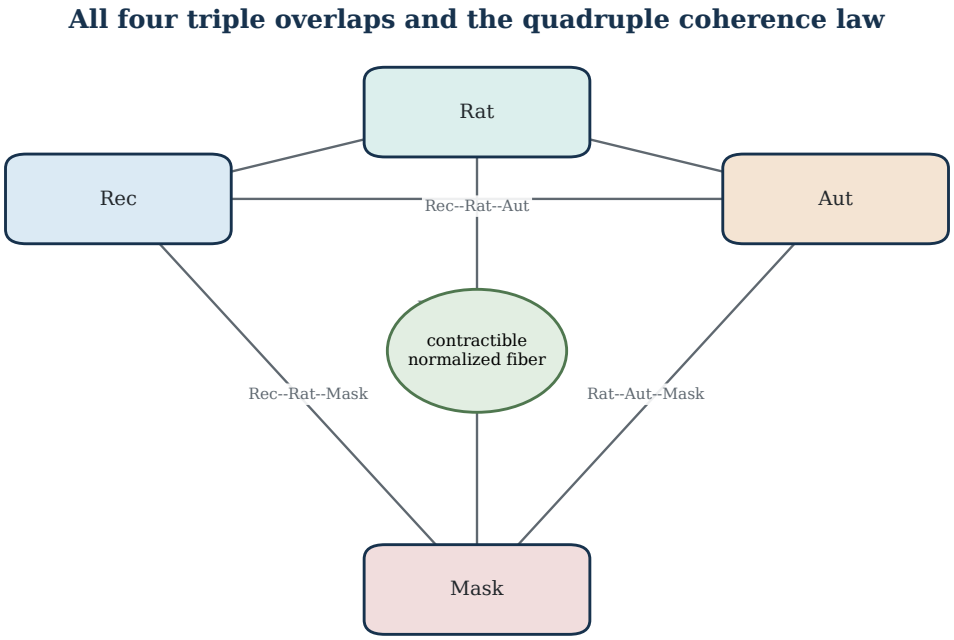


Figure 13: The coherence tetrahedron. Contractibility of every normalized semantic fiber makes each comparison cell unique, so all triple and quadruple coherence equations hold automatically after the bridge constructions have been typed.

9.5 Effective descent for objects and morphisms

Let $\mathfrak{R}_p^{\mathrm{per}}$ denote the four-chart pseudofunctor defined by the preceding passports and comparison cells.

Theorem 9.10 (First heterogeneous effective-descent theorem)**.** *The semantic pseudonatural transformation induces an equivalence*

$$\mathrm{Desc}(\mathfrak{R}_p^{\mathrm{per}}) \simeq \mathbf{Per}_p^{\times,\mathrm{per}}. \tag{15}$$

In particular, descent is effective for both objects and morphisms.

Proof. Given $a \in \mathbf{Per}_p^{\times,\mathrm{per}}$, form its minimal recurrence, reduced rational OGF, canonical minimal p-kernel automaton, and reduced rational mask. The explicit bridges supply a descent object. This defines a functor

$$N : \mathbf{Per}_p^{\times,\mathrm{per}} \longrightarrow \mathrm{Desc}(\mathfrak{R}_p^{\mathrm{per}}).$$

Conversely, any descent object has local presentations whose semantic images agree on every overlap. Since the semantic target is discrete, they determine one sequence a. Normalization and lemma 9.8 identify the local presentations with those constructed by $N(a)$. Thus N is essentially surjective and fully faithful.

For morphisms, a compatible local family lies over one equality of semantic sequences. Each normalized fiber has at most one morphism, so the local family glues uniquely. Identities, composition, equality, and invertibility are therefore detected componentwise and descend. □

Corollary 9.11 (A verified instance of the assembly theorem)**.** *The atlas $\mathfrak{R}_p^{\mathrm{per}}$ satisfies realization, semantic adequacy, typed overlaps, higher coherence, effective descent, and preservation for the finite algebraic constructions*

used in this section. Therefore theorem 6.3 applies without a remaining categorical hypothesis on this restricted domain.

The conclusion is intentionally local to the selected atlas. It does not imply effective descent for arbitrary automatic sequences, unrestricted rational masks, general local analytic objects, or the full eight-stratum registry.

9.6 A compatible complete invariant family

For $a \in \mathbf{Per}_p^{\times,\mathrm{per}}$, define the periodic resolved invariant package

$$\mathcal{J}_p(a) = \big(T(a), w(a), m_a(x), P_a(z)/Q_a(z), \mathcal{K}_p(a), \mathcal{A}_p(a), R_a(z)\big), \tag{16}$$

where $\mathcal{A}_p(a)$ is the canonical minimal digital automaton. Its comparison identities include

$$\begin{aligned} \deg m_a &= \deg Q_a = \operatorname{rank} H(a) = \dim \mathbf{State}_{\min}(a), \\ T(a) &= \min\{N : m_a \mid x^N - 1\} = \min\{N : Q_a \mid 1 - z^N\}, \\ |\mathcal{A}_p(a)| &= |\mathcal{K}_p(a)| \le T(a)\big(\nu_p(T(a)) + \operatorname{ord}_M(p)\big), \\ R_a(z) &= \frac{F_a(z)}{F_a(z^p)}, \qquad T(a) = p^{\nu_p(T(a))}M, \quad \gcd(M,p) = 1. \end{aligned} \tag{17}$$

Theorem 9.12 (Invariant descent and global completeness on the periodic atlas)**.** *The native invariants in the recurrence, rational, digital, and mask charts admit coherent comparison maps to $\mathcal{J}_p$. These maps descend to*

$$\mathcal{J}_p : \mathrm{Desc}(\mathfrak{R}_p^{\mathrm{per}}) \longrightarrow \mathbf{Inv}_p^{\mathrm{per}}.$$

The descended invariant is complete on the declared semantic category.

Proof. Every component of $\mathcal{J}_p(a)$ is reconstructed from the coefficient sequence by the explicit normalization procedures above. Hence every pairwise bridge preserves the package and the comparison cells are identities after semantic identification. Triple coherence follows from theorem 9.9. If $\mathcal{J}_p(a) = \mathcal{J}_p(b)$, then the primitive period words agree, so $a = b$. Under Equation (15), equality of semantic objects is exactly isomorphism in the descent category. □

The completeness claim has a narrow classification target: unit purely periodic sequences over $\mathbb{F}_p$. It is not a complete invariant for all C-finite, all automatic, or all rational functional-Mahler sequences.

9.7 Exact examples and a normalization boundary

Table 12: Named exact examples in the heterogeneous periodic atlas.

Example	Field / word	Minimal recurrence	Reduced OGF denominator	Reduced mask numerator	Kernel states	Compiler bound
Even indicator	$\mathbb{F}_2$ / 10	$1 + x^2$	$1 + z^2$	$1 + z^2$	3	8
Alternating unit	$\mathbb{F}_3$ / 12	$1 + x$	$1 + z$	$1 + 2z + z^2$	2	9
Period-three word	$\mathbb{F}_2$ / 110	$1 + x + x^2$	$1 + z + z^2$	$1 + z + z^2$	3	8

Even indicator over $\mathbb{F}_2$. For the word 10,

$$T = 2, \qquad m(x) = x^2 + 1 = (x+1)^2, \qquad F(z) = \frac{1}{1+z^2}, \qquad R(z) = 1 + z^2.$$

The repeated recurrence root is compatible with pure periodicity because the field has characteristic two. The canonical 2-kernel has three states: the original sequence, the constant-one sequence, and the constant-zero sequence. The rational-mask compiler gives the safe ambient bound $2^3 = 8$ before reachability restriction.

Alternating unit over $\mathbb{F}_3$. For the word 12,

$$T = 2, \qquad m(x) = x + 1, \qquad F(z) = \frac{1}{1+z}, \qquad R(z) = \frac{1+z^3}{1+z} = 1 + 2z + z^2.$$

The recurrence order is one although the minimal period is two. The 3-kernel consists of the sequence and its negative, giving two minimal digital states. The compiler bound is nine.

Period-three unit word over $\mathbb{F}_2$. For the word 110,

$$T = 3, \qquad m(x) = x^2 + x + 1, \qquad F(z) = \frac{1}{1+z+z^2}, \qquad R(z) = 1 + z + z^2.$$

The reachable pair model has six representatives, while equality of subsequence behavior collapses them to three kernel states. This is a concrete instance of the directional quotient in proposition 9.7.

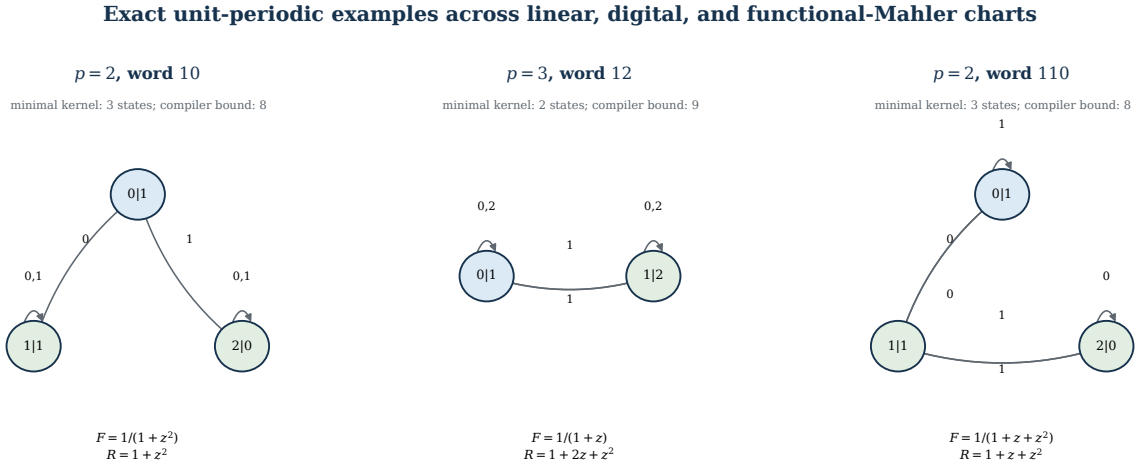


Figure 14: Three exact examples. A state label $s \mid o$ gives the state identifier and output. The compiler bounds are ambient theorem bounds before reachability restriction, not claims of minimal state count.

Negative control: a nonunit periodic sequence. Over $\mathbb{F}_2$, the word 01 gives

$$F(z) = \frac{z}{1+z^2}.$$

Then

$$\frac{F(z)}{F(z^2)} = \frac{1+z^2}{z},$$

which is Laurent rather than regular at the origin. The sequence remains purely periodic, C-finite, and automatic, but it is not an object of $\mathbf{Mask}_2^{\times,\mathrm{per}}$. This verifies that the unit overlap condition is structural rather than cosmetic.

9.8 Exact finite verification

Table 13: Exact finite validation of the Block 03 periodic heterogeneous atlas.

Check	Recorded result
Enumerated normalized objects	$p = 2$ through period 8, $p = 3$ through period 6; 583 primitive unit-periodic words.
Exact identities and automaton checks	39805 checks; 0 failures.
Arithmetic model	exact prime-field and polynomial arithmetic.
Overall status	PASS.
Boundary	Finite enumeration validates the declared ranges and implementation formulas. It does not replace the symbolic proofs or establish unrestricted automatic-to-Mahler equivalence.

The supplied verifier enumerates every primitive unit-periodic word over $\mathbb{F}_2$ through period eight and over $\mathbb{F}_3$ through period six. For each object it checks, using exact prime-field and polynomial arithmetic:

1. Berlekamp-Massey recovery of the minimal recurrence;
2. divisibility by $x^T - 1$ and $1 - z^T$;
3. recovery of the least period from both normalized linear charts;
4. the reduced rational functional-Mahler identity;
5. the exact reachable pair model and its behavioral quotient;
6. the multiplier-orbit formula Equation (7);
7. coefficient output of the minimal digit automaton;
8. the Cartier compiler state bound Equation (14).

The finite computation supports the implementation and examples. The symbolic theorems above do not depend on the enumeration.

Section result. The manuscript now contains a fully declared finite atlas spanning three genuinely different strata: minimal finite-linear recurrences, minimal digital automata, and rational functional-Mahler masks. Every reversible bridge has a complete passport. The raw Cartier compiler is retained as a directional constructor with an exact loss statement. All four triple overlaps and the quadruple coherence equation are proved through rigidity of normalized fibers. Objects and morphisms descend effectively, and the compatible periodic invariant package is globally complete on the declared unit-periodic semantic category. This is the first theorem in the manuscript that satisfies the finite heterogeneous completion criterion rather than only the one-stratum cross-presentation criterion.

10 Rational-mask digital descent beyond pure periodicity

Periodic atlas verified a four-chart atlas on unit purely periodic sequences. That domain made the linear, rational, digital, and functional-Mahler charts meet simultaneously, but it left open the larger

overlap between rational functional-Mahler masks and automatic sequences. This section removes the periodicity hypothesis while keeping the normalization and bridge standards unchanged.

Fix a prime p. Let

$$\mathbf{RMah}_p^{\times} = \left\{ F(z) \in 1 + z\mathbb{F}_p[[z]] : \frac{F(z)}{F(z^p)} \in \mathbb{F}_p(z) \right\}$$

be the discrete category of unit formal series with rational canonical functional-Mahler mask. This category contains the periodic objects of Section 9, but for odd p it also contains genuinely nonrational and non-C-finite automatic series.

10.1 Frobenius-radical characterization

For a unit formal series, define its canonical local mask by

$$R_F(z) = \frac{F(z)}{F(z^p)}. \tag{18}$$

The coefficientwise infinite product

$$\mathcal{M}_p[R](z) = \prod_{j \geq 0} R\left(z^{p^j}\right) \tag{19}$$

is well defined for every $R \in 1 + z\mathbb{F}_p[[z]]$, because only finitely many factors contribute to a fixed coefficient.

Theorem 10.1 (Frobenius-radical characterization)**.** *Let $F \in 1 + z\mathbb{F}_p[[z]]$. Then*

$$R_F(z) = F(z)^{1-p}. \tag{20}$$

Consequently, the following are equivalent.

(i) *The canonical mask R_F is rational.*

(ii) *The power F^{p-1} belongs to $\mathbb{F}_p(z)$.*

(iii) *There are coprime polynomials $A, B \in \mathbb{F}_p[z]$ with*

$$A(0) = B(0) = 1$$

and

$$B(z)F(z) = A(z)F(z^p). \tag{21}$$

For the reduced mask $R_F = A/B$, one has

$$F(z)^{p-1} = \frac{B(z)}{A(z)}. \tag{22}$$

The ratio and product constructions are strict inverses:

$$F \longmapsto R_F, \qquad R \longmapsto \mathcal{M}_p[R].$$

In particular, every object of $\mathbf{RMah}_p^{\times}$ is algebraic over $\mathbb{F}_p(z)$ of degree at most $p-1$.

Proof. In characteristic p, Frobenius gives

$$F(z^p) = F(z)^p.$$

Substitution into Equation (18) gives Equation (20). Thus R_F is rational exactly when $F^{p-1} = R_F^{-1}$ is rational. Writing the reduced rational mask as A/B gives Equation (21) and Equation (22).

Conversely, a reduced rational mask A/B with unit constant terms defines a unique formal solution of Equation (21): the product in Equation (19). Telescoping gives

$$\frac{\mathcal{M}_p[R](z)}{\mathcal{M}_p[R](z^p)} = R(z),$$

while uniqueness follows by comparing coefficients recursively. Hence the ratio and product constructions are strict inverses. Finally, Equation (22) is an algebraic equation of degree $p-1$. □

Boundary 10.2 (Functional Mahler versus Newton-Mahler). The word "Mahler" continues to name two distinct constructions. This section concerns the functional equation

$$F(z) = R(z)F(z^p).$$

It does not identify rational masks with Newton-Mahler expansions

$$f(x) = \sum_{r\geq 0} c_r \binom{x}{r}$$

or with p-adic analytic interpolation. Any bridge to that local analytic chart still requires its own passport and proof.

10.2 The rational-collapse obstruction

Let $R = A/B$ be reduced and normalized. For each monic irreducible polynomial $\pi \in \mathbb{F}_p[z]$, define

$$\omega_\pi(R) = \big(v_\pi(B) - v_\pi(A)\big) \bmod (p-1).$$

Include the valuation at infinity

$$\omega_\infty(R) = \big(\deg A - \deg B\big) \bmod (p-1).$$

The finite-support family

$$\omega_{p-1}(R) = \big(\omega_v(R)\big)_v \tag{23}$$

is the *rational-collapse obstruction*.

Theorem 10.3 (Exact rational-collapse criterion)**.** *Let $F = \mathcal{M}_p[A/B]$. The following are equivalent.*

(i) $F \in \mathbb{F}_p(z)$.

(ii) B/A *is a* $(p-1)$*-st power in* $\mathbb{F}_p(z)$.

(iii) $\omega_{p-1}(A/B) = 0$.

When these conditions hold, there is a unique rational function $G \in \mathbb{F}_p(z)$ with $G(0) = 1$ and

$$G^{p-1} = B/A,$$

and $F = G$.

If $p = 2$, every rational mask collapses to a rational series. If $p > 2$ and $\omega_{p-1}(A/B) \neq 0$, then F is nonrational, non-C-finite, and not eventually periodic.

Proof. The equivalence of (i) and (ii) follows from Equation (22). If $B/A = G^{p-1}$, every valuation of B/A is divisible by $p-1$, so (ii) implies (iii).

Conversely, suppose all valuation residues vanish. Since A and B are coprime, every irreducible exponent in each polynomial is divisible by $p-1$. Factoring into monic irreducibles therefore gives

$$A = C^{p-1}, \qquad B = D^{p-1}$$

for polynomials C, D after a constant normalization. The unit conditions $A(0) = B(0) = 1$ remove the remaining scalar obstruction. Thus $B/A = (D/C)^{p-1}$. Multiplying D/C by its inverse constant term does not change its $(p-1)$-st power, and gives the unique root with constant term one. The formal solution is unique, so $F = G$.

For $p = 2$, the exponent $p-1$ is one, hence the obstruction group is trivial and $F = B/A$. Over a finite field, a rational coefficient series is C-finite and eventually periodic, while an eventually periodic sequence has a rational generating function. Therefore a nonzero obstruction forces all three failures stated in the theorem. □

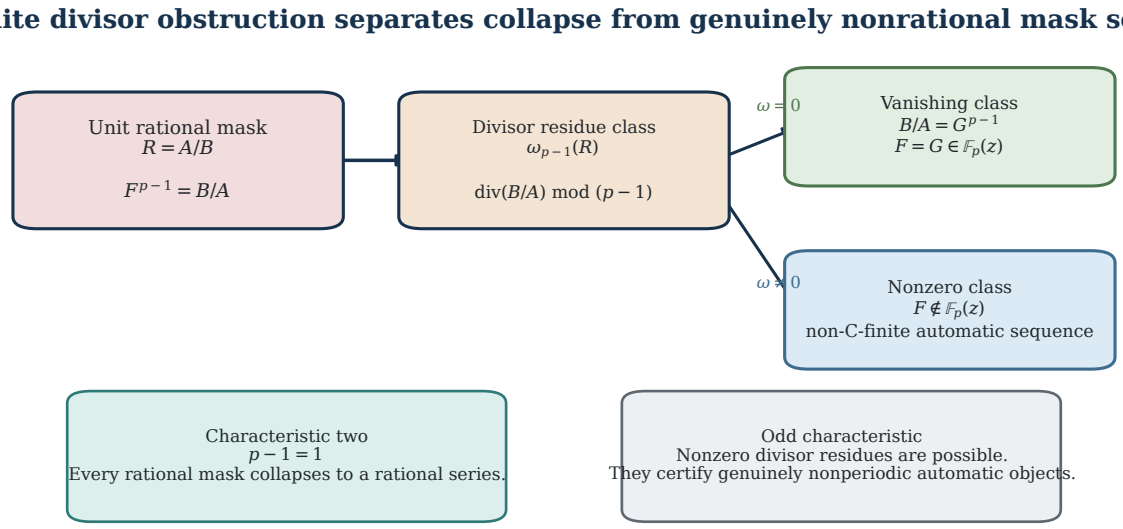


Figure 15: The rational-collapse obstruction. A rational mask always gives an automatic series, but it gives a rational or C-finite series exactly on the vanishing divisor-residue locus.

10.3 Denominator stabilization and the canonical Cartier module

Write

$$a = \deg A, \qquad b = \deg B, \qquad H(z) = A(z)B(z)^{p-2},$$

and define

$$D(A,B) = \max\left\{b, \left\lceil \frac{a+(p-2)b}{p-1} \right\rceil \right\}. \tag{24}$$

Let $\mathcal{P}_D = \mathbb{F}_p[z]_{\le D}$. For $0 \le d < p$, define

$$T_d(P) = \Lambda_d(PH), \tag{25}$$

where Λ_d selects the coefficients in degrees congruent to d modulo p.

Theorem 10.4 (Fixed-denominator Cartier realization)**.** *Let $F = \mathcal{M}_p[A/B]$ and $D = D(A,B)$. Then:*

(i) *every T_d preserves $\mathcal{P}_D$;*

(ii) *the linear map*

$$\Theta_{A,B} : \mathcal{P}_D \longrightarrow \mathbb{F}_p[[z]], \qquad \Theta_{A,B}(P) = \frac{P}{B}F, \tag{26}$$

is injective;

(iii) *the transitions intertwine exactly:*

$$\Lambda_d \Theta_{A,B}(P) = \Theta_{A,B}(T_d(P)); \tag{27}$$

(iv) *starting from $P_0 = B$, the coefficient of index*

$$n = d_0 + d_1 p + \cdots + d_{\ell-1}p^{\ell-1}$$

is

$$[z^n]F = \epsilon_0\big(\Theta_{A,B}\big(T_{d_{\ell-1}} \cdots T_{d_0} B\big)\big), \tag{28}$$

where $\epsilon_0(G) = [z^0]G$.

Proof. If $\deg P \le D$, then

$$\deg(PH) \le D + a + (p-2)b.$$

Cartier selection divides the retained degrees by p. The definition of D gives

$$\deg T_d(P) \le \left\lfloor \frac{D + a + (p-2)b}{p} \right\rfloor \le D.$$

This proves (i).

The series F/B is a unit in $\mathbb{F}_p[[z]]$. Multiplication by a unit is injective, so $\Theta_{A,B}$ is injective. The rational-mask Cartier identity imported from [8] is precisely Equation (27). Iterating it from $\Theta(B) = F$ and taking constant terms proves Equation (28). □

For a digit word $w = d_0 \cdots d_{\ell-1}$, read least significant digit first and write

$$\Lambda_w = \Lambda_{d_{\ell-1}} \cdots \Lambda_{d_0}.$$

Define the Cartier kernel and its linear span by

$$\mathcal{K}_p(F) = \{\Lambda_w F : w \in \{0,\dots,p-1\}^*\}, \qquad \mathcal{V}_p(F) = \operatorname{span}_{\mathbb{F}_p} \mathcal{K}_p(F). \tag{29}$$

Definition 10.5 (Canonical Cartier module and Cartier rank)**.** The canonical Cartier module of F is

$$\mathsf{C}_p(F) = \left(\mathcal{V}_p(F), (\Lambda_d|_{\mathcal{V}_p(F)})_{0\le d<p}, F, \epsilon_0\right). \tag{30}$$

Its dimension

$$r_p(F) = \dim_{\mathbb{F}_p} \mathcal{V}_p(F) \tag{31}$$

is the *Cartier rank*.

Theorem 10.6 (Minimality of the canonical Cartier module)**.** *For every $F \in \mathbf{RMah}_p^{\times}$, the module $\mathsf{C}_p(F)$ is finite dimensional, reachable, and observable. It is therefore a minimal digit-linear realization of the coefficient function. Every other reachable-observable digit-linear realization of the same coefficient function is uniquely similar to $\mathsf{C}_p(F)$.*

Proof. By theorem 10.4, every section $\Lambda_w F$ lies in the $(D+1)$-dimensional image of $\Theta_{A,B}$. Hence $\mathcal{V}_p(F)$ is finite dimensional. Reachability is built into its definition as the span of the orbit of F.

For observability, let $G \in \mathcal{V}_p(F)$ satisfy

$$\epsilon_0(\Lambda_w G) = 0$$

for every digit word w. For any $n \ge 0$, choose the base-p digits of n. The section identity gives

$$[z^n]G = \epsilon_0(\Lambda_w G) = 0.$$

Thus $G = 0$, so the observation family separates points.

For uniqueness, let $(V, (L_d), v, \lambda)$ be another reachable-observable realization of the same digit function. Send the reachable vector $L_w v$ to $\Lambda_w F$. If a linear combination of reachable vectors vanishes, applying every continuation and then λ shows that the corresponding combination of Cartier sections has every coefficient zero. Observability makes the assignment well defined. Reachability makes it surjective, and the symmetric construction gives an inverse. The intertwining, initial-vector, and output identities are immediate. Any such similarity is determined on the reachable spanning set, so it is unique. □

10.4 The exact reachable compiler core

Let $\mathcal{R}(A,B) \subseteq \mathcal{P}_D$ be the finite set of numerator states reachable from B under the transitions T_d, and let

$$W(A,B) = \operatorname{span}_{\mathbb{F}_p} \mathcal{R}(A,B).$$

Theorem 10.7 (Reachable numerator core)**.** *The map $\Theta_{A,B}$ restricts to*

$$\mathcal{R}(A,B) \xrightarrow{\sim} \mathcal{K}_p(F) \tag{32}$$

and to a linear isomorphism

$$W(A,B) \xrightarrow{\sim} \mathcal{V}_p(F). \tag{33}$$

The reachable deterministic output automaton of the fixed-denominator compiler is already minimal. No nontrivial output-equivalence quotient remains after reachability restriction.

Proof. For every word w, repeated use of Equation (27) gives

$$\Theta_{A,B}(T_w B) = \Lambda_w F.$$

Thus $\Theta_{A,B}$ maps the reachable numerator set onto the Cartier kernel. Injectivity of $\Theta_{A,B}$ makes the map bijective, and taking linear spans gives Equation (33).

If two reachable numerators $P \neq Q$ had identical future output behavior, the two series $\Theta(P)$ and $\Theta(Q)$ would have identical coefficients and would therefore be equal. Injectivity would force $P = Q$, a contradiction. Hence all reachable states are behaviorally distinct. Reachability and distinguishability prove minimality. □

The preceding theorem sharpens the generic reduction language used in Periodic atlas. A behavioral quotient remains a safe general construction for arbitrary output automata, but it is trivial for the normalized fixed-denominator numerator compiler.

Let

$$\kappa_p(F) = |\mathcal{K}_p(F)|. \tag{34}$$

Corollary 10.8 (Exact state-complexity hierarchy)**.** *For a reduced rational mask A/B,*

$$r_p(F) \le \kappa_p(F) \le p^{r_p(F)} \le p^{D(A,B)+1}. \tag{35}$$

Moreover,

$$r_p(F) = \dim W(A,B), \qquad \kappa_p(F) = |\mathcal{R}(A,B)|.$$

Proof. The Cartier kernel spans $\mathcal{V}_p(F)$, so its cardinality is at least the dimension. It is a subset of the $p^{r_p(F)}$ vectors in $\mathcal{V}_p(F)$. Finally, $\mathcal{V}_p(F)$ embeds into the $(D+1)$-dimensional compiler space by theorem 10.7. □

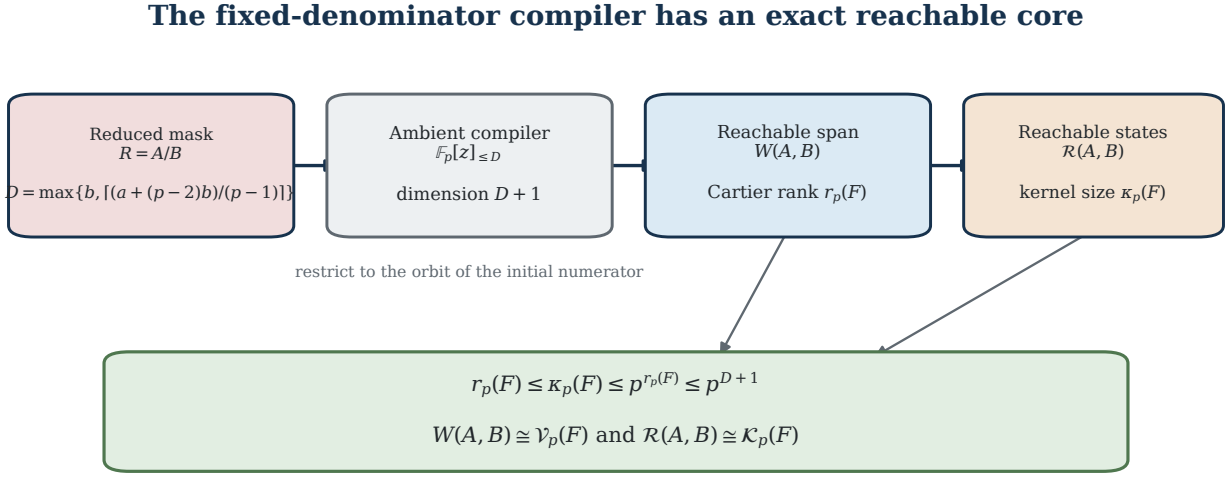


Figure 16: The exact reachable core of the fixed-denominator compiler. The linear rank and deterministic state count are different invariants, but both are bounded by the explicit mask degree profile.

10.5 Three normalized charts

The rational-mask domain has three normalized charts.

Reduced mask chart. An object of $\mathbf{Mask}_p^{\mathrm{rat}}$ is a reduced rational function

$$R(z) = \frac{A(z)}{B(z)}, \qquad A(0) = B(0) = 1.$$

Morphisms preserve the rational function. The semantics is $R \mapsto \mathcal{M}_p[R]$. The inverse chart map is the canonical ratio $F \mapsto F/F(z^p)$.

Minimal Cartier-module chart. An object of $\mathbf{Cart}_p^{\mathrm{min,rat}}$ is a finite-dimensional tuple

$$(V, (L_d)_{d=0}^{p-1}, v, \lambda)$$

that is reachable and observable, whose digit coefficient function belongs to $\mathbf{RMah}_p^{\times}$. Morphisms are similarities preserving every L_d, the initial vector, and the output functional. Its normalized representative is $\mathsf{C}_p(F)$.

Minimal digital chart. An object of $\mathbf{Aut}_p^{\mathrm{min,rat}}$ is a reachable minimal least-significant-digit-first p-DFAO whose coefficient series belongs to $\mathbf{RMah}_p^{\times}$. Its canonical representative has state set $\mathcal{K}_p(F)$, transition $G \mapsto \Lambda_d G$, start state F, and output $G \mapsto G(0)$.

Table 14: Bridge passports for the rational-mask digital atlas.

Bridge	Class	Preserved data	Exact boundary
Mask ↔ series	strict inverse pair	every coefficient and the canonical mask	the mask must be rational and unit normalized
Series ↔ Cartier module	normalized equivalence	all Cartier sections, linear relations, initial vector, output	the module is restricted to reachable-observable realizations
Series ↔ minimal DFAO	normalized equivalence	the complete p-kernel and coefficient function	the automaton uses the fixed radix and LSD-first convention
Mask → ambient compiler	directional constructor	exact coefficients and fixed-denominator transitions	adds unreachable ambient numerator states
Ambient compiler → normalized core	reachable restriction	all reachable series and all future outputs	discards only states outside the orbit of B

Theorem 10.9 (Rational-mask three-chart equivalence)**.** *The semantic functors induce equivalences of groupoids*

$$\mathbf{Mask}_p^{\mathrm{rat}} \simeq \mathbf{Cart}_p^{\mathrm{min,rat}} \simeq \mathbf{Aut}_p^{\mathrm{min,rat}} \simeq \mathbf{RMah}_p^{\times}. \tag{36}$$

The mask-series bridge is a strict inverse pair. The Cartier and digital bridges are normalized equivalences with unique comparison isomorphisms.

Proof. The strict mask-series equivalence is theorem 10.1. The canonical Cartier construction gives an object of the second chart, and theorem 10.6 gives a unique similarity from every other minimal realization of the same series. The canonical p-kernel automaton is finite by theorem 10.4. It is reachable by construction. Distinct kernel series differ in some coefficient, and the digit word for that coefficient distinguishes their future outputs, so the automaton is minimal. The usual behavioral map gives the unique isomorphism from any other reachable minimal DFAO for the same series. □

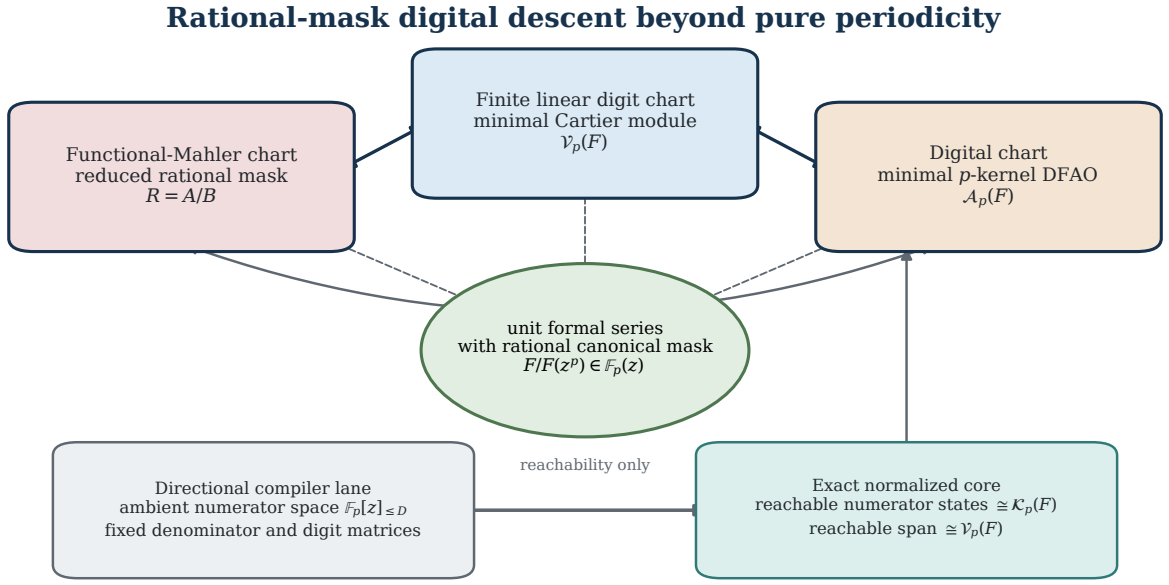


Figure 17: The Rational-mask atlas atlas. The normalized reversible core contains a rational mask, a minimal Cartier module, and a minimal digital automaton. The full polynomial compiler remains a larger directional presentation.

10.6 Coherence and effective descent

Lemma 10.10 (Rigidity of rational-mask normalized fibers)**.** *Fix $F \in \mathbf{RMah}_p^{\times}$. The groupoid of normalized presentations of F in each of the three charts is contractible.*

Proof. The reduced canonical mask is unique. Any automorphism of a reachable Cartier module that fixes the initial vector and commutes with all digit operators fixes a spanning set, hence is the identity. Any automorphism of a reachable deterministic automaton that fixes the start state and commutes with every digit transition fixes every state, hence is the identity. Existence and uniqueness of isomorphisms between two normalized realizations follow from theorems 10.6 and 10.9. □

Let $\mathfrak{R}_p^{\mathrm{rm}}$ denote the three-chart pseudofunctor defined by the passports above.

Theorem 10.11 (Rational-mask coherence and effective descent)**.** *Choose every pairwise bridge by the canonical constructions and every comparison cell by the unique normalized isomorphism over the common series. Then the triple-overlap coherence law holds, and*

$$\mathrm{Desc}(\mathfrak{R}_p^{\mathrm{rm}}) \simeq \mathbf{RMah}_p^{\times}. \tag{37}$$

Descent is effective for objects and morphisms.

Proof. On the only nonempty triple overlap, both composites are isomorphisms between normalized presentations of the same series. By lemma 10.10, the isomorphism is unique, so the comparison diagram commutes.

Given $F \in \mathbf{RMah}_p^{\times}$, its reduced mask, canonical Cartier module, and minimal p-kernel automaton form descent data. Conversely, compatible local objects have one common semantic series. Fiber rigidity identifies them uniquely with its normalized presentations. The same argument applies to morphisms. Hence the normalization functor is fully faithful and essentially surjective. □

This theorem is strictly broader than theorem 9.10: no periodicity or C-finiteness hypothesis is imposed. It is also strictly narrower than an atlas of all automatic sequences, because an automatic series may have a nonrational canonical mask.

10.7 Compatible invariants and collapse stratification

For $F \in \mathbf{RMah}_p^{\times}$, define

$$\mathcal{J}_p^{\mathrm{rm}}(F) = \big(R_F, \omega_{p-1}(R_F), D(R_F), \mathcal{V}_p(F), r_p(F), \mathcal{K}_p(F), \kappa_p(F), \mathcal{A}_p(F)\big)\,. \tag{38}$$

Theorem 10.12 (Invariant descent on the rational-mask atlas)**.** *The native mask, Cartier-module, and automaton invariants admit coherent comparison maps to $\mathcal{J}_p^{\mathrm{rm}}$. They descend to the category in Equation (37). The descended invariant is complete on $\mathbf{RMah}_p^{\times}$.*

Proof. The fixed-denominator intertwining theorem identifies the reachable numerator span with $\mathcal{V}_p(F)$ and the reachable numerator set with $\mathcal{K}_p(F)$. Hence all components of the package agree under the bridges. Coherence follows from theorem 10.11. If two packages agree, their reduced canonical masks agree. The coefficientwise mask product then reconstructs the same formal series, so the semantic objects are equal. □

The obstruction ω_{p-1} divides the semantic category into two exact loci:

$$\mathbf{RMah}_p^{\times} = \mathbf{RMah}_{p,\mathrm{col}}^{\times} \sqcup \mathbf{RMah}_{p,\mathrm{ncol}}^{\times}.$$

The collapse locus consists of rational, hence ultimately periodic, series. The noncollapse locus consists of non-C-finite automatic series. The unit-periodic atlas of Periodic atlas is the persistent, zero-transient subatlas inside $\mathbf{RMah}_{p,\mathrm{col}}^{\times}$.

10.8 Exact examples and strict boundaries

Cantor-digit indicator. Over $\mathbb{F}_3$, let

$$R(z) = 1 + z.$$

Then

$$F(z) = \prod_{j \geq 0} (1 + z^{3^j})$$

has coefficient one exactly when the ternary expansion of the index uses only the digits zero and one. The functional equation is

$$F(z) = (1+z)F(z^3), \qquad F(z)^2 = \frac{1}{1+z}.$$

Table 15: Named exact examples in the rational-mask digital atlas.

Example	Field	Reduced mask A/B	First coefficients	r_p/κ_p	Collapse status / compiler bound
Cantor-digit indicator	$\mathbb{F}_3$	$(1+z)/(1)$	110110000110110000	1/2	non-collapse compiler bound 9
Square-root denominator	$\mathbb{F}_3$	$(1)/(1+z)$	121121000121121000	2/4	non-collapse compiler bound 9
Alternating collapse	$\mathbb{F}_3$	$(1+2z+z^2)/(1)$	1212121212121212	1/2	collapse compiler bound 9
Finite transient	$\mathbb{F}_2$	$(1)/(1+z+z^2)$	1110000000000000	3/4	collapse compiler bound 8

The factor $1+z$ has odd divisor multiplicity, so the collapse obstruction is nonzero. Thus the sequence is automatic but not C-finite or eventually periodic. The canonical automaton has two states, while the Cartier rank is one.

A genuine denominator example. Over $\mathbb{F}_3$, take

$$R(z) = \frac{1}{1+z}.$$

Then

$$F(z)^2 = 1+z.$$

The reduced denominator remains fixed under every Cartier section. The ambient compiler has dimension two, its reachable numerator set is

$$\{0, 1, 2, 1+z\},$$

and its reachable span has dimension two. Hence

$$r_3(F) = 2, \qquad \kappa_3(F) = 4.$$

The nonzero collapse obstruction again proves that the automatic coefficient sequence is not eventually periodic.

Collapse back to the periodic atlas. For

$$R(z) = (1+z)^2 = 1+2z+z^2$$

over $\mathbb{F}_3$, one has

$$F(z)^2 = \frac{1}{(1+z)^2}, \qquad F(z) = \frac{1}{1+z}.$$

The coefficients alternate between one and two. This object lies in the rational-collapse locus and in the unit-periodic atlas of Periodic atlas.

Characteristic-two transient. Over $\mathbb{F}_2$, choose

$$R(z) = \frac{1}{1+z+z^2}.$$

Since $p-1=1$,

$$F(z) = 1+z+z^2.$$

Its coefficient sequence is $1,1,1,0,0,\ldots$. It is ultimately periodic but not purely periodic. This is the simplest exact witness that Rational-mask atlas extends the semantic domain even in characteristic two.

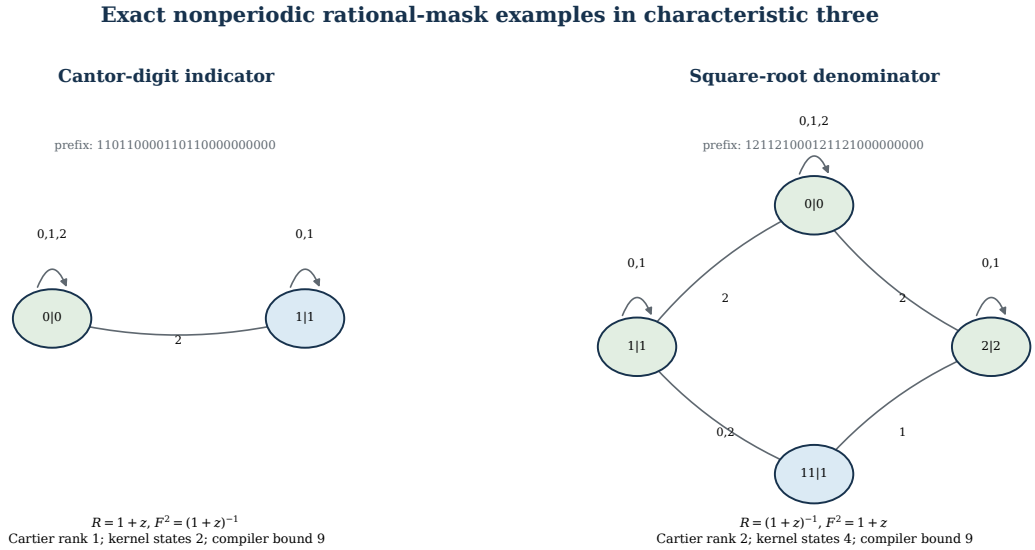


Figure 18: Two exact nonperiodic objects in the rational-mask atlas. State labels give the reachable numerator polynomial followed by its output.

Negative control: automatic does not imply rational mask. Let $t_n \in \mathbb{F}_2$ be the parity of the binary digit sum of n, and put

$$u_n = 1 + t_n.$$

The sequence u is unit and 2-automatic. It is not eventually periodic. Indeed, eventual period T would propagate backwards through the identities

$$t_{2^k+n} = 1 + t_n \qquad (0 \le n < 2^k)$$

and force a pure period T. Writing $T = 2^\nu q$ with q odd, the binary digit count of $2^m - T$ has parity

$$m - \nu - s_2(q-1)$$

for all sufficiently large m. Choosing the parity of m makes

$$t_{2^m-T} \neq t_{2^m},$$

contradicting pure periodicity.

Let $U(z) = \sum_{n\ge 0} u_n z^n$. If its canonical mask were rational, then theorem 10.3 with $p=2$ would make U rational, hence eventually periodic. Therefore $U/U(z^2)$ is not rational. The rational-mask atlas is a proper normalized subatlas of the unit automatic stratum.

Table 16: Exact finite validation of the Block 04 rational-mask digital atlas.

Check	Recorded result
Enumerated reduced unit masks	$p = 2$ with $\deg A, \deg B \leq 3$, $p = 3$ with $\deg A, \deg B \leq 2$, $p = 5$ with $\deg A, \deg B \leq 1$; 125 masks in total.
Rational-collapse split	51 collapse masks and 74 genuinely nonrational mask series.
Exact identities and state checks	11391 checks; 0 failures.
Arithmetic model	exact polynomial, finite-field, Cartier-transition, and formal-series arithmetic.
Overall status	PASS.
Boundary	Finite enumeration checks the compiler, reachable Cartier module, state hierarchy, examples, and collapse classification on the declared ranges. It does not replace the unbounded proofs or classify all automatic sequences.

10.9 Exact finite verification

The Rational-mask atlas verifier enumerates every reduced unit mask with

$$\deg A, \deg B \leq 3 \quad (p = 2), \qquad \deg A, \deg B \leq 2 \quad (p = 3), \qquad \deg A, \deg B \leq 1 \quad (p = 5).$$

For every mask it checks, using exact finite-field and polynomial arithmetic:

1. the invariant degree bound on the full ambient numerator space;
2. the functional-equation coefficient recursion against compiler output;
3. transition closure and fixed-denominator intertwining;
4. distinguishability of every pair of reachable numerator states;
5. the identities
$$W(A, B) \cong \mathcal{V}_p(F), \qquad \mathcal{R}(A, B) \cong \mathcal{K}_p(F);$$
6. the state hierarchy in Equation (35);
7. characteristic-two collapse and the named collapse or noncollapse examples.

The finite calculation validates the implementation on the recorded ranges. It is not used as a substitute for the unbounded proofs.

Section result. The verified heterogeneous atlas now extends beyond pure periodicity. Every unit rational functional-Mahler mask over $\mathbb{F}_p$ has a finite canonical Cartier module and a finite canonical p-kernel automaton. The three normalized charts satisfy coherence and effective descent. The fixed-denominator compiler has an exact reachable core: reachable numerator states are already the minimal digital states, and their span is the minimal Cartier module. The rational-collapse obstruction separates the ultimately periodic sublocus from genuinely non-C-finite automatic series. The resulting invariant package is globally complete on the declared rational-mask semantic category, but no claim is made for automatic sequences whose canonical mask is nonrational.

11 Fixed-radix regular normalization and the automatic finite-orbit boundary

Rational-mask atlas closes a substantial automatic slice, but its semantic domain is still defined by the existence of a rational functional-Mahler mask. The next digital question is more basic and more general:

what is the canonical normalized representation of a sequence whose complete fixed-radix section orbit spans a finite-dimensional vector space, even when that orbit itself is infinite?

This section answers that question over a field. It separates two invariants that coincide only in special cases:

linear section rank and cardinality of the section orbit.

The first controls regularity and minimal digit-linear dimension. The second controls automaticity and minimal deterministic-state cardinality. Their exact relationship is the automatic–regular boundary developed below.

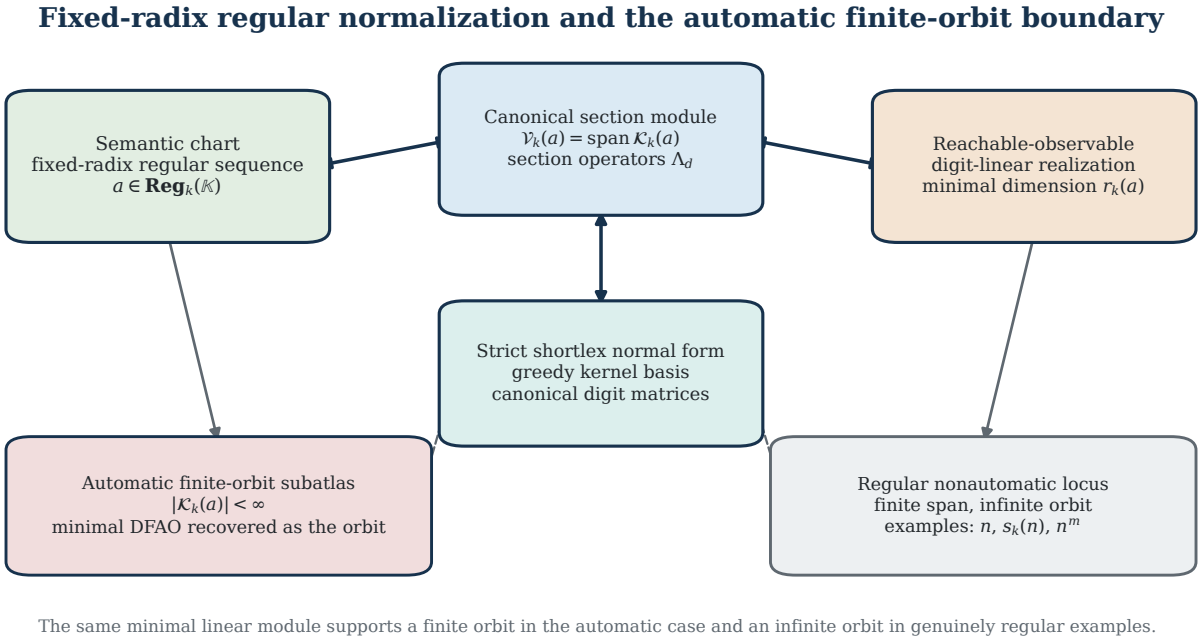


Figure 19: The fixed-radix regular atlas. The canonical section module and its strict shortlex matrix form exist throughout the regular locus. The minimal DFAO chart appears exactly on the finite-orbit sublocus.

11.1 Section operators and the canonical kernel module

Fix an integer radix $k \geq 2$ and a field K. Let

$$\Sigma_k = \{0, 1, \ldots, k-1\}.$$

Words are read from least significant digit to most significant digit. For

$$w = d_0 d_1 \cdots d_{\ell-1} \in \Sigma_k^*,$$

define

$$|w| = \ell, \qquad [w]_k = \sum_{j=0}^{\ell-1} d_j k^j.$$

The empty word ε has length zero and value zero.

For a sequence $a = (a_n)_{n \geq 0} \in K^{\mathbb{N}}$, define the digit-section operators

$$(\Lambda_d a)(n) = a(kn + d), \qquad 0 \leq d < k.$$

For a word $w = d_0 \cdots d_{\ell-1}$, put

$$\Lambda_w = \Lambda_{d_{\ell-1}} \cdots \Lambda_{d_1} \Lambda_{d_0}.$$

Then

$$(\Lambda_w a)(n) = a\big(k^{|w|} n + [w]_k\big). \tag{39}$$

Definition 11.1 (Fixed-radix kernel and section module)**.** The k-kernel and its linear span are

$$\mathcal{K}_k(a) = \{\Lambda_w a : w \in \Sigma_k^*\}, \qquad \mathcal{V}_k(a) = \operatorname{span}_K \mathcal{K}_k(a).$$

The sequence is *k-regular over K* when $\mathcal{V}_k(a)$ is finite dimensional. Its regular rank is

$$r_k(a) = \dim_K \mathcal{V}_k(a).$$

It is *k-automatic in the present reading convention* when $\mathcal{K}_k(a)$ is finite. In that case set

$$\kappa_k(a) = |\mathcal{K}_k(a)|.$$

The terminology records two different finiteness statements. Regularity asks for a finite-dimensional linear hull. Automaticity asks for a finite orbit under the section semigroup itself.

Definition 11.2 (Digit-linear realization)**.** A digit-linear realization of a is a tuple

$$\mathsf{L} = (W, A_0, \dots, A_{k-1}, v, \lambda),$$

where W is a finite-dimensional K-vector space, $A_d \in \operatorname{End}_K(W)$, $v \in W$, and $\lambda \in W^*$, such that for every word $w = d_0 \cdots d_{\ell-1}$,

$$a_{[w]_k} = \lambda A_w v, \qquad A_w = A_{d_{\ell-1}} \cdots A_{d_0}. \tag{40}$$

Because different words can represent the same integer after terminal high zeros are appended, equation (40) includes the required leading-zero consistency.

Theorem 11.3 (Section-module realization theorem)**.** *For a sequence $a \in K^{\mathbb{N}}$, the following are equivalent:*

1. *a is k-regular over K;*
2. *a admits a finite-dimensional digit-linear realization.*

When these conditions hold, the tuple

$$\mathsf{C}_k(a) = \big(\mathcal{V}_k(a), \Lambda_0, \dots, \Lambda_{k-1}, a, \epsilon_0\big), \qquad \epsilon_0(f) = f(0), \tag{41}$$

is a canonical realization of dimension $r_k(a)$.

Proof. Assume first that $\mathcal{V}_k(a)$ is finite dimensional. It is invariant under every Λ_d by construction. Choose any basis, represent the section operators by matrices, take the initial vector to be a, and take output to be evaluation at zero. For every word w,

$$\epsilon_0(\Lambda_w a) = (\Lambda_w a)(0) = a_{[w]_k},$$

so the resulting realization satisfies (40).

Conversely, suppose a has a realization (W, A_d, v, λ). For $x \in W$, define its represented behavior by

$$\mathcal{B}(x)(n) = \lambda A_{\mathrm{dig}_k(n)} x,$$

where $A_{\mathrm{dig}_k(n)}$ is the product associated with the canonical least-significant-digit-first expansion of n. Equation (40) and concatenation of digit words give

$$\Lambda_w a = \mathcal{B}(A_w v).$$

Hence every kernel section belongs to $\mathcal{B}(W)$, so

$$\dim_K \mathcal{V}_k(a) \le \dim_K W.$$

Thus a is regular. The realization in (41) has dimension exactly $r_k(a)$. □

Proposition 11.4 (Reachability and observability)**.** *The canonical section realization $\mathsf{C}_k(a)$ is linearly reachable and observable:*

$$\operatorname{span}_K\{\Lambda_w a : w \in \Sigma_k^*\} = \mathcal{V}_k(a),$$

and

$$\bigcap_{w\in\Sigma_k^*} \ker(\epsilon_0 \Lambda_w) = \{0\}.$$

Proof. Reachability is the definition of $\mathcal{V}_k(a)$. If $f \in \mathcal{V}_k(a)$ satisfies $(\epsilon_0\Lambda_w)(f) = 0$ for every word w, then

$$f([w]_k) = 0$$

for every word. Every nonnegative integer has such an expansion, so $f = 0$. □

11.2 Word-Hankel rank and minimal digit-linear dimension

The canonical module can be read directly from a Hankel-type matrix indexed by digit words.

Definition 11.5 (Word-Hankel matrix)**.** The fixed-radix word-Hankel matrix of a is the infinite matrix

$$H_a(u, v) = a_{[u]_k + k^{|u|}[v]_k} = (\Lambda_u a)([v]_k), \qquad u, v \in \Sigma_k^*. \tag{42}$$

Theorem 11.6 (Regular rank theorem)**.** *Let a be k-regular over K. Then*

$$\boxed{r_k(a) = \operatorname{rank}_K H_a = \min\{\dim_K W : W \text{ supports a digit-linear realization of } a\}.} \tag{43}$$

Moreover:

1. *there exist finite word sets $U, V \subset \Sigma_k^*$, each of cardinality $r_k(a)$, for which the minor $H_a[U, V]$ is nonsingular;*
2. *every reachable-observable realization of dimension $r_k(a)$ is isomorphic to $\mathsf{C}_k(a)$;*
3. *any two minimal reachable-observable realizations are related by a unique similarity intertwining all digit operators, the initial vector, and the output functional.*

Proof. The row indexed by u in H_a is the behavior of the section $\Lambda_u a$ evaluated on all continuation words. By observability in proposition 11.4, evaluation on all words is injective on $\mathcal{V}_k(a)$. Therefore the row span of H_a is naturally isomorphic to $\mathcal{V}_k(a)$, and

$$\operatorname{rank}_K H_a = r_k(a).$$

For an arbitrary realization $(W, A_d, v_{\mathrm{in}}, \lambda)$, concatenation gives the factorization

$$H_a(u, v) = \lambda A_v A_u v_{\mathrm{in}}.$$

Thus H_a factors through W, so rank $H_a \leq \dim W$. The canonical realization has dimension $r_k(a)$, proving the minimality formula.

An infinite matrix of finite rank $r_k(a)$ has a nonsingular finite $r_k(a) \times r_k(a)$ minor, which gives the first assertion. For the remaining assertions, define the behavior map of any realization by

$$\mathcal{B}_W(x)(n) = \lambda A_{\mathrm{dig}_k(n)} x.$$

Observability makes this map injective. Reachability and equality of the represented sequence make its image equal to $\mathcal{V}_k(a)$. In minimal dimension it is therefore an isomorphism intertwining the digit operators, initial vector, and output. Composing the two behavior isomorphisms gives the unique similarity between any two minimal reachable-observable realizations. □

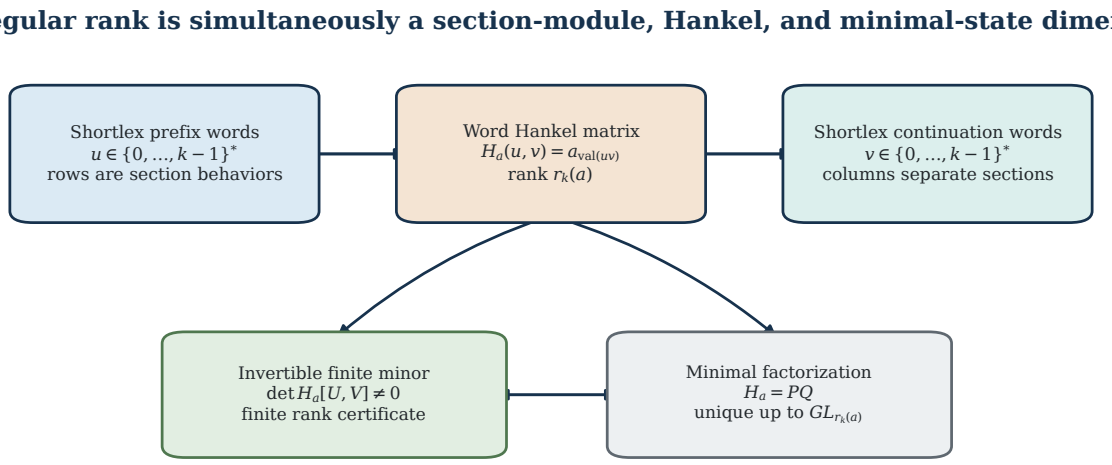


Figure 20: Three exact meanings of the regular rank. The finite nonsingular minor is a certificate rather than an approximation: it proves a lower bound that is attained by the canonical section module.

Remark 11.7 (Finite certification). The theorem is infinite in statement but finite in certificate. A proposed rank r can be certified by:

1. an explicit r-dimensional section-stable span containing a, which proves $r_k(a) \leq r$;
2. an explicit nonsingular $r \times r$ word-Hankel minor, which proves $r_k(a) \geq r$.

When both are supplied, the regular rank is exact.

11.3 Strict shortlex normalization

Minimal realizations are unique up to similarity, but the stratified atlas needs a literal normalized chart rather than an unnamed similarity class. The actual section functions supply a canonical basis choice.

Order Σ_k^* by shortlex order: shorter words precede longer words, and words of equal length are ordered lexicographically. Starting from the empty list, scan the words in this order and retain a word precisely when its section is not in the span of the previously retained sections.

Definition 11.8 (Greedy shortlex kernel basis)**.** For a regular sequence a, let

$$U_a = (u_1, \dots, u_{r_k(a)})$$

be the unique ordered list produced by the greedy shortlex rule, and put

$$b_i = \Lambda_{u_i} a.$$

For the zero sequence, the list is empty.

Because membership in a linear span is intrinsic, the resulting ordered basis depends only on the represented sequence, the field, the radix, and the fixed word order.

Construction 11.9 (Strict shortlex realization)**.** Write

$$\Lambda_d b_j = \sum_{i=1}^{r_k(a)} (M_d(a))_{ij} b_i, \qquad a = \sum_i v_i(a) b_i, \qquad \ell_i(a) = b_i(0).$$

The strict shortlex record is

$$\mathsf{SL}_{k,K}(a) = \big(U_a, M_0(a), \dots, M_{k-1}(a), v(a), \ell(a)\big). \tag{44}$$

Theorem 11.10 (Strict fixed-radix normal form)**.** *The record $\mathsf{SL}_{k,K}(a)$ is uniquely determined by a and reconstructs a exactly through*

$$a_{[w]_k} = \ell(a) M_w(a) v(a).$$

It has dimension $r_k(a)$. Two regular sequences have the same strict shortlex record if and only if they are equal.

Proof. The greedy word list is uniquely determined by the actual section functions. Once the ordered basis is fixed, the coefficients of every digit section, the initial vector, and the output functional are unique. The reconstruction identity is the coordinate form of the canonical section realization. Equality of records therefore implies equality of every coefficient, while equality of sequences gives the same sections and hence the same greedy record. □

Theorem 11.11 (Fixed-radix regular normalized atlas)**.** *Let $\mathbf{Reg}_k(K)$ be the discrete semantic category of K-valued k-regular sequences. Let $\mathbf{Sec}_k^{\mathrm{can}}(K)$ be the category of canonical section-module records (41), and let $\mathbf{Lin}_{k,K}^{\mathrm{sl}}$ be the category of strict shortlex records (44), with morphisms required to preserve the represented sequence. Then*

$$\boxed{\mathbf{Reg}_k(K) \simeq \mathbf{Sec}_k^{\mathrm{can}}(K) \simeq \mathbf{Lin}_{k,K}^{\mathrm{sl}}.} \tag{45}$$

Every normalized fiber over a fixed sequence is contractible. Consequently the associated finite presentation atlas has effective object and morphism descent.

Proof. A sequence determines its canonical section module and its strict shortlex record functorially. Conversely, either record reconstructs every coefficient. The two composites are identities because the section module is defined from the represented sequence and the shortlex basis is intrinsic. The uniqueness in theorem 11.10 leaves no nontrivial automorphism in a normalized fiber. Compatible local objects and morphisms therefore glue uniquely. □

Definition 11.12 (Fixed-radix regular invariant)**.** The normalized fixed-radix invariant is

$$\mathcal{J}^{\mathrm{reg}}_{k,K}(a) = \big(r_k(a), \mathsf{C}_k(a), \mathsf{SL}_{k,K}(a), [H_a[U,V]]\big), \tag{46}$$

where $[H_a[U,V]]$ denotes any recorded nonsingular rank certificate together with its word addresses. The strict normal form makes this package complete on $\mathbf{Reg}_k(K)$.

The finite Hankel minor is not canonical unless an ordering rule is imposed on candidate column sets. It is therefore treated as a certificate attached to the canonical invariant, not as the invariant itself.

11.4 The automatic finite-orbit subatlas

The automatic locus is not a second unrelated theory. It is the locus on which the orbit of the distinguished initial vector inside the canonical regular module is finite.

Theorem 11.13 (Kernel automaton theorem)**.** *Assume $\mathcal{K}_k(a)$ is finite. Define a deterministic automaton with output by*

$$\text{state set } \mathcal{K}_k(a), \qquad f \xrightarrow{d} \Lambda_d f, \qquad \text{initial state } a, \qquad \text{output } f(0).$$

Then:

1. *the automaton represents a under least-significant-digit-first reading;*
2. *every state is reachable;*
3. *distinct states are behaviorally distinguishable;*
4. *it is the unique minimal DFAO up to state renaming.*

Proof. After reading a word w, the state is $\Lambda_w a$, whose output is $a_{[w]_k}$. Reachability is immediate from the definition of the kernel. If $f \neq g$, then there is an integer n with $f(n) \neq g(n)$. Reading a digit word for n from either state distinguishes them. Thus no two states can be merged, and the reachable automaton is minimal. The usual residual-behavior argument then gives uniqueness up to renaming. □

Theorem 11.14 (Automatic–regular finite-orbit equivalence)**.** *For a k-regular sequence a, the following are equivalent:*

1. *a is k-automatic;*
2. *the orbit $\{\Lambda_w a : w \in \Sigma_k^*\}$ of the initial vector in $\mathsf{C}_k(a)$ is finite;*
3. *the orbit of $v(a)$ under the strict shortlex digit matrices is finite.*

On this locus, the minimal DFAO is recovered as the orbit itself. Hence

$$\mathbf{Aut}_k(K) \simeq \mathbf{FinOrbSec}_k(K) \simeq \mathbf{FinOrbLin}^{\mathrm{sl}}_{k,K} \simeq \mathbf{DFAO}^{\min}_{k,K}. \tag{47}$$

If $K = \mathbb{F}_q$ is finite, every k-regular K-valued sequence is automatic.

Proof. The first three conditions are the same orbit statement written in semantic, intrinsic-module, and coordinate form. The kernel automaton in theorem 11.13 recovers the deterministic chart, while its state-to-section map recovers the finite orbit inside the canonical module. These constructions are inverse on normalized objects. If $K = \mathbb{F}_q$ and $r_k(a) < \infty$, the vector space $\mathcal{V}_k(a)$ has only $q^{r_k(a)}$ elements, so the section orbit is automatically finite. □

Corollary 11.15 (State-rank hierarchy). *Let a be automatic with values in $\mathbb{F}_q$. Then*

$$\boxed{r_k(a) \le \kappa_k(a) \le q^{r_k(a)}.} \tag{48}$$

The first inequality can be strict: behavior-distinct kernel states may be linearly dependent.

Proof. The kernel states span $\mathcal{V}_k(a)$, so their number is at least its dimension. They form a subset of the $q^{r_k(a)}$-element vector space $\mathcal{V}_k(a)$, proving the upper bound. □

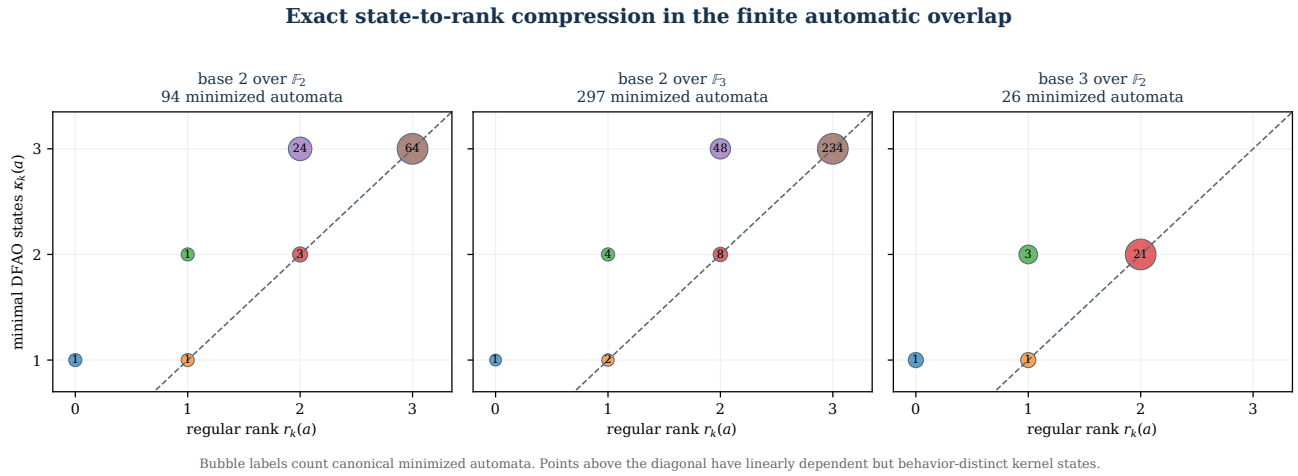


Figure 21: Exact bounded enumeration of the automatic overlap. Bubble labels count distinct minimized automata. Points above the diagonal are finite kernel orbits whose state behaviors are distinct but linearly dependent.

The corollary explains why deterministic state minimization and linear-state minimization must remain distinct coordinates. The minimal automaton records the finite orbit as a set with digit action. The regular rank records the dimension of its linear span. Neither number alone reconstructs the other.

11.5 Exact regular examples beyond automaticity

Over an infinite field, finite linear rank does not force a finite section orbit. The following families give exact, unbounded witnesses.

Theorem 11.16 (Polynomial regular rank in characteristic zero). *Let K have characteristic zero and let $P \in K[x]$ be nonzero of degree m. For the sequence $a_n = P(n)$,*

$$r_k(a) = m + 1.$$

If $m \ge 1$, its k-kernel is infinite.

Proof. Every section has the form

$$(\Lambda_w a)(n) = P(k^{|w|}n + [w]_k),$$

which is a polynomial in n of degree at most m. Hence the section span has dimension at most $m + 1$. Choose ℓ with $k^\ell > m$ and consider the $m + 1$ sections

$$P(k^\ell n + r), \qquad 0 \le r \le m.$$

When expanded in the basis $1, n, \dots, n^m$, their coefficient functions as functions of r have successive degrees $m, m-1, \dots, 0$, with nonzero leading coefficients. Evaluation at the distinct points $0, 1, \dots, m$ therefore gives an invertible generalized Vandermonde matrix. These sections are linearly independent, proving rank $m+1$.

For $m \geq 1$, the sections $P(k^\ell n + r)$ are distinct as r varies through arbitrarily large complete residue ranges, so the kernel orbit is infinite. □

Corollary 11.17. *For $a_n = n$, one has $r_k(a) = 2$ and exactly k^ℓ distinct sections at depth ℓ. For $a_n = n^2$, one has $r_k(a) = 3$ and again exactly k^ℓ distinct sections at depth ℓ.*

Theorem 11.18 (Digit-sum regularity)**.** *Let $s_k(n)$ be the sum of the base-k digits of n, regarded as a sequence over a characteristic-zero field. Then*

$$r_k(s_k) = 2,$$

and

$$\mathcal{K}_k(s_k) = \{s_k + c : c \in \mathbb{Z}_{\geq 0}\}. \tag{49}$$

At exact depth ℓ, the number of distinct sections is

$$(k-1)\ell + 1.$$

Thus s_k is k-regular but not k-automatic over a characteristic-zero field.

Proof. The digit identity

$$s_k(kn + d) = s_k(n) + d$$

gives

$$\Lambda_d s_k = s_k + d.$$

Iterating shows that every section is $s_k + c$, so the span is contained in $\operatorname{span}\{1, s_k\}$. The two sequences are independent, giving rank two. At depth ℓ, the additive constant is the digit sum of an ℓ-digit residue. Every integer from 0 through $(k-1)\ell$ occurs, giving exactly $(k-1)\ell + 1$ states. □

Table 17: Named examples at the automatic–regular boundary.

Example	Base / field	Regular rank	Kernel or states	Classification
Ternary Cantor-digit indicator	3 / $\mathbb{F}_3$	1	2	automatic and regular
Thue-Morse parity	2 / $\mathbb{F}_2$	2	2	automatic and regular
Identity sequence n	2 / $\mathbb{Q}$	2	infinite	regular, not automatic
Binary digit sum $s_2(n)$	2 / $\mathbb{Q}$	2	infinite	regular, not automatic
Square sequence n^2	2 / $\mathbb{Q}$	3	infinite	regular, not automatic

11.6 Exact finite verification

The Fixed-radix regular atlas verifier performs two complementary exact audits.

First, it exhaustively enumerates bounded deterministic automata in three configurations:

$$(k, q, s_{\max}) = (2, 2, 3), \qquad (2, 3, 3), \qquad (3, 2, 2).$$

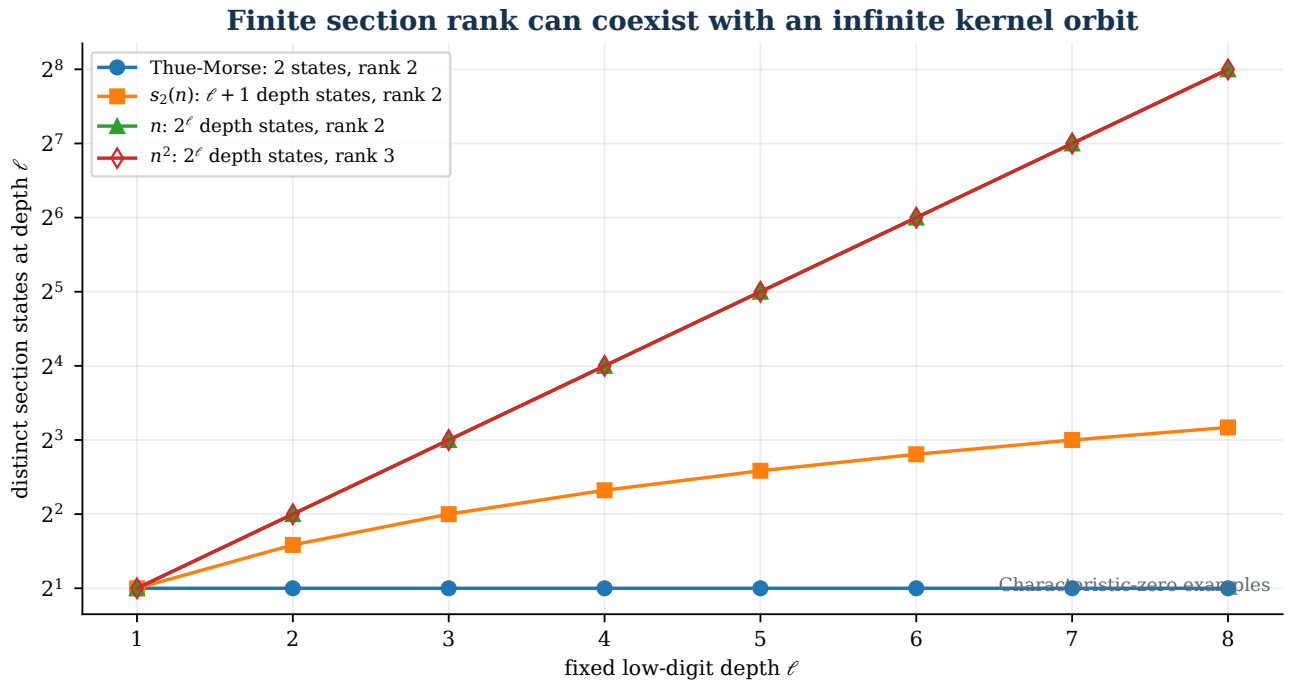


Figure 22: Fixed section rank does not bound orbit cardinality over an infinite field. The Thue–Morse orbit is finite, while n, n^2, and the digit sum have finite linear rank but unboundedly many distinct sections.

Table 18: Exact finite validation of the Fixed-radix regular atlas fixed-radix regular atlas.

Check	Recorded result
Exhaustive finite-state enumeration	base 2 over $\mathbb{F}_2$ through 3 raw states: 5,898 raw candidates, 2,830 leading-zero-consistent candidates, 94 distinct minimized automata; base 2 over $\mathbb{F}_3$ through 3 raw states: 19,830 raw candidates, 6,990 leading-zero-consistent candidates, 297 distinct minimized automata; base 3 over $\mathbb{F}_2$ through 2 raw states: 258 raw candidates, 170 leading-zero-consistent candidates, 26 distinct minimized automata.
Automatic-to-linear checks	86,207 exact checks; 83 minimized automata with strict state-to-rank compression.
Unbounded regular examples	Polynomial sequences n^m for bases 2 and 3 through degree 6, plus digit sums in bases 2, 3, and 4; 2,438 exact checks.
Combined Fixed-radix regular atlas checks	89,055 checks; 0 failures.
Overall status	PASS.
Boundary	The computation validates the bounded automata and named exact examples. It does not prove an all-radix, all-ring, semiring, or all-moduli minimality theorem.

Every candidate is restricted to reachable states, minimized by behavioral equivalence, and filtered by the terminal-high-zero consistency required by integer semantics. For every resulting normalized automaton the verifier checks:

1. reachability and pairwise behavioral distinguishability;
2. construction of the canonical linear quotient from continuation-output rows;
3. transition and output equivariance of the state embedding;
4. the inequalities in Equation (48);
5. equality of word-Hankel rank and regular rank;
6. exact equality between the finite linear orbit and the minimized automaton state set;
7. coefficient agreement on a fixed exact test range.

Second, it checks unbounded symbolic families using exact integer and rational matrices. Polynomial sequences n^m are checked for bases two and three through degree six, and digit sums are checked in bases two, three, and four, including their exact depth-orbit growth formulas.

The finite audit records 89,055 exact checks and no failures. It supports the implementation, examples, and bounded state-rank distribution. The unbounded theorems above are proved symbolically and do not depend on the enumeration.

Exact boundary of Fixed-radix regular atlas. The normalization theorem is fixed-radix and field-valued. It does not yet give one integral normal form valid simultaneously modulo every prime power, a semiring minimality theorem, or a canonical all-radix comparison invariant. Over a general commutative ring, the section module can have torsion, fail to be free, and admit inequivalent minimal generating systems. Saturation, Smith/Fitting data, prime-power reduction towers, and local completion therefore remain separate arithmetic obligations rather than being hidden inside the field theorem.

Section result. Every field-valued fixed-radix regular sequence has a canonical section module, a minimal reachable-observable digit-linear realization, an exact word-Hankel rank, and a strict shortlex normal form. These charts satisfy effective normalized descent and yield a complete invariant on the declared fixed-radix regular category. The automatic locus is exactly the finite-orbit locus inside the same module, where the minimal DFAO is recovered as the orbit of the distinguished initial vector. Over finite fields regularity and automaticity coincide, while over characteristic-zero fields polynomial and digit-sum examples have finite rank and infinite section orbit. This closes fixed-radix linear-state minimality over fields, not the all-moduli or integral regular-sequence problem.

12 Integral saturation, Smith–Fitting defects, and prime-power effectivity

Fixed-radix regular atlas proves a complete fixed-radix normalization theorem over a field. The next question is not obtained by replacing the coefficient field with $\mathbb{Z}$ symbol by symbol. Over $\mathbb{Z}$, two canonical lattices appear inside the same rational section space. Their finite-index difference is invisible over $\mathbb{Q}$, but it controls reduction modulo every integer, the appearance of ghost coordinates, and the distinction between an effective p-adic lift and an effective integral lift.

Throughout this section, $k \geq 2$ is fixed and

$$a : \mathbb{N} \longrightarrow \mathbb{Z}$$

is an integer-valued sequence whose rational section space

$$V_{\mathbb{Q}}(a) = \operatorname{span}_{\mathbb{Q}} \mathcal{K}_k(a) \subseteq \mathbb{Q}^{\mathbb{N}}$$

has finite dimension

$$\rho_k(a) = \dim_{\mathbb{Q}} V_{\mathbb{Q}}(a).$$

Thus a is k-regular after extension of scalars to $\mathbb{Q}$.

12.1 The raw section lattice and its canonical saturation

Definition 12.1 (Raw and saturated section lattices)**.** The *raw section lattice* and the *saturated section lattice* are

$$M_k(a) = \operatorname{span}_{\mathbb{Z}} \mathcal{K}_k(a), \qquad L_k(a) = V_{\mathbb{Q}}(a) \cap \mathbb{Z}^{\mathbb{N}}.$$

The intersection defining $L_k(a)$ is taken inside $\mathbb{Q}^{\mathbb{N}}$.

The raw lattice is generated by the actual radix sections. The saturated lattice contains every integer-valued sequence lying in their rational span. These lattices agree over $\mathbb{Q}$, but need not agree over $\mathbb{Z}$.

Lemma 12.2 (Finite coordinate detection)**.** *Let $V \subseteq \mathbb{Q}^{\mathbb{N}}$ be a vector space of finite dimension ρ. There exist indices*

$$0 \leq n_1 < \cdots < n_\rho$$

such that the evaluation map

$$E_{\mathbf{n}} : V \longrightarrow \mathbb{Q}^{\rho}, \qquad f \longmapsto \big(f(n_1), \ldots, f(n_\rho)\big)$$

is an isomorphism.

Proof. The coordinate functionals on $\mathbb{Q}^{\mathbb{N}}$ separate points of V. Choose n_1 for which some element of V has nonzero value. If $n_1, \ldots, n_j$ have been chosen and the joint evaluation map has rank $j < \rho$, its kernel contains a nonzero f. Choose n_{j+1} with $f(n_{j+1}) \neq 0$. The rank increases. After ρ steps, the evaluation map has full rank. □

Theorem 12.3 (Canonical finite saturated section lattice)**.** *The groups $M_k(a)$ and $L_k(a)$ are free abelian of rank $\rho_k(a)$. They are stable under every digit section Λ_d, and*

$$M_k(a) \subseteq L_k(a)$$

has finite index. Moreover, $L_k(a)$ is the unique maximal subgroup of $V_{\mathbb{Q}}(a)$ whose elements are integer-valued.

Proof. Choose a coordinate-detection set $\mathbf{n}$ by lemma 12.2. Evaluation embeds $L_k(a)$ into $\mathbb{Z}^\rho$, so $L_k(a)$ is a finitely generated free abelian group of rank at most ρ. The raw lattice $M_k(a)$ is contained in $L_k(a)$, and its rational span is all of $V_{\mathbb{Q}}(a)$. Hence $M_k(a)$ has rank ρ, and therefore $L_k(a)$ also has rank ρ. A subgroup of a finitely generated free abelian group is finitely generated and free, so $M_k(a)$ is free of rank ρ. Equal-rank lattice inclusion has finite index.

Every Λ_d preserves $V_{\mathbb{Q}}(a)$ and preserves integer-valued sequences. Hence it preserves $L_k(a)$. It also maps each radix section to another radix section, so it preserves $M_k(a)$. Finally, any integer-valued subgroup of $V_{\mathbb{Q}}(a)$ lies in $V_{\mathbb{Q}}(a) \cap \mathbb{Z}^{\mathbb{N}} = L_k(a)$. □

Definition 12.4 (Integral saturation defect)**.** The finite abelian group

$$C_k(a) = L_k(a)/M_k(a)$$

is the *integral saturation defect*. Its order

$$\delta_k(a) = [L_k(a) : M_k(a)]$$

is the *saturation index*.

By the Smith normal form, there are uniquely determined integers

$$1 \le d_1(a) \mid d_2(a) \mid \cdots \mid d_\rho(a)$$

such that

$$C_k(a) \cong \bigoplus_{i=1}^{\rho} \mathbb{Z}/d_i(a)\mathbb{Z}.$$

Factors with $d_i = 1$ are harmless but are retained so that the list has length ρ.

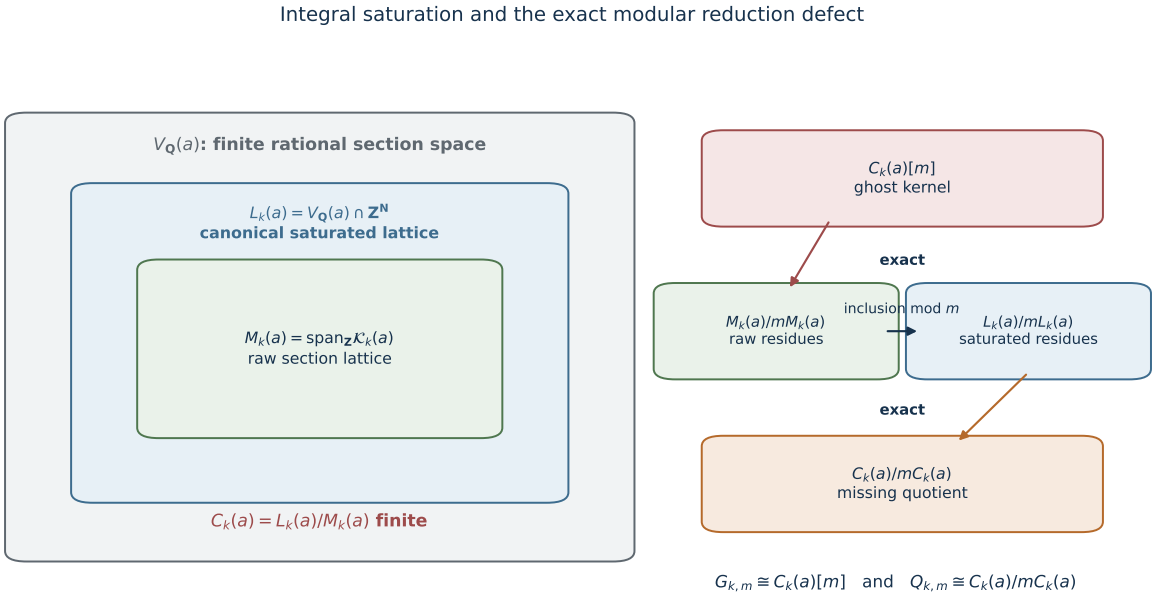


Figure 23: The rational section space contains the raw section lattice and its canonical integer-valued saturation. The finite quotient is invisible over $\mathbb{Q}$, but it controls every modular reduction.

Example 12.5 (A rank-two nontrivial defect)**.** Let $s_k(n)$ be the base-k digit-sum sequence and let $a = 2s_k$. Since

$$\Lambda_d s_k = s_k + d,$$

the rational section space is spanned by s_k and the constant sequence 1. The actual sections of $2s_k$ generate

$$M_k(2s_k) = 2\mathbb{Z}\, s_k \oplus 2\mathbb{Z}\, 1,$$

whereas

$$L_k(2s_k) = \mathbb{Z}\, s_k \oplus \mathbb{Z}\, 1.$$

Thus

$$C_k(2s_k) \cong (\mathbb{Z}/2\mathbb{Z})^2.$$

The rational regular rank is two, but reduction modulo two loses the entire semantic lattice.

12.2 The exact reduction sequence

For an integer $m \geq 2$, the raw coefficient reduction is

$$M_k(a)/mM_k(a),$$

while the semantic reduction inside the saturated lattice is

$$\overline{M}_{k,m}(a) = \operatorname{im}\left(M_k(a)/mM_k(a) \longrightarrow L_k(a)/mL_k(a)\right).$$

These need not agree.

Definition 12.6 (Ghost and missing reduction modules)**.** Define

$$G_{k,m}(a) = \ker\left(M_k(a)/mM_k(a) \longrightarrow L_k(a)/mL_k(a)\right)$$

and

$$Q_{k,m}(a) = \operatorname{coker}\left(M_k(a)/mM_k(a) \longrightarrow L_k(a)/mL_k(a)\right).$$

The first records raw residue coordinates that represent the zero semantic sequence. The second records saturated residue classes not reached by raw sections.

Theorem 12.7 (Smith–Tor reduction sequence)**.** *For every $m \geq 2$, tensoring*

$$0 \longrightarrow M_k(a) \longrightarrow L_k(a) \longrightarrow C_k(a) \longrightarrow 0$$

with $\mathbb{Z}/m\mathbb{Z}$ gives a canonical digit-equivariant exact sequence

$$0 \longrightarrow C_k(a)[m] \longrightarrow M_k(a)/mM_k(a) \longrightarrow L_k(a)/mL_k(a) \longrightarrow C_k(a)/mC_k(a) \longrightarrow 0.$$

Consequently,

$$G_{k,m}(a) \cong C_k(a)[m], \qquad Q_{k,m}(a) \cong C_k(a)/mC_k(a).$$

Proof. Both $M_k(a)$ and $L_k(a)$ are free abelian. The long exact Tor sequence therefore begins

$$0 \longrightarrow \operatorname{Tor}_1^{\mathbb{Z}}(C_k(a), \mathbb{Z}/m\mathbb{Z}) \longrightarrow M_k(a) \otimes \mathbb{Z}/m\mathbb{Z} \longrightarrow L_k(a) \otimes \mathbb{Z}/m\mathbb{Z}.$$

For any abelian group C,

$$\operatorname{Tor}_1^{\mathbb{Z}}(C, \mathbb{Z}/m\mathbb{Z}) \cong C[m], \qquad C \otimes \mathbb{Z}/m\mathbb{Z} \cong C/mC.$$

The digit-section maps preserve the original short exact sequence, so naturality of Tor makes the five-term sequence digit-equivariant. □

Corollary 12.8 (Exact Smith formulas)**.** *Let $g_i(m) = \gcd(m, d_i(a))$. Then*

$$G_{k,m}(a) \cong Q_{k,m}(a) \cong \bigoplus_{i=1}^{\rho} \mathbb{Z}/g_i(m)\mathbb{Z},$$

and

$$\left|\overline{M}_{k,m}(a)\right| = \prod_{i=1}^{\rho} \frac{m}{g_i(m)}.$$

The reduction map is an isomorphism if and only if

$$\gcd\big(m, d_\rho(a)\big) = 1.$$

Equivalently, every prime dividing a reduction defect divides the saturation index.

Proof. Choose Smith bases in which the inclusion $M_k(a) \hookrightarrow L_k(a)$ is the diagonal matrix

$$D = \operatorname{diag}(d_1, \dots, d_\rho).$$

Modulo m, the kernel and cokernel of multiplication by d_i are both cyclic of order $\gcd(m, d_i)$, and the image has order $m/\gcd(m, d_i)$. The map is invertible in every coordinate exactly when every d_i is a unit modulo m, which is equivalent to $\gcd(m, d_\rho) = 1$. □

Proposition 12.9 (Fitting-ideal certificate)**.** *For $0 \le j < \rho$,*

$$\operatorname{Fitt}_j(C_k(a)) = \left(d_1(a)d_2(a)\cdots d_{\rho-j}(a)\right),$$

and

$$\operatorname{Fitt}_j(C_k(a)) = \mathbb{Z} \qquad (j \ge \rho).$$

Thus the Smith profile and the Fitting filtration contain the same reduction-defect data at different resolutions.

Proof. Use the diagonal presentation $D : \mathbb{Z}^\rho \to \mathbb{Z}^\rho$. The j-th Fitting ideal is generated by the minors of size $\rho - j$. Since $d_1 \mid \cdots \mid d_\rho$, the product of the first $\rho - j$ diagonal entries divides every other such minor and is itself one of the minors. □

Boundary 12.10 (What the defect classifies). The group $C_k(a)$ classifies the finite-index failure of the raw section lattice to be saturated. It does not classify the whole integral digit module, the sequence, or its section orbit. Two nonisomorphic digit modules can have the same Smith profile.

12.3 Digit equivariance and the prime-power tower

Choose Smith bases for $M_k(a)$ and $L_k(a)$, and let

$$D = \operatorname{diag}(d_1, \dots, d_\rho)$$

be the inclusion matrix. Let A_d and B_d be the integer matrices of Λ_d on the raw and saturated lattices.

Proposition 12.11 (Intertwining across saturation)**.** *For every digit d,*

$$B_d D = D A_d.$$

Consequently, the raw modules, semantic images, ghost modules, and missing modules at every modulus form compatible digit modules.

Proof. The inclusion $M_k(a) \hookrightarrow L_k(a)$ commutes with Λ_d. The displayed matrix identity is that commutative square in the chosen bases. Reduction modulo m, passage to kernel, image, and cokernel preserve the induced digit action. □

Fix a prime p and write

$$\nu_i = \nu_{p,i}(a) = v_p(d_i(a)).$$

Theorem 12.12 (Prime-power defect profile)**.** *For $r \ge 1$,*

$$G_{k,p^r}(a) \cong Q_{k,p^r}(a) \cong \bigoplus_{i=1}^{\rho} \mathbb{Z}/p^{\min(r,\nu_i)}\mathbb{Z},$$

and

$$\left|\overline{M}_{k,p^r}(a)\right| = p^{\sum_{i=1}^{\rho}(r-\min(r,\nu_i))}.$$

The minimal number of generators of $\overline{M}_{k,p^r}(a)$ *as a* $\mathbb{Z}/p^r\mathbb{Z}$*-module is*

$$\mu_{p^r}(a) = \#\{i : \nu_i < r\}.$$

In particular, p *is a good reduction prime exactly when every* $\nu_i = 0$*.*

Proof. Apply corollary 12.8 to $m = p^r$. The i-th semantic image is the ideal

$$p^{\min(r,\nu_i)}\mathbb{Z}/p^r\mathbb{Z}.$$

It is nonzero and cyclic exactly when $\nu_i < r$. Over the local ring $\mathbb{Z}/p^r\mathbb{Z}$, the minimal number of generators of a direct sum of cyclic ideals is the number of nonzero summands. □

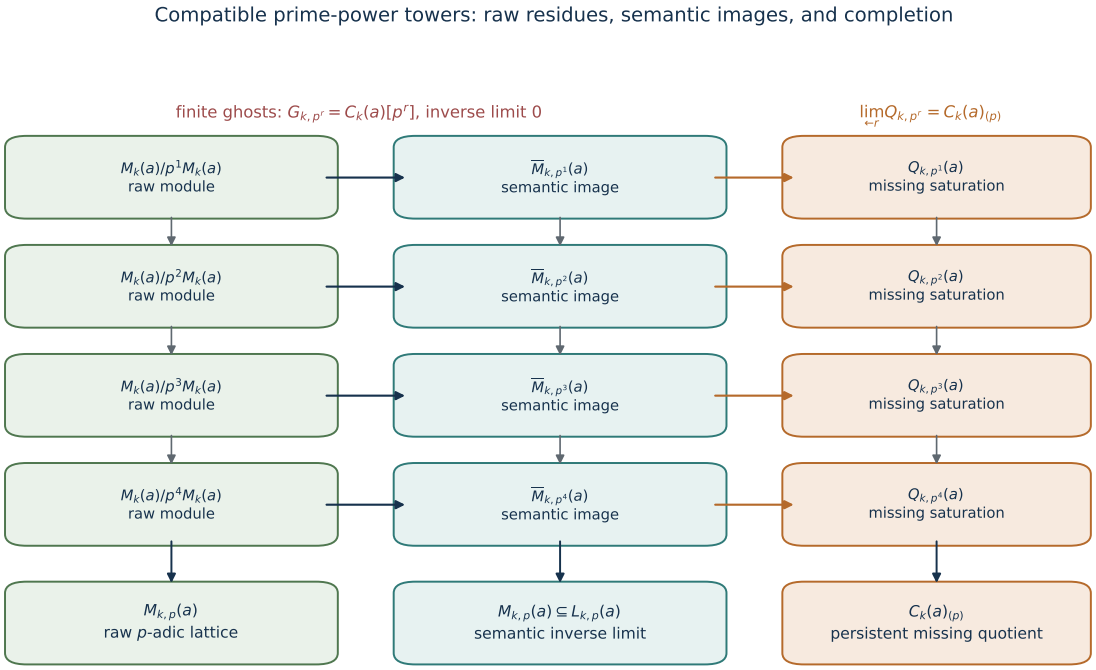


Figure 24: The raw residue tower maps onto the semantic prime-power tower. Ghost coordinates occur at finite depth when the raw lattice is not p-saturated. The compatible semantic tower converges to the embedded p-adic raw lattice.

Definition 12.13 (Prime-power section automaton)**.** The semantic p^r-kernel is

$$\mathcal{K}^{\mathrm{sem}}_{k,p^r}(a) = \{\Lambda_w a \bmod p^r L_k(a) : w \in \Sigma_k^*\} \subseteq \overline{M}_{k,p^r}(a).$$

Its cardinality is denoted

$$\kappa^{\mathrm{sem}}_{k,p^r}(a).$$

Theorem 12.14 (Finite prime-power automata)**.** *For every* $r \geq 1$*, the semantic* p^r*-kernel is finite and is the state set of the unique minimal least-significant-digit-first deterministic output automaton for the coefficient sequence* $a \bmod p^r$*. Moreover,*

$$\kappa^{\mathrm{sem}}_{k,p^r}(a) \leq \left|\overline{M}_{k,p^r}(a)\right| = p^{\sum_i(r-\min(r,\nu_i))}.$$

Reduction induces surjective digit-equivariant maps

$$\mathcal{K}^{\mathrm{sem}}_{k,p^{r+1}}(a) \twoheadrightarrow \mathcal{K}^{\mathrm{sem}}_{k,p^r}(a).$$

Proof. The semantic section module is finite, so its orbit under the digit maps is finite. As in the field-valued proof of theorem 11.13, every orbit state is reachable. Distinct section sequences have a continuation on which they differ, so they are behaviorally distinguishable. This gives the minimal deterministic automaton. The cardinality bound is the ambient module bound. The same section word at level $r+1$ reduces to the corresponding section word at level r, so the transition map on orbit states is surjective. □

Corollary 12.15 (Good-prime transfer)**.** *If $p \nmid \delta_k(a)$, then for every $r \geq 1$,*

$$M_k(a)/p^r M_k(a) \xrightarrow{\sim} \overline{M}_{k,p^r}(a) = L_k(a)/p^r L_k(a).$$

At a good prime, the raw and saturated prime-power towers agree exactly.

Table 19: Canonical integral defect profiles for named fixed-radix families.

Family	Smith profile	Bad primes	Interpretation
Digit sum s_k	$(1,1)$	none	Saturated rank-two lattice; no modular defect.
Doubled digit sum $2s_k$	$(2,2)$	2	Rank-two defect $(\mathbb{Z}/2)^2$; total semantic collapse modulo 2.
Constant sequence 2	(2)	2	One raw ghost coordinate modulo 2; one semantic state.
Constant sequence 6	(6)	2, 3	Bad primes 2 and 3; good reduction at every other prime.

12.4 p-adic completion and the disappearance of finite ghosts

Define

$$M_{k,p}(a) = M_k(a) \otimes_{\mathbb{Z}} \mathbb{Z}_p, \qquad L_{k,p}(a) = L_k(a) \otimes_{\mathbb{Z}} \mathbb{Z}_p.$$

Theorem 12.16 (p-adic saturation sequence)**.** *There is a digit-equivariant short exact sequence*

$$0 \longrightarrow M_{k,p}(a) \longrightarrow L_{k,p}(a) \longrightarrow C_k(a)_{(p)} \longrightarrow 0,$$

where $C_k(a)_{(p)}$ is the p-primary subgroup of $C_k(a)$. In Smith coordinates,

$$M_{k,p}(a) = \bigoplus_{i=1}^{\rho} p^{\nu_i}\mathbb{Z}_p e_i \subseteq \bigoplus_{i=1}^{\rho} \mathbb{Z}_p e_i = L_{k,p}(a).$$

Proof. The $\mathbb{Z}$-module $\mathbb{Z}_p$ is torsion free and hence flat. Tensor the integral short exact sequence with $\mathbb{Z}_p$. For a finite abelian group C,

$$C \otimes_{\mathbb{Z}} \mathbb{Z}_p \cong C_{(p)}.$$

The Smith-coordinate formula follows by tensoring the diagonal inclusion. □

Theorem 12.17 (Raw and semantic inverse limits)**.** *There are canonical digit-equivariant identifications*

$$\varprojlim_r M_k(a)/p^r M_k(a) \cong M_{k,p}(a)$$

and

$$\varprojlim_r \overline{M}_{k,p^r}(a) \cong M_{k,p}(a) \subseteq L_{k,p}(a).$$

Furthermore,

$$\varprojlim_r G_{k,p^r}(a) = 0, \qquad \varprojlim_r Q_{k,p^r}(a) \cong C_k(a)_{(p)}.$$

Proof. The first identification is the definition of the p-adic completion of a finite free lattice. In Smith coordinate i, the semantic image at level r is

$$p^{\min(r,\nu_i)}\mathbb{Z}/p^r\mathbb{Z}.$$

Its inverse limit is $p^{\nu_i}\mathbb{Z}_p$, giving the second identification.

A compatible ghost tower represents an element $x \in \mathbb{Z}_p$ satisfying $p^{\nu_i}x = 0$. Since $\mathbb{Z}_p$ is torsion free, $x = 0$. Hence the inverse limit of the ghost modules vanishes. The missing quotient stabilizes coordinatewise at $\mathbb{Z}/p^{\nu_i}\mathbb{Z}$, whose product is the p-primary Smith quotient. □

The finite ghosts are therefore real finite-level defects, but they do not survive as nonzero points of the completed raw lattice. By contrast, the missing saturated quotient survives exactly as the p-primary saturation defect.

12.5 Orbit closure and three levels of effectivity

Let

$$\mathcal{O}_k(a) = \{\Lambda_w a : w \in \Sigma_k^*\} \subseteq M_k(a)$$

be the actual section orbit, and let

$$\mathcal{O}^{\mathrm{raw}}_{k,p^r}(a)$$

be its image in $M_k(a)/p^r M_k(a)$.

Theorem 12.18 (Raw orbit-closure descent)**.** *The reduction maps*

$$\mathcal{O}^{\mathrm{raw}}_{k,p^{r+1}}(a) \twoheadrightarrow \mathcal{O}^{\mathrm{raw}}_{k,p^r}(a)$$

are surjective, and

$$\overline{\mathcal{O}_k(a)}^{\,p} \xrightarrow{\sim} \varprojlim_r \mathcal{O}^{\mathrm{raw}}_{k,p^r}(a),$$

where the closure is taken in $M_{k,p}(a)$.

Proof. A section word at level r is lifted by the same word at level $r+1$, proving surjectivity. Every p-adic limit of orbit points gives a compatible tower. Conversely, choose for each compatible residue state an orbit representative x_r. Compatibility gives

$$x_{r+1} - x_r \in p^r M_k(a),$$

so (x_r) is Cauchy in $M_{k,p}(a)$. Its limit lies in the orbit closure and maps to the prescribed tower. □

Definition 12.19 (Three effectivity obstructions)**.** Let $x \in M_{k,p}(a)$.

1. The *p-adic effectivity condition* is simply $x \in M_{k,p}(a)$; every compatible full semantic module tower satisfies it by theorem 12.17.
2. The *integral lattice obstruction* is
$$\epsilon_{\mathrm{int}}(x) = [x] \in M_{k,p}(a)/M_k(a).$$
It vanishes exactly when x is represented by one integral lattice element.
3. The *section-orbit obstruction* is membership in
$$M_k(a) \setminus \mathcal{O}_k(a).$$
Even after integral effectivity holds, the lift need not be an actual radix section.

Boundary 12.20 (Semantic orbit towers). The inverse limit of the full semantic modules is always $M_{k,p}(a)$. For the smaller semantic orbit sets, surjectivity of reduction remains automatic, but lifting to an actual orbit closure can be obscured by finite ghost fibers. The unconditional orbit-closure theorem is therefore stated for the raw orbit tower. A semantic-orbit effectivity theorem requires a compatible choice of raw lifts or an equivalent fiberwise Mittag–Leffler condition.

Example 12.21 (Digit sum separates all three levels)**.** For the base-k digit sum s_k,
$$\mathcal{O}_k(s_k) = \{s_k + c : c \in \mathbb{Z}_{\geq 0}\},$$
and
$$M_k(s_k) = L_k(s_k) = \mathbb{Z}s_k \oplus \mathbb{Z}1.$$
Hence
$$\overline{\mathcal{O}_k(s_k)}^{\,p} = \{s_k + c : c \in \mathbb{Z}_p\}.$$
The partial sums
$$c_N = \sum_{j=0}^{N} p^{2j}$$
are nonnegative integers and therefore produce actual section states $s_k + c_N$. They converge in $\mathbb{Z}_p$ to
$$c_\infty = \frac{1}{1-p^2},$$
which is not an ordinary integer. Thus $s_k + c_\infty$ is p-adically effective and lies in the orbit closure, but its integral obstruction is nonzero.

By contrast, $s_k - 1$ is an integral lattice point and lies in the p-adic orbit closure because $-1 \in \mathbb{Z}_p$ is a limit of nonnegative integers. It is not an actual section, since every actual offset is nonnegative. Integral effectivity therefore does not imply section-orbit effectivity.

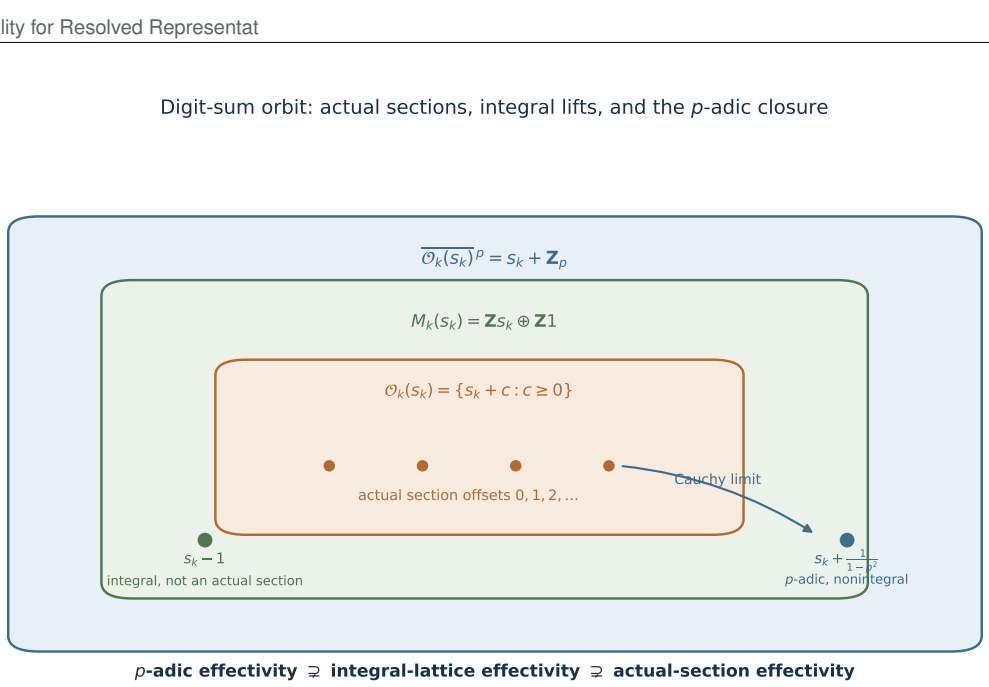


Figure 25: For the digit-sum sequence, actual section states form a nonnegative integral ray. Its p-adic closure contains all p-adic offsets, including nonintegral limits and integral points that are not actual sections.

12.6 All-moduli completion and the integral effectivity obstruction

Let

$$\widehat{M_k(a)} = \varprojlim_m M_k(a)/mM_k(a)$$

be the profinite completion over divisibility of moduli.

Theorem 12.22 (Profinite saturation sequence)**.** *There are canonical digit-equivariant identifications*

$$\widehat{M_k(a)} \cong \prod_p M_{k,p}(a), \qquad \widehat{L_k(a)} \cong \prod_p L_{k,p}(a),$$

and a short exact sequence

$$0 \longrightarrow \widehat{M_k(a)} \longrightarrow \widehat{L_k(a)} \longrightarrow C_k(a) \longrightarrow 0.$$

A compatible all-moduli raw tower has an ordinary integral lift exactly when its class in

$$\widehat{M_k(a)}/M_k(a)$$

vanishes.

Proof. Finite free abelian groups have profinite completion equal to the product of their p-adic completions. In Smith coordinates, the inclusion is multiplication by d_i in each profinite coordinate. Its cokernel is

$$\widehat{\mathbb{Z}}/d_i\widehat{\mathbb{Z}} \cong \mathbb{Z}/d_i\mathbb{Z}.$$

Taking the direct sum over i gives $C_k(a)$. The final assertion is the definition of belonging to the diagonal image of the integral lattice inside its profinite completion. □

Corollary 12.23 (Local-global separation)**.** *Every compatible prime-power tower is effective in a p-adic completion. A compatible family over all primes is effective in the profinite completion. Neither statement alone implies an ordinary integral lift. Integral effectivity is the separate diagonal condition*

$$\widehat{x} \in M_k(a) \subseteq \widehat{M_k(a)}.$$

This is the precise sense in which prime-power effectivity and integral effectivity differ. Completion solves the Cauchy problem in the declared topology. It does not force the resulting point to lie in the original discrete lattice.

12.7 Canonical integral record and directional reduction atlas

Definition 12.24 (Canonical integral regular record)**.** The canonical integral record of a is

$$\mathcal{I}_k(a) = (V_{\mathbb{Q}}(a), M_k(a), L_k(a), C_k(a), (\Lambda_d)_{0\le d<k}, a, \mathrm{ev}_0)\,.$$

Its Smith–Fitting profile is

$$\mathsf{SF}_k(a) = \big(d_1, \dots, d_\rho; \mathrm{Fitt}_0(C_k(a)), \dots, \mathrm{Fitt}_{\rho-1}(C_k(a))\big)\,.$$

The record is canonical but not a single field-valued normal form. It retains both the raw generative lattice and the maximal integer-valued lattice, together with the exact quotient between them.

Theorem 12.25 (Integral normalization and directional local reduction)**.** *For fixed k, the assignment*

$$a \longmapsto \mathcal{I}_k(a)$$

is functorial for digit-module morphisms preserving the distinguished sequence and evaluation. It is uniquely determined by the semantic sequence. For every prime p, reduction defines a directional bridge

$$\mathcal{I}_k(a) \longmapsto \Big(\overline{M}_{k,p^r}(a), \mathcal{K}^{\mathrm{sem}}_{k,p^r}(a)\Big)_{r\ge 1}$$

whose exact information-loss passport is:

1. *finite-depth ghost kernel $C_k(a)[p^r]$;*
2. *finite-depth missing quotient $C_k(a)/p^r C_k(a)$;*
3. *completed essential image $M_{k,p}(a) \subseteq L_{k,p}(a)$;*
4. *integral effectivity obstruction $M_{k,p}(a)/M_k(a)$;*
5. *actual-section obstruction inside the integral lattice.*

Proof. Every component of $\mathcal{I}_k(a)$ is defined from the semantic sequence and its digit sections. A compatible morphism carries actual sections to actual sections and integer-valued rational combinations to integer-valued rational combinations, so it induces maps on $M_k(a)$, $L_k(a)$, and the quotient. The local reduction statements are theorems 12.7 and 12.17 and definition 12.19. □

Section result. For every integer-valued fixed-radix sequence that is regular over $\mathbb{Q}$, the actual sections generate a canonical finite free lattice $M_k(a)$, and the integer-valued part of the rational

section space is a canonical finite free saturation $L_k(a)$. Their Smith–Fitting quotient controls every modular reduction exactly. At each prime power, the coefficient sequence has a finite minimal semantic section automaton. The compatible tower reconstructs the p-adic raw lattice, while finite ghost coordinates vanish in the inverse limit and the missing saturated quotient survives as the p-primary defect. p-adic, profinite, integral-lattice, and actual-section effectivity are distinct and are separated by explicit obstruction quotients.

Table 20: Exact finite validation of the Integral prime-power atlas integral and prime-power layer.

Check family	Recorded result
Smith and modular profiles	1984 divisibility chains of ranks one through four; moduli 2 through 24; 45632 chain-modulus cases.
Fitting and prime-power profiles	8840 Fitting cases and 39680 prime-power profile cases over $p = 2, 3, 5, 7$.
Digit-equivariant systems	320 exact Smith-square systems with integral and modular intertwining checks.
Orbit and completion controls	Digit-sum state formulas over four primes and 1681 bounded ghost-contraction samples.
Total exact checks	697958; 0 failures.
Overall status	PASS.
Boundary	Bounded exact verification supports the printed formulas and examples. It does not replace the symbolic proofs or establish unrestricted integral recognition.

Exact boundary of Integral prime-power atlas. The results classify saturation and reduction defects for integer-valued sequences that are already known to have finite rational section rank. They do not prove finite-data recognition of that hypothesis, a minimal realization theorem over arbitrary rings or semirings, a base-independent integral normal form, or compatibility across different radices. The prime-power tower is a directional local reduction system, not an equivalence between integral and modular presentations.

13 Filtered Euler products, prime-power seeds, and the divisor–ghost bridge

The next native-invariant layer has two different index monoids. Ordinary generating functions use the additive monoid of coefficient addresses: Cauchy multiplication combines indices by addition. Arithmetic functions use the multiplicative monoid of positive integers: Dirichlet convolution combines divisor addresses by multiplication. These structures meet exactly in selected local and logarithmic-derivative coordinates, but they are not globally interchangeable. This section constructs the fixed-basis filtered-product normal form, the prime-power normal form for multiplicative arithmetic functions, and one rigorously typed cross-stratum bridge between them.

13.1 Filtered products and coefficient stabilization

Definition 13.1 (Coefficientwise filtered product)**.** Let R be a commutative ring. A filtered factor family is a sequence

$$F_m(z) \in 1 + z^{w(m)} R[[z]], \qquad w(m) \longrightarrow \infty.$$

Its coefficientwise product is

$$\prod_{m\geq 1} F_m(z),$$

where the coefficient of z^N is computed from the finitely many factors satisfying $w(m) \leq N$.

Proposition 13.2 (Filtered product completion)**.** *Every filtered factor family has a unique coefficientwise product in* $1 + zR[[z]]$*. For every* N*, the truncation modulo* z^{N+1} *is equal to the product of the finitely many factors with* $w(m) \leq N$*.*

Proof. Fix N. Every factor with $w(m) > N$ is congruent to 1 modulo z^{N+1}, so only finitely many factors can affect the coefficient prefix through degree N. These finite products are compatible as N increases. Their inverse limit in the z-adic filtration is the required series. Uniqueness follows because every coefficient is fixed at a finite stage. □

The proposition supplies existence and exact finite-prefix evaluation. It does not assert that an arbitrary factor family is structurally compressed. A finite grammar requires an effective rule for the factors and a proof of stabilization; otherwise the factor list can merely store the coefficient data in another form.

13.2 Canonical Euler coordinates

Let

$$\mathcal{U}_{\mathbb{Z}} = 1 + z\mathbb{Z}[[z]]$$

be the multiplicative group of integral unit series, and let

$$\mathcal{E}_{\mathbb{Z}} = \mathbb{Z}^{\mathbb{N}_{\geq 1}}$$

be the additive group of integer exponent sequences.

Definition 13.3 (Euler-coordinate map)**.** For $b = (b_m)_{m\geq 1} \in \mathcal{E}_{\mathbb{Z}}$, define

$$\mathcal{E}(b)(z) = \prod_{m\geq 1} (1 - z^m)^{-b_m}.$$

The product is coefficientwise filtered because the mth factor is congruent to 1 modulo z^m.

Theorem 13.4 (Integral Euler-coordinate classification)**.** *The map*

$$\mathcal{E} : (\mathcal{E}_{\mathbb{Z}}, +) \longrightarrow (\mathcal{U}_{\mathbb{Z}}, \cdot)$$

is a group isomorphism. Consequently, every $A(z) \in 1 + z\mathbb{Z}[[z]]$ *has a unique representation*

$$\boxed{A(z) = \prod_{m\geq 1} (1 - z^m)^{-b_m}, \qquad b_m \in \mathbb{Z}.}$$

Proof. Filtered multiplication gives

$$\mathcal{E}(b + b') = \mathcal{E}(b)\mathcal{E}(b').$$

To prove surjectivity and uniqueness, construct the exponents recursively. Suppose $b_1, \ldots, b_{n-1}$ have been chosen so that

$$A(z) \prod_{m<n} (1 - z^m)^{b_m} \equiv 1 + r_n z^n \pmod{z^{n+1}}.$$

Because

$$(1-z^n)^{-r_n} \equiv 1 + r_n z^n \pmod{z^{n+1}},$$

the unique choice $b_n = r_n$ cancels the degree-n residual. Every r_n is integral because all preceding operations occur in $\mathbb{Z}[[z]]$. Induction produces one exponent sequence with the required product. If two exponent sequences differed first at n, their quotient would have nonzero coefficient $(b_n - b'_n)z^n$ modulo z^{n+1}, contradicting equality. Hence the representation is unique. □

Corollary 13.5 (Finite-prefix triangularity)**.** *For each N, the coefficient prefix*

$$(a_1, \dots, a_N) \quad \textit{of} \quad A(z) = 1 + \sum_{n \geq 1} a_n z^n$$

and the exponent prefix $(b_1, \dots, b_N)$ determine one another by integral triangular recursions. No exponent b_m with $m > N$ affects the coefficient prefix through degree N.

13.3 Ghost coordinates and logarithmic-derivative recurrence

Definition 13.6 (Euler ghost coordinates)**.** For $b \in \mathcal{E}_{\mathbb{Z}}$, define

$$c_n(b) = \sum_{d|n} d\, b_d, \qquad n \geq 1.$$

Let $\mathcal{G}_{\mathbb{Z}}$ be the set of integer sequences $c = (c_n)$ satisfying

$$\frac{1}{n} \sum_{d|n} \mu\Big(\frac{n}{d}\Big)\, c_d \in \mathbb{Z} \qquad (n \geq 1).$$

Theorem 13.7 (Ghost-coordinate equivalence)**.** *The divisor ghost map*

$$\mathcal{W} : \mathcal{E}_{\mathbb{Z}} \longrightarrow \mathcal{G}_{\mathbb{Z}}, \qquad b \longmapsto c(b),$$

is an additive group isomorphism. Its inverse is

$$\boxed{b_n = \frac{1}{n} \sum_{d|n} \mu\Big(\frac{n}{d}\Big)\, c_d.}$$

If $A(z) = \mathcal{E}(b)(z)$, then

$$\boxed{\frac{zA'(z)}{A(z)} = \sum_{n \geq 1} c_n z^n.}$$

Proof. Formal logarithmic differentiation gives

$$\frac{zA'}{A} = \sum_{m \geq 1} b_m \frac{mz^m}{1 - z^m} = \sum_{m \geq 1} \sum_{r \geq 1} m b_m z^{mr}.$$

The coefficient at z^n is $\sum_{m|n} m b_m = c_n$. Möbius inversion gives the displayed inverse formula. The definition of $\mathcal{G}_{\mathbb{Z}}$ is exactly the integrality condition for that inverse. □

Theorem 13.8 (Exact coefficient recurrence)**.** *Let*

$$A(z) = 1 + \sum_{n\geq 1} a_n z^n$$

and let c_n be its Euler ghost coordinates. Then

$$\boxed{na_n = \sum_{j=1}^{n} c_j a_{n-j} \qquad (n \geq 1),}$$

with $a_0 = 1$. Conversely, any $c \in \mathcal{G}_{\mathbb{Z}}$ and this recurrence reconstruct the integral unit series corresponding to its Euler exponents.

Proof. The logarithmic-derivative identity is equivalent to

$$zA'(z) = \left(\sum_{j\geq 1} c_j z^j\right) A(z).$$

Comparing coefficients gives the recurrence. Conversely, the recurrence uniquely determines the coefficient sequence over $\mathbb{Q}$. Since $c \in \mathcal{G}_{\mathbb{Z}}$ reconstructs an integral exponent sequence, the resulting series is the Euler product from theorem 13.4 and therefore has integral coefficients. □

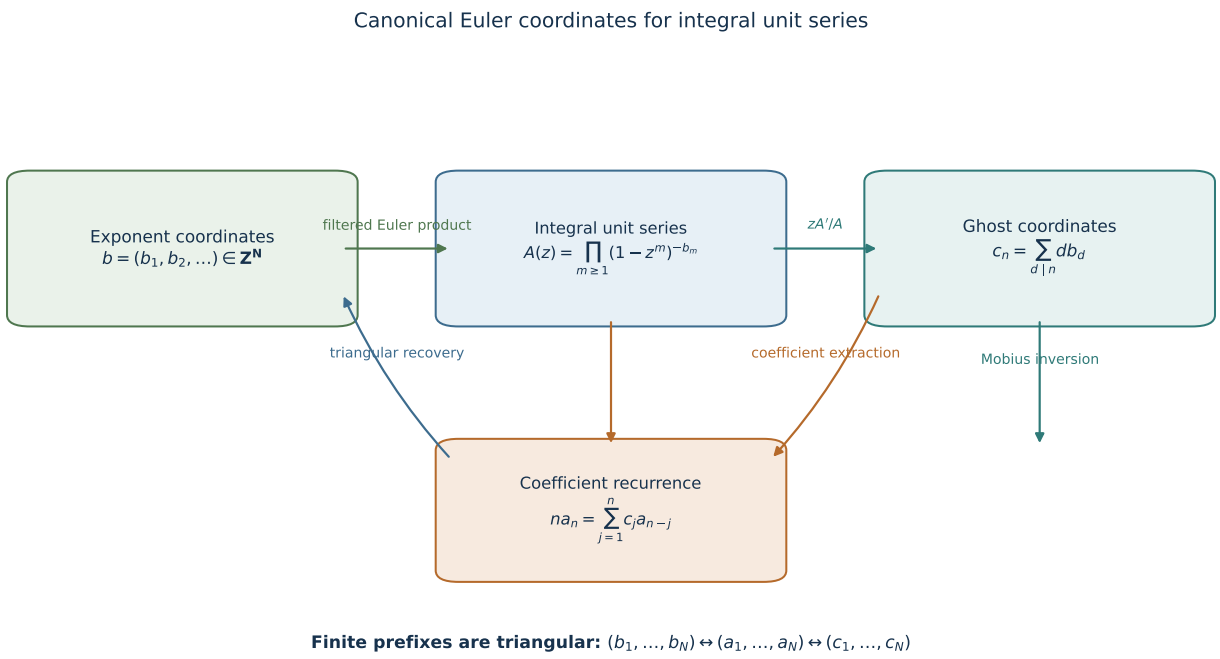


Figure 26: The fixed Euler basis gives three exact coordinate charts: integral exponents, coefficients, and divisor ghosts. Every finite prefix is triangular, while the ghost chart is restricted by its explicit Möbius divisibility conditions.

Example 13.9 (Partition and power families)**.** The following fixed-basis exponent functions give familiar filtered products:

b_n	$A_b(z)$	c_n
1	$\prod_{n\geq 1}(1-z^n)^{-1}$	$\sigma_1(n)$
n	$\prod_{n\geq 1}(1-z^n)^{-n}$	$\sigma_2(n)$
n^r	$\prod_{n\geq 1}(1-z^n)^{-n^r}$	$\sigma_{r+1}(n)$.

The first row is the ordinary partition series, and the second is the plane-partition series. More generally, the power family converts the exponent rule n^r into the divisor-sum ghost σ_{r+1}.

Example 13.10 (Distinct parts and normalization)**.** The product

$$\prod_{m\geq 1}(1+z^m)=\prod_{m\geq 1}\frac{1-z^{2m}}{1-z^m}$$

has normalized Euler exponent

$$b_n=\begin{cases}1, & n \text{ odd},\\ 0, & n \text{ even}.\end{cases}$$

Thus the distinct-part and odd-part products become the same object after normalization in the fixed Euler basis. This is an exact equality inside the declared product grammar, not a claim that arbitrary product grammars possess one preferred factorization.

Boundary 13.11 (Coordinate universality versus compact structural realization)*.* Theorem 13.4 gives a unique coordinate sequence for every integral unit series. It does not make every such series finitely describable. If the exponent sequence is merely an arbitrary list encoding all coefficients, the representation is storage universality. A compact seed grammar additionally requires an effective finite rule for the exponent sequence or factor family.

13.4 Multiplicative arithmetic functions as prime-power descent

For arithmetic functions $f, g : \mathbb{N}_{\geq 1} \to \mathbb{Z}$, write

$$(f *_D g)(n)=\sum_{d\mid n} f(d)g(n/d)$$

for Dirichlet convolution. Its identity is $\varepsilon(1)=1$ and $\varepsilon(n)=0$ for $n>1$.

Definition 13.12 (Integral multiplicative category)**.** Let $\mathbf{Mult}_{\mathbb{Z}}$ be the discrete category whose objects are multiplicative arithmetic functions $f : \mathbb{N}_{\geq 1} \to \mathbb{Z}$ satisfying

$$f(1)=1, \qquad f(mn)=f(m)f(n) \quad \text{when } (m,n)=1.$$

The monoidal operation is Dirichlet convolution.

Proposition 13.13 (Dirichlet inverse and multiplicative closure)**.** *If $f(1)=1$, then f has a unique integral Dirichlet inverse. If f and g are multiplicative, then $f *_D g$ and f^{-1_D} are multiplicative.*

Proof. The coefficient equation at n is triangular:

$$g(1) = 1, \qquad g(n) = -\sum_{\substack{d \mid n \\ d<n}} f(n/d)g(d).$$

Hence the inverse exists integrally and uniquely. The standard coprime-divisor factorization of a divisor of mn proves closure of multiplicative functions under convolution; uniqueness of the inverse then proves multiplicativity of f^{-1_D}. □

Definition 13.14 (Prime-power seed series)**.** For a multiplicative function f and a prime p, define

$$F_{f,p}(T) = \sum_{e \geq 0} f(p^e)T^e \in 1 + T\mathbb{Z}[[T]].$$

Theorem 13.15 (Prime-power descent equivalence)**.** *The assignment*

$$f \longmapsto \big(F_{f,p}\big)_p$$

is a group isomorphism

$$\boxed{(\mathbf{Mult}_{\mathbb{Z}}, *_D) \simeq \prod_{p \text{ prime}} (1 + T\mathbb{Z}[[T]], \cdot).}$$

The inverse reconstructs

$$\boxed{f(n) = \prod_{p^e \| n} f(p^e).}$$

Under this correspondence,

$$F_{f *_D g,p}(T) = F_{f,p}(T)F_{g,p}(T).$$

Proof. Unique factorization writes every positive integer as a finite product of pairwise coprime prime powers, so multiplicativity gives the reconstruction formula. Conversely, an arbitrary family of local unit series defines a multiplicative function by the same finite product. For $n = p^e$, Dirichlet convolution becomes

$$(f *_D g)(p^e) = \sum_{r=0}^{e} f(p^r)g(p^{e-r}),$$

which is exactly the coefficient of T^e in the local Cauchy product. □

13.5 Prime-Euler and local ghost coordinates

Apply the Euler-coordinate theorem independently at each prime:

$$F_{f,p}(T) = \prod_{m \geq 1} (1 - T^m)^{-\beta_{p,m}(f)}.$$

Define local ghosts

$$\gamma_{p,e}(f) = \sum_{d \mid e} d\, \beta_{p,d}(f).$$

Theorem 13.16 (Complete prime-Euler coordinates)**.** *For integral multiplicative functions, the following normalized data are equivalent and complete:*

$$f, \qquad \big(F_{f,p}\big)_p, \qquad \big(\beta_{p,m}(f)\big)_{p,m}, \qquad \big(\gamma_{p,e}(f)\big)_{p,e}.$$

Dirichlet convolution is linearized by the last two charts:

$$\beta_{p,m}(f *_D g) = \beta_{p,m}(f) + \beta_{p,m}(g),$$

$$\gamma_{p,e}(f *_D g) = \gamma_{p,e}(f) + \gamma_{p,e}(g).$$

The ghost chart consists exactly of the prime-indexed sequences satisfying the local Möbius divisibility conditions

$$\frac{1}{e}\sum_{d|e} \mu\Big(\frac{e}{d}\Big)\, \gamma_{p,d} \in \mathbb{Z}.$$

Proof. Theorem 13.4 and Theorem 13.7 apply to each local unit series $F_{f,p}$. The prime-power descent theorem then reconstructs the global function. Local multiplication corresponds to Dirichlet convolution, while Euler exponents and ghosts add under multiplication. □

Theorem 13.17 (Valuation logarithmic derivative)**.** *Fix a prime p. Let $\Gamma_{f,p}$ be the arithmetic function supported on powers of p with*

$$\Gamma_{f,p}(1) = 0, \qquad \Gamma_{f,p}(p^e) = \gamma_{p,e}(f) \quad (e \geq 1).$$

Then

$$\boxed{(v_p \cdot f)(n) = (\Gamma_{f,p} *_D f)(n) \qquad (n \geq 1).}$$

Equivalently,

$$\frac{TF'_{f,p}(T)}{F_{f,p}(T)} = \sum_{e\geq 1} \gamma_{p,e}(f) T^e.$$

Proof. The local logarithmic-derivative recurrence gives

$$ef(p^e) = \sum_{j=1}^{e} \gamma_{p,j}(f) f(p^{e-j}).$$

Write $n = p^e u$ with $(u, p) = 1$ and multiply by $f(u)$. Since $\Gamma_{f,p}$ is supported on p-powers, the resulting sum is exactly the Dirichlet convolution at n. □

Proposition 13.18 (Multiplicative residue-packet law)**.** *For $N \geq 1$, let $P_{N,r} f$ be the arithmetic function equal to $f(n)$ when $n \equiv r \pmod N$ and zero otherwise. Then*

$$\boxed{P_{N,k}(f *_D g) = \sum_{rs \equiv k\,(N)} (P_{N,r} f) *_D (P_{N,s} g).}$$

Thus the packet address algebra for Dirichlet convolution is the finite multiplicative monoid $\mathbb{Z}/N\mathbb{Z}$, including its zero divisors.

Proof. In each divisor pair $n = d(n/d)$, the residue of n modulo N is the product of the two divisor residues. Group the divisor sum by those two residues. □

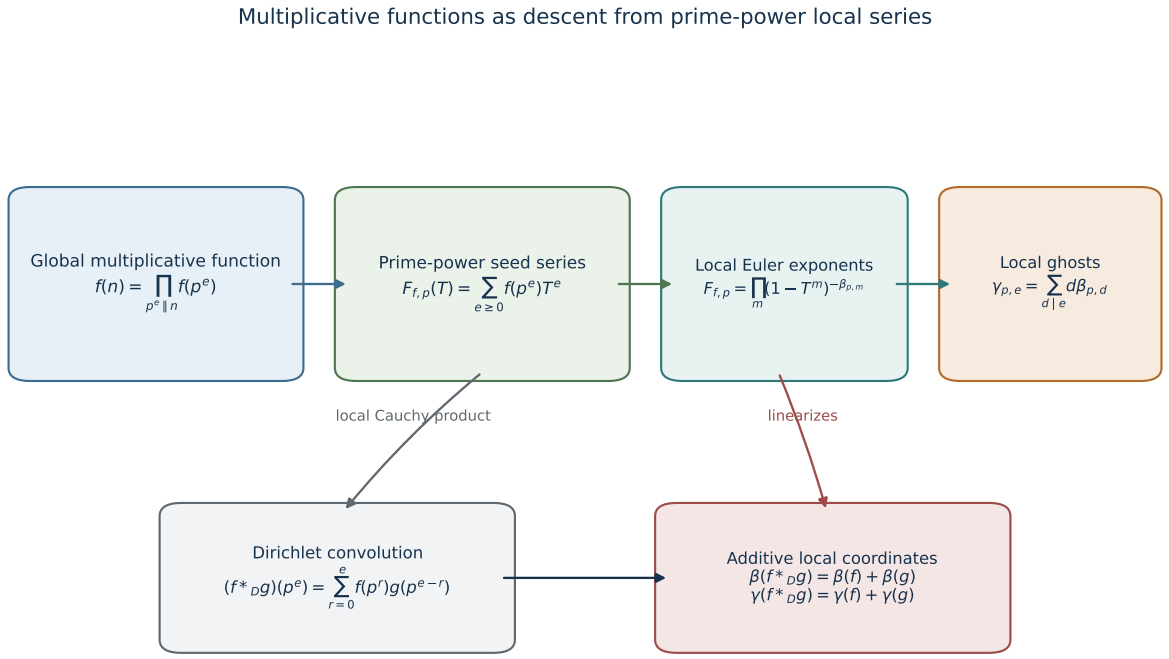


Figure 27: A multiplicative arithmetic function is effective descent data over its prime-power local series. Local Euler exponents and local ghosts turn Dirichlet convolution into addition.

Table 21: Named prime-power seed families in local Euler coordinates.

Arithmetic function	Nonzero local Euler exponents	Prime-power seed	Local ghost
Constant-one function $\mathbf{1}$	$\beta_1 = 1$	$1, 1, 1, 1, 1, 1, \ldots$	$1, 1, 1, 1, 1, 1, \ldots$
Möbius μ	$\beta_1 = -1$	$1, -1, 0, 0, 0, 0, \ldots$	$-1, -1, -1, -1, -1, -1, \ldots$
Liouville λ	$\beta_1 = -1, \beta_2 = 1$	$1, -1, 1, -1, 1, -1, \ldots$	$-1, 1, -1, 1, -1, 1, \ldots$
Squarefree indicator μ^2	$\beta_1 = 1, \beta_2 = -1$	$1, 1, 0, 0, 0, 0, \ldots$	$1, -1, 1, -1, 1, -1, \ldots$
Three-fold divisor function τ_3	$\beta_1 = 3$	$1, 3, 6, 10, 15, 21, \ldots$	$3, 3, 3, 3, 3, 3, \ldots$

13.6 The divisor–ghost cross-stratum bridge

The Euler ghost transform can itself be written in the Dirichlet-convolution language. Let $\mathbf{1}(n) = 1$, let $\mathrm{id}(n) = n$, and let pointwise multiplication be denoted by a centered dot.

Definition 13.19 (Divisor–ghost transform)**.** For an arithmetic exponent function $b : \mathbb{N}_{\geq 1} \to \mathbb{Z}$, define

$$\mathfrak{G}(b) = \mathbf{1} *_D (\mathrm{id} \cdot b),$$

so that

$$\mathfrak{G}(b)(n) = \sum_{d|n} d\, b(d).$$

Theorem 13.20 (Exact divisor–ghost bridge)**.** *The map $b \mapsto c = \mathfrak{G}(b)$ is injective, with inverse*

$$\boxed{b(n) = \frac{1}{n}(\mu *_D c)(n).}$$

Its integral image consists exactly of the arithmetic functions c satisfying

$$n \mid (\mu *_D c)(n) \qquad (n \geq 1).$$

For the Euler product

$$A_b(z) = \prod_{n \geq 1} (1 - z^n)^{-b(n)},$$

one has

$$\frac{z A_b'(z)}{A_b(z)} = \sum_{n \geq 1} c(n) z^n.$$

Proof. The first identity is the convolution form of the divisor sum. Since $\mu *_D \mathbf{1} = \varepsilon$,

$$\mu *_D c = \mathrm{id} \cdot b.$$

Division by n gives the inverse and its exact integrality condition. The logarithmic-derivative identity is Theorem 13.7 with the exponent sequence regarded as an arithmetic function. □

Theorem 13.21 (Multiplicativity is preserved and reflected)**.** *Assume $b(1) = 1$ and $c = \mathfrak{G}(b)$. Then*

$$\boxed{b \text{ is multiplicative} \iff c \text{ is multiplicative.}}$$

On this locus, both b and c are reconstructed from prime-power seeds. For every prime p and $e \geq 1$,

$$\boxed{c(p^e) = \sum_{j=0}^{e} p^j b(p^j),}$$

$$\boxed{b(p^e) = \frac{c(p^e) - c(p^{e-1})}{p^e}.}$$

Proof. If b is multiplicative, then $\mathrm{id} \cdot b$ is multiplicative, and Dirichlet convolution with $\mathbf{1}$ preserves multiplicativity. Conversely, if c is multiplicative, then $\mu *_D c$ is multiplicative. Dividing pointwise by the completely multiplicative function id preserves multiplicativity over $\mathbb{Q}$; the integral image condition ensures that the resulting values are integers. The prime-power formulas follow by listing the divisors of p^e and subtracting consecutive sums. □

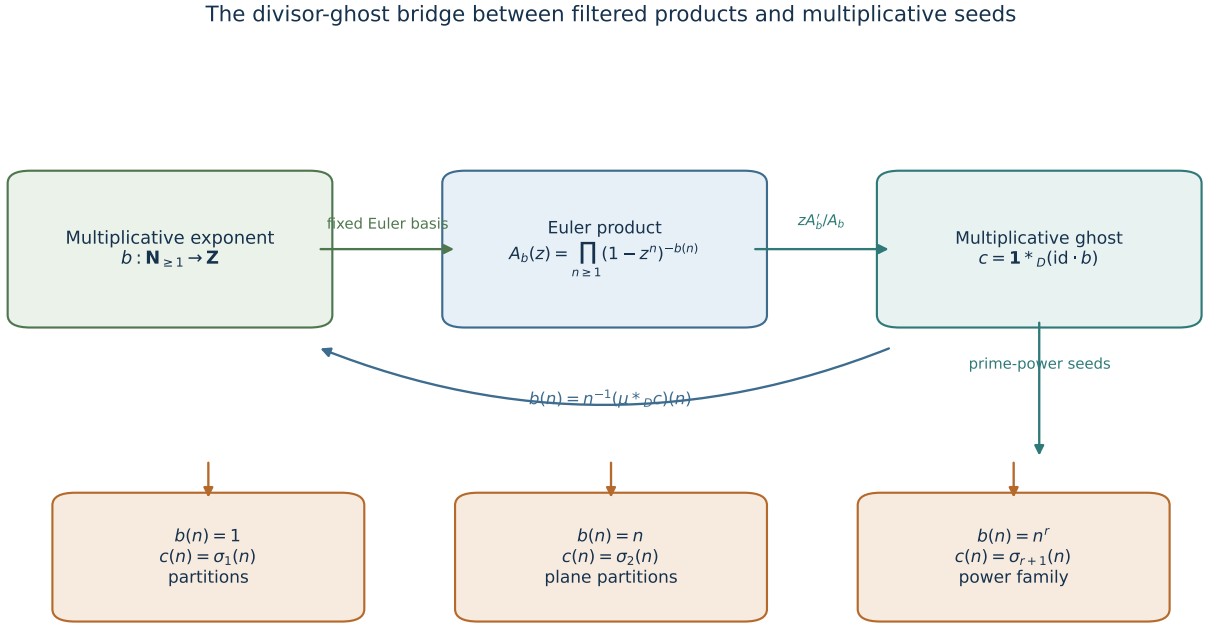


Figure 28: The divisor–ghost transform gives a reversible invariant bridge on the multiplicative exponent locus. It connects a fixed-basis ordinary Euler product to multiplicative prime-power data without identifying the two global convolution algebras.

13.7 A restricted product–multiplicative atlas

Definition 13.22 (Multiplicative Euler-overlap category)**.** Let $\mathbf{EulMult}_{\mathbb{Z}}$ be the discrete category of multiplicative exponent functions

$$b : \mathbb{N}_{\geq 1} \to \mathbb{Z}, \qquad b(1) = 1.$$

Associate to b the four normalized charts

$$\mathsf{B}(b) = b, \qquad \mathsf{A}(b) = A_b(z), \qquad \mathsf{C}(b) = \mathfrak{G}(b), \qquad \mathsf{P}(b) = \big(b(p^e), c(p^e)\big)_{p,e}.$$

Theorem 13.23 (Effective descent on the divisor–ghost overlap)**.** *The exponent, fixed Euler-product, multiplicative ghost, and prime-power seed charts form a finite normalized atlas*

$$\mathfrak{R}_{\mathrm{DG}}.$$

Every bridge is reversible on the declared overlap, every multiple-overlap comparison is the unique identity over the represented exponent function, and

$$\boxed{\mathrm{Desc}(\mathfrak{R}_{\mathrm{DG}}) \simeq \mathbf{EulMult}_{\mathbb{Z}}.}$$

Descent is effective for objects and morphisms.

Proof. The Euler-product chart is uniquely inverted by Theorem 13.4. The ghost chart is uniquely inverted by Theorem 13.20. The prime-power chart is uniquely inverted by multiplicativity and unique

factorization. Hence every normalized fiber over b is a singleton. All comparison cells and all multiple-overlap coherence laws are therefore forced to be identities. A compatible local family glues to its common exponent function, and morphism effectivity is automatic because the chart categories are discrete. □

Theorem 13.24 (Complete descended prime-power invariant)**.** *The prime-power package*

$$\boxed{\mathcal{J}_{\mathrm{DG}}(b) = (b(p^e), c(p^e))_{p \text{ prime},\, e\geq 0}}$$

subject to

$$p^e b(p^e) = c(p^e) - c(p^{e-1}) \quad (e \geq 1)$$

is complete on $\mathbf{EulMult}_{\mathbb{Z}}$*. It reconstructs* b*,* c*, the Euler product* A_b*, and the coefficient recurrence of* A_b*.*

Proof. Prime-power values reconstruct the multiplicative functions b and c. Theorem 13.4 reconstructs A_b, and Theorem 13.8 reconstructs its coefficient sequence from c. Therefore equal invariant packages imply equality in every chart. □

Example 13.25 (Power bridge)**.** For $b(n) = n^r$, which is completely multiplicative, the divisor ghost is

$$c(n) = \sum_{d|n} d^{r+1} = \sigma_{r+1}(n).$$

At a prime power,

$$\sigma_{r+1}(p^e) = 1 + p^{r+1} + \cdots + p^{e(r+1)}.$$

Thus one simple multiplicative exponent rule simultaneously produces a filtered ordinary product, a multiplicative divisor-sum ghost, and a one-dimensional geometric local rule at every prime.

13.8 Why the bridge is restricted

Proposition 13.26 (The multiplicative exponent locus is not a product subgroup)**.** *The set of multiplicative Euler exponent functions is not closed under the addition of exponent coordinates. Consequently, multiplication of the corresponding ordinary Euler products does not define an internal group law on* $\mathbf{EulMult}_{\mathbb{Z}}$*.*

Proof. The constant function $b(n) = 1$ is multiplicative. But $b + b$ has value 2 at 1 and therefore is not multiplicative under the normalization $f(1) = 1$. Since ordinary Euler-product multiplication adds exponents, the multiplicative exponent locus is not closed under that operation. □

Proposition 13.27 (Euler coefficients are not multiplicative semantics)**.** *The coefficient sequence of* $A_b(z)$ *need not be a multiplicative arithmetic function even when* b *is multiplicative. For example,* $b(n) = 1$ *gives the partition numbers, and*

$$p(6) = 11 \neq p(2)p(3) = 6.$$

Boundary 13.28 (Prime-power data require multiplicativity). Two arbitrary arithmetic functions can agree on every prime power and differ on a composite supported at more than one prime. Prime-power seeds are complete only after the multiplicativity axiom has been declared. A compact local seed rule also does not remove the computational cost of factoring the input integer.

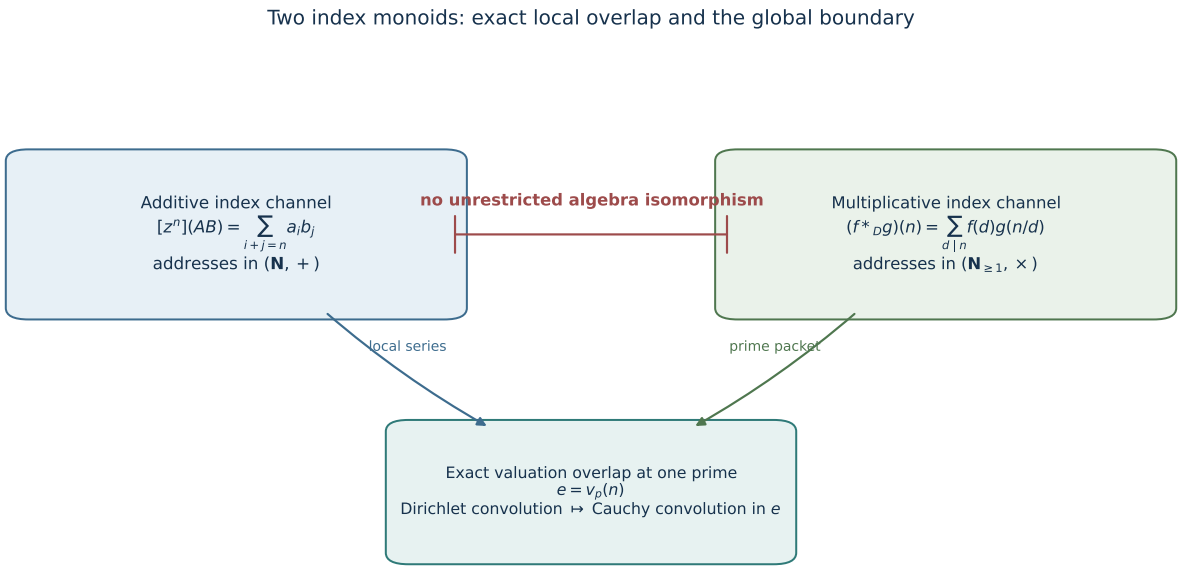


Figure 29: Cauchy and Dirichlet convolution use different global address monoids. They overlap exactly after localization at one valuation coordinate, but no unrestricted algebra equivalence follows.

Boundary 13.29 (No universal additive–multiplicative equivalence). The divisor–ghost transform is an exact invariant bridge on a named essential image. It is not an algebra isomorphism between ordinary Cauchy convolution and Dirichlet convolution, does not identify the coefficients of an Euler product with a multiplicative function, and does not produce a canonical factorization across arbitrary product grammars. Any larger bridge must name its transform, kernel, essential image, and information-loss profile.

13.9 Exact finite verification

Table 22: Exact finite validation of the Product-multiplicative atlas filtered-product and multiplicative layer.

Check family	Recorded result
Euler-coordinate atlas	15625 exponent prefixes and 3125 arbitrary integral unit-series prefixes.
Prime-power multiplicative atlas	96 deterministic local systems and 48 Dirichlet-convolution pairs.
Divisor-ghost overlap	128 multiplicative exponent systems and 6 power families.
Named local coordinates	5 arithmetic families and 6 geometric necklace modes.
Total exact checks	595964; 0 failures.
Overall status	PASS.
Boundary	Bounded exact verification supports the printed formulas and examples. It does not establish compactness or minimality for arbitrary product grammars, nor a global Cauchy–Dirichlet algebra isomorphism.

The Product-multiplicative atlas verifier performs four exact audits. First, it exhaustively checks the coefficient, exponent, and ghost triangular maps for every exponent prefix in $\{-2,-1,0,1,2\}^6$ and every unit coefficient prefix in $\{-2,-1,0,1,2\}^5$. Second, it constructs deterministic prime-power systems, reconstructs global multiplicative functions, and verifies local Euler inversion, local logarithmic

derivatives, and Dirichlet convolution. Third, it checks the divisor–ghost bridge and the power family $b(n) = n^r$. Fourth, it verifies the named local Euler coordinates printed in Table 21. The recorded 595,964 exact checks have no failures.

Exact boundary of Product-multiplicative atlas. The fixed Euler basis gives a canonical coordinate system for integral unit series, and prime-power seed series give a canonical coordinate system for integral multiplicative functions. The divisor–ghost transform produces one complete cross-stratum overlap on multiplicative exponent functions. The block does not prove compactness for arbitrary exponent data, a minimal product compiler, closure of the multiplicative exponent slice under ordinary product multiplication, a global Cauchy–Dirichlet algebra isomorphism, or a canonical factorization across unrestricted product grammars.

Section result. Integral unit series admit unique fixed-basis Euler exponents and unique divisor ghosts, with exact logarithmic-derivative recurrence and finite-prefix triangularity. Integral multiplicative functions are effective descent from prime-power local series; local Euler exponents and local ghosts linearize Dirichlet convolution and form complete invariants. On the multiplicative exponent locus, the divisor–ghost transform gives a reversible four-chart atlas linking exponent functions, ordinary filtered Euler products, multiplicative ghosts, and prime-power seeds. This closes one explicit product–multiplicative invariant bridge while preserving the global distinction between additive and multiplicative index monoids.

14 Local Newton–Mahler coordinates, affine polyphase descent, and precision ladders

Local Mahler-polyphase atlas develops the local stratum promised by the work program. Its purpose is not to merge every use of the word "Mahler." It freezes one precise local semantic doctrine, constructs its normalized coefficient and polyphase charts, types the characteristic-zero, integral, and residue-level affine channels separately, and identifies the exact data needed to pass from finite modular dynamics to certified local analytic packets.

14.1 Two Mahler languages and the declared local doctrine

Two constructions in the Library use the same historical name but have different variables and different semantics.

1. A *functional-Mahler equation*
$$F(z) = R(z)F(z^p)$$
is an identity for a formal coefficient series in the variable z. Cartier operators act on the coefficient index. the periodic and rational-mask atlases use this channel over finite fields.

2. A *Newton–Mahler expansion*
$$f(x) = \sum_{r\geq 0} c_r \binom{x}{r}$$
is a coordinate expansion for a function of a p-adic input variable x. Affine substitutions $x = Mq + s$ produce polyphase branches of the input domain.

This section treats the second channel. Fix a prime p. Let

$$\mathcal{C}_p = C(\mathbb{Z}_p, \mathbb{Z}_p)$$

with the uniform norm, and let

$$c_0(\mathbb{Z}_p) = \{c = (c_r)_{r\geq 0} : c_r \in \mathbb{Z}_p,\ c_r \longrightarrow 0\}$$

with the coefficient sup norm.

Theorem 14.1 (Mahler coordinate theorem)**.** *The synthesis map*

$$\mathcal{M}_p : c_0(\mathbb{Z}_p) \longrightarrow \mathcal{C}_p, \qquad c \longmapsto \left(x \longmapsto \sum_{r\geq 0} c_r \binom{x}{r}\right),$$

is an isometric isomorphism of $\mathbb{Z}_p$-modules. Its inverse is

$$\boxed{c_r = \Delta^r f(0).}$$

In particular, the Newton–Mahler coefficient sequence is a canonical complete coordinate for a continuous $\mathbb{Z}_p$-valued function.

Proof. This is the classical Mahler coordinate theorem [14]. Finite Newton interpolation gives the coefficient identity on nonnegative integers. Continuity implies that the forward-difference coefficients tend to zero, so the binomial series converges uniformly on $\mathbb{Z}_p$. Conversely, every coefficient sequence tending to zero gives a uniformly convergent continuous function. Uniqueness follows from the finite-difference formula. □

Definition 14.2 (Exact valuation profile and certified class)**.** For $f = \mathcal{M}_p(c)$, define

$$\nu_f(r) = v_p(c_r) \in \mathbb{Z}_{\geq 0} \cup \{\infty\}.$$

If $w : \mathbb{N} \to \mathbb{Z}_{\geq 0}$ is nondecreasing and tends to infinity, put

$$\mathcal{M}_w = \{c \in c_0(\mathbb{Z}_p) : v_p(c_r) \geq w(r)\}, \qquad \mathcal{C}_w = \mathcal{M}_p(\mathcal{M}_w).$$

The function w is a declared precision certificate. The class $\mathcal{C}_w$ is not asserted to be a characterization of every possible notion of local analyticity.

14.2 Complete clopen packets and effective local descent

For $a \geq 1$ and $0 \leq s < p^a$, define the branch

$$\mathcal{B}_{a,s} f(q) = f(p^a q + s), \qquad q \in \mathbb{Z}_p,$$

and the complete packet

$$\mathcal{B}_a f = (\mathcal{B}_{a,s} f)_{0\leq s<p^a}.$$

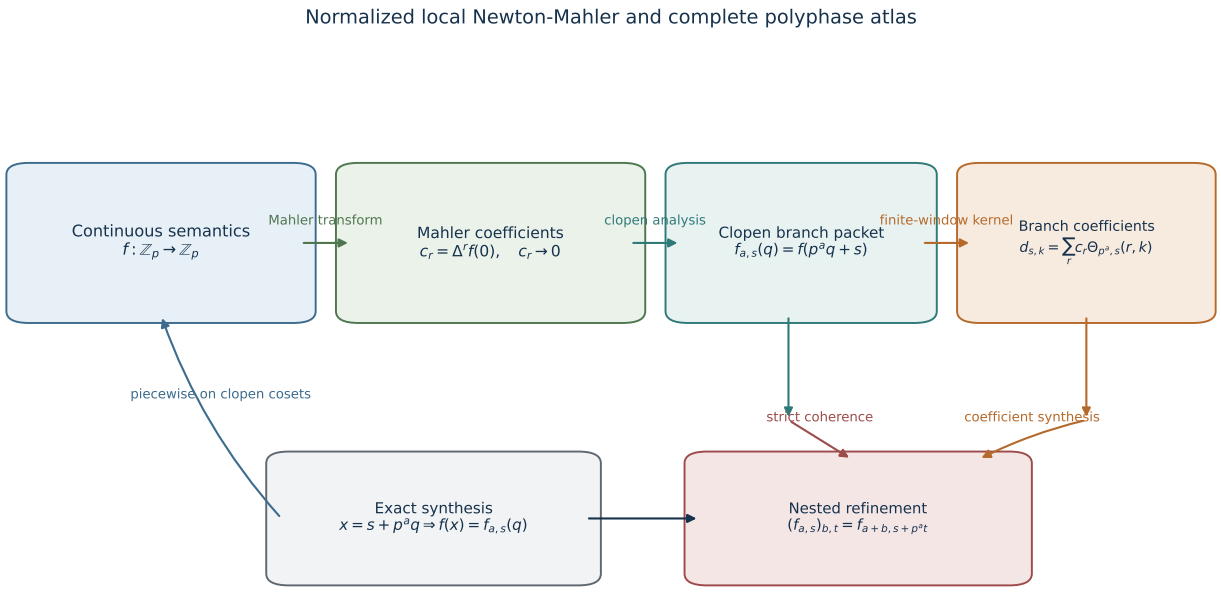


Figure 30: The normalized local atlas. Continuous semantics, canonical Mahler coefficients, complete clopen branch packets, and branchwise Mahler coefficients are equivalent charts. A single continuous branch remains directional.

Theorem 14.3 (Complete clopen packet equivalence)**.** *The analysis map*

$$\mathcal{B}_a : \mathcal{C}_p \longrightarrow \prod_{s=0}^{p^a-1} \mathcal{C}_p$$

is an isometric isomorphism. Its inverse is the piecewise synthesis

$$\boxed{(\mathcal{S}_a g)(x) = g_s\left(\frac{x-s}{p^a}\right) \quad \text{for } x \in s + p^a\mathbb{Z}_p.}$$

Thus the clopen residue cover of $\mathbb{Z}_p$ has effective descent for both objects and equality morphisms.

Proof. The cosets $s + p^a\mathbb{Z}_p$ are pairwise disjoint, clopen, and cover $\mathbb{Z}_p$. Restriction followed by the displayed synthesis is the identity. Conversely, synthesis followed by restriction recovers every component. Continuity is checked separately on each clopen component, and the uniform norm is the maximum of the branch norms. □

Theorem 14.4 (Strict nested-branch law)**.** *For $a, b \geq 1$, $0 \leq s < p^a$, and $0 \leq t < p^b$,*

$$\boxed{(f_{a,s})_{b,t} = f_{a+b,s+p^a t}.}$$

All multiple-overlap comparison cells for the nested p-power branch atlas are identities. Hence its higher coherence is strict rather than merely up to a chosen natural isomorphism.

Proof. Direct substitution gives

$$f_{a,s}(p^b q + t) = f(p^a(p^b q + t) + s) = f(p^{a+b} q + s + p^a t).$$

The address is uniquely determined in the interval $0 \le s + p^a t < p^{a+b}$. □

A single branch $\mathcal{B}_{a,s}$ is not injective on $\mathcal{C}_p$ when $a \ge 1$: any nonzero continuous function supported on another clopen coset lies in its kernel. The complete packet is therefore the reversible object. This observation will be contrasted below with the finite-degree characteristic-zero polynomial chart, where one affine branch is invertible.

14.3 The affine Mahler kernel and mixed-radix coherence

For $M \ge 1$, $s \ge 0$, and $r, k \ge 0$, define

$$\Phi_M(z) = (1+z)^M - 1$$

and

$$\boxed{\Theta_{M,s}(r,k) = [z^r](1+z)^s \Phi_M(z)^k.}$$

Equivalently,

$$\boxed{\Theta_{M,s}(r,k) = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} \binom{Mj+s}{r}.}$$

The support is finite:

$$\Theta_{M,s}(r,k) = 0 \quad \text{unless} \quad k \le r \le Mk + s.$$

Theorem 14.5 (Affine Newton–Mahler transfer)**.** *Let*

$$f(x) = \sum_{r \ge 0} c_r \binom{x}{r}$$

be a convergent Mahler expansion and put $g_s(q) = f(Mq + s)$. Then

$$g_s(q) = \sum_{k \ge 0} d_{s,k} \binom{q}{k},$$

where

$$\boxed{d_{s,k} = \sum_{r=k}^{Mk+s} c_r \Theta_{M,s}(r,k).}$$

Every output coefficient depends on a finite source window.

Proof. Expand

$$(1+z)^{Mq+s} = (1+z)^s \big(1 + \Phi_M(z)\big)^q = \sum_{k \ge 0} \binom{q}{k} (1+z)^s \Phi_M(z)^k.$$

Taking $[z^r]$ gives the affine expansion of $\binom{Mq+s}{r}$. Substitution into the Mahler series gives the formula. The finite support of $\Theta_{M,s}$ makes each coefficient sum finite. □

Let $\mathcal{T}_{M,s}$ denote this coefficient operator. Conjugating the clopen packet theorem by $\mathcal{M}_p$ gives the following normalized coefficient atlas.

Corollary 14.6 (Complete coefficient-packet equivalence). *For $M = p^a$, the map*

$$\mathcal{T}_a : c_0(\mathbb{Z}_p) \longrightarrow \prod_{s=0}^{p^a-1} c_0(\mathbb{Z}_p), \qquad c \longmapsto (\mathcal{T}_{p^a,s}c)_s,$$

is a topological isomorphism. It is the coefficient conjugate of complete clopen analysis.

Theorem 14.7 (Mixed-radix kernel composition). *For $M, N \geq 1$,*

$$\boxed{\sum_{k\geq 0} \Theta_{M,s}(r,k)\Theta_{N,t}(k,\ell) = \Theta_{MN,Mt+s}(r,\ell).}$$

Equivalently,

$$\boxed{\mathcal{T}_{N,t}\mathcal{T}_{M,s} = \mathcal{T}_{MN,Mt+s}.}$$

When $\gcd(M,N) = 1$, the two Chinese-remainder refinement paths to a common address agree exactly. Their holonomy is the identity.

Proof. Apply the affine substitution $x = Mq + s$, then $q = Ny + t$. The composite is $x = MNy + Mt + s$. Uniqueness of Mahler coefficients yields the kernel identity. The Chinese-remainder diamond is the same equality written along two factorizations of one affine address. □

Normalized local atlas. For every fixed prime p and depth a, the following normalized charts are equivalent:

$$\boxed{\mathbf{Loc}_p^{\mathrm{cts}} \simeq \mathbf{Mah}_p^{c_0} \simeq \mathbf{Branch}_{p,a}^{\mathrm{cts}} \simeq \mathbf{BranchMah}_{p,a}^{c_0}.}$$

The first chart contains continuous functions on $\mathbb{Z}_p$, the second their unique Mahler coefficients, the third the complete clopen branch packet, and the fourth the corresponding branchwise coefficient packet. Every normalized fiber is contractible. The descent category is therefore equivalent to the discrete semantic category $\mathbf{Loc}_p^{\mathrm{cts}}$.

14.4 Finite-degree bridge ladder and collective recovery

The same integer kernel matrix has different categorical status after changing the coefficient doctrine. Fix $d \geq 0$ and write

$$\mathcal{M}_d(R) = R^{d+1}$$

for the formal truncated Mahler coefficient module. Let

$$A_{M,s}^{(d)} = \big[\Theta_{M,s}(r,k)\big]_{0\leq k,r\leq d},$$

where the target degree k is the row index.

Theorem 14.8 (Typed affine-channel ladder). *For every $M \geq 1$, $s \geq 0$, and $d \geq 0$:*

1. *The matrix $A_{M,s}^{(d)}$ is upper triangular with diagonal*

$$1, M, M^2, \ldots, M^d.$$

Hence

$$\boxed{\det A_{M,s}^{(d)} = M^{d(d+1)/2}.}$$

Every characteristic-zero polynomial chart is invertible.

2. *On the integral lattice $\mathbb{Z}^{d+1}$, one channel is injective. If $M > 1$ and $d \geq 1$, its image has index $M^{d(d+1)/2}$, so the channel has no integer linear left inverse.*

3. *Let p be prime and put $\beta_p = p^{v_p(M)}$. Reduction modulo p has exact rank*

$$\boxed{\operatorname{rank}_{\mathbb{F}_p} \overline{A}^{(d)}_{M,s} = \left\lfloor \frac{d}{\beta_p} \right\rfloor + 1.}$$

   A local channel loses residual rank exactly at primes dividing M.

4. *A complete neighboring packet can recover the source integrally. The canonical prefix of $\min(M, d+1)$ standard shifts has an integer left inverse, valid after base change to every commutative ring.*

Proof. The diagonal follows from the leading term of $\binom{Mq+s}{r}$, or from the generating identity for $\Theta_{M,s}$. The determinant gives the integral index statement. Modulo p, the first nonzero source degree in target row k is $\beta_p k$, which gives the rank formula. For the prefix packet, choose from channel s and target row k the equation indexed by $n = Mk + s$. The selected square coefficient matrix is unit lower triangular, so its inverse is integral and remains a left inverse after every base change. □

Theorem 14.9 (Sharp standard packet size)**.** *Define*

$$B(M) = \max\left(\{1\} \cup \{p^{v_p(M)} : p \mid M\}\right).$$

The least number of standard affine channels whose stacked map has an integer left inverse is

$$\boxed{L_{\mathbb{Z}}(M,d) = \min(d+1, B(M)).}$$

The prefix $\{0, 1, \ldots, L_{\mathbb{Z}}(M,d) - 1\}$ attains the minimum.

For a fixed modulus $N \geq 2$, set

$$B_N(M) = \max\left(\{1\} \cup \{p^{v_p(M)} : p \mid \gcd(M,N)\}\right).$$

The least packet size for a left inverse over $\mathbb{Z}/N\mathbb{Z}$ is

$$\boxed{L_N(M,d) = \min(d+1, B_N(M)).}$$

The exponents $v_p(N)$ do not change the minimum; the first residue layer already detects the obstruction.

Proof. This is the primary-rank and Smith/Hermite criterion of Core-Transfer Addendum 04. At a prime $p \mid M$, selected shifts are visible only through their residues modulo $p^{v_p(M)}$. A prefix of the stated length has full rank at every obstructing prime, hence a unimodular-row left inverse. Any smaller packet fails at a prime attaining the largest relevant primary component. A nonzero first-layer kernel vector lifts to a kernel vector at every higher prime-power depth. □

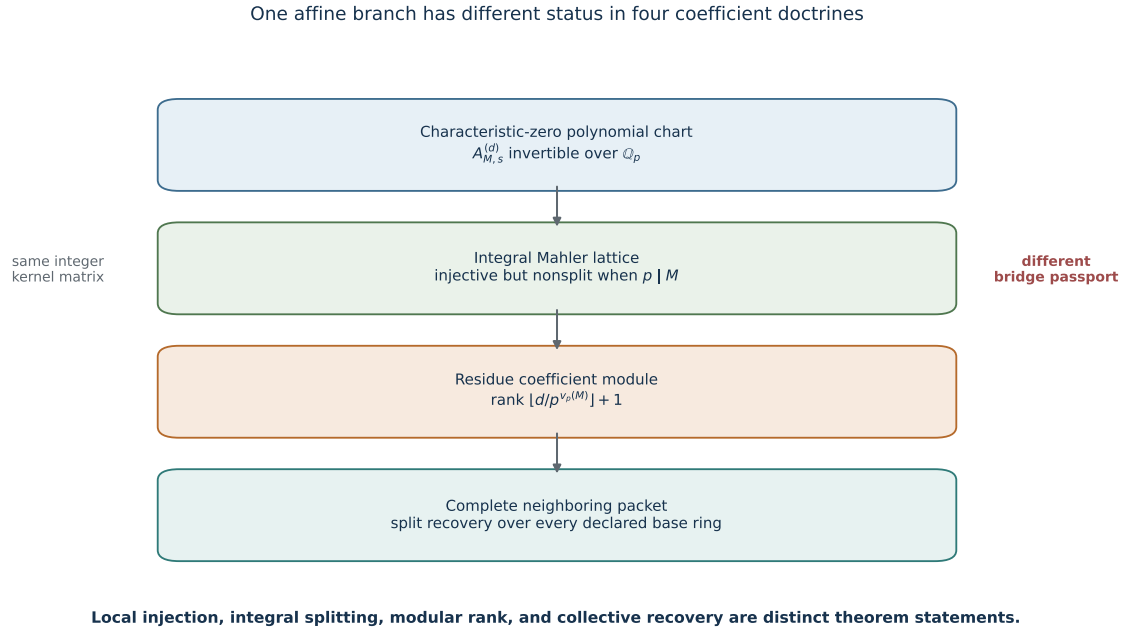


Figure 31: The same affine Mahler kernel has four distinct bridge passports. Characteristic-zero invertibility, integral injection, modular rank, and complete-packet splitting must not be conflated.

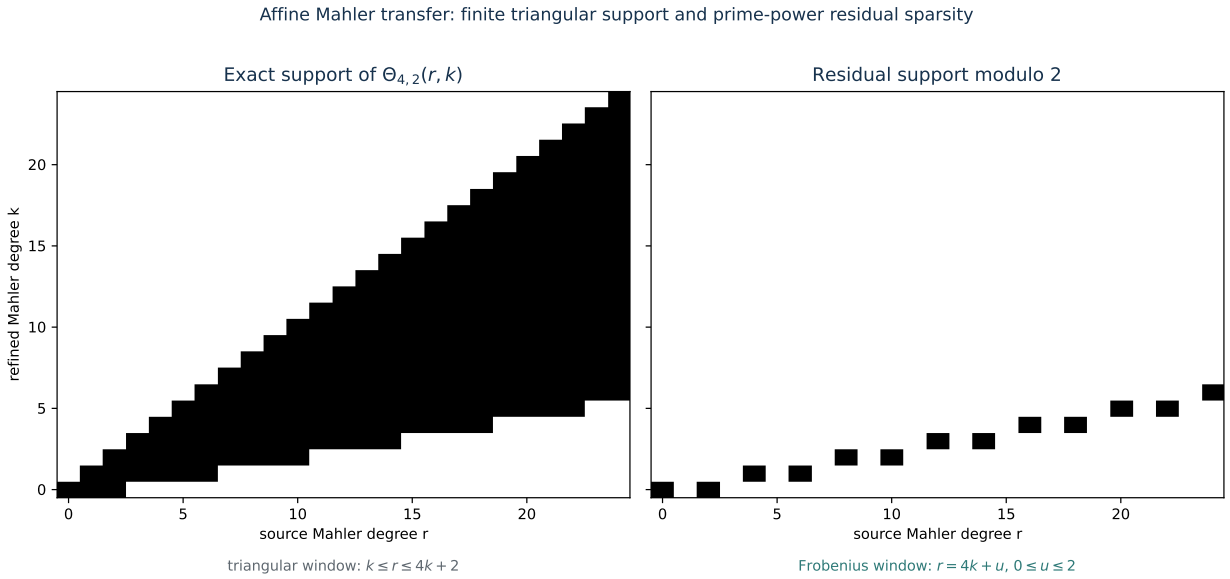


Figure 32: The finite triangular support of the affine kernel and its Frobenius-sparse residue pattern at a prime-power radix.

14.5 Valuation ledgers and Frobenius sparsity

The coefficient transform carries a precise lower-bound calculus.

Theorem 14.10 (Exact branchwise valuation ledger)**.** *For every refined coefficient,*

$$\boxed{v_p(d_{s,k}) \geq \min_{k \leq r \leq Mk+s} \left(v_p(c_r) + v_p(\Theta_{M,s}(r,k)) \right).}$$

If $v_p(c_r) \geq w(r)$ for a nondecreasing weight w, then

$$\boxed{v_p(d_{s,k}) \geq w(k).}$$

Thus every declared monotone Mahler precision ladder survives affine polyphase refinement.

Proof. The transfer sum is finite. Apply the non-Archimedean triangle inequality. Since $r \geq k$, monotonicity gives $w(r) \geq w(k)$, and every kernel entry is integral. □

Define the min-plus certificate transform

$$\boxed{(\mathfrak{W}_{M,s} w)(k) = \min_{k \leq r \leq Mk+s} \left(w(r) + v_p(\Theta_{M,s}(r,k)) \right).}$$

This transform is a certified lower bound. It need not be the exact output valuation, because cancellation can increase the valuation beyond the termwise minimum.

Theorem 14.11 (Prime-power Frobenius sparsification)**.** *For $M = p^a$,*

$$\boxed{\Theta_{p^a,s}(r,k) \equiv \binom{s}{r - p^a k} \pmod{p},}$$

where the binomial coefficient is zero outside $0 \leq r - p^a k \leq s$. Therefore

$$\boxed{d_{s,k} \equiv \sum_{u=0}^{s} \binom{s}{u} c_{p^a k + u} \pmod{p}.}$$

In particular,

$$\boxed{d_{0,k} \equiv c_{p^a k} \pmod{p}.}$$

Proof. In characteristic p,

$$(1+z)^{p^a} - 1 \equiv z^{p^a}.$$

Substitute this into the generating definition of $\Theta_{p^a,s}$ and compare coefficients. □

14.6 Carmichael–Mahler state constructors and precision amplification

Finite modular state dynamics provide a directional constructor into the local Mahler chart. Let

$$v_{n+1} = C v_n, \qquad a_n = w^{\mathsf{T}} C^n v_0,$$

with $C \in M_d(\mathbb{Z}_p)$ invertible modulo p.

Theorem 14.12 (Matrix Carmichael–Mahler constructor)**.** *Assume*

$$C^H = I + p^L B, \qquad L \geq 1.$$

For every address j,

$$\boxed{a_{Hx+j} = \sum_{r\geq 0} p^{Lr} \binom{x}{r} w^{\mathsf{T}} C^j B^r v_0,}$$

where the series converges for $x \in \mathbb{Z}_p$. Its Mahler coefficients satisfy

$$\boxed{v_p(c_{j,r}) \geq Lr,}$$

and

$$\boxed{\Delta_H^r a_n \equiv 0 \pmod{p^{Lr}}.}$$

Proof. Use

$$C^{Hx+j} = C^j (I + p^L B)^x$$

and expand the principal-unit matrix power by the convergent binomial series. One forward difference in x contributes the factor $p^L B$. □

Theorem 14.13 (Principal-unit precision amplification)**.** *If $X = I + p^L B$, then for every $a \geq 0$ there exists $B_a \in M_d(\mathbb{Z}_p)$ such that*

$$\boxed{X^{p^a} = I + p^{L+a} B_a.}$$

Consequently, for $H_a = Hp^a$, every branch $x \mapsto a_{H_a x+j}$ has Mahler coefficients satisfying

$$\boxed{v_p(c_{a,j,r}) \geq (L+a)r.}$$

Each additional p-power refinement contributes at least one further digit of precision per Mahler depth.

Proof. Induct on a. If $X_a = I + p^{L+a} B_a$, the binomial expansion of X_a^p has linear term divisible by p^{L+a+1}, and every higher term has at least the same valuation. Apply theorem 14.12 at the refined jump. □

Example 14.14 (Refined dyadic Fibonacci packets)**.** For the Fibonacci companion matrix

$$C = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},$$

one has

$$C^{12} = I + 8 \begin{pmatrix} 29 & 18 \\ 18 & 11 \end{pmatrix}.$$

Hence

$$\boxed{\Delta_{12\cdot 2^a}^r F_n \equiv 0 \pmod{2^{(a+3)r}}.}$$

The state-to-local construction is directional. A local branch can forget unreachable state directions, nonminimal coordinates, or modes annihilated by the output functional. Recovering a state presentation requires a separate recognition or minimality theorem and is not part of the Carmichael–Mahler constructor itself.

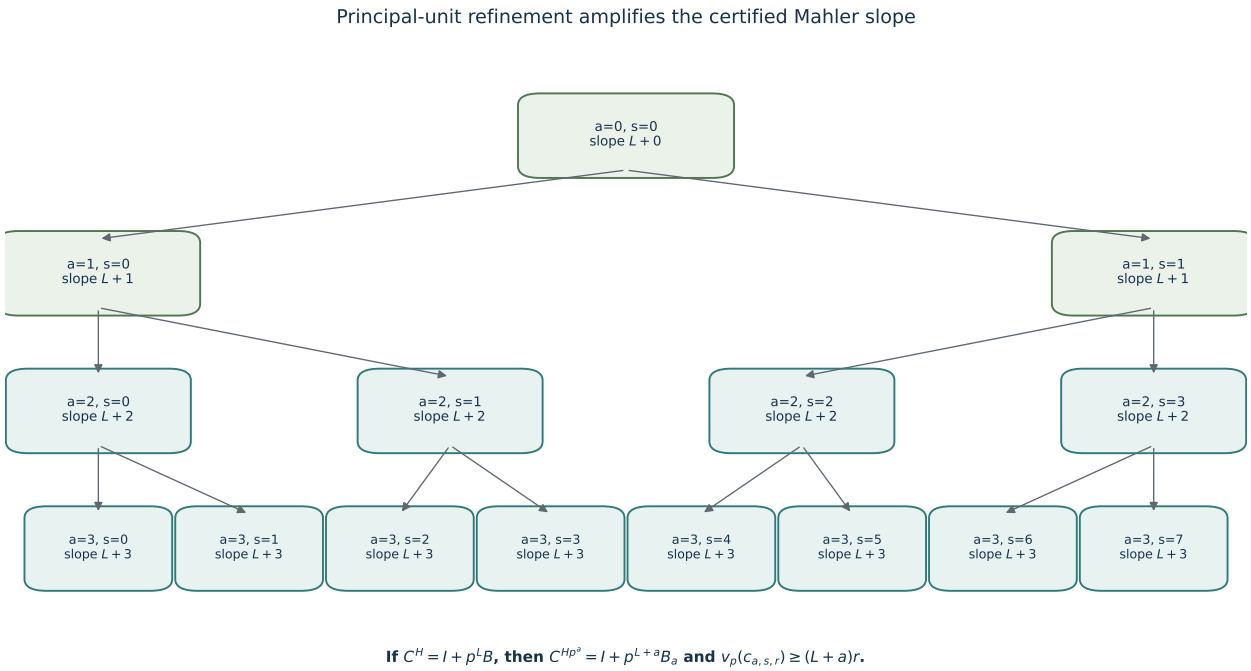


Figure 33: A principal-unit state packet acquires one additional guaranteed p-adic digit per Mahler depth at each p-power refinement.

14.7 The exact boundary with functional-Mahler masks

Proposition 14.15 (No canonical unrestricted bridge between the two Mahler channels)**.** *Functional-Mahler mask data over $\mathbb{F}_p$ does not determine a unique Newton–Mahler chart over $\mathbb{Z}_p$, and a continuous Newton–Mahler function does not in general determine a rational functional-Mahler mask.*

Proof. A residue sequence $(\bar{a}_n)$ admits many integral lifts. If (a_n) is one lift, then $(a_n + pn)$ is another with the same reduction modulo p. Their Newton–Mahler coefficients differ by the nonzero coefficient sequence of pn. Thus the residue-series or functional-mask presentation cannot select one local interpolation without a declared lifting rule.

Conversely, a continuous function $f : \mathbb{Z}_p \to \mathbb{Z}_p$ canonically determines its Newton–Mahler coefficients, but rationality of an ordinary generating series or of the mask $F(z)/F(z^p)$ is an additional property of a separate coefficient-series presentation. Continuity alone does not supply it. □

Exact local boundary. Local Mahler-polyphase atlas proves a complete normalized atlas for continuous $\mathbb{Z}_p$-valued functions and their complete p^a-branch packets. It proves exact affine coefficient transport, mixed-radix coherence, finite-degree characteristic-zero invertibility, integral split recovery, modular rank loss, valuation preservation, and principal-unit precision amplification under the printed hypotheses. It does not identify every continuous function as locally analytic, compute exact higher valuations of every $\Theta_{p^a,s}(r,k)$, choose an optimal mixed-radix refinement tree, or construct the combined spectral–local invariant requested by the broader program.

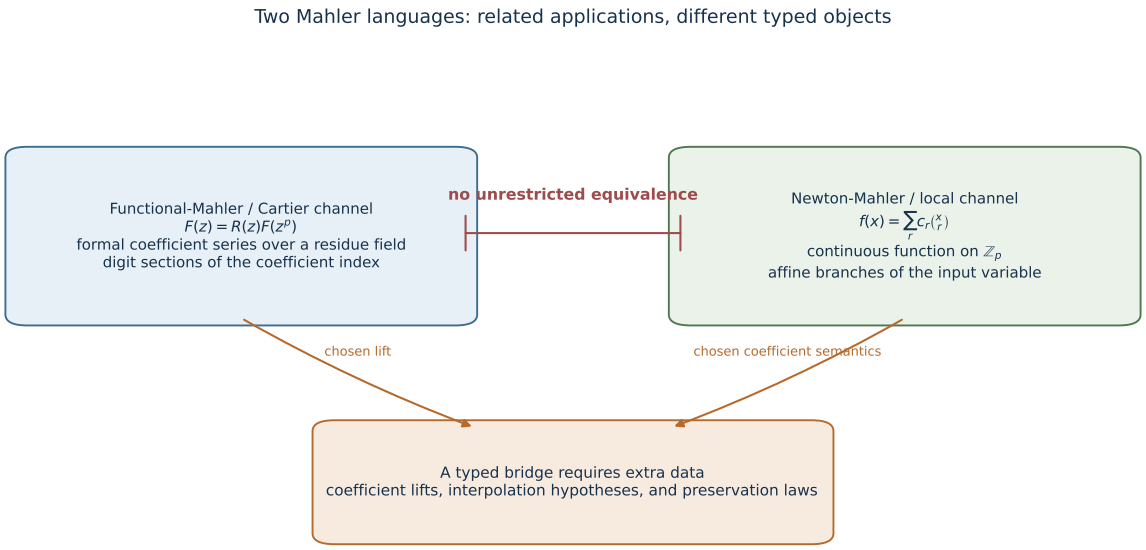


Figure 34: Functional-Mahler and Newton–Mahler constructions require different variables, coefficient doctrines, and normalization data. A bridge exists only after those extra choices are declared.

14.8 Exact examples and finite verification

For $f(x) = x^2$, the source Mahler vector is

$$(c_0, c_1, c_2) = (0, 1, 2).$$

The two dyadic branches are

$$f(2q) = 4\binom{q}{1} + 8\binom{q}{2},$$

and

$$f(2q+1) = 1 + 8\binom{q}{1} + 8\binom{q}{2}.$$

This example exhibits exact finite-window coefficient transport.

For a principal unit $u = 1 + p$,

$$u^x = \sum_{r \geq 0} p^r \binom{x}{r}.$$

The branch $x = pq$ has coefficients

$$\left(u^p - 1\right)^k,$$

whose valuation is at least $2k$. The branch therefore improves the original slope-one certificate.

At $p = 2$, $M = 2$, and $d = 3$, each single reduced channel has rank two, while the two-channel packet has rank four. This is the smallest visible instance of collective recovery.

The verifier checks the affine generating identity, finite polynomial transport, triangular determinants, mixed-radix composition, Chinese-remainder diamonds, nested branch addresses, prime-power Frobenius sparsity, exact local ranks, packet minimality, integer decoders, composite primary minima, valuation ledgers, scalar principal-unit amplification, and the refined Fibonacci matrix congruences. It also includes a negative control showing that equal residue data can have distinct Newton–Mahler lifts.

Table 23: Exact local Mahler and polyphase examples used in Local Mahler-polyphase atlas.

Example	Source Mahler data	Refined data or rank
x^2, branch 2q+0	$(0,1,2)$	$(0,4,8)$
x^2, branch 2q+1	$(0,1,2)$	$(1,8,8)$
$(1+2)^x$, branch 2q	$c_r = 2^r,\ 0 \le r \le 4$	$d_k = 8^k,\ 0 \le k \le 4$
$(1+3)^x$, branch 3q	$c_r = 3^r,\ 0 \le r \le 4$	$d_k = 63^k,\ 0 \le k \le 4$
$(1+5)^x$, branch 5q	$c_r = 5^r,\ 0 \le r \le 4$	$d_k = 7775^k,\ 0 \le k \le 4$
mod-2 rank distribution, M=2,d=3	$(2,2)$	$(4,4)$

Table 24: Local Mahler-polyphase atlas exact verification summary.

Validation class	Recorded result
Affine-kernel rows	312 exact row cases
Finite coefficient transports	936 source/chart cases
Mixed-radix compositions and CRT diamonds	2,102 exact diagrams
Prime-power rank and sparsity cases	95 degree/radix cases
Integral decoder and composite-minimum cases	156 cases
Valuation-ledger cases	5,100 coefficient ladders
Scalar and matrix precision cases	923 cases
Total exact checks	**229,743**
Failures	0

Local Mahler–polyphase result. The local continuous, Newton–Mahler coefficient, complete clopen branch, and branch-coefficient presentations form an effective normalized atlas. Affine polyphase kernels compose strictly and transport declared valuation ladders. The finite-degree restriction exposes a four-level bridge ladder: reversible characteristic-zero charts, nonsplit integral channels, rank-deficient residue channels, and split neighboring packets. Carmichael–Mahler transfer embeds finite principal-unit state dynamics into the local atlas with a certified precision slope that improves under p-power refinement. Functional-Mahler masks remain a distinct typed channel unless coefficient lifts and interpolation data are supplied.

15 Holonomic Ore charts, multisection modules, and a Fuchsian packet theorem

Holonomic atlas develops the next native stratum in the declared work program. The general holonomic problem has two logically separate parts. First, a D-finite series and its P-recursive coefficient sequence require normalized operator charts that remember the finite initial data lost by tail recurrences. Second, arithmetic multisection preserves holonomicity but can change minimal order through root-orbit aliasing, resonance, and address-dependent cancellation. The first part is closed here on the ordinary-point locus. The second is organized in general and solved exactly for a nontrivial regular-singular family.

Throughout this section, K is a characteristic-zero field. When local monodromy is discussed, K is embedded in $\mathbb{C}$ and all local exponents are taken over an algebraic closure.

15.1 Ore annihilators and normalized holonomic charts

Let

$$\mathcal{D}_K = K(z)\langle D\rangle, \qquad Dq = qD + q',$$

be the differential Ore algebra, and let

$$\mathcal{S}_K = K(n)\langle S\rangle, \qquad Sq(n) = q(n+1)S,$$

be the forward-shift Ore algebra. Both are Euclidean skew-polynomial rings in their operator variable, so their nonzero left ideals have unique monic generators [11].

Definition 15.1 (Minimal differential annihilator)**.** For a nonzero D-finite series $F \in K[[z]]$, define

$$\mathrm{Ann}_{\mathcal{D}}(F) = \{L \in \mathcal{D}_K : LF = 0\}.$$

Its unique monic generator is denoted

$$\boxed{L_F.}$$

The differential order

$$r_D(F) = \mathrm{ord}_D\, L_F$$

is the minimal scalar differential order of F over $K(z)$.

Definition 15.2 (Minimal tail recurrence)**.** For a sequence $a = (a_n)_{n\ge 0}$, define its tail annihilator by

$$\mathrm{Ann}_{\mathrm{tail}}(a) = \{R \in \mathcal{S}_K : (Ra)_n = 0 \text{ for all sufficiently large } n\}.$$

For a non-eventually-zero P-recursive sequence, its unique monic generator is denoted

$$\boxed{R_a.}$$

If the sequence is eventually zero, the tail generator is 1 and has order zero.

Write

$$R_a = S^s + \sum_{j=0}^{s-1} r_j(n) S^j.$$

Let ν_a be the least index from which R_a annihilates the sequence, and define N_a to be the least integer $N \ge \nu_a$ larger than every nonnegative pole of the rational functions $r_j(n)$. Then

$$\mathfrak{R}(a) = \big(R_a, N_a, a_0, \dots, a_{N_a+s-1}\big)$$

is a canonical complete recurrence chart. The operator controls the tail, while the finite prefix crosses every singular index and fixes the exact sequence rather than only its eventual equivalence class.

Proposition 15.3 (Derivative-module rank)**.** *For a nonzero D-finite series F,*

$$\boxed{r_D(F) = \dim_{K(z)} \mathrm{span}_{K(z)}\{F, DF, D^2F, \dots\}.}$$

Proof. A differential operator of order r gives a linear dependence among $F, DF, \dots, D^rF$. Minimality of L_F says that no dependence exists among the first r derivative vectors. Therefore they form a basis of the derivative span. □

15.2 The exact coefficient bridge

The classical equivalence between P-recursive sequences and D-finite ordinary generating functions is effective in both directions [12]. The following formulas make the finite residual explicit.

Let

$$F(z) = \sum_{n \geq 0} a_n z^n$$

and

$$L = \sum_{j=0}^{r} q_j(z) D^j, \qquad q_j(z) = \sum_{m=0}^{d_j} q_{j,m} z^m.$$

Theorem 15.4 (Differential-to-recurrence coefficient formula)**.** *For every $n \geq 0$,*

$$\boxed{[z^n]LF = \sum_{j=0}^{r} \sum_{m=0}^{\min(d_j,n)} q_{j,m}(n-m+j)^{\underline{j}} a_{n-m+j},}$$

where

$$x^{\underline{j}} = x(x-1)\cdots(x-j+1).$$

Consequently, every polynomial-coefficient differential equation yields an explicit polynomial recurrence after a finite reindexing.

Proof. The coefficient of z^{n-m} in $D^j F$ is

$$(n-m+j)^{\underline{j}} a_{n-m+j}.$$

Multiplication by $q_{j,m} z^m$ shifts that contribution to degree n. Summing over j and m gives the formula. □

Conversely, suppose

$$\sum_{s=0}^{q} p_s(n) a_{n+s} = 0 \qquad (n \geq N),$$

where $p_s \in K[n]$, and put

$$\theta = zD.$$

Theorem 15.5 (Recurrence residual as an Euler-operator polynomial)**.** *Define*

$$\boxed{Q = \sum_{s=0}^{q} z^{q-s} p_s(\theta - s).}$$

Then

$$\boxed{[z^{n+q}]QF = \sum_{s=0}^{q} p_s(n) a_{n+s}.}$$

Hence QF is a polynomial of degree less than $N+q$, and

$$\boxed{D^{N+q} QF = 0.}$$

Proof. The operator $p_s(\theta - s)$ multiplies the coefficient of z^k by $p_s(k-s)$. After multiplication by z^{q-s}, the coefficient at z^{n+q} is $p_s(n)a_{n+s}$. The recurrence kills every coefficient from degree $N+q$ onward. Differentiating more times than the residual degree gives a homogeneous annihilator. □

The polynomial residual is not noise. It is the exact finite-prefix certificate that distinguishes a recurrence valid only in the tail from an identity in the full formal-series chart.

15.3 The ordinary-point holonomic atlas

Definition 15.6 (Ordinary-point locus)**.** A nonzero D-finite series $F \in K[[z]]$ belongs to

$$\mathbf{Hol}_K^{\mathrm{ord}}$$

when its monic minimal operator can be written

$$L_F = D^r + \sum_{j=0}^{r-1} \ell_j(z) D^j$$

with every ℓ_j regular at $z = 0$.

The normalized jet is

$$J_0(F) = \big(F(0), F'(0), \dots, F^{(r-1)}(0)\big).$$

Theorem 15.7 (Ordinary-point reconstruction)**.** *Let*

$$L = D^r + \sum_{j=0}^{r-1} \ell_j(z) D^j$$

with all ℓ_j regular at zero. For every jet

$$(y_0, \dots, y_{r-1}) \in K^r$$

there is a unique formal series $F \in K[[z]]$ satisfying

$$LF = 0, \qquad F^{(j)}(0) = y_j.$$

Proof. Write $F(z) = \sum c_n z^n$. At coefficient degree n, the monic term $D^r F$ contributes

$$\frac{(n+r)!}{n!} c_{n+r},$$

while every lower derivative involves only $c_0, \dots, c_{n+r-1}$. Characteristic zero makes the displayed coefficient invertible. The equation therefore determines c_{n+r} recursively from the jet. □

Theorem 15.8 (Normalized ordinary holonomic atlas)**.** *On the ordinary-point locus, the following normalized presentations determine one another exactly:*

$$\boxed{\begin{array}{l} F \in K[[z]], \\ (L_F, J_0(F)), \\ \operatorname{span}_{K(z)}\{D^j F : j \geq 0\} \text{ with cyclic vector } F, \\ (R_a, N_a, a_0, \dots, a_{N_a+s-1}), \qquad a_n = [z^n]F. \end{array}}$$

With equality morphisms, these charts form equivalent normalized groupoids, and every fiber over a fixed semantic series is contractible.

Proof. The series gives its unique minimal Ore operators and finite data. The derivative module recovers L_F as the first monic dependence among consecutive derivatives. The operator and jet recover F by theorem 15.7. The recurrence chart reconstructs every coefficient after its singular prefix. The two explicit coefficient formulas transport between the differential and recurrence channels. Minimal monic normalization removes coordinate automorphisms, so each normalized fiber has only the identity. □

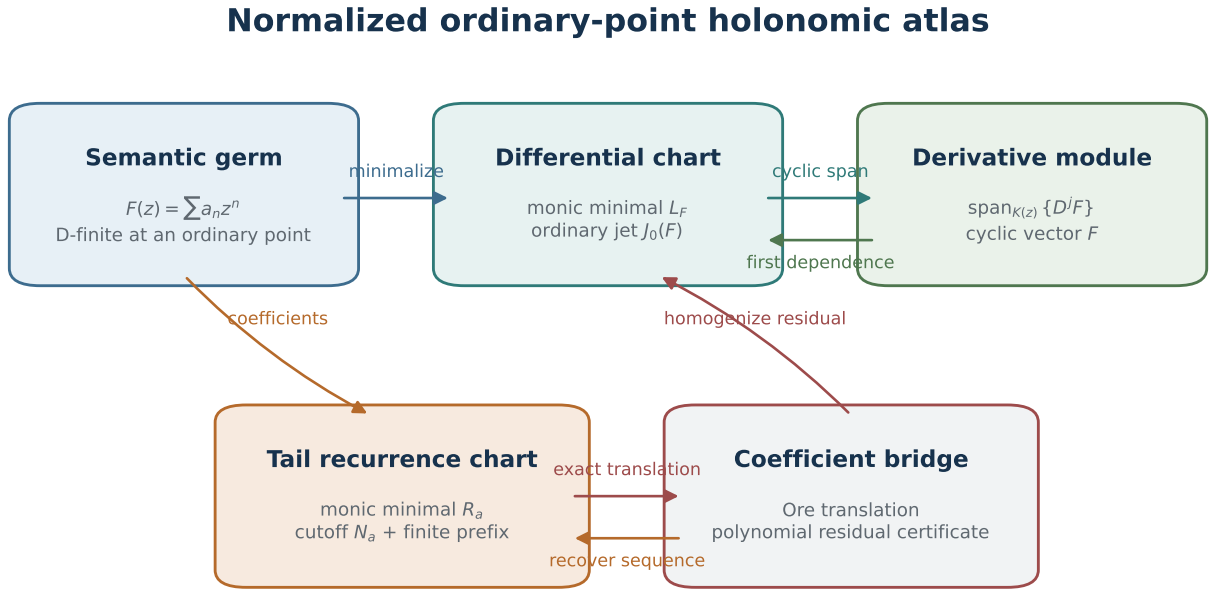


Figure 35: The ordinary holonomic atlas keeps the semantic series, minimal differential operator and jet, cyclic derivative module, and minimal tail recurrence with singular prefix as separate normalized charts.

15.4 General arithmetic multisection

For $N \geq 1$ and $0 \leq j < N$, define

$$\mathfrak{M}_{N,j}F(x) = \sum_{m \geq 0} a_{Nm+j} x^m.$$

If ω_N is a primitive Nth root of unity and $x = t^N$, then

$$\boxed{t^j \mathfrak{M}_{N,j}F(t^N) = \frac{1}{N} \sum_{q=0}^{N-1} \omega_N^{-jq} F(\omega_N^q t).}$$

This is the character projector for the cyclic rotation action.

Theorem 15.9 (Holonomic multisection closure)**.** *If F is D-finite, then every $\mathfrak{M}_{N,j}F$ is D-finite. Equivalently, arithmetic sections of P-recursive sequences are P-recursive.*

Proof. The root-filter formula uses finitely many algebraic substitutions and a finite linear combination. D-finite series are closed under those operations. The resulting section has coefficients in the original base field, so the operator descends. □

Closure does not determine the smallest operator. A separate exact state construction gives a useful bound.

Theorem 15.10 (Nonautonomous state jump)**.** *Suppose a P-recursive sequence is represented, beyond its singular prefix, by*

$$v_{n+1} = C(n)v_n, \qquad a_n = \lambda(n)v_n,$$

where $v_n \in K^s$ and $C(n)$ is rational. Then the jth N-section has the jump system

$$\boxed{w_{m+1} = J_{N,j}(m)w_m,}$$

where

$$\boxed{J_{N,j}(m) = C(Nm + j + N - 1) \cdots C(Nm + j).}$$

After denominator clearing, every scalar output has a polynomial recurrence of order at most s.

Proof. Compose the N consecutive one-step transfers. The $s + 1$ consecutive scalar output rows are linearly dependent over $K(m)$, producing a recurrence of order at most s. □

The root filter also gives a construction-independent order ledger before the algebraic descent from t to $x = t^N$. Work over a finite extension E/K containing ω_N. Put

$$F_q(t) = F(\omega_N^q t), \qquad G_{N,j}(t) = \frac{1}{N} \sum_{q=0}^{N-1} \omega_N^{-jq} F_q(t).$$

Let $L_q \in E(t)\langle D_t \rangle$ be the monic minimal annihilator of F_q. Rotation is an Ore-algebra automorphism, so every L_q has order $r = \operatorname{ord} L_F$. Let

$$L_N^{\mathrm{orb}} = \mathrm{LCLM}(L_0, \ldots, L_{N-1}), \qquad r_N^{\mathrm{orb}} = \operatorname{ord} L_N^{\mathrm{orb}},$$

and, when $G_{N,j} \neq 0$, let $L_{N,j}^{\mathrm{char}}$ be its monic minimal annihilator with order $r_{N,j}^{\mathrm{char}}$. For a zero character packet put $r_{N,j}^{\mathrm{char}} = 0$.

Definition 15.11 (Root-filter alias and address-cancellation defects)**.** Define

$$\mathfrak{a}_N(F) = Nr - r_N^{\mathrm{orb}}, \qquad \mathfrak{c}_{N,j}(F) = r_N^{\mathrm{orb}} - r_{N,j}^{\mathrm{char}}.$$

The first quantity measures dependence already present among the rotated holonomic modules. The second measures the additional loss caused by the jth character projector after the rotated orbit has been assembled.

Theorem 15.12 (Exact two-stage root-filter defect identity)**.** *The defects in definition 15.11 are nonnegative and satisfy*

$$\boxed{Nr - r_{N,j}^{\mathrm{char}} = \mathfrak{a}_N(F) + \mathfrak{c}_{N,j}(F).}$$

Thus orbit aliasing occurs before address selection, while address cancellation occurs after the Fourier character has been applied.

Proof. The least common left multiple has order at most the sum of the orders of its inputs, hence $r_N^{\mathrm{orb}} \le Nr$. It annihilates every rotated series and therefore annihilates their character sum $G_{N,j}$. By minimality, L_N^{orb} belongs to the principal left annihilator ideal generated by $L_{N,j}^{\mathrm{char}}$ whenever the character sum is nonzero, so $r_{N,j}^{\mathrm{char}} \le r_N^{\mathrm{orb}}$. The zero case is immediate. The displayed identity is the telescoping decomposition

$$Nr - r_{N,j}^{\mathrm{char}} = (Nr - r_N^{\mathrm{orb}}) + (r_N^{\mathrm{orb}} - r_{N,j}^{\mathrm{char}}).$$

□

Root-filter order versus packet order. The preceding theorem is exact in the rotated t-coordinate over $E(t)$. The arithmetic packet is the series $\mathfrak{M}_{N,j}F(x)$ over the base field, related by

$$G_{N,j}(t) = t^j \mathfrak{M}_{N,j}F(t^N).$$

Algebraic pullback, base extension, gauge multiplication, and descent may change the minimal scalar operator visible in the x-chart. That comparison is a separately typed descent problem. The exact binomial–Fuchsian family below computes both sides and therefore resolves this additional step in a nontrivial model case.

Let $Q_{N,j}^{\mathrm{cl}}$ be an annihilator produced by a declared closure or elimination pipeline, and let $L_{N,j}$ be the monic minimal operator of the section. Since every annihilator belongs to the principal left ideal generated by $L_{N,j}$,

$$Q_{N,j}^{\mathrm{cl}} = A_{N,j} L_{N,j}$$

for some Ore operator $A_{N,j}$. The nonnegative quantity

$$\delta_{N,j}^{\mathrm{cl}} = \operatorname{ord} Q_{N,j}^{\mathrm{cl}} - \operatorname{ord} L_{N,j}$$

measures cancellation relative to that construction. It is not intrinsic until the construction has been fixed.

15.5 Operator singularities and local exponent packets

Multiply L_F by a common denominator, divide the polynomial coefficients by their greatest common divisor, and normalize the leading coefficient to be monic. This gives a unique primitive polynomial representative

$$\widetilde{L}_F = \sum_{j=0}^{r} q_j(z) D^j.$$

Its operator singular support is

$$\Sigma_{\mathrm{op}}(F) = \{q_r(z) = 0\} \cup \{\infty\}.$$

Apparent singularities remain part of this operator coordinate until an explicit desingularization has been performed.

Let α be a finite point and put

$$v_j = v_\alpha(q_j), \qquad \mu = \min_j (v_j - j).$$

Write the first nonzero local coefficient of q_j as q_{j,v_j}. The indicial polynomial is

$$\boxed{I_{\alpha,L}(\rho) = \sum_{v_j - j = \mu} q_{j,v_j} \rho^{\underline{j}}.}$$

At a regular singular point, its roots form the local exponent packet. Over $\mathbb{C}$, the corresponding local monodromy eigenvalues are

$$e^{2\pi i\rho}.$$

Integer differences between exponents signal resonance. Exponent eigenvalues alone do not determine logarithmic extension data or the full monodromy conjugacy class.

Proposition 15.13 (Singularity orbit under multisection)**.** *Assume F is an analytic germ at zero with analytic continuation away from a finite set Σ. Then every finite nonzero analytic singularity of $\mathfrak{M}_{N,j}F$ lies in*

$$\boxed{\Sigma^{[N]} = \{\alpha^N : \alpha \in \Sigma\}.}$$

Distinct singularities in one fiber of $\alpha \mapsto \alpha^N$ are aliased by the multisection map. A character projection may remove some aliased local contributions.

Proof. Away from $\Sigma^{[N]}$ and zero, choose a local analytic branch $t = x^{1/N}$. Every rotated argument $\omega_N^q t$ avoids Σ, so the root-filter expression is analytic. A singularity can therefore occur only when one rotated preimage meets Σ. □

At nonzero regular singular points, the map $t \mapsto t^N$ is unramified. It preserves local exponents before character projection. At zero and infinity it is ramified, so exponent scaling must be recorded separately.

15.6 An exact binomial–Fuchsian packet family

The general minimal multisection problem remains open. The family

$$\boxed{F_{\lambda,\alpha}(z) = (1 - \alpha z)^{-\lambda}}$$

provides a nontrivial exact model that contains irreducible Fuchsian packets, rational collapse, finite-support aliasing, and zero-address cancellation in one parameter space.

Its coefficients are

$$a_n = \frac{(\lambda)_n}{n!}\alpha^n.$$

For $0 \le j < N$, put

$$G_{N,j}^{\lambda,\alpha}(x) = \sum_{m\ge 0} \frac{(\lambda)_{Nm+j}}{(Nm+j)!}\alpha^{Nm+j}x^m.$$

Define the upper and lower parameter packets

$$A_{N,j}(\lambda) = \left\{\frac{\lambda + j + r}{N} : 0 \le r < N\right\},$$

$$B_{N,j} = \left\{\frac{j + 1 + r}{N} : 0 \le r < N\right\} \setminus \{1\}.$$

Theorem 15.14 (Exact hypergeometric packet form)**.** *If $(\lambda)_j \neq 0$, then*

$$\boxed{G_{N,j}^{\lambda,\alpha}(x) = \frac{(\lambda)_j\alpha^j}{j!}\,{}_NF_{N-1}\left(\begin{matrix}A_{N,j}(\lambda)\\ B_{N,j}\end{matrix}; \alpha^N x\right).}$$

Its coefficients satisfy the first-order polynomial recurrence

$$\boxed{\prod_{r=0}^{N-1}(Nm+j+1+r)u_{m+1} = \alpha^N \prod_{r=0}^{N-1}(Nm+j+\lambda+r)u_m.}$$

The series is annihilated by

$$\boxed{\mathcal{H}_{N,j}^{\lambda,\alpha} = \theta \prod_{b\in B_{N,j}}(\theta+b-1) - \alpha^N x \prod_{a\in A_{N,j}(\lambda)}(\theta+a),}$$

where $\theta = xD_x$.

Proof. Split the rising factorials and ordinary factorials into their N residue factors. The multiplication formula gives the displayed hypergeometric coefficients. Taking the ratio of consecutive packet coefficients gives the recurrence. Coefficient comparison with the standard ${}_NF_{N-1}$ operator gives the differential equation. □

Theorem 15.15 (Minimal differential order phase diagram)**.** *Let*

$$r_D(N,j;\lambda) = r_D\big(G_{N,j}^{\lambda,\alpha}\big), \qquad \alpha \neq 0.$$

Then:

1. *If* $\lambda \notin \mathbb{Z}$, *then*
$$\boxed{r_D(N,j;\lambda) = N.}$$
2. *If* $\lambda = L \in \mathbb{Z}_{>0}$, *then every packet is a nonzero rational function with exact denominator*
$$(1-\alpha^N x)^L,$$
and
$$\boxed{r_D(N,j;L) = 1.}$$
3. *If* $\lambda = -M \in \mathbb{Z}_{\leq 0}$, *then* $G_{N,j}^{-M,\alpha}$ *is zero when* $j > M$. *Otherwise it is a nonzero polynomial of degree*
$$\left\lfloor \frac{M-j}{N} \right\rfloor$$
and
$$\boxed{r_D(N,j;-M) = \begin{cases} 1, & j \leq M, \\ 0, & j > M. \end{cases}}$$

Proof. For nonintegral λ, no difference between an upper parameter and a completed lower parameter is an integer. The classical irreducibility criterion for generalized hypergeometric operators therefore makes $\mathcal{H}_{N,j}^{\lambda,\alpha}$ irreducible of order N [13]. Since the packet is nonzero, its minimal operator has order N.

For $L > 0$, the coefficient is a polynomial of degree $L-1$ in m times α^{Nm+j}. Its generating function is rational with denominator $(1-\alpha^N x)^L$, and the leading polynomial term prevents cancellation of the highest denominator power. Every nonconstant rational function has a first-order logarithmic-derivative annihilator.

For $\lambda = -M$, the source series is the polynomial $(1-\alpha z)^M$. Selecting one residue class leaves exactly the displayed finite support. A nonzero polynomial has differential order one, while the zero function has order zero. □

The tail recurrence order and differential order are different invariants. A nonzero finite-support coefficient sequence has tail recurrence order zero but its polynomial generating function has differential order one.

15.7 Orbit aliasing and address cancellation

Let ω_N be primitive and consider the rotated orbit

$$\mathcal{O}_N(F_{\lambda,\alpha}) = \left\{ (1-\alpha\omega_N^r t)^{-\lambda} : 0 \le r < N \right\}.$$

Let r_{orb} be its dimension over the constant field.

Theorem 15.16 (Exact orbit rank)**.** *For the binomial family,*

$$\boxed{r_{\mathrm{orb}}(N;\lambda) = \begin{cases} N, & \lambda \notin \mathbb{Z}_{\le 0}, \\ \min(N, M+1), & \lambda = -M \in \mathbb{Z}_{\le 0}. \end{cases}}$$

Proof. If λ is nonintegral, each rotated branch has a branch singularity at a different point. If λ is a positive integer, each has a pole at a different point. A constant linear relation would therefore have zero coefficient at every branch.

If $\lambda = -M$, expand

$$(1-\alpha\omega_N^r t)^M = \sum_{n=0}^{M} \binom{M}{n} (-\alpha)^n \omega_N^{rn} t^n.$$

The rotation matrix is a truncated discrete Fourier matrix. Its rank is the number of available residue classes, namely $\min(N, M+1)$. □

Define the alias and cancellation deficits by

$$\boxed{A_N(\lambda) = N - r_{\mathrm{orb}}(N;\lambda),}$$

$$\boxed{C_{N,j}(\lambda) = r_{\mathrm{orb}}(N;\lambda) - r_D(N,j;\lambda).}$$

Then

$$\boxed{N - r_D(N,j;\lambda) = A_N(\lambda) + C_{N,j}(\lambda).}$$

Corollary 15.17 (Exact defect formulas)**.** *The three parameter regimes have the following signatures.*

1. *If $\lambda \notin \mathbb{Z}$, then*
$$A_N(\lambda) = 0, \qquad C_{N,j}(\lambda) = 0.$$
2. *If $\lambda = L > 0$, then*
$$A_N(L) = 0, \qquad C_{N,j}(L) = N - 1.$$

3. *If* $\lambda = -M \le 0$, *then*

$$A_N(-M) = \max(0, N - M - 1),$$

and

$$C_{N,j}(-M) = \begin{cases} \min(N, M+1) - 1, & j \le M, \\ \min(N, M+1), & j > M. \end{cases}$$

This separates two mechanisms that are easily conflated. Orbit aliasing occurs before an address is selected. Address cancellation occurs after the Fourier projector has been applied.

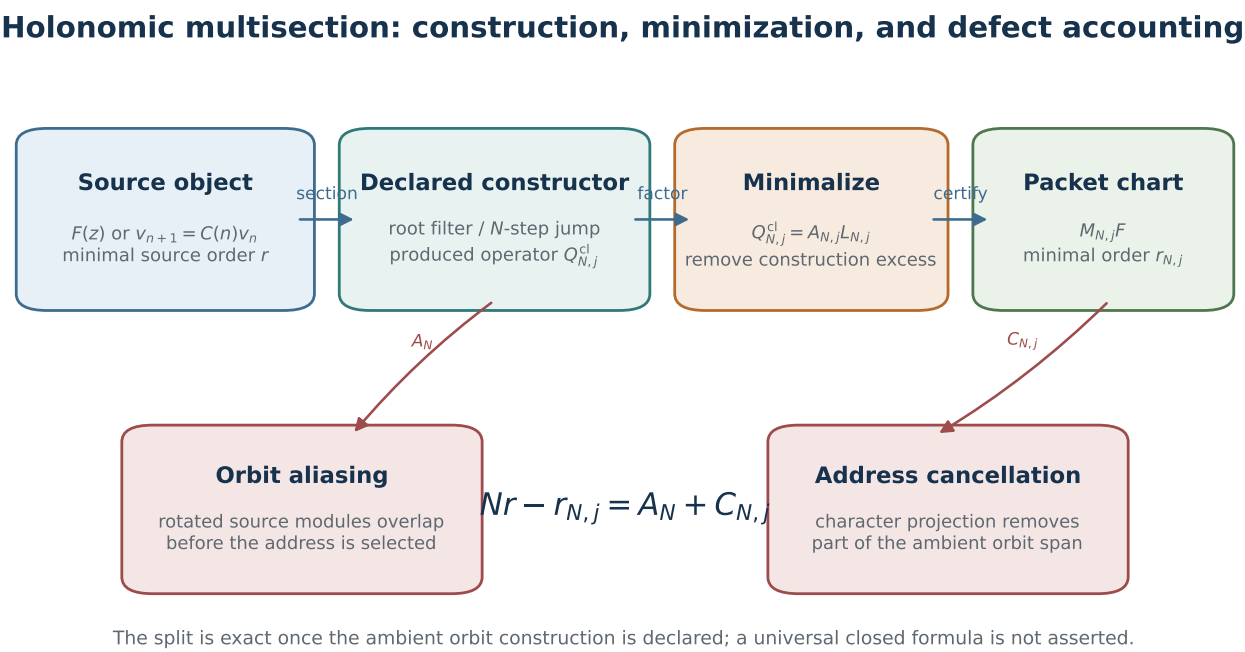


Figure 36: The declared holonomic multisection pipeline. General closure constructs an annihilator but does not prove minimality. The binomial–Fuchsian family admits an exact orbit-rank and address-cancellation decomposition.

15.8 Local exponent and monodromy packet

For $\lambda \notin \mathbb{Z}$, the hypergeometric operator in theorem 15.14 is Fuchsian with singularities

$$\boxed{0, \qquad \alpha^{-N}, \qquad \infty.}$$

Its exponent packets are

$$\boxed{E_0 = \{0\} \cup \{1 - b : b \in B_{N,j}\},}$$

$$\boxed{E_{\alpha^{-N}} = \{0, 1, \dots, N-2, -\lambda\},}$$

and

$$\boxed{E_\infty = A_{N,j}(\lambda).}$$

Modulo integers, E_0 is the complete N-torsion set

$$\left\{0, \frac{1}{N}, \dots, \frac{N-1}{N}\right\},$$

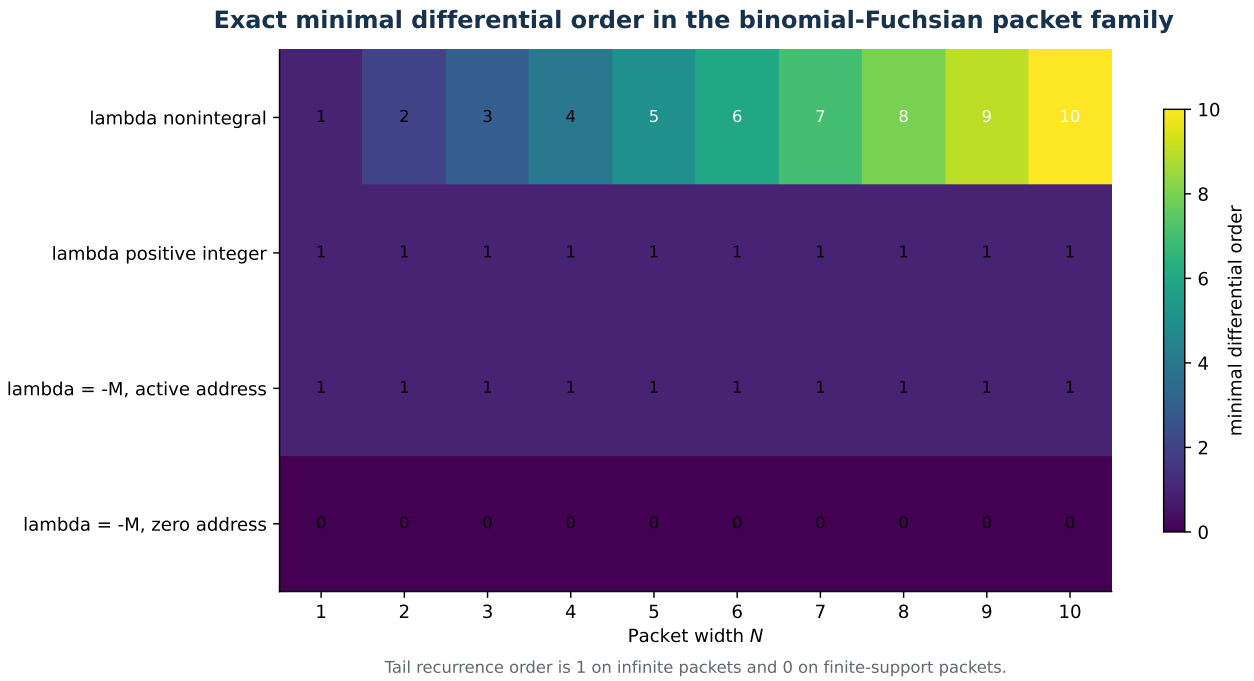


Figure 37: Minimal differential order in the binomial packet family. Nonintegral exponents retain order N, positive integers collapse to rational order one, and nonpositive integers terminate with explicit aliasing and zero-address packets.

and the pairwise differences in E_∞ are also N-torsion.

Proposition 15.18 (Local monodromy ledger)**.** *For nonintegral rational λ:*

1. *the monodromy eigenvalues at zero form the complete group of Nth roots of unity;*
2. *the ratios of monodromy eigenvalues at infinity are Nth roots of unity;*
3. *the distinguished nontrivial eigenvalue at α^{-N} is*

$$e^{-2\pi i\lambda}.$$

The local eigenvalues are finite-order when $\lambda \in \mathbb{Q}$. This does not imply that the global monodromy group is finite. Connection matrices and logarithmic extension data remain additional invariants.

15.9 Marked normalized packet atlas

The order theorem is most naturally a theorem about marked packets. Without the mark, distinct choices of source exponent, scale, width, and address can represent the same zero packet or the same rational function.

Definition 15.19 (Marked binomial–Fuchsian packet)**.** A marked packet is a tuple

$$\mathfrak{b} = (\alpha, \lambda, N, j; G),$$

where $\alpha \neq 0$, $N \geq 1$, $0 \leq j < N$, and

$$G(x) = G_{N,j}^{\lambda,\alpha}(x).$$

The parameters (α, λ, N, j) are part of the semantic object rather than metadata to be forgotten.

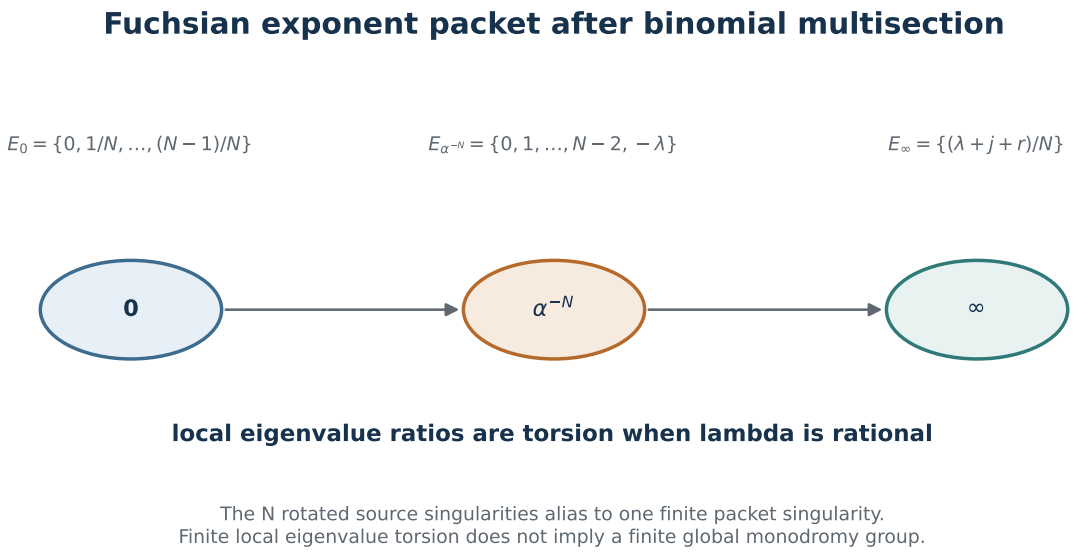


Figure 38: The exact Fuchsian exponent packet of a nonintegral binomial multisection. Root-orbit aliasing moves N rotated source singularities into one finite singularity while the complete N-torsion exponent sets remain visible at zero and infinity.

For a nonzero marked packet, consider the following normalized charts:

(i) the marked coefficient series G;

(ii) the primitive first-order tail recurrence obtained from the exact coefficient ratio, together with its finite initial record;

(iii) the monic minimal differential operator from theorem 15.15, together with the distinguished power-series solution selected by the marked coefficient record;

(iv) the hypergeometric parameter packet $\left(A_{N,j}(\lambda), B_{N,j}; \alpha^N\right)$ and leading coefficient;

(v) the primitive minimal-operator singular and indicial record

$$\mathscr{E}(G) = \left(\Sigma_{\mathrm{op}}(G), \left(I_{\xi,L_G}\right)_{\xi\in\Sigma_{\mathrm{op}}(G)}\right),$$

together with the orbit-rank and defect data

$$\left(r_{\mathrm{orb}}, r_D, A_N, C_{N,j}\right).$$

In the nonintegral Fuchsian regime, $\mathscr{E}(G)$ is the explicit three-point packet

$$\left(E_0, E_{\alpha^{-N}}, E_\infty\right)$$

computed above. In the integral regimes it is computed from the corresponding minimal first-order operator. The zero packets in the terminated regime are retained as separately marked zero objects with the zero-operator convention.

Theorem 15.20 (Effective marked binomial–Fuchsian atlas)**.** *The five normalized presentations above determine one another on the declared marked family. With equality morphisms they form equivalent discrete groupoids, every normalized fiber is contractible, and objects and morphisms descend effectively. The package*

$$\boxed{\mathcal{J}_{\mathrm{BF}} = \big(\alpha, \lambda, N, j, R_{N,j}, L_{N,j}, A_{N,j}(\lambda), B_{N,j}, \mathscr{E}(G), r_{\mathrm{orb}}, r_D, A_N, C_{N,j}\big)}$$

is complete on this marked category.

Proof. The mark constructs the coefficient packet directly. Consecutive coefficient ratios give the normalized recurrence chart, and the multiplication formula gives the hypergeometric chart. Theorem 15.15 identifies the monic minimal differential operator in every parameter regime. The primitive polynomial representative of that operator determines $\mathscr{E}(G)$ by the indicial construction in Section 15.5. In the nonintegral regime, proposition 15.18 gives its explicit Fuchsian eigenvalue ledger. Theorem 15.16 and corollary 15.17 compute the orbit and cancellation data in every regime. Conversely, the displayed invariant retains the complete mark and therefore reconstructs the source packet and every normalized chart. Uniqueness of monic Ore normalization and of the marked coefficient record leaves no nontrivial automorphism in a fiber. □

15.10 Examples and exact verification

The parity sections of

$$\frac{2}{1-z^2}$$

give the smallest singularity-alias example. The source singularities 1 and -1 both map to 1 under squaring. The even address is $2/(1-x)$, while the odd address vanishes.

For

$$2\cosh z,$$

the even packet is nonzero and the odd packet is zero. There is no finite singularity, so the cancellation occurs at the level of the character-projected solution rather than by finite singularity collision.

For the Catalan sequence, every arithmetic section remains hypergeometric in the packet index. Its recurrence order is one, while its ordinary generating function remains a second-order algebraic D-finite function. This again shows that recurrence order and scalar differential order are different coordinates.

For the Apéry numbers, the known two-dimensional nonautonomous state jumps to every arithmetic section. The jump theorem gives an exact order-two recurrence bound, but the present holonomic theorem does not claim a closed minimality formula for all Apéry addresses.

Table 25: Representative exact order profiles in the binomial–Fuchsian packet family.

Source	Packet	r_{tail}	r_D	r_{orb}	Alias	Cancel	Interpretation
$(1-z)^{-1/2}$	$N=2,\ j=0$	1	2	2	0	0	irreducible nonintegral packet
$(1-z)^{-1/3}$	$N=3,\ j=2$	1	3	3	0	0	finite local exponent torsion
$(1-z)^{-2}$	$N=3,\ j=1$	1	1	3	0	2	rational collapse
$(1-z)^{2}$	$N=4,\ j=1$	0	1	3	1	2	finite-support polynomial
$(1-z)^{2}$	$N=4,\ j=3$	0	0	3	1	3	zero address packet

The Holonomic atlas verifier checks the binomial packet recurrence, the generalized hypergeometric differential coefficient identity, nonresonance of the irreducible parameter packets, zero and infinity

Table 26: Exact Holonomic atlas verification counts.

Validation class	Exact checks
Nonintegral binomial packets	110,880
Positive-integral rational collapse	27,720
Nonpositive-integral aliasing	11,492
Root-filter and order ledger	24,848
Holonomic atlas total	174,940
Cumulative the preceding certified atlas program total	1,838,856
Failures	0

exponent torsion classes, positive-integral rational collapse, nonpositive-integral orbit rank and address support, and root-filter orthogonality. It records 174,940 exact checks with no failures.

Exact holonomic boundary. The holonomic section proves normalized minimal Ore charts on the ordinary-point locus and an exact multisection minimality theorem for the binomial–Fuchsian packet family. It does not prove a universal formula for minimal multisection operators of arbitrary D-finite series. Irregular singularities, Stokes data, apparent-singularity removal, resonant logarithmic extensions, general differential-module cyclicity after character projection, and a finite all-address signature for unrestricted holonomic systems remain open.

Section result. P-recursive and D-finite presentations now form an explicit normalized atlas on the ordinary-point locus, with minimal Ore annihilators, finite prefix and jet certificates, and exact coefficient transport. General arithmetic multisection has a typed root-filter and state-jump construction, together with an explicit construction-relative minimality defect. The binomial–Fuchsian family supplies the first exact holonomic packet phase diagram: nonintegral exponents retain differential order N, positive integers collapse to rational order one, nonpositive integers terminate with a closed orbit-alias and address-cancellation formula, and the nonintegral local exponent packet exposes the complete finite torsion at zero and infinity.

16 Selected algebraic branches, packet Weil restriction, and collision invariants

Algebraic atlas develops the algebraic stratum after the fixed-radix, product, local, and holonomic milestones. The central issue is not algebraicity alone. A polynomial equation may have several formal branches, and a packet elimination can contain branch tuples that do not belong to the selected semantic germ. A normalized algebraic chart must therefore retain both a minimal polynomial and a finite branch selector. Arithmetic packets are then compiled as a finite polynomial system. The generic packet scheme is a Weil restriction of the source branch scheme, which gives a uniform degree bound. A separate collision ledger explains why the degree of one selected packet may be smaller than that bound.

Throughout this section, K is a characteristic-zero field and $\overline{K}$ is an algebraic closure. Algebraic degree always means degree over the displayed rational-function field. All assertions about formal branches are made inside $\overline{K}[[z]]$ unless a different local field is named.

16.1 Minimal polynomials and finite branch selectors

Let $F \in K[[z]]$ be algebraic over $K(z)$. Its monic irreducible polynomial is

$$P_F(z, Y) \in K(z)[Y], \qquad P_F(z, F(z)) = 0.$$

Write

$$d(F) = \deg_Y P_F.$$

After multiplying by a common denominator and dividing by the greatest common divisor of the polynomial coefficients, one obtains a primitive representative in $K[z, Y]$, unique up to a nonzero scalar. We normalize that scalar by fixing the first nonzero coefficient in graded lexicographic order to be one.

A polynomial alone does not select a branch. Let

$$\mathrm{Br}_0(P_F) = \{G \in \overline{K}[[z]] : P_F(z, G(z)) = 0\}.$$

This set has cardinality at most $d(F)$.

Definition 16.1 (Branch-separation index)**.** For $F \in \mathrm{Br}_0(P_F)$, define

$$\sigma(P_F, F) = \begin{cases} 0, & \mathrm{Br}_0(P_F) = \{F\}, \\ 1 + \max\limits_{G \in \mathrm{Br}_0(P_F),\, G \neq F} \mathrm{ord}_z(F - G), & \text{otherwise.} \end{cases}$$

The corresponding branch selector is the finite coefficient jet

$$\boxed{J_{\mathrm{alg}}(F) = (a_0, a_1, \ldots, a_{\sigma-1}), \qquad F(z) = \sum_{n \geq 0} a_n z^n.}$$

When $\sigma = 0$, the selector is empty.

Theorem 16.2 (Finite branch separation)**.** *The pair*

$$\boxed{(P_F, J_{\mathrm{alg}}(F))}$$

selects F uniquely among the formal power-series branches of P_F at the origin.

Proof. There are at most $d(F)$ formal branches. The difference of two distinct formal series has finite z-adic order. By definition of σ, every other branch differs from F in one of the coefficients recorded by $J_{\mathrm{alg}}(F)$. □

The simplest case is the ordinary implicit-function locus.

Corollary 16.3 (Simple-branch normalization)**.** *If*

$$P_F(0, a_0) = 0, \qquad \partial_Y P_F(0, a_0) \neq 0,$$

then a_0 alone selects a unique formal branch. Equivalently, $\sigma(P_F, F) \leq 1$.

Proof. Coefficient comparison in $P_F(z, F(z)) = 0$ determines each next coefficient linearly, with diagonal coefficient $\partial_Y P_F(0, a_0)$. □

The finite selector is semantic data. It cannot be dropped merely because the polynomial is irreducible over $K(z)$. Irreducibility controls the generic function-field extension, while the jet chooses one local embedding of that extension into formal series.

16.2 The normalized selected-branch atlas

Let

$$A_F = K(z)[Y]/(P_F),$$

and let $\bar{Y}$ denote the residue class of Y. In the basis

$$1, \bar{Y}, \dots, \bar{Y}^{d-1},$$

multiplication by $\bar{Y}$ is represented by the companion matrix M_F. Its characteristic and minimal polynomials are both P_F.

Theorem 16.4 (Normalized selected-branch algebraic atlas)**.** *The following normalized presentations determine one another exactly:*

$$\boxed{\begin{aligned}&F \in K[[z]],\\ &(P_F, J_{\mathrm{alg}}(F)),\\ &(A_F, \bar{Y}, J_{\mathrm{alg}}(F)),\\ &(M_F, J_{\mathrm{alg}}(F)).\end{aligned}}$$

With equality morphisms, these presentations form equivalent normalized groupoids. Every fiber over a fixed semantic branch is contractible.

Proof. The semantic branch gives its unique monic minimal polynomial and its finite branch selector. The polynomial gives the finite field algebra and its companion multiplication matrix. Conversely, the companion matrix recovers P_F as its minimal polynomial. The branch selector then recovers the selected formal root by theorem 16.2. Every normalized presentation is determined literally, so a normalized automorphism fixing the semantic branch is the identity. □

16.3 The exact residue-packet polynomial compiler

Fix $N \geq 1$. For

$$F(z) = \sum_{n \geq 0} a_n z^n,$$

define the arithmetic packets

$$U_j(x) = \sum_{m \geq 0} a_{Nm+j} x^m, \qquad 0 \leq j < N.$$

They satisfy the exact synthesis identity

$$\boxed{F(z) = \sum_{j=0}^{N-1} z^j U_j(z^N).}$$

Let $P(z, Y) \in K[z, Y]$ be the normalized primitive representative of P_F. Introduce indeterminates

$$\mathbf{U} = (U_0, \dots, U_{N-1})$$

and substitute

$$Y = \sum_{j=0}^{N-1} z^j U_j.$$

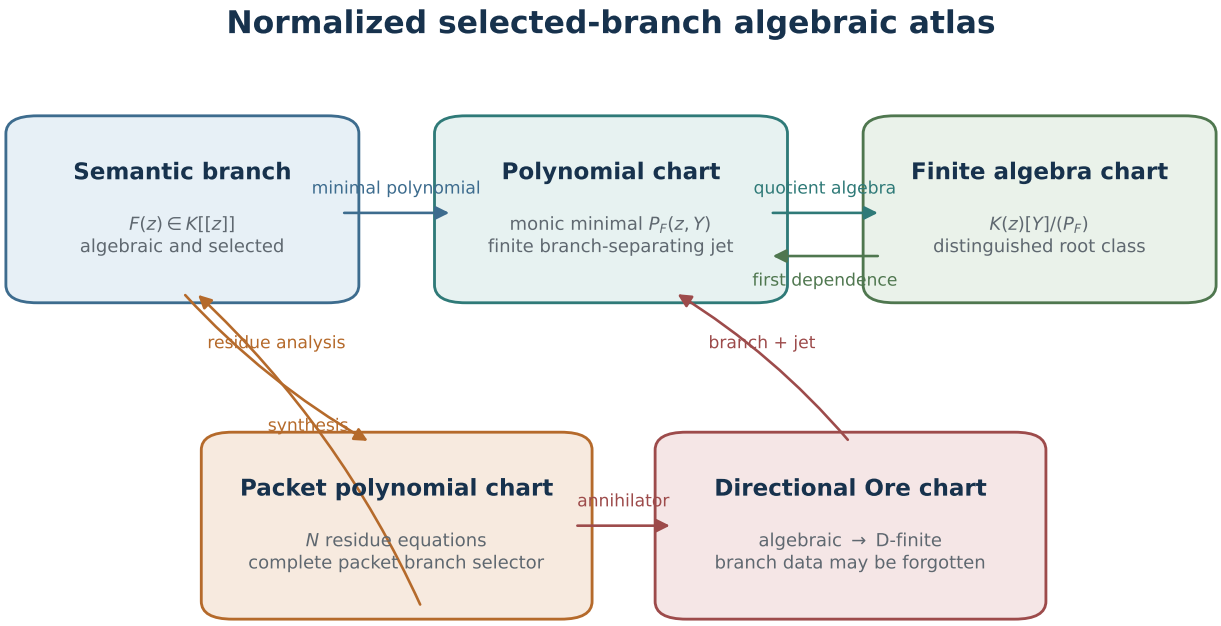


Figure 39: The selected algebraic branch, its minimal polynomial and finite selector, its finite quotient algebra, and its complete packet system are reversible normalized charts. The Ore chart is directional because an annihilator can forget branch data.

Every polynomial in z has a unique decomposition modulo the relation $z^N = x$:

$$P\left(z, \sum_{j=0}^{N-1} z^j U_j\right) = \sum_{r=0}^{N-1} z^r \Phi_{N,r}(x, \mathbf{U}).$$

The polynomials

$$\boxed{\Phi_{N,r}(x, \mathbf{U}) \in K[x, \mathbf{U}]}$$

are the algebraic packet equations.

Theorem 16.5 (Exact algebraic packet system)**.** *Let $d = \deg_Y P_F$. Then:*

1. *the semantic packet tuple $(U_0, \dots, U_{N-1})$ satisfies*
$$\Phi_{N,r}(x, \mathbf{U}) = 0 \qquad (0 \le r < N);$$
2. *every $\Phi_{N,r}$ has total degree at most d in the packet variables;*
3. *a formal packet tuple satisfying all N equations synthesizes a formal branch of P_F;*
4. *the packet tuple selected by the transported coefficient jet reconstructs F uniquely.*

Proof. Substituting the semantic synthesis formula into $P_F(z, F(z)) = 0$ gives the first statement. Substitution does not increase degree in the U_j, so the second follows. If all residue polynomials vanish, then the synthesized series satisfies $P_F = 0$. The source branch selector is distributed among the packets by residue class. The resulting packet jet synthesizes the original jet, so theorem 16.2 identifies the selected branch. □

If the source selector records $a_0, \dots, a_{\sigma-1}$, the transported packet selector records precisely

$$[x^m]U_j = a_{Nm+j} \qquad \text{for } Nm + j < \sigma.$$

No information is added or lost. The coefficient prefix is merely rewritten in packet addresses.

16.4 Weil-restricted packet degree

Put

$$R = K(x), \qquad E_N = R[z]/(z^N - x).$$

Characteristic zero makes $z^N - x$ separable over R. Interpret the monic polynomial $P_F(z, Y)$ as a polynomial in $E_N[Y]$, and define

$$B_{N,F} = E_N[Y]/(P_F).$$

The packet polynomial system is the coordinate presentation of the finite Weil restriction

$$\mathrm{Res}_{E_N/R} \operatorname{Spec} B_{N,F}.$$

Theorem 16.6 (Algebraic packet degree bound)**.** *Let F have algebraic degree d over $K(z)$. For every packet width N:*

1. *the generic geometric degree of the complete packet scheme is*

$$\boxed{d^N;}$$

2. *the joint field generated by a selected complete packet satisfies*

$$\boxed{[K(x)(U_0, \dots, U_{N-1}) : K(x)] \le d^N;}$$

3. *every individual packet satisfies*

$$\boxed{[K(x)(U_j) : K(x)] \le d^N.}$$

Proof. Over an algebraic closure of $K(x)$, the equation $z^N = x$ has N distinct roots. The algebra E_N therefore splits into a product of N copies of the algebraic closure. At each address, the separable degree-d polynomial P_F has d geometric roots. The Weil-restricted fiber is their Cartesian product and has d^N points. A selected packet tuple is one point of this finite scheme, so its residue field and every coordinate subfield have degree at most the scheme degree. □

This bound is constructive. The quotient of

$$K(x)[U_0, \dots, U_{N-1}]$$

by the packet equations is a finite algebra of generic rank d^N. Multiplication by U_j is therefore a matrix of size d^N. Its characteristic polynomial annihilates U_j by Cayley–Hamilton.

Corollary 16.7 (Constructive packet eliminant)**.** *For each j, one can compute an annihilating polynomial*

$$Q_{N,j}(x, T) \in K(x)[T]$$

with

$$\deg_T Q_{N,j} \le d^N$$

by multiplication matrices, resultants, or a Gröbner elimination of the exact packet system.

The bound is a universal degree ceiling for the minimal-polynomial grammar. It does not assert that every coordinate attains degree d^N or that one elimination algorithm returns the minimal polynomial without factor selection.

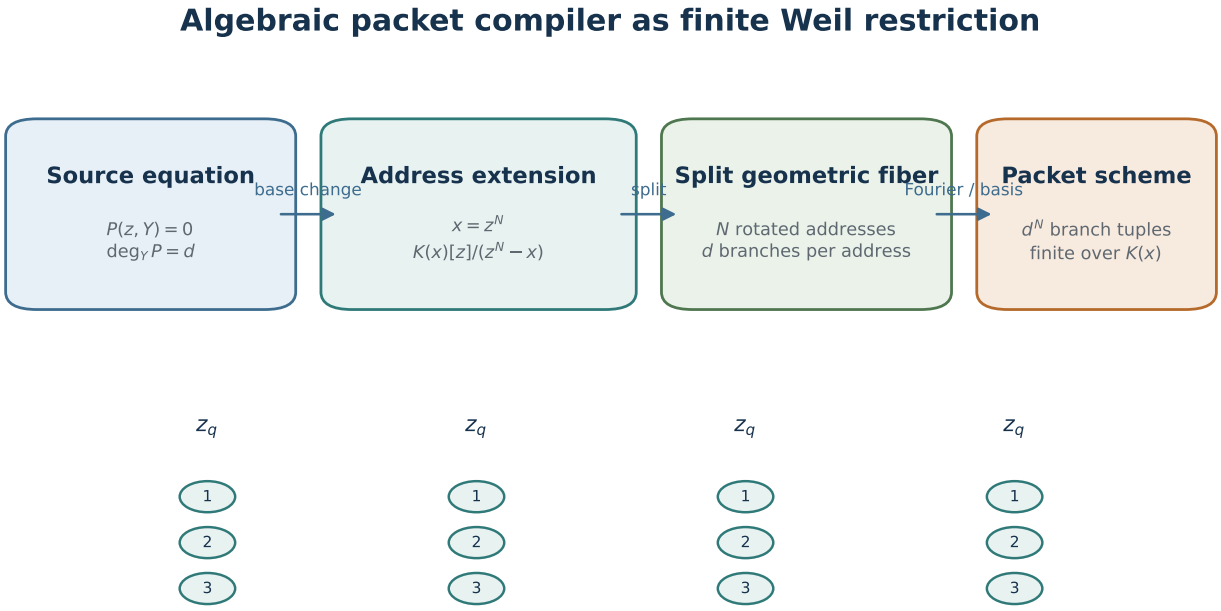


Figure 40: After the address extension $x = z^N$ splits, each of the N addresses carries d algebraic branches. The complete packet scheme is their Cartesian product and has generic degree d^N.

16.5 Tuple collision and selected-orbit loss

Let Ω be a normal splitting field of the generic packet scheme over $K(x)$, and let

$$\Gamma = \mathrm{Gal}(\Omega/K(x)).$$

After splitting E_N, the geometric packet branches form a set

$$\mathcal{B}_N$$

of cardinality d^N. Let

$$\pi_j : \mathcal{B}_N \to \Omega$$

be the jth packet coordinate, and set

$$\mathcal{C}_{N,j} = \pi_j(\mathcal{B}_N).$$

For the selected semantic packet U_j, separability gives

$$[K(x)(U_j) : K(x)] = |\Gamma \cdot U_j|.$$

Definition 16.8 (Algebraic packet collision defects)**.** Define

$$\mathfrak{c}_{N,j}^{\mathrm{tuple}} = d^N - |\mathcal{C}_{N,j}|,$$

and

$$\mathfrak{s}_{N,j}^{\mathrm{Gal}} = |\mathcal{C}_{N,j}| - |\Gamma \cdot U_j|.$$

The first records collisions caused by projection from complete branch tuples to one packet coordinate. The second records the remaining loss caused by the Galois stabilizer of the selected coordinate.

Theorem 16.9 (Exact algebraic degree-defect identity)**.** *The two defects are nonnegative and satisfy*

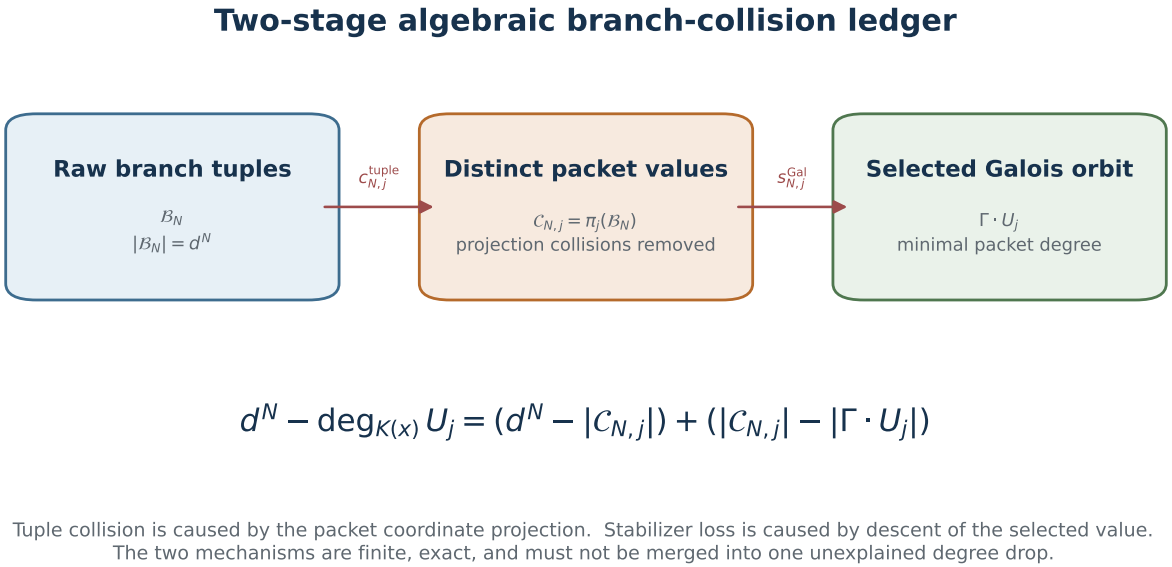
$$\boxed{d^N - [K(x)(U_j) : K(x)] = \mathfrak{c}_{N,j}^{\mathrm{tuple}} + \mathfrak{s}_{N,j}^{\mathrm{Gal}}.}$$

Proof. The coordinate image $\mathcal{C}_{N,j}$ has at most d^N values. The Galois orbit of the selected value is contained in that image. The displayed identity is the telescoping decomposition

$$d^N - |\Gamma \cdot U_j| = (d^N - |\mathcal{C}_{N,j}|) + (|\mathcal{C}_{N,j}| - |\Gamma \cdot U_j|).$$

□

This signature separates two mechanisms that a raw resultant merges. Repeated projected values are a coordinate collision. A smaller selected orbit is a descent or stabilizer phenomenon. They are not the same obstruction.



Tuple collision is caused by the packet coordinate projection. Stabilizer loss is caused by descent of the selected value. The two mechanisms are finite, exact, and must not be merged into one unexplained degree drop.

Figure 41: The universal degree d^N can drop first because several branch tuples have the same packet coordinate, then because the selected coordinate has a nontrivial Galois stabilizer.

16.6 A sharp Catalan parity theorem

Let

$$C(z) = 1 + zC(z)^2$$

be the Catalan branch with $C(0) = 1$, and write

$$C(z) = E(z^2) + zO(z^2).$$

Put $x = z^2$.

Theorem 16.10 (Exact Catalan packet system)**.** *The parity packets satisfy*

$$\boxed{E = 1 + 2xEO, \qquad O = E^2 + xO^2.}$$

Elimination gives

$$\boxed{4xE^4 - E^2 + 1 = 0}$$

and

$$\boxed{4x^3O^4 - 8x^2O^3 + 5xO^2 - O + 1 = 0.}$$

Both packets have algebraic degree four over $K(x)$. Hence the universal bound

$$d^N = 2^2 = 4$$

is sharp.

Proof. Substitute $C = E + zO$ into $C = 1 + zC^2$ and separate even and odd powers. The two resultants give the displayed quartics.

For the even packet, put $T = E^2$. Then

$$4xT^2 - T + 1 = 0.$$

Its discriminant $1 - 16x$ is not a square in $K(x)$, so $[K(x)(T) : K(x)] = 2$. The norm of T from this quadratic field is $1/(4x)$, which is not a square in $K(x)$. Therefore T is not a square in its quadratic field and E has degree four.

The first packet equation gives

$$O = \frac{E - 1}{2xE}, \qquad E = \frac{1}{1 - 2xO}.$$

Thus $K(x)(E) = K(x)(O)$, and the odd packet also has degree four. □

The Catalan parity packet has no tuple-collision or selected-orbit defect. Its one-coordinate degree already sees all four geometric branch tuples of the quadratic source over the two addresses.

16.7 A quadratic-radical collision phase diagram

Let

$$F_{\alpha,\beta}(z) = \sqrt{(1 - \alpha z)(1 - \beta z)}, \qquad F_{\alpha,\beta}(0) = 1,$$

and assume first that $\alpha \neq \beta$, so the normalized source degree is two. Write

$$F_{\alpha,\beta}(z) = E(x) + zO(x), \qquad x = z^2,$$

and put

$$s = \alpha + \beta, \qquad p = \alpha\beta.$$

Theorem 16.11 (Quadratic-radical parity packets)**.** *The packets satisfy*

$$\boxed{E^2 + xO^2 = 1 + px, \qquad 2EO = -s.}$$

Their eliminants are

$$\boxed{E^4 - (1 + px)E^2 + \frac{s^2 x}{4} = 0}$$

and

$$\boxed{xO^4 - (1 + px)O^2 + \frac{s^2}{4} = 0.}$$

If

$$s \neq 0, \qquad \alpha^2 \neq \beta^2,$$

then both selected packets have degree four over $K(x)$.

Proof. The two equations follow by adding and subtracting the squares of

$$F(z) = E + zO, \qquad F(-z) = E - zO.$$

Elimination gives the quartics. Their quadratic discriminant is

$$\Delta(x) = (1 - \alpha^2 x)(1 - \beta^2 x),$$

which is not a square under the stated hypotheses. Hence E^2 generates a quadratic extension. Its norm is $s^2 x/4$, which is not a square in $K(x)$, so E has degree four. Since $EO = -s/2$, the fields generated by E and O coincide. □

The symmetric case produces an exact collision.

Corollary 16.12 (Antisymmetric branch collision)**.** *If $\beta = -\alpha \neq 0$, then*

$$\boxed{O = 0, \qquad E^2 = 1 - \alpha^2 x.}$$

For the even coordinate,

$$|\mathcal{B}_2| = 4, \qquad |\mathcal{C}_{2,0}| = 3, \qquad |\Gamma \cdot E| = 2.$$

Thus

$$\mathfrak{c}_{2,0}^{\mathrm{tuple}} = 1, \qquad \mathfrak{s}_{2,0}^{\mathrm{Gal}} = 1.$$

For the selected odd coordinate,

$$|\mathcal{C}_{2,1}| = 3, \qquad |\Gamma \cdot O| = 1,$$

so

$$\mathfrak{c}_{2,1}^{\mathrm{tuple}} = 1, \qquad \mathfrak{s}_{2,1}^{\mathrm{Gal}} = 2.$$

Proof. Now $F(z) = F(-z) = \sqrt{1 - \alpha^2 z^2}$ for the selected branch, so the odd packet vanishes. Over the split branch tuple set, the two signs are chosen independently. The even projection has values $E, 0, -E$, while the odd projection has values $0, E/z, -E/z$. The selected Galois orbits have sizes two and one, respectively. □

If $\alpha = \beta$, the quadratic equation

$$Y^2 - (1 - \alpha z)^2 = 0$$

is not the normalized source polynomial. The selected branch is rational:

$$F_{\alpha,\alpha}(z) = 1 - \alpha z,$$

so the correct source degree is one. This is a negative control against using a nonminimal defining equation in the degree ledger.

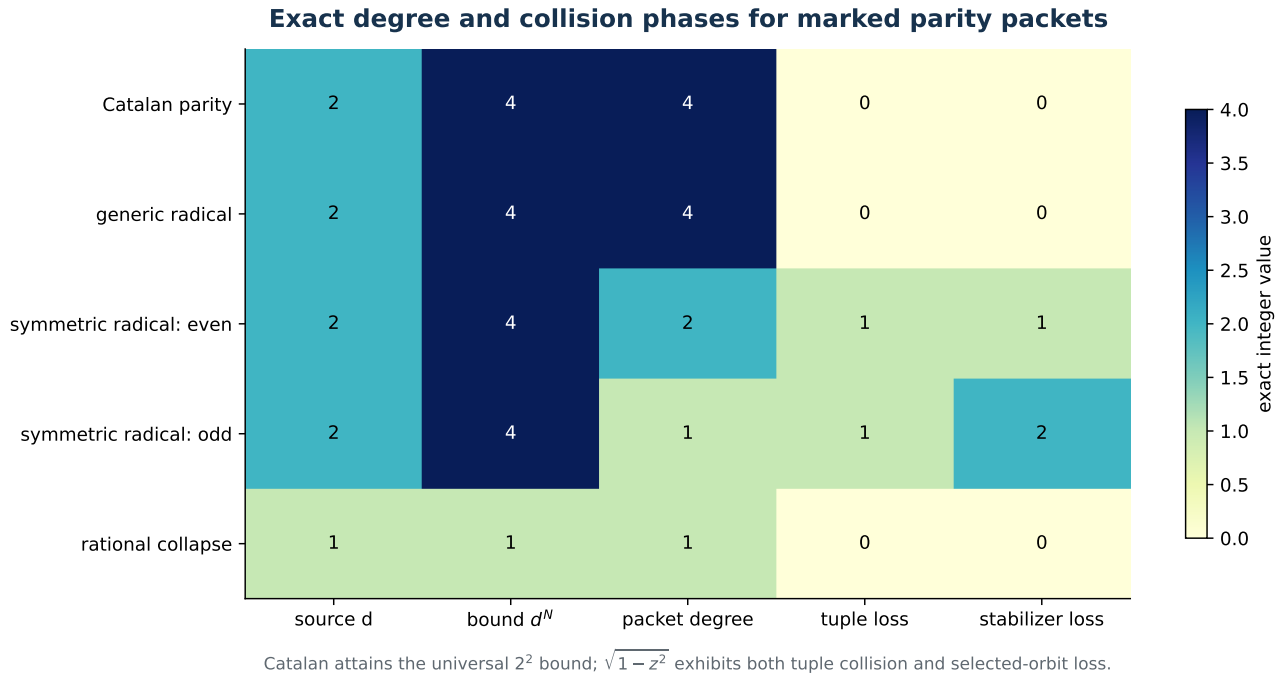


Figure 42: Catalan parity packets attain the universal quadratic bound. Generic radicals do the same, while the antisymmetric radical separates tuple collision from selected-orbit loss. A repeated linear factor must first be reduced to its minimal source polynomial.

16.8 The directional algebraic-to-holonomic bridge

Algebraic functions are D-finite, but this bridge is directional unless branch data is retained.

Proposition 16.13 (Algebraic derivative closure)**.** *Let F have separable minimal polynomial P_F of degree d. Then*

$$F' = -\frac{\partial_z P_F(z,F)}{\partial_Y P_F(z,F)} \in K(z,F).$$

Every higher derivative lies in the same degree-d function field. Consequently, F has a differential annihilator of order at most d.

Proof. Differentiate $P_F(z, F(z)) = 0$. Separability makes $\partial_Y P_F(z,F)$ nonzero in the function field. The field is closed under the induced derivation. The $d+1$ elements $F, F', \ldots, F^{(d)}$ are linearly dependent over $K(z)$. □

The annihilator can forget both the polynomial relation and the selected branch. For example,

$$\sqrt{1+z} \quad \text{and} \quad -\sqrt{1+z}$$

satisfy the same first-order differential operator. Their initial values distinguish them. More generally, a D-finite equation may have transcendental solutions, so an Ore chart does not reconstruct an algebraic minimal polynomial without additional recognition data.

Exact algebraic boundary. Algebraic atlas proves finite selected-branch normalization, an exact packet polynomial compiler, the generic d^N Weil-restriction degree, a constructive coordinate eliminant, and a two-stage collision invariant. It proves sharp packet degrees for Catalan parity and a complete collision phase for a marked quadratic-radical family. It does not prove that every elimination output is minimal without factor selection, one canonical minimal polynomial-system dimension across arbitrary multivariate grammars, an optimal packet-degree bound using all source symmetries, or unrestricted equivalence between algebraic and D-finite presentations.

16.9 Exact examples and finite verification

The verifier independently constructs packet equations by substitution and reduction modulo $z^N = x$. It checks Catalan, Motzkin, and marked radical branches at several packet widths. It verifies the sharp Catalan quartics, the quadratic-radical trace and norm systems, the generic and antisymmetric collision phases, the rejection of nonminimal quadratic equations, and finite branch-selector certificates.

Section result. A selected algebraic branch has a canonical minimal polynomial and a finite branch-separating jet. Complete arithmetic packets are the solutions of an explicit N-equation polynomial system obtained by residue reduction. The generic packet scheme is a finite Weil restriction of degree d^N, giving a constructive degree bound for every packet coordinate. Degree loss splits exactly into tuple projection collisions and Galois stabilizer loss. Catalan parity attains the universal quadratic bound, while the marked radical family exhibits generic sharpness, antisymmetric cancellation, and rational collapse. The algebraic-to-holonomic bridge remains directional unless the minimal polynomial and branch selector are retained.

Table 27: Representative exact degree and collision profiles in the Algebraic atlas algebraic packet atlas.

Source	Packet	d	d^N	degree	tuple loss	stabilizer loss	Interpretation
$C = 1 + zC^2$	parity (E, O)	2	4	4	0	0	sharp Weil-restriction bound
$\sqrt{(1-z)(1-2z)}$	parity even/odd	2	4	4	0	0	generic biquadratic packet
$\sqrt{1-z^2}$	even packet	2	4	2	1	1	rotation alias plus descent stabilizer
$\sqrt{1-z^2}$	odd packet	2	4	1	1	2	exact zero-address cancellation
$\sqrt{(1-z)^2} = 1-z$	normalized source	1	1	1	0	0	nonminimal quadratic equation rejected

Table 28: Exact finite validation of the Algebraic atlas algebraic packet layer.

Validation class	Checks	Scope
General Packet Compiler	3,690	Catalan, Motzkin, and marked quadratic radicals at packet widths two through four
Catalan Sharp Parity	7,426	exact parity system, quartic eliminants, and irreducibility ledger
Quadratic Radical Collision Phase	43,900	generic degree four, antisymmetric cancellation, and rational collapse
Finite Branch Selectors	16,632	finite jets separating explicitly colliding formal branches
Collision Ledger Combinatorics	1,440	finite tuple-collision and selected-orbit defect identities
Algebraic atlas total	**73,088**	**0 failures**
Cumulative the preceding certified atlas program	1,911,944	frozen prior records plus recomputed Algebraic atlas

17 Nonlinear polynomial fixed points, compressed packet composition, and finite tree signatures

Nonlinear tree atlas develops the nonlinear and rooted-tree stratum after the algebraic milestone. The existing nonlinear sources already prove exact packet jets, the packet Faà di Bruno law, colored operadic composition, refinement coherence, rooted-tree reconstruction, analytic majorant estimates, and local near-identity inversion [3, 2, 9]. The remaining problem is not whether nonlinear maps can be expanded. It is whether a declared nonlinear family admits a normalized finite presentation and a finite signature that controls a named packet complexity.

This section gives three restricted answers. First, a marked polynomial fixed-point system is organized into equivalent system, packet-hypergraph, operadic-tree, and exact packet-equation charts. Second, repeated-direction packet Faà di Bruno sums admit an exact orbit compression indexed by colored integer partitions rather than all labeled set partitions. Third, simply generated plane-tree equations possess a finite arity-gcd signature that determines every packet-support pattern and, under the classical smooth implicit schema, the complete dominant-singularity orbit after arithmetic packetization. A dependency condensation isolates the exact cyclic nonlinear core of a marked packet system. None of these results is promoted to a universal nonlinear normal form.

Throughout the analytic part, X is a Banach space over $\mathbb{R}$ or $\mathbb{C}$. A finite packet system on X is a family of bounded projections

$$P_i P_j = \delta_{ij} P_i, \qquad \sum_{i \in I} P_i = I_X.$$

All sums over packet addresses are finite. The formal-series part is valid over any characteristic-zero field unless positivity or singularity analysis is explicitly invoked.

17.1 Marked polynomial fixed-point systems

Let

$$A_m : X^m \longrightarrow X, \qquad 2 \le m \le D,$$

be bounded symmetric multilinear maps, and let $s \in X$. Consider the marked equation

$$\boxed{u = s + \Phi(u), \qquad \Phi(u) = \sum_{m=2}^{D} A_m[u, \dots, u].} \tag{11.1}$$

The word *marked* means that the source term, every multilinear operation, the packet system, and the selected convergence or filtration certificate are retained as part of the presentation. The fixed point u alone is not assumed to reconstruct those data.

Put

$$a = \|s\|, \qquad b_m = \|A_m\|, \qquad \varphi(t) = \sum_{m=2}^{D} b_m t^m.$$

For a radius $R > 0$, define

$$\kappa_R = \varphi'(R) = \sum_{m=2}^{D} m b_m R^{m-1}.$$

Theorem 17.1 (Selected polynomial fixed point and tree expansion)**.** *Assume*

$$a + \varphi(R) \le R, \qquad \kappa_R < 1.$$

Then equation (11.1) *has a unique solution in the closed ball $\overline{B}_R$. Moreover, that solution is the absolutely convergent sum*

$$\boxed{u = \sum_{T \in \mathcal{T}_{\mathcal{A}}} \mathrm{ev}(T),} \tag{11.2}$$

where $\mathcal{A} = \{m : A_m \neq 0\}$, the leaves of T evaluate to s, and an internal vertex of arity m evaluates by A_m applied to the ordered values of its children.

Proof. The map

$$\Psi(x) = s + \Phi(x)$$

preserves $\overline{B}_R$ because

$$\|\Psi(x)\| \le a + \sum_{m=2}^{D} b_m R^m \le R.$$

For $x, y \in \overline{B}_R$, multilinearity gives

$$A_m[x^m] - A_m[y^m] = \sum_{r=1}^{m} A_m[y^{r-1}, x - y, x^{m-r}],$$

so

$$\|\Psi(x) - \Psi(y)\| \le \kappa_R \|x - y\|.$$

The contraction theorem gives the unique fixed point.

Let $u^{(0)} = 0$ and $u^{(h+1)} = \Psi(u^{(h)})$. Expanding recursively shows that $u^{(h)}$ is the sum of the evaluations of all allowed plane rooted trees of height at most $h - 1$. If $r_0 = 0$ and

$$r_{h+1} = a + \varphi(r_h),$$

then the sum of the norms of those tree evaluations is at most r_h. The scalar iterates increase and remain bounded by R, hence converge. Monotone convergence of the nonnegative norm majorant proves absolute convergence of the full tree sum. Regrouping trees by the arity of the root gives equation (11.1). □

Remark 17.2 (Formal filtered version). If X is replaced by a complete filtered module, s begins in positive filtration, and every A_m raises total filtration, then the same tree expansion is coefficientwise finite at each filtration depth. It therefore defines the unique formal solution without a Banach norm. The analytic and formal doctrines are related but are not silently identified.

Boundary 17.3 (The selected object is a marked system). Two different polynomial systems can have the same fixed point. Conversely, a single polynomial system can have several fixed points outside the contraction ball. The normalized object in this section is therefore

$$(s, A_2, \dots, A_D; \mathcal{P}; \text{certificate}),$$

not an unmarked semantic vector u.

17.2 Packet jets, hypergraphs, and the marked nonlinear atlas

For $i, j_1, \dots, j_m \in I$, define the resolved multilinear block

$$A_{m;i|j_1,\dots,j_m} = P_i A_m(P_{j_1}\cdot, \dots, P_{j_m}\cdot).$$

Completeness of the packet family gives the exact reconstruction

$$\boxed{A_m = \sum_{i,j_1,\dots,j_m} A_{m;i|j_1,\dots,j_m}.} \tag{11.3}$$

Write

$$u_i = P_i u, \qquad s_i = P_i s.$$

Then equation (11.1) is equivalent to the complete packet system

$$\boxed{u_i = s_i + \sum_{m=2}^{D} \sum_{j_1,\dots,j_m} A_{m;i|j_1,\dots,j_m}[u_{j_1}, \dots, u_{j_m}].} \tag{11.4}$$

The active nonlinear transfer hypergraph has packet addresses as colors and one typed hyperedge

$$(j_1, \dots, j_m) \longrightarrow i$$

for every nonzero block $A_{m;i|j_1,\dots,j_m}$. Rooted substitution of these hyperedges is precisely the colored symmetric operad already present in the nonlinear core sources [2, 9].

Theorem 17.4 (Normalized atlas of a marked polynomial system)**.** *Fix a finite packet system and a certificate satisfying the hypotheses of theorem 17.1, or a declared complete filtration satisfying the formal version. The following presentations determine one another exactly:*

the marked polynomial system $(s, A_2, \dots, A_D)$,
the complete family of packet blocks $A_{m;i\|j_1,\dots,j_m}$,
the colored operadic tree grammar with its actual vertex operations,
the exact packet fixed-point system (11.4).

Each chart reconstructs the same selected fixed point. With literal normalized data and equality morphisms, every semantic fiber is contractible.

Proof. Equation (11.3) reconstructs each multilinear operation from its complete block family. The active hypergraph together with the actual block maps is simply another indexing of the same data. Rooted operadic substitution generates exactly the tree evaluations of theorem 17.1. Applying the output projections to the fixed-point equation gives (11.4), and synthesis recovers the original equation. Once the packet system, block maps, source term, and certificate are normalized literally, an automorphism fixing the semantic object and all marked data is the identity. □

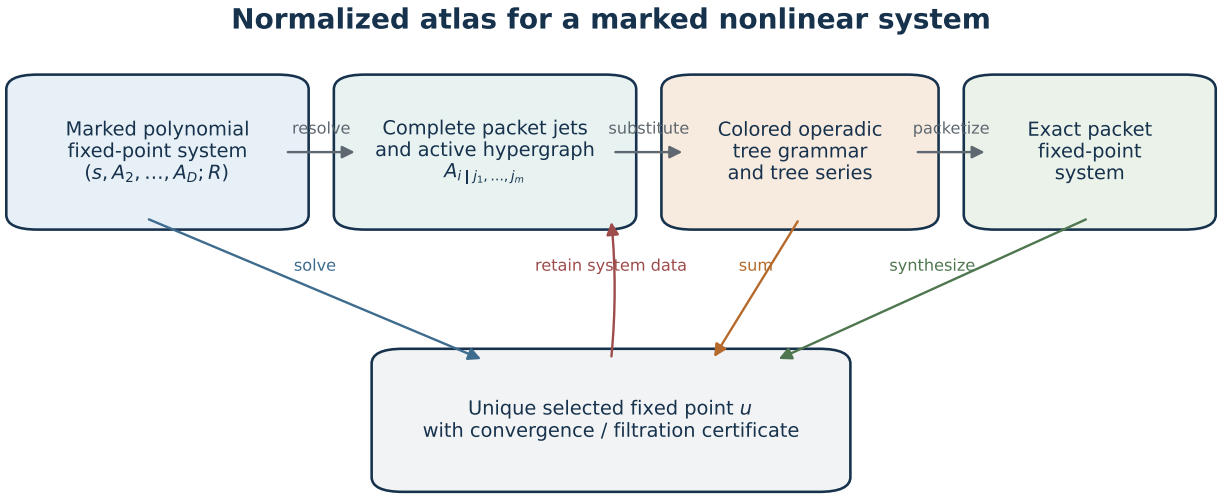


Figure 43: The marked system, its complete packet jets, its operation-labeled tree grammar, and its exact packet equations are reversible charts. The fixed point by itself is directional because it forgets the generating system.

17.3 Exact diagonal packet Faà di Bruno compression

The packet Faà di Bruno theorem from the nonlinear source expands an nth derivative over all set partitions and all intermediate packet labels [3, 2, 9]. When all directions are equal, the symmetric derivative tensors make many labeled terms identical. They can be grouped exactly by block-size and packet-color multiplicities.

Let $F : Y \to Z$ and $G : X \to Y$ be n times differentiable. Fix an input packet β, an output packet δ, a point x, and a direction $h \in P_\beta X$. For every intermediate packet γ and every $r \geq 1$, put

$$g_{r,\gamma} = Q_\gamma D^r G(x)[h, \ldots, h].$$

A colored partition profile is a finite family

$$\mu = (\mu_{r,\gamma})_{r\geq 1,\, \gamma \in \Gamma}$$

of nonnegative integers satisfying

$$\sum_{r,\gamma} r\mu_{r,\gamma} = n.$$

Let

$$k(\mu) = \sum_{r,\gamma} \mu_{r,\gamma}.$$

Theorem 17.5 (Compressed diagonal packet Faà di Bruno formula)**.** *With the notation above,*

$$\boxed{\begin{aligned}&P_\delta D^n(F\circ G)(x)[h,\dots,h]\\&=\sum_{\mu}\frac{n!}{\prod_{r,\gamma}(r!)^{\mu_{r,\gamma}}\mu_{r,\gamma}!}\,P_\delta D^{k(\mu)}F(Gx)\left[(g_{r,\gamma})^{\mu_{r,\gamma}}\right]_{r,\gamma}.\end{aligned}} \tag{11.5}$$

Here the last line means that $g_{r,\gamma}$ is inserted $\mu_{r,\gamma}$ times; symmetry makes the insertion order irrelevant.

Proof. Start with the packet Faà di Bruno sum over a set partition π of $\{1,\dots,n\}$ and a packet color attached to every block. Because every input direction is h, the value of one term depends only on the number $\mu_{r,\gamma}$ of blocks of size r carrying color γ. The symmetric group acts transitively on all colored set partitions with one fixed profile. The stabilizer permutes the r entries inside each block and permutes the indistinguishable blocks of a fixed type (r,γ). Its order is

$$\prod_{r,\gamma}(r!)^{\mu_{r,\gamma}}\mu_{r,\gamma}!,$$

so the orbit has the coefficient printed in (11.5). Summing one representative per orbit gives the result. □

Corollary 17.6 (Colored integer-partition term count)**.** *If Y has M intermediate packets, the number of compressed diagonal profiles at order n is*

$$\boxed{C_{n,M}=[x^n]\prod_{r\ge1}(1-x^r)^{-M}.} \tag{11.6}$$

Before compression, the number of packet-labeled set-partition terms is

$$N_{n,M}=\sum_{k=1}^{n}S(n,k)M^k.$$

Proof. For each block size r, the multiplicities among M colors contribute

$$(1-x^r)^{-M}.$$

Multiplying over all r and taking the coefficient of x^n counts exactly the profiles satisfying the weighted-size constraint. The raw count is the existing packet term-count formula, grouped by the number of blocks. □

Boundary 17.7 (Scope of the compression theorem). Equation (11.5) closes the repeated-direction, single-input-packet case. For fully mixed directions, blocks carry the types of their actual input subsets; the relevant orbit type is finer and can approach the full set-partition expansion. No universal mixed-direction compression ratio is asserted.

17.4 The cyclic nonlinear core of a packet system

For the packet system (11.4), define the directed dependency graph G_Φ on I by placing an edge

$$j\longrightarrow i$$

Table 29: Raw and compressed diagonal Faà di Bruno term counts for three intermediate packets.

Order n	Raw packet terms	Colored profiles	Compression factor
1	3	3	1.0
2	12	9	1.3
3	57	22	2.6
4	309	51	6.1
5	1,866	108	17.3
6	12,351	221	55.9
7	88,563	429	206.4
8	681,870	810	841.8
9	5,597,643	1,479	3784.7
10	48,718,569	2,640	18454.0

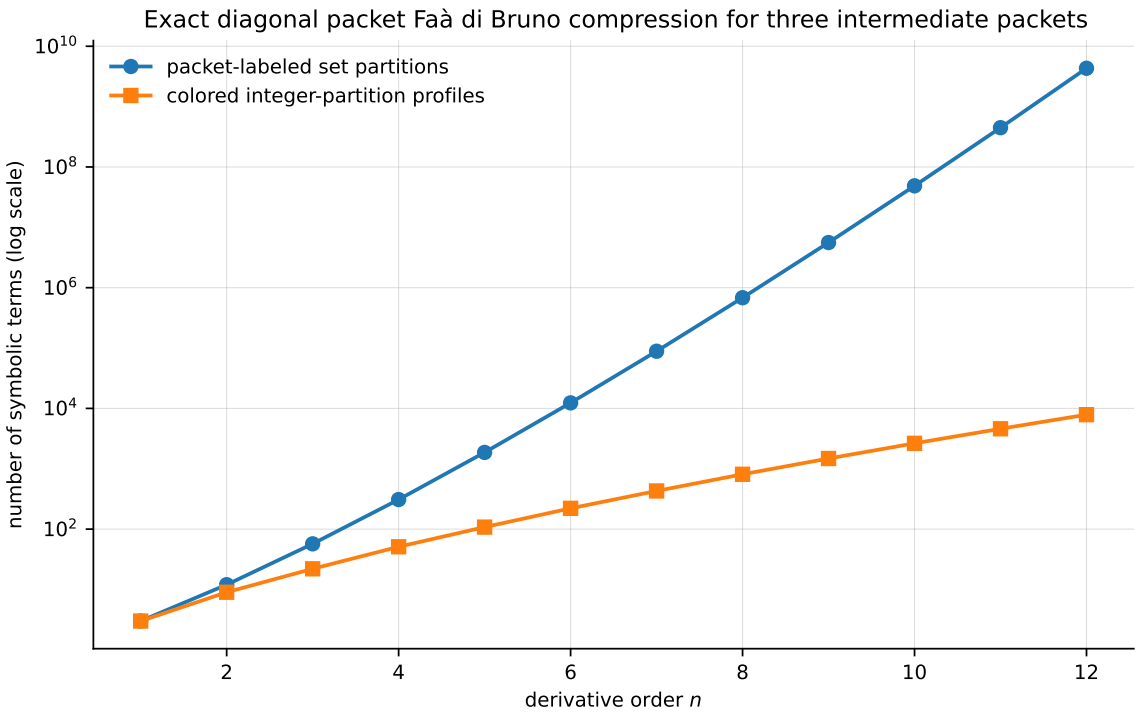


Figure 44: For repeated directions, packet-labeled set partitions collapse to colored integer-partition profiles. The compression is exact, not asymptotic.

when some active block contributing to the equation for u_i contains u_j as an input. Let

$$\mathrm{SCC}(G_\Phi)$$

be its strongly connected components. A component is called cyclic when it contains more than one address or is a singleton carrying a self-loop.

Definition 17.8 (Address-preserving cyclic dimension)**.** The cyclic packet dimension of the marked system is

$$\boxed{\mathrm{cdim}_{\mathcal{P}}(\Phi) = \sum_{C \text{ cyclic SCC}} |C|.} \tag{11.7}$$

It measures the number of packet variables that remain in coupled nonlinear components after exact elimination of the acyclic singleton components.

Theorem 17.9 (Exact SCC reduction)**.** *Order the SCC condensation graph topologically. Every acyclic singleton component can be substituted exactly once its predecessors have been determined. Repeating this elimination gives a solution-preserving bijection between:*

1. *solutions of the full packet system, and*
2. *solutions of the subsystem supported on the cyclic SCCs, followed by deterministic reconstruction along the condensation graph.*

Consequently, the coupled nonlinear core has exactly $\mathrm{cdim}_{\mathcal{P}}(\Phi)$ *packet variables.*

Proof. A singleton SCC without a self-loop has an equation in which its own variable does not occur. Once all predecessor components are known, that equation computes the variable directly. Substituting this expression into every successor equation is an algebraic identity and gives a bijection of solution sets. The condensation graph is acyclic, so repeated topological elimination terminates. Components containing a directed cycle cannot be removed by this one-pass address-preserving substitution because each contains a variable depending, directly or transitively, on itself. □

Boundary 17.10 (Not a universal nonlinear state minimum). The quantity $\mathrm{cdim}_{\mathcal{P}}$ is invariant under relabeling of the declared packet system and exact elimination within its dependency grammar. A nonlinear change of variables, hidden-variable introduction, algebraic elimination, or another packet family can change it. It is therefore a certified address-preserving system dimension, not a universal minimum over all nonlinear presentations.

17.5 A finite residue signature for simply generated trees

Consider the scalar marked equation

$$\boxed{u(z) = z + \Phi(u(z)), \qquad \Phi(y) = \sum_{m \in \mathcal{A}} w_m y^m,} \tag{11.8}$$

where $\mathcal{A} \subseteq \{2, 3, \dots\}$ is finite and every w_m is a positive integer. Its unique formal solution belongs to $z\mathbb{Z}[[z]]$ and counts plane rooted trees whose internal arities lie in $\mathcal{A}$, weighted by the w_m.

Define the arity period

$$\boxed{d_{\mathcal{A}} = \gcd\{m - 1 : m \in \mathcal{A}\}.} \tag{11.9}$$

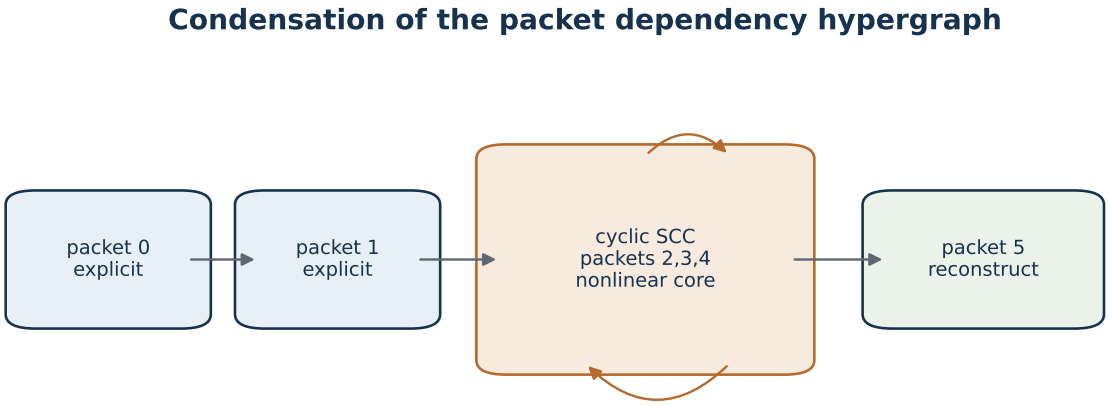


Figure 45: Condensation separates exact one-pass reconstruction from the cyclic nonlinear core. The resulting dimension is relative to the declared packet coordinates.

Every tree with internal vertices v has leaf count

$$\ell(T) = 1 + \sum_{v} (\deg(v) - 1).$$

Theorem 17.11 (Finite tree-address signature)**.** *Fix a packet width N and put*

$$g_N = \gcd(N, d_{\mathcal{A}}).$$

For the arithmetic packets

$$U_j(x) = \sum_{r \geq 0} [z^{Nr+j}] u(z)\, x^r,$$

the following are equivalent:

1. $U_j \neq 0$;
2. $j - 1$ *belongs to the subgroup of* $\mathbb{Z}/N\mathbb{Z}$ *generated by* $\{m - 1 : m \in \mathcal{A}\}$;
3. $$\boxed{j \equiv 1 \pmod{g_N}.}$$

Hence the exact number of active packet addresses is

$$\boxed{\#\mathrm{Supp}_N(u) = \frac{N}{g_N}.} \tag{11.10}$$

Proof. Replacing one leaf by an m-ary vertex increases the leaf count by $m - 1$. Starting from the one-leaf tree, every nonnegative sum of allowed increments is realized by a sequence of leaf replacements. Positivity of the weights prevents cancellation, so the coefficient support is exactly

$$1 + \langle m - 1 : m \in \mathcal{A} \rangle_{\mathbb{N}}.$$

Modulo N, the monoid generated by the increments equals the subgroup they generate because every element of a finite cyclic subgroup has a nonnegative representative. That subgroup is $g_N\mathbb{Z}/N\mathbb{Z}$, which proves the congruence and the count. □

Corollary 17.12 (Finite bottom-up residue automaton)**.** *Label every leaf by* $1 \in \mathbb{Z}/N\mathbb{Z}$ *and every* m*-ary vertex by the sum of its child labels. The reachable state set is exactly*

$$\{j : j \equiv 1 \pmod{g_N}\}.$$

Thus the finite pair

$$\boxed{\mathfrak{T}_N(\mathcal{A}) = (d_{\mathcal{A}}, g_N, N/g_N)} \tag{11.11}$$

classifies packet support for the marked simply generated family.

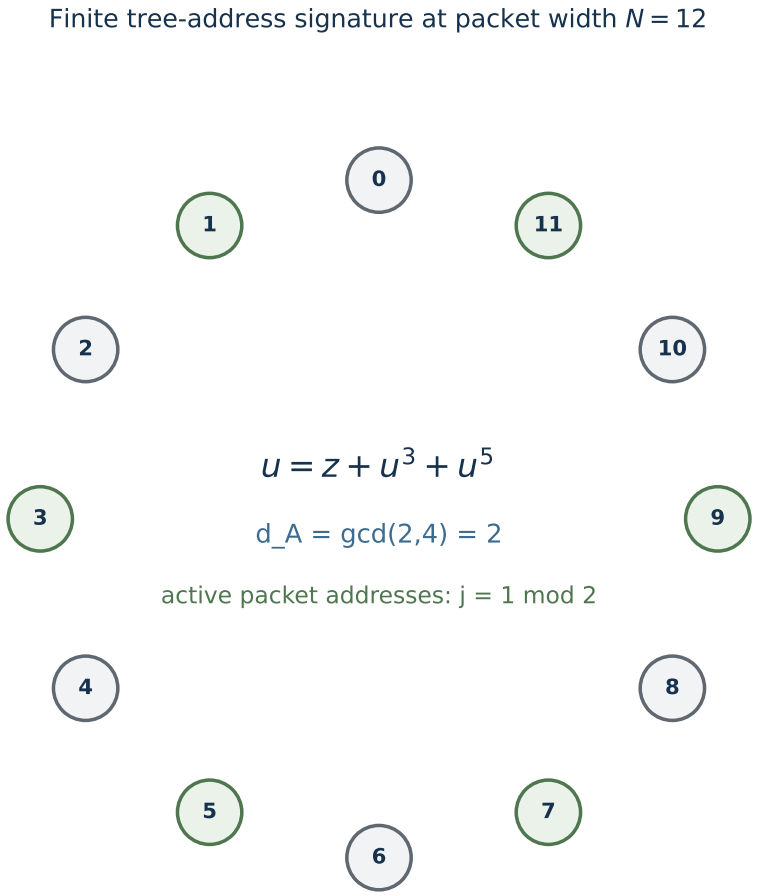


Figure 46: For arities three and five, the increment gcd is two. At width twelve, exactly the six addresses congruent to one modulo two are active.

17.6 Dominant singularity orbits

Assume now that the coefficients w_m are positive real numbers. The classical smooth implicit-function schema for simply generated trees applies when there is a positive solution τ of

$$\Phi'(\tau) = 1$$

and

$$\rho = \tau - \Phi(\tau) > 0.$$

Under its standard nondegeneracy hypotheses, the reduced aperiodic series has a square-root singularity at its positive radius [15]. The arity period restores the complete source orbit.

Theorem 17.13 (Complete dominant-orbit signature)**.** *Under the smooth implicit schema, the dominant singularities of $u(z)$ form*

$$\boxed{\rho\,\mu_{d_{\mathcal{A}}},} \tag{11.12}$$

where μ_d is the group of dth roots of unity. After packetization with $x = z^N$, the union of dominant singularities of the complete packet tuple is

$$\boxed{\rho^N\,\mu_{h_N}, \qquad h_N = \frac{d_{\mathcal{A}}}{\gcd(N, d_{\mathcal{A}})}.} \tag{11.13}$$

Therefore the same finite arity signature determines both the number of active packet addresses and the size of the complete dominant-singularity orbit.

Proof. By theorem 17.11, the coefficient support is contained in $1 + d_{\mathcal{A}}\mathbb{N}$ and the reduced exponent set is aperiodic by maximality of the gcd. Hence

$$u(z) = zV(z^{d_{\mathcal{A}}})$$

with V aperiodic. The smooth implicit schema gives one positive dominant singularity for V, and pulling back through $z^{d_{\mathcal{A}}}$ gives (11.12). The map $z \mapsto z^N$ sends $\mu_{d_{\mathcal{A}}}$ onto a root group of cardinality

$$d_{\mathcal{A}} / \gcd(N, d_{\mathcal{A}}).$$

No packet can introduce a new dominant image outside this set. Conversely, if every packet were analytic at one image, the exact synthesis

$$u(z) = \sum_{j=0}^{N-1} z^j U_j(z^N)$$

would make u analytic at every source point above that image, contradicting (11.12). Thus the union of packet singularities is exactly (11.13). □

Boundary 17.14 (Individual-packet cancellation). The theorem identifies the union of dominant singularities of the complete packet tuple. One individual packet can lose some orbit points through address cancellation. The finite signature does not, by itself, determine every individual local amplitude or cancellation coefficient.

17.7 The sharp full q-ary family

Take

$$\boxed{u = z + u^q, \qquad q \geq 2.} \tag{11.14}$$

Lagrange inversion gives the exact coefficient formula

$$\boxed{[z^n]u = \begin{cases} \dfrac{1}{(q-1)m+1}\dbinom{qm}{m}, & n = 1 + (q-1)m, \\ 0, & n \not\equiv 1 \pmod{q-1}. \end{cases}} \tag{11.15}$$

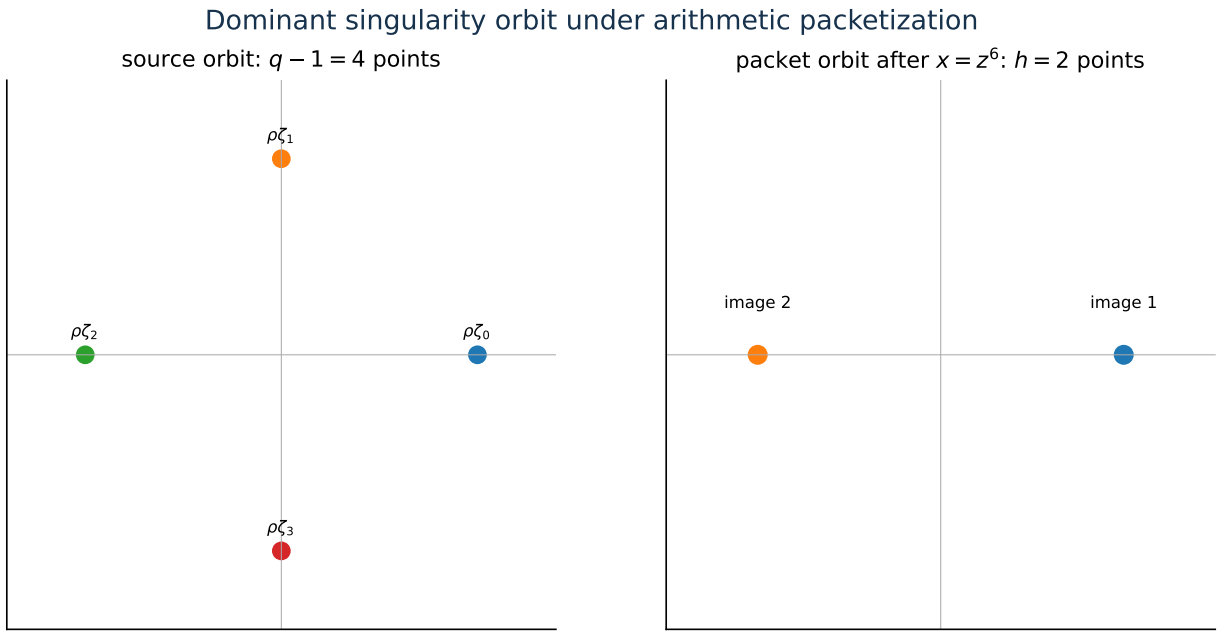


Figure 47: Arithmetic packetization maps the source root-of-unity singularity orbit through $z \mapsto z^N$. Distinct source singularities can alias to one packet-variable singularity.

The nonzero coefficients are the Fuss–Catalan numbers of full plane q-ary trees.

Write

$$u(z) = \sum_{j=0}^{N-1} z^j U_j(z^N), \qquad x = z^N.$$

Theorem 17.15 (Exact q-ary packet phase theorem)**.** *Put*

$$g = \gcd(N, q-1), \qquad h = \frac{q-1}{g}.$$

Then:

1. *the active packet addresses are exactly*

$$\boxed{j \equiv 1 \pmod{g};}$$

2. *the complete packet system is*

$$\boxed{U_j(x) = \delta_{j,1} + \sum_{\substack{0 \le r_1,\dots,r_q < N \\ r_1+\cdots+r_q \equiv j \pmod N}} x^{(r_1+\cdots+r_q-j)/N} \prod_{a=1}^{q} U_{r_a}(x);} \tag{11.16}$$

3. *the smooth critical values are*

$$\tau = q^{-1/(q-1)}, \qquad \rho = (q-1)q^{-q/(q-1)};$$

4. *the source dominant orbit has $q-1$ points, while the complete packet orbit has exactly h points.*

The phase is therefore determined by the finite tuple

$$\boxed{(q, N, g, N/g, h).} \tag{11.17}$$

Proof. Formula (11.15) follows from Lagrange inversion applied to

$$z = u(1 - u^{q-1}).$$

The support and active-address statements are theorem 17.11 with $d_{\mathcal{A}} = q - 1$. Expanding u^q in complete packets and collecting the exponent carry

$$r_1 + \cdots + r_q = j + Nc$$

gives (11.16). The critical equations

$$q\tau^{q-1} = 1, \qquad \rho = \tau - \tau^q$$

give the printed values, and theorem 17.13 gives the orbit count. □

Table 30: Exact packet signatures for the marked full q-ary tree family $u = z + u^q$.

q	N	$g = \gcd(N, q-1)$	Active packets	Packet orbit	Active addresses
2	5	1	5	1	$\{0,1,2,3,4\}$
3	4	2	2	1	$\{1,3\}$
3	5	1	5	2	$\{0,1,2,3,4\}$
4	4	1	4	3	$\{0,1,2,3\}$
4	6	3	2	1	$\{1,4\}$
5	6	2	3	2	$\{1,3,5\}$
5	7	1	7	4	$\{0,1,2,3,4,5,6\}$
6	10	5	2	1	$\{1,6\}$

17.8 Verification, status, and exact boundary

The Nonlinear tree atlas verifier performs four independent exact audits. It enumerates colored integer-partition profiles and compares their orbit-weighted Faà di Bruno values with the full set-partition expansion. It constructs exact fixed-point prefixes for marked polynomial tree families and checks the finite arity-gcd support theorem over many packet widths. It verifies Fuss–Catalan formulas and the complete carry-form packet equations for the full q-ary family. Finally, it checks dependency-graph reductions against exhaustive bounded graph instances.

Section result. Marked polynomial fixed-point systems admit exact packet, hypergraph, operadic-tree, and packet-equation charts. The repeated-direction packet Faà di Bruno expansion compresses exactly from colored set partitions to colored integer-partition profiles. The packet dependency graph has an exact SCC reduction whose cyclic vertices form the address-preserving nonlinear core. For positive simply generated tree systems, the arity gcd is a finite signature determining every arithmetic packet-support pattern and the complete dominant-singularity orbit. The full q-ary family supplies a closed Fuss–Catalan phase theorem and exact packet equations.

Exact boundary of Nonlinear tree atlas. The atlas classifies marked polynomial systems, not semantic fixed points independently of their generators. The compression theorem treats repeated directions; arbitrary mixed-direction jets can require a finer orbit type. The SCC dimension is relative to the declared packet coordinates. The tree signature classifies support and the complete dominant-

Table 31: Exact finite verification for Nonlinear tree atlas.

Validation class	Exact checks
Colored profile counts	100
Orbit-size identity	48
Exact compressed Faà di Bruno values	126
Polynomial fixed-point prefixes	300
Finite tree-address signatures	70,458
Fuss-Catalan formulas	2,892
Exact q-ary packet systems	37,567
Dominant-orbit combinatorics	176
Dependency-core reductions	4,054
Nonlinear tree atlas total	**115,721**
Cumulative the preceding certified atlas program total	2,027,665
Failures	0

singularity orbit for the marked simply generated family. It does not classify algebraic degree, universal minimal system dimension, every individual packet amplitude, arbitrary sign-canceling weights, noncontractive global dynamics, or all nonlinear operadic systems.

18 Finite Galois, profinite, and Tannakian resolved symmetry

Galois-Tannakian atlas develops the final declared stratum in the present eight-stratum program: finite semisimple Galois charts, cohomological descent, profinite refinement, and the tensor-fiber boundary at which a symmetry group can be reconstructed. The strongest Galois source already contains Reynolds projectors, the subgroup–fixed-field correspondence, normality and quotient compression, conjugate projector atlases, Mackey double-coset packets, ramification details, Haar projectors, Hilbert 90, the Brauer obstruction, finite étale permutation resolutions, and neutral Tannakian reconstruction [6]. The purpose here is to place those results into the same passport, bridge, obstruction, and effective-descent language used by the preceding sections.

The scope is deliberately semisimple or tensor-exact. For finite group averaging, the coefficient field has characteristic zero or characteristic prime to the relevant group order. Linear semilinear descent is taken over a finite Galois extension. Projective descent is typed separately from vector descent. Tannakian reconstruction is asserted only after a faithful exact tensor fiber functor has been supplied. Modular fixed points, wild integral descent, nonneutral fiber functors, and derived or flat replacements for failed averaging remain outside this block.

18.1 Finite subgroup, fixed-field, and Reynolds-projector charts

Let L/K be a finite Galois extension with group

$$G = \mathrm{Gal}(L/K).$$

Let V be a finite-dimensional representation of G over a characteristic-zero field, or a unitary complex realization obtained after scalar extension. For every subgroup $H \leq G$, define

$$\boxed{P_H = \frac{1}{|H|} \sum_{h \in H} U_h.} \tag{12.1}$$

Theorem 18.1 (Reynolds fixed packet and reverse order)**.** *The operator P_H is the projector onto the fixed packet*

$$V^H = \{v : U_h v = v \text{ for all } h \in H\}.$$

If $H \leq J$, then

$$\boxed{P_H P_J = P_J P_H = P_J.} \tag{12.2}$$

Consequently, subgroup inclusion reverses the fixed-projector order.

Proof. The average is invariant under left and right multiplication by elements of H. Therefore $P_H^2 = P_H$, and its range is exactly the fixed subspace. If $H \leq J$, then $V^J \subseteq V^H$. The two projections are nested and multiply to the projector with smaller range. □

In the faithful regular realization, or in the field realization supplied by a normal basis, the projector determines the subgroup. Indeed, the support of the group-algebra idempotent

$$e_H = \frac{1}{|H|} \sum_{h \in H} h$$

is exactly H.

Theorem 18.2 (Normalized finite Galois chart)**.** *Fix the finite Galois extension L/K. The following data determine one another exactly:*

$$\boxed{H \quad \longleftrightarrow \quad L^H \quad \longleftrightarrow \quad P_H} \tag{12.3}$$

where the projector is taken in a faithful field or regular realization. The correspondence is order reversing:

$$H \leq J \quad \Longleftrightarrow \quad L^H \supseteq L^J \quad \Longleftrightarrow \quad P_H \geq P_J.$$

Thus the subgroup poset, the intermediate-field poset, and the faithful fixed-projector poset form three normalized charts of one finite semisimple Galois object.

Proof. The finite Galois correspondence gives $H = \mathrm{Gal}(L/L^H)$ and $E = L^{\mathrm{Gal}(L/E)}$. The range of P_H is L^H after faithful scalar extension, and the support of e_H recovers H. All three assignments are therefore mutually inverse on the declared charts. □

18.2 Compatibility, normality, and conjugate atlases

For two subgroups $H, J \leq G$, the common fixed packet is always

$$V^H \cap V^J = V^{\langle H, J \rangle}.$$

The two averaging projectors need not commute.

Theorem 18.3 (Permutable subgroups and exact common refinement)**.** *In a faithful regular realization, the following are equivalent:*

1. *$HJ = JH$;*
2. *HJ is a subgroup of G;*
3. *$P_H P_J = P_J P_H$.*

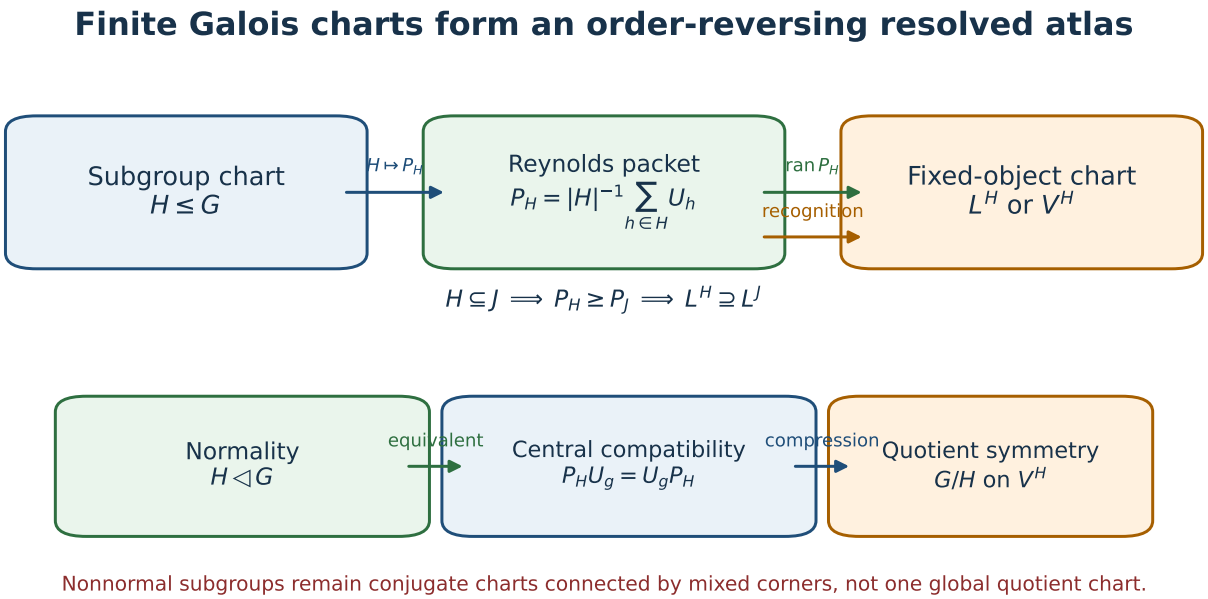


Figure 48: The finite subgroup, Reynolds-projector, and fixed-field charts are reversible in a faithful semisimple realization. Normality is the additional condition required for quotient compression.

When these conditions hold,

$$\boxed{P_H P_J = P_{HJ}.} \tag{12.4}$$

If they fail, $P_H P_J$ and $P_J P_H$ are mixed atlas corners rather than projectors.

Proof. If $HJ = JH$, every element of HJ has $|H \cap J|$ representations as hj. Multiplying the two group averages therefore gives the average over HJ. Conversely, the support of $e_H e_J$ is HJ, while the support of $e_J e_H$ is JH. Equality in the regular group algebra forces equality of the product sets. □

Theorem 18.4 (Normality and quotient compression)**.** *For a subgroup $H \leq G$, the following conditions are equivalent:*

1. *$H \lhd G$;*
2. *e_H is central in the group algebra;*
3. *$P_H U_g = U_g P_H$ for every $g \in G$;*
4. *the fixed packet V^H is G-invariant;*
5. *the fixed field L^H is Galois over K.*

When they hold, the compressed action on V^H factors through G/H, and

$$\boxed{\mathrm{Gal}(L^H/K) \cong G/H.} \tag{12.5}$$

Proof. Conjugation sends e_H to $e_{gHg^{-1}}$. Centrality is therefore equivalent to normality. The represented identity

$$U_g P_H U_g^{-1} = P_{gHg^{-1}}$$

gives the operator conditions. The field and quotient statements are the normal-subgroup part of the finite Galois correspondence. □

If H is not normal, its conjugates form a genuine atlas rather than one quotient chart. Put

$$\mathrm{Core}_G(H) = \bigcap_{g\in G} gHg^{-1}, \qquad \mathrm{Ncl}_G(H) = \left\langle gHg^{-1} : g \in G \right\rangle.$$

Theorem 18.5 (Canonical envelopes of a nonnormal Galois atlas)**.** *Let $M = L^H$. Then*

$$\boxed{L^{\mathrm{Core}_G(H)} = \prod_{g\in G} g(M)} \tag{12.6}$$

is the smallest Galois extension inside L containing every conjugate chart, while

$$\boxed{L^{\mathrm{Ncl}_G(H)} = \bigcap_{g\in G} g(M)} \tag{12.7}$$

is the largest Galois subfield contained in every chart. The core and normal-closure projectors are the two canonical global envelopes of the conjugate atlas.

Proof. Composita of intermediate fields correspond to intersections of subgroups, while intersections of fields correspond to generated subgroups. The core is the largest normal subgroup inside H, and the normal closure is the smallest normal subgroup containing H. □

Example 18.6 (The S_3 nonnormal atlas)**.** Let $G = S_3$ and let H be generated by one transposition. There are three conjugate order-two subgroups and hence three conjugate cubic fixed-field charts. Their subgroup core is trivial, while their normal closure is all of S_3. By contrast, the order-three subgroup A_3 is normal and its fixed chart carries the quotient symmetry $S_3/A_3 \cong C_2$.

18.3 Double cosets as exact transition packets

Restriction and induction connect subgroup charts that need not commute. Their exact address set is the double-coset space

$$K\backslash G/H.$$

Theorem 18.7 (Mackey transition-packet decomposition)**.** *Let $H, K \le G$, let W be a finite-dimensional representation of H, and choose representatives x for $K\backslash G/H$. Then*

$$\boxed{\mathrm{Res}^G_K \mathrm{Ind}^G_H W \cong \bigoplus_{x\in K\backslash G/H} \mathrm{Ind}^K_{K\cap xHx^{-1}} \left({}^x\mathrm{Res}^H_{H\cap x^{-1}Kx} W\right).} \tag{12.8}$$

Each double coset is one exact transfer packet between the H- and K-charts.

Proof. The induced representation is a function space on G with H-covariance. Functions supported on different double cosets are invariant direct summands under K. Identifying the summand on KxH with an induced representation from $K \cap xHx^{-1}$ gives the decomposition. □

The dimension identity for the trivial H-representation is

$$\boxed{[G:H] = \sum_{x\in K\backslash G/H} [K : K\cap xHx^{-1}].} \tag{12.9}$$

This is the finite packet-count law audited by the Galois-Tannakian atlas verifier.

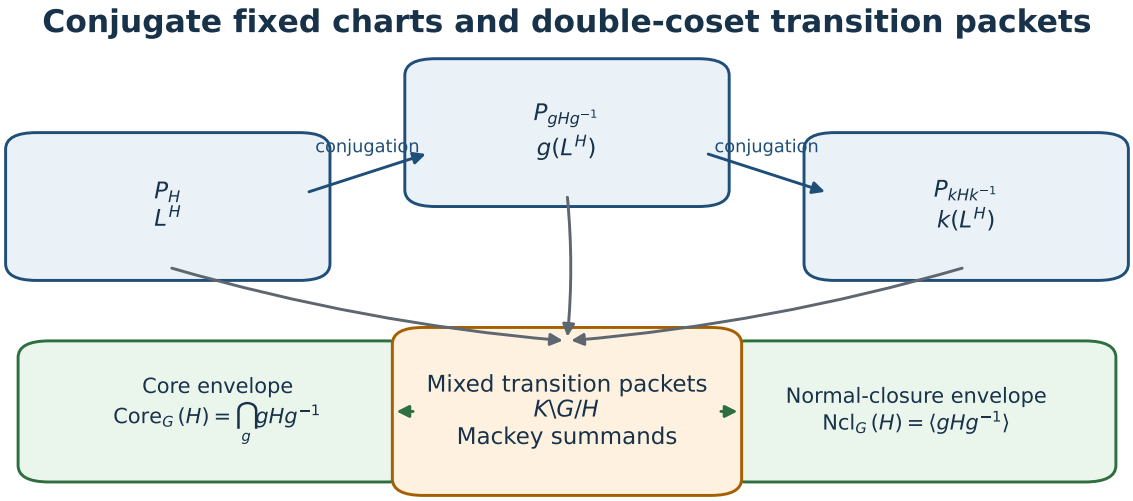


Figure 49: A nonnormal fixed object is an atlas of conjugate charts. Core and normal closure give two global envelopes, while double cosets give exact mixed transition packets.

Table 32: Finite group models used in the Galois-Tannakian atlas subgroup-projector audit.

Group	Order	Subgroups	Normal	Permutable ordered pairs	Noncommuting ordered pairs	Double-coset packets audited
C2	2	2	2	4	0	5
C3	3	2	2	4	0	6
C4	4	3	3	9	0	15
C5	5	2	2	4	0	8
C6	6	4	4	16	0	30
C8	8	4	4	16	0	37
V4	4	5	5	25	0	37
S3	6	6	3	30	6	65
D4	8	10	6	92	8	195
A4	12	10	3	64	36	250

18.4 Ramification as a nested resolution

Let L/K be a finite Galois extension of local fields. Write

$$G_0 \supseteq G_1 \supseteq G_2 \supseteq \cdots$$

for the lower ramification filtration after inertia, and let V be a finite-dimensional complex representation of G. Define

$$P_i = \frac{1}{|G_i|} \sum_{g \in G_i} U_g.$$

The ramification groups are nested and normal, so

$$P_0 \leq P_1 \leq P_2 \leq \cdots \leq I.$$

Put

$$D_i = P_{i+1} - P_i.$$

Theorem 18.8 (Orthogonal ramification details)**.** *The operators D_i are mutually orthogonal projectors, and*

$$\boxed{V = V^{G_0} \oplus \bigoplus_i D_i V.} \tag{12.10}$$

Every detail packet is stable under the quotient action permitted by the corresponding normal subgroup.

Proof. Nested orthogonal projectors commute, and the difference of two nested projectors is a projector onto the orthogonal difference of their ranges. The decomposition follows by telescoping. □

Theorem 18.9 (Artin conductor as weighted unresolved rank)**.** *The Artin and Swan conductors satisfy*

$$\boxed{a(V) = \sum_{i \geq 0} \frac{|G_i|}{|G_0|} \operatorname{Tr}(I - P_i),} \tag{12.11}$$

$$\boxed{\operatorname{Sw}(V) = \sum_{i \geq 1} \frac{|G_i|}{|G_0|} \operatorname{Tr}(I - P_i).} \tag{12.12}$$

If $d_j = \operatorname{rank} D_j$, then

$$\boxed{a(V) = \sum_j d_j \sum_{i=0}^{j} \frac{|G_i|}{|G_0|}.} \tag{12.13}$$

Thus the conductor is the weighted total rank that remains unresolved at each ramification scale.

Proof. The classical conductor formula uses $\operatorname{codim} V^{G_i}$. Since P_i projects onto V^{G_i}, this codimension is $\operatorname{Tr}(I - P_i)$. Also

$$\operatorname{codim} V^{G_i} = \sum_{j \geq i} d_j.$$

Reversing the finite sums gives the detail formula. □

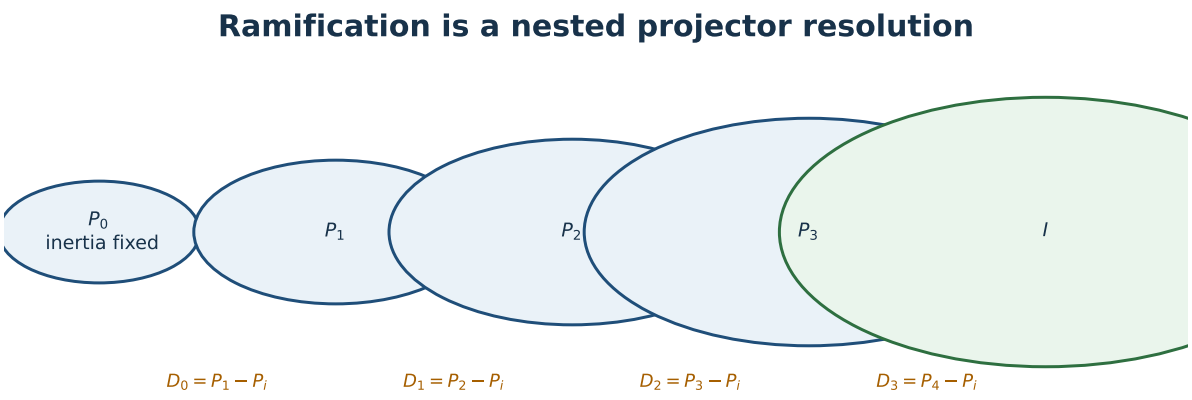


Figure 50: Ramification subgroups give nested fixed projectors. Their orthogonal differences are the resolved tame and wild detail packets, and the conductor is their weighted unresolved rank.

18.5 Profinite completion and finite quotient effectivity

Let G be a compact group and let $U : G \to \mathcal{U}(\mathcal{H})$ be strongly continuous. For a closed subgroup H, normalized Haar measure defines

$$\boxed{P_H = \int_H U_h \, d\mu_H(h).} \tag{12.14}$$

The same averaging proof as in the finite case shows that P_H is the orthogonal projector onto $\mathcal{H}^H$.

Theorem 18.10 (Shrinking-subgroup reconstruction)**.** *Let G be profinite and let*

$$U_1 \supseteq U_2 \supseteq \cdots$$

be a neighborhood basis of the identity consisting of open normal subgroups. Then

$$\boxed{P_{U_n} \longrightarrow I} \tag{12.15}$$

strongly in every strongly continuous unitary representation. More generally, if $N \triangleleft G$ is closed and U_n/N shrinks to the identity in G/N, then $P_{U_n} \to P_N$ strongly.

Proof. For each $v \in \mathcal{H}$, strong continuity gives

$$\sup_{u \in U_n} \|U_u v - v\| \longrightarrow 0.$$

Averaging this estimate over U_n proves the first statement. Apply it to the quotient action on the N-fixed packet for the second. □

Corollary 18.11 (Finite-dimensional representations occur at finite level)**.** *Every continuous finite-dimensional complex representation of a profinite group has finite image and factors through a finite quotient.*

Proof. The image is a compact totally disconnected subgroup of $\mathrm{GL}_n(\mathbb{C})$. After conjugation it lies in a compact Lie group. A compact totally disconnected Lie group is finite. □

This finite-level factorization is a strong effectivity statement for finite-dimensional complex representations. It is not a statement that every integral, modular, or derived representation becomes semisimple at finite level.

18.6 Semilinear vector descent and Hilbert 90

Let L/K be finite Galois with group G. A semilinear G-object is an L-vector space V with maps

$$\rho_\sigma : V \to V, \qquad \rho_\sigma(\lambda v) = \sigma(\lambda)\rho_\sigma(v),$$

satisfying the group law. Scalar extension and fixed vectors define functors

$$L \otimes_K - : \mathrm{Vect}_K \longrightarrow \mathrm{Semilin}_G(L),$$

$$(-)^G : \mathrm{Semilin}_G(L) \longrightarrow \mathrm{Vect}_K .$$

Theorem 18.12 (Effective finite Galois vector descent)**.** *The two functors above are quasi-inverse equivalences:*

$$\boxed{\mathrm{Vect}_K \simeq \mathrm{Semilin}_G(L).} \tag{12.16}$$

For every semilinear object V, the fixed packet $W = V^G$ satisfies

$$\boxed{W \otimes_K L \cong V.} \tag{12.17}$$

The equivalence is effective on both objects and morphisms.

Proof. This is finite Galois descent. After an L-basis is chosen, the semilinear action is represented by a cocycle

$$a_{\sigma\tau} = a_\sigma \sigma(a_\tau), \qquad a_\sigma \in \mathrm{GL}_n(L).$$

Nonabelian Hilbert 90 gives $B \in \mathrm{GL}_n(L)$ with

$$a_\sigma = B^{-1}\sigma(B).$$

The columns of B^{-1} applied to a K-basis are fixed by the twisted action. They form an n-dimensional fixed packet, and scalar extension reconstructs V. An equivariant L-linear map restricts to a K-linear map on fixed packets, and scalar extension recovers it. □

A useful constructive form is obtained by choosing any matrix $C \in M_n(L)$ and setting

$$T(C) = \sum_{\sigma \in G} a_\sigma \sigma(C).$$

Then

$$a_\tau \tau(T(C)) = T(C).$$

Whenever $T(C)$ is invertible,

$$B = T(C)^{-1}$$

is a Hilbert 90 trivialization. Standard descent guarantees that such a C exists.

Definition 18.13 (Framed semilinear descent chart)**.** A framed descent chart retains an ordered K-basis of the fixed packet W. Its scalar extension gives a distinguished L-basis of V, and hence a literal cocycle matrix family.

Proposition 18.14 (Gauge and intrinsic automorphisms remain distinct)**.** *For fixed rank n, nonabelian Hilbert 90 implies that every $\mathrm{GL}_n(L)$-cocycle is gauge equivalent to the trivial cocycle. The unframed descent groupoid is not contractible: its automorphism group is $\mathrm{GL}_n(K)$. After a global K-frame is retained, the normalized fiber becomes contractible.*

Proof. Hilbert 90 gives one isomorphism class. Automorphisms of the descended K-space are exactly $\mathrm{GL}_n(K)$. A chosen ordered basis kills those automorphisms. □

18.7 Projective descent and the Brauer obstruction

A projective cocycle takes values in $\mathrm{PGL}_n(L)$. Choose lifts

$$A_\sigma \in \mathrm{GL}_n(L).$$

Their cocycle failure is scalar:

$$\boxed{c(\sigma,\tau) = A_\sigma \sigma(A_\tau) A_{\sigma\tau}^{-1} \in L^\times.} \tag{12.18}$$

Changing the lifts alters c by a scalar coboundary.

Theorem 18.15 (Exact projective descent obstruction)**.** *The scalar class*

$$\boxed{[c] = \delta([\alpha]) \in H^2(G, L^\times) = \mathrm{Br}(L/K)} \tag{12.19}$$

is independent of the chosen lifts. It vanishes exactly when the projective cocycle lifts to a linear semilinear cocycle. In that case theorem 18.12 produces a descended vector space. If it does not vanish, the descended object is a module over a central simple K-algebra split by L, not an ordinary global vector packet.

Proof. Apply the connecting map in nonabelian cohomology to

$$1 \longrightarrow L^\times \longrightarrow \mathrm{GL}_n(L) \longrightarrow \mathrm{PGL}_n(L) \longrightarrow 1.$$

Vanishing is equivalent to rescaling the lifts so that the scalar defect is removed. The lifted cocycle then descends by Hilbert 90. The nonzero classes are the relative Brauer classes of central simple algebras split by L. □

Example 18.16 (The quaternionic obstruction over $\mathbb{C}/\mathbb{R}$)**.** Let $G = \{1, \sigma\}$, where σ is complex conjugation, and put

$$A_\sigma = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Then

$$A_\sigma \sigma(A_\sigma) = -I.$$

The projective cocycle condition holds because $-I$ is scalar. Its boundary is represented by -1 in

$$\mathbb{R}^\times / N_{\mathbb{C}/\mathbb{R}}(\mathbb{C}^\times).$$

Every complex norm is positive, so -1 is nontrivial. The descended central simple algebra is the Hamilton quaternion algebra. Replacing A_σ by

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

gives scalar defect $+1$ and hence split vector descent.

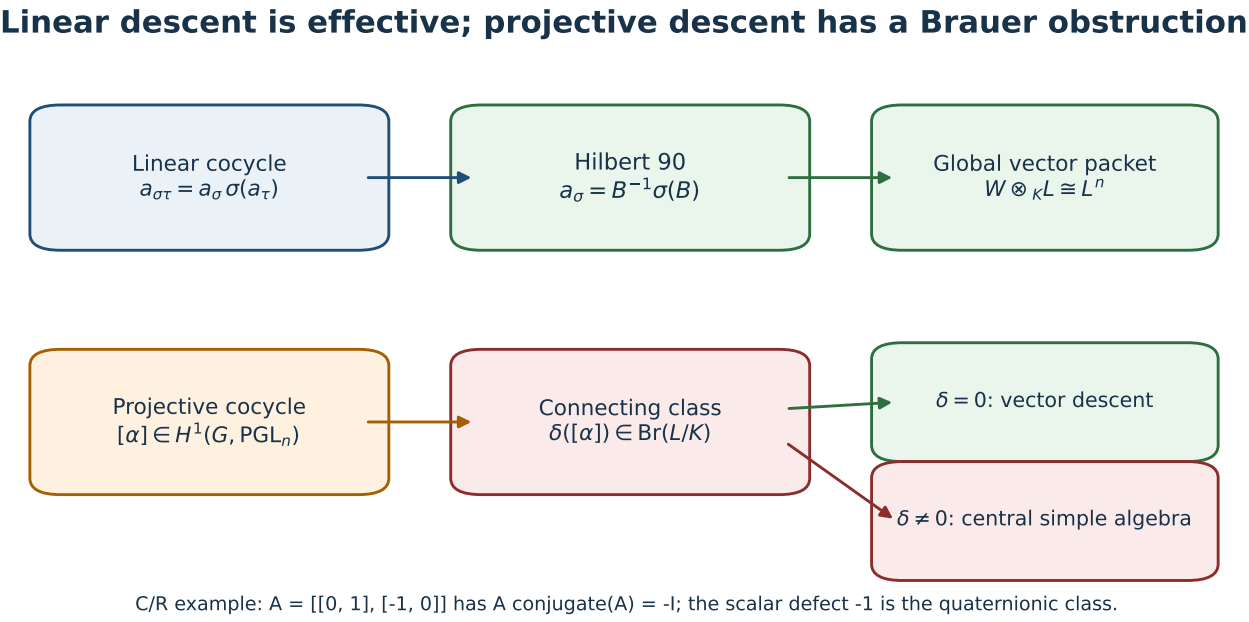


Figure 51: Linear cocycles are removed by Hilbert 90. Projective cocycles can satisfy every visible PGL overlap law while retaining a nonzero degree-two Brauer obstruction.

18.8 Finite étale and neutral Tannakian symmetry

For a connected scheme X with geometric point $\bar{x}$, the fiber functor on finite étale covers identifies that category with finite continuous $\pi_1^{\text{ét}}(X, \bar{x})$-sets. Every finite fiber S produces a permutation resolution on $\ell^2(S)$: coordinate projectors are indexed by points, orbit projectors by transitive components, and covering morphisms give equivariant packet transfers. The profinite fundamental group is the automorphism group of this fiber functor.

The tensor-linear version requires more structure.

Theorem 18.17 (Neutral Tannakian reconstruction)**.** *Let $\mathcal{C}$ be a neutral Tannakian category over k, and let*

$$\omega : \mathcal{C} \longrightarrow \mathrm{Vect}_k$$

be a faithful exact tensor fiber functor. Then

$$\boxed{\mathcal{G} = \mathrm{Aut}^{\otimes}(\omega)} \tag{12.20}$$

is an affine group scheme, and

$$\boxed{\mathcal{C} \simeq \mathrm{Rep}_k(\mathcal{G})} \tag{12.21}$$

as tensor categories. Declared packet idempotents and refinement maps preserved by ω become tensor-compatible concrete projector resolutions in every fiber.

Proof. The group scheme and tensor equivalence are the neutral Tannakian reconstruction theorem. Additivity and exact tensor preservation transport the declared idempotent and refinement equations to the fibers. □

Boundary 18.18 (Faithful corner realization is not a fiber functor). A faithful additive or Hilbert-space realization can preserve packet idempotents, transfer blocks, and operator equations without preserving tensor products, duals, exact sequences, or the tensor unit. Such a realization does not by itself determine $\mathrm{Aut}^{\otimes}(\omega)$. Tannakian symmetry begins only after the faithful exact tensor gate has been verified.

Proposition 18.19 (Finite group model of the tensor boundary)**.** *Let G be finite and let $\omega : \mathrm{Rep}_k(G) \to \mathrm{Vect}_k$ be the forgetful functor. The regular representation is faithful, the action on tensor products is diagonal, the unit is fixed, and the dual action is inverse transpose. These data distinguish every $g \in G$ and realize the finite constant-group case of theorem 18.17.*

Proof. Left translation on the regular representation is faithful. Functoriality of the tensor product, unit, and dual actions is immediate from the representation axioms. Since different group elements act differently on the regular basis, the induced tensor automorphisms are distinct. □

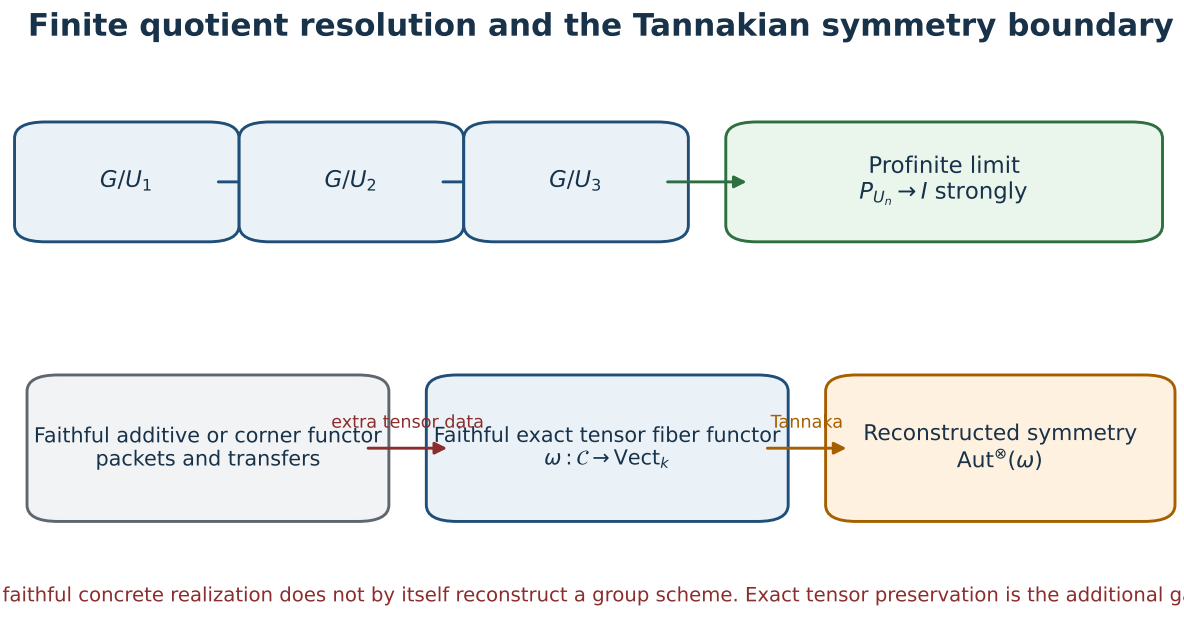


Figure 52: Finite quotient projectors converge strongly in profinite realizations. The lower row isolates the additional exact tensor structure required before a symmetry group scheme can be reconstructed.

18.9 The assembled finite Galois resolved atlas

The preceding results can be collected into one typed finite atlas. The reversible charts are:

1. subgroups $H \leq G$, fixed fields L^H, and faithful Reynolds projectors P_H;

2. finite-dimensional K-vector spaces and semilinear L-spaces with effective descent data;
3. finite étale fibers and their continuous permutation actions;
4. neutral Tannakian categories equipped with a specified faithful exact tensor fiber functor.

Directional or obstructed bridges include:

1. restriction and induction between subgroup charts, decomposed by Mackey packets;
2. passage from a nonnormal chart to its core or normal-closure envelope;
3. projectivization from linear to projective cocycles;
4. forgetting tensor structure from a Tannakian fiber functor to an additive or corner realization;
5. passage from finite quotients to a profinite inverse system.

Theorem 18.20 (Finite semisimple Galois and Tannakian assembly)**.** *Under the hypotheses printed in this section:*

1. *the subgroup, fixed-field, and faithful fixed-projector charts form an order-reversing normalized atlas;*
2. *normal charts admit quotient compression, while nonnormal charts form conjugate projector atlases with canonical core and normal-closure envelopes;*
3. *double cosets give exact transition packets for induction and restriction;*
4. *ramification filtrations give nested fixed projectors whose weighted unresolved ranks are the Artin and Swan conductors;*
5. *continuous finite-dimensional profinite representations are detected at a finite quotient level;*
6. *linear semilinear descent is effective on objects and morphisms;*
7. *projective descent is effective exactly when its relative Brauer class vanishes;*
8. *neutral Tannakian symmetry is reconstructed exactly when a faithful exact tensor fiber functor is supplied.*

These statements form a complete assembly theorem for the declared finite semisimple and neutral tensor strata. They do not imply effective descent for modular, wild, nonneutral, or arbitrary heterogeneous atlases.

Proof. Items 1–5 are theorems 18.2, 18.4, 18.5, 18.7 and 18.9 and corollary 18.11. Item 6 is theorem 18.12. Item 7 is theorem 18.15. Item 8 is theorem 18.17. The final boundary follows because the hypotheses used by Reynolds averaging, Hilbert 90, and neutral Tannakian reconstruction are not automatic in the excluded settings. □

A complete chart record on the finite subgroup locus is

$$\mathcal{J}_{L/K}(H) = \left(H, L^H, P_H, \mathrm{Core}_G(H), \mathrm{Ncl}_G(H), \{K\backslash G/H\}_{K\leq G}\right).$$

The first three entries already classify the normalized object, while the remaining entries record its atlas position and transition structure. For a framed semilinear vector object, the descended K-space with its frame is complete. For a projective object, the Brauer class is an obstruction coordinate, not a complete invariant of all central simple algebras without the remaining algebra data.

18.10 Exact finite verification

The Galois-Tannakian atlas verifier uses exact rational group-algebra arithmetic, exact finite permutation combinatorics, and exact Gaussian-rational matrix algebra. It audits cyclic groups, V_4, S_3, D_4, and A_4; every subgroup Reynolds idempotent; subgroup inclusion and projector order; normality and centrality; core and normal closure; commuting and noncommuting subgroup pairs; double-coset partitions and the Mackey dimension law; cyclic ramification-style filtrations; explicit Hilbert 90 coboundaries over $\mathbb{C}/\mathbb{R}$; the split and quaternionic projective examples; and faithful regular tensor actions.

Table 33: Exact verification of the Galois-Tannakian atlas finite Galois and Tannakian models.

Validation class	Checks
Finite subgroup and Reynolds-projector identities	1,020
Normality, core, and normal-closure identities	192
Double-coset and Mackey packet identities	1,590
Nested ramification-style projector filtrations	39
Exact Hilbert 90 cocycles over Gaussian rationals	884
Projective and Brauer obstruction examples	44
Finite faithful tensor-model checks	7,426
Galois-Tannakian atlas exact checks	**11,195**
Failures	**0**

The verifier records 11,195 exact checks and no failures. Together with the preceding certified atlas program, the cumulative exact record is

$$\boxed{2{,}038{,}860}$$

checks with no recorded failures.

Exact boundary of Galois-Tannakian atlas. The finite subgroup/projector identities, semilinear vector descent, relative Brauer obstruction, profinite finite-level resolution, and neutral Tannakian tensor gate are established under their printed hypotheses. The block does not prove exactness of fixed points when the group order is not invertible, derived replacement of failed Reynolds projectors, integral wild descent, classification of all central simple algebras from the Brauer class alone, existence of a neutral fiber functor, or effective descent for the entire heterogeneous Komorebi atlas.

Section result. Finite semisimple Galois theory now appears as a fully typed resolved stratum. Subgroups, fixed fields, and faithful fixed projectors are reversible charts. Nonnormality produces conjugate atlases and mixed Mackey transition packets. Ramification is a nested detail resolution. Hilbert 90 makes linear semilinear descent effective, while the relative Brauer class is the exact obstruction to projective vector descent. Profinite quotient charts reconstruct continuous finite-dimensional symmetry, and a faithful exact tensor fiber functor crosses the separate Tannakian boundary to reconstruct an affine group scheme. This closes the declared finite Galois and neutral tensor stratum, not modular, wild, nonneutral, or unrestricted global descent.

Part III

Certified global synthesis and scope

19 Global invariant descent on the certified finite atlas

Global synthesis atlas performs the assembly step requested after the native stratum invariants. The preceding sections do not justify one unrestricted equivalence among all eight broad strata. They do, however, supply several finite connected systems of normalized charts for which the semantic object, overlap categories, bridge directions, information loss, coherence, and local completeness are explicit. This section freezes those systems into one authoritative certified atlas, constructs the invariant comparison maps, verifies their higher coherence, and proves the strongest global completeness theorem supported by the assembled results.

The global object in this section is deliberately a finite disjoint union of connected semantic components. Each component contains nontrivial reversible bridges across different kinds of presentation. The component tag is retained as part of the invariant. No bridge is invented between two components merely because their formulas use similar words or symbols. In particular, the functional-Mahler coefficient component and the continuous Newton–Mahler interpolation component remain distinct even though both belong to the broad local stratum.

19.1 The authoritative bridge matrix

Write

$$\mathsf{E}, \quad \mathsf{FF}, \quad \mathsf{D}, \quad \mathsf{P}, \quad \varnothing$$

for an equivalence on a named normalized overlap, a fully faithful inclusion, an information-losing directional functor, a partial or hypothesis-dependent bridge, and the absence of a certified broad bridge, respectively. The matrix below records only relations already established in the preceding sections or in the audited source concordance. A symbol carrying a subscript refers to the named overlap rather than to the whole broad stratum.

Table 34: Authoritative broad-stratum bridge matrix for the certified global synthesis atlas. Every decorated equivalence is restricted to the named normalized overlap; the undecorated broad categories are not thereby identified.

Source \ target	LIN	HOL	DIG	ALG	TREE	PROD	LOC	GAL
LIN	E	FF	E_{per}	–	–	–	E_{per}	–
HOL	P	E	–	E_{Fuss}/P	E_{Fuss}/P	–	–	–
DIG	E_{per}	P	E	P	–	–	E_{rm}	–
ALG	–	E_{Fuss}/D	P	E	E_{Fuss}/P	–	–	–
TREE	–	E_{Fuss}/D	–	E_{Fuss}/D	E	–	–	–
PROD	D	–	–	–	–	E	P	–
LOC	E_{per}	–	E_{rm}	–	–	P	E	–
GAL	–	–	–	–	–	–	–	E

Legend. E: equivalence; FF: fully faithful inclusion; D: information-losing directional functor; P: partial, constructor-dependent, or certificate-dependent bridge; –: no certified bridge. The subscripts per, rm, and Fuss denote the unit-periodic, rational functional-Mahler, and marked weighted Fuss overlaps. A cell of the form E_{Fuss}/D means equivalence on the marked Fuss slice and directional loss on the ambient bridge. Functional-Mahler coefficient equations and Newton–Mahler interpolation remain distinct despite sharing the LOC row.

The matrix has three immediate consequences.

1. The reversible core is strictly smaller than the directed transformation category.
2. A bridge may become an equivalence after restricting to a normalized overlap without becoming invertible on the ambient stratum.
3. Empty cells are mathematical data. They prevent later arguments from silently transporting an invariant through an uncertified route.

The entries E_{per}, E_{rm}, and E_{mark} refer, respectively, to the unit-periodic overlap, the rational functional-Mahler overlap, and the marked polynomial fixed-point overlap. In particular, the functional-Mahler and Newton–Mahler languages remain separate outside their explicitly declared interfaces, and faithful corner realization remains separate from neutral Tannakian reconstruction.

The full object-level bridge ledger, including every normalized source, target, overlap, invariant comparison, and loss or gate, is retained as a machine-generated source artifact. The printed matrix and the component table below contain the proof-relevant summary; a relation absent from the ledger is not used in the descent nerve.

19.2 Certified connected components

Definition 19.1 (Certified Global synthesis atlas atlas)**.** Fix a prime p, a characteristic-zero field K, and a finite Galois extension L/K for the Galois component. The certified atlas $\mathfrak{A}_{13}$ is the finite coproduct of the following five connected profiled atlases.

(C1) $\mathfrak{A}_{\mathrm{DRLM},p}$, the fixed-prime digital–regular component with rational functional-Mahler and periodic linear subcharts;

(C2) $\mathfrak{A}_{\mathrm{Loc},p}$, the continuous Newton–Mahler and complete clopen-branch component;

(C3) $\mathfrak{A}_{\mathrm{Fuss}}$, the marked Fuss–Catalan nonlinear–algebraic–holonomic component;

(C4) $\mathfrak{A}_{\mathrm{EM}}$, the fixed-Euler-basis and prime-uniform multiplicative component;

(C5) $\mathfrak{A}_{\mathrm{Gal},L/K}$, the finite semisimple Galois component for the fixed extension L/K.

The reversible arrows are exactly the normalized equivalences printed below. Directional constructions, reductions, forgetful maps, and obstructed lifts are attached as satellites but do not enter the maximal subgroupoid used for invariant descent.

Table 35: Certified connected components of the Global synthesis atlas atlas.

Component	Semantic domain and normalized charts	Complete coordinate and reversible overlap	Directional satellites or exact boundary
DRLM	fixed-prime regular sequences; canonical section module, strict shortlex record, and minimal DFAO; rational-mask/Cartier and periodic linear subcharts	strict shortlex pointed digit-linear record, refined on the rational-mask and unit-periodic full replete subcategories	integral reductions and Newton–Mahler interpolation are directional; many automatic sequences have no rational mask
LOC	continuous $\mathbb{Z}_p$-valued functions; Newton–Mahler coefficients, complete clopen branch packets, and affine branch coordinates	canonical Mahler coefficient sequence; complete branch analysis and synthesis at each declared depth	functional-Mahler equations require a separately declared coefficient lift and interpolation constructor
Fuss	marked $U = z + \alpha U^q$ tree system, selected algebraic branch, and compressed P-recursive coefficient chart	parameter pair (q, α) with its support and singularity-orbit package	forgetting the branch mark or generator grammar is information-losing
EM	fixed-basis Euler exponents, divisor ghosts, integral unit series, and prime-uniform multiplicative local factors	Euler exponent sequence on the fixed-basis, prime-uniform essential image	arbitrary product grammars and nonuniform prime-local fields remain outside the reversible overlap
Gal	subgroup, fixed field, and faithful Reynolds projector for one fixed finite Galois extension	actual subgroup $H \leq G$ with functorial core, normal-closure, and double-coset data	projective Brauer obstruction, modular or wild failure, and the tensor-fiber gate remain separately typed

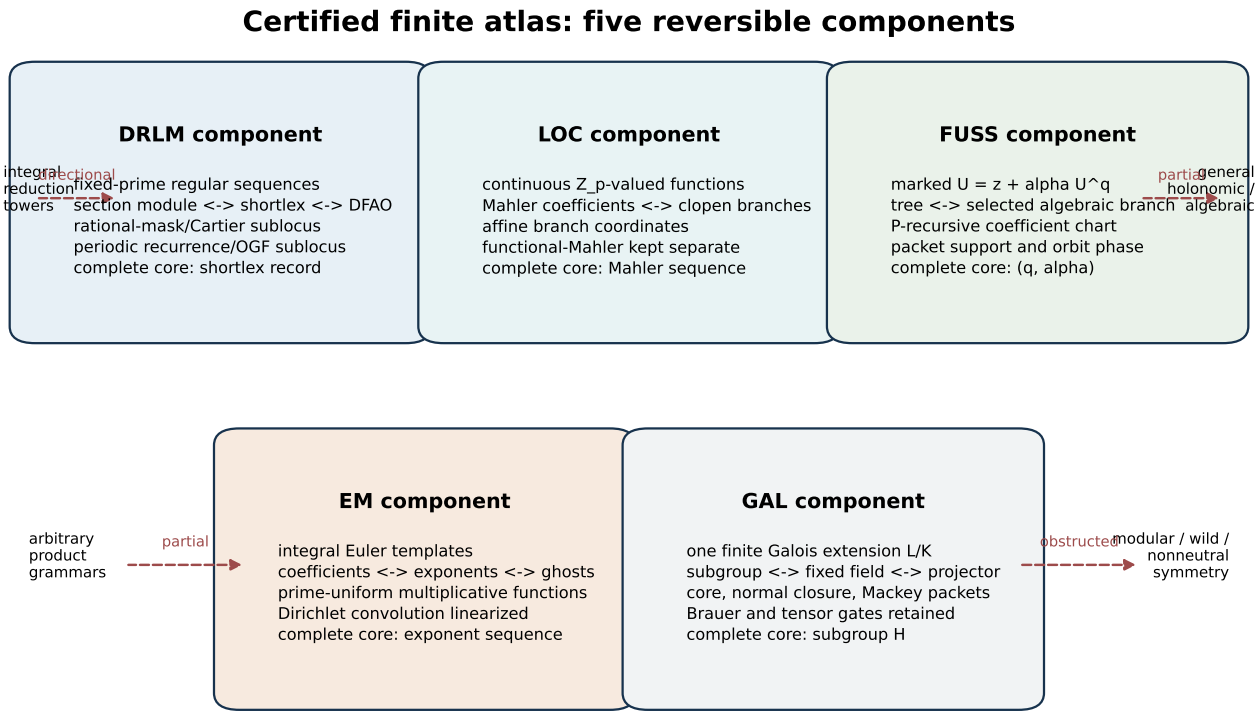


Figure 53: The certified finite atlas. Each box is a nontrivial connected system of normalized charts. Dashed external arrows are directional, partial, or obstructed satellites and are not promoted to equivalences.

The semantic category of the total atlas is

$$\mathcal{O}_{13} = \mathcal{O}_{\mathrm{DRLM}} \sqcup \mathcal{O}_{\mathrm{Loc}} \sqcup \mathcal{O}_{\mathrm{Fuss}} \sqcup \mathcal{O}_{\mathrm{EM}} \sqcup \mathcal{O}_{\mathrm{Gal}}.$$

The coproduct symbol is literal. It does not assert that a finite-field regular sequence, a continuous p-adic function, a characteristic-zero marked tree branch, an integral Euler template, and a finite Galois subgroup are objects of one untyped category. The tag records which semantic doctrine is in force.

19.3 Normalizer-induced higher coherence

Every connected component in definition 19.1 has a canonical semantic normalizer. For a chart $\mathcal{C}_{\lambda,a}$ in component λ, write

$$N_{\lambda,a} : \mathcal{C}_{\lambda,a} \longrightarrow \mathcal{O}_{\lambda}$$

for the semantic normalization functor and choose its canonical inverse chart

$$Q_{\lambda,a} : \mathcal{O}_{\lambda} \longrightarrow \mathcal{C}_{\lambda,a}.$$

The certified bridge from chart a to chart b is

$$\Phi_{ab} = Q_{\lambda,b} N_{\lambda,a}.$$

Theorem 19.2 (Normalizer-induced coherence)**.** *Assume that every normalized fiber of $N_{\lambda,a}$ is contractible. Then the comparison isomorphism*

$$\eta_{abc} : \Phi_{bc}\Phi_{ab} \Longrightarrow \Phi_{ac}$$

is unique. Every triple-overlap diagram commutes, every quadruple-overlap coherence equation holds, and the composite around every finite bridge cycle is the identity transformation of the normalized chart.

Proof. Both $\Phi_{bc}\Phi_{ab}(X)$ and $\Phi_{ac}(X)$ are normalized presentations of the same semantic object $N_{\lambda,a}(X)$. Contractibility of the normalized fiber gives a unique isomorphism between them. Any two composites occurring in a coherence diagram are isomorphisms between the same two normalized presentations, so uniqueness makes them equal. The cycle statement follows by applying the same argument to the starting chart and itself. □

This theorem is not a substitute for constructing the normalizer. It applies here because the preceding sections have already supplied unique reduced masks, reachable-observable Cartier modules, minimal automata, canonical section modules, marked fixed-point systems, selected algebraic branches, fixed Euler exponents, prime-power local factors, fixed subgroups, fixed fields, and faithful Reynolds projectors on the declared domains.

19.4 Invariant descent theorem

For every chart $\mathcal{C}_{\lambda,a}$, let

$$I_{\lambda,a} : \mathcal{C}_{\lambda,a} \longrightarrow \mathcal{V}_{\lambda,a}$$

be its native normalized invariant. A bridge comparison is a natural isomorphism

$$\kappa_{ab} : J_{ab} I_{\lambda,a} \Longrightarrow I_{\lambda,b} \Phi_{ab},$$

where J_{ab} transports invariant coordinates between their chart-specific target categories.

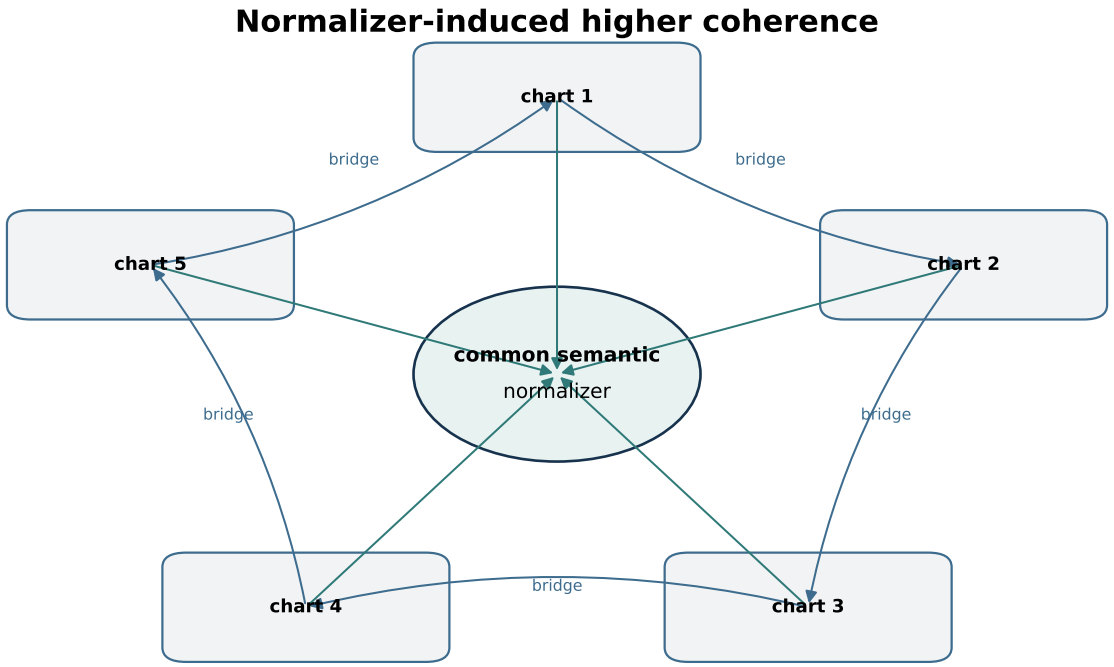


Figure 54: Every reversible chart in one certified component normalizes to the same semantic object. The unique normalized comparison forces all finite bridge cycles to commute.

Definition 19.3 (Coherent invariant family)**.** The family $(I_{\lambda,a}, \kappa_{ab})$ is coherent when the two induced comparison maps from a to c agree on every triple overlap and the corresponding quadruple-overlap equation holds.

Theorem 19.4 (Certified finite invariant descent)**.** *Let $\mathfrak{A}_{13}$ be the atlas of definition 19.1. Suppose that, in every connected component:*

1. *the local invariant is complete on every normalized chart;*
2. *the bridge comparisons are coherent;*
3. *invariant-compatible local isomorphisms satisfy effective descent.*

Then the local invariants induce a functor

$$\mathcal{I}_{13} : \mathrm{Desc}(\mathfrak{A}_{13}) \longrightarrow \mathrm{Desc}(\mathfrak{V}_{13})$$

that reflects isomorphisms. If each semantic component is discrete, equality of descended invariants is equivalent to equality of the represented semantic objects.

Proof. The local comparison maps form a pseudonatural invariant family, so theorem 6.3 gives the descended functor. Equality or isomorphism of the descended invariant gives compatible local invariant matches. Local completeness supplies local object isomorphisms. Effective isomorphism descent glues those local isomorphisms to an isomorphism in the descent category. On a discrete semantic component, every isomorphism is equality. □

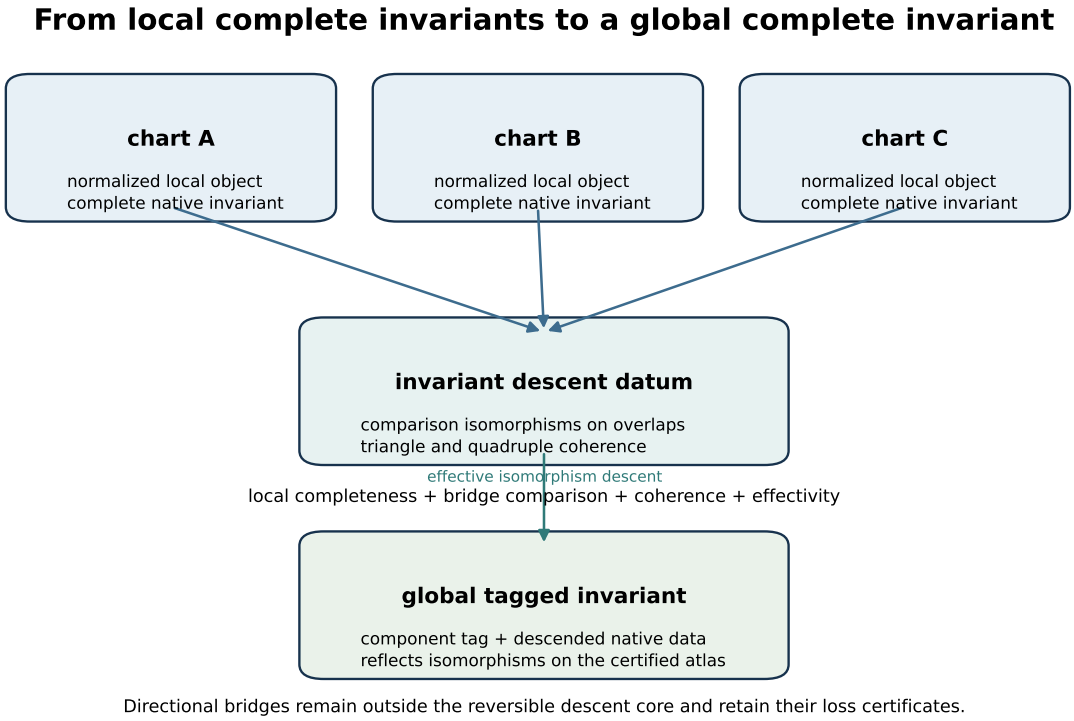


Figure 55: The global invariant theorem has four separate requirements. Local completeness alone is insufficient without comparison maps, higher coherence, and effective descent of the resulting isomorphisms.

19.5 Digital–regular–linear–functional-Mahler component

Fix the prime p and let

$$\mathcal{O}_{\mathrm{DRLM}} = \mathbf{Reg}_p(\mathbb{F}_p)$$

be the discrete semantic category of p-regular sequences over $\mathbb{F}_p$. By the finite-field fixed-radix theorem, every object is also p-automatic. Its globally defined normalized charts are the canonical section module, the strict shortlex pointed digit-linear record, and the minimal least-significant-digit-first p-DFAO.

Two full replete subcategories carry additional charts.

1. On the rational-mask locus $\mathbf{RMah}_p^{\times}$, the reduced functional-Mahler mask and the minimal reachable-observable Cartier module are equivalent to the digital charts.
2. On the unit purely periodic locus, the minimal recurrence, reduced rational OGF, minimal finite linear state, primitive period word, and rational-mask chart are all equivalent.

For $a \in \mathcal{O}_{\mathrm{DRLM}}$, define

$$\mathcal{J}_{\mathrm{DRLM},p}(a) = \left(\mathsf{SL}_{p,\mathbb{F}_p}(a), \mathcal{P}_{\mathrm{rm}}(a), \mathcal{P}_{\mathrm{per}}(a)\right),$$

where $\mathsf{SL}_{p,\mathbb{F}_p}(a)$ is the strict shortlex record, $\mathcal{P}_{\mathrm{rm}}(a)$ is empty off the rational-mask locus and otherwise contains the reduced mask, Cartier module, collapse obstruction, and minimal automaton coordinates, and $\mathcal{P}_{\mathrm{per}}(a)$ is empty off the purely periodic locus and otherwise contains the primitive period word, exact period, minimal recurrence, reduced OGF, and periodic packet profile.

Theorem 19.5 (Digital–regular–linear–Mahler comparison theorem)**.** *The invariant $\mathcal{J}_{\mathrm{DRLM},p}$ is preserved by every reversible bridge in $\mathfrak{A}_{\mathrm{DRLM},p}$. Its comparison cells satisfy all higher coherence equations. The strict shortlex*

coordinate reconstructs the represented regular sequence, so the descended invariant is complete on $\mathbf{Reg}_p(\mathbb{F}_p)$. *The mask and periodic packages are compatible exact refinements on their respective subloci.*

Proof. The canonical section module, strict shortlex record, and minimal orbit automaton normalize to the same coefficient sequence by the Fixed-radix regular atlas effective-descent theorem. On the rational-mask locus, fixed-denominator Cartier intertwining identifies the reachable numerator span with the canonical section module and the reachable numerator set with the minimal automaton states. On the periodic locus, the recurrence, rational, state, automaton, and mask charts normalize to the same primitive coefficient word. Every comparison therefore factors through the same semantic normalizer, and theorem 19.2 applies. Completeness follows from the strict shortlex reconstruction formula. □

Table 36: Representative cross-stratum invariant comparisons.

Example	Period	Recurrence order	Regular rank	DFAO states	Mask compiler degree
$\mathbb{F}_2$/ word 10	2	2	2	3	2
$\mathbb{F}_2$/ word 110	3	2	2	3	2
$\mathbb{F}_3$/ word 12	2	1	1	2	1

The table gives representative objects for which recurrence order, regular rank, automaton size, and mask-compiler dimension are different coordinates of the same normalized semantic sequence. Equality of one displayed scalar alone is not a complete invariant; the complete records are the chart objects described above.

The inclusions of the rational-mask and periodic loci are not essentially surjective. Over a finite field every regular sequence is automatic, but many automatic sequences have no rational canonical mask, and many rational-mask sequences are not periodic. The optional fields in $\mathcal{J}_{\mathrm{DRLM},p}$ record these exact domain boundaries rather than forcing one chart to exist everywhere.

19.6 Continuous Newton–Mahler component

Let

$$\mathcal{O}_{\mathrm{Loc}} = C(\mathbb{Z}_p, \mathbb{Z}_p)$$

with equality morphisms. Its canonical coordinate is the Mahler sequence

$$\mathcal{M}_p(f) = (c_r(f))_{r\geq 0}, \qquad c_r(f) = \Delta^r f(0), \qquad c_r(f) \longrightarrow 0.$$

At every declared depth p^a, the complete clopen branch packet

$$\big(f(p^a q + s)\big)_{0\leq s<p^a}$$

and the corresponding branchwise Mahler coefficients are reversible by explicit piecewise synthesis. Affine branch kernels and mixed-radix refinements are comparison charts of the same semantic function.

Theorem 19.6 (Continuous local invariant comparison)**.** *The Mahler coefficient sequence*

$$\mathcal{J}_{\mathrm{Loc},p}(f) = (c_r(f))_{r\geq 0}$$

is a complete invariant of $\mathfrak{A}_{\mathrm{Loc},p}$. *Complete branch packets, branchwise Mahler coordinates, affine coefficient kernels, and nested refinements admit coherent comparison maps to* $\mathcal{J}_{\mathrm{Loc},p}$.

Proof. Mahler analysis and synthesis are mutually inverse on $C(\mathbb{Z}_p, \mathbb{Z}_p)$. Complete clopen analysis and piecewise synthesis are mutually inverse at each depth, and the nested branch law is strict. Conjugating those maps by Mahler analysis gives the affine coefficient comparisons, whose mixed-radix composition was proved in Local Mahler-polyphase atlas. All normalized charts therefore return the same function and the same Mahler sequence, so theorem 19.2 gives higher coherence. □

Functional-Mahler equations for coefficient series are not included in this component. A finite-state or functional-Mahler object enters the Newton–Mahler component only through a separately declared lift and interpolation constructor carrying its own precision and information-loss certificate.

19.7 A genuine nonlinear–algebraic–holonomic overlap

Let K be a characteristic-zero field. For an integer $q \geq 2$ and $\alpha \in K^\times$, let

$$U_{q,\alpha}(z) \in zK[[z]]$$

be the unique selected branch satisfying

$$U = z + \alpha U^q. \tag{50}$$

The derivative of

$$P_{q,\alpha}(z, Y) = Y - z - \alpha Y^q$$

with respect to Y is one at $(0, 0)$, so the branch is simple and selected by $U(0) = 0$.

Put

$$x = z^{q-1}, \qquad U_{q,\alpha}(z) = zC_{q,\alpha}(x).$$

Then

$$C = 1 + \alpha x C^q. \tag{51}$$

Lagrange inversion gives

$$[x^n]C_{q,\alpha}(x) = c_n(q, \alpha) = \frac{\alpha^n}{(q-1)n+1}\binom{qn}{n}. \tag{52}$$

Define

$$A_q(n) = (n+1)\prod_{r=2}^{q}((q-1)n + r),$$

$$B_q(n) = \prod_{r=1}^{q}(qn + r).$$

Then the coefficient sequence satisfies the first-order polynomial recurrence

$$\boxed{A_q(n)c_{n+1} = \alpha B_q(n)c_n.} \tag{53}$$

After dividing the polynomial gcd of A_q and B_q, this is the normalized first-order tail recurrence with initial value $c_0 = 1$.

Theorem 19.7 (Fuss–Catalan three-stratum equivalence)**.** *The following normalized marked presentations determine one another exactly:*

1. *the polynomial tree system $U = z + \alpha U^q$;*

2. *the selected algebraic branch* $(P_{q,\alpha}, U(0) = 0)$*;*
3. *the compressed P-recursive coefficient chart* $(\text{Equation (53)}, c_0 = 1)$ *together with the support embedding* $U(z) = zC(z^{q-1})$.

Every normalized fiber is contractible. All triangle comparisons return the same parameter pair (q, α).

Proof. The marked tree equation is the selected algebraic equation. Simplicity at the origin gives its unique formal branch. Substitution $U = zC(z^{q-1})$ gives Equation (51). Lagrange inversion gives Equation (52), and taking the ratio of consecutive coefficients gives Equation (53). Conversely, the recurrence and initial value reconstruct C, the support embedding reconstructs U, and the reconstructed series satisfies the selected algebraic equation. A normalized automorphism fixes q, α, and the unique branch, hence is the identity. □

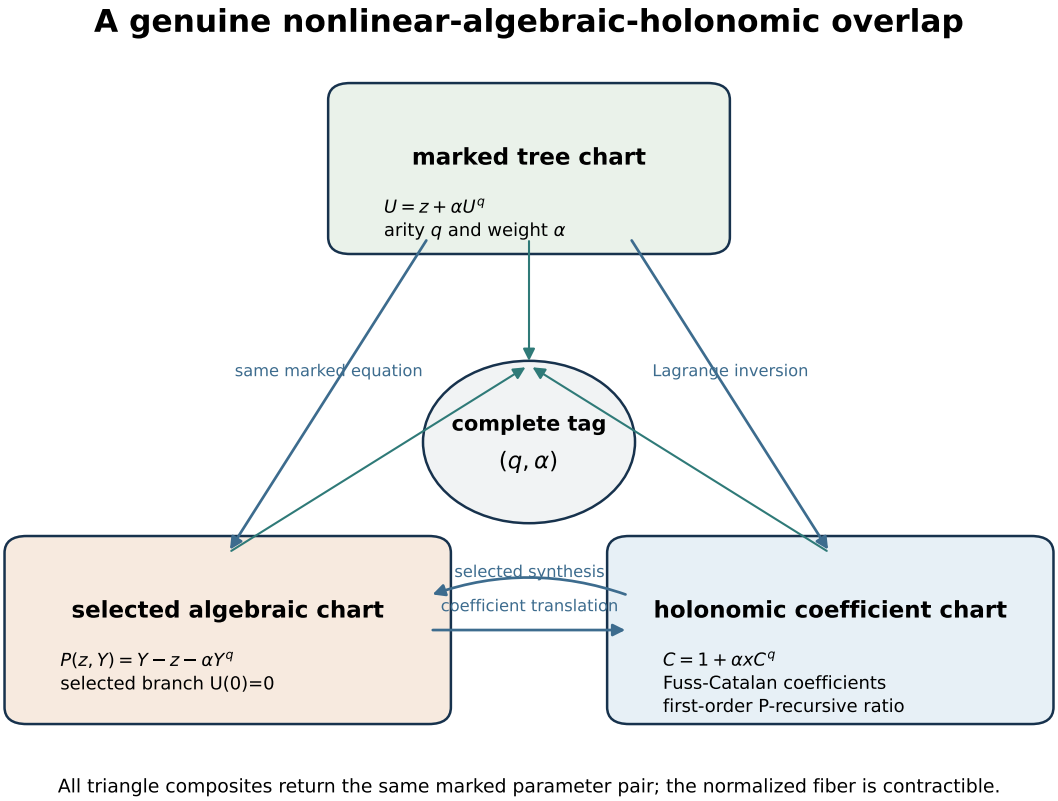


Figure 56: The marked Fuss–Catalan family gives a genuine overlap among the nonlinear tree, selected algebraic, and holonomic recurrence strata. Every path returns the same marked pair (q, α).

The tree arity period is $q - 1$. For packet modulus N, set

$$g_N = \gcd(N, q - 1).$$

The nonzero coefficient addresses of $U_{q,\alpha}$ are exactly

$$j \equiv 1 \pmod{g_N},$$

and the complete dominant singularity orbit has size

$$\frac{q-1}{g_N}$$

under the positive analytic hypotheses of theorem 17.15. These packet signatures agree in the tree and algebraic charts because both are computed from the same selected branch.

Define

$$\mathcal{J}_{\mathrm{Fuss}}(q,\alpha) = \left(q, \alpha, P_{q,\alpha}, \{c_n(q,\alpha)\}_{n\geq 0}, \{g_N\}_{N\geq 1}, \left\{\frac{q-1}{g_N}\right\}_{N\geq 1}\right).$$

Corollary 19.8 (Complete cross-stratum invariant on the marked Fuss family)**.** *The nonlinear, algebraic, holonomic, packet-support, and dominant-orbit invariants admit coherent comparison maps to $\mathcal{J}_{\mathrm{Fuss}}$. The package is complete on the declared marked family.*

Proof. Coherence follows from theorems 19.2 and 19.7. The first two entries reconstruct the marked fixed-point equation and therefore its unique selected branch. □

Table 37: Named weighted q-ary examples in the cross-stratum atlas.

q	λ	Support period	First supported coefficients	Active addresses at $N=6$	Packet-orbit size
2	1	1	$(1,1,2,5,\dots)$	$\{0,1,2,3,4,5\}$	1
3	2	2	$(1,2,12,96,\dots)$	$\{1,3,5\}$	1
4	-1	3	$(1,-1,4,-22,\dots)$	$\{1,4\}$	1
5	$1/2$	4	$(1,1/2,5/4,35/8,\dots)$	$\{1,3,5\}$	2

The examples show that the finite invariant retains the weight parameter as well as the arity. Support period alone recovers only $q-1$; the coefficient $[z^q]U = \alpha$ supplies the second coordinate.

This is not a classification of arbitrary algebraic branches, arbitrary P-recursive sequences, or unmarked nonlinear fixed points. It is the first explicit complete invariant on a connected overlap of those three strata.

19.8 Euler–multiplicative component

Let

$$A_b(z) = \prod_{m\geq 1}(1-z^m)^{-b_m}$$

be an integral unit series in the fixed Euler basis, and let

$$c_n = \sum_{d\mid n} d\, b_d$$

be its divisor-ghost sequence. Define the prime-uniform multiplicative function

$$f_b(n) = \prod_{p^e \| n} [z^e] A_b(z).$$

Every prime has the same local factor $A_b(T)$. By theorems 13.4 and 13.23, the coefficient, exponent, ghost, and prime-uniform multiplicative charts are reversible on this declared locus.

Theorem 19.9 (Euler–multiplicative invariant comparison)**.** *The exponent sequence*

$$\mathcal{J}_{\mathrm{EM}}(A_b) = b$$

is a complete invariant of the certified Euler–multiplicative component. In every chart it is recovered by the same Möbius formula

$$b_n = \frac{1}{n}\sum_{d\mid n}\mu(n/d)c_d,$$

and Dirichlet convolution corresponds to exponent addition.

Proof. Unique Euler factorization reconstructs A_b from b. Prime-power reconstruction then reconstructs f_b. The logarithmic-derivative identity gives c, and Möbius inversion recovers b. Every bridge therefore factors through the same exponent normalizer, so higher coherence follows from theorem 19.2. □

The theorem remains confined to the fixed Euler basis and the prime-uniform multiplicative essential image. It does not identify arbitrary Cauchy-product grammars with arbitrary Dirichlet-convolution systems.

19.9 Finite Galois component

Fix a finite Galois extension L/K in the semisimple setting. For each subgroup $H \leq G$, define

$$\mathcal{J}_{\mathrm{Gal}}(H) = \left(H, L^H, P_H, \mathrm{Core}_G(H), \mathrm{Ncl}_G(H), \{J\backslash G/H\}_{J\leq G}\right).$$

Theorem 19.10 (Finite Galois invariant comparison)**.** *The subgroup, fixed-field, and faithful Reynolds-projector charts have coherent comparison maps to $\mathcal{J}_{\mathrm{Gal}}$. The invariant is complete on the normalized subgroup atlas for the fixed extension L/K.*

Proof. The first three entries determine one another by theorem 18.2. Core, normal closure, and double-coset packets are functorial constructions from the subgroup. The subgroup coordinate itself is complete, and the strict subgroup normalizer forces coherence. □

The Brauer class and Tannakian group scheme are not added to this complete subgroup invariant as if they were automatic coordinates. They belong to separately typed projective and tensor extensions whose hypotheses and obstructions were stated in Galois-Tannakian atlas.

19.10 The global tagged invariant

Let

$$\Lambda_{13} = \{\mathrm{DRLM}, \mathrm{Loc}, \mathrm{Fuss}, \mathrm{EM}, \mathrm{Gal}\}$$

be the component-tag set. Define

$$\mathcal{J}_{13}(X) = \big(\lambda(X), \mathcal{J}_{\lambda(X)}(X)\big).$$

Theorem 19.11 (Certified global completeness theorem)**.** *The invariant $\mathcal{J}_{13}$ descends across every reversible bridge in $\mathfrak{A}_{13}$ and is complete on*

$$\mathcal{O}_{13} = \mathcal{O}_{\mathrm{DRLM}} \sqcup \mathcal{O}_{\mathrm{Loc}} \sqcup \mathcal{O}_{\mathrm{Fuss}} \sqcup \mathcal{O}_{\mathrm{EM}} \sqcup \mathcal{O}_{\mathrm{Gal}}.$$

More precisely,

$$\boxed{\mathcal{J}_{13}(X) = \mathcal{J}_{13}(Y) \quad \Longleftrightarrow \quad X = Y}$$

inside the declared discrete semantic category. The associated invariant functor reflects isomorphisms on the descent category.

Proof. Within each component, completeness is theorems 19.5, 19.6, 19.9 and 19.10 and corollary 19.8. Their comparison maps are coherent by theorem 19.2. The component tag prevents an object in one semantic doctrine from being identified with an object in another. Apply theorem 19.4 componentwise and then take the finite coproduct. □

Remark 19.12 (What is genuinely global). The theorem is not the tautology that every object is classified by itself. Each component invariant is computed in one native chart and transported through several nonidentical presentation categories. The global assertion is that these transported coordinates agree coherently and classify the descent object. The coproduct tag is nevertheless essential: the current Library does not provide reversible bridges joining all five components.

19.11 Directional satellites and exact loss records

The certified reversible core is surrounded by directional bridges that remain useful for theorem transport and computation. Their failure to enter $\mathcal{J}_{13}$ is controlled by explicit loss data.

Table 38: Directional satellites of the certified global invariant atlas.

Directional bridge	Retained structure	Exact loss or obstruction
Integral regular lattice → residue section module	digit transitions and residue semantics	Smith–Tor ghost kernel and missing quotient
General regular sequence ⇢ rational mask	coefficient semantics on the essential image	rational-mask existence and collapse obstruction
Algebraic branch → holonomic annihilator	D-finite coefficient semantics	selected branch and minimal polynomial may be forgotten
Marked nonlinear system → semantic fixed point	the represented fixed point	generator labels, arities, and tree grammar may be forgotten
Finite-state dynamics → Newton–Mahler chart	congruence branch values and precision ladder	depends on a chosen lift and principal-unit hypothesis
Projective Galois cocycle ⇢ linear descent	projective action	relative Brauer class
Faithful corner realization ⇢ Tannakian symmetry	additive packet and operator equations	exact tensor, dual, unit, and fiber-functor data are absent

Proposition 19.13 (No completeness transport through an unreflected directional bridge)**.** *Let $T : \mathcal{C} \to \mathcal{D}$ be a directional bridge. A complete invariant on $\mathcal{D}$ pulls back to a complete invariant on $\mathcal{C}$ only if T reflects isomorphisms on the declared source subcategory. In particular, any nontrivial object-collapse relation or obstruction fiber prevents automatic completeness transport.*

Proof. If T does not reflect isomorphisms, there exist nonisomorphic X, Y with $T(X) \cong T(Y)$. Every invariant depending only on the target object then takes the same value on X and Y, so its pullback cannot be complete. □

19.12 Exact finite verification

Two complementary exact suites audit Global synthesis atlas. The first authenticates the frozen preceding verification records and checks the broad bridge matrix, periodic and product–multiplicative overlap replays, directional-loss controls, finite Čech holonomy models, and the five completeness gates. The second checks the weighted q-ary tree–algebraic–holonomic overlap, its coefficient and packet equations, parameter recovery, detailed bridge ledger, and tagged component separation.

Table 39: Exact verification of the Global synthesis atlas invariant-descent and weighted q-ary cross-stratum models.

Validation class	Checks
Bridge-matrix structural checks	172
Weighted q-ary coefficient identities	1,856
Exact support-period identities	8,192
Formal fixed-point identities	8,192
Minimal-polynomial relation checks	8,192
Finite parameter recovery	128
Tree/algebraic/holonomic comparison paths	64
Supported first-order recurrence checks	1,792
Packet support and orbit-phase checks	2,112
Exact packet synthesis checks	44,352
Address-carry packet equations	44,352
Tagged component-separation checks	49
Global synthesis exact checks	**119,453**
Failures	**0**

Table 40: Exact verification of the Global synthesis atlas global-invariant and cross-stratum synthesis layer.

Validation suite	Exact assertions
Bridge matrix, overlap replay, directional-loss, and finite Čech suite	154,209
Weighted q-ary tree–algebraic–holonomic and tagged-invariant suite	119,453
Regression/authentication assertions already represented in the certified native-stratum program	39,846
New Global synthesis atlas comparison and coherence assertions	**233,816**
All assertions executed by the two Global synthesis atlas suites	273,662
Failures	0

The two suites execute 273, 662 exact assertions with no failures. Of these, 39, 846 authenticate or replay assertions already represented in the preceding cumulative record. The global synthesis therefore contributes 233, 816 nonduplicative comparison, coherence, packet, and loss-control assertions. The blockwise cumulative record is

$$\boxed{2,272,676}$$

exact assertions with no recorded failures. Counting every regression assertion executed during the Global synthesis atlas replay gives 2, 312, 522 total executed assertions. Neither number is a count of independent theorems.

The bridge-synthesis suite replays primitive unit-periodic words over finite prime fields, compares their recurrence, OGF, automaton, section-module, and functional-Mahler coordinates, verifies the prime-

uniform Euler–multiplicative overlap, and tests finite nontrivial Čech cycle models. The weighted q-ary suite checks the fixed-point identity, Lagrange coefficients, first-order P-recursive recurrence, support phases, exact packet synthesis, address-carry equations, and all triangle normalizers over a declared finite parameter grid. Both machine-readable records are included in the source package.

Exact boundary of Global synthesis atlas. The global invariant theorem is proved for the five-component certified finite atlas in definition 19.1. It does not assert a complete invariant for every object in the eight broad strata, a reversible bridge between the five semantic components, effective descent for every directional satellite, cross-radix or all-ring normalization, modular or wild Galois effectivity, nonneutral Tannakian reconstruction, or classification of unmarked general holonomic, algebraic, and nonlinear systems.

Section result. The manuscript now contains an authoritative broad-stratum bridge matrix, a finite certified atlas with five nontrivial connected components, explicit invariant comparison maps, a normalizer-induced higher-coherence theorem, and a complete tagged global invariant on the declared semantic union. The new Fuss–Catalan triangle gives a genuine nonlinear–algebraic–holonomic overlap. The digital–regular–functional-Mahler, continuous Newton–Mahler, Euler–multiplicative, and finite Galois components consolidate the exact cross-presentation results of the preceding stratum sections without conflating the two Mahler theories. Directional bridges remain outside the reversible core and retain their precise loss or obstruction records. This closes the finite global-invariant milestone on a substantial atlas, not the unrestricted eight-stratum stack.

20 Universal lane transport at finite profile

The preceding parts of the manuscript establish packet coordinates in an additive idempotent-complete setting and verify them separately in several native strata. This section isolates the common finite theorem. The word “lane” will mean a packet together with its address, its chosen splitting, and its position in a declared refinement system. Thus a lane is more than a subsequence or a suggestive decomposition. It carries the exact data needed for analysis, synthesis, transfer, refinement, and reconstruction.

The theorem is universal only in a profiled sense. Its algebraic core applies to every additive realization that preserves the finite sums, compositions, split idempotents, and refinement identities used in the proof. Kernel, quotient, tensor, topological, metric, and nonlinear conclusions require the additional hypotheses stated where they occur.

20.1 Finite profiled lane systems

Definition 20.1 (Finite resolution shape)**.** A finite resolution shape is a finite partially ordered set R. We write $s \succeq r$ when s is finer than r. The shape contains no address sets: object-specific addresses and their refinement maps are declared in the lane system below.

Definition 20.2 (Finite profiled lane system)**.** Let $\mathcal{C}$ be an additive idempotent-complete category, let $X \in \mathcal{C}$, and let R be a finite resolution shape. A finite profiled lane system $\mathcal{L}_X$ consists, for every $r \in \mathsf{R}$, of a finite address set $I_{X,r}$ and a family

$$\mathcal{P}_X^r = (p_{X,i}^r)_{i \in I_{X,r}} \subseteq \mathrm{End}_{\mathcal{C}}(X)$$

with

$$(p_{X,i}^r)^2 = p_{X,i}^r, \qquad p_{X,i}^r p_{X,j}^r = 0 \quad (i \neq j), \qquad \sum_{i \in I_{X,r}} p_{X,i}^r = 1_X,$$

together with chosen splittings

$$X_i^r \xrightarrow{\iota_{X,i}^r} X \xrightarrow{\pi_{X,i}^r} X_i^r, \qquad \iota_{X,i}^r \pi_{X,i}^r = p_{X,i}^r, \qquad \pi_{X,i}^r \iota_{X,i}^r = 1_{X_i^r}.$$

For $s \succeq r$, the system also contains a surjection

$$\rho_{sr}^X : I_{X,s} \twoheadrightarrow I_{X,r}$$

such that

$$p_{X,i}^r = \sum_{\rho_{sr}^X(j)=i} p_{X,j}^s, \qquad \rho_{tr}^X = \rho_{sr}^X \rho_{ts}^X.$$

Here $\rho_{rr}^X = 1_{I_{X,r}}$. The address sets and refinement maps are declared data; they are not inferred from their cardinalities. Its verified profile contains finite biproducts, addition of morphisms, composition, the displayed splittings, and these projector-sum refinement identities. Any further constructor is included only after its preservation has been verified.

At resolution r, write

$$\mathcal{A}_X^r := (\pi_{X,i}^r)_i : X \longrightarrow \bigoplus_i X_i^r, \qquad \mathcal{S}_X^r := (\iota_{X,i}^r)_i : \bigoplus_i X_i^r \longrightarrow X.$$

For profiled lane systems on X and Y, define the coordinate block of $f : X \to Y$ at resolutions r, u by

$$[f]_{ai}^{u,r} := \pi_{Y,a}^u f \iota_{X,i}^r : X_i^r \longrightarrow Y_a^u.$$

Its ambient corner is

$$p_{Y,a}^u f p_{X,i}^r = \iota_{Y,a}^u [f]_{ai}^{u,r} \pi_{X,i}^r.$$

20.2 The universal finite transport theorem

Theorem 20.3 (Universal Lane Transport Theorem, finite profiled form)**.** *Let $\mathcal{C}$ be an additive idempotent-complete category. Give each of the objects $X_0, \dots, X_m$ a finite profiled lane system. Then all of the following statements hold.*

(L1) ***Exact analysis and synthesis.*** *At every declared resolution,*

$$\mathcal{S}_X^r \mathcal{A}_X^r = 1_X, \qquad \mathcal{A}_X^r \mathcal{S}_X^r = 1_{\oplus_i X_i^r}.$$

Hence $X \cong \bigoplus_i X_i^r$.

(L2) ***Unique lane matrix.*** *Every morphism $f : X \to Y$ is recovered uniquely from its lane blocks:*

$$f = \sum_{a,i} p_{Y,a}^u f p_{X,i}^r = \sum_{a,i} \iota_{Y,a}^u [f]_{ai}^{u,r} \pi_{X,i}^r.$$

Equivalently, the assignment

$$f \longmapsto \left([f]_{ai}^{u,r}\right)_{a,i}$$

is a bijection from $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ to the group of rectangular matrices with entries in $\mathrm{Hom}_{\mathcal{C}}(X_i^r, Y_a^u)$.

(L3) ***Composition and path law.*** *If $f : X \to Y$ and $g : Y \to Z$, with the same resolution u used on the intermediate object, then*

$$[gf]_{ci}^{v,r} = \sum_{a \in I_{Y,u}} [g]_{ca}^{v,u} [f]_{ai}^{u,r}.$$

More generally, a block of a composable string is the finite sum of the products along all compatible intermediate address paths.

(L4) ***Projector-sum refinement.*** *If $s \succeq r$ on X and $v \succeq u$ on Y, then*

$$p_{Y,a}^u f p_{X,i}^r = \sum_{\rho_{vu}^Y(b) = a} \sum_{\rho_{sr}^X(j) = i} p_{Y,b}^v f p_{X,j}^s.$$

Coarsening a fine composite gives the same ambient morphism as composing at the coarse level.

(L5) ***Selectors and named residuals.*** *For $E \subseteq I_{Y,u} \times I_{X,r}$, put*

$$C_E^{u,r}(f) := \sum_{(a,i) \in E} p_{Y,a}^u f p_{X,i}^r, \qquad R_E^{u,r}(f) := f - C_E^{u,r}(f).$$

As additive endomorphisms of $\mathrm{Hom}_{\mathcal{C}}(X, Y)$,

$$C_E^2 = C_E, \qquad C_E C_F = C_{E \cap F}, \qquad R_E = 1 - C_E, \qquad f = C_E(f) + R_E(f).$$

Thus deletion of a lane block is never silent. Its complement is an exact, named residual.

(L6) ***Support transport.*** *If $f = C_E(f)$ and $g = C_F(g)$, then*

$$gf = C_{F \circ E}(gf),$$

where $F \circ E$ is relational composition through the intermediate addresses. For arbitrary f, g, writing $f_E = C_E(f)$, $r_f = R_E(f)$, $g_F = C_F(g)$, and $r_g = R_F(g)$, one has the exact identity

$$gf - g_F f_E = r_g f_E + g_F r_f + r_g r_f.$$

(L7) ***Additive realization.*** *Let $F : \mathcal{C} \to \mathcal{D}$ be an additive functor to another additive idempotent-complete category. If the verified profile of F contains the chosen splittings and refinement identities, then*

$$F([f]^{u,r}_{ai}) = [F(f)]^{u,r}_{ai}$$

after the declared splitting comparisons. The functor carries analysis, synthesis, matrix multiplication, projector-sum refinement, selectors, and named residuals to their corresponding constructions in $\mathcal{D}$.

(L8) ***Exact fold loss.*** *Let E be a finite set of retained lane positions and set*

$$B_e := \mathrm{Hom}_{\mathcal{C}}(X^r_i, Y^u_a) \quad \text{for } e = (a,i) \in E.$$

Let T be a finite set, let $\kappa : E \to T$, let C_t be abelian groups, and let $\tau_e : B_e \to C_{\kappa(e)}$ be homomorphisms. With $C = \bigoplus_{t\in T} C_t$ and coordinate inclusions $j_t : C_t \to C$, the fold

$$\Phi_{\kappa,\tau} : \bigoplus_{e\in E} B_e \longrightarrow C, \qquad (b_e)_e \longmapsto \sum_{e\in E} j_{\kappa(e)}\tau_e(b_e)$$

has exact information-loss group $\ker \Phi_{\kappa,\tau}$. In particular, two retained lane matrices have the same folded record exactly when their difference belongs to this kernel.

Suppose, more strongly, that every τ_e is injective and that there are globally typed selectors $Q_e \in \mathrm{End}(C)$ satisfying, for every $e, e' \in E$ and every $b_{e'} \in B_{e'}$,

$$Q_e j_{\kappa(e')}\tau_{e'}(b_{e'}) = \delta_{e,e'} j_{\kappa(e')}\tau_{e'}(b_{e'}).$$

Then $\Phi_{\kappa,\tau}$ is injective. If there are also homomorphisms

$$r_e : C \longrightarrow B_e, \qquad r_e j_{\kappa(e)}\tau_e = 1_{B_e},$$

then

$$L := \big(r_e Q_e\big)_{e\in E} : C \longrightarrow \bigoplus_{e\in E} B_e$$

is a left inverse of $\Phi_{\kappa,\tau}$, so the fold is split lossless.

Proof. For (L1), the first composite is

$$\sum_i \iota^r_{X,i}\pi^r_{X,i} = \sum_i p^r_{X,i} = 1_X.$$

The (i,j)-entry of the other composite is $\pi^r_{X,i}\iota^r_{X,j}$. It is the identity when $i = j$, and it is zero otherwise because the corresponding idempotents annihilate one another.

For (L2), insert the two reconstruction identities on the two sides of f. Applying $\pi^u_{Y,a}$ on the left and $\iota^r_{X,i}$ on the right proves uniqueness of every entry. For (L3), insert $1_Y = \sum_a \iota^u_{Y,a}\pi^u_{Y,a}$ between g and f. Iteration gives the path formula.

For (L4), substitute the two projector-sum refinement identities into the coarse ambient corner and distribute the finite sums. The assertion about composition follows from the same substitution at the intermediate object.

For (L5), each selector either fixes an ambient block or kills it. Applying two selectors therefore retains precisely the intersection. The residual identities follow by subtraction. For (L6), expand each block of gf through the intermediate resolution. A surviving summand supplies an intermediate address in the relational product. The residual formula is the finite expansion of $(g_F + r_g)(f_E + r_f) - g_F f_E$.

For (L7), additivity preserves zero maps and finite sums, functoriality preserves identities and composition, and the splitting comparisons identify the image of every source lane object with the declared target lane object. The profile hypothesis is needed precisely for those identifications and for the refinement equations.

For (L8), equality of two fold values is equivalent to the difference of the two input families lying in the kernel. Under the selector hypothesis, if $\Phi_{\kappa,\tau}((b_e)_e) = 0$, applying Q_e gives $j_{\kappa(e)}\tau_e(b_e) = 0$. Both $j_{\kappa(e)}$ and τ_e are injective, hence $b_e = 0$ for every e. The chosen retractions compose with the Q_e to give the displayed left inverse: its e-coordinate on the fold is $r_e Q_e j_{\kappa(e)}\tau_e = 1_{B_e}$, while strict separation kills every other coordinate. □

Remark 20.4 (Why the fold selectors live in one group). The endomorphisms Q_e in (L8) all act on the single group $C = \bigoplus_t C_t$. Writing separate selectors on unrelated codomains would make the Kronecker relation ill-typed and would not prove injectivity of the combined fold.

Proposition 20.5 (Exact visible-support loss ledger)**.** *Use the finite fold data of (L8) and fix an input family $b = (b_e)_{e\in E}$. Define*

$$S_0 := \{e \in E : b_e \neq 0\}, \qquad A_0 := \{e \in S_0 : \tau_e(b_e) = 0\}, \qquad S_1 := S_0 \setminus A_0,$$

and let

$$\Gamma := \kappa(S_1) \subseteq T.$$

For $t \in T$, write the folded coordinate as

$$\overline{b}_t := \sum_{\substack{e\in S_1\\ \kappa(e)=t}} \tau_e(b_e) \in C_t,$$

and put

$$\operatorname{supp}(\overline{b}) := \{t \in T : \overline{b}_t \neq 0\}, \qquad K_0 := \{t \in \Gamma : \overline{b}_t = 0\}.$$

Then

$$\boxed{|S_0| - |\operatorname{supp}(\overline{b})| = |A_0| + \big(|S_1| - |\Gamma|\big) + |K_0|.}$$

The three terms record, respectively, annihilation by the local maps τ_e, collision of surviving lane addresses under κ, and cancellation among folded values at a common address. If every τ_e is injective, then $A_0 = \varnothing$. If globally typed selectors satisfy the strict separation law of (L8), then $K_0 = \varnothing$.

Proof. The first partition gives $|S_0| = |A_0| + |S_1|$. The definition of K_0 gives the disjoint partition

$$\Gamma = \operatorname{supp}(\overline{b}) \sqcup K_0,$$

so $|\Gamma| = |\operatorname{supp}(\overline{b})| + |K_0|$. Subtracting the second equality from the first proves the formula.

Injectivity of τ_e makes $\tau_e(b_e) \neq 0$ for every $e \in S_0$, so there is no annihilation. For strict selectors, suppose $t \in K_0$ and choose $e \in S_1$ with $\kappa(e) = t$. Applying Q_e to

$$j_t\overline{b}_t = \sum_{\kappa(e')=t} j_t\tau_{e'}(b_{e'}) = 0$$

leaves $j_t\tau_e(b_e) = 0$, contrary to $e \in S_1$. Hence $K_0 = \varnothing$. □

20.3 Certified change of resolution and exact loss sequences

For resolutions $s \succeq r$ on X and $v \succeq u$ on Y, let

$$\mathsf{B}^{v,s}(X,Y) := \bigoplus_{b \in I_{Y,v}} \bigoplus_{j \in I_{X,s}} \mathrm{Hom}_{\mathcal{C}}(X_j^s, Y_b^v),$$

and define $\mathsf{B}^{u,r}(X,Y)$ similarly. Analysis of a morphism gives an isomorphism

$$\mathfrak{a}_{X,Y}^{v,s} : \mathrm{Hom}_{\mathcal{C}}(X,Y) \xrightarrow{\sim} \mathsf{B}^{v,s}(X,Y),$$

whose inverse is lane synthesis.

Proposition 20.6 (Certified coarsening is an exact coordinate change)**.** *The change from fine to coarse block coordinates, canonical relative to the chosen splittings, is*

$$\Gamma_{(v,s)}^{(u,r)} := \mathfrak{a}_{X,Y}^{u,r} \big(\mathfrak{a}_{X,Y}^{v,s}\big)^{-1}.$$

For a fine matrix $h = (h_{bj})$, its (a,i)-entry is

$$\big(\Gamma_{(v,s)}^{(u,r)} h\big)_{ai} = \sum_{b,j} \big(\pi_{Y,a}^{u} \iota_{Y,b}^{v}\big) h_{bj} \big(\pi_{X,j}^{s} \iota_{X,i}^{r}\big).$$

Only descendants of a and i can contribute. The inverse is the corresponding fine analysis after coarse synthesis. Consequently, changing a complete lane resolution loses no information. Loss occurs only after a subsequent selector, quotient, summary, or fold is applied.

Proof. The displayed definition is a composite of two isomorphisms from theorem 20.3. Expanding coarse analysis after fine synthesis gives the entry formula. If b is not a descendant of a, then $p_{Y,a}^{u} p_{Y,b}^{v} = 0$, hence $\pi_{Y,a}^{u} \iota_{Y,b}^{v} = 0$. The source factor vanishes by the same argument when j is not a descendant of i. Reversing analysis and synthesis produces the inverse. □

Corollary 20.7 (Functorial coarsening)**.** *For $t \succeq s \succeq r$ and compatible target resolutions, the certified coordinate changes compose strictly. They also intertwine matrix multiplication. Thus a finite refinement diagram gives a genuine diagram of equivalent matrix-coordinate categories, not merely a family of numerically similar arrays.*

Theorem 20.8 (Abelian fold-loss exact sequence)**.** *Let $\mathcal{A}$ be an abelian category. Suppose a finite family of lane objects $(B_e)_{e \in E}$, target objects $(C_t)_{t \in T}$, and morphisms $\tau_e : B_e \to C_{\kappa(e)}$ is given. Let $q_e : \bigoplus_{e' \in E} B_{e'} \to B_e$ be the coordinate projection and define*

$$\Phi := \sum_{e \in E} j_{\kappa(e)} \tau_e q_e : \bigoplus_{e \in E} B_e \longrightarrow \bigoplus_{t \in T} C_t.$$

Then there are canonical short exact sequences

$$0 \longrightarrow \ker \Phi \longrightarrow \bigoplus_{e} B_e \longrightarrow \operatorname{im} \Phi \longrightarrow 0$$

and

$$0 \longrightarrow \operatorname{im} \Phi \longrightarrow \bigoplus_{t} C_t \longrightarrow \operatorname{coker} \Phi \longrightarrow 0.$$

The first sequence is the exact fold-loss certificate. Suppose every τ_e is monic and there are endomorphisms $Q_e \in \mathrm{End}(\bigoplus_t C_t)$ satisfying

$$Q_e j_{\kappa(e')} \tau_{e'} = \delta_{e,e'} j_{\kappa(e')} \tau_{e'} \quad \text{as morphisms } B_{e'} \longrightarrow \bigoplus_{t} C_t.$$

Then Φ is monic. If, in addition, there are morphisms

$$r_e : \bigoplus_t C_t \longrightarrow B_e, \qquad r_e j_{\kappa(e)} \tau_e = 1_{B_e},$$

then $L = (r_e Q_e)_e$ is a left inverse of Φ. Thus the first exact sequence splits, $\ker \Phi = 0$, and the image inclusion in the second exact sequence also splits.

Proof. The two sequences are the kernel–image and image–cokernel factorizations in an abelian category. The selector argument in (L8), interpreted internally in $\mathcal{A}$, makes Φ monic. The displayed morphism L is a left inverse by strict separation. Hence the first sequence splits; moreover, ΦL is an idempotent with image $\operatorname{im} \Phi$, so the image inclusion in the second sequence splits as well. □

Boundary 20.9 (Coordinate coarsening versus statistic coarsening). The complete fine and coarse block matrices are isomorphic coordinate records. Replacing each coarse block by a dimension, norm, valuation, degree, support bit, or color is a different map. Its kernel or fiber must be computed in the native doctrine. The lane theorem does not make such a statistic complete.

20.4 The matrix category carried by a lane system

Definition 20.10 (Lane matrix category at a fixed resolution)**.** Let $\mathcal{L}_{\mathsf{R}}$ be the category whose objects are objects of $\mathcal{C}$ equipped with full finite profiled lane systems over R, with all underlying $\mathcal{C}$-morphisms allowed. For $r \in \mathsf{R}$, define $\mathrm{Mat}_r(\mathcal{L}_{\mathsf{R}})$ to have the tagged finite families $[X; (X_i^r)_{i \in I_{X,r}}]$ as objects and rectangular matrices

$$(h_{ai}), \qquad h_{ai} : X_i^r \to Y_a^r,$$

as morphisms. Composition is matrix contraction and identities are diagonal identity matrices.

For $s \succeq r$, define

$$\Gamma_{sr} : \mathrm{Mat}_s(\mathcal{L}_{\mathsf{R}}) \longrightarrow \mathrm{Mat}_r(\mathcal{L}_{\mathsf{R}})$$

by replacing each object tag with its r-tag and sending a fine matrix to the certified source-and-target coordinate change of proposition 20.6.

Theorem 20.11 (Fixed-resolution matrix-category equivalence)**.** *The functor*

$$\mathfrak{M}_r : \mathcal{L}_{\mathsf{R}} \longrightarrow \mathrm{Mat}_r(\mathcal{L}_{\mathsf{R}}), \qquad X \longmapsto [X; (X_i^r)_i], \qquad f \longmapsto ([f]_{ai}^{r,r}),$$

is an equivalence. A quasi-inverse returns the lane-decorated object named by the tag and sends a matrix to

$$\sum_{a,i} \iota_{Y,a}^r h_{ai} \pi_{X,i}^r.$$

For every $s \succeq r$, the coordinate changes are functors and satisfy

$$\Gamma_{sr} \mathfrak{M}_s = \mathfrak{M}_r.$$

For $t \succeq s \succeq r$, they also satisfy

$$\Gamma_{tr} = \Gamma_{sr} \Gamma_{ts}.$$

Thus the equivalences are compatible with every certified change of resolution.

Proof. The unique lane matrix and composition clauses of theorem 20.3 show that $\mathfrak{M}_r$ is fully faithful. Every object of the displayed matrix category is, by definition, the lane family of an object of $\mathcal{L}_{\mathsf{R}}$, so it is essentially surjective. Analysis and synthesis furnish the unit and counit isomorphisms. Each Γ_{sr} is conjugation by exact analysis and synthesis, so it preserves identities and matrix composition. The two displayed identities follow from cancellation of adjacent analysis–synthesis pairs, equivalently from functorial coarsening in corollary 20.7. □

Remark 20.12 (No dimension claim). The number $|I_{X,r}|$ counts declared addresses. A lane object can have arbitrary dimension, rank, length, measure, or entropy, including zero where the ambient doctrine permits it. Matrix equivalence therefore does not identify address count with any of those invariants.

20.5 Multilinear and specified-polynomial transport

The additive theorem does not automatically govern arbitrary nonlinear maps. There is, however, an exact extension in the category of modules when the multilinear pieces are supplied as data.

Theorem 20.13 (Module multilinear lane expansion)**.** *Let $\Bbbk$ be a commutative ring. Let*

$$X^{(1)}, \dots, X^{(d)}, Y$$

be $\Bbbk$-modules with finite split lane systems, and let

$$T : X^{(1)} \times \cdots \times X^{(d)} \longrightarrow Y$$

be a specified $\Bbbk$-multilinear map. Define

$$T_{a;i_1,\dots,i_d} := \pi_{Y,a} T(\iota_{1,i_1} -, \dots, \iota_{d,i_d} -).$$

Then

$$T(x_1, \dots, x_d) = \sum_{a,i_1,\dots,i_d} \iota_{Y,a} T_{a;i_1,\dots,i_d}(\pi_{1,i_1} x_1, \dots, \pi_{d,i_d} x_d).$$

Substitution of multilinear maps is governed by finite contraction over every intermediate lane address.

Separately, let X, Y be $\Bbbk$-modules with finite split lane systems and suppose a polynomial map is presented with specified data

$$P(x) = T_0 + T_1(x) + T_2(x,x) + \cdots + T_D(x, \dots, x),$$

where

$$T_0 \in Y, \qquad T_n : X^n \longrightarrow Y \quad (1 \le n \le D)$$

and every T_n is a supplied multilinear map. Then

$$T_0 = \sum_a \iota_{Y,a} \pi_{Y,a}(T_0),$$

and adding this degree-zero decomposition to the displayed multilinear expansions gives an exact lane expansion of P.

Proof. Insert $x_\ell = \sum_{i_\ell} p_{\ell,i_\ell} x_\ell$ in every input, use multilinearity to distribute the finite sums, and insert $1_Y = \sum_a p_{Y,a}$ in the output. The coordinate formula follows from the chosen splittings. Repeating the calculation after substitution gives the contraction law. The degree-zero identity is output synthesis applied to T_0; the polynomial statement is then a finite sum of the specified homogeneous pieces. □

Boundary 20.14 (The nonlinear doctrine must be specified). Over a general commutative ring, a polynomial function need not determine a unique symmetric multilinear polarization. Infinite analytic expansions also require a declared topology and convergence theorem. Therefore theorem 20.13 is stated for supplied multilinear maps and supplied finite polynomial data. It does not assert a canonical lane calculus for every nonlinear category.

20.6 Factorial lanes and residue packets

The factorial construction is an exact arithmetic realization of the abstract lane mechanism. It is useful because the addresses, refinement maps, Fourier projectors, and hypergeometric coordinates can all be written explicitly.

For integers $N \geq 1$ and $n \geq 0$, define the step factorial

$$(a)_{n;N} := \prod_{k=0}^{n-1}(a + Nk), \qquad (a)_{0;N} := 1.$$

For a commutative ring A and a formal series $F(z) = \sum_{m\geq 0} a_m z^m \in A[[z]]$, define its residue lane

$$\mathcal{P}_{N,j}F(z) := \sum_{\substack{m\geq 0 \\ m\equiv j \pmod N}} a_m z^m, \qquad 0 \leq j < N.$$

Theorem 20.15 (Exact factorial–residue realization). *Let $N, c, n \in \mathbb{Z}$ with $N \geq 1$, $c \geq 0$, and $n \geq 0$. Then*

$$\boxed{\frac{(Nn+c)!}{c!} = \prod_{r=c+1}^{c+N}(r)_{n;N}.}$$

Each integer factor from $c+1$ through $c+Nn$ occurs in exactly one lane. For every commutative ring A, on the A-module $A[[z]]$ the operators $(\mathcal{P}_{N,j})_{0\leq j<N}$ satisfy

$$\mathcal{P}_{N,j}^2 = \mathcal{P}_{N,j}, \qquad \mathcal{P}_{N,j}\mathcal{P}_{N,k} = 0 \ (j \neq k), \qquad \sum_{j=0}^{N-1}\mathcal{P}_{N,j} = 1.$$

If $M \in \mathbb{Z}$ and $M \geq 1$, their exact refinement law is

$$\mathcal{P}_{N,j} = \sum_{\ell=0}^{M-1}\mathcal{P}_{NM,j+\ell N}.$$

Suppose further that K is a field, that N is invertible in K, and that K contains a primitive N-th root of unity ω. Then for every $F \in K[[z]]$,

$$\mathcal{P}_{N,j}F(z) = \frac{1}{N}\sum_{\ell=0}^{N-1}\omega^{-j\ell}F(\omega^\ell z).$$

If, in addition, K has characteristic zero, the exponential and hypergeometric series below are defined in $K[[z]]$. Put

$$E_{N,j}(z) := \mathcal{P}_{N,j}e^z = \sum_{n\geq 0}\frac{z^{Nn+j}}{(Nn+j)!}.$$

If

$$\mathbf{b}_{N,j} := \left\{\frac{j+r}{N} : 1 \leq r \leq N,\ r \neq N-j\right\},$$

then

$$E_{N,j}(z) = \frac{z^j}{j!}{}_0F_{N-1}\left(\begin{matrix}-\\ \mathbf{b}_{N,j}\end{matrix}; \frac{z^N}{N^N}\right).$$

Thus coefficient-residue, root-of-unity, and hypergeometric descriptions are exact additive coordinate realizations of one residue-address decomposition. The step-factorial identity is an exact multiplicative companion with the same address arithmetic; it is not asserted to be an additive idempotent decomposition of the factorial.

Proof. Every integer m with $c < m \le c + Nn$ has a unique representation $m = r + Nk$ with $c + 1 \le r \le c + N$ and $0 \le k < n$. Multiplying these disjoint factors proves the factorial identity.

The projector laws follow coefficient by coefficient. The residue classes modulo NM lying over $j \pmod N$ are precisely $j, j + N, \dots, j + (M-1)N$, which proves refinement. Character orthogonality gives the Fourier formula. Finally,

$$(Nn + j)! = j!\, N^{Nn} \prod_{r=1}^{N} \left(\frac{j+r}{N}\right)_n.$$

Exactly one factor in the product is $(1)_n = n!$. Removing it leaves the parameter set $\mathbf{b}_{N,j}$ and gives the hypergeometric series. □

Corollary 20.16 (Factorial-lane genealogy)**.** *The step-factorial split supplies an exact multiplicative partition of the factor set, while the residue projectors supply an exact additive partition of the coefficient series. The residue projectors instantiate theorem 20.3 directly. The step-factorial product is realized by the supplied finite multilinear pointwise multiplication map on the module of lane sequences, as in theorem 20.13. Their common address refinement is divisibility of the modulus. Hence the additive transport law and the multiplicative companion share exact address-refinement combinatorics, without identifying a product factorization with an additive idempotent splitting.*

20.7 Typed native specializations

The next statements identify concrete realizations and record the hypotheses that make each use of the theorem legal. They do not identify the surrounding strata with one another.

20.7.1 Fourier and polyphase lanes

Proposition 20.17 (Typed polyphase realization)**.** *Let K be a field, let $N \in \mathbb{Z}$ with $N \ge 1$ be invertible in K, and suppose K contains a primitive N-th root of unity. For $F(z) = \sum_{n\ge 0} a_n z^n \in K[[z]]$, set*

$$F_j(w) := \sum_{n\ge 0} a_{Nn+j} w^n.$$

Then

$$F(z) = \sum_{j=0}^{N-1} z^j F_j(z^N)$$

is exact analysis and synthesis. The Fourier residue projectors and the polyphase components satisfy the explicit coordinate relation

$$\mathcal{P}_{N,j} F(z) = z^j F_j(z^N).$$

For every integer $M \ge 1$, refinement from N to NM is the projector-sum law of theorem 20.15.

Proof. Every exponent has a unique quotient and remainder on division by N. The Fourier formula in theorem 20.15 identifies the same coefficient sections with character projectors. □

20.7.2 Cartier and fixed-radix lanes

Proposition 20.18 (Typed Cartier realization)**.** *Let K be a field and let $k \geq 2$. Work with K-valued formal series and fix the least-significant-digit-first convention. Define*

$$\Lambda_d\left(\sum_{n\geq 0} a_n z^n\right) := \sum_{n\geq 0} a_{kn+d} z^n, \qquad S_d(G) := z^d G(z^k), \quad 0 \leq d < k.$$

Then

$$\Lambda_d S_e = \delta_{de} 1, \qquad \sum_{d=0}^{k-1} S_d \Lambda_d = 1.$$

For a digit word $w = d_0 d_1 \cdots d_{a-1}$, the iterated Cartier lane is

$$\Lambda_{d_{a-1}} \cdots \Lambda_{d_1} \Lambda_{d_0} F.$$

It selects coefficients with index

$$k^a n + [w]_k, \qquad [w]_k = d_0 + k d_1 + \cdots + k^{a-1} d_{a-1}.$$

Concatenation gives strict radix refinement and the lane path law.

Proof. The first identities follow by coefficient extraction and unique quotient and remainder. Iterating in the stated reading convention gives the displayed address, and concatenation composes the affine index maps. □

20.7.3 Holonomic lanes

Proposition 20.19 (Typed holonomic realization)**.** *Let K be a characteristic-zero field and let the K-valued sequence (u_n) satisfy a linear recurrence with polynomial coefficients. Beyond a finite singular prefix, write the recurrence as a finite-dimensional rational nonautonomous state model*

$$v_{n+1} = A(n) v_n, \qquad A(n) \in \mathrm{Mat}_d(K(n)).$$

For a fixed integer $N \geq 1$ and residue $0 \leq j < N$, the lane state $w_m = v_{Nm+j}$ evolves by the exact N-step rational transfer

$$w_{m+1} = A(Nm + j + N - 1) \cdots A(Nm + j) w_m$$

whenever the displayed factors are defined. Equivalently, residue multisection of a D-finite generating series gives D-finite lanes. The finite singular prefix and any exceptional denominators remain explicit data.

Proof. Iterate the state equation N times. Closure of D-finite series under scaling the variable, finite sums, and root-of-unity multisection gives the equivalent generating-series statement after a field extension containing the needed roots. No autonomous finite-state claim is used. □

20.7.4 Local Newton–Mahler lanes

Proposition 20.20 (Typed local clopen realization)**.** *Fix a prime p and an integer $a \geq 0$. For $f \in C(\mathbb{Z}_p, \mathbb{Z}_p)$, with $q \in \mathbb{Z}_p$ the branch coordinate, define*

$$\mathcal{A}_{p^a}(f) := \big(q \mapsto f(p^a q + s)\big)_{0 \leq s < p^a}.$$

Piecewise synthesis on the disjoint clopen cosets $s + p^a\mathbb{Z}_p$ is its inverse. Refining from p^a to p^{a+b}, for every integer $b \geq 0$, gives the exact child-address and mixed-radix composition law. Let

$$\mathcal{M} := \mathcal{M}_p^{-1} : C(\mathbb{Z}_p, \mathbb{Z}_p) \xrightarrow{\sim} c_0(\mathbb{Z}_p)$$

be Mahler analysis from theorem 14.1, and write $R_{p^a,s} f(q) := f(p^a q + s)$. Conjugation gives the exact coefficient operator

$$\mathcal{T}_{p^a,s} := \mathcal{M} R_{p^a,s} \mathcal{M}^{-1} : c_0(\mathbb{Z}_p) \longrightarrow c_0(\mathbb{Z}_p),$$

whose affine kernel is computed in theorem 14.5. Its mixed-radix composition is theorem 14.7.

The assertion of complete clopen analysis is restricted to these prime-power packets. More general affine coefficient kernels may still compose according to their declared mixed-radix law, but an arbitrary integer modulus does not automatically partition $\mathbb{Z}_p$ into that many disjoint clopen branches.

Proof. The p^a cosets are disjoint, clopen, and cover $\mathbb{Z}_p$. Restricting after piecewise synthesis and synthesizing after restriction are identities. Writing a child address as $s + p^a t$ proves refinement. The cited Mahler isomorphism makes the displayed conjugation exact, and the two cited kernel theorems give its coefficient formula and composition law. □

20.7.5 Nonlinear tree lanes

Proposition 20.21 (Typed Fuss–Catalan realization)**.** *Let K be a characteristic-zero field, let $q \in \mathbb{Z}$ with $q \geq 2$, and let $\alpha \in K^\times$. The uniquely marked formal solution of*

$$U = z + \alpha U^q, \qquad U(0) = 0,$$

has support

$$[z^{1+(q-1)n}]U = \frac{1}{(q-1)n+1}\binom{qn}{n}\alpha^n \neq 0 \qquad (n \geq 0),$$

and every other coefficient is zero. Thus its support is exactly the set of degrees congruent to $1 \pmod{q-1}$. For an integer packet width $N \geq 1$, set $g = \gcd(N, q-1)$. Exactly N/g residue addresses are active, namely the congruence class $1 \pmod{g}$. Packetizing the specified multilinear map $(U_1, \ldots, U_q) \mapsto U_1 \cdots U_q$ gives the exact finite carry-convolution system supplied by theorem 20.13. On a cyclic orbit of order $q-1$, the power map used for packet synthesis has

$$\frac{q-1}{\gcd(N, q-1)}$$

distinct images. If a fixed embedding $K \hookrightarrow \mathbb{C}$ sends α to a positive real number and the marked branch is interpreted in the smooth implicit analytic schema, its critical points and values satisfy

$$q\alpha\tau^{q-1} = 1, \qquad \rho = \tau - \alpha\tau^q = \frac{q-1}{q}\tau.$$

The resulting orbit $\rho\mu_{q-1}$ is precisely the dominant singularity orbit, and packetization leaves $(q-1)/g$ distinct images.

Proof. Lagrange inversion gives the displayed coefficient formula and zero at every other degree. Characteristic zero and $\alpha \neq 0$ make every displayed coefficient nonzero. Their residues modulo N form the coset of the subgroup generated by $q-1$, whose size is N/g. Multilinear lane expansion of the

product gives the packet equation, including the power of the packet variable produced by the quotient in residue addition. On the cyclic group of order $q-1$, the kernel of multiplication by N has order g, so the image has order $(q-1)/g$. The critical equations give the displayed values, while theorem 17.13 identifies their full orbit as the dominant one under the stated analytic hypotheses. □

20.7.6 Algebraic branch lanes

Proposition 20.22 (Typed cyclic-radical realization)**.** *Let K be a characteristic-zero field containing μ_d and a primitive N-th root of unity ω, let $d, m, N \in \mathbb{Z}_{\geq 1}$, and choose the formal branch*

$$F(z) = (1 - \alpha z^m)^{-1/d}, \qquad \alpha \in K^\times.$$

With $g = \gcd(N, m)$, its nonzero coefficients occupy exactly the subgroup of $\mathbb{Z}/N\mathbb{Z}$ generated by m, which has N/g addresses. The rotated radicands

$$1 - \alpha\omega^{jm} z^m, \qquad 0 \leq j < N,$$

also have exactly N/g distinct values.

If, in addition, their distinct Kummer classes generate a subgroup isomorphic to $(\mathbb{Z}/d\mathbb{Z})^{N/g}$ in the Kummer quotient

$$K(z)^\times / K(z)^{\times d},$$

then the compositum of the corresponding radical branch fields has degree

$$d^{N/g}$$

over $K(z)$. Without that independence hypothesis, the lane orbit count remains exact but the degree formula is only an upper bound.

Proof. The binomial series is supported at exponents mn. The cyclic subgroup calculation gives both address and rotation counts. The degree statement is the standard Kummer independence calculation for the explicitly listed classes. A relation among the classes produces a field collision not detected by address count alone. □

20.7.7 Finite Galois lanes

Proposition 20.23 (Typed semisimple Galois realization)**.** *Let G be finite and let K be a characteristic-zero splitting field for G. The primitive central idempotents*

$$e_\chi = \frac{\chi(1)}{|G|} \sum_{g \in G} \chi(g^{-1}) g \in K[G]$$

for $\chi \in \mathrm{Irr}_K(G)$ form a complete pairwise-annihilating lane system on every finite-dimensional semisimple $K[G]$-module V:

$$V = \bigoplus_{\chi \in \mathrm{Irr}_K(G)} e_\chi V.$$

Every G-equivariant map is block diagonal in these isotypic lanes, while an arbitrary K-linear map has the full lane matrix of theorem 20.3.

For a subgroup $H \leq G$, the Reynolds operator

$$e_H = \frac{1}{|H|} \sum_{h \in H} h$$

is an idempotent projector onto V^H; equivalently, $(e_H, 1 - e_H)$ is a two-lane system on the underlying K-vector space. It is not in general a Boolean selector formed from the isotypic projectors above. Projectors belonging to unrelated subgroups need not annihilate one another, so they do not automatically form one lane system. Transport through induction, restriction, or subgroup refinement requires the corresponding Mackey or double-coset data.

Proof. Maschke semisimplicity and the character idempotent relations give the first decomposition. An equivariant map commutes with the center of $K[G]$, hence with every e_χ. Averaging proves that e_H is idempotent with image V^H. Two subgroup averages generally fail the pairwise-annihilation axiom, which proves the stated boundary. □

20.8 Descent of lane systems through a finite atlas

The local theorem becomes global only when the addresses and splittings glue without address holonomy. The following statement makes those additional hypotheses explicit.

Theorem 20.24 (Finite atlas descent for lane transport)**.** *Let $\mathfrak{R}$ be a finite profiled atlas whose participating chart categories are additive and idempotent-complete, with additive transition functors $F_{UV} : \mathfrak{R}(U) \to \mathfrak{R}(V)$ on its declared overlaps. Let*

$$X = (X_U, \theta^X_{UV}), \qquad Y = (Y_U, \theta^Y_{UV}), \qquad Z = (Z_U, \theta^Z_{UV})$$

be three compatible descent objects. Thus the transition isomorphisms satisfy the identity and cocycle equations. Let

$$f = (f_U) : X \longrightarrow Y, \qquad g = (g_U) : Y \longrightarrow Z$$

be compatible descended morphisms.

Fix one finite resolution shape R*. Suppose every local object A_U, for $A \in \{X, Y, Z\}$, carries a finite profiled lane system over* R*. For every overlap, every $r \in$ R*, and every such A, suppose there is an address bijection*

$$\lambda^{A,r}_{UV} : I_{A_U,r} \xrightarrow{\sim} I_{A_V,r}$$

such that

$$\lambda^{A,r}_{UU} = 1, \qquad \lambda^{A,r}_{UT} = \lambda^{A,r}_{VT} \lambda^{A,r}_{UV},$$

on every triple overlap (U, V, T), and, for $s \succeq r$,

$$\rho^{A_V}_{sr} \lambda^{A,s}_{UV} = \lambda^{A,r}_{UV} \rho^{A_U}_{sr}.$$

Require trivial address holonomy: for every A and r, there is a global finite address set $I_{A,r}$, a global refinement map $\rho^A_{sr} : I_{A,s} \twoheadrightarrow I_{A,r}$, and bijections

$$\mu^{A,r}_U : I_{A_U,r} \xrightarrow{\sim} I_{A,r}$$

such that

$$\mu^{A,r}_V \lambda^{A,r}_{UV} = \mu^{A,r}_U, \qquad \mu^{A,r}_U \rho^{A_U}_{sr} = \rho^A_{sr} \mu^{A,s}_U.$$

Assume also that the transition isomorphisms intertwine the local projectors:

$$\theta^A_{UV} F_{UV}(p^r_{A_U,i}) = p^r_{A_V, \lambda^{A,r}_{UV}(i)} \theta^A_{UV},$$

and that the verified transition profiles preserve the chosen splittings, finite sums, composition, and refinement equations, with coherent splitting comparisons on triple overlaps.

Then:

(D1) *after relabeling by the $\mu_U^{A,r}$, each local lane and its chosen splitting define a lane summand in* $\mathrm{Desc}(\mathfrak{R})$*, and the global addresses carry the declared projector-sum refinement;*

(D2) *the local matrices $([f_U]_{ai})$, $([g_U]_{ca})$, and $([g_U f_U]_{ci})$ glue after the prescribed address relabelings and splitting comparisons;*

(D3) *the descended matrices satisfy*

$$[gf]_{ci} = \sum_a [g]_{ca}[f]_{ai};$$

(D4) *every compatible family of support selectors and named residuals descends, and every compatible refinement square remains a projector-sum refinement square in the descent category;*

(D5) *every fold whose source objects, target objects, and structure maps are compatible descent data defines a fold in* $\mathrm{Desc}(\mathfrak{R})$*. Compatible selectors and left inverses descend, so a split-losslessness certificate descends directly from its left-inverse identity. Moreover, suppose the relevant local kernels $K_U = \ker(\Phi_U)$ exist, every transition functor preserves them, and there are coherent comparison isomorphisms*

$$\eta_{UV} : F_{UV}(K_U) \xrightarrow{\sim} K_V$$

compatible with the kernel inclusions and satisfying the triple-overlap cocycle. Then the global kernel exists and is computed chartwise:

$$\big(\ker_{\mathrm{Desc}(\mathfrak{R})} \Phi\big)_U \cong \ker_{\mathfrak{R}(U)}(\Phi_U).$$

Proof. For a global address $i \in I_{A,r}$, use the local projector indexed by $(\mu_U^{A,r})^{-1}(i)$. The trivialization identity makes these projectors correspond under the transition isomorphisms on every overlap; its refinement identity makes their local projector-sum laws agree. The coherent splitting comparisons make the corresponding inclusions and projections descent morphisms. Finite biproducts in the descent category are computed chartwise, so local analysis and synthesis glue and remain mutually inverse.

Compatibility of f and g, followed by the splitting comparisons, shows that their local coordinate blocks agree on overlaps after applying the specified address relabelings. Hence the matrices descend. Matrix multiplication is a finite sum of compositions, both computed chartwise, so the local path law glues to the displayed global law. The same argument applies to projector sums, selectors, and residual differences.

Finally, a compatible fold is a morphism in the relevant descent category. Compatible selectors and left inverses glue, and their identities can be checked chartwise. Under the additional kernel hypotheses, the objects (K_U, η_{UV}) form a descent object; compatibility with the local kernel inclusions gives a global morphism into the fold source. Its universal property is checked in each chart, which proves the displayed chartwise kernel identity. The split case needs no kernel-preservation hypothesis: the compatible local left inverses glue directly. □

Corollary 20.25 (Certified atlas lane transport)**.** *Let $\mathfrak{A}_{\mathrm{cert}}$ be the certified finite atlas of this manuscript. On every subatlas whose passports include additive split-lane systems, compatible refinement maps, coherent address relabelings with trivial address holonomy, and the required preservation profile, the Universal Lane Transport Theorem holds in the descent category. Kernel claims additionally require the kernel-preservation and coherence data in theorem 20.24. In particular, any of the typed specializations above may be transported across a certified reversible overlap after its native hypotheses, address trivializations, and splitting cocycles have been checked.*

Proof. Apply theorem 20.24 to the verified subatlas. The conclusion is deliberately restricted to the overlaps whose passports contain all of the listed data. □

Boundary 20.26 (Descent is conditional on address coherence). An equivalence of underlying semantic objects does not by itself identify two lane systems. Without the bijections $\lambda_{UV}^{A,r}$, their cocycle law, compatible global trivializations $\mu_U^{A,r}$, refinement compatibility, and coherent splitting comparisons, the local matrices need not glue as ordinary globally addressed lanes. Nontrivial address holonomy can permute lanes around a loop; then only invariant orbit sums, or lanes in a separately defined local-system-valued category, may descend. The theorem above verifies ordinary descent under trivial holonomy. It does not manufacture missing lane passports for an arbitrary heterogeneous atlas.

20.9 Sharp boundaries and a finite counterexample

Proposition 20.27 (Endpoint selection does not implement refinement)**.** *Let K be a field and let $X = K^2$. At the coarse resolution use the single projector $p = 1_X$. At the fine resolution use*

$$p_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad p_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad p = p_1 + p_2.$$

For

$$f = p_2,$$

the unique coarse block is $pfp = f \neq 0$, whereas selecting the single fine endpoint (p_1, p_1) gives $p_1 f p_1 = 0$. Therefore a coarse block is the sum over all compatible descendant source-target pairs. It cannot in general be represented by choosing one fine endpoint.

Proof. The displayed matrix multiplication gives the two values immediately. The full refinement identity recovers $f = \sum_{a,i \in \{1,2\}} p_a f p_i$, whose only nonzero term here is $p_2 f p_2$. □

Boundary 20.28 (Finite profiled meaning of "universal"). The theorem proves a common calculus for finite split lanes in additive categories and for the explicitly stated profile extensions. It does not assert any of the following:

1. that every mathematical object has a canonical or minimal lane system;
2. that countably infinite lane sums converge without a topology and a convergence theorem;
3. that arbitrary nonlinear, probabilistic, operator-algebraic, or homotopical constructions preserve additive lane identities;
4. that address count equals vector-space dimension, module rank, information, entropy, algebraic degree, or computational complexity;
5. that a support pattern, heatmap, norm profile, valuation profile, or orbit count is a complete invariant;
6. that native invariants in different strata are equal merely because their formulas contain the same gcd or residue operation;
7. that the theorem supplies historical novelty evidence or external validation.

Each stronger conclusion requires a native theorem and a preservation certificate.

Remark 20.29 (Classical core and present synthesis). Finite biproduct coordinates, Peirce decompositions, block-matrix multiplication, residue filters, and polyphase identities have classical antecedents. The theorem packages that core with explicit address refinement, profile-controlled realization, named residuals, exact fold kernels, and typed descent. This manuscript treats the resulting assembly as an internal mathematical contribution. It does not infer a priority claim without a separate literature study.

20.10 Exact finite validation and interpretation

The accompanying verifier is an exact finite regression suite for selected implementation clauses. It checks integer and rational block reconstruction, matrix composition, refinement aggregation, support and residual identities, fold collision kernels, factorial-lane partitions, residue-projector refinement, and explicit negative controls. The proofs above do not depend on finite sampling. Conversely, the symbolic proofs do not certify that a particular program implements the formulas correctly. The two records serve different purposes.

Table 41: Exact finite regression for selected implementation clauses of the Universal Lane Transport theorem. This table is not an exhaustive proof or an external validation.

Validation family	Exact assertions
Integer/unimodular lane matrices	28,490
Dense rational three-level refinement	16,674
Abstract additive folds and strict selectors	1,000
Actual-support loss decomposition	300
Factorial and residue-lane realization	150,630
Deliberately invalid negative controls	5
Total selected implementation assertions	**197,099**
Recorded failures	0

Universal Lane Transport result. Finite split lane systems admit exact analysis and synthesis, unique transfer matrices, path composition, strict projector-sum refinement, Boolean support selectors, named residuals, and exact fold-loss kernels. Complete changes of resolution are coordinate equivalences. In abelian targets, fold loss sits in a canonical short exact sequence. In module targets, specified multilinear and finite polynomial operations expand lane by lane. Residue/Fourier, polyphase, Cartier, holonomic, local Newton–Mahler, Fuss–Catalan, cyclic radical, and semisimple Galois systems realize these clauses under their printed hypotheses. The step-factorial product is an exact multiplicative companion transported through a specified multilinear multiplication map, not an additive idempotent splitting. Compatible address trivializations, refinement, and splitting data make the calculus descend through a finite certified atlas when address holonomy is trivial. No claim is made beyond that finite profiled domain.

21 Holonomy-twisted lane transport and orbit descent

The finite atlas theorem in theorem 20.24 imposed trivial address holonomy in order to obtain one globally fixed set of lane labels. That hypothesis is sufficient for ordinary globally addressed lanes, but it is stronger than the hypothesis needed for transport itself. When addresses are permuted around atlas loops, the local lane matrices still glue. They glue as sections of a finite matrix local system rather than as one matrix with a fixed global row and column labeling.

This section isolates that distinction. The local analysis, synthesis, composition, refinement, selector, and residual laws survive arbitrary finite address holonomy. The holonomy orbits then give the finest ordinary global coarsening obtainable by summing the original lane projectors. Thus nontrivial holonomy is an obstruction to untwisted labels, not an obstruction to the finite transport calculus.

21.1 Holonomy-profiled lane data

Let $\mathscr{G}$ be a finite connected groupoid of charts. For an arrow $\gamma : U \to V$, let

$$F_\gamma : \mathcal{C}_U \longrightarrow \mathcal{C}_V$$

be an additive equivalence between additive idempotent-complete categories. The transition functors carry specified identity and composition comparisons. To keep the formulas readable, those comparison isomorphisms are suppressed; every displayed cocycle equation is understood after inserting the declared comparisons.

A descent object

$$A = (A_U, \theta^A_\gamma)$$

consists of objects $A_U \in \mathcal{C}_U$ and isomorphisms

$$\theta^A_\gamma : F_\gamma(A_U) \xrightarrow{\sim} A_V$$

satisfying the identity and cocycle equations. A descended morphism $f : X \to Y$ is a compatible family $(f_U)_U$ satisfying

$$\theta^Y_\gamma F_\gamma(f_U) = f_V \theta^X_\gamma.$$

Definition 21.1 (Holonomy-profiled finite lane datum)**.** Fix a finite resolution shape R. A holonomy-profiled finite lane datum on a descent object A consists of the following information.

For every chart U and resolution $r \in \mathsf{R}$, the object A_U has a finite profiled lane system

$$\left(I_{A,U,r}, p^r_{A,U,i}, A^r_{U,i}, \iota^r_{A,U,i}, \pi^r_{A,U,i}\right)_{i \in I_{A,U,r}}$$

in the sense of definition 20.2. For every arrow $\gamma : U \to V$, there is an address bijection

$$\lambda^A_{\gamma,r} : I_{A,U,r} \xrightarrow{\sim} I_{A,V,r}$$

and, for every address i, a splitting comparison

$$\beta^A_{\gamma,r,i} : F_\gamma(A^r_{U,i}) \xrightarrow{\sim} A^r_{V,\lambda^A_{\gamma,r}(i)}.$$

They satisfy the address and splitting cocycle laws and the two compatibility identities

$$\theta^A_\gamma F_\gamma(\iota^r_{A,U,i}) = \iota^r_{A,V,\lambda^A_{\gamma,r}(i)} \beta^A_{\gamma,r,i},$$

$$\beta^{A}_{\gamma,r,i} F_\gamma(\pi^{r}_{A,U,i}) = \pi^{r}_{A,V,\lambda^{A}_{\gamma,r}(i)} \theta^{A}_{\gamma}.$$

For $s \succeq r$, the refinement maps are equivariant:

$$\rho^{A,V}_{sr} \lambda^{A}_{\gamma,s} = \lambda^{A}_{\gamma,r} \rho^{A,U}_{sr}.$$

No global address set and no global trivialization of the address bijections is assumed.

Definition 21.2 (Matrix local system)**.** Let X, Y carry holonomy-profiled lane data. At source resolution r and target resolution u, define the matrix group over a chart U by

$$\mathscr{M}^{u,r}_{Y,X}(U) := \bigoplus_{a \in I_{Y,U,u}} \bigoplus_{i \in I_{X,U,r}} \mathrm{Hom}_{\mathcal{C}_U}(X^{r}_{U,i}, Y^{u}_{U,a}).$$

For $h = (h_{ai}) \in \mathscr{M}^{u,r}_{Y,X}(U)$, define transport along $\gamma : U \to V$ by

$$(\gamma_* h)_{\lambda^{Y}_{\gamma,u}(a), \lambda^{X}_{\gamma,r}(i)} := \beta^{Y}_{\gamma,u,a} F_\gamma(h_{ai}) \left(\beta^{X}_{\gamma,r,i}\right)^{-1}.$$

The group of compatible sections is

$$\Gamma\left(\mathscr{G}, \mathscr{M}^{u,r}_{Y,X}\right) := \left\{ (h_U)_U : \gamma_* h_U = h_V \text{ for every } \gamma : U \to V \right\}.$$

21.2 The holonomy-twisted transport theorem

Theorem 21.3 (Holonomy-Twisted Universal Lane Transport Theorem)**.** *Let X, Y, Z be descent objects over $\mathscr{G}$, each equipped with a holonomy-profiled finite lane datum over the same finite resolution shape. Then the following statements hold.*

(H1) ***Matrix local system.*** *The maps γ_* of definition 21.2 are additive isomorphisms and satisfy the identity and cocycle laws. Hence $\mathscr{M}^{u,r}_{Y,X}$ is a local system of abelian groups with finitely many matrix positions in every chart.*

(H2) ***Exact twisted analysis and synthesis.*** *Lane analysis induces an isomorphism*

$$\mathfrak{A}^{u,r}_{Y,X} : \mathrm{Hom}_{\mathrm{Desc}(\mathscr{G})}(X, Y) \xrightarrow{\sim} \Gamma\left(\mathscr{G}, \mathscr{M}^{u,r}_{Y,X}\right), \qquad f \longmapsto \left(\left([f_U]^{u,r}_{ai} \right)_{a,i} \right)_U.$$

Its inverse is chartwise lane synthesis:

$$(h_U)_U \longmapsto \left(\sum_{a,i} \iota^{u}_{Y,U,a} h_{U,ai} \pi^{r}_{X,U,i} \right)_U.$$

Thus nontrivial address holonomy does not destroy the complete matrix record. It changes the global type of that record from a constantly labeled matrix to a compatible section of a matrix local system.

(H3) ***Twisted matrix composition.*** *For*

$$h \in \Gamma(\mathscr{G}, \mathscr{M}^{u,r}_{Y,X}), \qquad k \in \Gamma(\mathscr{G}, \mathscr{M}^{v,u}_{Z,Y}),$$

define chartwise contraction by

$$(k \star h)_{U,ci} := \sum_{a \in I_{Y,U,u}} k_{U,ca} h_{U,ai}.$$

Then $k \star h$ is a compatible section of $\mathscr{M}^{v,r}_{Z,X}$, and

$$\mathfrak{A}^{v,r}_{Z,X}(gf) = \mathfrak{A}^{v,u}_{Z,Y}(g) \star \mathfrak{A}^{u,r}_{Y,X}(f).$$

The diagonal identity matrices are compatible sections. More precisely, fix one declared resolution r_A for each lane-decorated descent object A. The category with the same objects, morphism groups

$$\Gamma\left(\mathscr{G}, \mathscr{M}^{r_B,r_A}_{B,A}\right),$$

and composition $\star$ is isomorphic, by analysis and synthesis, to the corresponding category of lane-decorated descent objects and their underlying descended morphisms.

(H4) ***Equivariant refinement.*** *Every certified fine-to-coarse coordinate change from proposition 20.6 commutes with the transition maps γ_*. It therefore defines an isomorphism of matrix local systems. These isomorphisms compose strictly over chains of declared resolutions and intertwine twisted matrix contraction.*

(H5) ***Holonomy-compatible selectors and residuals.*** *Let*

$$E_U \subseteq I_{Y,U,u} \times I_{X,U,r}$$

be a family satisfying

$$E_V = (\lambda^Y_{\gamma,u} \times \lambda^X_{\gamma,r})(E_U)$$

for every arrow $\gamma : U \to V$. Then the local selectors

$$C_{E_U}(h_U)_{ai} = \begin{cases} h_{U,ai}, & (a,i) \in E_U, \\ 0, & (a,i) \notin E_U \end{cases}$$

assemble to an idempotent endomorphism C_E of the section group. For two compatible support families E, F,

$$C_E C_F = C_{E \cap F}, \qquad R_E := 1 - C_E, \qquad h = C_E h + R_E h.$$

Hence named residuals remain exact under arbitrary finite address holonomy, provided the retained support itself is holonomy compatible.

Proof. For (H1), additivity and invertibility follow from the corresponding properties of F_γ and the splitting comparisons β. The address and splitting cocycle laws give $(\delta\gamma)_* = \delta_*\gamma_*$ after the declared transition comparisons.

For (H2), let $f = (f_U)_U$ be descended. For a transition $\gamma : U \to V$, the transported (a,i)-block is

$$\begin{aligned} &\beta^Y_{\gamma,u,a} F_\gamma\left(\pi^u_{Y,U,a} f_U \iota^r_{X,U,i}\right) \left(\beta^X_{\gamma,r,i}\right)^{-1} \\ &\quad = \pi^u_{Y,V,\lambda^Y(a)} \theta^Y_\gamma F_\gamma(f_U) (\theta^X_\gamma)^{-1} \iota^r_{X,V,\lambda^X(i)} \\ &\quad = \pi^u_{Y,V,\lambda^Y(a)} f_V \iota^r_{X,V,\lambda^X(i)}, \end{aligned}$$

which is the corresponding block of f_V. Thus analysis produces a compatible section.

Conversely, let $h = (h_U)_U$ be a compatible section and synthesize

$$f_U := \sum_{a,i} \iota^u_{Y,U,a} h_{U,ai} \pi^r_{X,U,i}.$$

Transporting this sum, applying the splitting comparison identities, and reindexing through the address bijections gives f_V. Thus $(f_U)_U$ is a descended morphism. Chartwise analysis and synthesis are mutually inverse by theorem 20.3; therefore their restrictions to compatible families are mutually inverse.

For (H3), the local path law gives the displayed contraction formula in every chart. Applying γ_* to a contracted block distributes over the finite sum and carries each product to the product of the transported blocks. Since $\lambda^Y_{\gamma,u}$ is a bijection, reindexing the intermediate address shows

$$\gamma_*(k_U \star h_U) = (\gamma_* k_U) \star (\gamma_* h_U) = k_V \star h_V.$$

The identity section is compatible by the splitting cocycle. The equivalence then follows from (H2).

For (H4), the local fine-to-coarse coordinate map is analysis after synthesis. Both operations commute with transition by (H2), so their composite does as well. The strict composition and contraction identities are the corresponding chartwise identities from corollary 20.7 and theorem 20.11.

For (H5), support compatibility implies $\gamma_* C_{E_U} = C_{E_V} \gamma_*$, so the selectors preserve compatible sections. Every Boolean and residual identity is then checked chartwise. □

Corollary 21.4 (Base-fiber invariant normal form)**.** *Fix a base chart U_0. Evaluation at U_0 induces an isomorphism*

$$\Gamma\left(\mathscr{G}, \mathscr{M}^{u,r}_{Y,X}\right) \xrightarrow{\sim} \mathscr{M}^{u,r}_{Y,X}(U_0)^{\mathrm{Aut}_{\mathscr{G}}(U_0)},$$

where the right-hand side is the subgroup fixed by every loop-transport map. Consequently,

$$\mathrm{Hom}_{\mathrm{Desc}(\mathscr{G})}(X, Y) \cong \mathscr{M}^{u,r}_{Y,X}(U_0)^{\mathrm{Aut}_{\mathscr{G}}(U_0)}.$$

After choosing one arrow $\tau_U : U_0 \to U$ for each chart, the inverse sends a fixed base matrix h_0 to the section $h_U = (\tau_U)_* h_0$. The result is independent of the chosen arrows because two choices differ by loop transport at U_0.

Proof. A compatible section evaluates to a loop-fixed element at U_0. Conversely, for an arrow $\gamma : U \to V$, the two arrows $\gamma\tau_U$ and τ_V differ by a loop at U_0; loop invariance therefore gives $\gamma_* h_U = h_V$. Evaluation and this reconstruction are mutually inverse. Compose with the analysis isomorphism in (H2). □

Remark 21.5 (What has and has not been removed). The theorem removes only the global address trivializations $\mu^{A,r}_U$ from theorem 20.24. It does not remove the address cocycle, refinement naturality, projector compatibility, splitting comparisons, or additive preservation hypotheses. Without those data, even the matrix transition maps are undefined or fail their cocycle law.

21.3 Monodromy orbits and the finest untwisted coarsening

Fix a base chart U_0. Every loop $\gamma \in \mathrm{Aut}_{\mathscr{G}}(U_0)$ acts on an address set by $i \mapsto \lambda^A_{\gamma,r}(i)$. Write

$$H^r_A := \mathrm{im}(\mathrm{Aut}_{\mathscr{G}}(U_0) \longrightarrow \mathrm{Sym}(I_{A,U_0,r}))$$

for the resulting finite monodromy group. For a source-target pair, the same loop acts diagonally by

$$(a, i) \longmapsto \left(\lambda^Y_{\gamma,u}(a), \lambda^X_{\gamma,r}(i)\right).$$

Its image in the symmetric group of $I_{Y,U_0,u} \times I_{X,U_0,r}$ is denoted $H^{u,r}_{Y,X}$. This is the simultaneous monodromy image; it need not be the direct product of the two separate monodromy groups.

Theorem 21.6 (Orbit descent and the finest globally labeled coarsening)**.** *Under the hypotheses of theorem 21.3, the following statements hold.*

(O1) *For every orbit $\mathcal{O} \subseteq I_{A,U_0,r}$ of H^r_A, path transport defines a path-independent orbit subset $\mathcal{O}_U \subseteq I_{A,U,r}$ in every chart. The local projector sums*

$$P^r_{A,U,\mathcal{O}} := \sum_{i \in \mathcal{O}_U} p^r_{A,U,i}$$

descend to a split idempotent on A. As $\mathcal{O}$ ranges over the monodromy orbits, these projectors are pairwise annihilating and sum to 1_A.

(O2) *The orbit projectors form the finest ordinary globally labeled coarsening whose local projectors are sums of the original lane projectors. More precisely, any globally labeled projector of the form*

$$Q_U = \sum_{i \in S_U} p^r_{A,U,i}$$

with $S_V = \lambda^A_{\gamma,r}(S_U)$ and with the same global label preserved around every loop has S_{U_0} equal to a union of H^r_A-orbits.

(O3) *A particular local lane descends as an individually labeled global lane exactly when its address is fixed by H^r_A. All original lane labels descend simultaneously exactly when the monodromy action on the address set is trivial.*

(O4) *Holonomy-compatible support families in (H5) are in bijection with unions of $H^{u,r}_{Y,X}$-orbits in the base pair-address set. The minimal nonempty compatible supports are therefore the single pair orbits. Their associated selectors need not act nontrivially on the section group: a single-orbit selector is nonzero exactly when that orbit supports a nonzero compatible section.*

(O5) *If $s \succeq r$, equivariance of the refinement map sends every fine monodromy orbit onto a coarse monodromy orbit. For a coarse orbit $\mathcal{O}$,*

$$P^r_{A,\mathcal{O}} = \sum_{\substack{\mathcal{Q} \text{ a fine orbit} \\ \mathcal{Q} \subseteq (\rho^A_{sr})^{-1}(\mathcal{O})}} P^s_{A,\mathcal{Q}}.$$

Thus orbit descent and projector-sum refinement commute exactly.

Proof. Choose a path from U_0 to U and transport $\mathcal{O}$ along its address bijection. Two path choices differ by a loop at U_0, which preserves $\mathcal{O}$ setwise. Hence $\mathcal{O}_U$ is path independent. Projector compatibility gives

$$\theta^A_\gamma F_\gamma(P^r_{A,U,\mathcal{O}}) = P^r_{A,V,\mathcal{O}} \theta^A_\gamma,$$

so the orbit sums descend. Their local splittings are the finite direct sums $\bigoplus_{i \in \mathcal{O}_U} A^r_{U,i}$; the direct sums of the β-maps, with the induced permutation of summands, provide their descent data. This proves (O1).

For (O2), a globally fixed label must return to itself after every loop, so S_{U_0} is invariant under H^r_A. Every invariant subset of a finite group action is a union of orbits. Therefore every such globally labeled coarsening is coarser than the orbit partition. Statement (O3) is the singleton case.

The same path-independence argument on the pair-address set shows that a compatible support family is determined by an invariant base subset, and every invariant base subset is a union of pair orbits. A selector acts as the identity on compatible sections supported in its orbit and vanishes on the complementary support. It is therefore nonzero precisely when the orbit supports a nonzero compatible section. This proves (O4).

Finally, equivariance of ρ_{sr}^{A} implies that the image of a fine orbit is the orbit of the image of any one of its points. Hence the preimage of a coarse orbit is a union of fine orbits. Summing the local projector-refinement identity over all addresses in the coarse orbit gives the formula in (O5). □

Remark 21.7 (Object orbits and matrix orbits differ). The orbit lanes on an object are controlled by $H_A^r \backslash I_{A,r}$. Transport selectors are controlled by the generally finer orbit set

$$H_{Y,X}^{u,r} \backslash (I_{Y,u} \times I_{X,r}).$$

Even when an object has only one transitive address orbit, its endomorphism matrices can have several pair orbits. Holonomy can therefore erase individual object labels while retaining a nontrivial global transport algebra.

21.4 Permutation modules and the orbital path algebra

The one-chart groupoid with automorphism group H gives a concrete exact model. It also identifies the algebra carried by the pair-orbit selectors.

Theorem 21.8 (Permutation-module orbital lane calculus)**.** *Let $\Bbbk$ be a commutative ring, let a finite group H act on finite sets I, J, K, and let $\Bbbk[I], \Bbbk[J], \Bbbk[K]$ be the corresponding permutation modules. For an orbit $\mathcal{O} \subseteq J \times I$ under the diagonal action, let $B_{\mathcal{O}} : \Bbbk[I] \to \Bbbk[J]$ be the matrix whose (j,i)-entry is one when $(j,i) \in \mathcal{O}$ and zero otherwise. Then*

$$\mathrm{Hom}_{\Bbbk H}(\Bbbk[I], \Bbbk[J]) = \bigoplus_{\mathcal{O} \in H\backslash(J\times I)} \Bbbk B_{\mathcal{O}}.$$

In particular, an equivariant matrix is exactly a matrix constant on every pair orbit.

Let

$$\mathcal{P} \subseteq J \times I, \qquad \mathcal{Q} \subseteq K \times J, \qquad \mathcal{R} \subseteq K \times I$$

be diagonal H-orbits. For any $(k,i) \in \mathcal{R}$, define

$$c_{\mathcal{Q},\mathcal{P}}^{\mathcal{R}} := \#\{j \in J : (k,j) \in \mathcal{Q}, (j,i) \in \mathcal{P}\}.$$

This integer is independent of the chosen $(k,i) \in \mathcal{R}$, and

$$B_{\mathcal{Q}} B_{\mathcal{P}} = \sum_{\mathcal{R} \in H\backslash(K\times I)} c_{\mathcal{Q},\mathcal{P}}^{\mathcal{R}} B_{\mathcal{R}}.$$

Thus the lane path law descends to an exact orbit basis with nonnegative integer structure constants.

Proof. Write an arbitrary $\Bbbk$-linear map as a matrix $A = (A_{ji})$. The intertwining equations for the permutation actions are equivalent to

$$A_{hj,hi} = A_{ji} \qquad (h \in H).$$

Hence the entries are constant on diagonal orbits. The orbit-indicator matrices have disjoint supports, so they are linearly independent over every commutative ring and span all invariant matrices.

The (k,i)-entry of $B_{\mathcal{Q}} B_{\mathcal{P}}$ is the displayed cardinality. If $(k',i') = (hk, hi)$, then $j \mapsto hj$ is a bijection between the two sets being counted. The entry is therefore constant on $\mathcal{R}$, which gives the orbit-basis expansion. □

Corollary 21.9 (Orbital endomorphism algebra)**.** *For one finite H-set I, the pair-orbit matrices form a basis of* $\mathrm{End}_{\Bbbk H}(\Bbbk[I])$*. Matrix multiplication in this basis is exactly the finite compatible-path contraction of theorem 21.3, with the preceding intersection numbers as structure constants.*

Example 21.10 (The smallest nontrivial holonomy)**.** Let $H = C_2 = \{1, \sigma\}$ act on $I = \{1, 2\}$ by swapping the two addresses. On $X = \Bbbk^2$, let

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad p_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad p_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

The individual projectors do not descend because $Sp_1S^{-1} = p_2$, while their orbit sum $p_1 + p_2 = 1$ does descend. The pair-address set has two orbits,

$$\Delta = \{(1,1), (2,2)\}, \qquad \Omega = \{(1,2), (2,1)\}.$$

Accordingly,

$$\mathrm{End}_{\Bbbk C_2}(\Bbbk^2) = \left\{ \begin{pmatrix} a & b \\ b & a \end{pmatrix} : a, b \in \Bbbk \right\} = \Bbbk B_\Delta \oplus \Bbbk B_\Omega.$$

The singleton endpoint selector $\{(1,1)\}$ is not holonomy compatible and sends the equivariant identity matrix to p_1, which is not equivariant. The orbit selectors C_Δ and C_Ω are compatible and split every equivariant endomorphism into its diagonal-orbit and off-diagonal-orbit parts.

21.5 Cyclotomic orbit lanes over the rational field

The split cyclic group algebra supplies an arithmetic realization in which the address monodromy and its orbit coarsening are explicit.

Proposition 21.11 (Rational descent of split cyclic Fourier lanes)**.** *Let $N \geq 1$, let $C_N = \langle g \rangle$, and let $K = \mathbb{Q}(\zeta_N)$. In $K[C_N]$, define*

$$e_j := \frac{1}{N} \sum_{\ell=0}^{N-1} \zeta_N^{-j\ell} g^\ell, \qquad j \in \mathbb{Z}/N\mathbb{Z}.$$

These are the primitive split Fourier idempotents. For $\sigma_a \in \mathrm{Gal}(K/\mathbb{Q})$, where $\sigma_a(\zeta_N) = \zeta_N^a$,

$$\sigma_a(e_j) = e_{aj}.$$

The Galois orbits are indexed by divisors $q \mid N$, with

$$\mathcal{O}_q = \left\{ j \in \mathbb{Z}/N\mathbb{Z} : \frac{N}{\gcd(N, j)} = q \right\}.$$

Their orbit sums

$$E_q := \sum_{j \in \mathcal{O}_q} e_j$$

belong to $\mathbb{Q}[C_N]$ and satisfy

$$E_q E_{q'} = \delta_{q,q'} E_q, \qquad \sum_{q \mid N} E_q = 1.$$

More explicitly,

$$\boxed{E_q = \frac{1}{N} \sum_{\ell=0}^{N-1} c_q(\ell) g^\ell,}$$

where

$$c_q(\ell) := \sum_{\substack{1\le a\le q\\ \gcd(a,q)=1}} e^{2\pi i a\ell/q} = \sum_{d\mid \gcd(q,\ell)} d\,\mu(q/d) \in \mathbb{Z}$$

is the Ramanujan sum. The projectors $(E_q)_{q|N}$ form the finest $\mathbb{Q}$-rational coarsening obtainable by summing the split Fourier idempotents, and

$$\mathbb{Q}[C_N]E_q \cong \mathbb{Q}(\zeta_q).$$

Proof. Character orthogonality gives the primitive idempotent relations for the e_j. Applying σ_a to their coefficients gives $\sigma_a(e_j) = e_{aj}$. The unit group $(\mathbb{Z}/N\mathbb{Z})^\times$ acts transitively on elements of $\mathbb{Z}/N\mathbb{Z}$ having the same additive order, so its orbits are the sets $\mathcal{O}_q$.

Write every $j \in \mathcal{O}_q$ uniquely as $j = (N/q)a$ with $a \in (\mathbb{Z}/q\mathbb{Z})^\times$. The coefficient of g^ℓ in E_q is then

$$\frac{1}{N}\sum_{a\in(\mathbb{Z}/q\mathbb{Z})^\times} \zeta_q^{-a\ell} = \frac{c_q(\ell)}{N},$$

where the last equality uses the integrality and evenness of the Ramanujan sum. Hence $E_q \in \mathbb{Q}[C_N]$. Orthogonality and completeness follow by grouping the primitive e_j by their disjoint Galois orbits. Any rational sum of split primitive idempotents must be Galois invariant, so its support is a union of these orbits; this proves fineness. Finally,

$$\mathbb{Q}[C_N] \cong \mathbb{Q}[x]/(x^N-1) \cong \prod_{q|N}\mathbb{Q}[x]/(\Phi_q(x)) \cong \prod_{q|N}\mathbb{Q}(\zeta_q),$$

and E_q is the identity projector of the q-th factor. □

Boundary 21.12 (Cyclic character lanes are not coefficient-residue lanes). The preceding proposition concerns primitive character idempotents in the split group algebra $K[C_N]$ and their rational Galois descent. The coefficient-residue projectors $\mathcal{P}_{N,j}$ on formal series from theorem 20.15 are defined coefficientwise over every commutative ring and already have fixed residue labels. A chosen rotation representation relates the two constructions after scalar extension, but their descent data are not identical and should not be conflated. Nothing in orbit descent turns the factorial step-product into an additive idempotent splitting.

21.6 Exact finite validation and scope

The accompanying verifier tests finite permutation actions, pair-orbit bases, equivariant matrix composition, orbit-compatible and deliberately incompatible selectors, equivariant refinement, and the exact Ramanujan-sum projectors in $\mathbb{Q}[C_N]$. It uses exact integer arithmetic after clearing the common cyclotomic denominators. These tests are an implementation regression for the displayed formulas, not a substitute for the proofs and not an external validation record.

Holonomy-twisted lane transport result. Finite nontrivial address holonomy does not invalidate exact lane transport. The complete global coordinate object is a matrix local system, and descended morphisms are exactly its compatible sections. Composition remains finite path contraction, complete resolution changes remain isomorphisms, and every holonomy-compatible selector retains an exact named residual. Monodromy orbits give the finest ordinary globally labeled coarsening of object lanes, while simultaneous pair orbits give the atomic global transport selectors. In permuta-

Table 42: Exact finite regression for selected holonomy-twisted lane clauses. The table is an implementation record, not a proof or external validation.

Validation family	Exact assertions
Group actions and object-orbit descent	7,780
Pair-orbit partitions and intertwiner bases	3,537
Orbital path composition and structure constants	1,212,963
Compatible selectors and negative controls	994
Equivariant refinement and orbit-projector sums	10,162
Rational cyclotomic orbit projectors	109,374
Theorem-source contract	13
Total selected implementation assertions	1,344,823
Recorded failures	0

tion modules these pair orbits form an exact basis of the intertwiner space with integer path-count structure constants.

Boundary 21.13 (Finite twisted scope). The theorem remains finite and profiled. It assumes invertible chart transitions, finite address sets, coherent address and splitting cocycles, and profile preservation of the operations used. It does not prove convergence for countably infinite lane sums, descent across noninvertible bridges, higher stack effectivity, or a canonical orbit system independent of the initially declared lane presentation. Orbit count is not identified with dimension, rank, entropy, information, or complexity. The permutation-module and cyclotomic formulas have classical algebraic antecedents; their role here is to provide exact native realizations of the twisted lane doctrine.

Part IV

Relative semantic quotients and changes of doctrine

22 Relative semantic quotients and behavioral cores

The preceding parts fix resolved signatures, profile-preserving realizations, normalized strata, typed bridges, and descent data. A second question remains inside each such setting: which distinctions must a representation retain for a declared family of future observations? This section formalizes that question without replacing the existing preservation-profile machinery. The central object is a behavioral quotient computed relative to a carrier doctrine, a continuation grammar, and an observation category.

22.1 Semantic slices and resolved observational systems

Definition 22.1 (Semantic slice)**.** A *semantic slice* is a tuple

$$\Sigma = (\mathcal{A}, \mathcal{G}, \mathcal{V}, \mathcal{O})$$

consisting of:

(i) a carrier doctrine $\mathcal{A}$, which specifies how alternatives, routes, or packets combine;

(ii) a continuation grammar $\mathcal{G}$, presented here by a monoid M or by a chosen generating family of admissible endomorphisms;

(iii) an observation category $\mathcal{V}$, such as **Set**, R-**Mod**, or K-**Vect**;

(iv) a declared family $\mathcal{O} = (o_j)_{j\in J}$ of observations in $\mathcal{V}$.

The slice belongs to the stratum passport. It does not replace the preservation profile Π. The profile specifies which constructions a bridge preserves, while the slice specifies which future distinctions define the current semantic task.

Definition 22.2 (Resolved observational system)**.** Let X be an object equipped with an action

$$\rho : M \longrightarrow \mathrm{End}(X)$$

of a continuation monoid and with observations

$$o_j : X \longrightarrow Y_j \qquad (j \in J).$$

The tuple

$$\mathsf{S} = (X, M, \rho, (o_j)_{j\in J})$$

is a *resolved observational system*. When the indicated product exists, its behavior map is

$$\mathrm{Beh}_{\mathsf{S}} : X \longrightarrow \prod_{(m,j)\in M\times J} Y_j, \qquad x \longmapsto \big(o_j(\rho(m)x)\big)_{m,j}.$$

The *relative behavioral object* is the coimage

$$Q_{\mathsf{S}} := \mathrm{Coim}(\mathrm{Beh}_{\mathsf{S}})$$

whenever that coimage exists.

In **Set**, the behavior map identifies states exactly when every declared future observation agrees. In a module category, the same object can be constructed directly from the intersection of kernels, without assuming that the possibly infinite product is part of the ambient profile.

Definition 22.3 (Behavioral congruence and linear behavioral kernel)**.** In **Set** or a variety of algebras, put

$$x \equiv_{\mathsf{S}} y \quad\Longleftrightarrow\quad o_j(\rho(m)x) = o_j(\rho(m)y) \quad \text{for every } m \in M,\ j \in J.$$

Write Θ_{S} for this relation. In R-**Mod**, define

$$N_{\mathsf{S}} := \bigcap_{m\in M}\bigcap_{j\in J} \ker(o_j\rho(m)).$$

Then

$$Q_{\mathsf{S}} = X/N_{\mathsf{S}}.$$

Theorem 22.4 (Universal behavioral coimage)**.** *Assume the relevant effective quotient or module quotient exists. Then Q_{S} preserves every declared future observation. Moreover, if*

$$q : X \twoheadrightarrow Z$$

is any surjective representation through which every map $o_j\rho(m)$ factors, there is a unique map

$$\overline{q} : Z \longrightarrow Q_{\mathsf{S}}$$

with $\overline{q}\,q : X \to Q_{\mathsf{S}}$ equal to the canonical quotient map. Thus Q_{S} is the coarsest exact representation for the declared semantic slice.

Proof. In the relational setting, the kernel pair of q is contained in Θ_S, because equality under q forces equality under every factorized future observation. The universal property of the quotient by Θ_S gives the unique map. In the module setting,

$$\ker q \subseteq \bigcap_{m,j} \ker(o_j \rho(m)) = N_S,$$

so the ordinary quotient universal property gives the same conclusion. □

Boundary 22.5 (No observer-independent minimum). The phrase "minimal representation" is meaningful only after the carrier, continuation grammar, observation category, and task have been declared. Changing any one of these data can change the quotient. In particular, equality of residual functions in **Set** and linear dependence of residual functions in K-**Vect** are different minimization problems.

22.2 The maximal stable invisible core

Let

$$K_0 := \bigcap_j \mathrm{KerPair}(o_j)$$

in the relational setting and

$$N_0 := \bigcap_j \ker(o_j)$$

in the linear setting.

Theorem 22.6 (Behavioral core theorem)**.** *The behavioral relation Θ_S is the greatest continuation-stable congruence contained in K_0. In the module setting, N_S is the greatest $\rho(M)$-invariant submodule contained in N_0.*

Proof. The identity continuation gives $\Theta_S \subseteq K_0$ and $N_S \subseteq N_0$. If $x \equiv_S y$, then for every $n, m \in M$ and every j,

$$o_j\rho(m)\rho(n)x = o_j\rho(mn)x = o_j\rho(mn)y = o_j\rho(m)\rho(n)y,$$

so the relation is stable. The same calculation proves invariance of N_S.

Conversely, let $R \subseteq K_0$ be continuation stable. If xRy, then $\rho(m)x\, R\, \rho(m)y$ for every m, and membership in K_0 forces

$$o_j\rho(m)x = o_j\rho(m)y$$

for every j. Hence $R \subseteq \Theta_S$. The submodule statement is identical with relations replaced by differences. □

Remark 22.7 (Comparison with holonomy). The maximal path-flat core in Section 4 intersects fixed spaces of closed transport words. The behavioral core intersects kernels of future observation words. Both constructions extract the greatest subobject on which a declared family of transformations becomes semantically invisible. They answer different questions and are not identified without a typed comparison map.

22.3 Greatest-fixed-point construction and finite recognition

Assume M is generated by a family A. On congruences or submodules define

$$\Phi(R) := K_0 \cap \bigcap_{a \in A} \rho(a)^{-1}(R),$$

with the corresponding submodule formula in the linear setting.

Theorem 22.8 (Coinductive characterization)**.** *The behavioral core is the greatest fixed point of Φ:*

$$\Theta_S = \nu R.\Phi(R), \qquad N_S = \nu N.\Phi(N).$$

Proof. Every fixed point is invisible and stable under every generator, hence under the generated monoid, so it is contained in the behavioral core by theorem 22.6. Conversely, the behavioral core is a fixed point: stability gives its inclusion in its image under Φ, while membership in Φ of that core makes the identity observation invisible and makes every first-generator successor behaviorally invisible, hence makes every continuation word invisible. Its maximality therefore proves the asserted greatest-fixed-point identities. □

Corollary 22.9 (Finite certification)**.** *For finite X, descending partition refinement stabilizes after finitely many strict refinements. For finite-dimensional X over a field, the chain*

$$N^{(0)} = N_0, \qquad N^{(r+1)} = N_0 \cap \bigcap_{a \in A} \rho(a)^{-1}(N^{(r)})$$

stabilizes after at most $\dim X$ *strict dimension drops. At stabilization it equals N_S.*

Proof. The iteration is descending and every strict inclusion of finite-dimensional subspaces lowers dimension by at least one. The behavioral core lies in $N^{(0)}$, and monotonicity of Φ, together with its fixed-point property, preserves this inclusion at every stage. Thus the stabilized fixed point contains the behavioral core. Greatest-fixed-point maximality gives the reverse inclusion, proving equality. □

22.4 The task–congruence Galois connection

Fix a universe $\mathcal{Q}(X)$ of admissible queries on X. For $\mathcal{Q} \subseteq \mathcal{Q}(X)$, define

$$\mathrm{Ind}(\mathcal{Q}) := \bigcap_{q \in \mathcal{Q}} \mathrm{KerPair}(q).$$

For an equivalence relation or congruence R, define

$$\mathrm{Def}(R) := \{q \in \mathcal{Q}(X) : R \subseteq \mathrm{KerPair}(q)\}.$$

Theorem 22.10 (Task–congruence Galois connection)**.** *For every query family $\mathcal{Q}$ and relation R,*

$$\mathcal{Q} \subseteq \mathrm{Def}(R) \quad \Longleftrightarrow \quad R \subseteq \mathrm{Ind}(\mathcal{Q}).$$

Consequently,

$$\mathrm{Cl}(\mathcal{Q}) := \mathrm{Def}(\mathrm{Ind}(\mathcal{Q}))$$

is a closure operator on query families, and

$$\mathrm{Sat}(R) := \mathrm{Ind}(\mathrm{Def}(R))$$

is the observational saturation of R relative to the admitted query universe.

Proof. Both conditions say that every query in $\mathcal{Q}$ is constant on every R-class. The closure laws follow from the standard identities associated with an antitone Galois connection. □

Definition 22.11 (Extensional task)**.** An *extensional task* is a closed query theory, equivalently a query family modulo equality of the closure Cl. When the admitted query universe separates the desired quotients, an extensional task can equivalently be represented by its observationally definable congruence.

Corollary 22.12 (Task strengthening)**.** *Adding continuations or observations can only refine behavioral indistinguishability. Thus*

$$M \subseteq M', \qquad \mathcal{O} \subseteq \mathcal{O}' \quad \Longrightarrow \quad \Theta_{M',\mathcal{O}'} \subseteq \Theta_{M,\mathcal{O}},$$

and there is a canonical forgetting map

$$Q_{M',\mathcal{O}'} \longrightarrow Q_{M,\mathcal{O}}.$$

Boundary 22.13 (Definable tasks form a closure system). All continuation-stable congruences form a complete lattice. Congruences definable by the chosen query universe are always closed under arbitrary meets. Closure under joins requires an additional richness or quotient-completeness hypothesis on the query universe and is not asserted automatically.

22.5 Semantic quotient lattice

Theorem 22.14 (Semantic quotient classification, classical)**.** *Let X be an algebra in a variety. Surjective semantic maps*

$$q : X \twoheadrightarrow Q$$

up to isomorphism of codomains are classified by congruences on X through

$$q \longmapsto \ker q.$$

Moreover, a quotient q maps to a quotient r exactly when $\ker q \subseteq \ker r$.

Proof. This is the first isomorphism theorem and the universal property of quotient algebras. It is included to fix the direction of the information order used below. □

Corollary 22.15 (Combined and shared semantics)**.** *For semantic quotients with kernels N and M:*

(i) *the least common refinement has kernel $N \cap M$;*

(ii) *the greatest common coarsening has kernel the generated congruence join $N \vee M$;*

(iii) *in a module category, $N \vee M = N + M$.*

22.6 The two-view semantic diamond

Let X be an R-module and let

$$s : X \twoheadrightarrow S, \qquad t : X \twoheadrightarrow T$$

be surjective linear semantic maps. Put

$$N = \ker s, \qquad M = \ker t, \qquad J = X/(N \cap M), \qquad C = X/(N + M).$$

Theorem 22.16 (Semantic diamond and exact compatibility sequence)**.** *The canonical square*

$$\begin{array}{ccc} J & \longrightarrow & T \\ \downarrow & & \downarrow \\ S & \longrightarrow & C \end{array}$$

is a pullback in R-**Mod**. *Equivalently,*

$$J \cong S \times_C T.$$

There is also a canonical short exact sequence

$$0 \longrightarrow X/(N \cap M) \xrightarrow{\alpha} X/N \oplus X/M \xrightarrow{\beta} X/(N+M) \longrightarrow 0,$$

where

$$\alpha(x + N \cap M) = (x + N, x + M)$$

and

$$\beta(a + N, b + M) = a - b + (N + M).$$

Proof. The map α is injective because its kernel is $N \cap M$. The formula for β is well defined and surjective. Clearly $\beta\alpha = 0$. If $\beta(a + N, b + M) = 0$, then

$$a - b = n + m$$

for some $n \in N$ and $m \in M$. Set $x = a - n = b + m$. Then

$$(a + N, b + M) = (x + N, x + M) = \alpha(x + N \cap M).$$

Thus $\ker \beta = \operatorname{im} \alpha$, proving exactness. The kernel description of the fiber product is the same calculation. □

Definition 22.17 (Relative semantic residual)**.** The information needed to augment S so that it also retains T is

$$R_{T|S} := \ker(J \to S) \cong N/(N \cap M) \cong (N + M)/M.$$

Corollary 22.18 (Factorization criterion)**.** *The following are equivalent:*

$$t \text{ factors through } s, \qquad N \subseteq M, \qquad R_{T|S} = 0, \qquad J \to S \text{ is an isomorphism.}$$

Moreover,

$$0 \longrightarrow R_{T|S} \longrightarrow J \longrightarrow S \longrightarrow 0$$

is exact.

Boundary 22.19 (Residuals need not split). The exact sequence above does not canonically identify J with $S \oplus R_{T|S}$. Vector-space extensions split noncanonically. Module, integral, semiring, filtered, or topological extensions may fail to split. A residual packet presented as an independent coordinate therefore requires an explicit splitting theorem or constructor passport.

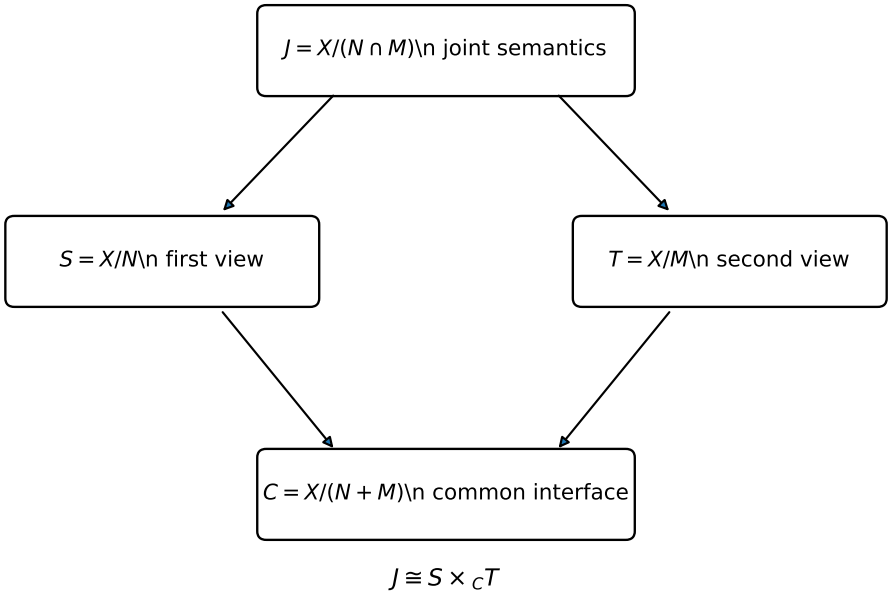


Figure 57: The semantic diamond. The joint representation J retains both views. The common quotient C is the compatibility interface over which the separate coordinates glue.

22.7 Exact change of task

Theorem 22.20 (Task-change exact sequence)**.** *Let a stronger task have behavioral kernel N_s, and let a weaker task have behavioral kernel N_w, with*

$$N_s \subseteq N_w.$$

Then

$$0 \longrightarrow N_w/N_s \longrightarrow X/N_s \longrightarrow X/N_w \longrightarrow 0$$

is exact. Thus N_w/N_s is the exact semantic information discarded when the task is weakened.

Proof. Apply the third isomorphism theorem to the chain $N_s \subseteq N_w \subseteq X$. □

Corollary 22.21 (Composition of task loss)**.** *For*

$$N_1 \subseteq N_2 \subseteq N_3,$$

there is an exact sequence

$$0 \longrightarrow N_2/N_1 \longrightarrow N_3/N_1 \longrightarrow N_3/N_2 \longrightarrow 0.$$

Hence task loss composes as extension data rather than as a canonically split sum.

Semantic quotient result. Inside a fixed doctrine, declared future tasks are represented by continuation-stable congruences or subobjects. Behavioral minimization extracts the maximal stable invisible core. Task strengthening induces canonical forgetting maps, joint tasks form semantic diamonds, and relative task loss is measured by exact quotient objects. The construction is relative to the semantic slice and does not define an observer-independent minimum.

23 Carrier change, observer base change, and provenance

A behavioral quotient is computed inside an observation category, but the represented computation also has a carrier algebra. Additive convolution, Boolean support, max-plus aggregation, and convex support geometry do not combine routes by the same law. This section separates carrier translation from observer minimization and gives exact obstruction objects for both.

23.1 Carrier factorization and joint semantic completion

Let $\mathcal{R}$ be a raw algebra of expressions or histories for a fixed operation signature. Let

$$s : \mathcal{R} \longrightarrow S, \qquad t : \mathcal{R} \longrightarrow T$$

be homomorphisms into two semantic carriers.

Theorem 23.1 (Carrier factorization gate)**.** *There is a unique homomorphism*

$$\varphi : \mathrm{im}(s) \longrightarrow T$$

with $t = \varphi s$ if and only if

$$\mathrm{KerPair}(s) \subseteq \mathrm{KerPair}(t).$$

When the condition fails, any pair $x, y \in \mathcal{R}$ satisfying

$$s(x) = s(y), \qquad t(x) \neq t(y)$$

is an exact carrier-change obstruction witness.

Proof. Factorization immediately implies constancy of t on every fiber of s. Conversely, define

$$\varphi(s(x)) := t(x).$$

The fiber condition makes this well defined. Since both s and t preserve every primitive operation, so does φ. Uniqueness follows because s is surjective onto its image. □

Definition 23.2 (Joint semantic completion)**.** Define

$$j = (s, t) : \mathcal{R} \longrightarrow S \times T, \qquad J_{s,t} := \mathrm{im}(j).$$

The projection

$$p_s : J_{s,t} \longrightarrow \mathrm{im}(s)$$

is the *relative completion map* from the joint semantics to the source semantics.

Theorem 23.3 (Universal joint completion)**.** *The object $J_{s,t}$ is the coarsest quotient of $\mathcal{R}$ preserving both s and t. More precisely, if*

$$c : \mathcal{R} \twoheadrightarrow C$$

admits maps $a : C \to S$ and $b : C \to T$ with $s = ac$ and $t = bc$, then there is a unique map

$$C \longrightarrow J_{s,t}$$

compatible with both semantics. Furthermore,

$$t \text{ factors through } s \iff p_s \text{ is an isomorphism.}$$

Proof. The pair (a, b) factors j through c, and the image universal property gives the map to $J_{s,t}$. If $t = \varphi s$, then $J_{s,t}$ is the graph of φ, so p_s is invertible. Conversely, if p_s is invertible, define $\varphi = p_t p_s^{-1}$. □

23.2 Behavioral bridges and exact loss

Let $f : X \to Y$ be a linear bridge between observational systems, with behavioral kernels $N_X \subseteq X$ and $N_Y \subseteq Y$.

Definition 23.4 (Task-compatible bridge)**.** The bridge is *task compatible* when

$$f(N_X) \subseteq N_Y.$$

Equivalently,

$$N_X \subseteq f^{-1}(N_Y).$$

Then there is a unique induced map

$$\overline{f} : X/N_X \longrightarrow Y/N_Y.$$

Theorem 23.5 (Bridge-loss exact sequence)**.** *For every task-compatible linear bridge, there is an exact sequence*

$$0 \longrightarrow \frac{f^{-1}(N_Y)}{N_X} \longrightarrow \frac{X}{N_X} \xrightarrow{\overline{f}} \frac{Y}{N_Y} \longrightarrow \frac{Y}{N_Y + f(X)} \longrightarrow 0.$$

Thus

$$L_f := \frac{f^{-1}(N_Y)}{N_X}$$

measures source distinctions collapsed by the bridge, while

$$E_f := \frac{Y}{N_Y + f(X)}$$

measures target semantic states outside the essential image.

Proof. The kernel of $\overline{f}$ consists of classes $x + N_X$ for which $f(x) \in N_Y$, hence equals $f^{-1}(N_Y)/N_X$. The cokernel of $\overline{f}$ is the quotient of Y/N_Y by $(f(X) + N_Y)/N_Y$, which is $Y/(N_Y + f(X))$. □

Corollary 23.6 (Exact semantic equivalence gate)**.** *The induced bridge $\overline{f}$ is an isomorphism exactly when*

$$L_f = 0 \qquad \textit{and} \qquad E_f = 0.$$

Faithfulness on the declared task and essential surjectivity on the declared task are therefore independent obligations.

Theorem 23.7 (Loss-composition extension)**.** *For induced quotient maps*

$$Q_X \xrightarrow{\overline{f}} Q_Y \xrightarrow{\overline{g}} Q_Z,$$

there is a natural exact sequence

$$0 \longrightarrow \ker \overline{f} \longrightarrow \ker(\overline{g}\,\overline{f}) \longrightarrow \ker \overline{g} \cap \operatorname{im} \overline{f} \longrightarrow 0.$$

If $\overline{f}$ is surjective, this reduces to

$$0 \longrightarrow L_f \longrightarrow L_{gf} \longrightarrow L_g \longrightarrow 0.$$

Proof. Map $x \in \ker(\overline{g}\,\overline{f})$ to $\overline{f}(x)$. Its image is exactly $\ker \overline{g} \cap \operatorname{im} \overline{f}$, and its kernel is $\ker \overline{f}$. Surjectivity of $\overline{f}$ identifies the intersection with $\ker \overline{g}$. □

Boundary 23.8 (Loss is extension data)*.* The preceding sequence need not split. Consequently, composite information loss is not in general a scalar sum or a canonical direct sum of stagewise losses. A numerical loss budget may be useful, but it is a further invariant of the exact extension rather than a replacement for it.

23.3 Carrier-indexed observer packets

Theorem 23.9 (Carrier-indexed observer packet theorem)**.** *Let S be a finite continuation-state set, let $(A, \oplus, 0)$ be a commutative monoid, let B be a finite branch alphabet, and let*

$$\delta : S \times B \longrightarrow S$$

be a state transition. Suppose every branch $b \in B$ acts by a monoid homomorphism

$$L_b : A \longrightarrow A.$$

Take an empty aggregation to be 0. For a state packet $a = (a_s)_{s \in S}$, define

$$a'_t := \bigoplus_{\delta(s,b)=t} L_b(a_s).$$

If a finite family of raw histories has already been aggregated by their state using $\oplus$, then this update gives exactly the same result as advancing every raw history separately and aggregating afterward.

Proof. Expand the raw history update and group all terms by their predecessor state and new branch. Associativity and commutativity permit the regrouping. The identity

$$L_b\left(\bigoplus_i x_i\right) = \bigoplus_i L_b(x_i)$$

then moves aggregation through the branch action. The displayed formula is the resulting statewise sum. □

Two important realizations have the same state skeleton but different carriers.

1. For convex value semantics, A is the family of compact convex sets, including the empty set, $\oplus$ is convex hull of union with identity $\varnothing$, and $L_b(C) = A_b^{\mathsf{T}} C$.
2. For linear polyphase routing, A is an additive convolution algebra, $\oplus$ is coefficient addition, and L_b multiplies route kernels by the declared filter or transition weight.

The theorem transports the observer skeleton only after the carrier law and the homomorphism obligations are specified.

23.4 Carrier no-go mechanisms

Theorem 23.10 (Cancellation barrier)**.** *Let R be a nonzero ring and let S be a nontrivial zerosumfree semiring, so*

$$a + b = 0 \Longrightarrow a = b = 0.$$

There is no unital semiring homomorphism $R \to S$.

Proof. A unital homomorphism φ would give

$$0 = \varphi(1 + (-1)) = 1_S + \varphi(-1).$$

Zerosumfreeness forces $1_S = 0$, contradicting nontriviality. □

Theorem 23.11 (Idempotence barrier)**.** *Let S be a nontrivial unital semiring with idempotent addition, and let R be an additively cancellative nontrivial semiring. There is no unital semiring homomorphism $S \to R$.*

Proof. In S, $1 + 1 = 1$. Its image in R would satisfy $1_R + 1_R = 1_R$, and additive cancellation would imply $1_R = 0$. □

Example 23.12 (Haar cancellation)**.** For the normalized Haar bank,

$$\ell = \frac{x_0 + x_1}{\sqrt{2}}, \qquad d = \frac{x_0 - x_1}{\sqrt{2}},$$

and

$$x_0 = \frac{\ell + d}{\sqrt{2}}, \qquad x_1 = \frac{\ell - d}{\sqrt{2}}.$$

At $x = (1, 0)$, the two route contributions to x_0 are $1/2, 1/2$, while those to x_1 are $1/2, -1/2$. Additive reconstruction returns $(1, 0)$. Replacing route addition by maximum returns $(1/2, 1/2)$. The convex/max carrier is therefore not an exact replacement for the signed additive carrier.

23.5 Observer base change and the relation module

Let $E \subseteq K^M$ be the set of distinct residual future functions of a scalar series or observational system. Set minimization retains the elements of E as distinct states. Free linearization produces the vector space $K^{(E)}$, while linear minimization retains only their span.

Theorem 23.13 (Observer base-change exact sequence)**.** *The evaluation map*

$$\pi : K^{(E)} \longrightarrow \operatorname{span}_K(E), \qquad \sum_{e \in E} a_e [e] \longmapsto \sum_{e \in E} a_e e$$

is surjective and fits into the exact sequence

$$0 \longrightarrow \operatorname{Rel}_K(E) \longrightarrow K^{(E)} \xrightarrow{\pi} \operatorname{span}_K(E) \longrightarrow 0,$$

where

$$\operatorname{Rel}_K(E) := \ker \pi$$

is the module of linear relations among distinct discrete residual states. Hence Set minimization commutes with free linearization exactly when E is linearly independent.

Proof. Surjectivity is the definition of the linear span. The sequence is the kernel-cokernel sequence of π. The kernel vanishes exactly when the basis symbols $[e]$ remain linearly independent after evaluation. □

Theorem 23.14 (Exact polyphase separation)**.** *Fix $N \geq 1$ and $z \neq 0$. Let S_z be the cyclic shift on K^N satisfying*

$$S_z e_j = e_{j+1} \quad (j < N - 1), \qquad S_z e_{N-1} = z e_0,$$

so $S_z^N = zI$. Put

$$M_N = I + S_z, \qquad f_N(n) = e_0^{\mathsf{T}} M_N^n e_0.$$

Then the $N \times N$ Hankel block

$$H_{ij} = f_N(i + j), \qquad 0 \leq i, j < N,$$

has determinant

$$\det H = \pm z^{N-1}.$$

Consequently its linear residual space has dimension N. If $z > 0$, the residual functions $k \mapsto f_N(n + k)$ are pairwise distinct, so the exact discrete future quotient is infinite while the linear continuation dimension is N.

Proof. Factor $H = OC$, where row i of O is $e_0^{\mathsf{T}} M_N^i$, and column j of C is $M_N^j e_0$. For $j < N$, the binomial expansion of $(I + S_z)^j e_0$ is a unit upper-triangular change from $(e_0, \dots, e_{N-1})$, hence $\det C = 1$. The rows $e_0^{\mathsf{T}} S_z^k$, $0 \le k < N$, are e_0^{T} followed by $z e_{N-k}^{\mathsf{T}}$, so their determinant is $\pm z^{N-1}$. The binomial row change is also unit triangular, proving the determinant formula.

For positive z,

$$f_N(n) = \sum_{r=0}^{\lfloor n/N \rfloor} \binom{n}{Nr} z^r.$$

The sequence is strictly increasing from index $N-1$ onward. Prefixes below N are distinguished by choosing the first suffix that crosses index N, and later prefixes differ already at suffix zero. Thus all residual functions are distinct. □

The interpretation of finite Hankel rank as minimal weighted-linear realization dimension is classical weighted-automata theory [20, 21]. The displayed family is retained here because it gives an exact internal example of an infinite discrete quotient and a finite linear quotient.

23.6 Observer-conditioned convex packets

Let S be a finite observer-state set, let B be a finite branch alphabet, and let $C_s \subseteq \mathbb{R}^d$ be a compact convex packet for each state, with empty packets allowed. Use the conventions

$$h_\varnothing(x) = -\infty, \qquad \operatorname{conv}(\varnothing) = \varnothing, \qquad \max \varnothing = -\infty.$$

For branch matrices $(A_b)_{b \in B}$ and a transition $\delta : S \times B \to S$, put

$$C'_t = \operatorname{conv} \bigcup_{\delta(s,b)=t} A_b^{\mathsf{T}} C_s.$$

Theorem 23.15 (Observer-conditioned convex packet transport)**.** *For every state $t \in S$ and direction x,*

$$h_{C'_t}(x) = \max_{\delta(s,b)=t} h_{C_s}(A_b x).$$

Repeated updates compose exactly with branch words. Within each fixed observer state, replacing a finite atom family by any other finite generating family of the same convex hull preserves all future value queries generated by linear branch actions.

Proof. For a linear map A,

$$h_{A^{\mathsf{T}} C}(x) = h_C(Ax).$$

If the incoming fiber is empty, both sides of the formula are $-\infty$. Otherwise, the support function of the convex hull of a finite union is the maximum of the component support functions, with the same conventions for empty components. These identities give the formula. Finite unions of compact sets have compact convex hulls in finite dimension, so every updated packet stays in the declared carrier. Iteration follows from composition of the linear maps and of δ. Equality of convex hulls is preserved by linear images and convex hull of union, proving future value equivalence. □

Proposition 23.16 (Singular resurrection)**.** *A provenance state carried only by a strictly interior atom may be inactive for every current nonzero linear query and yet become active after an admissible singular continuation. In particular, if the zero map is admissible, every atom is sent to the same point and all provenance states tie.*

Proposition 23.17 (Invertible hiding)**.** *Assume the total convex body is full-dimensional, query directions are nonzero, and every admissible branch matrix is invertible. A strictly interior atom remains strictly interior under every continuation and therefore cannot maximize a nonzero linear functional. Its provenance remains permanently hidden from the declared value query.*

Proof. Invertible linear maps preserve relative interior and full dimension. A nonzero linear functional cannot attain its maximum at a strict interior point of a compact full-dimensional convex body. □

Boundary 23.18 (Value, witness, and history are distinct tasks). Exact convex geometry determines value semantics and its derived normal-fan structure. It does not determine branch identity, maximizing multiplicity, or raw history. Value-only, one-witness, all-witness, and complete-history tasks therefore have different minimal quotients. No provenance deletion rule is valid without the admissible continuation class and the downstream query family.

Carrier and observer result. Exact carrier replacement is equivalent to semantic factorization. When it fails, the joint image is the universal completion preserving both semantics. Behavioral bridges carry independent kernel-loss and essential-image defects. Observer minimization then depends on the observation category: discrete residual equality and linear residual span are related by an exact relation module, not by automatic identification.

24 Admissibility change, geometric obstruction, and doctrine lifting

Task strengthening changes which distinctions must be retained. Enlarging an admissible realization class changes a different quantity: which completions or normal forms are allowed. The two variances should not be compressed into one undifferentiated order. This section formalizes admissibility monotonicity, separates gap from nonattainment, and places enrichment problems inside the existing bridge-passport architecture.

24.1 Admissible realization classes and optimized defect

Definition 24.1 (Class-relative defect)**.** Let X be a represented system, let Γ be a declared class of admissible auxiliary structures, and let

$$d(X, g) \in [0, \infty]$$

be a defect functional. Define

$$\delta_\Gamma(X) := \inf_{g \in \Gamma} d(X, g).$$

The pair consisting of the infimum and its attainment status is the *class-relative defect profile*.

Theorem 24.2 (Admissibility monotonicity)**.** *If* $\Gamma \subseteq \Gamma'$*, then*

$$\delta_{\Gamma'}(X) \leq \delta_\Gamma(X).$$

Thus an obstruction in a restricted realization class need not remain an obstruction after the class is enlarged.

Proof. The infimum on the left is taken over a superset of the candidates used on the right. □

Definition 24.3 (Completion, gap, and escape)**.** Assume $d \geq 0$. The defect profile is:

(i) a *completion* when $\delta_\Gamma(X) = 0$ and the infimum is attained;

(ii) a *gap obstruction* when $\delta_\Gamma(X) > 0$;

(iii) an *escape obstruction* when $\delta_\Gamma(X) = 0$ but the infimum is not attained.

The trichotomy separates approximation from realization. A sequence of increasingly good coordinates may approach a missing completion only by leaving every compactly controlled part of the admissible parameter space.

Proposition 24.4 (Compactness certificate for escape)**.** *Let Γ be a Hausdorff parameter space, and let $d(X,-)$ be continuous. Suppose every sublevel set satisfying a declared conditioning bound is compact. If $\delta_\Gamma(X) = 0$ is not attained, then every near-minimizing sequence must eventually violate every such uniform conditioning bound.*

Proof. Fix a declared conditioning bound c, write Γ_c for the parameters satisfying it, and let

$$K_c = \{g \in \Gamma_c : d(X,g) \le 1\}.$$

By hypothesis K_c is compact. If K_c is empty, every near-minimizing sequence eventually has defect below 1 and therefore lies outside Γ_c. Otherwise continuity gives an attained minimum

$$a_c = \min_{g \in K_c} d(X,g).$$

Since the zero infimum is not attained anywhere in Γ, $a_c > 0$. Every near-minimizing sequence eventually has defect below $\min\{1, a_c\}$, and therefore cannot belong to Γ_c. Since c was arbitrary, each fixed conditioning bound is eventually violated; in particular no subsequence satisfying one fixed bound can be near-minimizing. □

24.2 Metric packets and intrinsic distortion

Let $T : (V, Q_s) \to (W, Q_t)$ be invertible between real or complex spaces of common finite dimension $n \ge 1$ with positive-definite metrics. Choose factors $Q_s = R_s^* R_s$ and $Q_t = R_t^* R_t$, and put

$$B_Q(T) := R_t T R_s^{-1}.$$

Define

$$D_Q(T) := \frac{\sigma_{\max}(B_Q(T))}{\sigma_{\min}(B_Q(T))} \ge 1, \qquad \Delta_Q(T) := \log D_Q(T).$$

Proposition 24.5 (Coordinate covariance and composition)**.** *Simultaneous coordinate transport of T, Q_s, Q_t leaves $D_Q(T)$ invariant. For composable invertible maps,*

$$D_Q(T_2 T_1) \le D_Q(T_2) D_Q(T_1),$$

so

$$\Delta_Q(T_2 T_1) \le \Delta_Q(T_2) + \Delta_Q(T_1).$$

For a closed word H,

$$D_Q(H) \ge \frac{\max_{\lambda \in \mathrm{spec}(H)} |\lambda|}{\min_{\lambda \in \mathrm{spec}(H)} |\lambda|}.$$

Proof. Coordinate transport changes $B_Q(T)$ by left and right unitary factors. Singular-value condition numbers are submultiplicative. The spectral lower bound follows because every eigenvalue modulus lies between the smallest and largest singular values. □

Theorem 24.6 (Conformal metric-completion criterion)**.** *Let a connected finite graph carry real or complex vector spaces of common dimension $n \geq 1$ and invertible edge transports T_e. Reverse traversal of an edge uses T_e^{-1}. At a chosen root, determinant-normalized closed holonomy words are*

$$\widehat{H} := |\det H|^{-1/n} H.$$

They form the determinant-normalized holonomy representation. Positive-definite vertex metrics and positive edge scales satisfying

$$T_e^* Q_{t(e)} T_e = \lambda_e^2 Q_{s(e)}$$

exist if and only if the determinant-normalized holonomy representation preserves a positive-definite metric at one root vertex. Equivalently, the normalized holonomy is unitarizable. In finite dimension, this is equivalent to relative compactness of its image.

Proof. A compatible family makes every closed root holonomy H conformal for the root metric Q_o. Taking determinants fixes its positive scale as $|\det H|^{1/n}$, so $\widehat{H}$ preserves Q_o.

Conversely, suppose every normalized root holonomy preserves Q_o. Choose spanning-tree maps G_v from the root to each vertex and put

$$Q_v = (G_v^{-1})^* Q_o G_v^{-1}.$$

For an edge $e : u \to v$, the root word $H_e = G_v^{-1} T_e G_u$ satisfies

$$H_e^* Q_o H_e = |\det H_e|^{2/n} Q_o.$$

Consequently

$$T_e^* Q_v T_e = |\det H_e|^{2/n} Q_u, \qquad \lambda_e = |\det H_e|^{1/n} > 0.$$

This gives the required compatible family directly. A group preserving a positive-definite metric is conjugate into a compact orthogonal or unitary group. Conversely, averaging an inner product over the compact closure of a relatively compact image gives an invariant positive-definite metric. □

24.3 Exact convex-versus-quadratic separation

Consider

$$A_1 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix}.$$

Theorem 24.7 (Sharp geometry-class separation)**.** *The norm $\|x\|_\infty$ satisfies*

$$\max\{\|A_1 x\|_\infty, \|A_2 x\|_\infty\} = \|x\|_\infty$$

for every $x \in \mathbb{R}^2$. In contrast, for every positive-definite quadratic metric Q,

$$\max\{\|A_1\|_Q, \|A_2\|_Q\} \geq \sqrt{2}.$$

Equality holds for scalar multiples of the Euclidean metric. Hence the exact optimal convex/Finsler bound is 1, while the exact optimal quadratic bound is $\sqrt{2}$.

Proof. The two images are

$$A_1(x_1, x_2) = (x_1, x_1), \qquad A_2(x_1, x_2) = (x_2, -x_2),$$

so the first identity is immediate.

Write

$$Q = \begin{pmatrix} a & c \\ c & b \end{pmatrix}, \qquad d = ab - c^2 > 0.$$

The squared induced norms are

$$u := \|A_1\|_Q^2 = \frac{b(a+b+2c)}{d}, \qquad v := \|A_2\|_Q^2 = \frac{a(a+b-2c)}{d}.$$

A direct calculation gives

$$ab\big((a+b)^2 - 4c^2\big) - 4(ab-c^2)^2 = ab(a-b)^2 + 4c^2(ab-c^2) \geq 0.$$

Thus $uv \geq 4$, and therefore

$$\max\{\sqrt{u}, \sqrt{v}\} \geq (uv)^{1/4} \geq \sqrt{2}.$$

Equality forces $a = b$ and $c = 0$, and then $u = v = 2$. □

This example is a sharp instance of the established gap between general extremal norms and common quadratic Lyapunov functions [24, 23]. Its role here is not a novelty claim. It proves that obstruction status must record the admissible geometry class.

24.4 Escape without positive gap

Proposition 24.8 (Unipotent escape)**.** *Let*

$$H = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

There is no positive-definite metric Q with $H^*QH = Q$. Nevertheless, for

$$Q_s = \mathrm{diag}(s^2, 1),$$

we have

$$D_{Q_s}(H) \longrightarrow 1 \qquad (s \downarrow 0),$$

while

$$\kappa_2(Q_s) = s^{-2} \longrightarrow \infty.$$

Thus the metric-completion defect has zero infimum but is not attained.

Proof. Writing $Q = \left(\begin{smallmatrix} a & c \\ c & b \end{smallmatrix}\right)$, the equation $H^*QH = Q$ forces $a = 0$, contradicting positive definiteness. For Q_s, the metric-normalized matrix is

$$\begin{pmatrix} s & 0 \\ 0 & 1 \end{pmatrix} H \begin{pmatrix} s^{-1} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix},$$

whose singular-value condition number tends to one. The conditioning formula is immediate. □

24.5 Forgetful descent and enrichment lifting

Suppose richer categories $\widetilde{\mathcal{C}}$ and $\widetilde{\mathcal{D}}$ carry forgetful functors

$$U : \widetilde{\mathcal{C}} \to \mathcal{C}, \qquad V : \widetilde{\mathcal{D}} \to \mathcal{D}.$$

Proposition 24.9 (Forgetful descent and enhancement lifting)**.** *A rich bridge*

$$\widetilde{B} : \widetilde{\mathcal{C}} \to \widetilde{\mathcal{D}}$$

automatically determines an underlying bridge

$$V\widetilde{B} : \widetilde{\mathcal{C}} \to \mathcal{D}.$$

Given only a bare bridge $B : \mathcal{C} \to \mathcal{D}$, constructing $\widetilde{B}$ together with a comparison

$$V\widetilde{B} \cong BU$$

is a lifting problem. Its existence, uniqueness, and residual automorphisms are additional passport fields and may carry typed obstruction objects.

Proof. The first statement is composition. The second is not an existence theorem but a precise reformulation of the enhancement problem: a lift is extra structure over the specified commutative square. Any failure must therefore be recorded relative to the chosen forgetful functors and comparison data. □

The pattern contains several earlier obstruction mechanisms:

Underlying structure	Requested enrichment	Typed gate or obstruction
Projective semilinear cocycle	Linear semilinear descent	Relative Brauer class
Invertible transport graph	Compatible metric packet	Unitarizability of normalized holonomy
Faithful additive realization	Exact tensor fiber functor	Strong monoidal and exactness obligations
Max-plus value	Finite-temperature sum semantics	Residual partition mass
Bare quotient coordinates	Metric-decorated quotient	Compatible quotient metric and conditioning boundary

No single obstruction object is asserted for all rows. The common statement concerns the shape of the lifting problem.

24.6 Two variances and the indexed architecture

Task changes and admissibility changes have opposite-looking effects. If a task is strengthened, its behavioral kernel shrinks and its quotient retains more information:

$$N_{\text{strong}} \subseteq N_{\text{weak}}, \qquad Q_{\text{strong}} \twoheadrightarrow Q_{\text{weak}}.$$

If an admissible realization class is enlarged, the optimized defect can only decrease:

$$\Gamma \subseteq \Gamma' \quad \Longrightarrow \quad \delta_{\Gamma'} \leq \delta_{\Gamma}.$$

These are different orders and should not be merged into one scalar notion of a stronger contract.

Definition 24.10 (Doctrine-indexed task fiber)**.** For a doctrine D and represented object X, let

$$\mathsf{Task}_D(X)$$

denote the ordered family of continuation-stable semantic quotients admitted by the declared query universe. Inside one fiber, kernel or congruence calculus governs task change. A change of doctrine

$$D \longrightarrow D'$$

requires a bridge passport specifying carrier transport, preserved constructions, comparison maps, and any lifting obstruction.

Theorem 24.11 (Relative stratified semantic synthesis)**.** *Within the finite certified setting of this manuscript, the following statements hold.*

1. *Every declared semantic slice has a task-relative behavioral quotient whenever the required effective quotient exists.*
2. *Inside a module-valued slice, joint tasks, common coarsenings, bridge loss, essential-image failure, and task weakening are governed by the exact sequences of Sections 22 and 23.*
3. *Carrier changes are legal exactly on the factorization loci certified by theorem 23.1; otherwise the joint semantic image is the canonical exact completion preserving both views.*
4. *Observer changes may introduce additional relation objects such as* $\mathrm{Rel}_K(E)$*; they are not inferred from carrier compatibility alone.*
5. *Enlarging an admissible realization class weakens optimized defect but need not produce an attained completion.*
6. *Crossing to richer structure is a typed lifting problem and does not follow from the underlying bridge without an enhancement certificate.*

Proof. The clauses are respectively theorems 22.4, 22.16, 23.1, 23.5, 23.13 and 24.2 and proposition 24.9. No clause identifies distinct carriers, observer categories, or admissibility classes without the corresponding comparison data. □

Boundary 24.12 (Scope of the second-order calculus). The indexed architecture does not establish a complete double category of all mathematical doctrines, nor does it prove that every task quotient or enrichment is effectively recognizable. The fiberwise quotient and exact-sequence results are proved in the stated algebraic settings. Cross-doctrine coherence remains conditional on explicit carrier, topology, metric, tensor, convergence, or higher-categorical passports.

Admissibility and lifting result. Obstructions are relative to both the requested enrichment and the admissible realization class. Enlarging that class can remove a positive gap, while zero defect may still fail to be attained. Rich structure descends under forgetting, but reconstructing it from a bare representation is a lifting problem with its own typed gate. The resulting architecture is a family of task-quotient fibers connected only by certified doctrine bridges.

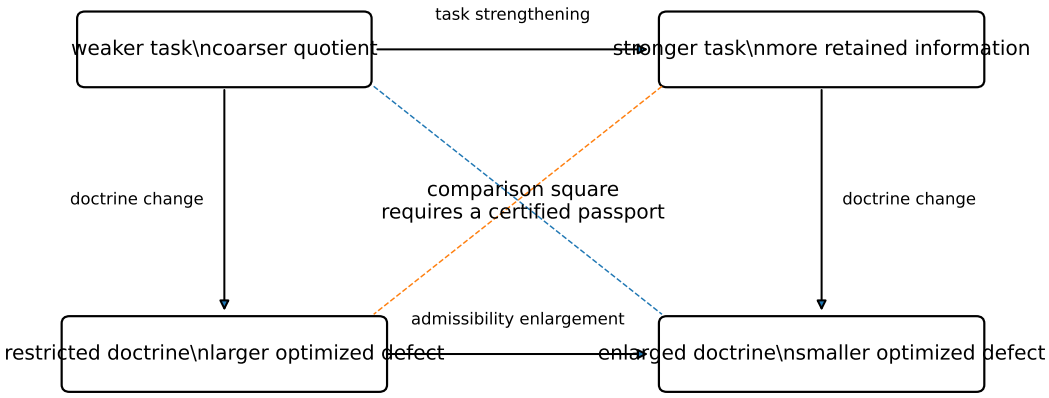


Figure 58: Two independent contract directions. Horizontal task strengthening retains more semantic information. Vertical admissibility enlargement allows more realizations and can weaken obstruction. A commuting square is a theorem-level comparison, not an automatic identification.

25 Scope, classification targets, and open boundaries

The global classification theorem is componentwise because the certified reversible core has five connected semantic components. The component tag is part of the invariant. Omitting it would compare objects for which no reversible semantic bridge has been proved.

The lane theorems have a different scope. The finite profiled Universal Lane Transport Theorem is exact inside a declared additive, idempotent-complete setting with complete split packet systems. Its realization, kernel, multilinear, fold, and descent clauses apply only under their additional hypotheses. Trivial address holonomy is required only for one fixed global address set. With a coherent finite address cocycle, complete matrices instead descend as sections of a matrix local system, and monodromy orbits give the finest ordinary global projector coarsening. The word “universal” refers to this formal lane calculus. It does not claim that every object has a finite lane presentation or that address count equals dimension, rank, entropy, information, or computational complexity.

The semantic quotient theorem is relative in another direction. It fixes a carrier doctrine, continuation grammar, observation category, and query family. The resulting quotient is universal only for that semantic slice. A different observation category can identify different linear combinations of residual states. A different carrier can change the legal aggregation law. A different continuation class can expose provenance that was previously invisible. There is therefore no observer-independent minimum hidden behind the notation.

The admissibility theory records a third boundary. A positive defect in a quadratic, integral, finite, or neutral class need not survive enlargement to a convex, rational, completed, or nonneutral class. Conversely, zero optimized defect does not imply that a completion exists. Attainment and conditioning remain part of the defect profile.

Table 43: Certified classification components and exact boundaries.

Component	Semantic domain	Complete descended coordinates	Boundary
Digital–regular–linear–Mahler	Fixed-prime regular sequences with rational-mask and periodic subloci	Strict shortlex state, Cartier/mask data where defined, periodic linear data where defined	No all-radix or unrestricted functional/Newton-Mahler equivalence
Continuous local	$C(\mathbb{Z}_p, \mathbb{Z}_p)$	Canonical Mahler coefficients and complete clopen branch packets	No universal local-analytic normal form or optimal mixed-radix tree
Marked Fuss	$U = z + \alpha U^q$	Marked parameters (q, α), algebraic branch, recurrence, tree support and orbit data	Complete only on the marked nonlinear–algebraic–holonomic family
Euler–multiplicative	Fixed Euler basis and prime-uniform multiplicative locus	Euler exponent, divisor ghost, common local factor, multiplicative function	No arbitrary product-grammar or additive–multiplicative equivalence
Finite Galois	One finite semisimple Galois extension	Subgroup, fixed field, Reynolds projector, quotient and Mackey data	No modular/wild or nonneutral tensor reconstruction

25.1 Directional satellites

Several useful constructions remain adjacent to the reversible core without becoming equivalences:

- reduction from integral section lattices to finite residue systems, with Smith–Tor ghosts and missing saturated coordinates;
- passage from a selected algebraic branch to a differential annihilator, which can forget the branch and its minimal polynomial;
- functional-Mahler masks and Newton–Mahler interpolation, which use different variables and require additional lifting and interpolation data;
- projective Galois descent, whose lift to an ordinary vector realization is governed by a relative Brauer class;
- faithful additive or Hilbert-space realizations that do not supply the exact tensor fiber functor required for Tannakian reconstruction;
- incomplete lane selections and noninjective folds, which retain explicit residuals or collision kernels;
- Boolean or max-valued summaries of signed additive computation, where cancellation obstructs carrier replacement;
- Set-valued future quotients and weighted-linear residual spaces, whose difference is measured by a relation module;
- geometric or provenance compactions that are valid only for the declared continuation class and downstream task.

These constructions remain in the directed transformation category with their exact loss, completion, or obstruction records.

25.2 Open mathematical obligations

The remaining foundational obligations are more sharply localized than in earlier editions.

1. Extend the certified atlases across radices, coefficient rings, semirings, and mixed local–global completions without losing normalization or effectivity.
2. Develop modular and wild analogues of finite semisimple Galois projector theory when Reynolds averaging fails.
3. Replace marked model-family results in the holonomic, algebraic, and nonlinear strata by broader native minimality and collision invariants.
4. Construct nonneutral and integral tensor-descent variants with their gerbe or torsor data retained explicitly.
5. Prove finite-data recognition theorems only on declared subgrammars where identifiability and certification are available.
6. Extend finite split and finite holonomy-twisted lane transport to infinite, completed, nonsplit, noninvertible, or genuinely nonadditive resolutions only after specifying the required convergence, exactness, and coherence doctrines.

7. Characterize which query universes are rich enough that every continuation-stable congruence is observationally definable, and determine when the definable task family is closed under joins.
8. Extend the module exact-sequence calculus to exact, regular, homological, enriched, and higher-categorical settings without silently importing additive arguments.
9. Develop effective algorithms for joint semantic completion, relation-module reduction, and task-relative bridge loss beyond finite matrices and finite automata.
10. Construct a fully coherent doctrine-indexed comparison object only after the required horizontal, vertical, and higher comparison cells have been verified.

No global promotion. Completion of the finite atlas, lane theorems, semantic quotient calculus, or class-relative defect theory does not promote absent bridges, incompatible carriers, noninjective folds, unresolved enrichments, or broad-stratum normal forms to equivalences. Every claimed transport remains relative to its grammar, profile, semantic slice, admissible realization class, and proof-status record. The detailed obligation and claim ledgers are retained in Appendices D and E.

26 Conclusion

The paper establishes a finite, typed, and auditable form of stratified universality. Its common object is not one formula and not one global coordinate system. It is a resolved syntax together with profile-preserving realizations, normalized local charts, typed bridges, task-relative quotients, and declared descent and enrichment doctrines.

The first proof architecture concerns representation. Packet and refinement data are literal in the additive–Karoubi envelope. Representation groupoids are separated from information-losing transformations. Bridge passports identify what is preserved, reflected, forgotten, or only partially defined. Holonomy, cokernel, Brauer, and higher-coherence defects record different failures of global trivialization. Effective descent turns compatible local presentations into global semantic objects. Compatible native invariants descend and become complete on the five declared normalized components.

The finite profiled Universal Lane Transport Theorem isolates the algebraic mechanism inside that architecture. Complete finite packet systems give exact analysis and synthesis. Every morphism becomes a unique lane-to-lane block matrix, composition becomes contraction over intermediate addresses, and certified coarsening aggregates fine blocks without changing the underlying morphism. Selectors split retained transport from named residuals. The holonomy-twisted extension shows that chart loops may permute lane labels without destroying the calculus. The matrices then form a local system, descended morphisms are its compatible sections, object-lane orbit sums give the finest ordinary global coarsening, and pair orbits carry exact path composition.

The second proof architecture concerns semantic sufficiency. For a fixed carrier, continuation grammar, observation category, and query family, the behavioral quotient is the coarsest representation preserving every future observation. Its kernel is the maximal stable invisible core. Task strengthening, joint semantics, common coarsening, relative residuals, bridge loss, essential-image failure, and task weakening become exact quotient constructions. This replaces an informal search for a universally smallest packet by a family of precise minima indexed by declared semantic tasks.

The third architecture concerns changes between those tasks and doctrines. Carrier replacement is a factorization theorem, not a vocabulary choice. Observer base change can introduce a relation module

even when the carrier is unchanged. Convex value, additive reconstruction, provenance, and weighted-linear state are therefore distinct semantic objects until a comparison theorem is supplied. Enlarging an admissible geometry can remove a positive obstruction, while zero infimum can still escape every well-conditioned realization. Rich structure descends under forgetting, but recovering it from a bare representation is a lifting problem.

The resulting theory is neither storage universality nor a claim that all mathematics is categorically equivalent. It is a calculus for asking, with explicit types, what survives a change of representation, what is lost, what obstructs reconstruction, what a declared task can safely forget, and what extra structure must be supplied to return. New strata, task fibers, or doctrines enter the theory only through normalizations, transport laws, effectivity proofs, comparison maps, exact loss records, and claim boundaries. Historical novelty and independent mathematical review remain separate from this internal formal synthesis.

A Theorem concordance and proof-status index

The machine-readable concordance in the source package preserves, for all 75 components, the selected strongest source, exact theorem coordinate, hypotheses, proof class, imported input, and remaining obligation. The publication PDF retains the compact index below. This avoids duplicating the full theorem prose while preserving a complete auditable record in `Komorebi_Theorem_Concordance_Final_2026-10-06.json` and `final_generated/concordance_entries.tex`.

Table 44: Compact index to the theorem concordance. The detailed hypotheses and obligations appear in Appendix A.

ID	Component	Status	Strongest source
FND-01	Typed linear syntax and module semantics	proved internally	`PSC_I_Symbolic_Core.tex`
FND-02	Finite free seed presentations	proved internally	`PSC_I_Symbolic_Core.tex`
FND-03	Seed rank and functorial inequalities	proved internally	`PSC_I_Symbolic_Core.tex`
FND-04	Closed seed-rank formulas	proved internally	`PSC_I_Symbolic_Core.tex`
FND-05	Exact packet coordinates and reconstruction	proved internally	`PSC_I_Symbolic_Core.tex`
FND-06	Block algebra and support calculus	proved internally	`PSC_I_Symbolic_Core.tex`
FND-07	Coordinate transport and transfer naturality	proved internally	`PSC_I_Symbolic_Core.tex`
FND-08	Subobject, quotient, and packet exactness	proved internally	`PSC_I_Symbolic_Core.tex`
FND-09	Preservation profiles and theorem transport	proved internally	`PSC_I_Symbolic_Core.tex`
FND-10	Finite resolved transfer and refinement calculus	proved internally	`Resolution_Theory_Final_Journal_Manuscript.tex`

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ID	Component	Status	Strongest source
FND-11	Universal corner extension and faithful realization	proved internally	Resolution_Theory_Final_Journal_Manuscript.tex
FND-12	Recognition and finite faithful dimension	proved internally	Resolution_Theory_Final_Journal_Manuscript.tex
FND-13	Karoubi, countable, W*, and affiliated completion	proved internally	Resolution_Theory_Final_Journal_Manuscript.tex
FND-14	Nonlinear packet operad and refinement coherence	proved internally	Resolution_Theory_Final_Journal_Manuscript.tex;Nonlinear_Transfer_Operad.texasspecializedsource
FND-15	Local analytic nonlinear inversion	proved internally	Resolution_Theory_Final_Journal_Manuscript.tex
FND-16	Universal Lane Transport Theorem, finite profiled form	proved internally	block15_section.tex,UniversalLaneTransportTheoremandcorollaries
FND-17	Holonomy-Twisted Universal Lane Transport and Orbit Descent	proved internally	block16_section_final.tex,holonomy-twistedtransport,orbitdescent,orbitalcomposition,andcyclotomicrealization
CAT-01	Representation groupoids separated from directional transformations	framework only	Abstraction_as_a_Resolved_Practice(1).tex;PSCPaperIcategoryofseededpacketsystems
GLU-01	Finite graph holonomy and global-frame criterion	proved internally	PSC_Paper_III_Strengthened_Annals_Edition.tex
GLU-02	Gauge classification of finite transports	proved internally	PSC_Paper_III_Strengthened_Annals_Edition.tex
GLU-03	Maximal path-flat core	proved internally	Affine_Mahler_Packets_and_Periodic_Test_Sets.tex (Core-TransferAddendum04)
GLU-04	Split analysis as a genuine two-chart atlas	proved internally	Affine_Mahler_Packets_and_Periodic_Test_Sets.tex (Addendum04)
GLU-05	Exact affine Mahler chart composition and trivial CRT holonomy	proved internally	Affine_Mahler_Packets_and_Periodic_Test_Sets.tex (Addendum04)
DSC-01	Descent commutes with complete resolution	proved conditionally	Projector_Calculus_Final_Current.tex
OBS-01	Canonical obstruction relative to declared correction data	proved internally	PSC_Paper_III_Strengthened_Annals_Edition.tex

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ID	Component	Status	Strongest source
OBS-02	Obstruction naturality and packet decomposition	proved internally	PSC_Paper_III_Strengthened_Annals_Edition.tex
OBS-03	Relative/cohomological obstruction	proved internally	PSC_Paper_III_Strengthened_Annals_Edition.tex
OBS-04	Nonlinear correction and complete filtered lifting	proved internally	PSC_Paper_III_Strengthened_Annals_Edition.tex
GAL-01	Finite Galois projector realization	classical input	Resolution_Theory_Galois_Expanded_Journal_Manuscript.tex
GAL-02	Effective linear semilinear descent	classical input	Resolution_Theory_Galois_Expanded_Journal_Manuscript.tex
GAL-03	Brauer obstruction to projective descent	classical input	Resolution_Theory_Galois_Expanded_Journal_Manuscript.tex
GAL-04	Tannakian reconstructed symmetry	classical input	Resolution_Theory_Galois_Expanded_Journal_Manuscript.tex
LIN-01	C-finite recurrence $\leftrightarrow$ rational OGF $\leftrightarrow$ finite state	classical input	Projector_Seed_Calculus_for_Integer_Sequences_Local_Polyphase_Extension.tex
LIN-02	Minimal linear dimension and similarity normalization	classical input	Projector_Seed_Calculus_for_Integer_Sequences_Local_Polyphase_Extension.tex
LIN-03	Complete persistent C-finite packet invariant	proved internally	Projector_Seed_Calculus_for_Integer_Sequences_Local_Polyphase_Extension.tex
HOL-01	P-recursive $\leftrightarrow$ D-finite coefficient translation	classical input	Projector_Seed_Calculus_for_Integer_Sequences_Local_Polyphase_Extension.tex
DIG-01	Finite digit realization $\leftrightarrow$ finite k-kernel	proved internally	Projector_Seed_Calculus_for_Integer_Sequences_Local_Polyphase_Extension.tex
DIG-02	Fixed-radix automaton normal form	classical input	Projector_Seed_Calculus_for_Integer_Sequences_Local_Polyphase_Extension.tex

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ID	Component	Status	Strongest source
ALG-01	Selected algebraic branch → packet polynomial system	proved internally	`Projector_Seed_Calculus_for_Integer_Sequences_Local_Polyphase_Extension.tex`
TRN-01	Invertible transform ↔ transported packet resolution	proved internally	`Projector_Seed_Calculus_for_Integer_Sequences_Local_Polyphase_Extension.tex`
PRD-01	Euler-product exponents ↔ logarithmic-derivative recurrence	proved internally	`Projector_Seed_Calculus_for_Integer_Sequences_Local_Polyphase_Extension.tex`
MUL-01	Prime-power seeds reconstruct multiplicative functions	proved internally	`Projector_Seed_Calculus_for_Integer_Sequences_Local_Polyphase_Extension.tex`
LOC-01	Finite Newton–Mahler reconstruction and local transfer	proved internally	`Projector_Seed_Calculus_for_Integer_Sequences_Local_Polyphase_Extension.tex`
LOC-02	Exact Mahler ↔ affine polyphase coordinate transport	proved internally	`Projector_Seed_Calculus_for_Integer_Sequences_Local_Polyphase_Extension.tex;Addendum04strengthensit`
LOC-03	Sharp minimal integral packet and universal decoder	proved internally	`Affine_Mahler_Packets_and_Periodic_Test_Sets.tex (Addendum04)`
FMH-01	Rational functional-Mahler mask → Cartier representation	proved internally	`Komorebi_Custom_GF_Cartier_Padic_Research_Report_0.3.pdf`
FMH-02	Functional-Mahler finite state → automaton	proved internally	`Komorebi_Custom_GF_Cartier_Padic_Research_Report_0.3.pdf`
FMH-03	Finite-mask Cartier transitions ↔ carry matrices	proved internally	`Komorebi_Custom_GF_Cartier_Padic_Research_Report_0.3.pdf`
ASM-01	One explicit profile-indexed resolved envelope	conjectural	`ControllinghandoffÂğ4GapA;localingredientsinPaperI,ResolutionTheory,andnonlinearoperadsources`

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ID	Component	Status	Strongest source
ASM-02	Authoritative declared stratum atlas	framework only	ControllinghandoffÂğ4GapB
ASM-03	Bridge semantic, packet, refinement, and residual naturality	conjectural	ControllinghandoffÂğ4GapC;localnaturalityinputsFND-07/FND-09/OBS-02
ASM-04	Triple-overlap and quadruple coherence	conjectural	ControllinghandoffÂğ2.6andÂğ4GapC
ASM-05	Effective descent for the heterogeneous atlas	conjectural	ControllinghandoffÂğ4GapD
ASM-06	Native normalization and complete invariants beyond existing slices	conjectural	ControllinghandoffÂğ4GapE
ASM-07	Coherent invariant descent and conditional global completeness	conjectural	ControllinghandoffÂğ4GapF
ASM-08	Theorem concordance and edition control	proved internally	ThisWorkPackage0;Master_Theorem_Migration_Ledger.csv;latermigrationmaps
ASM-09	Finite Profiled Resolved Atlas Assembly Theorem	proved conditionally	ThisWorkPackage0,Âğ6andcompanionLaTeXtheoremfile
SEM-01	Universal behavioral coimage and task-relative minimum	proved internally	block17_semantic_quotients_final.tex
SEM-02	Maximal stable invisible core	proved internally	block17_semantic_quotients_final.tex
SEM-03	Greatest-fixed-point construction and finite certification	proved internally	block17_semantic_quotients_final.tex
SEM-04	Task–congruence Galois connection	proved internally	block17_semantic_quotients_final.tex
SEM-05	Semantic quotient classification by congruences	classical input	block17_semantic_quotients_final.tex
SEM-06	Semantic diamond and exact compatibility sequence	proved internally	block17_semantic_quotients_final.tex
SEM-07	Task-change exact sequence and extension-valued loss	proved internally	block17_semantic_quotients_final.tex
BRG-01	Task-compatible bridge and five-term exact loss sequence	proved internally	block18_carrier_observer_change_final.tex
BRG-02	Extension law for composite semantic loss	proved internally	block18_carrier_observer_change_final.tex
CAR-01	Carrier factorization gate and joint semantic completion	classical input	block18_carrier_observer_change_final.tex

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ID	Component	Status	Strongest source
CAR-02	Carrier-indexed observer packet theorem	proved internally	`block18_carrier_observer_change_final.tex`
CAR-03	Cancellation and idempotence barriers	classical input	`block18_carrier_observer_change_final.tex`
OBC-01	Observer base-change relation module	proved internally	`block18_carrier_observer_change_final.tex`
OBC-02	Polyphase infinite-discrete versus finite-linear separation	proved internally	`block18_carrier_observer_change_final.tex`
CVX-01	Observer-conditioned convex packet transport and provenance boundary	proved internally	`block18_carrier_observer_change_final.tex`
ADM-01	Admissibility monotonicity and completion/gap/escape stratification	proved internally	`block19_admissibility_lifting_final.tex`
ADM-02	Sharp convex-versus-quadratic packet separation	classical input	`block19_admissibility_lifting_final.tex`
DOC-01	Forgetful descent, enhancement lifting, and doctrine-indexed task fibers	framework only	`block19_admissibility_lifting_final.tex`

B Edition-control ledger

Table 45: Edition-control ledger. Full hashes are retained so later strengthened editions can supersede specific rows without losing provenance.

Source family	Selected file	SHA-256	Decision
Paper I canonical symbolic core	`PSC_I_Symbolic_Core.tex`	`39c869f3bd16cbef102fe68dcad51125ab4511077cc45de6d91562c0748fa95c`	Primary theorem-ID and proof-status spine. Use the Expanded Clarified Edition for exposition, but preserve these stable labels/RecordIDs.
Paper I expanded clarified edition	`PSC_Paper_I_Expanded_Clarified_Edition.tex`	`e91239446e2e149bbf571c5bbc7e71a76a6f9a351d68963963cfe87f5fbf3c0f`	Co-primary expository edition, not a replacement for the canonical symbolic RecordID spine.
Resolution Theory final journal manuscript	`Resolution_Theory_Final_Journal_Manuscript.tex`	`3c0f37a5132c04d0715b0b85a88e34020ffdfc299bf6dd79b7fb4c4bdb32ecec`	Primary integrated source for finite resolved systems, corner representation/recognition, countable completion, and nonlinear operadic structure.
Paper III strengthened Annals edition	`PSC_Paper_III_Strengthened_Annals_Edition.tex`	`20452d11443835ca9c0e0484250c01fa51e0456574c7f7c7f5549fbd0f934c89`	Supersedes older Paper III status language on the relative universal obstruction: coker(D), naturality, quotient projectors, relative exact sequence, and mapping cones are theorem-level. The choice-free cross-problem assignment of D remains open.
Core-Transfer Addendum 04	`Affine_Mahler_Packets_and_Periodic_Test_Sets.tex`	`d864bebe02eac16037f17dc677dee9078fbc02a093c7b491b31d2b0208bb91e1`	Strongest source for maximal directed cores and the exact finite Mahler/polyphase atlas, composite rank law, sharp channel minimum, and stabilization.
Resolution Theory Galois expanded manuscript	`Resolution_Theory_Galois_Expanded_Journal_Manuscript.tex`	`f3c89f23db5f6e1ee8da7c396c0b30d8274154b8d1fca25d18415ec5dfe54480`	Primary source for Galois projector strata and classical descent/reconstruction inputs.

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Source family	Selected file	SHA-256	Decision
Sequence local/polyphase extension	`Projector_Seed_Calculus_for_Integer_Sequences_Local_Polyphase_Extension.tex`	`e310e5fbec56954e4490615acdeecb04122bad9bd9243e79d84a08174039038a`	Primary source for the broad sequence grammar, pairwise bridges, persistent C-finite complete packet invariant, and local/polyphase exact transfers.
Functional-Mahler/Cartier report 0.3	`Komorebi_Custom_GF_Cartier_Padic_Research_Report_0.3.pdf`	`4480786c49abcb60fb39e07acf0e250c9a3af68c38ac7b4801f34bdaefd5a936`	Primary source for the rational functional-Mahler → Cartier → automaton direction and finite-mask Cartier/carry identification.
Nonlinear transfer operad specialized manuscript	`Nonlinear_Transfer_Operad.tex`	`051970594d8b34c2ec5af67e20e063963b7c8684b75a439f62ed2c5f0e70d3a8`	Specialized secondary source. Resolution Theory Final is the integrated primary source for overlapping operadic results.
Abstraction as a Resolved Practice	`Abstraction_as_a_Resolved_Practice(1).tex`	`78bad0bd370fec069f61b5b616dbc5898277229429213a227772a2df4f96861d`	Conceptual source for representation groupoids, broader directed categories, and atlas language; not the theorem source for heterogeneous effective descent.
Projector Calculus final current	`Projector_Calculus_Final_Current.tex`	`cc313296416b08ccbe73d893e1744ae840b6419fc5b6676805ef722ab0401b39`	Selected exact source for descent–limit interchange and cofinal resolution equivalence.
Master migration ledger	`Master_Theorem_Migration_Ledger.csv`	`dceabf345ac09e507a754a928ca716ed2f797ca9aff8d777e104b49ab73dbdd1`	Historical dependency/migration evidence only; it does not override a strengthened theorem edition.
Paper III migration/expansion map	`MIGRATION_AND_EXPANSION_MAP(1).md`	`5f42dc03e5989b8883ce934752c3885ce58ae923b900a02943c86c93e4ca2242`	Authoritative supersession note for the strengthened Paper III rewrite.
Remaining open questions after two resolutions	`Remaining_Open_Questions_and_Conjectures_After_Two_Resolutions.pdf`	`13e41d43b9bbeaf0d4e5ed6dd223f6adfaf46c63ebc51987a118d9ec40843c90`	Historical open-question snapshot. Every "open" entry must be checked against later strengthened editions before reuse.
Universal Lane Transport Theorem, finite profiled form	`block15_section.tex`	`c1ae8aa2472eda9df467bbbf20b3c861f1261333364af06ffaf1cd455b5ab026`	Selected controlling source for FND-16 and the finite lane-transport theorem family.

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Source family	Selected file	SHA-256	Decision
Holonomy-Twisted Universal Lane Transport and Orbit Descent	`block16_section_final.tex`	`2b086cc6d806c3cf25b443a6ecefa871adba282d913deffaafabcadad2c0258a`	Selected controlling source for FND-17 and the finite nontrivial-address-holonomy extension. Final publication copy includes explicit 6 October 2026 local amendments; inherited source retained.
Part IV semantic quotient calculus	`block17_semantic_quotients_final.tex`	`ef408f87352057761b6bf55742f848612f585632691f71488d510393749b0735`	Primary source for behavioral cores, task congruences, semantic diamonds, residuals, and task-change exactness. Final publication copy includes the explicit 6 October 2026 coinduction proof clarification; inherited source retained.
Part IV carrier and observer calculus	`block18_carrier_observer_change_final.tex`	`6babbf13d884e5bdf86174ce89729ccf69feb694cf84e03c0f52c32038775157`	Primary source for carrier factorization, joint completion, bridge loss, observer base change, and provenance. Final publication copy includes explicit 6 October 2026 local amendments; inherited source retained.
Part IV admissibility and lifting calculus	`block19_admissibility_lifting_final.tex`	`038f352d1e9a82984963af0e90a14b104b8f17f7b2e8e4df2c8da627bc089799`	Primary source for optimized defect, completion/gap/escape, geometry-class separation, and enrichment lifting. Final publication copy includes explicit 6 October 2026 local amendments; inherited source retained.
Part IV exact symbolic validation	`results/block17_semantic_extension_validation.json`	`218123b49533a702126e8eea6f2e5f5172d4e9802902c6acb2460d39c0d334d1`	Finite exact implementation audit retained separately from universal proofs.

C Publication audit and development provenance

The consolidated edition retains a compact theorem-suite map and a development-provenance ledger so that editorial reordering does not erase proof dependencies or claim boundaries. The tables in this appendix are inherited records: their page counts and validation results refer to the original artifacts and the dated version 2.0.0 mathematical base. The final-edition build and publication checks are recorded separately in the release validation report.

Table 46: Publication-source audit for the Finite lane-transport theorem consolidated edition.

Audit area	Record	Status
Source graph	55 TeX files	PASS
Abstract length	316 words	PASS
Labels and references	519 labels; 131 references	PASS
Citation keys	14 cited keys	PASS
Graphics	57 included assets	PASS
Lane verifier	197,099 exact assertions	PASS
Publication structure audit	77 assertions	PASS

Table 47: Publication-source and release-integrity audit for the Block 16 holonomy-twisted extension.

Audit area	Record	Status
Source graph	60 TeX files	PASS
Abstract	346 words	PASS
Labels and references	542 labels; 138 references	PASS
Citation keys	14 cited keys	PASS
Graphics	57 included assets	PASS
Block 16 verifier	1,344,823 exact assertions	PASS
Concordance	57 synchronized entries through FND-17	PASS
Compiled PDF	239 pages	PASS
Publication audit	85 checks	PASS

Table 48: Final formal-foundations publication audit.

Audit surface	Recorded result
Source graph	65 recursively included TeX files; 620 labels; 153 source references
Concordance	75 components with synchronized status and domain counts
Inherited validation	Block 15 PASS; Block 16 PASS with 1,344,823 assertions
Part IV validation	3,022 exact symbolic assertions; zero recorded failures
Compiled artifact	247 pages; metadata, theorem-layer text, references, citations, and graphics checked
Audit result	77 checks; 0 failures
Boundary	Internal publication integrity only; not independent peer review or historical-priority validation.

Table 49: Development provenance retained after journal-facing reorganization.

Stage	Mathematical contribution
Foundation	Typed resolved syntax, packet coordinates, preservation profiles, additive-Karoubi envelope
Atlas typing	Normalized strata, representation groupoids, bridge passports, information loss, obstruction types
Finite effectivity	Periodic and rational-mask heterogeneous atlases
Digital normalization	Fixed-radix regular, automatic finite-orbit, and integral prime-power layers
Product/local normalization	Euler–multiplicative and Newton-Mahler–polyphase atlases
Holonomic, algebraic, nonlinear	Marked model-family minimality, collision, and tree signatures
Symmetry	Finite Galois, Brauer, profinite, and neutral Tannakian slices
Global synthesis	Authoritative bridge matrix, higher coherence, invariant descent, tagged completeness
Lane transport	Complete finite-resolution analysis, synthesis, block transport, refinement, selection, residual accounting, and typed specializations
Holonomy transport	Matrix local systems, exact twisted synthesis, base-fiber invariants, monodromy-orbit coarsening, and orbital composition
Publication consolidation	Theorem spine, proof-order reorganization, scope compression, and source audit

D Detailed open-obligation ledger

D.1 Correct dependency order

The remaining heterogeneous work has a strict topological order:

$$\boxed{\begin{array}{c}\text{profiled envelope} \longrightarrow \text{declared atlas} \longrightarrow \text{typed bridge laws} \\ \longrightarrow \text{higher coherence} \longrightarrow \text{effective descent} \longrightarrow \text{compatible invariants}\end{array}}$$

The first three arrows cannot be skipped. Pairwise formulas do not themselves define a pseudofunctor. Local completeness does not itself construct compatible global isomorphisms.

D.2 What this manuscript closes

This paper closes the following finite parts of the program:

- the typed finite resolved signature, exact finite packet/refinement/residual calculus, preservation-profile gate, and additive split-idempotent resolved envelope;
- the formal finite atlas assembly theorem;
- explicit profiled stratum passports, normalized subcategories, representation groupoids, and the typed directed transformation category;
- bridge passports, bridge classes, information-loss profiles, overlap categories, bridge-law certificates, and a concrete higher-coherence defect;
- the split-bridge reversible-core theorem, the maximal holonomy-fixed path-flat core, and the exact obstruction-rank formula;
- the relative cokernel, coarsening exact sequence, mapping-cone extension, Hilbert-90 effectivity, Brauer obstruction, and Tannakian boundary in one typed comparison;
- a precise isomorphism-effectivity condition for global invariant completeness;
- two exact finite affine-Mahler atlas models;
- the normalized C-finite recurrence-rational-state atlas with effective object and morphism descent;
- cross-presentation descent of the exact C-finite packet-complexity profile;
- a four-chart atlas spanning linear, digital, and functional-Mahler strata on the unit purely periodic domain;
- complete bridge passports for that atlas, including the directional Cartier compiler lane and its exact loss profile;
- all four triple-overlap laws, the quadruple coherence equation, effective object and morphism descent, and a globally complete compatible invariant package;
- the Frobenius-radical characterization of rational canonical masks and the exact divisor-residue obstruction to rational collapse;
- the canonical finite Cartier module, its reachable-observable minimality theorem, and the exact identification of reachable compiler states with the p-kernel;

- effective descent across reduced mask, minimal Cartier-module, and minimal DFAO charts on the full unit rational-mask domain;
- a globally complete rational-mask invariant package and a strict negative control showing that the rational-mask atlas is a proper subatlas of the automatic stratum;
- the canonical section-module, strict shortlex, minimal digit-linear, and word-Hankel normalization theorem for every fixed-radix field-valued regular sequence;
- the automatic finite-orbit theorem over finite fields, including exact state-to-rank bounds and a strict compression corpus;
- the canonical raw and saturated integral section lattices for every integer-valued sequence of finite rational section rank;
- the digit-equivariant Smith–Tor reduction sequence, exact Smith and Fitting formulas, and good-prime criterion;
- finite minimal prime-power semantic section automata with explicit ambient module bounds and compatible reduction maps;
- the p-adic and profinite completion sequences, disappearance of finite ghost coordinates, persistence of the missing saturation quotient, and the separate integral and actual-section effectivity obstructions;
- the integral fixed-Euler-basis classification, divisor-ghost integrality locus, exact logarithmic-derivative recurrence, and finite-prefix triangular normalization;
- prime-power effective descent for integral multiplicative arithmetic functions, complete local Euler and ghost coordinates, valuation logarithmic derivatives, and multiplicative residue packets;
- the restricted divisor–ghost product–multiplicative atlas with effective object and morphism descent and a complete prime-power invariant;
- the continuous Newton–Mahler coefficient theorem as a normalized local chart, complete clopen branch analysis and synthesis, and strict nested branch coherence;
- exact affine Mahler coefficient transport, mixed-radix kernel composition, Chinese-remainder diamonds, and the complete branch-coefficient packet equivalence;
- the finite-degree bridge ladder, sharp integral and fixed-modulus packet sizes, explicit integer decoders, and prime-power rank-loss formulas;
- branchwise valuation ledgers, Frobenius sparsification, matrix Carmichael–Mahler state constructors, and principal-unit precision amplification;
- canonical monic minimal differential and tail-recurrence Ore charts, together with the finite transient ledger needed for exact reconstruction;
- the general arithmetic-multisection defect decomposition into orbit aliasing and address cancellation, with singularity-orbit and local-exponent passports;
- the exact hypergeometric packet equation, minimal differential-order phase diagram, orbit rank, and complete marked invariant for the binomial–Fuchsian model family;
- finite selected-branch algebraic normalization, exact residue-packet polynomial systems, the generic d^N packet-degree theorem, and explicit tuple/stabilizer collision signatures;

- the marked polynomial fixed-point atlas across complete packet jets, colored operation trees, and exact packet equations;
- exact repeated-direction packet Faà di Bruno compression by colored integer-partition profiles;
- exact SCC elimination to the address-preserving cyclic nonlinear core;
- the finite arity-gcd tree signature controlling arithmetic packet support and the complete dominant-singularity orbit, together with the full q-ary phase theorem;
- the normalized finite subgroup–fixed-field–Reynolds atlas, exact normality and quotient-compression criterion, conjugate nonnormal atlas envelopes, and Mackey double-coset transition packets;
- nested ramification projectors and the weighted unresolved-rank formulas for Artin and Swan conductors;
- effective finite Galois vector descent, the framed Hilbert-90 chart, the exact relative Brauer obstruction, profinite finite-level reconstruction, and the neutral Tannakian tensor-fiber gate;
- an authoritative bridge matrix with named restricted equivalence loci, directional-loss entries, partial constructors, and explicit absent bridges;
- normalizer-induced triple and quadruple coherence for every reversible cycle in the certified finite atlas;
- coherent invariant comparison maps and effective invariant descent on five connected components;
- a complete tagged global invariant on the digital–regular–functional-Mahler, continuous Newton–Mahler, marked Fuss, Euler–multiplicative, and finite Galois semantic union;
- the weighted Fuss–Catalan nonlinear–algebraic–holonomic equivalence and its complete finite parameter invariant.
- the finite profiled Universal Lane Transport Theorem: exact analysis and synthesis, unique transfer blocks, compositional contraction, compatible coarsening, selectors and residuals, profiled realizations, conditional kernel and fold-loss clauses, factorial residue realization, and overlap-compatible descent under an explicit address cocycle.
- the finite holonomy-twisted lane extension: matrix-local-system analysis and synthesis, twisted path contraction, equivariant resolution change, holonomy-compatible selectors and residuals, the finest monodromy-orbit projector coarsening, pair-orbit transport atoms, orbital structure constants, and rational cyclic Fourier-orbit descent.

D.3 What remains

Table 50: General extension obligations after the formal foundations v2.0 milestone.

ID	Obligation	Current status	Remaining completion criterion
PO-01	Extend the profile-indexed resolved envelope	proved conditionally	The additive-Karoubi core is proved. Add nonlinear, completion, monoidal, or integral doctrines only for newly selected strata and prove the corresponding profile-indexed universal property.
PO-02	Freeze explicit finite atlases	proved internally	Closed for the unit-periodic, rational-mask, fixed-radix regular, product-multiplicative, continuous local Newton–Mahler, marked binomial–Fuchsian, selected algebraic, marked nonlinear-tree, and finite semisimple Galois atlases, and assembled into the five-component certified Block 13 atlas. Every further enlargement must again declare its semantic domain, chart categories, overlap categories, and component tag.
PO-03	Define morphisms and normalized subcategories	proved internally	Closed for fixed-radix regular sequences, the canonical integral raw/saturated lattice record, fixed-basis Euler products, integral multiplicative functions, and continuous Newton–Mahler functions with complete branch packets. The marked binomial–Fuchsian holonomic chart, selected one-variable algebraic branch chart, marked polynomial fixed-point chart, finite semisimple subgroup chart, framed semilinear vector chart, and neutral Tannakian chart are closed; unrestricted holonomic, multivariate algebraic, unmarked nonlinear, semiring, cross-radix, unrestricted-product, modular, wild, and nonneutral enlargements remain separate passport obligations.
PO-04	Type each bridge	proved internally	Closed for the authoritative Block 13 matrix on the certified domains, including equivalences on named overlaps, fully faithful inclusions, directional losses, constructors, partial bridges, and explicit absent cells. Every extension must retain the same bridge passport and information-loss discipline.

Continued on next page

ID	Obligation	Current status	Remaining completion criterion
PO-05	Prove bridge naturality	proved internally	Closed for digit sections, behavior quotients, Hankel factorization, Smith reduction, Euler logarithmic derivatives, local Dirichlet products, divisor-ghost inversion, Mahler analysis and synthesis, affine polyphase transport, mixed-radix composition, valuation certificates, coefficient/Ore translation, hypergeometric packet synthesis, the marked alias/cancellation ledger, packet-polynomial residue reduction, Weil-restriction degree transport, the algebraic collision ledger, diagonal packet Faà di Bruno profile compression, SCC elimination, marked tree-address transport, subgroup inclusion, Reynolds averaging, normal quotient compression, Hilbert-90 gauge trivialization, and projective Brauer boundary transport on the declared overlaps.
PO-06	Prove higher coherence	proved internally	Closed on every reversible cycle of the five-component certified atlas by the Block 13 normalizer-induced coherence theorem. Unframed vector descent retains intrinsic GL automorphisms, and broader atlases may have nontrivial comparison automorphisms or nonvanishing higher defects.
PO-07	Prove object effectivity	proved internally	Closed for the verified digital, regular, integral, fixed-Euler-basis, multiplicative, divisor-ghost, and continuous Newton–Mahler domains, the marked binomial–Fuchsian domain, the selected algebraic branch domain, the marked nonlinear-system domain, the finite subgroup/projector domain, and framed finite Galois vector descent by explicit normalization from the represented semantic object and retained generator or symmetry data.

Continued on next page

ID	Obligation	Current status	Remaining completion criterion
PO-08	Prove morphism effectivity	proved internally	Closed for the verified atlases because their normalized fibers are contractible or discrete; the complete clopen local packet also glues equality morphisms by explicit piecewise synthesis; the marked holonomic, selected algebraic, and marked nonlinear fibers are discrete; finite Galois vector morphisms descend by fixed points and scalar extension. Unframed symmetry groupoids retain intrinsic automorphisms and require genuine groupoid descent.
PO-09	Construct native comparison maps	proved internally	Closed on the five certified components. Block 13 packages the digital-regular-mask-periodic, continuous Newton–Mahler, marked Fuss, Euler–multiplicative, and finite Galois native coordinates into coherent comparison families. Comparisons for unrestricted broad strata remain separate obligations.
PO-10	Control isomorphism descent	proved internally	Closed for the five-component certified atlas: normalized fibers are discrete or contractible, comparison cells are coherent, and compatible local isomorphisms descend. In broader atlases identify or kill holonomy, cokernel, Brauer, nonneutral tensor, or higher-coherence obstructions.
PO-11	Broaden automatic and regular invariants	proved conditionally	Fixed-radix field normalization, canonical integral saturation, Smith–Fitting reduction profiles, and compatible prime-power semantic towers are closed under the Block 06 hypotheses. Cross-radix comparison, arbitrary coefficient rings and semirings, and effective recognition from finite data remain open.
PO-12	Normalize product and multiplicative strata	proved internally	Closed for the fixed integral Euler basis, integral multiplicative functions, and the multiplicative divisor-ghost overlap. Minimal compilers, unrestricted product grammars, monoidal closure of the overlap, and any global Cauchy-Dirichlet equivalence remain open.

Continued on next page

ID	Obligation	Current status	Remaining completion criterion
PO-13	Broaden local Mahler theory	proved internally	Closed for continuous $\mathbb{Z}_p$-valued functions, canonical Newton–Mahler coefficients, complete clopen and coefficient packets, finite-degree affine channels, sharp standard packet sizes, and principal-unit precision ladders. Exact higher kernel valuations, optimal mixed-radix trees, nonlinear local systems, state recognition, and a combined spectral–local signature remain open.
PO-14	Broaden holonomic packet minimality	proved conditionally	Canonical minimal Ore charts and the marked binomial–Fuchsian packet-order theorem are closed. General minimal multisection operators, canonical desingularization, monodromy-extension data, differential-module cyclicity, and a finite signature beyond the marked family remain open.
PO-15	Broaden algebraic packet minimality	proved internally	Finite selected-branch normalization, exact packet polynomial systems, the generic d^N Weil-restriction bound, and marked collision signatures are closed. Minimal multivariate system dimension, optimal symmetry-reduced degree bounds, ramified Puiseux passports, and universal branch-recognition algorithms remain open.
PO-16	Broaden nonlinear and tree signatures	proved internally	Closed for marked polynomial fixed points, complete packet jets, repeated-direction Faà di Bruno profile compression, address-preserving SCC reduction, and positive simply generated tree support and dominant-orbit signatures. Arbitrary mixed directions, coordinate-free minimum nonlinear system dimension, sign-canceling tree weights, unmarked generators, noncontractive global dynamics, and a universal operadic normal form remain open.
PO-17	Extend modular, wild, and nonneutral symmetry	proved conditionally	Finite semisimple Reynolds charts, framed Hilbert-90 descent, the projective Brauer obstruction, profinite finite-level reconstruction, and neutral Tannakian symmetry are closed. Derived fixed points, integral wild ramification, group-scheme torsion, nonneutral fiber functors, and gerbe-valued descent remain open.
PO-18	Maintain theorem and edition provenance	proved conditionally	Ongoing. Every new claim must cite the selected strongest edition, exact theorem, hypotheses, imported input, and superseded predecessor.

Continued on next page

ID	Obligation	Current status	Remaining completion criterion
PO-19	Separate internal proof from external validation	proved conditionally	Ongoing. Keep direct proof, exact computation, imported classical theorem, conjecture, and independent peer review in distinct status fields.
PO-20	Extend the global invariant beyond the certified spine	proved conditionally	Work Package 6 is closed on the five-component finite atlas. Remaining targets are cross-radix comparison, semiring and all-ring normalization, unrestricted holonomic and multivariate algebraic invariants, unmarked nonlinear systems, modular and wild symmetry, nonneutral tensor descent, and any genuine bridges joining the current component tags.
PO-21	Extend lane transport beyond the finite profiled theorem	proved conditionally	The complete finite split-idempotent calculus and its finite nontrivial-address-holonomy extension are closed under their printed hypotheses. Complete matrices descend as matrix-local-system sections, monodromy orbits give the finest ordinary projector coarsening, and pair orbits govern compatible selectors. Infinite resolutions, non-split decompositions, noninvertible transitions, general nonlinear doctrines, higher-stack effectivity, historical novelty, and unrestricted atlas globalization remain separate obligations.
PO-22	Recognize task quotients effectively	proved conditionally	The fiberwise behavioral quotient and finite-dimensional stabilization theorems are proved. Effective recognition for infinite systems, symbolic query languages, enriched observations, and undecidable continuation grammars remains open.
PO-23	Extend exact semantic loss beyond modules	proved conditionally	The semantic diamond, bridge-loss sequence, and extension law are proved in module categories. Exact, regular, homological, semiring, topological, and higher-categorical analogues require their own quotient and exactness doctrines.
PO-24	Classify carrier and observer comparison maps	proved conditionally	Kernel factorization and the joint semantic completion are closed for algebraic carriers, and the relation module is closed over fields. Classification of lawful changes among noncommutative, measure-valued, completed, probabilistic, and nonlinear carriers remains open.

Continued on next page

ID	Obligation	Current status	Remaining completion criterion
PO-25	Control admissibility and attainment	proved conditionally	Admissibility monotonicity and the completion/gap/escape trichotomy are proved. Effective optimal-geometry construction, attainment criteria, and conditioning rates remain doctrine-specific.
PO-26	Build doctrine-level coherence only from certified cells	framework only	The doctrine-indexed task-fiber architecture is stated and several concrete lifting gates are proved. A complete double category, bicategory, or higher stack of doctrines is not yet constructed and must not be inferred from the fiberwise theorems.

D.4 Finite completion criterion

A rigorous first heterogeneous stratified-universality theorem should be declared complete only after the following finite milestone is met:

1. a profile-indexed envelope is constructed for every profile used by the selected atlas;
2. at least two genuinely different normalized strata are fully specified as categories;
3. every overlap bridge is typed and its information loss is stated;
4. every nonempty triple-overlap coherence equation is proved;
5. descent for objects and morphisms is effective;
6. one nontrivial native invariant family has coherent comparison maps and a proved global completeness theorem on the normalized descent category;
7. the theorem concordance and source-edition ledger are synchronized with the final manuscript.

The C-finite theorem completes a three-chart presentation atlas within one stratum. Theorems 9.10 and 9.12 meet all seven items for a restricted atlas spanning three strata. Theorems 10.11 and 10.12 enlarge the digital and functional-Mahler overlap from unit pure periodicity to every unit series with rational canonical mask. Theorems 11.6, 11.10 and 11.11 close fixed-base field-valued regular normalization. Theorems 12.3, 12.7 and 12.25 close the canonical integral saturation and prime-power reduction package for integer-valued sequences already known to have finite rational section rank. Theorems 13.4, 13.15, 13.23 and 13.24 now close fixed-basis product normalization, multiplicative prime-power normalization, and one explicit cross-stratum invariant overlap. Theorems 14.1, 14.3, 14.7, 14.9 and 14.13 now close the continuous Newton–Mahler chart, complete local polyphase descent, finite-degree collective recovery, and the principal-unit precision ladder. Theorems 15.8, 15.9, 15.12, 15.15 and 15.20 now close canonical Ore normalization, the general two-stage multisection defect ledger, and one exact marked regular-singular model case. Theorems 16.4 to 16.6 closes the selected one-variable algebraic chart. Theorems 17.4, 17.5, 17.9, 17.11, 17.13 and 17.15 now close the marked polynomial-system atlas, diagonal composition compression, address-preserving cyclic-core reduction, and one finite tree-signature family. Theorems 18.2, 18.12, 18.15 and 18.20 close the finite semisimple subgroup atlas, effective linear semilinear descent, exact projective obstruction, and the neutral tensor reconstruction gate. Theorems 19.2, 19.4,

19.7 and 19.11 now complete the finite global-invariant milestone: five certified connected components, coherent comparison maps, effective componentwise descent, and a complete tagged invariant on the declared semantic coproduct. The remaining program is therefore a cross-radix, semiring, modular and wild symmetry, nonneutral tensor, unrestricted holonomic, multivariate algebraic, general nonlinear, stronger local-valuation, optimal mixed-radix, and unrestricted product-grammar compatibility problem rather than a missing elementary Mahler-coordinate, Ore-normalization, rooted-tree expansion, finite Galois descent, or finite certified invariant-descent theorem.

Block 15 closes the common complete finite-resolution lane calculus under its printed hypotheses. Block 16 removes trivial address holonomy as a requirement for transport itself: complete coordinates descend as matrix-local-system sections and ordinary object lanes descend at monodromy-orbit resolution. Infinite or nonsplit lane doctrines, noninvertible transition theories, higher-stack effectivity, historical novelty review, and globalization over the unrestricted heterogeneous atlas remain open. The remaining mathematical program still includes cross-radix and semiring compatibility, modular and wild symmetry, nonneutral tensor descent, unrestricted holonomic and multivariate algebraic normalization, general nonlinear transport, stronger local valuations, optimal mixed-radix refinement, and unrestricted product grammars.

E Detailed claim-boundary ledger

E.1 Claims supported at manuscript level

The following statements are now supportable under the printed hypotheses:

- Komorebi Math has a substantial common resolved syntax and many proved local realization theories.
- The finite additive-Karoubi resolved envelope exists and has the stated universal property.
- Invertible representation changes and noninvertible transformations can be separated formally through representation groupoids, bridge passports, and a typed directed category.
- Information loss can be recorded separately as morphism collapse, object collapse, essential-image restriction, and loss of declared constructors or certificates.
- A finite profiled atlas has a descent, theorem-transport, and invariant-assembly theorem once its hypotheses are verified.
- Exact finite atlas models exist for graph holonomy, semilinear Galois descent, affine Mahler chart composition, split Mahler packet cores, and the normalized C-finite recurrence-rational-state triangle.
- Finite graph transport has a unique maximal path-flat core, while full frame trivialization is equivalent to identity fundamental holonomy.
- A canonical obstruction object exists relative to declared correction data; coarse vanishing can erase a nonzero fine obstruction.
- Under the stated classical hypotheses, Hilbert 90 gives effective vector descent and the Brauer class is the exact projective-to-linear lifting obstruction.
- Restricted complete invariants exist in several strata. The persistent C-finite slice has a complete global-to-packet minimality transfer theorem.
- The unit purely periodic atlas over $\mathbb{F}_p$ is a verified heterogeneous atlas across linear, digital, and functional-Mahler strata. Its objects and morphisms descend effectively, and its periodic resolved invariant package is complete on the declared normalized semantic category.
- The unit rational-mask atlas over $\mathbb{F}_p$ is a verified heterogeneous atlas across reduced functional-Mahler masks, minimal Cartier modules, and minimal digital automata. Its exact collapse obstruction separates the ultimately periodic locus from the genuinely non-C-finite automatic locus, and its descended invariant package is complete on the declared semantic category.
- For a fixed radix and field, every regular sequence has a canonical minimal section module, a strict shortlex coordinate record, and a word-Hankel rank certificate. These charts satisfy effective descent, and the pointed section module is complete on the declared regular category.
- Over a finite field, regularity is equivalent to automaticity and the minimal automaton is the finite section orbit. Over characteristic-zero fields, finite regular rank can coexist with an infinite kernel orbit.
- Every integer-valued fixed-radix sequence of finite rational section rank has a canonical finite free raw section lattice and a canonical finite free integer-valued saturation of the same rank.

- The Smith invariants and Fitting ideals of the finite saturation quotient determine the ghost kernel, semantic image size, missing quotient, good moduli, and prime-power generator profile exactly.
- Every prime-power coefficient reduction has a finite minimal semantic section automaton, and the compatible full module tower reconstructs the embedded p-adic raw lattice.
- p-adic completion, profinite completion, ordinary integral lifting, and actual section realization are distinct effectivity levels with explicit obstruction quotients.
- Every integral unit series has unique fixed Euler exponents and unique divisor ghosts, with exact Möbius integrality conditions and coefficient recurrence.
- Every integral multiplicative arithmetic function is reconstructed exactly from prime-power local series, and local Euler exponents and ghosts form complete additive coordinates for Dirichlet convolution.
- On the multiplicative exponent locus, the divisor–ghost transform gives an effective four-chart atlas and a complete prime-power descended invariant.
- Continuous $\mathbb{Z}_p$-valued functions, their canonical Mahler coefficients, complete clopen branch packets, and branchwise Mahler coefficients form an effective normalized local atlas.
- Affine Mahler kernels have finite triangular support, compose by exact mixed-radix substitution, and satisfy strict Chinese-remainder coherence.
- Finite-degree affine channels have distinct characteristic-zero, integral, modular, and collective-packet bridge passports, with sharp minimal packet sizes under the printed hypotheses.
- Declared monotone Mahler valuation profiles survive affine polyphase refinement, and principal-unit state packets acquire a certified linearly improving precision slope under p-power refinement.
- Every nonzero D-finite series and non-eventually-zero P-recursive sequence over a characteristic-zero field has a canonical monic minimal Ore annihilator; a finite transient ledger upgrades the tail operator to an exact semantic chart.
- Arithmetic multisection preserves holonomicity, and every fixed closure pipeline has an exact excess-order certificate relative to the unique minimal packet operator.
- In the marked binomial–Fuchsian family, nonintegral exponent gives minimal packet differential order N, positive integral exponent gives rational order one, and nonpositive integral exponent gives the exact finite-support alias/cancellation phase diagram.
- For a finite Galois extension in the semisimple setting, subgroups, fixed fields, and faithful Reynolds projectors form an order-reversing normalized atlas.
- Nonnormal intermediate fields form conjugate projector atlases with canonical core and normal-closure envelopes, while Mackey double cosets give exact induction–restriction transition packets.
- Ramification groups form a nested projector resolution, and the Artin and Swan conductors are weighted unresolved-rank sums.
- Finite Galois vector descent is effective on objects and morphisms; projective descent lifts to an ordinary vector packet exactly when its relative Brauer class vanishes.
- Every continuous finite-dimensional complex representation of a profinite group factors through a finite quotient, and shrinking open normal-subgroup projectors converge strongly to the identity.

- A faithful exact tensor fiber functor reconstructs an affine group scheme by neutral Tannakian duality; faithful corner realization alone does not cross this tensor boundary.
- The certified global-synthesis bridge matrix distinguishes reversible named overlaps from fully faithful inclusions, directional loss, partial constructors, and absent bridges.
- On the five declared connected components, local complete invariants have coherent comparison maps and descend to a complete tagged invariant that reflects isomorphisms.
- The marked weighted Fuss family gives a genuine reversible overlap among tree, selected algebraic, and P-recursive presentations, classified by the finite pair (q, α).
- At every declared complete finite resolution in an additive idempotent-complete category, analysis and synthesis are inverse, morphisms have unique lane-transfer matrices, composition is block contraction, and compatible refinement aggregates those matrices functorially.
- Finite lane selectors split retained transport from an exact residual. Under the separately printed abelian or strict-fold hypotheses, the discarded or collided part is measured by the corresponding kernel-exact sequence.
- Factorial residue lanes realize the formal theorem exactly, and the stated polyphase, Cartier, holonomic, local, nonlinear-tree, algebraic, and Galois cases are valid only with their declared native hypotheses.
- Finite nontrivial address holonomy does not obstruct complete lane transport when the address and splitting cocycles remain coherent: lane matrices descend as compatible sections of a matrix local system, and composition remains exact path contraction.
- Monodromy-orbit sums are the finest ordinary globally labeled coarsening obtainable from the original object-lane projectors, while simultaneous source-target pair orbits are the minimal nonempty holonomy-compatible Boolean transport supports; their selectors are nonzero only on orbits supporting nonzero compatible sections.
- For finite permutation modules, pair-orbit indicator matrices form an exact intertwiner basis with nonnegative integer path-count structure constants. For cyclic Fourier idempotents, rational Galois orbit sums are indexed by divisors and have exact Ramanujan-sum coefficients.

E.2 Claims not established

The following statements are not proved here:

- every mathematical object has a finite Komorebi presentation;
- all strata are equivalent categories;
- every bridge is invertible;
- one global canonical minimal presentation exists across arbitrary grammars;
- one finite invariant classifies every stratum;
- equivalence of arbitrary presentations is decidable;
- the full eight-stratum heterogeneous atlas already satisfies effective descent, or the five certified global-synthesis components are reversibly connected to one another;

- every automatic sequence has a rational functional-Mahler mask, or functional Mahler equations and Newton-Mahler interpolation are unrestrictedly equivalent;
- one base-independent or all-ring canonical regular normal form, one minimal semiring realization theorem, or one cross-radix invariant compatible with every base;
- the integral lattice theorem automatically extends to arbitrary coefficient rings, non-Noetherian modules, semirings, all radices at once, or finite-data recognition of rational regularity;
- unique Euler coordinates imply finite structural complexity, a minimal product compiler, or a canonical factorization across arbitrary product grammars;
- ordinary Cauchy convolution and Dirichlet convolution are globally isomorphic, or prime-power seeds classify arithmetic functions without the multiplicativity axiom;
- the multiplicative Euler-exponent locus is closed under ordinary Euler-product multiplication;
- functional-Mahler coefficient equations and Newton–Mahler p-adic interpolation are canonically or unrestrictedly equivalent;
- every continuous p-adic function is locally analytic, every affine-kernel valuation lower bound is always sharp, or one optimal mixed-radix refinement tree has been constructed;
- the local atlas already carries a complete combined spectral–local invariant or a universal state-recognition theorem;
- arbitrary D-finite multisections have a proved finite torsion signature, a closed minimal-order formula, or a canonical apparent-singularity removal algorithm;
- local exponent eigenvalues alone determine the full monodromy representation, logarithmic extension data, or global differential Galois group;
- the marked binomial–Fuchsian invariant classifies unmarked holonomic series or recovers an algebraic branch from an annihilator;
- the d^N algebraic packet bound is always minimal, or every resultant automatically selects the intended branch factor;
- selected algebraic branches admit one canonical minimal multivariate system dimension across arbitrary polynomial grammars;
- algebraic and D-finite presentations are equivalent without retained branch data;
- Reynolds averaging remains exact when the group order is not invertible, or finite semisimple Galois charts automatically extend to modular and wild integral representations;
- every projective cocycle descends to a vector space, or the Brauer class alone classifies the full central simple algebra without its remaining algebra data;
- every tensor category has a neutral fiber functor, or a faithful additive or corner representation automatically reconstructs a Tannakian group scheme;
- finite Galois effectivity proves effective descent for the unrestricted heterogeneous Komorebi atlas;
- internal manuscript proof status implies independent peer review;
- the name Universal Lane Transport establishes historical priority or novelty for classical residue filters, polyphase decompositions, Peirce block matrices, Cartier sections, or other imported ingredients;

- every object admits a complete finite split lane system, every useful resolution is additive, or the finite theorem and its finite holonomy-twisted extension automatically extend to infinite, completed, nonsplit, noninvertible, higher-stack, or arbitrary nonlinear settings;
- the number of lane addresses equals dimension, rank, minimal state size, entropy, independent information, or computational complexity without a separate theorem;
- an incomplete selector or noninjective fold is reversible, or its residual and collision kernel can be discarded without changing the represented transport.

Final status of the present paper. The project now has an explicit typed and packet-resolved common foundation, a proved additive-Karoubi envelope, profiled stratum and bridge passports, a directed information-loss calculus, a typed comparison of four obstruction theories, a proved finite assembly theorem, two exact affine-Mahler mechanisms, one fully effective normalized C-finite presentation atlas, and two fully effective heterogeneous digital/Mahler atlases. The fixed-radix regular stage adds the full fixed-radix field-valued regular atlas, a canonical word-Hankel and short-lex normal form, and the automatic finite-orbit theorem. The integral prime-power stage adds the canonical integral raw/saturated lattice pair, the exact Smith–Fitting reduction defect, finite prime-power semantic automata, and the p-adic/profinite effectivity hierarchy. The product-multiplicative stage adds canonical fixed-basis Euler and ghost coordinates, prime-power effective descent for multiplicative functions, and the complete restricted divisor–ghost atlas. The local Mahler-polyphase stage adds the normalized continuous Newton–Mahler atlas, strict complete polyphase descent, sharp finite-degree channel packets, valuation-profile transport, and principal-unit precision amplification. The holonomic stage adds canonical minimal Ore charts with finite transients, the general multisection alias/cancellation ledger, and an exact marked binomial–Fuchsian packet-minimality theorem. The algebraic stage adds finite selected-branch normalization, exact packet polynomial systems, a generic d^N packet-degree theorem, and collision signatures with sharp Catalan and radical model cases. The nonlinear tree stage adds a marked polynomial fixed-point atlas, exact diagonal packet Faà di Bruno compression, an address-preserving cyclic-core dimension, and finite simply-generated-tree support and singularity signatures. The Galois-Tannakian stage adds the normalized finite subgroup–fixed-field–projector atlas, conjugate and Mackey transition packets, ramification details, effective Hilbert-90 descent, the exact relative Brauer obstruction, profinite finite-level resolution, and the neutral Tannakian tensor-fiber gate. The global synthesis stage adds the authoritative bridge matrix, normalizer-induced higher coherence, a five-component certified invariant atlas, the marked Fuss nonlinear–algebraic–holonomic triangle, and a complete tagged global invariant on the declared semantic union. The finite lane-transport stage adds the complete finite-resolution analysis, synthesis, block-composition, refinement, selector, residual, conditional fold-loss, factorial-realization, and certified-descent theorem family. The holonomy-twisted stage then removes global address triviality as a transport requirement, replacing fixed labels by matrix-local-system sections and recovering canonical ordinary lanes at monodromy-orbit resolution, with pair-orbit composition and rational cyclic orbit descent as exact realizations. The broad heterogeneous claim remains conditional on cross-radix and semiring compatibility, modular and wild symmetry, nonneutral tensor descent, unrestricted holonomic and multivariate algebraic normalization, general nonlinear normal forms and mixed-direction compression, stronger local valuation and optimal mixed-radix theory, unrestricted product grammars, and effective invariant comparison beyond the normalized domains proved here.

F Validation record

The validation summarized below concerns internal package integrity and exact finite verification. It is not an external authenticity or peer-review certificate. The thematic verifiers record 39,805 checks for the unit-periodic atlas, 11,391 for rational masks, 89,055 for fixed-radix regular systems, 697,958 for integral saturation and prime-power towers, 595,964 for Euler-product and multiplicative coordinates, 229,743 for local Newton–Mahler and polyphase transport, 174,940 for holonomic packets, 73,088 for selected algebraic packets, 115,721 for nonlinear tree systems, and 11,195 for finite Galois and tensor models. The two global-synthesis suites execute 154,209 bridge assertions and 119,453 weighted Fuss and tagged-invariant assertions. They include 39,846 authenticated replay assertions already counted in the preceding records and contribute 233,816 new comparison, coherence, packet, and loss-control assertions. The nonduplicated cumulative record is 2,272,676 exact assertions with no failures; the raw executed total, including replay, is 2,312,522. The Block 15 verifier separately executes 197,099 exact finite regression assertions for selected lane-analysis, block-composition, refinement, fold, support, and factorial implementation clauses. The Block 16 verifier executes 1,344,823 exact assertions for finite group actions, object and pair orbits, orbital structure constants, compatible and incompatible selectors, equivariant refinement, rational cyclotomic orbit projectors, and theorem-source contracts, with 0 recorded failures. Detailed tables and machine-readable JSON records are included in the source package. The Part IV semantic-diamond audit additionally executes exact rational linear-algebra trials for quotient dimensions, factorization, relative residuals, and compatibility sequences. Its role is implementation validation; the universal statements are proved in the text.

Table 51: Internal validation record for Core-Transfer Addendum 04.

Check	Recorded result
Package verifier	Overall pass: true; 383 ledger entries; 0 missing; 0 extra; 0 mismatched.
Exact theorem verifier	48 of 48 gates passed; arithmetic used exact integers/fractions, finite-field elimination, integer/composite-modulus split decoding.
Full unit-test attempt	The full test script did not finish within two bounded runs (120 seconds and 600 seconds). The completed tests were passing when the second run timed out. No claim of a full-suite pass is made here.
Boundary	Internal consistency and exact finite verification only; no external authenticity or peer-review claim.

Table 52: Exact finite and symbolic audit for the Part IV semantic quotient extension.

Audit family	Assertions
Semantic diamond dimension	1,000
Relative residual dimension	1,000
Carrier factorization gate	1,000
Quadratic/convex symbolic identity	1
Polyphase Hankel determinants	8
Theorem-source contracts	13
Total	**3,022**
Recorded failures	0

A legacy all-project unit-test script was attempted twice during an earlier consolidation stage, including a 600-second bounded run. The completed tests were passing when the second run timed out, but that

historical script did not finish within the bound. This statement does not qualify or replace the named thematic and Block 15 verifier records. No pass is claimed for that separate legacy all-project invocation.

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Exact reduction, hidden memory, and certified reconstruction

Traditional paper

Komorebi P.

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REALIZATION CALCULUS

Reading scope

Reading set. This traditional paper is accompanied by a visual guide, a Lean guide, and the complete source ZIP. The ZIP includes the `.md` files, editable manuscript sources, Lean sources, Python implementations, example data, and verification records.

Scope. Exact linear representation and behavioral reduction; bounded discrete and continuous dynamics; real Hilbert reconstruction and regularization; certified discrepancy rules; and calibration-based operator allowances. Truth allowances and feasibility assumptions are stated explicitly.

Abstract

A representation is sufficient relative to an observation and a family of permitted continuations. This paper develops that principle through exact analysis and synthesis, hidden-return defects, behavioral quotients, finite certificates, quantitative approximation, and certified inverse problems. Split compression fails to preserve composition by an explicit hidden excursion; for finite-dimensional hidden state, finitely many feedback moments decide all-depth closure. Behavioral reduction retains precisely the distinctions that some permitted future observation can detect. In real Hilbert settings, compatibility, realizability, and source identifiability produce separate obstructions and conditioning requirements. Canonical reconstruction, Tikhonov regularization, discrepancy selection, inexact solves, changing operators, and redundant calibration frames are assembled into a common chain of contracts. Classical linear algebra and analysis provide the foundations. The contribution of the assembled calculus is the explicit relation between these contracts, their reusable certificates, and their sharp boundaries. The paper consolidates sixteen development blocks and maps its principal claims to existing Lean declarations. Finite rational implementations are companion evidence and are not asserted to be formally verified.

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1 The problem and its mathematical contracts

Sufficiency depends on what a representation must support. A compression that preserves today's observation can lose a distinction needed after a later operation. A coordinate enlargement can be redundant yet exact. A compatible local datum can still fail to have a global source. A small fitting residual can coexist with large source error. These are distinct phenomena with different mathematical witnesses.

The paper follows two connected lines. The first is the reduction of known linear dynamics relative to declared observations. The second is the reconstruction of a source from bounded linear data with explicit error allowances. The first yields an exact semantic state quotient. The second yields metric reconstruction and stability contracts. The present formalization does not yet identify every behaviorally minimal state with a stably recoverable target in an inverse problem. That bridge is an open extension, stated in the final section.

1.1 Conventions and domains

Operator multiplication applies the right factor first: $STx = S(Tx)$. A word $w = [i_1, \dots, i_n]$ is chronological, so $T_w = T_{i_n} \cdots T_{i_1}$; $T_\varnothing = I$. Concatenating u then v gives $T_{u{+}{+}v} = T_v T_u$. Empty sums are zero and empty products are one. All equalities of operators are equalities on the entire stated domain.

The exact redundant-coordinate theory works for modules over a semiring. Subtraction and compression defects use modules over a commutative ring. Dimension and annihilating-polynomial arguments use a field. Norm estimates use bounded linear maps on normed spaces. Continuous flow results use bounded, time-independent real generators on Banach spaces. Reconstruction and regularization use real Hilbert spaces unless a weaker projection hypothesis is printed explicitly. These changes of domain are part of the contracts.

For a subspace L admitting orthogonal projection, P_L denotes that actual projection. On a Hilbert space, closed subspaces admit it. The notation $\operatorname{dist}(v, L) = \inf_{\ell \in L} \|v - \ell\|$ always refers to the actual subspace, not a replacement chosen to make a theorem true. Closure is written $\operatorname{cl} L$.

1.2 How to read the formal evidence

The stable identifiers SR-01 through SR-64 label consolidated theorem groups. They are editorial units, with hypotheses and proofs printed here. A separate CSV/JSON concordance records their exact source declarations and source hashes. Several formal lemmas can support one printed theorem, and one source lemma can serve more than one explanatory purpose. The source inventory contains 998 theorem/lemma declarations in 93 production modules, including helper lemmas and examples. It is not a count of independent new mathematical results.

The arguments below reconstruct the mathematics of the supplied Lean sources. Classical inputs such as Cayley–Hamilton, orthogonal projection, the open mapping theorem, Lax–Milgram, and the intermediate value theorem enter through the source's Mathlib dependencies. This paper does not assert a new historical priority assessment for those inputs. The Lean guide records exactly which compilations and kernel replays are archived or freshly performed during this release. Python certificates are tested implementations of mathematical contracts; their parsers and algorithms are not Lean-verified.

2 Exact coordinates, compression, and behavioral state

The first contract supplies two maps rather than identifying a coordinate space with the object by convention. Let $a_X : X \to V_X$ and $s_X : V_X \to X$ be linear with $s_X a_X = I_X$. The ambient space V_X may have redundant coordinates. Define $E_X = a_X s_X$. For another such resolution of Y, put

$$\mathcal{K}(T) = a_Y T s_X, \qquad \mathcal{D}(K) = s_Y K a_X.$$

We call a proposed coordinate operator compatible when $E_Y K E_X = K$.

Hypotheses. Semiring modules; supplied linear analysis/synthesis pairs with s_X a_X=I and s_Y a_Y=I.

Theorem 2.1 (SR-01 Exact redundant representation)**.** *$E_X^2 = E_X$, and $E_X v = v$ exactly when $v \in \operatorname{Ran} a_X$. Moreover*

$$\mathcal{D}\mathcal{K}(T) = T, \qquad \mathcal{K}\mathcal{D}(K) = E_Y K E_X.$$

Consequently a compatible K is the encoding of a unique map $T : X \to Y$. Encoding preserves composition. Its identity on resolved coordinates is E_X.

Proof. The splitting gives $E_X^2 = a_X(s_X a_X)s_X = a_X s_X$. If $E_X v = v$, then $v = a_X(s_X v)$; conversely $E_X a_X x = a_X x$. Substituting the definitions yields

$$s_Y(a_Y T s_X)a_X = T, \qquad a_Y(s_Y K a_X)s_X = E_Y K E_X.$$

For compatible K, take $T = \mathcal{D}(K)$. If another T' encodes to K, decoding proves $T' = \mathcal{D}(K)$. Finally, an intermediate resolution of Y has $s_Y a_Y = I_Y$, so $\mathcal{K}(S)\mathcal{K}(T) = a_Z S(s_Y a_Y)T s_X = a_Z S T s_X$. Encoding I_X gives $a_X s_X = E_X$, which acts as identity on compatible kernels. No inverse on all of V_X has been assumed.

□

Formal source: `Resolution.projector_idempotent;Resolution.fixed_iff_realizable;Resolution.compatible_iff_existsUnique;Resolution.encode_comp;Resolution.encode_id`.

Example 2.2 (Redundant coordinates)**.** For $X = \mathbb{R}$ and $V_X = \mathbb{R}^2$, take $a(x) = (x, x)$ and $s(u, v) = (u + v)/2$. Then $E(u, v) = ((u + v)/2, (u + v)/2)$. The encoded scalar transformation $x \mapsto cx$ has matrix $\frac{c}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Its redundant coordinate identity is E, while the ambient identity matrix also acts on an unrealizable anti-diagonal sector. The distinction is exact.

Hypotheses. Commutative-ring modules; linear $C : X \to Z$, $R : Z \to X$ with $CR = I_Z$; arbitrary endomorphisms S, T, U of X.

Theorem 2.3 (SR-02 Hidden-return defect and coherence)**.** *Set $P = RC$, $Q = I - P$, $\Phi(T) = CTR$, and $\Delta(S, T) = \Phi(ST) - \Phi(S)\Phi(T)$. Then*

$$\Delta(S, T) = CSQTR,$$

and associativity gives

$$\Delta(S, T)\Phi(U) + \Delta(ST, U) = \Phi(S)\Delta(T, U) + \Delta(S, TU).$$

For two successive split compressions,

$$\Delta_{21}(S, T) = \Phi_2(\Delta_1(S, T)) + \Delta_2(\Phi_1(S), \Phi_1(T)).$$

Proof. Insert $I = P + Q$ between S and T: $CSTR = CS(P+Q)TR = (CSR)(CTR) + CSQTR$. This proves the factorization. Expanding either side of the coherence identity cancels the terms $\Phi(S)\Phi(T)\Phi(U)$ and leaves $\Phi(STU) - \Phi(S)\Phi(TU)$ together with the same compensating term. For the two-stage formula, add and subtract $\Phi_2(\Phi_1(S)\Phi_1(T))$ in the composite defect. Linearity of Φ_2 identifies the first difference with $\Phi_2(\Delta_1(S,T))$; the second is the second-stage defect. Thus a later representation transports the earlier error and can introduce another one. It does not remove the first term by definition.

□

Formal source: `SplitCompression.defect_factorization;SplitCompression.defect_coherence;SplitCompression.defect_then`.

Hypotheses. Commutative-ring modules and a linear presentation $G \xrightarrow{A} V \xrightarrow{\delta} W$ with $\delta A = 0$; a compatible vector $v \in \ker \delta$.

Theorem 2.4 (SR-03 Actual globalization obstruction)**.** *For $\delta A = 0$, define $K = \ker \delta$, $R = \operatorname{Ran} A \subseteq K$, and $\mathcal{H} = K/R$. A compatible v has class zero exactly when $v = Ag$ for some g. All classes vanish exactly when $R = K$. If A is injective, a zero class has a unique source.*

Commuting presentation maps $LA = A'F$ and $\delta' L = M\delta$ induce a map $\overline{L} : \mathcal{H} \to \mathcal{H}'$ by $[v] \mapsto [Lv]$. These maps compose and preserve zero classes. Reflection of zero follows when $\overline{L}$ is injective.

Proof. The chain condition puts the actual range of A inside K. The zero class criterion for a quotient is membership in the submodule divided out, giving $[v] = 0 \iff v \in R \iff \exists g, Ag = v$. Every compatible class is zero exactly when $K \subseteq R$; the reverse inclusion is already known. Injectivity of A gives uniqueness of any preimage.

For transport, if $v \in K$, then $\delta' Lv = M\delta v = 0$. If $v = Ag$, then $Lv = A'Fg$ lies in R'. Hence L descends to the actual quotients. Evaluating on representatives proves the composition law. A linear map sends zero to zero. If its induced quotient map is injective, $\overline{L}[v] = 0 = \overline{L}[0]$ implies $[v] = 0$. The commuting squares alone provide no such injectivity and do not assert that either obstruction quotient vanishes.

□

Formal source: `Globalization.obstructionClass_zero_iff;Globalization.all_obstructions_zero_iff_exact;Globalization.unique_realization_of_zero;Globalization.PresentationMap.obstructionMap_compose;Globalization.PresentationMap.reflects_zero_obstruction`.

Hypotheses. Commutative-ring modules; a linear observation $O : X \to Y$ and an arbitrary family of allowed endomorphisms $T_i : X \to X$; every finite chronological word is included.

Theorem 2.5 (SR-04 The all-future invisible quotient)**.** *Let $N = \bigcap_w \ker(OT_w)$. It is the largest submodule contained in $\ker O$ and invariant under every T_i. The operations and observation descend to X/N, preserving every future observation. In particular,*

$$[x] = [y] \iff OT_w x = OT_w y \quad \textit{for every finite word } w.$$

If $q : X \to Z$ satisfies $qT_i = U_i q$ and $Vq = O$, then $\ker q \subseteq N$.

Proof. An intersection of kernels is a submodule. If $x \in N$, then $OT_w T_i x = OT_{i::w} x = 0$, so $T_i x \in N$. If an invariant submodule $L \subseteq \ker O$ contains x, induction on word length gives $T_w x \in L$; therefore $x \in N$. This establishes maximality.

Invariance defines quotient operations $[x] \mapsto [T_i x]$; containment in $\ker O$ defines $[x] \mapsto Ox$. Induction gives their all-word fidelity. Equality $[x] = [y]$ means $x - y \in N$, which is precisely equality of the displayed observations by linearity. In an exact competing model, induction gives $qT_w = U_w q$ and hence $OT_w = VU_w q$. If $qx = 0$, every future observation of x is zero. Thus a competing model can retain extra distinctions, but cannot erase any distinction visible to the specified future task.

□

Formal source: `Behavioral.invisible_invariant;Behavioral.invariant_submodule_le_invisible;Behavioral.all_future_observations_preserved;Behavioral.quotient_eq_iff_same_future;Behavioral.exact_model_kernel_le`.

3 Discrete hidden memory and finite closure

Let V be visible state and H hidden state. Over a commutative ring, specify four maps and form the actual full dynamics

$$T(y, z) = (Ay + Bz, Gy + Dz).$$

Put $C(y, z) = y$, $Ry = (y, 0)$, $E_n = CT^n R$, and $M_k = BD^k G$. The return M_k contains entry into H, k hidden transitions, and return to V. Its first effect occurs after $k + 2$ full transitions.

Hypotheses. Commutative-ring modules; actual powers of T(y,z)=(Ay+Bz,Gy+Dz); arbitrary initial y_0,z_0.

Theorem 3.1 (SR-05 Exact hidden elimination)**.** *For the trajectory* $(y_n, z_n) = T^n(y_0, z_0)$,

$$z_n = D^n z_0 + \sum_{j=0}^{n-1} D^{n-1-j} G y_j,$$

and

$$y_{n+1} = Ay_n + BD^n z_0 + \sum_{j=0}^{n-1} M_{n-1-j} y_j.$$

Prepared evolution ($z_0 = 0$) *satisfies* $E_0 = I$, $E_1 = A$, *and*

$$E_{n+1} = AE_n + \sum_{j=0}^{n-1} M_{n-1-j} E_j.$$

Proof. At $n = 0$, the first formula is $z_0 = z_0$ with an empty sum. If it holds at n, the actual recurrence $z_{n+1} = Gy_n + Dz_n$ gives the same formula at $n + 1$ after distributing D and appending the newest forcing term. Substituting this expression into $y_{n+1} = Ay_n + Bz_n$ proves the visible memory equation. For a prepared input y_0, each y_j equals $E_j y_0$. The pointwise memory equation is therefore an operator identity because it holds for every y_0. The equations are derived from powers of T; the trajectory recurrence is not an extra certificate assumption.

□

Formal source: `BlockDynamics.hidden_elimination;BlockDynamics.discrete_memory_equation;BlockDynamics.prepared_recurrence`.

Hypotheses. Same block dynamics; no finite-dimensional or orthogonality assumption.

Theorem 3.2 (SR-06 Prepared closure and the first return)**.** *The following conditions are equivalent:*

$$E_n = A^n \ (\forall n), \qquad M_k = 0 \ (\forall k), \qquad E_{m+n} = E_m E_n \ (\forall m, n).$$

More precisely,

$$E_n = A^n \ (n \le N+1) \iff M_k = 0 \ (k < N).$$

If $M_j = 0$ for $j < k$, then $E_{k+2} - A^{k+2} = M_k$. Closure for every initial hidden state is equivalent to $B = 0$.

Proof. If all memory maps vanish, the recurrence in SR-05 reduces to $E_{n+1} = AE_n$, proving $E_n = A^n$ by induction. Conversely, suppose exact powers hold. Induct on k. In the recurrence for E_{k+2}, every term except $M_k E_0$ involves a previously vanishing memory map. Thus $E_{k+2} = AE_{k+1} + M_k$, forcing $M_k = 0$. The same induction restricted to $k < N$ proves the finite-prefix equivalence and, before setting the last term to zero, the first-return identity.

Exact powers imply the semigroup law. The law conversely gives $E_{n+1} = E_n E_1 = E_n A$, so it implies exact powers. For arbitrary hidden initialization, evaluating closure at $n = 1$, $y_0 = 0$ gives $Bz_0 = 0$ for every z_0, hence $B = 0$. That condition also suffices directly from the visible recurrence. Vanishing $BD^k G$ imposes a weaker prepared-state requirement.

□

Formal source: `BlockDynamics.prepared_powers_iff_memory_zero;BlockDynamics.prepared_semigroup_iff_memory_zero;BlockDynamics.first_return_discrepancy;BlockDynamics.prepared_prefix_iff_memory_prefix;BlockDynamics.all_state_closure_iff_B_zero`.

Hypotheses. Field modules; supplied monic annihilating polynomial p(D)=0, or finite-dimensional hidden H with d=dim H.

Theorem 3.3 (SR-07 Finite all-depth closure certificate)**.** *If a monic polynomial p annihilates D, vanishing of M_k for $k <$ deg p implies vanishing for all k. If* dim $H = d < \infty$*, then*

$$E_n = A^n \ (\forall n) \iff M_k = 0 \ (0 \le k < d) \iff E_n = A^n \ (0 \le n \le d+1).$$

The visible space need not be finite-dimensional. The case $d = 0$ is included.

Proof. Divide X^n by the monic p to obtain a remainder q_n of degree below deg p. Evaluation at D gives $D^n = q_n(D)$ because $p(D) = 0$. Writing $q_n(D)$ as a finite linear combination of the powers D^k below deg p shows $BD^n G = 0$. For finite-dimensional H, Cayley–Hamilton provides the monic characteristic polynomial of D with degree d. SR-06 then yields both displayed equivalences. In dimension zero, all maps through H vanish and the empty list of memory tests is sufficient. If exact closure fails, the contrapositive gives a nonzero memory map with index below d. These tests concern complete linear maps, not sampled state vectors or approximate floating-point agreement.

□

Formal source: `BlockDynamics.memory_zero_of_annihilator;BlockDynamics.prepared_powers_iff_finrank_memory;BlockDynamics.prepared_powers_iff_finite_horizon;BlockDynamics.nonclosure_has_finite_feedback_witness`.

Hypotheses. Commutative-ring split C,R with CR=I; hidden H=ker C; finite-H corollary over a field.

Theorem 3.4 (SR-08 Canonical hidden state of a split compression). *For $C : X \to Z$, $R : Z \to X$, $CR = I$, let $Q = I - RC$ and $H = \ker C$. The maps*

$$x \longmapsto (Cx, Qx), \qquad (y, z) \longmapsto Ry + z$$

are inverse linear maps between X and $Z \times H$. An endomorphism T is conjugate to block dynamics with

$$A = CTR, \quad B = CT\iota_H, \quad G = QTR, \quad D = QT\iota_H.$$

Their prepared maps equal CT^nR. Hence exact compressed powers are equivalent to zero canonical memory. If $\dim\ker C = d$, complete compressed map tests through $d + 1$ decide all depths.

Proof. $CQ = C - CRC = 0$, so Qx belongs to H. Synthesis after analysis gives $RCx + Qx = x$. Conversely, for $z \in \ker C$, one has $C(Ry + z) = y$ and $Q(Ry + z) = z$, using $QR = 0$ and $Qz = z$. Conjugating T by these inverse maps gives the four displayed blocks; in the hidden blocks, Q supplies the required codomain. Induction transports powers under conjugacy. The lift of y is $(y, 0)$ in these coordinates, so the prepared visible map is precisely CT^nR. Apply SR-06 and SR-07. The construction uses the actual discarded subspace and imposes no orthogonality on the chosen splitting.

□

Formal source: `SplitCompression.block_synthesis_analysis;SplitCompression.block_analysis_synthesis;SplitCompression.block_powers_eq;SplitCompression.exact_powers_iff_canonical_memory_zero;SplitCompression.exact_powers_iff_canonical_finite_horizon.`

Example 3.5 (Delayed return and disconnected return). On $\mathbb{Q}^3$, set $T(y, z_1, z_2) = (z_1, z_2, y)$ and retain y. Then $A = 0$, $M_0 = 0$, $M_1 = I$, and $E_1 = E_2 = 0$ while $E_3 = I$. Two shallow exact checks cannot certify all depths without the dimension budget. For another system, let $A = D = I$, $B(z_1, z_2) = z_1$, and $G(y) = (0, y)$. Both couplings are nonzero but every BD^kG vanishes. Prepared closure is exact, while an arbitrary hidden initial first coordinate affects the next visible state.

4 Finite behavioral models and minimality

Now work over a field and, for dimension statements, assume $\dim X < \infty$. The allowed operations need not commute. Define

$$N_r = \bigcap_{|w| \le r} \ker(OT_w), \qquad k = \dim\ker O.$$

A finite horizon means bounded word length. It gives a finite enumeration of constraints only when the operation alphabet is finite.

Hypotheses. Finite-dimensional X over a field; arbitrary alphabet; all bounded-length words including empty word.

Theorem 4.1 (SR-09 Exact horizon and plateau certificate). $N_0 = \ker O$ *and*

$$N_{r+1} = \ker O \cap \bigcap_i T_i^{-1} N_r \subseteq N_r.$$

One plateau $N_{r+1} = N_r$ implies $N_r = N$. In all cases $N_k = N$. Thus equality of future observations through word length k is equivalent to equality for every finite word.

Proof. The empty word enforces $Ox = 0$. A nonempty word $i :: w$ of length at most $r + 1$ enforces $OT_w T_i x = 0$, yielding the recursion. At a plateau, that recursion shows N_r is invariant under every T_i and lies in $\ker O$. By SR-04, $N_r \subseteq N$, while $N \subseteq N_r$ always holds.

Before reaching N, each successive inclusion must be strict, since an equality would already certify the final subspace. Every strict inclusion reduces dimension by at least one. Starting from dimension k allows at most k such reductions; if N_k has dimension zero it is already N, and otherwise an earlier plateau has occurred. Applying the kernel identity to $x - y$ proves the all-future comparison. No word-order commutativity is used.

□

Formal source: `FiniteHorizon.visibleKernel_succ;FiniteHorizon.visibleKernel_eq_invisible_of_plateau;FiniteHorizon.visibleKernel_stabilizes;FiniteHorizon.same_future_iff_bounded`.

Hypotheses. Finite-dimensional X over a field; exact linear competitor q with qT_i=U_iq and Vq=O; finite Z only for final ambient comparison.

Theorem 4.2 (SR-10 Minimum exact state dimension)**.** *The model X/N is observable. Every exact competitor induces a unique surjection* $\operatorname{Ran} q \to X/N$*, taking qx to $[x]$. Therefore*

$$\dim(X/N) \le \dim \operatorname{Ran} q,$$

with equality exactly when $\ker q = N$*. In that case* $\operatorname{Ran} q \cong X/N$*. If Z is finite-dimensional, also* $\dim \operatorname{Ran} q \le \dim Z$*. Surjectivity of q onto Z is not required.*

Proof. If $qx = qy$, SR-04 gives $x - y \in N$, so $qx \mapsto [x]$ is well-defined. It is linear and surjective because every class has a representative x. Its values on the actual range force uniqueness. Rank-nullity gives $\dim \operatorname{Ran} q = \dim X - \dim \ker q$ and $\dim(X/N) = \dim X - \dim N$. Since $\ker q \subseteq N$, this proves the inequality. Equality of dimensions forces equality of the nested finite-dimensional subspaces; conversely their equality makes the comparison injective and hence an equivalence. In the quotient model, an invisible class $[x]$ has every $OT_w x = 0$, so $x \in N$ and $[x] = 0$. This establishes observability. Unused coordinates in Z do not change the reachable-image comparison.

□

Formal source: `BehavioralMinimality.quotient_is_observable;BehavioralMinimality.imageComparison;BehavioralMinimality.finrank_state_le_image;BehavioralMinimality.image_dimension_minimal_iff;BehavioralMinimality.minimalImageEquiv`.

Hypotheses. Finite-dimensional X over a field; finite operation alphabet; k=dim ker O; target Y can be infinite-dimensional.

Theorem 4.3 (SR-11 Realizable finite observation tables)**.** *Let* $H_k x = (OT_w x)_{|w| \le k}$ *and* $Z_k = \operatorname{Ran} H_k$*. Then* $\ker H_k = N$ *and* $X/N \cong Z_k$*. The formulas*

$$U_i(H_k x) = H_k T_i x, \qquad V(H_k x) = Ox$$

define actual linear maps on Z_k and preserve every finite future word. This model has minimum exact dimension.

Proof. The recursively enumerated words are exactly those of length at most k. SR-09 identifies the kernel of their combined observation map with N. The first isomorphism theorem gives $X/N \cong \operatorname{Ran} H_k$.

If $H_k x = H_k y$, then $x - y \in N$; its invariance gives $H_k T_i x = H_k T_i y$, and $N \subseteq \ker O$ gives $Ox = Oy$. Hence the displayed operations and readout are well-defined on realizable tables. Induction proves $U_w H_k x = H_k T_w x$ and $V U_w H_k x = O T_w x$ for words of any length. Apply SR-10 to get minimality. An arbitrary table in the ambient coordinate product need not be realizable and is not part of this constructed state space.

□

Formal source: `FiniteObservableModel.mem_wordsUpTo;FiniteObservableModel.ker_complete_observation;FiniteObservableModel.observableEquiv;FiniteObservableModel.coordinate_all_future_fidelity;FiniteObservableModel.coordinate_dimension_le_exact_model`.

Hypotheses. Field modules, not necessarily finite-dimensional; finite operation alphabet and finite horizon observation map H_r; actual maps q=F H_r, qT_i=U_iq, Vq=O.

Theorem 4.4 (SR-12 Finite minimality certificate)**.** *For a finite operation alphabet, suppose a proposed representation satisfies*

$$q = FH_r, \qquad qT_i = U_i q, \qquad Vq = O.$$

Then $\ker q = N$*, regardless of whether* r *reaches the a priori horizon. If* q *is surjective, it gives an equivalence* $Z \cong X/N$ *preserving the observation and intertwining every operation. Without surjectivity, the equivalence applies to* $\operatorname{Ran} q$*.*

Proof. If $x \in N$, then every coordinate of $H_r x$ is zero, so $qx = FH_r x = 0$. This gives $N \subseteq \ker q$. Exact intertwining and output give the reverse inclusion by SR-04. The equality makes $qx \mapsto [x]$ injective on the range and surjective onto the quotient. Its inverse is the corresponding quotient map induced by q. Evaluating both sides on qx proves that the equivalence intertwines U_i with the quotient operations and preserves the readout. The certificate has two roles: generation by observations prevents retention of invisible distinctions, while exact dynamics and output prevent loss of future-visible distinctions. Finite dimension is needed to guarantee that such a certificate can be found at a bounded depth, not for its soundness.

□

Formal source: `FiniteCertificate.certificate_kernel_eq;FiniteCertificate.certificate_same_future_iff;FiniteCertificate.certificateEquiv;FiniteCertificate.certificateEquiv_dynamics;FiniteCertificate.certificateEquiv_observation`.

Example 4.5 (A noncommuting minimal model)**.** On $\mathbb{Q}^3$, let $O(x,y,h) = x$, $T_A(x,y,h) = (y,0,h)$, and $T_B(x,y,h) = (0,x,h)$. The two operations do not commute: on $(1,0,0)$ their two compositions give $(1,0,0)$ and $(0,0,0)$. The map $q(x,y,h) = (x,y)$ is generated by O and OT_A, intertwines both operations, and factors O. It therefore has a depth-one certificate. The hidden coordinate h is invisible to every future word, and the minimum exact state dimension is two. The current observation alone has dimension one and misses the distinction between $(0,1,0)$ and zero, which one application of T_A reveals.

5 Quantitative discrete sufficiency

Let $q : X \to Z$ be a proposed representation, with true operations T_i, reduced operations U_i, and readouts $O : X \to Y$, $V : Z \to Y$. Define the step and word defects

$$\delta_i = qT_i - U_i q, \qquad E_w = qT_w - U_w q.$$

Algebraic identities require only the stated commutative-ring structure. Quantitative estimates below use continuous linear maps over a nontrivially normed field and their whole-map operator norms.

Hypotheses. Commutative-ring modules; arbitrary chronological words and linear operations/representation/readouts.

Theorem 5.1 (SR-13 Exact word-defect transport)**.** *One has* $E_\varnothing = 0$*,*

$$E_{i::w} = E_w T_i + U_w \delta_i, \qquad E_{u++v} = E_v T_u + U_v E_u.$$

For $w = [i_1, \dots, i_n]$ *this yields*

$$E_w = \sum_{j=1}^{n} U_{[i_{j+1},\dots,i_n]} \delta_{i_j} T_{[i_1,\dots,i_{j-1}]}.$$

The output defect is $(O - Vq)T_w + VE_w$*.*

Proof. The recursive identity follows by adding and subtracting $U_w q T_i$ in $qT_w T_i - U_w U_i q$. For concatenation, add and subtract $U_v q T_u$; chronological order gives $T_{u++v} = T_v T_u$ and the analogous identity for U. Inducting on the word expands one term per letter: each step defect is transported by its true prefix and reduced suffix. The empty word contributes no terms. Finally $OT_w - VU_w q = (O - Vq)T_w + V(qT_w - U_w q)$. Neither output mismatch nor accumulated dynamics mismatch is discarded.

□

Formal source: `ApproximateAlgebra.wordDefect_cons;ApproximateAlgebra.wordDefect_append;ApproximateAlgebra.wordDefect_eq_sum;ApproximateAlgebra.output_defect_decomposition`.

Hypotheses. Normed spaces over nontrivially normed field; bounded maps; nonnegative letter bounds a_i,b_i,epsilon_i and eta$\geq \|$O-Vq$\|$.

Theorem 5.2 (SR-14 Heterogeneous operator budgets)**.** *Suppose* $\|T_i\| \le a_i$*,* $\|U_i\| \le b_i$*,* $\|\delta_i\| \le \epsilon_i$*. Let* A_w, B_w *be the products of the respective letter bounds and define*

$$D_\varnothing = 0, \qquad D_{i::w} = D_w a_i + B_w \epsilon_i.$$

Then $\|T_w\| \le A_w$*,* $\|U_w\| \le B_w$*,* $\|E_w\| \le D_w$*, and*

$$D_{u++v} = D_v A_u + B_v D_u.$$

If $\|O - Vq\| \le \eta$*, then*

$$\|OT_w - VU_w q\| \le \eta A_w + \|V\| D_w.$$

For arbitrary initial $x \in X$*,* $z \in Z$*,* $\|qT_w x - U_w z\| \le D_w \|x\| + B_w \|qx - z\|$*.*

Proof. Induction and submultiplicativity bound the two word operators by the products. Apply the triangle inequality and submultiplicativity to the first identity in SR-13 to obtain exactly the recursion for D_w. Induction on u proves the displayed scalar concatenation identity; it tracks the same algebra as the exact defect identity, with nonnegative factors preventing an inequality reversal. The output bound follows from its exact decomposition in SR-13. For initialization, write $qT_w x - U_w z = E_w x + U_w(qx - z)$ and bound both terms. These are operator budgets; agreement on a finite set of sample vectors does not supply them.

□

Formal source: `ApproximateBounds.norm_wordError_le_budget;ApproximateBounds.budget_append;ApproximateBounds.norm_outputError_le;ApproximateBounds.norm_state_error_le`.

Hypotheses. Uniform nonnegative whole-map bounds a,b,epsilon; all finite words with length n; eta bound on output mismatch.

Theorem 5.3 (SR-15 Two-rate response and equal-rate limit)**.** *For* $H_0(a,b) = 0$ *and* $H_{n+1}(a,b) = aH_n(a,b) + b^n$,

$$H_n(a,b) = \sum_{j=0}^{n-1} a^j b^{n-1-j}, \qquad (a-b)H_n = a^n - b^n.$$

If $a \neq b$, $H_n = (a^n - b^n)/(a-b)$*; for equal rates,* $H_{n+1}(a,a) = (n+1)a^n$*. Uniform step bounds give*

$$\|E_w\| \leq \epsilon H_n(a,b), \qquad \|OT_w - VU_w q\| \leq \eta a^n + \|V\|\,\epsilon H_n(a,b).$$

Proof. Induction proves the finite-sum formula: multiplying the old sum by a and adjoining b^n generates every term of the new sum. Multiplication by $a-b$ telescopes, leaving $a^n - b^n$. Division is used only when the rates are distinct. If they coincide, all $n+1$ terms of H_{n+1} equal a^n, including the zero-rate boundary with the usual $a^0 = 1$. For uniform bounds, the heterogeneous budget becomes $D_{i::w} = aD_w + b^{|w|}\epsilon$, so induction gives $D_w = \epsilon H_{|w|}(a,b)$. Apply SR-14. The empty word has zero dynamics defect and is handled by $H_0 = 0$, with no negative exponent required.

□

Formal source: `ScalarBudgets.response_eq_sum;ScalarBudgets.response_mul_sub;ScalarBudgets.response_of_ne;ScalarBudgets.response_same_succ;StableApproximation.norm_wordError_uniform;StableApproximation.norm_outputError_uniform`.

Hypotheses. Bounded maps on the normed spaces of SR-14; $\|T_i\| \leq 1$ for every true operation; $\|U_i\| \leq 1$ in the nonexpansive case or a common $0 \leq b < 1$ with $\|U_i\| \leq b$ in the contractive case; $\|qT_i - U_i q\| \leq \epsilon$ and $\|O - Vq\| \leq \eta$ with $\epsilon, \eta \geq 0$.

Theorem 5.4 (SR-16 Nonexpansive and contractive continuations)**.** *If both families are nonexpansive,* $\|E_w\| \leq n\epsilon$ *and the output defect is at most* $\eta + \|V\|\,n\epsilon$*. If the reduced family has a common bound* $0 \leq b < 1$*, then*

$$\|E_w\| \leq \frac{\epsilon}{1-b}, \qquad \|OT_w - VU_w q\| \leq \eta + \|V\|\,\frac{\epsilon}{1-b}.$$

For arbitrary reduced initialization,

$$\|OT_w x - VU_w z\| \leq \left(\eta + \|V\|\,\frac{\epsilon}{1-b}\right)\|x\| + \|V\|\,b^n\,\|qx - z\|.$$

Proof. For $a = b = 1$, the response sum has n unit terms. For $a = 1$ and $b < 1$, it is $1 + b + \cdots + b^{n-1} \leq 1/(1-b)$. SR-15 gives the operator estimates. For initialization, add $VU_w(qx - z)$ to the operator output error applied to x, and use $\|U_w\| \leq b^n$. The common bound concerns every allowed operation in the chosen norm. Individual spectral radii do not imply a common product bound for a noncommuting family. The nonexpansive estimate is an upper bound and does not state that every such system accumulates error linearly.

□

Formal source: `StableApproximation.norm_wordError_nonexpansive;StableApproximation.norm_outputError_nonexpansive;StableApproximation.norm_wordError_contractive;StableApproximation.norm_output_initial_contractive`.

6 Continuous flow and exact hidden memory

Let L be a bounded real operator on a Banach space. Its flow is the actual operator exponential e^{tL}. Boundedness allows differentiation of the norm-convergent exponential series and integration of continuous operator curves. Equalities in this section hold for all real t, using oriented interval integrals when $t < 0$.

Hypotheses. Real Banach space; bounded autonomous D; continuous forcing f; differentiable actual z with z′=Dz+f at every real time.

Theorem 6.1 (SR-17 Actual flow and variation of constants). *$\frac{d}{dt}e^{tD} = De^{tD} = e^{tD}D$, $e^{(s+t)D} = e^{sD}e^{tD}$, and e^{-tD} is the inverse flow. Every actual curve satisfying $z' = Dz + f$ with continuous f obeys*

$$z(t) = e^{tD}z(0) + \int_0^t e^{(t-s)D}f(s)\,ds.$$

The initial value uniquely determines the homogeneous solution; two forced solutions with the same forcing and initial value also agree.

Proof. Termwise differentiation of the exponential series gives both derivative orders, since all powers of D commute with D. The exponential addition identity for commuting multiples of one bounded operator gives the flow law and inverse. For fixed t, differentiate $e^{(t-s)D}z(s)$. The product rule produces $-De^{(t-s)D}z(s) + e^{(t-s)D}(Dz(s) + f(s))$; commutation cancels the generator terms. The remaining derivative is the continuous integrand $e^{(t-s)D}f(s)$. The fundamental theorem of calculus on the oriented interval gives the displayed identity. Setting $f = 0$ yields the homogeneous formula. Subtracting two forced solutions reduces uniqueness to that case. The theorem starts from the differential equation, not from an assumed integral representation.

□

Formal source: `ContinuousFlow.hasDerivAt_flow;ContinuousFlow.flow_add;ContinuousFlow.variation_of_constants;ContinuousFlow.homogeneous_unique;ContinuousFlow.forced_unique`.

Hypotheses. Bounded real generator algebras and bounded linear observation maps; norm-convergent exponentials; all real times or all natural powers.

Theorem 6.2 (SR-18 Power identities and exponential identities). *For bounded linear observation maps K, J on bounded generator algebras,*

$$K(e^{tL}) = J(e^{tD})\ (\forall t) \iff K(L^n) = J(D^n)\ (\forall n).$$

In particular, $Be^{tD}G = 0$ for every t exactly when $BD^nG = 0$ for every n. For $CR = I$, $A = CLR$, one has

$$Ce^{tL}R = e^{tA}\ (\forall t) \iff CL^nR = A^n\ (\forall n),$$

and, with $Q = I - RC$,

$$Ce^{(s+t)L}R - (Ce^{sL}R)(Ce^{tL}R) = Ce^{sL}Qe^{tL}R.$$

Proof. The nth derivative at zero of e^{tL} is L^n. Bounded linear maps commute with these derivatives, so equality for all times implies equality of every observed power. Conversely bounded linear maps commute with the sum of the norm-convergent exponential series: $K(e^{tL}) = \sum_n (t^n/n!)K(L^n)$. Equality term by term gives equality of the sums. Sandwiching and compression are bounded linear maps on the corresponding operator spaces, so they give the specializations. For the last identity, use the flow law and insert $I = RC + Q$ between the two flows. The defect factorization itself does not need $CR = I$; that condition makes Q the discarded-state projector and ensures $E(0) = I$.

□

Formal source: `ContinuousFlow.mapped_flow_eq_iff_powers_eq;ContinuousFlow.sandwich_flow_zero_iff;ContinuousFlow.compressed_flow_iff_powers;ContinuousFlow.compressed_flow_defect`.

Hypotheses. Real Banach visible/hidden spaces; bounded autonomous blocks A,B,G,D; actual exponential full trajectory; arbitrary hidden initial state.

Theorem 6.3 (SR-19 Continuous hidden elimination)**.** *For* $L(y,z) = (Ay + Bz, Gy + Dz)$ *and* $(y(t), z(t)) = e^{tL}(y_0, z_0)$,

$$z(t) = e^{tD}z_0 + \int_0^t e^{(t-s)D}Gy(s)\,ds.$$

Writing $M(t) = Be^{tD}G$, *the visible dynamics satisfy*

$$y'(t) = Ay(t) + Be^{tD}z_0 + \int_0^t M(t-s)y(s)\,ds.$$

Prepared initialization removes the term involving z_0.

Proof. Projecting the derivative of e^{tL} onto the visible and hidden components gives $y' = Ay + Bz$ and $z' = Gy + Dz$. The visible curve is continuous, so Gy is continuous forcing for the hidden equation. SR-17 yields the hidden elimination formula. Bounded B commutes with the interval integral, giving the memory kernel in the visible equation. Taking $z_0 = 0$ removes only the initialization term. No converse construction of a full trajectory from an arbitrary Volterra solution is asserted here, and no unbounded generator domain is present.

□

Formal source: `ContinuousBlockDynamics.hasDerivAt_visible;ContinuousBlockDynamics.hidden_elimination;ContinuousBlockDynamics.continuous_memory_equation;ContinuousBlockDynamics.prepared_memory_equation`.

Hypotheses. Bounded real Banach block system; finite-dimensional hidden H only for finite certificates, or supplied monic annihilator.

Theorem 6.4 (SR-20 Continuous closure, finite moments, and jets)**.** *Let* $E(t) = Ce^{tL}R$. *The following are equivalent:*

$$E(t) = e^{tA}\ (\forall t), \quad M(t) = 0\ (\forall t), \quad BD^kG = 0\ (\forall k), \quad E(s+t) = E(s)E(t)\ (\forall s,t).$$

If $\dim H = d$, *each is equivalent to* $BD^kG = 0$ *for* $k < d$, *or to* $E^{(n)}(0) = A^n$ *for* $n \le d+1$. *If the first k return moments vanish,*

$$E^{(k+2)}(0) - A^{k+2} = BD^kG.$$

Proof. SR-18 converts prepared flow equality into prepared power equality and memory flow vanishing into memory moment vanishing. SR-06 equates the corresponding algebraic conditions, and SR-07 supplies the finite moment criterion. Exact prepared flow gives its semigroup law. Conversely, if E is a semigroup, differentiating $E(s+t) = E(s)E(t)$ in s at zero gives $E'(t) = AE(t)$ because $E'(0) = CLR = A$ and $E(0) = I$. The bounded-generator uniqueness theorem yields $E(t) = e^{tA}$.

Actual iterated derivatives of the exponential satisfy $E^{(n)}(0) = CL^nR$. Applying the finite algebraic horizon and first-return identity proves the jet statements. The leading Taylor coefficient has the additional denominator $(k+2)!$; the derivative itself does not. The visible Banach space can be infinite-dimensional. The finite tests are whole-operator jet identities, not a finite list of sampled times.

□

Formal source: `ContinuousBlockDynamics.prepared_exact_iff_memory_zero;ContinuousBlockDynamics.prepared_semigroup_iff_memory_zero;ContinuousBlockDynamics.prepared_exact_iff_finite_memory;ContinuousBlockDynamics.prepared_exact_iff_finite_derivatives;ContinuousBlockDynamics.first_return_derivative`.

Example 6.5 (A linear continuous memory kernel)**.** Take $V = \mathbb{R}$, $H = \mathbb{R}^2$, $A = 0$, $B(z_1, z_2) = z_1$, $G(y) = (0, y)$, and $D(z_1, z_2) = (z_2, 0)$. Since $D^2 = 0$, $e^{tD}(z_1, z_2) = (z_1 + tz_2, z_2)$ and $M(t) = tI$. The first two prepared jets agree with the reduced generator, while the third derivative differs by I. Changing D to zero leaves both boundary couplings nonzero but makes the memory kernel identically zero. Arbitrary hidden initialization still affects the visible initial derivative.

7 Continuous approximate sufficiency

For bounded real generators L on X and A on Z, and bounded $q : X \to Z$, write $\delta = qL - Aq$ and $E(t) = qe^{tL} - e^{tA}q$. Algebraic evolution identities hold for all real times. Norm bounds in this section restrict to $t \geq 0$, with growth assumptions on the nonnegative-time flows.

Hypotheses. Real Banach source/reduced spaces; bounded autonomous generators; bounded q and second representation p; all real times.

Theorem 7.1 (SR-21 Duhamel transport and composition)**.** *One has the operator identity*

$$E(t) = \int_0^t e^{(t-s)A}\delta e^{sL}\, ds,$$

and $\delta = 0$ exactly when $E(t) = 0$ for every t. Also

$$E(s+t) = E(s)e^{tL} + e^{sA}E(t).$$

For a second representation $p : Z \to W$ with generator B,

$$\delta_{pq} = p\delta_q + \delta_p q, \qquad E_{pq}(t) = pE_q(t) + E_p(t)q.$$

Proof. Differentiate $F(s) = e^{(t-s)A}qe^{sL}$ using the product rule. Its derivative is $e^{(t-s)A}(qL - Aq)e^{sL}$. Integrating from zero to t gives $F(t) - F(0) = E(t)$ and proves the formula. Zero generator defect therefore forces zero evolution defect; differentiating E at zero gives the converse. The addition identity follows by adding and subtracting $e^{sA}qe^{tL}$ and using the two flow laws. For a composite representation, add and subtract pAq in its generator defect and $pe^{tA}q$ in its evolution defect. Both stages' contributions remain explicit. Boundedness guarantees the continuous operator integrand.

□

Formal source: `ContinuousDefect.duhamel;ContinuousDefect.generatorDefect_zero_iff;ContinuousDefect.evolutionDefect_add;ContinuousDefect.generatorDefect_composite;ContinuousDefect.evolutionDefect_composite`.

Hypotheses. Real Banach spaces X, Z; bounded autonomous real generators L, A and bounded representation/readouts q, O, V; real rates a, b of either sign; $t \geq 0$ and $M_L, M_A, \epsilon, \eta \geq 0$. For every $s \geq 0$, $\left\|e^{sL}\right\| \leq M_L e^{as}$ and $\left\|e^{sA}\right\| \leq M_A e^{bs}$; also $\|qL - Aq\| \leq \epsilon$ and $\|O - Vq\| \leq \eta$.

Theorem 7.2 (SR-22 Growth envelopes and scalar response)**.** *For $\|\delta\| \leq \epsilon$, put*

$$R(a,b,t) = \int_0^t e^{b(t-s)} e^{as}\, ds = \begin{cases} (e^{at} - e^{bt})/(a-b), & a \neq b, \\ te^{at}, & a = b. \end{cases}$$

Then

$$\|E(t)\| \leq M_L M_A \epsilon R(a,b,t).$$

If $\|O - Vq\| \leq \eta$, the output defect $F(t) = Oe^{tL} - Ve^{tA}q$ obeys

$$\|F(t)\| \leq \eta M_L e^{at} + \|V\|\, M_L M_A \epsilon R(a,b,t).$$

Proof. Apply the integral norm inequality and submultiplicativity to SR-21. For $0 \leq s \leq t$, the integrand norm is at most $M_A e^{b(t-s)} \epsilon M_L e^{as}$. Integration gives the first estimate. Factoring e^{bt} and integrating $e^{(a-b)s}$ gives the distinct-rate formula; for equal rates the integrand is constant e^{at}. The output decomposition is $F(t) = (O - Vq)e^{tL} + VE(t)$, giving the stated bound. The same argument works with continuous nonnegative envelope functions u, v on the interval, replacing the scalar response by $\int_0^t v(t-s)u(s)\, ds$. In particular these assumptions concern actual flow norms, not just eigenvalues of the generators.

□

Formal source: `ExponentialBudgets.response_of_ne;ExponentialBudgets.response_self;ContinuousApproximation.norm_evolution_le_response;ContinuousApproximation.norm_output_le_response`.

Hypotheses. True flow bounded by M_L; reduced flow bounded by M_A exp(-rt), r>0, for t≥0; generator/readout allowances; arbitrary initial reduced z.

Theorem 7.3 (SR-23 Uniform continuous stability and initialization)**.** *If $\left\|e^{tL}\right\| \leq M_L$ and $\left\|e^{tA}\right\| \leq M_A e^{-rt}$ with $r > 0$, then*

$$\|E(t)\| \leq \frac{M_L M_A \epsilon}{r}, \qquad \|F(t)\| \leq \eta M_L + \|V\| \frac{M_L M_A \epsilon}{r}.$$

For arbitrary x, z,

$$\begin{aligned} \left\|Oe^{tL}x - Ve^{tA}z\right\| \leq &\left(\eta M_L + \|V\| \frac{M_L M_A \epsilon}{r}\right) \|x\| \\ &+ \|V\|\, M_A e^{-rt} \|qx - z\|. \end{aligned}$$

Composition through a second representation propagates allowances with its explicit map norm, as in SR-21.

Proof. Use SR-22 with $a = 0$, $b = -r$. Its response equals $(1 - e^{-rt})/r \leq 1/r$ for $t \geq 0$. This gives the evolution and output operator bounds. The initialization error decomposes into $F(t)x$ plus $Ve^{tA}(qx - z)$, which proves the displayed estimate. The first allowance must be multiplied by $\|x\|$ because it is an

operator estimate. For another representation p with its own evolution defect, SR-21 gives $\|E_{pq}(t)\| \leq \|p\| \|E_q(t)\| + \|E_p(t)\| \|q\|$. This retains error introduced at both stages, rather than identifying their budgets with a single unqualified small quantity.

□

Formal source: `ContinuousStability.norm_evolution_uniform;ContinuousStability.norm_output_uniform;ContinuousStability.norm_initialization_uniform;ContinuousStability.norm_composite_evolution`.

Hypotheses. Actual real scalar bounded generators, q=I; t≥0; b≤a for equality and epsilon>0 for counterexample.

Theorem 7.4 (SR-24 Sharp response and small-defect failure)**.** *For scalar generators $a \geq b$,*

$$|e^{at} - e^{bt}| = (a-b)R(a,b,t),$$

so the response estimate is attained. For every $\epsilon > 0$, choosing $L = \epsilon I$, $A = 0$, $q = I$ gives

$$\|\delta\| = \epsilon, \qquad \|E(t)\| = e^{\epsilon t} - 1 \geq \epsilon t.$$

Its evolution defect exceeds every fixed bound at some nonnegative time.

Proof. The scalar exponential is the actual flow by the homogeneous ODE uniqueness theorem. For $a > b$, the response formula immediately gives the equality; for $a = b$, both sides vanish. In the second example, the generator defect is multiplication by ϵ, while the evolution defect is multiplication by $e^{\epsilon t} - 1$. The elementary exponential inequality $e^u \geq 1 + u$ for $u \geq 0$ gives the lower bound. Taking t sufficiently large makes it exceed any prescribed real bound. Thus a small generator defect alone does not supply uniform accuracy. The hypothesis of a bounded true flow in SR-23 fails here.

□

Formal source: `ContinuousApproxExamples.scalar_response_sharp;ContinuousApproxExamples.unstable_generator_exact;ContinuousApproxExamples.unstable_error_exact;ContinuousApproxExamples.unstable_error_unbounded`.

8 Orthogonal globalization and conditioning

Let $R \subseteq K$ be subspaces of a real inner-product space V, both admitting orthogonal projections. Define the actual maps

$$r(v) = P_R v, \quad o(v) = (P_K - P_R)v, \quad i(v) = (I - P_K)v.$$

They distinguish realized data, compatible obstruction, and incompatibility. In a globalization presentation use $R = \operatorname{Ran} A$, $K = \ker \delta$ with $\delta A = 0$.

Hypotheses. Real inner-product V; nested subspaces R≤K both with orthogonal projections; arbitrary v and represented candidate u∈R.

Theorem 8.1 (SR-25 Unique three-component geometry)**.** *The components lie in R, $K \cap R^\perp$, and $K^\perp$, respectively, are pairwise orthogonal, and give the unique decomposition $v = r(v) + o(v) + i(v)$. Furthermore,*

$$\|v\|^2 = \|r\|^2 + \|o\|^2 + \|i\|^2, \qquad \operatorname{dist}(v,R)^2 = \|o\|^2 + \|i\|^2.$$

For any $u \in R$,

$$\|v - u\|^2 = \operatorname{dist}(v,R)^2 + \|r(v) - u\|^2.$$

Thus $r(v)$ is the unique nearest represented vector.

Proof. Nested projections satisfy $P_R P_K = P_R$. Hence $P_K v - P_R v$ lies in K and is orthogonal to R; the last component is orthogonal to K. Their definitions sum to v and yield pairwise orthogonality. In any competing decomposition, projection onto R selects its first term and projection onto K selects the first two, so all components are forced. Pythagoras gives the norm identities. For $u \in R$, the residual $v - r(v)$ is orthogonal to $r(v) - u$, so the candidate identity follows. It proves optimality and uniqueness, and taking the infimum over actual $u \in R$ proves the distance formula. Each component map is an orthogonal projection onto its displayed sector and consequently is nonexpansive.

□

Formal source: `OrthogonalResolution.components_sum;OrthogonalResolution.unique_components;OrthogonalResolution.norm_sq_components;OrthogonalResolution.candidate_error_sq;OrthogonalResolution.infDist_sq_decomposition`.

Hypotheses. Real linear presentation δA=0; projections on actual R=range A and K=kerδ; source G only needs real module structure.

Theorem 8.2 (SR-26 Harmonic realization of the obstruction quotient)**.** *The same quotient $\mathcal{H} = K/R$ from SR-03 is linearly equivalent to $K \cap R^\perp$ by*

$$[v] \longmapsto v - P_R v \quad (v \in K).$$

It is inverse to the quotient map restricted to the harmonic subspace. For compatible data, the class is zero exactly when $o(v) = 0$; equality of classes is equality of harmonic representatives. The best image $P_R v$ has an actual source under A.

Proof. Adding a vector of R leaves $v - P_R v$ unchanged, so the map descends to K/R. Its values lie in $K \cap R^\perp$ by SR-25. For harmonic h, $P_R h = 0$, so its class maps back to h. For compatible v, the difference $v - (v - P_R v) = P_R v$ lies in R, so the reverse roundtrip returns $[v]$. The inverse laws prove the equivalence and both class criteria. Since $P_R v$ belongs to the actual range, some g satisfies $Ag = P_R v$. This is image existence; uniqueness of g requires the separate injectivity condition in SR-03. No norm isometry is inferred from linear equivalence alone; the metric statement is proved in SR-33.

□

Formal source: `HilbertGlobalization.obstructionEquivHarmonic;HilbertGlobalization.class_zero_iff_component_zero;HilbertGlobalization.same_class_iff_same_harmonic;HilbertGlobalization.exists_best_global_image`.

Hypotheses. Same projection presentation; bounded δ; sigma>0 with sigma$\|$x$\| \le \|\delta$x$\|$ on K$\perp$; arbitrary data.

Theorem 8.3 (SR-27 Residual bounds retain conditioning and obstruction)**.** *If $\sigma \|x\| \le \|\delta x\|$ for $x \in K^\perp$ and $\sigma > 0$, then*

$$\delta i(v) = \delta v, \qquad \|i(v)\| \le \frac{\|\delta v\|}{\sigma},$$

and

$$\|o(v)\| \le \operatorname{dist}(v, R), \qquad \operatorname{dist}(v, R)^2 \le \|o(v)\|^2 + \left(\frac{\|\delta v\|}{\sigma}\right)^2.$$

For compatible data, $\operatorname{dist}(v, R) = \|o(v)\|$.

Proof. The component $P_K v$ is in $\ker \delta$, so $\delta(v - P_K v) = \delta v$. Apply coercivity to $i(v) \in K^\perp$ and divide by the positive σ. SR-25 gives the exact sum of two nonnegative squares, proving both the lower bound and conditioned upper bound. If $\delta v = 0$, then $v \in K$ and $i(v) = 0$, giving equality. In particular, a compatibility residual bounds only inconsistency. A residual-only global bound needs a vanished obstruction or exactness $R = K$. The condition on σ is independent of the size of a particular measured residual and cannot be inferred from that measurement.

□

Formal source: `HilbertConditioning.inconsistency_residual;HilbertConditioning.inconsistency_le_residual;HilbertConditioning.conditioned_distance_sq;HilbertConditioning.obstruction_lower_bound;HilbertConditioning.compatible_distance`.

Hypotheses. Finite-dimensional real V for existence/characterization; fixed metric and K with projected R0≤R1≤K for refinement.

Theorem 8.4 (SR-28 Finite conditioning and realizable enlargement)**.** *In finite dimension some $\sigma > 0$ satisfies SR-27. Under $\delta A = 0$,*

$$\exists c > 0\ \forall v : \operatorname{dist}(v, \operatorname{Ran} A) \le c\,\|\delta v\| \iff \operatorname{Ran} A = \ker \delta.$$

For projected $R_0 \subseteq R_1 \subseteq K$ in a fixed metric, put $g(v) = (P_{R_1} - P_{R_0})v$. Then

$$\|o_0(v)\|^2 = \|o_1(v)\|^2 + \|g(v)\|^2, \quad \operatorname{dist}(v, R_0)^2 = \operatorname{dist}(v, R_1)^2 + \|g(v)\|^2.$$

No uniform conditioning constant exists across all nonzero scalar compatibility maps.

Proof. Restrict δ to $K^\perp$. It is injective: a zero image lies in both K and $K^\perp$. A finite-dimensional injective linear map has a positive lower norm bound, giving σ; the zero domain is harmless and permits any positive choice. If the residual-only inequality holds, every $v \in K$ has zero distance to $\operatorname{Ran} A$. Finite-dimensional ranges are closed, so $v \in \operatorname{Ran} A$. Conversely exactness removes $o(v)$ in SR-27.

Nested orthogonal projections split $o_0(v) = o_1(v) + g(v)$ into orthogonal sectors, and the inconsistency component stays fixed. Pythagoras gives both refinement identities. For the scalar family $\delta_s(x) = sx$, $s > 0$, take $R = 0$ and $v = 1$. The reconstruction distance is one and the residual is s. For any proposed common c, choosing s sufficiently small violates $1 \le cs$. The theorem guarantees a constant per presentation, not a common constant across degenerating presentations.

□

Formal source: `HilbertConditioning.exists_coercivity_finite;FiniteGlobalization.finite_residual_bound_iff_exact;GlobalizationRefinement.obstruction_refinement_sq;GlobalizationRefinement.distance_refinement_sq;GlobalizationExamples.no_uniform_conditioning_over_scalar_family`.

Example 8.5 (A complete error split)**.** For $A(g) = (g, 0, 0)$, $\delta(x, y, z) = 2z$, and $v = (3, 4, 5)$, the components are $(3, 0, 0)$, $(0, 4, 0)$, and $(0, 0, 5)$. The squared reconstruction distance is $41 = 16 + 25$. The compatibility residual has squared norm 100, and $\sigma = 2$ is sharp on $K^\perp$. Thus the conditioned upper bound is attained: $16 + 100/4 = 41$.

9 Canonical minimum-norm reconstruction

Let $A : G \to V$ be bounded between real Hilbert spaces, with actual closed range $R = \operatorname{Ran} A$. Write $N_A = \ker A$ and $S = N_A^\perp$. Closedness is a substantive hypothesis for a bounded exact inverse. It will be removed for regularized recovery in the following sections.

Hypotheses. Bounded real Hilbert operator A with actual closed range; core needs complete source/range and orthogonal projections.

Theorem 9.1 (SR-29 Constructed bounded canonical inverse)**.** *Restriction $A_S : S \to R$ is a bounded linear bijection with bounded inverse. Define*

$$B = \iota_S A_S^{-1} P_R.$$

Then $AB = P_R$, $BA = P_S$, $ABA = A$, and $BAB = B$. Its two projections are symmetric, $\ker B = R^\perp$*, and* $\operatorname{Ran} B = S$*. The best-image identity $AB = P_R$ and the condition* $\operatorname{Ran} B \subseteq S$ *uniquely determine B.*

Proof. The restriction is injective because $S \cap \ker A = 0$. Every $Ag \in R$ equals $AP_S g$, so it is surjective. Closed R and closed S are Banach spaces; the bounded inverse theorem supplies A_S^{-1} as an actual continuous linear inverse. The definition then gives $AB = P_R$. For g, BAg is the unique element of S with image Ag, namely $P_S g$. The two inverse identities imply the generalized inverse laws. Orthogonal projections are symmetric, proving the symmetry conditions. The definition and injectivity of A_S^{-1} identify the kernel and range. Finally, if B' has the same best image and values in S, then $B'v - Bv \in S \cap \ker A$ for each v, hence it is zero. This construction is the classical Moore–Penrose inverse with the actual range and source ambiguity retained.

□

Formal source: `MinimumNormReconstruction.restrictionEquiv;MinimumNormReconstruction.image_comp_recover;MinimumNormReconstruction.recover_comp_image;MinimumNormReconstruction.recover_unique;ClosedRangeReconstruction.closedRecover`.

Hypotheses. Same real Hilbert closed-range constructor; arbitrary data v, candidate g and arbitrary deterministic noise e.

Theorem 9.2 (SR-30 All least-squares sources and unique minimum norm)**.** *For every source g,*

$$\|v - Ag\|^2 = \operatorname{dist}(v, R)^2 + \|A(g - Bv)\|^2 .$$

All least-squares minimizers are $Bv + k$ with $k \in \ker A$. Among them Bv is the unique minimum-norm source. Exact source recovery $BAg = g$ holds exactly for $g \in S$. For data $Ag + e$,

$$\|B(Ag + e) - g\|^2 = \|Be\|^2 + \|P_{N_A} g\|^2 .$$

Proof. $ABv = P_R v$, so SR-25's candidate identity with $u = Ag$ proves the first formula. Equality to the minimum image error is equivalent to $A(g - Bv) = 0$, yielding exactly the affine family of minimizers. Since $Bv \in S$ and $k \in N_A$, Pythagoras gives $\|Bv + k\|^2 = \|Bv\|^2 + \|k\|^2$. It proves minimum norm and uniqueness. The identity $BA = P_S$ proves the exact recovery criterion. For noisy data, linearity gives $B(Ag + e) - g = Be - P_{N_A} g$. The first term belongs to S and the second to N_A, so they are orthogonal. This last identity does not assume a statistical noise model. It shows explicitly which source content the operator cannot identify.

□

Formal source: `MinimumNormOptimality.candidate_error_decomposition;MinimumNormOptimality.all_least_squares_sources;MinimumNormOptimality.unique_minimum_norm_least_squares;MinimumNormOptimality.exact_source_recovery_iff;MinimumNormOptimality.noisy_source_error_sq`.

Hypotheses. Closed-range A; alpha>0 with alpha‖g‖ ≤ ‖Ag‖ on S; computed source h may contain a kernel component.

Theorem 9.3 (SR-31 Source conditioning and solver ambiguity)**.** *Some $\alpha > 0$ satisfies $\alpha \|g\| \le \|Ag\|$ on S, and any supplied such constant gives $\|B\| \le 1/\alpha$. Therefore*

$$\|B(Ag+e)-g\|^2 \le (\|e\|/\alpha)^2 + \|P_{N_A} g\|^2 .$$

For an arbitrary computed source h,

$$\|h-Bv\|^2 \le \left(\frac{\|Ah-P_R v\|}{\alpha}\right)^2 + \|P_{N_A} h\|^2 .$$

If also $\delta A = 0$, a compatibility constant σ bounds the local error as in SR-27. The constants α and σ have separate roles.

Proof. The bounded inverse on S gives $\|g\| \le \|B\|\,\|Ag\|$. Taking, for example, $\alpha = 1/(\|B\|+1) > 0$ handles also the zero complement and proves existence. For a supplied constant, apply it to $Bv \in S$: $\alpha\|Bv\| \le \|ABv\| = \|P_R v\| \le \|v\|$. Take the operator norm and combine with SR-30 for the noisy bound. For a computed h, split $h - Bv = (P_S h - Bv) + P_{N_A} h$ orthogonally. Coercivity bounds the first term by $\|Ah - P_R v\|/\alpha$ because $AP_S h = Ah$. The compatibility operator bounds distance of local data to realizability; the forward operator bounds source amplification. Neither condition establishes the other. A small fit without control of the computed kernel component does not certify the minimum-norm source.

□

Formal source: `SourceConditioning.exists_source_coercivity;SourceConditioning.recover_opNorm_bound;SourceConditioning.noisy_source_bound_sq;SourceConditioning.candidate_source_bound_sq;ReconstructionGlobalization.two_condition_reconstruction`.

Hypotheses. Bounded A between real Hilbert spaces; actual range closedness; bounded recovery BA=P_S or off-kernel coercivity.

Theorem 9.4 (SR-32 Closed range is equivalent to bounded recovery)**.** *The following conditions are equivalent:*

$$\text{Ran } A \text{ is closed;} \qquad \exists \alpha > 0 : \alpha\|g\| \le \|Ag\| \ (g \in S); \qquad \exists B \text{ bounded} : BA = P_S .$$

Thus a nonclosed range admits no bounded exact identifiable-source recovery. Finite-dimensional source or target supplies closed range. There is no common inverse-sensitivity constant across all nonzero real scalar forward maps.

Proof. Closed range gives the bounded constructor and coercivity by SR-29–SR-31. Conversely suppose coercivity holds. If Ag_n converges in V, replace g_n by $P_S g_n$ without changing its image. Coercivity makes this source sequence Cauchy. Completeness gives a limit $g \in S$, and continuity gives the image limit $Ag \in \text{Ran } A$. Hence the range is closed. If bounded B satisfies $BA = P_S$, then for $g \in S$, $\|g\| = \|BAg\| \le \|B\|\,\|Ag\|$. The same positive choice $1/(\|B\|+1)$ gives coercivity. Finite-dimensional

image subspaces are closed in a normed space. Finally $A_c g = cg$, $c > 0$, has canonical inverse $B_c v = v/c$. At $v = c$, data tend to zero with c while the recovered source remains one, ruling out a common constant as $c \downarrow 0$.

□

Formal source: `ClosedRangeReconstruction.closed_range_iff_coercivity;ClosedRangeReconstruction.bounded_recovery_iff_closed_range;ClosedRangeReconstruction.nonclosed_no_bounded_recovery;ReconstructionExamples.no_uniform_source_conditioning`.

Example 9.5 (Source and image are different optima)**.** Let $A(g_1, g_2) = (g_1 + g_2, 0, 0)$ and $v = (3, 4, 5)$. The best image is $(3, 0, 0)$, with the same squared local error 41. The minimum-norm source is $(3/2, 3/2)$ of squared norm $9/2$. The source $(3, 0)$ gives the same best image but has squared norm 9. Their difference $(3/2, -3/2)$ is kernel ambiguity, not local obstruction.

10 Intrinsic metric obstruction and transport

Retain the globalization presentation $\delta A = 0$, the actual projected subspaces $R = \operatorname{Ran} A \subseteq K = \ker \delta$, and the same algebraic quotient $\mathcal{H} = K/R$. Its seminorm is the quotient seminorm induced by the ambient metric. We prove the harmonic equivalence respects that preexisting seminorm, rather than defining a new norm to match its answer.

Hypotheses. Real inner-product V; R=range A,K=kerδ both projected; δ bounded, δA=0; native quotient seminorm.

Theorem 10.1 (SR-33 Native quotient norm and isometry)**.** *The equivalence $K/R \cong K \cap R^{\perp}$ in SR-26 is a linear isometry. For compatible v, w,*

$$\|[v]\| = \|o(v)\| = \operatorname{dist}(v, R), \qquad \operatorname{dist}([v], [w]) = \operatorname{dist}(v - w, R).$$

The quotient norm detects zero classes. Its standard quotient inner product is preserved when the induced projection structure in K is used.

Proof. Let h be the harmonic representative of a class z. For any compatible representative v of z, its harmonic component is h, and the nonexpansive harmonic projection gives $\|h\| \leq \|v\|$. Taking the infimum over representatives proves $\|h\| \leq \|z\|$. Conversely h itself represents z, so the quotient map's norm bound gives $\|z\| \leq \|h\|$. This proves isometry against the existing quotient seminorm. The compatible distance formula follows from SR-25, and distances of classes follow by applying it to $v - w$. Zero norm implies zero harmonic vector and hence zero class by the inverse laws. Orthogonal projection onto R restricts to K, constructing the quotient inner-product structure there. A real linear isometry preserves its inner product by the polarization identity.

□

Formal source: `MetricObstruction.quotientToHarmonic_norm;MetricObstruction.obstructionIsometry;MetricObstruction.norm_class_eq_infDist;MetricObstruction.distance_classes;MetricObstruction.quotient_inner_preserved`.

Hypotheses. Same projected real presentation; arbitrary v for projected class c(v)=[P_Kv]; complete/closed source range only when canonical inverse B used.

Theorem 10.2 (SR-34 Minimum representatives and intrinsic class of data)**.** *Every class z has a unique compatible minimum-norm representative $s(z)$, and every compatible representative v satisfies*

$$\|v\|^2 = \|z\|^2 + \|v - s(z)\|^2 .$$

For arbitrary data define $c(v) = [P_K v]$. This is nonexpansive and

$$\operatorname{dist}(v, R)^2 = \|c(v)\|^2 + \|i(v)\|^2 .$$

The datum is realizable exactly when both $c(v) = 0$ and $\delta v = 0$. When the canonical inverse exists,

$$\|v - Ag\|^2 = \|c(v)\|^2 + \|i(v)\|^2 + \|A(g - Bv)\|^2 .$$

Proof. Take $s(z)$ to be its harmonic representative from SR-33. The difference between any other compatible representative and $s(z)$ lies in R, which is orthogonal to the harmonic sector. Pythagoras gives the first formula, including unique attainment of the minimum. The composition of P_K and the quotient map is bounded by one; its harmonic representative is $o(v)$. SR-25 and SR-33 give the intrinsic distance identity. If both displayed tests vanish, then $v \in K$ and its obstruction class is zero, so SR-03 supplies a source. Realizability implies both tests directly. Finally combine the candidate identity of SR-30 with the intrinsic distance split. The minimum representative of an obstruction class belongs to local data space; it is a different construction from the minimum-norm global source Bv.

□

Formal source: `MetricRepresentatives.unique_minimum_representative;MetricRepresentatives.representative_error_energy;IntrinsicGlobalization.projectedClass_opNorm_le;IntrinsicGlobalization.intrinsic_distance_sq;IntrinsicGlobalization.realizable_iff_class_and_residual_zero;IntrinsicGlobalization.intrinsic_candidate_total_error`.

Hypotheses. Commuting linear presentation maps; whole local-map norm bound C≥0; reverse presentation with bound M>0 and induced left inverse for two-sided estimate.

Theorem 10.3 (SR-35 Bounded quotient transport and reflection)**.** *A presentation map with $\|Lv\| \le C\,\|v\|$ induces the same quotient map as in SR-03, now satisfying*

$$\left\|\overline{L}z\right\| \le C\,\|z\| .$$

If a reverse presentation has bound $M > 0$ and its quotient map is a left inverse of $\overline{L}$, then

$$\frac{\|z\|}{M} \le \left\|\overline{L}z\right\| \le C\,\|z\| .$$

The corresponding class distances satisfy the same bounds.

Proof. For any representative v of z, the quotient norm of $[Lv]$ is at most $\|Lv\| \le C\,\|v\|$. Infimizing over representatives gives the upper bound and supplies a bounded linear map with exactly the algebraic action of SR-03. Alternatively, the minimum representative from SR-34 realizes the infimum and yields it immediately. Apply the reverse bound to $\overline{L}z$: the left-inverse identity gives $\|z\| \le M\left\|\overline{L}z\right\|$. Division by positive M proves the lower bound. Apply the bounds to $z - w$ for distances. Forward boundedness alone does not prove reflection; the reverse identity is genuine additional data.

□

Formal source: `MetricObstructionTransport.obstructionMap_norm_le;MetricObstructionTransport.boundedObstructionMap_opNorm;MetricObstructionTransport.reverse_bound;MetricObstructionTransport.two_sided_bound`.

Hypotheses. Projected nested R0≤R1≤K in fixed real metric for energy; arbitrary submodule R in real normed space for nonclosed seminorm boundary.

Theorem 10.4 (SR-36 Metric refinement and nonclosed boundary)**.** *Range enlargement induces a surjective contraction $F : K/R_0 \to K/R_1$. For the old minimum representative $s_0(z)$,*

$$\|z\|^2 = \|Fz\|^2 + \|(P_{R_1} - P_{R_0})s_0(z)\|^2.$$

Without a closedness or projection hypothesis, a general quotient instead has

$$\|[v]\| = 0 \iff v \in \operatorname{cl} R, \qquad [v] = 0 \iff v \in R.$$

Thus its seminorm detects zero exactly when R is closed. Nonclosed range can leave a nonzero class of zero seminorm.

Proof. The larger realization submodule contains the smaller, so the identity on K descends to a surjection of quotients. Every old representative is a new representative, proving contraction by the quotient infimum. For the energy identity apply the orthogonal refinement identity in SR-28 to $s_0(z)$, whose old norm is $\|z\|$. The new harmonic component represents Fz and has norm $\|Fz\|$ by SR-33.

For any submodule, the native quotient seminorm equals distance to that submodule. Zero distance is equivalent to membership in its closure by the metric closure criterion; algebraic quotient zero remains actual membership. If R is nonclosed, choose $v \in \operatorname{cl} R \setminus R$. Its class is nonzero with zero seminorm. Conversely closedness makes zero distance imply membership. This is why the projection hypotheses in the preceding metric theory cannot be discarded silently.

□

Formal source: `MetricRefinement.refinementMap_surjective;MetricRefinement.refinementMap_norm_le;MetricRefinement.quotient_refinement_energy;QuotientMetricBoundaries.zero_norm_iff_closure;QuotientMetricBoundaries.nonclosed_nonzero_zero_norm;QuotientMetricBoundaries.norm_detects_zero_iff_closed`.

Example 10.5 (A metric must survive its coordinate change)**.** For $K = \{(x, y, 0)\}$ and $R = \operatorname{span}\{(1, 0, 0)\}$, take the compatible basis $b_1 = (1, 0, 0)$, $b_2 = (1, 2, 0)$. Its Gram matrix is $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 5 \end{smallmatrix}\right)$. The vector $(3, 4, 0)$ has coordinates $(1, 2)$; its harmonic representative $(0, 4, 0)$ has coordinates $(-2, 2)$ and squared norm 16 in that Gram metric. Computing with an untransported coordinate Euclidean metric would change the problem. Redundant or nonorthogonal coordinates do not license such a change.

11 Tikhonov construction, bias, and perturbation

Fix a bounded operator $A : G \to V$ between real Hilbert spaces, without assuming closed range. For $\lambda > 0$, put

$$N_\lambda = A^*A + \lambda I, \qquad R_\lambda = N_\lambda^{-1}, \qquad B_\lambda = R_\lambda A^*.$$

The inverse is constructed from coercivity. It is not supplied as a certificate premise. Here A^* is the actual Hilbert adjoint.

Hypotheses. Real Hilbert G,V; actual bounded A and lambda>0; no finite dimension or closed range.

Theorem 11.1 (SR-37 Constructed normal inverse and energy)**.** *N_λ is a bounded bijection with bounded inverse R_λ. For $x_\lambda = B_\lambda v$,*

$$N_\lambda x_\lambda = A^* v, \qquad A^*(Ax_\lambda - v) + \lambda x_\lambda = 0,$$

and

$$\|Ax_\lambda\|^2 + \lambda \|x_\lambda\|^2 = \langle v, Ax_\lambda \rangle .$$

Proof. The bilinear form $b(x,y) = \langle Ax, Ay \rangle + \lambda \langle x, y \rangle$ is bounded and satisfies $b(x,x) \geq \lambda \|x\|^2$. Lax–Milgram gives a continuous linear equivalence from the source to itself under the Riesz identification. The adjoint identity identifies its map with $A^*A + \lambda I$. The two inverse laws give an actual resolvent. Applying one inverse law to A^*v yields the normal equation. Taking its inner product with x_λ and using $\langle A^*v, x_\lambda \rangle = \langle v, Ax_\lambda \rangle$ gives the energy identity. Strictly positive λ supplies coercivity even if A has a nontrivial kernel or a nonclosed range.

□

Formal source: `TikhonovConstruction.normalForm_coercive;TikhonovConstruction.normalEquiv_apply;TikhonovConstruction.normal_resolvent;TikhonovConstruction.resolvent_normal;TikhonovConstruction.normal_equation;TikhonovConstruction.solution_energy`.

Hypotheses. Same real Hilbert bounded A with lambda>0; arbitrary source candidate x and data v; actual normal residual.

Theorem 11.2 (SR-38 Objective optimality and sharp fixed-parameter bounds)**.** *For $F_\lambda(v,x) = \|Ax - v\|^2 + \lambda \|x\|^2$,*

$$F_\lambda(v,x) = F_\lambda(v,x_\lambda) + \|A(x - x_\lambda)\|^2 + \lambda \|x - x_\lambda\|^2 .$$

Thus x_λ is the unique minimizer and lies in $(\ker A)^\perp$. The whole-map bounds are

$$\|R_\lambda\| \leq 1/\lambda, \qquad \|B_\lambda\| \leq \frac{1}{2\sqrt{\lambda}}, \qquad \|AB_\lambda v\| \leq \|v\| , \quad \|AB_\lambda v - v\| \leq \|v\| .$$

For $r = N_\lambda x - A^*v$, one has $x - x_\lambda = R_\lambda r$ and $\|x - x_\lambda\| \leq \|r\| / \lambda$.

Proof. Expand both squared norms around x_λ. Their cross terms combine to

$$2 \langle A^*(Ax_\lambda - v) + \lambda x_\lambda, x - x_\lambda \rangle ,$$

which vanishes by SR-37. The remaining positive terms prove the increment identity and uniqueness. Taking the normal equation against a kernel vector k yields $\lambda \langle x_\lambda, k \rangle = 0$, proving source orthogonality.

For $y = R_\lambda r$, its energy gives $\lambda \|y\|^2 \leq \langle r, y \rangle \leq \|r\| \|y\|$, hence the resolvent bound. For x_λ, set $u = \|Ax_\lambda\|$. The energy gives $u^2 + \lambda \|x_\lambda\|^2 \leq \|v\| \, u$. Completing the square in u yields $4\lambda \|x_\lambda\|^2 \leq \|v\|^2$ and also $u \leq \|v\|$. Comparing the minimum objective with $F_\lambda(v,0) = \|v\|^2$ bounds the fitting residual. Subtract the exact normal equation from the computed one and apply the resolvent for the correction identity.

□

Formal source: `TikhonovOptimality.objective_increment;TikhonovOptimality.unique_minimizer;TikhonovBounds.resolvent_opNorm_bound;TikhonovBounds.regularized_opNorm_bound;TikhonovBounds.approximate_solver_bound;TikhonovBounds.source_kernel_orthogonal`.

Hypotheses. Same positive Tikhonov construction; arbitrary g,e; source-coercivity alpha>0 only for conditioned bound; g=A*Aw only for source-condition rate.

Theorem 11.3 (SR-39 Bias, noise, and kernel content)**.** *Write $p = P_{(\ker A)^\perp} g$, $k = P_{\ker A} g$. Then*

$$B_\lambda A g + \lambda R_\lambda g = g, \quad B_\lambda(Ag + e) - g = -\lambda R_\lambda g + B_\lambda e,$$

and

$$\|B_\lambda(Ag + e) - g\|^2 = \|-\lambda R_\lambda p + B_\lambda e\|^2 + \|k\|^2 .$$

If A has source coercivity $\alpha > 0$, this is at most

$$\left(\frac{\lambda}{\alpha^2 + \lambda}\|p\| + \frac{\|e\|}{2\sqrt{\lambda}}\right)^2 + \|k\|^2 .$$

For a supplied source condition $g = A^* A w$, $\|B_\lambda A g - g\| \le \lambda \|w\|$ without requiring closed range.

Proof. The inverse identity $R_\lambda(A^*A + \lambda I) = I$ gives the filter decomposition. Linearity adds the propagated noise. On $\ker A$, N_λ acts as λI, so $\lambda R_\lambda k = k$. The resolvent preserves the orthogonal complement: pairing its normal equation with every kernel vector proves this directly. Thus the combined identifiable error is orthogonal to k, giving the exact squared formula. Bias and noise inside that first term need not be orthogonal to each other.

For $y = R_\lambda p \in (\ker A)^\perp$, source coercivity and its energy give $(\alpha^2 + \lambda)\|y\|^2 \le \|p\|\,\|y\|$. Multiply by λ and combine with the noise bound in SR-38. For $g = A^* A w$, the normal-range source filter is a contraction, so $\|R_\lambda A^* A w\| \le \|w\|$. Multiplication by λ proves the source-condition bias rate. The representation by w is supplied; no controlled w is asserted for every identifiable source.

□

Formal source: `TikhonovBias.filter_decomposition;TikhonovBias.noisy_source_error_squared;TikhonovBias.full_conditioned_error_squared;TikhonovBias.source_condition_bias_bound`.

Hypotheses. Real Hilbert bounded A,C; positive lambda,mu; actual closed range only for B=A-dagger and intrinsic globalization formulas.

Theorem 11.4 (SR-40 Resolvent sensitivity and irreducible local error)**.** *For two operators and parameters,*

$$R_{A,\lambda} - R_{C,\mu} = R_{A,\lambda}(N_{C,\mu} - N_{A,\lambda})R_{C,\mu}.$$

At fixed positive λ, with $\epsilon = \|A - C\|$,

$$\|B_{A,\lambda} - B_{C,\lambda}\| \le \frac{\epsilon}{\lambda} + \frac{(\|A\| + \|C\|)\epsilon\,\|C\|}{\lambda^2}.$$

If A has closed range and canonical inverse B, then $B_\lambda v - Bv = -\lambda R_\lambda B v$. For $\delta A = 0$,

$$\|v - AB_\lambda v\|^2 = \|c(v)\|^2 + \|i(v)\|^2 + \|A(B_\lambda v - Bv)\|^2 .$$

Proof. Multiply the difference of inverses on the left and right by their normal operators, or insert their inverse laws, to obtain the resolvent identity. SR-38 bounds its norm by $\left\|N_{C,\mu} - N_{A,\lambda}\right\| / (\lambda\mu)$. At equal parameters, $C^*C - A^*A = (C^* - A^*)C + A^*(C - A)$, whose norm is at most $(\|A\| + \|C\|)\epsilon$. Splitting $R_A A^* - R_C C^* = R_A(A^* - C^*) + (R_A - R_C)C^*$ proves the stated bound. It permits rank changes and is not uniform as λ vanishes.

Under closed range, A^* ignores the part of data orthogonal to Ran A, so $B_\lambda v = B_\lambda A B v$. Apply the filter identity to Bv for its bias. The intrinsic candidate identity SR-34 with $g = B_\lambda v$ yields the local error split. Regularization introduces image bias while retaining the same obstruction and inconsistency.

□

Formal source: `TikhonovPerturbation.resolvent_difference;TikhonovPerturbation.regularized_operator_difference_bound;TikhonovGlobalization.regularized_pseudoinverse_bias;TikhonovGlobalization.regularized_intrinsic_error`.

Example 11.5 (Exact scalar response and a two-coordinate example)**.** For $A_c x = cx$, $B_\lambda v = cv/(c^2+\lambda)$. At $c > 0$, $\lambda = c^2$, the value is $v/(2c)$, attaining the noise bound. For $A(g_1, g_2) = (g_1 + g_2, 0, 0)$, $v = (3, 4, 5)$, and $\lambda = 1$, the source is $(1, 1)$. Its squared fit is $42 = 16 + 25 + 1$, separating obstruction, inconsistency, and regularization image bias. Its objective is 44, whereas the unregularized source $(3/2, 3/2)$ has penalized objective $91/2$. Positive regularization solves its own stated objective.

12 Strong limits and the uniform boundary

The following limits fix one bounded A between real Hilbert spaces. Statements can be indexed by an arbitrary filter. When uniqueness of limits or a converse is used, the filter is required to be nontrivial. Closed range is unnecessary for strong convergence on each fixed source.

Hypotheses. Fixed bounded real Hilbert A; positive lambda_i→0 along a filter; arbitrary fixed g; nontrivial filter for exact-limit iff identifiability.

Theorem 12.1 (SR-41 Density and strong identifiable-source convergence)**.** *One has* $\operatorname{cl}\operatorname{Ran}(A^*A) = (\ker A)^\perp$*, and for every* g, w*,*

$$\|B_\lambda A g - g\| \le \|g - A^* A w\| + \lambda \|w\| .$$

As $\lambda \downarrow 0$*,*

$$B_\lambda A g \longrightarrow P_{(\ker A)^\perp} g, \qquad \|B_\lambda A g - g\| \longrightarrow \|P_{\ker A} g\| .$$

For a nontrivial parameter filter, convergence to g *itself holds exactly when* $g \in (\ker A)^\perp$*.*

Proof. If x is orthogonal to $\operatorname{Ran}(A^*A)$, pairing with A^*Ax gives $\|Ax\|^2 = 0$. Conversely a kernel vector is orthogonal to every A^*Aw by the adjoint identity. Thus $(\operatorname{Ran}(A^*A))^\perp = \ker A$, and taking orthogonal complements gives the closure identity. Decompose the bias into the bias of $g - A^*Aw$ and of A^*Aw. The first is bounded by its input norm because λR_λ is contractive; the second is bounded by $\lambda \|w\|$ from SR-39.

For identifiable p, density lets us choose w making the first term arbitrarily small, independently of λ. Then let λ tend to zero to remove the second. For arbitrary g, $Ag = Ap$ with $p = P_{(\ker A)^\perp} g$, yielding the claimed source limit. Subtracting g and taking norms gives the kernel-error limit. Uniqueness of limits on a nontrivial filter gives the final equivalence.

□

Formal source: `TikhonovDensity.gram_range_closure;TikhonovDensity.source_approximation_bound;TikhonovConvergence.tendsto_source;TikhonovConvergence.exact_limit_iff_identifiable;TikhonovConvergence.tendsto_error_norm`.

Hypotheses. Fixed real Hilbert bounded A; lambda_i>0→0; deterministic noise with ‖e_i‖/sqrt(lambda_i)→0; actual normal residual scale vanishes for computed sources.

Theorem 12.2 (SR-42 Noise schedules and inexact strong limits)**.** *If* $\lambda_i > 0$ *tends to zero and* $\|e_i\| / \sqrt{\lambda_i} \to 0$*, then*

$$B_{\lambda_i}(Ag + e_i) \longrightarrow P_{(\ker A)^\perp} g.$$

The same limit holds for computed x_i when additionally

$$\frac{\|(A^*A + \lambda_i I)x_i - A^*(Ag + e_i)\|}{\lambda_i} \to 0.$$

For example, $\lambda_i = s_i^2$, $s_i > 0 \to 0$, and $\|e_i\| \le s_i^2$ give the exact-solve consistency condition.

Proof. The source difference between noisy and exact-data reconstruction is $B_{\lambda_i} e_i$, whose norm is at most $\|e_i\| / (2\sqrt{\lambda_i})$ by SR-38. Combine its vanishing with the exact-source limit in SR-41. For computed sources, the normal-residual correction in SR-38 bounds their difference from the actual regularized source by precisely the second displayed scale. The triangle inequality then proves the same limit. In the square schedule, the noise ratio is at most s_i, hence vanishes. These schedules are sufficient conditions; they are not asserted necessary for every favorable operator or noise direction.

□

Formal source: `NoisyRegularization.noisy_source_tendsto;NoisyRegularization.noisy_inexact_solver_tendsto;NoisyRegularization.square_schedule_consistent`.

Hypotheses. Fixed real Hilbert bounded A; positive parameter family→0; nontrivial filter for iff; source conditioning alpha only for positive estimate.

Theorem 12.3 (SR-43 Operator-norm convergence exactly at closed range)**.** *Let $E_\lambda = B_\lambda A - P_{(\ker A)^\perp}$. Always $\|E_\lambda\| \le 1$. If $\operatorname{Ran} A$ is nonclosed, then $\|E_\lambda\| = 1$ for every positive parameter. For a positive family tending to zero along a nontrivial filter,*

$$E_\lambda \to 0 \text{ in operator norm} \iff \operatorname{Ran} A \text{ is closed}.$$

Under source coercivity $\alpha > 0$, $\|E_\lambda\| \le \lambda / (\alpha^2 + \lambda)$.

Proof. The projected bias bound from SR-39 gives the universal upper bound one. Suppose any bounded B' satisfies $c = \|B'A - P_S\| < 1$, where $S = (\ker A)^\perp$. For $x \in S$,

$$\|x\| \le \|x - B'Ax\| + \|B'Ax\| \le c\,\|x\| + \|B'\|\,\|Ax\|.$$

Therefore $\alpha = (1 - c)/(\|B'\| + 1) > 0$ supplies off-kernel coercivity, forcing closed range by SR-32. A nonclosed range thus cannot have $\|E_\lambda\| < 1$, proving equality. For closed range, source coercivity exists and the conditioned bias bound yields the displayed quantity tending to zero. Conversely convergence to zero contradicts the constant norm one on any nontrivial filter. Strong convergence in SR-41 remains valid in the nonclosed case because it fixes a source before taking the limit.

□

Formal source: `UniformRegularization.closed_range_of_approximate_recovery;UniformRegularization.nonclosed_sourceError_norm;UniformRegularization.conditioned_sourceError_norm;UniformRegularization.uniform_convergence_iff_closed_range`.

Hypotheses. Fixed real Hilbert bounded A; positive lambda_i→0; data noise→0 for fitted limit; nontrivial filter for impossible source limit.

Theorem 12.4 (SR-44 Fitted-data limit and absent source limit)**.** *For arbitrary data v,*

$$AB_\lambda^A v = B_\lambda^{A^*} A^* v \longrightarrow P_{\operatorname{cl}\operatorname{Ran} A} v.$$

Adding data noise $e_i \to 0$ preserves this fitted-data limit. If $v \in \operatorname{cl}\operatorname{Ran} A \setminus \operatorname{Ran} A$, fitted data converge to v but the corresponding reconstructed source family has no norm limit in G.

Proof. The identity $(AA^* + \lambda I)A = A(A^*A + \lambda I)$ and uniqueness of the positive normal inverse prove the dual formula. Apply SR-41 to the bounded operator A^* with source v. Its identifiable space is $(\ker A^*)^\perp = \operatorname{cl}\operatorname{Ran} A$, yielding the fitted limit. SR-38 shows AB_λ is contractive, so changing data by e_i changes the fitted value by at most $\|e_i\|$. If reconstructed sources converged in norm to some g, continuity of A would make their images converge to Ag. On a nontrivial filter, the already established image limit v would force $Ag = v$, contrary to $v \notin \operatorname{Ran} A$. This proves absence of a source norm limit without adding an unproved assertion that source norms diverge.

□

Formal source: `RegularizedDataLimit.fitted_adjoint_identity;RegularizedDataLimit.fitted_data_tendsto;RegularizedDataLimit.noisy_fitted_data_tendsto;RegularizedDataLimit.unrealizable_no_source_limit`.

13 Discrepancy selection and strong consistency

Still fix a known bounded A between real Hilbert spaces. Define the actual residual curve on positive parameters $r_v(\lambda) = \|AB_\lambda v - v\|$. Its parameter is mathematical data, not an assumed noise estimate. Let η be a justified true-data allowance and $\tau > 1$ a band multiplier. A total rule must include a small-data branch.

Hypotheses. Real Hilbert bounded A; lambda>0; strict floor dist(v,range A)<rho<‖v‖ for existence; actual positive parameter only.

Theorem 13.1 (SR-45 Actual residual curve and unique interior root)**.** *The residual is continuous and nondecreasing on positive parameters; the source norm is nonincreasing. It is strictly increasing when $A^*v \neq 0$, and constant at $\|v\|$ when $A^*v = 0$. For*

$$\operatorname{dist}(v, \operatorname{Ran} A) < \rho < \|v\|,$$

there is exactly one $\lambda > 0$ with $r_v(\lambda) = \rho$. The resulting exact real parameter is a noncomputable choice with a proved specification.

Proof. Compare the minimum objectives at $0 < \lambda < \mu$ using each minimizer as a competitor for the other. Adding the two inequalities gives $(\mu - \lambda)(\|x_\mu\|^2 - \|x_\lambda\|^2) \leq 0$, so source norms are antitone. Substitution back into the smaller-parameter comparison shows residual monotonicity. If the two residuals were equal, both norms and objective values would be equal, and unique minimization would give $x_\lambda = x_\mu$. Subtracting their normal equations gives $(\mu - \lambda)x_\lambda = 0$, hence $x_\lambda = 0$ and $A^*v = 0$. This proves strictness and the constant case.

The resolvent parameter identity gives, for $s, t \geq a > 0$,

$$|r_v(s) - r_v(t)| \leq \frac{\|A\| \, \|A^*v\|}{a^2} |s - t|,$$

proving continuity on positive compact intervals. The strict floor permits a comparator g with $\|Ag - v\| < \rho$; objective minimization at sufficiently small positive λ gives a lower residual bracket. For large λ, $\|B_\lambda v\| \leq \|A^*v\| / \lambda$, so the residual approaches $\|v\| > \rho$ and gives an upper bracket. The intermediate value theorem provides a root; strictness gives uniqueness. The floor condition itself excludes the constant orthogonal-data case.

□

Formal source: `DiscrepancyResidual.source_norm_antitone;DiscrepancyResidual.residual_strictMono;DiscrepancyResidual.residual_constant_of_adjoint_zero;DiscrepancyResidual.residual_difference_bound;DiscrepancySelection.exists_unique_parameter_of_distance`.

Hypotheses. Fixed bounded real Hilbert A; p∈(ker A)⊥; family norm bounds ‖x_i‖ ≤ ‖p‖ and true residual ‖Ax_i-Ap‖ →0.

Theorem 13.2 (SR-46 Norm domination converts residual convergence to source convergence)**.** *If $\|Ag - v\| \le r_v(\lambda)$, then $\|B_\lambda v\| \le \|g\|$. For any x, p, w with $\|x\| \le \|p\|$,*

$$\|x - p\|^2 \le 4\,\|p\|\,\|p - A^*w\| + 2\,\|w\|\,\|Ax - Ap\|\,.$$

Consequently, for identifiable fixed p, the conditions $\|x_i\| \le \|p\|$ and $\|Ax_i - Ap\| \to 0$ imply $x_i \to p$ strongly, without assuming closed range.

Proof. Objective minimality gives $r_v(\lambda)^2 + \lambda\,\|B_\lambda v\|^2 \le \|Ag - v\|^2 + \lambda\,\|g\|^2$. The residual ordering and positive λ imply norm domination. For the source estimate, expand $\|x - p\|^2$ and use $\|x\| \le \|p\|$ to obtain $\|x - p\|^2 \le 2\,\langle p, p - x\rangle$. Write $p = (p - A^*w) + A^*w$. The first pairing is at most $\|p - A^*w\|\,\|p - x\| \le 2\,\|p\|\,\|p - A^*w\|$; the adjoint identity bounds the second by $\|w\|\,\|Ap - Ax\|$. This yields the displayed estimate. For identifiable p, the density in SR-41 permits w with $p - A^*w$ arbitrarily small (take $w = Au$ from the normal range). Fixing this w, the residual term tends to zero. An epsilon argument then proves norm convergence. No common inverse bound is invoked.

□

Formal source: `DiscrepancyConvergence.norm_domination_of_residual;DiscrepancyConvergence.source_distance_estimate;DiscrepancyConvergence.norm_control_residual_tendsto`.

Hypotheses. Known bounded $A : G \to V$ between real Hilbert spaces and a fixed true source g; a valid pointwise allowance $\|Ag - v\| \le \eta$. Accepted-source consistency uses a fixed $\tau \ge 1$ and genuine bounds $\|Ag - v_i\| \le \eta_i$ with $\eta_i \to 0$. Total selector existence additionally requires $\eta > 0$ and $\tau > 1$.

Theorem 13.3 (SR-47 Accepted bands and the total small-data rule)**.** *Call x accepted when either*

$$\|v\| \le \tau\eta, \quad x = 0,$$

or $x = B_\lambda v$ for some $\lambda > 0$ with $\eta \le r_v(\lambda) \le \tau\eta$. With a valid noise allowance, every accepted source satisfies

$$\|x\| \le \left\|P_{(\ker A)^\perp} g\right\|, \qquad \|Ax - Ag\| \le (\tau + 1)\eta.$$

For accepted families with $\eta_i \to 0$,

$$x_i \to P_{(\ker A)^\perp} g, \qquad \|x_i - g\| \to \|P_{\ker A} g\|\,.$$

With $\eta > 0$, $\tau > 1$, the total selector chooses zero on small data and otherwise the root $r_v(\lambda) = \tau\eta$. If $g = A^*w$, accepted sources obey $\|x - g\|^2 \le 2\,\|w\|\,(\tau + 1)\eta$.

Proof. Let $p = P_{(\ker A)^\perp} g$, so $Ap = Ag$. In the positive branch, $\|Ap - v\| \le \eta \le r_v(\lambda)$ gives norm domination by SR-46; the triangle inequality gives the true residual. In the zero branch, norm domination is immediate and $\|Ag\| \le \|Ag - v\| + \|v\| \le (\tau + 1)\eta$. SR-46 now gives strong convergence to p, and continuity of the error norm gives the kernel limit. For large data, feasibility supplies $\operatorname{dist}(v, \operatorname{Ran} A) \le \eta < \tau\eta < \|v\|$, so SR-45 provides the exact root. For $g = A^*w$, the approximation term in SR-46 is zero,

and the residual bound proves the squared source-condition rate. General identifiable sources have no asserted common rate. The rule controls actual Tikhonov sources; numerical solve errors require the additional certificates below.

□

Formal source: `DiscrepancyRule.selectedSource_accepted;DiscrepancyRule.accepted_consistent;DiscrepancyRule.selectedSource_consistent;DiscrepancyRule.accepted_error_norm_limit;DiscrepancyRule.accepted_source_rate`.

Hypotheses. Known bounded real Hilbert A and positive lambda; real scalar identity case 0<rho<v.

Theorem 13.4 (SR-48 Discrepancy endpoints and scalar formulas)**.** *No positive parameter has residual larger than $\|v\|$. When $A^*v \neq 0$, none has residual equal to $\|v\|$. For every positive parameter, $r_v(\lambda) = 0$ exactly when $v = 0$. For the real scalar identity model and $0 < \rho < v$,*

$$\lambda = \frac{\rho}{v - \rho}, \qquad B_\lambda v = v - \rho$$

is the unique root at residual ρ.

Proof. The first assertion is the residual contraction in SR-38. Equality at $\|v\|$ would give $F_\lambda(v, x_\lambda) \leq F_\lambda(v, 0) = \|v\|^2$ while its fit term already equals $\|v\|^2$. Hence $x_\lambda = 0$, forcing $A^*v = 0$. If the residual is zero, then $Ax_\lambda = v$, and the normal equation reduces to $\lambda x_\lambda = 0$. Thus $x_\lambda = 0$ and $v = 0$. The converse is direct. For $A = I$, the normal equation gives $B_\lambda v = v/(1 + \lambda)$ and residual $\lambda v/(1 + \lambda)$ for $v > 0$. Substitution proves the formula and SR-45 supplies uniqueness. These endpoints explain the strict root inequalities and the separate small-data branch.

□

Formal source: `DiscrepancyBoundaries.no_parameter_above_data;DiscrepancyBoundaries.no_parameter_at_data_norm;DiscrepancyBoundaries.zero_residual_iff_zero_data;DiscrepancyBoundaries.identity_parameter_formula`.

14 Reliable discrepancy with inexact solves

The discrepancy rule of the preceding section selects an actual regularized source. A computed vector need not equal that source, and a displayed residual need not equal its exact norm. Those two errors require separate allowances. The normal equation provides the connection without assuming the computation has already solved the inverse problem.

Throughout this section $A : G \to V$ is bounded between real Hilbert spaces, $N_A = \ker A$, $S_A = N_A^\perp$, and $\lambda > 0$. Write

$$N_\lambda = A^*A + \lambda I, \qquad x_\lambda = N_\lambda^{-1} A^* v, \qquad r_\lambda(x) = N_\lambda x - A^* v.$$

The inverse is the actual positive-parameter constructor already established. The vector $r_\lambda(x)$ is the normal-equation residual of the computed source x. Its size measures source correction and image correction on different scales. A fitting residual $\|Ax - v\|$ alone does not supply either correction.

Hypotheses. Real Hilbert spaces; bounded A; lambda>0; actual normal residual.

Theorem 14.1 (SR-49 Normal residual, two correction scales)**.** *For every computed source x,*

$$x - x_\lambda = N_\lambda^{-1} r_\lambda(x), \qquad \|x - x_\lambda\| \le \frac{\|r_\lambda(x)\|}{\lambda}, \qquad \|Ax - Ax_\lambda\| \le \frac{\|r_\lambda(x)\|}{2\sqrt{\lambda}}.$$

Consequently

$$|\,\|Ax - v\| - \|Ax_\lambda - v\|\,| \le \frac{\|r_\lambda(x)\|}{2\sqrt{\lambda}}.$$

The bounds use the same actual residual and the existing Hilbert norms.

Proof. Subtract $N_\lambda x_\lambda = A^* v$ from the definition of the normal residual and apply the inverse. For $u = N_\lambda^{-1} r$, the energy identity gives

$$\|Au\|^2 + \lambda \|u\|^2 = \langle r, u\rangle \le \|r\| \, \|u\| .$$

Dropping the first term shows $\|u\| \le \|r\| / \lambda$, including $u = 0$. To control the image, complete the scalar square:

$$\|Au\|^2 \le \|r\| \, \|u\| - \lambda \|u\|^2 \le \frac{\|r\|^2}{4\lambda}.$$

Both sides of the desired norm inequality are nonnegative, so taking square roots gives the image bound. Set $r = r_\lambda(x)$ and use linearity of A. Finally the reverse triangle inequality bounds the difference of the two fitting norms by $\|A(x - x_\lambda)\|$.

The distinction matters when λ is small. A normal residual sufficient for a prescribed image tolerance may be insufficient for a prescribed source tolerance. If source correction must be at most κ and image correction at most ξ, a sound pair of conditions is $\|r_\lambda(x)\| \le \lambda\kappa$ and $\|r_\lambda(x)\| \le 2\sqrt{\lambda}\,\xi$. These are parameter-dependent requirements, not a uniform inverse bound as λ tends to zero.

□

Formal source: `InexactResidual.correction_identity;InexactResidual.correction_bound;InexactResidual.image_correction_bound;InexactResidual.fitting_residual_difference`.

Hypotheses. Same Hilbert setting; nonnegative source/image/evaluation budgets; positive parameter; genuine measurement enclosure.

Theorem 14.2 (SR-50 Protected measured discrepancy margins)**.** *Suppose m is a measured fitting norm with*

$$|m - \|Ax - v\|\,| \le \omega, \quad \|r_\lambda(x)\| \le \lambda\kappa, \quad \|r_\lambda(x)\| \le 2\sqrt{\lambda}\,\xi.$$

If

$$\eta + \omega + \xi \le m \le \tau\eta - \omega - \xi,$$

then the actual exact source x_λ has discrepancy in $[\eta, \tau\eta]$, and $\|x - x_\lambda\| \le \kappa$. A protected scalar window exists exactly when

$$2(\omega + \xi) \le (\tau - 1)\eta.$$

Its midpoint $(\tau + 1)\eta/2$ is then admissible.

Proof. SR-49 and the measurement enclosure imply

$$|m - \|Ax_\lambda - v\|\,| \le |m - \|Ax - v\|\,| + |\,\|Ax - v\| - \|Ax_\lambda - v\|\,| \le \omega + \xi.$$

Subtracting this total allowance from the lower protected inequality gives $\eta \le \|Ax_\lambda - v\|$. Adding it to the upper inequality gives $\|Ax_\lambda - v\| \le \tau\eta$. The source-correction bound follows from the other residual condition in SR-49. Hence the computed source is within κ of an actual accepted discrepancy source.

The scalar window is nonempty exactly when its lower endpoint is at most its upper endpoint. Rearrangement gives the displayed condition. The proposed midpoint is the mean of the endpoints; direct substitution gives both inequalities. No property of an approximate solver is hidden in this scalar calculation: its residual, measurement enclosure and correction allowance have already been supplied.

For example, let $\eta = 1/10$, $\tau = 2$, and $\omega = \xi = 1/100$. The total protected margin is $1/50$, and the interval is $[3/25, 9/50]$. Its midpoint is $3/20$. This allocation proves room for the measurement and image-correction errors; it does not prove that a particular computed source attains the interval or that $1/10$ is a true data-noise allowance.

□

Formal source: `InexactResidual.measured_band_transfer;InexactDiscrepancy.measured_certificate;InexactToleranceCertificates.protected_window_exists_iff;InexactToleranceCertificates.protected_midpoint`.

Hypotheses. Fixed bounded real Hilbert A; actual noise bounds; fixed tau; NearAccepted with source budget tending to zero.

Theorem 14.3 (SR-51 Near-accepted consistency and kernel leakage)**.** *Call x near accepted with budget κ when some exact accepted source y satisfies $\|x - y\| \le \kappa$. If $\|Ag - v_i\| \le \eta_i$, $\eta_i \to 0$, and near-accepted x_i have $\kappa_i \to 0$, then*

$$x_i \longrightarrow P_{S_A} g, \qquad \|x_i - g\| \longrightarrow \|P_{N_A} g\| .$$

At every index $\|P_{N_A} x_i\| \le \kappa_i$; therefore the computed true-kernel component tends to zero. No separate parameter schedule is assumed.

Proof. Choose accepted comparison sources y_i from the definition. Exact accepted consistency gives $y_i \to p = P_{S_A} g$, while $\|x_i - y_i\| \le \kappa_i \to 0$ gives $x_i - y_i \to 0$. Adding the two limits proves $x_i \to p$. Continuity of the norm gives the source-error limit, because $p - g = -P_{N_A} g$.

The exact accepted source belongs to S_A, including the small-data zero branch. Orthogonal projection is nonexpansive, so

$$\|P_{N_A} x_i\| = \|P_{N_A}(x_i - y_i)\| \le \|x_i - y_i\| \le \kappa_i.$$

This pointwise estimate only needs near acceptance, not a truth-noise premise. Source consistency additionally needs the stated true-noise allowance tending to zero.

The underlying exact consistency theorem has substantive content even for a nonclosed range. Accepted sources satisfy norm domination $\|y_i\| \le \|p\|$ and $\|Ay_i - Ap\| \to 0$. Since $p \in S_A = \operatorname{cl} \operatorname{Ran} A^*$, approximate p by $A^* w$ and use

$$\|y_i - p\|^2 \le 2 \langle p, p - y_i \rangle \le 2 \|p - A^* w\| \, \|p - y_i\| + 2 \|w\| \, \|A(p - y_i)\| .$$

The first factor $\|p - y_i\|$ is bounded by $2\|p\|$. First choose the approximation accurately, then let i tend along the filter. This proves strong convergence without a bounded inverse.

□

Formal source: `InexactDiscrepancy.NearAccepted;InexactDiscrepancy.near_kernel_leakage;InexactConsistency.near_consistent;InexactConsistency.near_error_norm_limit;InexactConsistency.near_kernel_tendsto_zero`.

Hypotheses. Real Hilbert spaces; positive lambda; nonnegative scalar certificate fields; squared inequalities against actual norms.

Theorem 14.4 (SR-52 Squared proof-carrying discrepancy certificate)**.** *Suppose a record supplies $\lambda > 0$, nonnegative $\rho, \xi, \kappa, m, \omega$, and*

$$\|r_\lambda(x)\|^2 \le \rho^2, \quad \rho^2 \le (\lambda\kappa)^2, \quad \rho^2 \le 4\lambda\xi^2,$$

$$\max(0, m-\omega)^2 \le \|Ax - v\|^2 \le (m+\omega)^2, \qquad \eta + \omega + \xi \le m \le \tau\eta - \omega - \xi.$$

Then x is near accepted with source budget κ. Families of such records give SR-51 when the genuine data allowances and source budgets tend to zero.

Proof. All quantities whose squares are compared are nonnegative. The first two inequalities therefore give $\|r_\lambda(x)\| \le \rho \le \lambda\kappa$. The third gives $\rho \le 2\sqrt{\lambda}\,\xi$ because $(2\sqrt{\lambda}\,\xi)^2 = 4\lambda\xi^2$.

Put $d = \|Ax - v\| \ge 0$. The squared fitting enclosure is equivalent to the unsquared enclosure $\max(0, m-\omega) \le d \le m + \omega$. Since $d \ge 0$, its lower part is equivalent to $m - \omega \le d$. Together these say $|m - d| \le \omega$. Thus every premise of SR-50 has been obtained from the supplied record, and its conclusion proves soundness. Apply SR-51 pointwise for the family statement.

The record does not establish the truth of the data-noise allowance. It establishes a relation between the computed source, actual normal residual and actual fitting norm. An implementation may check rational squared inequalities without numerical square roots, but that requires a faithful enclosure of these actual quantities. The generic Lean structure is proof-carrying; a serialized Python record and its parser are separate implementation evidence.

If the normal residual itself is evaluated approximately as $\widehat{r}$, the distinct assumptions $\|r_\lambda(x) - \widehat{r}\| \le \nu$ and $\|\widehat{r}\| \le \widehat{\rho}$ give the allowance $\widehat{\rho} + \nu$. Substituting that sum into both correction tests retains the evaluation error rather than silently identifying it with zero.

□

Formal source: `InexactToleranceCertificates.squared_enclosure_iff;InexactToleranceCertificates.squared_image_budget;InexactToleranceCertificates.PositiveCertificate;InexactToleranceCertificates.certificate_sound;InexactToleranceCertificates.certificate_consistent`.

15 Uncertain operators and changing identifiable models

Data noise, solver error and model error are different inputs. Let A be the true forward operator and C the available operator, acting between the same real Hilbert spaces. The premise $\|C - A\| \le \delta$ is a whole-operator allowance. A small residual for one source does not establish it. This section first uses a fixed cap M on a true comparison source. The next section removes that cap from the acceptance rule by making the allowance depend on the candidate itself.

Hypotheses. Same normed source/target; bounded A,C; whole operator allowance; genuine data allowance and source cap.

Theorem 15.1 (SR-53 Effective model-noise allowance)**.** *If $\|C - A\| \le \delta$, $\|p\| \le M$ and $\|Ap - v\| \le \eta$, then*

$$\|Cp - v\| \le \epsilon := \eta + \delta M.$$

For every candidate x,

$$|\,\|Ax - v\| - \|Cx - v\|\,| \le \delta\,\|x\|\,.$$

If $|m - \|Cx - v\|\,| \le \omega$, then $|m - \|Ax - v\|\,| \le \omega + \delta\,\|x\|$. The effective-noise allowance is sharp in scalar examples.

Proof. The definition of operator norm gives $\|Cp - Ap\| \le \delta\,\|p\|$. Because a norm bound implies $\delta \ge 0$, the source cap gives

$$\|Cp - v\| \le \|Cp - Ap\| + \|Ap - v\| \le \delta M + \eta.$$

For the fitting-norm comparison, apply the reverse triangle inequality to $Ax - v$ and $Cx - v$, whose difference is $(A - C)x$. Combining this with the measurement enclosure gives the last claim.

To see the possible equality, take real scalar maps $A = 1$, $C = 1 + \delta$, $p = M$ and $v = M - \eta$, with $\eta, \delta, M \ge 0$. Then $Cp - v = \delta M + \eta \ge 0$, so its absolute value is exactly ϵ. The source-size factor cannot generally be omitted: for fixed $\delta > 0$, the action difference $\delta|p|$ grows without bound as $|p|$ grows. Thus replacing δM by an operator-only constant would strengthen the contract incorrectly.

□

Formal source: `OperatorUncertainty.action_error;OperatorUncertainty.effective_data_noise;OperatorUncertainty.fitting_norm_difference;OperatorUncertainty.measured_true_fitting_envelope;OperatorUncertaintyExamples.inflated_noise_sharp`.

Hypotheses. Same real Hilbert spaces; bounded A,C; positive or arbitrary common normal parameter; whole norm error delta.

Theorem 15.2 (SR-54 Transport of actual normal residuals)**.** *For the normal residuals formed with the same parameter,*

$$r_\lambda^A(x) - r_\lambda^C(x) = (A^* - C^*)(Cx - v) + A^*(A - C)x.$$

If $\|C - A\| \le \delta$, then

$$\left\| r_\lambda^A(x) - r_\lambda^C(x) \right\| \le \delta\big(\|Cx - v\| + (\|C\| + \delta)\,\|x\|\big).$$

Hence a computed-model normal allowance $\left\|r_\lambda^C(x)\right\| \le \rho$ gives the corresponding true normal allowance by adding this expression.

Proof. Expand the two residuals. The λx terms cancel. Insert and subtract A^*Cx to obtain the displayed identity. Adjoint isometry gives $\|A^* - C^*\| = \|A - C\| \le \delta$ and $\|A^*\| = \|A\|$. Submultiplicativity therefore bounds its two terms by $\delta\,\|Cx - v\|$ and $\|A\|\,\delta\,\|x\|$. The triangle inequality gives $\|A\| \le \|C\| + \delta$, yielding a bound written in terms of available-model quantities and the allowance. Finally apply $\|r^A\| \le \|r^C\| + \|r^A - r^C\|$.

The identity itself does not require a positive parameter; positivity is needed when this transported residual is used in SR-49 to control an inverse correction. The model error remains distinct from the solver's normal residual. Both contributions are amplified by the corresponding positive-parameter correction scale. A small fixed model allowance is therefore not a uniform guarantee when λ tends to zero.

□

Formal source: `OperatorResidualTransport.normal_residual_difference;OperatorResidualTransport.normal_residual_available_bound;OperatorResidualTransport.true_normal_residual_bound`.

Hypotheses. Same real Hilbert spaces; fixed true bounded A, fixed real tau and fixed identifiable p; source cap M; true bounds and exact/near accepted uncertain-model sources; vanishing allowances.

Theorem 15.3 (SR-55 Strong consistency without matching model kernels)**.** *Suppose $p \in S_A$, $\|p\| \le M$, and*

$$\|C_i - A\| \le \delta_i, \qquad \|Ap - v_i\| \le \eta_i.$$

If x_i lies within κ_i of an exact accepted source for C_i, v_i with allowance $\epsilon_i = \eta_i + \delta_i M$, and $\eta_i, \delta_i, \kappa_i \to 0$, then $x_i \to p$. The model kernels and ranks may change. For arbitrary g, using $p = P_{S_A} g$ yields $\|x_i - g\| \to \|P_{N_A} g\|$ and $P_{N_A} x_i \to 0$.

Proof. Let y_i be the exact accepted comparison. SR-53 makes p feasible for the available model at allowance ϵ_i. In the positive branch, regularized optimality and the lower discrepancy bound give $\|y_i\| \le \|p\|$: compare the objective at y_i and p, then cancel the fitting terms using $\|C_i p - v_i\| \le \epsilon_i \le \|C_i y_i - v_i\|$. The small-data branch is zero and satisfies the same domination.

The upper discrepancy bound gives $\|C_i y_i - v_i\| \le \tau \epsilon_i$. Returning to the true operator,

$$\|Ay_i - Ap\| \le \delta_i \|y_i\| + \tau \epsilon_i + \eta_i \le (\tau + 1)\epsilon_i.$$

Thus y_i is norm dominated by the fixed identifiable p and has true image residual tending to zero. The density argument printed in SR-51 proves $y_i \to p$ without projecting onto the changing model kernels. The near-source difference then tends to zero, proving $x_i \to p$.

For arbitrary g, $AP_{S_A} g = Ag$, so the same true-data bound applies to $p = P_{S_A} g$. Continuity of norm and true-kernel projection gives the remaining limits. The fixed cap concerns this limiting identifiable source. It is not a claim that the minimum-norm sources for unrelated models have already converged.

□

Formal source: `ChangingDiscrepancy.norm_control;ChangingDiscrepancy.near_true_residual;ChangingConsistency.projected_consistent;ChangingConsistency.source_error_limit;ChangingConsistency.true_kernel_leakage_tendsto_zero`.

Hypotheses. Real Hilbert setting; operational checks use C; external true allowances and source cap; positive effective allowance and tau>1 for selection.

Theorem 15.4 (SR-56 Available-model certificate and selected-source soundness)**.** *A source is certified using C when either $\|v\| \le \tau\epsilon$ and $\|x\| \le \kappa$, or it has an SR-52 positive certificate for C with allowance $\epsilon = \eta + \delta M$. Such a source is near accepted for the available model. With the external truth premises of SR-55 and vanishing allowances, certified sources converge to the true identifiable source.*

If $\epsilon > 0$ and $\tau > 1$, those truth premises also give feasibility of the total exact discrepancy selector for C; the selected exact sources are consistent under the same vanishing data/operator bounds.

Proof. In the small-data branch compare x with the accepted zero source: $\|x - 0\| \le \kappa$. In the positive branch, SR-52 supplies an actual accepted comparison source at distance at most κ. This proves operational soundness without using A.

For the true conclusion, separately supply $\|C - A\| \le \delta$, the true-data bound, and $\|p\| \le M$. These hypotheses are not fields established by an observed fitting certificate. They allow SR-55 to be applied to every certified source.

SR-53 gives the feasible witness p for the available model. A positive effective allowance and $\tau > 1$ ensure a strict residual target above its attainable floor in the non-small-data case; the total discrepancy construction chooses an actual positive-parameter source there and zero in the other branch. Each selected source is exact accepted, so SR-55 applies with $\kappa = 0$. This selected object is noncomputable in the generic formalization. Its existence theorem and soundness are distinct from a finite numerical search implementation.

□

Formal source: `OperatorCertificates.Certified;OperatorCertificates.certified_near;OperatorCertificates.certified_consistent;OperatorCertificates.inflated_feasibility;OperatorCertificates.selected_from_allowances_consistent`.

16 Candidate-dependent discrepancy without a supplied source cap

The acceptance rule can avoid an externally supplied number M by using the candidate itself in the uncertainty allowance. Put

$$c_{\eta,\delta}(x) = \eta + \delta \|x\| .$$

A generalized accepted source is either zero in the small-data branch $\|v\| \le \tau\eta$, or an actual Tikhonov source whose residual lies between $c_{\eta,\delta}(x)$ and $\tau c_{\eta,\delta}(x)$. This definition contains $C, v, \eta, \delta, \tau, x$ and the positive parameter; it does not contain a true-source cap. Feasibility remains a separate question and retains a quantitative relation between model accuracy and the true noise scale.

Hypotheses. Real Hilbert spaces; bounded C; lambda>0; delta≥0; feasible comparison p for candidate noise; actual regularized candidate.

Theorem 16.1 (SR-57 Candidate-dependent self-bounding)**.** *Let x be the actual regularized source for C, v, λ. If*

$$\|Cp - v\| \le c_{\eta,\delta}(p), \qquad c_{\eta,\delta}(x) \le \|Cx - v\| ,$$

then $\|x\| \le \|p\|$. In particular, if $\|C - A\| \le \delta$ and $\|Ap - v\| \le \eta$, every generalized accepted source satisfies $\|x\| \le \|p\|$.

Proof. Suppose instead $\|x\| > \|p\|$. Since $\delta \ge 0$, candidate noise is monotone in source norm: $c_{\eta,\delta}(p) \le c_{\eta,\delta}(x)$. The two displayed hypotheses imply $\|Cp - v\| \le \|Cx - v\|$. Comparing Tikhonov objectives, the fitting-square term at p is no larger, whereas its positive weighted source-square term is strictly smaller. The total objective at p is strictly smaller than the objective at its actual unique minimizer x, a contradiction.

With true operator bounds, SR-53 applied using $M = \|p\|$ gives the required candidate-noise feasibility. The operator allowance implies $\delta \ge 0$. The positive accepted branch therefore satisfies the lemma; the small-data branch $x = 0$ satisfies the norm conclusion directly.

The lower discrepancy inequality is decisive. An upper residual bound alone can allow overly large regularized source norms. Here the lower bound grows with the candidate norm, and objective minimality prevents that growth from exceeding a feasible comparison source. The comparison source enters the proof of soundness, while the acceptance definition continues to have no externally supplied source cap.

□

Formal source: `GeneralizedDiscrepancy.regularized_norm_domination;GeneralizedDiscrepancy.accepted_norm_control`.

Hypotheses. Real Hilbert spaces; true whole-operator/data bounds; tau≥0; generalized accepted or near-accepted sources; vanishing allowances.

Theorem 16.2 (SR-58 Generalized true residual and strong consistency)**.** *A generalized accepted x obeys*

$$\|Ax - Ap\| \le (\tau + 1)c_{\eta,\delta}(p).$$

A source within κ of such an x obeys the same bound with $\|A\|\ \kappa$ added. For $p \in S_A$, vanishing data/operator allowances and near-source budgets imply strong convergence to p. If $p = A^*w$, an inexact accepted candidate also satisfies

$$\|x - p\|^2 \le 2\kappa^2 + 4\,\|w\|\,(\tau + 1)c_{\eta,\delta}(p).$$

Proof. SR-57 gives $\|x\| \le \|p\|$, so $c_{\eta,\delta}(x) \le c_{\eta,\delta}(p)$. The upper generalized discrepancy bound and $\tau \ge 0$ yield $\|Cx - v\| \le \tau c_{\eta,\delta}(p)$. Insert Cx and v into the true residual:

$$\|Ax - Ap\| \le \delta\,\|x\| + \|Cx - v\| + \eta \le (\tau + 1)c_{\eta,\delta}(p).$$

For a near candidate, split its image difference through the exact accepted comparison and apply the operator norm bound to the correction.

As $\eta_i, \delta_i \to 0$, the comparison-source noise $\eta_i + \delta_i\,\|p\|$ tends to zero. Exact candidates are norm dominated by p and have true image error tending to zero; the same density argument as SR-51 proves their strong convergence. Add a vanishing near-source difference for inexact candidates. No relative feasibility condition is needed in this implication once accepted candidates have been supplied.

For the source-condition rate, norm domination and $p = A^*w$ give

$$\|y - p\|^2 \le 2\,\langle p, p - y\rangle \le 2\,\|w\|\ \|A(p - y)\|$$

for an exact comparison y. Bound its image by the first conclusion. Then $\|x - p\| \le \|x - y\| + \|y - p\|$ and $(a + b)^2 \le 2a^2 + 2b^2$ prove the stated inexact rate.

□

Formal source: `GeneralizedDiscrepancy.accepted_true_residual;GeneralizedDiscrepancy.accepted_consistent;GeneralizedInexact.near_true_residual;GeneralizedInexact.near_consistent;GeneralizedInexact.near_source_condition_rate`.

Hypotheses. Real Hilbert spaces; eta>0,tau>1; true operator/data bounds; delta‖p‖<(tau-1)eta.

Theorem 16.3 (SR-59 Relative feasibility and noncomputable selection)**.** *If $\|C - A\| \le \delta$, $\|Ap - v\| \le \eta$ and*

$$\delta\,\|p\| < (\tau - 1)\eta,$$

then an accepted source exists with norm at most $\|p\|$. In the large-data case one can choose a positive parameter whose residual is exactly $\tau\eta$. A sufficient ratio condition is $\delta \le \gamma\eta$ and $\gamma\,\|p\| < \tau - 1$. Noncomputably selected families converge to $p \in S_A$ when the data/operator allowances vanish and the pointwise feasibility conditions hold.

Proof. The true bounds imply $\|Cp - v\| \le \eta + \delta \|p\| < \tau\eta$. If $\|v\| \le \tau\eta$, the zero source is accepted. Otherwise the strict attainable-floor bound and $\tau\eta < \|v\|$ give a positive exact discrepancy root for C with residual $\tau\eta$. Let x be its actual Tikhonov source. Objective comparison, or the fixed-noise domination lemma, gives $\|x\| \le \|p\|$. Consequently

$$c_{\eta,\delta}(x) \le \eta + \delta \|p\| < \tau\eta = \|Cx - v\| .$$

Because $\delta \ge 0$ and $\tau > 1$, $\tau\eta \le \tau c_{\eta,\delta}(x)$. Thus both generalized band inequalities hold.

Multiplying $\delta \le \gamma\eta$ by $\|p\| \ge 0$ and the strict margin $\gamma \|p\| < \tau - 1$ by $\eta > 0$ proves the ratio specialization. The chosen source is defined by classical choice from the existence theorem. Its raw definition takes A, p and the truth/feasibility proofs as arguments; it is a theorem-level witness rather than an extracted algorithm using only available measurements. Each chosen source is accepted, so SR-58 proves the family convergence. The selection theorem retains the feasibility premises even though the acceptance predicate has no supplied scalar source cap.

□

Formal source: `GeneralizedDiscrepancySelection.positive_parameter_source;GeneralizedDiscrepancySelection.exists_accepted;GeneralizedDiscrepancySelection.selectedSource;GeneralizedFeasibility.relative_accuracy_of_scaled_error;GeneralizedDiscrepancyConsistency.selected_consistent`.

Hypotheses. Same Hilbert setting; near exact generalized acceptance with vanishing correction budget; scalar boundary uses valid but nonvanishing model allowance.

Theorem 16.4 (SR-60 Inexact generalized sources and the zero-model boundary)**.** *Near-generalized acceptance is preserved by enlarging a source-correction budget and gives the consistency conclusion of SR-58 when that budget tends to zero. Nevertheless valid true-noise and operator allowances alone do not ensure generalized feasibility. With $A = 1$, $C = 0$, $p = v = 1$, $\eta = 1/10$, $\delta = 1$ and $\tau = 2$, no source is generalized accepted.*

Proof. Near acceptance means there exists an exact generalized accepted comparison y with $\|x - y\| \le \kappa$. Increasing κ preserves that inequality. Along a family, choosing the comparisons and adding the vanishing difference proves consistency by SR-58.

For the scalar boundary, the true data error is zero and $\|C - A\| = 1 \le \delta$, so both stated truth allowances are valid. The small-data branch would require $1 \le 2/10$, which is false. For every positive parameter, the actual regularized source of $C = 0$ is zero and its fitting residual is one. Its candidate noise is $1/10 + 1 \cdot 0 = 1/10$, so the upper generalized band would again require $1 \le 2/10$. Every branch fails. The relative feasibility condition also fails, since $1 < (2 - 1)/10$ is false. Thus the boundary refutes an unconditional existence interpretation, not the conditional theorem.

For comparison, take $A = C = 1$, $v = 1$, $\eta = 1/10$, $\delta = 0$, $\tau = 2$ and $\lambda = 1/4$. Then $x_\lambda = 4/5$ and the residual is $1/5$. It lies in $[1/10, 2/10]$, so the candidate is accepted, and $0 < (2 - 1)/10$ verifies the strict feasibility margin. These examples use the actual scalar normal inverse and exact norms, not a substituted approximate residual.

□

Formal source: `GeneralizedInexact.NearAccepted;GeneralizedInexact.near_consistent;GeneralizedDiscrepancyExamples.identity_candidate_accepted;GeneralizedDiscrepancyExamples.zero_model_not_accepted;GeneralizedDiscrepancyExamples.zero_model_valid_true_bounds;GeneralizedDiscrepancyExamples.zero_model_relative_accuracy_fails`.

17 Calibration generates an operator allowance

The final link connects justified calibration measurements with the whole-operator premise used above. It does not infer the calibration truth allowance from the model's own fit. Let K be a real normed calibration coordinate space, G a real normed source space and V a real normed response space. A split frame has bounded synthesis $U : K \to G$ and analysis $R : G \to K$ with $UR = I_G$. Its coordinates may be redundant: U need not be injective or square. The reconstruction identity requires coverage of every source direction.

The observed response $Y : K \to V$ is an operator, and the assumptions $\|Y - AU\| \le \eta_{\mathrm{cal}}$ and $\|Y - CU\| \le f$ are operator-norm enclosures. Individual calibration samples do not automatically give these enclosures. A mathematical or experimental justification of η_{cal} remains a separate input.

Hypotheses. Real normed spaces; bounded U,R with UR=I; whole calibration truth/model-fit operator bounds.

Theorem 17.1 (SR-61 Redundant split calibration identifies the operator)**.** *Every bounded source operator T factors as $T = (TU)R$. Therefore*

$$C - A = ((CU) - (AU))R, \qquad \|C - A\| \le (\eta_{\mathrm{cal}} + f)\,\|R\|\,.$$

If a supplied analysis bound is $\|R\| \le k$, the allowance is at most $(\eta_{\mathrm{cal}} + f)k$. Exact calibration $Y = AU = CU$ forces $A = C$.

Proof. For every source g, $URg = g$, so $Tg = TU(Rg)$. This proves the actual factorization, including its application to $C - A$. The triangle inequality through the common measured operator gives

$$\|CU - AU\| \le \|CU - Y\| + \|Y - AU\| \le f + \eta_{\mathrm{cal}}.$$

Submultiplicativity then yields the displayed allowance. Both measured bounds imply their scalar allowances are nonnegative; hence replacing $\|R\|$ by a larger k preserves the inequality. With exact calibration, $CU - AU = 0$ and the factorization gives $C - A = 0$.

Redundancy causes no logical problem. For instance, let $K = \mathbb{R}^2$, $G = \mathbb{R}$, $U(k_1, k_2) = k_1$ and $Rg = (g, 0)$, with the standard product norm. Then $UR = I$, $\|U\| = \|R\| = 1$, but U has a nontrivial kernel. The frame spans the source even though it has an unused coordinate. A true scalar map $A = a$ and model $C = c$ with measured row $Y(k_1, k_2) = yk_1$ give $|y - a| \le \eta_{\mathrm{cal}}$ and $|y - c| \le f$, hence $|c - a| \le \eta_{\mathrm{cal}} + f$. The same estimate comes from the general factorization rather than assuming the coordinates are square and invertible.

□

Formal source: `CalibrationFrame.Frame.factor_through_frame;CalibrationFrame.Frame.difference_factorization;CalibrationFrame.Frame.measured_operator_bound;CalibrationFrame.Frame.measured_operator_bound_of_analysis_bound;CalibrationFrame.Frame.exact_calibration_identifies`.

Hypotheses. Fixed split calibration frame; real Hilbert G,V; fixed bounded true A and tau≥0; fixed identifiable p; whole calibration truth/model fit and true data allowances; generalized acceptance; relative feasibility for choice.

Theorem 17.2 (SR-62 Calibrated recovery and conditional selected sources)**.** *For recovery assume G, V are real Hilbert spaces, A is a fixed bounded true operator, $\tau \ge 0$ is fixed, and $p \in (\ker A)^{\perp}$ is fixed. Retain the whole calibration bounds and the true data bounds $\|Ap - v_i\| \le \eta_i$. Put $\delta_i = (\eta_{\mathrm{cal},i} + f_i)\,\|R\|$. If the true data, calibration-truth and model-fit allowances tend to zero, then $\delta_i \to 0$. Generalized accepted sources using this*

generated allowance converge to a fixed identifiable true source p. For an arbitrary true source g, apply this with $p = P_{S_A} g$ and genuine data bounds $\|Ag - v_i\| \le \eta_i$; then the source-error norm tends to $\|P_{N_A} g\|$.

If additionally $\eta_i > 0$, $\tau > 1$ and $\delta_i \|p\| < (\tau - 1)\eta_i$ pointwise, accepted sources exist and the noncomputable selected sources have the same limit.

Proof. Apply SR-61 at every index to obtain the actual operator bound $\|C_i - A\| \le \delta_i$. Because the frame is fixed, its finite norm $\|R\|$ is a constant. The sum of the two vanishing calibration allowances tends to zero, so multiplying by that constant proves $\delta_i \to 0$. With the genuine data-noise bounds, every premise of the accepted-source consistency theorem SR-58 is now available.

For arbitrary g, use $p = P_{S_A} g$. Its forward image is Ag, so the same data bounds hold. The strong source limit is p, and continuity gives $\|x_i - g\| \to \|p - g\| = \|P_{N_A} g\|$. No calibration can remove source ambiguity in the true forward kernel without an additional identifying condition.

The relative margin feeds the generated allowance into SR-59 and proves pointwise existence and norm domination. Its chosen sources are accepted, so the first conclusion applies to them. The raw selection is still noncomputable and proof-dependent. Calibration has generated the operator-error premise; it has not eliminated the distinct truth, data, feasibility or selection obligations.

In particular, declining observed model-calibration fit alone is insufficient: the calibration-truth allowance must also vanish. If the reconstruction frame changes with the index, a growing analysis norm would require additional control. The printed family theorem uses the fixed frame it actually assumes.

□

Formal source: `CalibrationRecovery.generated_operator_bound;CalibrationRecovery.generatedDelta_tendsto_zero;CalibrationRecovery.calibrated_consistent;CalibrationRecovery.calibrated_error_norm_limit;CalibrationSelection.exists_accepted_from_calibration;CalibrationSelection.calibrated_selected_consistent`.

Hypotheses. Real normed spaces; bounded split stage frames; or bounded U,R with whole reconstruction defect at most rho<1.

Theorem 17.3 (SR-63 Hierarchical and approximate reconstruction frames)**.** *For split frames $(U_E, R_E) : J \rightleftarrows K$ and $(U_F, R_F) : K \rightleftarrows G$, the composite is split, with analysis $R_E R_F$ and $\|R_E R_F\| \le \|R_E\| \, \|R_F\|$. The corresponding calibration allowance is at most $(\eta_{\text{cal}} + f) \, \|R_E\| \, \|R_F\|$.*

More generally, if $\|I - UR\| \le \rho < 1$, then

$$\|T\| \le \frac{\|TU\| \, \|R\|}{1 - \rho}, \qquad \|C - A\| \le \frac{(\eta_{\text{cal}} + f) \, \|R\|}{1 - \rho}.$$

Proof. The composite synthesis is $U_F U_E$ and analysis is $R_E R_F$. Their product is $U_F (U_E R_E) R_F = U_F R_F = I_G$. Submultiplicativity bounds the composite analysis norm; applying SR-61 and that bound proves the hierarchical allowance. There is no requirement that either synthesis be square or injective.

For the approximate frame, use the exact identity $T = (TU)R + T(I - UR)$. It gives

$$\|T\| \le \|TU\| \, \|R\| + \|T\| \, \|I - UR\| \le \|TU\| \, \|R\| + \rho \, \|T\| .$$

Since $1 - \rho > 0$, absorb the final term and divide. For $T = C - A$, the measured calibration bounds give $\|TU\| \le \eta_{\text{cal}} + f$, producing the operator allowance. This argument only uses norm inequalities; it does not substitute an assumed exact inverse for the approximate reconstruction.

The denominator records the loss of control. At $\rho = 0$ the exact frame estimate is recovered. At $\rho = 1$ the absorption step gives no bound, and it is not legitimate to reuse the formula by division. Likewise the product of hierarchical analysis norms is a valid conservative bound, not generally an equality or the optimal sensitivity of the composite. These conditioning factors belong in the certificate.

□

Formal source: `CalibrationHierarchy.comp;CalibrationHierarchy.comp_analysis_norm_le;CalibrationHierarchy.hierarchical_operator_bound;ApproximateCalibration.approximate_factorization;ApproximateCalibration.approximate_frame_bound;ApproximateCalibration.measured_approximate_operator_bound`.

Hypotheses. Actual partial calibration from R to R^2 with product norm; bounded unseen scalar operators; arbitrary real norm threshold.

Theorem 17.4 (SR-64 A missed calibration direction permits unbounded error)**.** *Let $U(t) = (t, 0)$ and $C_s(x, y) = sy$. Then $C_s U = 0$ and $\|C_s\| = |s|$. For every real M there is an operator agreeing perfectly with zero on the entire calibration subspace yet satisfying $\|C_s\| > M$. The calibration synthesis is not surjective. Thus perfect agreement on a proper source subspace does not control whole-source operator accuracy.*

Proof. Direct substitution gives $C_s U(t) = s \cdot 0 = 0$. The second-coordinate projection has norm one, so scalar multiplication gives $\|C_s\| = |s|$; the vector $(0, 1)$ also verifies the lower bound. Choose $s = |M| + 1 > 0$. Then $\|C_s\| = |M| + 1 > M$ while its calibrated response remains zero. Taking the true operator and measured response both zero shows that both calibration residuals vanish, even though the whole operator discrepancy exceeds M.

The vector $(0, 1)$ is outside the range of U, proving nonsurjectivity. A split reconstruction $UR = I_{\mathbb{R}^2}$ would make every source a range vector, so no such reconstruction exists here. Nor can any approximate reconstruction for this synthesis have whole defect below one: the range of UR has second coordinate zero, and therefore $(I - UR)(0, 1)$ has second coordinate one and norm at least one. This last observation explains the direction-coverage role of the strict absorption threshold in SR-63.

The positive redundant example and this negative partial example differ in coverage, not coordinate count. Extra coordinates can coexist with complete reconstruction of every source. Calibration of fewer source directions leaves an operator component that all tested calibration responses fail to see. The theorem makes that obstruction explicit without claiming that an externally supplied truth allowance can be inferred from those responses.

□

Formal source: `CalibrationExamples.partial_not_surjective;CalibrationExamples.unseen_comp_partial;CalibrationExamples.unseen_norm;CalibrationExamples.arbitrarily_large_unseen_error;CalibrationExamples.perfect_partial_calibration_unbounded`.

18 What the assembled calculus establishes

The results support a precise progression. Exact resolutions identify which coordinate vectors and coordinate kernels describe actual objects and operations. Split compression identifies the excursion through discarded state that prevents preservation of composition. Memory maps characterize prepared closure; finite hidden dimension turns their universal vanishing into a finite whole-map certificate.

Behavioral quotients retain exactly the distinctions accessible to the declared future task. Normed defect budgets determine how approximate representations propagate their errors.

In inverse problems, compatibility, realizability, and source identifiability remain distinct. The local obstruction quotient measures compatible data that have no global source. The compatibility residual measures the inconsistent sector, with its own conditioning constant. A global source has separate kernel ambiguity and inverse sensitivity. Regularization controls an actual penalized objective. Its strong limits recover each fixed identifiable source, while uniform source convergence requires closed range. Discrepancy rules and calibrated changing models add operational acceptance contracts whose true error allowances remain visible.

The common theme is that changing a representation changes the obligations needed to use it. A certificate can discharge an obligation from concrete maps and inequalities. A residual can expose the remaining defect. An obstruction theorem can prove that an obligation is impossible under the available structure. These outcomes have different mathematical meanings.

18.1 Exact representation and information loss

A redundant exact coordinate space need not lose any information about its object, because synthesis after analysis is identity. Coarsening a packet address family can also remain exact if the entire contents of the merged packets are retained. Information loss is supplied by a separate map and its kernel. In the behavioral setting, only the largest invariant invisible subspace can be quotiented without changing any declared future observation. In source reconstruction, the unavoidable ambiguity is the true forward operator's kernel. Neither coordinate count nor packet count alone decides sufficiency.

18.2 Relations to established mathematics

The mathematics deliberately uses established foundations. Split idempotents, quotients, observability, block elimination, operator exponentials, orthogonal geometry, Moore–Penrose inversion, Tikhonov regularization, and discrepancy principles each contribute concrete constructions. The paper's organizing claim is that their interfaces can be written together, with exact transport, obstruction, approximation, and certificate obligations. Historical novelty of each component is a separate question. The formal inventory and finite test totals do not answer it.

18.3 Open bridges and extensions

The following problems specify further research beyond the printed contracts.

1. **Stable target-only recovery from behavioral minimality.** The exact quotient X/N preserves all specified future observations. The Hilbert theory stably recovers an identifiable source under its reconstruction or regularization assumptions. A general theorem connecting these two constructions must identify a task target, its quotient metric, the relevant forward map, and sufficient conditioning. It should recover that target without requiring reconstruction of the full original source. No such unrestricted bridge is asserted in this edition.
2. **Calibration truth allowances from individual samples.** The calibration theorems start from the whole-operator inequality $\|Y - AU\| \leq \eta_{\mathrm{cal}}$. Deriving it from individual noisy probes requires a specified sample model, a justified aggregation bound, and control of untested directions. A measured model fit supplies neither the true operator nor that truth allowance by itself.

3. **Verified numerical procedures.** The formalization proves the certificate contracts. The rational Python implementations compute and check records under explicit input limits. Their parsers, arithmetic helpers, row-space saturation, parameter bracketing, and stopping procedures are not formally verified. Verified extraction or an independent formal certificate importer would add a new layer of assurance.
4. **Additional admissible categories.** Minimum linear dimension does not establish a minimum positive, stochastic, integral, geometric, or nonlinear realization. Bounded autonomous generators do not settle unbounded-domain or time-dependent flow problems. Such extensions must print the additional admissibility and domain hypotheses rather than importing them through terminology.

19 Source distribution and verification boundaries

This edition contains a traditional paper, a visual guide, a Lean guide, and a complete source ZIP. The `.md` files are included in that ZIP, alongside editable `.tex`, the 93 production Lean modules, theorem guides, exact Python companions, example data, source inventories, and verification records. The original v0.16.0 archive is preserved separately for provenance. Publication sources and reconciliation records are additional files, not silent replacements of the mathematical source history.

The principal concordance uses SR-01 through SR-64 with explicit hypotheses, dependencies, exact declaration names, and hashes of the source files read for this assembly. These are consolidated theorem groups. The declaration inventory comprises 998 theorem/lemma declarations and 1159 enumerated named declarations in 93 production modules. Supporting identities, classical specializations, definitions, and checked examples contribute to these totals.

The archived v0.16.0 final verification includes two runs of the latest block's 31-module project dependency closure and audits of its 52 new named declarations. It does not record a fresh simultaneous compilation of every one of the 93 modules at that final checkpoint. The complete release verification record distinguishes archived Lean evidence, fresh compilation if any, source inspections, finite Python replay, document builds, and archive integrity. This manuscript makes no fresh Lean compilation claim merely from the existence of those archives.

The exact-rational companions cover observation saturation, discrete memory, quantitative budgets, orthogonal globalization, quotient metrics, reconstruction, positive-parameter regularization, discrepancy, inexact solves, changing operators, generalized feasibility, and calibration. Their finite checks are implementation evidence in the tested dimensions and inputs. They are not universal proofs and do not infer external truth allowances. The current release's exact run ranges and counts belong in its machine-readable verification records so that archived and fresh results cannot be conflated.

19.1 Rebuilding the reading set

The supplied build scripts invoke pdfLaTeX three times for stable cross-references and table-of-contents entries. This rebuild creates the documents. It does not compile Lean, rerun scientific checks, infer a noise allowance, or authenticate a physical calibration experiment. Separate scripts and instructions address those activities.

The standalone Markdown manuscript preserves the same statements and proofs for readers and tools that work directly with text. The concordance CSV/JSON provides a structured route from a mathematical claim to its formal declarations. The visual guide illustrates the same contracts with exact vector figures and checked rational examples. The Lean guide explains the production modules, the

formal domains, and the evidence needed to reproduce the recorded builds.

A Coverage of the sixteen development blocks

The block numbers below preserve the development history; they are not the section numbers of the reorganized paper. Each row identifies four consolidated theorem groups and the principal mathematical contract.

Block	Stable claims	Contract
1	SR-01–SR-04	Exact redundant representation; compression defects; actual globalization obstruction; behavioral quotient.
2	SR-05–SR-08	Discrete hidden elimination; closure equivalence; finite feedback certificate; canonical hidden-space construction.
3	SR-09–SR-12	Finite observation horizon; minimal image dimension; realizable observation tables; finite certificate soundness.
4	SR-13–SR-16	Exact word-defect transport; heterogeneous budgets; two-rate response; stable initialization bounds.
5	SR-17–SR-20	Actual bounded flows; powers-to-exponentials bridge; continuous memory; finite moment and jet certificates.
6	SR-21–SR-24	Duhamel defect transport; flow growth envelopes; uniform stability; scalar sharpness and instability boundary.
7	SR-25–SR-28	Unique orthogonal error split; harmonic obstruction; compatibility conditioning; finite exactness and refinement.
8	SR-29–SR-32	Constructed canonical inverse; minimum-norm source; source sensitivity; closed-range characterization.
9	SR-33–SR-36	Native quotient metric; minimum representatives; bounded obstruction transport; refinement and nonclosed seminorm boundary.
10	SR-37–SR-40	Coercive Tikhonov construction; objective optimality; bias/noise/kernel accounting; perturbation and globalization.
11	SR-41–SR-44	Normal-range density; strong source limits; noisy and inexact consistency; uniform boundary and fitted-data limits.
12	SR-45–SR-48	Actual residual curve; noncomputable root existence; norm-controlled strong consistency; accepted rule and endpoints.
13	SR-49–SR-52	Source/image solver scales; protected measurements; near-accepted consistency; squared certificate soundness.
14	SR-53–SR-56	Whole-operator uncertainty; model residual transport; changing-model consistency; operational certificates.
15	SR-57–SR-60	Candidate-dependent self-bounding; source-cap-free acceptance; feasibility-aware selection; inexact and impossible-model boundaries.
16	SR-61–SR-64	Redundant calibration frames; generated uncertainty and recovery; hierarchical/approximate frames; unseen-direction obstruction.

B Principal dependency paths

The exact-coordinate path is SR-01 to SR-02. The hidden-memory path is SR-05 to SR-06 to SR-07, with SR-08 supplying the canonical bridge from an arbitrary split compression. The behavioral path starts at SR-04 and passes through SR-09–SR-12. Quantitative discrete transport is SR-13–SR-16. Bounded analytic flow converts the algebraic memory path through SR-17–SR-20 and provides approximate transport in SR-21–SR-24.

The globalization path begins with the actual quotient SR-03 and its projected geometry SR-25–SR-28. Its intrinsic metric is proved in SR-33–SR-36. Canonical source reconstruction SR-29–SR-32 needs the additional source-space and closed-range hypotheses. Regularization SR-37–SR-40 relaxes closed range at positive parameter. Density and norm domination in SR-41 and SR-46 provide the common strong-consistency engine for the discrepancy rules and changing-model extensions.

Calibration SR-61–SR-63 supplies operator-error allowances from its own externally justified truth bounds and conditioning. The recovery chain then uses generalized acceptance and explicit parameter feasibility. The negative control SR-64 explains why source-direction coverage is part of that contract.

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Hypergraph Lumpability and Bayesian Refinement for Dynamical Systems with Projector Resolutions

Exact Interaction Quotients, Filtering Cocycles,
and Certified Coarse Models

Komorebi P.

Internal completion release, version 0.1.4
September 2026

Abstract

We study when a coarse description of a nonlinear system is exact rather than fitted after the fact. The fine and coarse state spaces are linked by a surjective linear map C. Each homogeneous term in a local analytic vector field is resolved into packet interactions. Linear terms give ordinary directed edges, while higher order terms give directed hyperedges. We prove that a multilinear interaction of order m descends exactly when its coarse output vanishes whenever any input lies in $\operatorname{Ker} C$. For block sum aggregations, this becomes a direct coefficient test. Once the fine output coefficients are summed within a coarse fiber, the resulting value must depend only on the coarse input fibers.

The quotient condition survives passage to the full analytic field, all of its derivatives, every rooted interaction tree, and the locally unique flow. We represent a Bayesian observation as positive likelihood transfer followed by normalization. This normalized update descends precisely when the fine likelihood is constant within each aggregation fiber, up to a single positive scale. Under those two conditions, the coarse filter agrees with the aggregate of the fine filter for every finite observation sequence, and the evidence factors differ only by the same scales.

When exact descent fails, we keep the sources of error separate. A Grönwall argument turns interaction mismatch into a bound on the flow, while variation of the likelihood inside a fiber creates its own normalization error. A recursive estimate carries both contributions to the final posterior. In the Hilbert Schmidt setting, an orthogonal residual identity also gives the best approximation that retains a prescribed number of hyperedges. Deterministic examples and the reference implementation check the finite formulas, the perturbation bounds, and negative controls. Exact theorems, numerical checks, and claims about historical priority remain distinct throughout.

Status and scope. This manuscript is complete for internal use. It contains the finite theorem chain, proofs, examples, algorithms, certificate code, and reproducibility files. Before submission, it still needs independent proof review and a focused literature review. The algebraic results concern spaces of finite dimension, and the flow results are local in time under the assumptions stated in the text. The numerical checks verify the examples and the implementation, not the proofs. We do not assume that useful packet resolutions always exist, that an optimal quotient is easy to find, that local flow theory guarantees global existence, or that this presentation settles priority across neighboring work.

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1 Introduction

1.1 The quotient problem

Let X_{f} be a fine state space, let X_{c} be a coarse state space, and let

$$C : X_{\mathrm{f}} \twoheadrightarrow X_{\mathrm{c}}$$

be a surjective linear map. A fine trajectory satisfies

$$\dot{x}(t) = F_{\mathrm{f}}(x(t)), \qquad x(0) = x_0, \tag{1}$$

and the coarse observable of interest is

$$z(t) = Cx(t).$$

The first question is whether $z(t)$ obeys a closed equation of its own. Equivalently, can we define a vector field F_{c} such that

$$CF_{\mathrm{f}} = F_{\mathrm{c}}C? \tag{2}$$

When this identity holds and both initial value problems have unique solutions, aggregation commutes with evolution:

$$C\Phi^{\mathrm{f}}_t = \Phi^{\mathrm{c}}_t C. \tag{3}$$

The coarse trajectory is then an exact quotient of the fine one. No fitting step is involved.

The linear versions of this question are classical. For finite Markov chains, they appear as lumpability and Markov function criteria [12, 14, 5]. Nearly decomposable systems lead instead to perturbative aggregation [15, 13]. Polynomial systems and reaction networks admit exact differential equivalences and reductions based on bisimulation [7, 8]. We isolate a quotient test for each order of a multilinear interaction and then follow it through nonlinear composition, continuous evolution, and Bayesian observation.

1.2 Why hypergraphs appear

Suppose a local analytic vector field has the expansion

$$F_{\mathrm{f}}(x) = \sum_{m \geq 0} A^{\mathrm{f}}_m[x, \ldots, x]. \tag{4}$$

The linear coefficient A^{f}_1 moves one input packet into one output packet. The quadratic coefficient A^{f}_2 combines two input packets before producing an output. In general, A^{f}_m combines m inputs. We record this incidence by the directed hyperedge

$$(j_1, \ldots, j_m) \longrightarrow i.$$

Nonlinear composition is not ordinary path concatenation. It inserts the output of one multilinear interaction into an input of another. Repeated insertion produces rooted trees and is governed by the Faà di Bruno formula [9]. The same tree structure appears in nonlinear evolution and in numerical integration [6, 10].

At order m, the central question is

$$\exists A^{\mathrm{c}}_m \quad \text{such that} \quad CA^{\mathrm{f}}_m = A^{\mathrm{c}}_m C^{\times m}? \tag{5}$$

If this question has an answer at every order, the quotient relation can be carried from individual interactions to the full field, its derivatives, its interaction trees, and its flow.

1.3 Why Bayesian observations add another nonlinear operation

Prediction is only one part of a filter. Given a probability vector μ and a positive likelihood λ, Bayesian conditioning is

$$\mathsf{B}_\lambda(\mu) = \frac{\lambda \odot \mu}{\lambda^T \mu}. \tag{6}$$

Likelihood multiplication is linear, but the evidence in the denominator couples all packets. The full update is therefore nonlinear even when the likelihood acts separately on each state.

Classical sufficiency theory asks whether a statistic preserves all information relevant to a chosen observation family [11, 4]. Here the operator question is

$$C\mathsf{B}_{\lambda^{\mathrm{f}}} = \mathsf{B}_{\lambda^{\mathrm{c}}} C. \tag{7}$$

When both the flow identity (3) and the Bayesian identity (7) hold at each observation time, the whole filtering sequence commutes with aggregation. Hidden Markov filtering is the linear unary case [3]. The same argument applies when the prediction step comes from a nonlinear quotient flow.

1.4 Main results

Eight results form the mathematical spine of the paper.

C1. A quotient criterion for every interaction order. For a surjective map C, a multilinear fine interaction descends exactly when CA^{f}_m vanishes whenever one input lies in $\operatorname{Ker} C$. Equivalently, its linearization on $X_{\mathrm{f}}^{\otimes m}$ factors through $C^{\otimes m}$.

C2. A concrete packet test. For block sum aggregation, exact descent is equivalent to constancy of the aggregated fine coefficients over each product of input fibers. These common values determine the coarse interaction uniquely.

C3. Compatibility across several resolutions. Exact descent is transitive. Approximate descent obeys an exact defect identity that can be iterated along a refinement tower.

C4. Naturality of trees and flows. Descent of the coefficients passes to local analytic fields, all derivatives, every rooted interaction tree, and every locally unique flow.

C5. Naturality after normalization. If a positive observation transfer respects the quotient up to a scalar, normalization removes that scalar and preserves the quotient. For diagonal likelihoods, this happens exactly when the fine likelihood is constant on each fiber, up to one global scale.

C6. Exact filtering cocycles. When both dynamics and observations descend, every finite sequence of predictions and updates agrees after aggregation. The evidence factors obey the same quotient law.

C7. A certificate for approximate filtering. Interaction mismatch first appears as flow error, while observation mismatch appears at normalization. A closed recurrence carries both contributions to the final posterior and to the accumulated log evidence.

C8. A separate pruning certificate. Discarded hyperedges remain in a named residual tensor. Orthogonal packet blocks give an exact residual identity and an optimal retention theorem for a fixed number of hyperedges.

The proof strategy can be summarized by the chain

$$\boxed{\begin{array}{c}\text{multilinear quotient} \Longrightarrow \text{interaction tree quotient} \Longrightarrow \text{flow quotient} \\ \Longrightarrow \text{Bayesian quotient} \Longrightarrow \text{filtering cocycle quotient} \Longrightarrow \text{posterior certificate.}\end{array}}$$

1.5 Four distinctions

The same partition can succeed at one stage of the problem and fail at another. Four distinctions prevent those cases from being conflated.

(i) A *resolution* states which packets are being distinguished.

(ii) An *exact quotient* proves that the coarse dynamics do not depend on hidden coordinates.

(iii) A *compression* deletes or approximates interaction blocks, so the omitted part must remain visible as a residual.

(iv) *Statistical sufficiency* concerns the information retained for a chosen observation family. Here it is expressed by naturality of the normalized transfer.

A partition may be exact for the dynamics yet lose information carried by the observations. It may preserve the observation model while failing to close the dynamics. It may also work through one interaction order and fail at the next. Each failure receives its own defect.

1.6 Organization

Section 2 introduces packet resolutions and interaction tensors. Section 3 proves the quotient criteria. Section 4 carries the result to analytic fields, rooted trees, and flows. Section 5 studies normalized positive transfer and Bayesian refinement. Section 6 combines prediction and observation into an exact filtering cocycle. Sections 7 to 9 develop the error bounds. Section 10 keeps track of residuals created by hyperedge removal. Sections 11 and 12 describe the finite algorithms, numerical checks, and worked models. The final sections compare the framework with nearby theories, then state the limitations and open problems.

2 Resolved interaction systems

2.1 Finite packet resolutions

The main algebraic results are stated for vector spaces of finite dimension over $\mathbb{R}$ or $\mathbb{C}$. The quotient and tensor factorizations are then purely algebraic. The later analytic results add the stated convergence conditions and the usual assumptions for local uniqueness.

Definition 2.1 (Packet resolution)**.** A packet resolution of a vector space X is a finite family of projections $(P_i)_{i\in I}$ satisfying

$$P_iP_j = \delta_{ij}P_i, \qquad \sum_{i\in I} P_i = I_X.$$

Thus

$$X = \bigoplus_{i\in I} X_i, \qquad X_i = P_iX.$$

When X is a Hilbert space, the principal norm identities will assume that the P_i are orthogonal.

Let $(P_j)_{j\in J}$ resolve a source X and $(Q_i)_{i\in I}$ resolve a target Y.

Definition 2.2 (Resolved multilinear block and hyperedge)**.** For a multilinear map $A : X^m \to Y$ of order m, define

$$A_{i|j_1,\dots,j_m} = Q_i A(P_{j_1}\cdot,\dots,P_{j_m}\cdot). \tag{8}$$

The ordered incidence

$$(j_1,\dots,j_m) \longrightarrow i$$

is the corresponding directed interaction hyperedge. We call the hyperedge active when the block in (8) is nonzero.

When $m = 1$, the active blocks form an ordinary directed transfer graph. When $m = 2$, each active block joins a pair of input packets. For general m, the block records one interaction with m inputs.

Theorem 2.3 (Exact multilinear packet reconstruction)**.** *Every multilinear map $A : X^m \to Y$ of order m satisfies*

$$A(x_1,\dots,x_m) = \sum_{i,j_1,\dots,j_m} A_{i|j_1,\dots,j_m}(P_{j_1}x_1,\dots,P_{j_m}x_m). \tag{9}$$

Proof. Insert $I_X = \sum_j P_j$ in every input slot and $I_Y = \sum_i Q_i$ after applying A. Multilinearity distributes all finite sums and gives (9). □

2.2 Coarse maps, sections, and hidden directions

Let

$$C : X_{\mathrm{f}} \twoheadrightarrow X_{\mathrm{c}}$$

be a surjective linear coarse map. Finite dimensionality guarantees a linear right inverse

$$E : X_{\mathrm{c}} \to X_{\mathrm{f}}, \qquad CE = I_{X_{\mathrm{c}}}.$$

Define

$$P_C = EC, \qquad Q_C = I_{X_{\mathrm{f}}} - EC. \tag{10}$$

Then

$$P_C^2 = P_C, \qquad Q_C^2 = Q_C, \qquad \operatorname{Ran} Q_C = \operatorname{Ker} C.$$

Every fine state has the exact decomposition

$$x = ECx + Q_C x. \tag{11}$$

The first term is the chosen fine representative of the coarse state. The second lies in $\operatorname{Ker} C$, the hidden subspace. Exact quotienting means that motion inside this subspace cannot change the aggregated interaction output.

Definition 2.4 (Compatible packet aggregation)**.** Suppose $(P_i)_{i\in I_{\mathrm{f}}}$ and $(\overline{P}_a)_{a\in I_{\mathrm{c}}}$ resolve the fine and coarse spaces. A surjection

$$\pi : I_{\mathrm{f}} \twoheadrightarrow I_{\mathrm{c}}$$

is compatible with C when

$$\overline{P}_a C = C \sum_{\pi(i)=a} P_i \qquad (a \in I_{\mathrm{c}}). \tag{12}$$

In the scalar coordinate model used later, each P_i selects one coordinate and C sums all coordinates in the same fiber of π.

2.3 Multilinear descent

Definition 2.5 (Exact interaction quotient)**.** Let $A_m^{\mathrm{f}} : X_{\mathrm{f}}^m \to X_{\mathrm{f}}$ and $A_m^{\mathrm{c}} : X_{\mathrm{c}}^m \to X_{\mathrm{c}}$ be multilinear maps of order m. We say that A_m^{f} *descends through* C to A_m^{c}, and write

$$A_m^{\mathrm{f}} \Downarrow_C A_m^{\mathrm{c}},$$

when

$$CA_m^{\mathrm{f}} = A_m^{\mathrm{c}} C^{\times m}. \tag{13}$$

For $m = 0$, descent means $CA_0^{\mathrm{f}} = A_0^{\mathrm{c}}$.

Here

$$C^{\times m}(x_1, \dots, x_m) = (Cx_1, \dots, Cx_m).$$

We write $C^{\otimes m}$ for the induced linear map between algebraic tensor powers.

Definition 2.6 (Interaction defect)**.** For a proposed coarse tensor B_m, define

$$\Delta_m[B_m] = CA_m^{\mathrm{f}} - B_m C^{\times m}. \tag{14}$$

Exact descent is the condition $\Delta_m[B_m] = 0$.

2.4 Norm conventions

For a multilinear map $A : X^m \to Y$ of order m, let

$$\|A\| = \sup_{\|x_r\| \le 1} \|A(x_1, \dots, x_m)\|.$$

If E is a fixed right inverse of C, then any exact coarse tensor necessarily satisfies

$$A_m^{\mathrm{c}}(z_1, \dots, z_m) = CA_m^{\mathrm{f}}(Ez_1, \dots, Ez_m), \tag{15}$$

so

$$\|A_m^{\mathrm{c}}\| \le \|C\| \left\|A_m^{\mathrm{f}}\right\| \|E\|^m.$$

The formula also makes the later uniqueness argument immediate.

3 Hypergraph lumpability at every interaction order

3.1 Invariant quotient criterion

The quotient problem has a simple algebraic answer.

Theorem 3.1 (Universal multilinear quotient criterion)**.** *Let $C : X_{\mathrm{f}} \twoheadrightarrow X_{\mathrm{c}}$ be surjective and let*

$$A_m^{\mathrm{f}} : X_{\mathrm{f}}^m \to X_{\mathrm{f}}$$

be a multilinear map of order m, where $m \ge 1$. The following statements are equivalent.

(i) There exists a multilinear map $A_m^{\mathrm{c}} : X_{\mathrm{c}}^m \to X_{\mathrm{c}}$ of order m such that

$$CA_m^{\mathrm{f}} = A_m^{\mathrm{c}} C^{\times m}.$$

(ii) For every slot $r \in \{1, \dots, m\}$,

$$CA_m^{\mathrm{f}}(x_1, \dots, x_{r-1}, k, x_{r+1}, \dots, x_m) = 0 \tag{16}$$

whenever $k \in \operatorname{Ker} C$.

(iii) The map CA_m^{f} is constant on products of fibers: if $Cx_r = Cx_r'$ for all r, then

$$CA_m^{\mathrm{f}}(x_1, \dots, x_m) = CA_m^{\mathrm{f}}(x_1', \dots, x_m'). \tag{17}$$

(iv) Let $\widehat{A}_m^{\mathrm{f}} : X_{\mathrm{f}}^{\otimes m} \to X_{\mathrm{c}}$ be the linearization of CA_m^{f}. Then

$$\operatorname{Ker}(C^{\otimes m}) \subseteq \operatorname{Ker}(\widehat{A}_m^{\mathrm{f}}). \tag{18}$$

When these conditions hold, the coarse tensor is unique and, for every right inverse E of C,

$$A_m^{\mathrm{c}}(z_1, \dots, z_m) = CA_m^{\mathrm{f}}(Ez_1, \dots, Ez_m). \tag{19}$$

The right side is independent of the chosen right inverse.

Proof. Assume (i). If $k \in \operatorname{Ker} C$, then inserting k into any input slot gives

$$CA_m^{\mathrm{f}}(x_1, \dots, k, \dots, x_m) = A_m^{\mathrm{c}}(Cx_1, \dots, 0, \dots, Cx_m) = 0,$$

so (ii) holds.

Assume (ii). If $Cx_r = Cx_r'$, then $x_r' - x_r \in \operatorname{Ker} C$. Replace the inputs one at a time. Multilinearity and (16) show that each replacement leaves CA_m^{f} unchanged. Hence (iii) holds.

Assume (iii). For $z_r \in X_{\mathrm{c}}$ choose arbitrary lifts x_r with $Cx_r = z_r$ and set

$$A_m^{\mathrm{c}}(z_1, \dots, z_m) = CA_m^{\mathrm{f}}(x_1, \dots, x_m).$$

Fiber constancy makes the definition independent of the lifts. Multilinearity is inherited from A_m^{f}, and the defining identity gives (i). Surjectivity of C also gives uniqueness: every coarse input tuple has a fine lift.

To compare (ii) and (iv), identify a multilinear map of order m with its linearization on the algebraic tensor product. Since C is surjective,

$$\operatorname{Ker}(C^{\otimes m}) = \sum_{r=1}^{m} X_{\mathrm{f}}^{\otimes(r-1)} \otimes \operatorname{Ker} C \otimes X_{\mathrm{f}}^{\otimes(m-r)}.$$

Choose a splitting $X_{\mathrm{f}} = E(X_{\mathrm{c}}) \oplus \operatorname{Ker} C$ and expand the tensor power. Vanishing of $\widehat{A}_m^{\mathrm{f}}$ on this sum is exactly the slot condition. Thus (ii) and (iv) are equivalent. Formula (19) follows by choosing $x_r = Ez_r$. □

Remark 3.2 (Nonlinear congruence interpretation). For a general map $F : X_{\mathrm{f}} \to X_{\mathrm{f}}$, descent through C is equivalent to

$$Cx = Cy \quad \Longrightarrow \quad CF(x) = CF(y).$$

Thus the relation $x \sim_C y$, defined by $Cx = Cy$, must be a congruence for the coarse output of F. Theorem 3.1 turns that statement into a condition on each multilinear coefficient.

Corollary 3.3 (Bounded coarse tensor)**.** *Under the hypotheses of Theorem 3.1, for every bounded right inverse E,*

$$\|A_m^{\mathrm{c}}\| \le \|C\| \left\|A_m^{\mathrm{f}}\right\| \|E\|^m .$$

Proof. Apply (19) to unit coarse inputs. □

3.2 Criterion in packet coordinates

For a block sum aggregation, the kernel condition can be read directly from the tensor coefficients.

Let

$$I_{\mathrm{f}} = \{1, \dots, n\}, \qquad I_{\mathrm{c}} = \{1, \dots, q\},$$

and let

$$\pi : I_{\mathrm{f}} \twoheadrightarrow I_{\mathrm{c}}.$$

Write (e_i) and (f_a) for the standard bases of $\mathbb{R}^n$ and $\mathbb{R}^q$. Define

$$Ce_i = f_{\pi(i)}, \qquad (Cx)_a = \sum_{\pi(i)=a} x_i. \tag{20}$$

Suppose

$$A_m^{\mathrm{f}}(e_{j_1}, \dots, e_{j_m}) = \sum_{i=1}^{n} a^{\mathrm{f},(m)}_{i;j_1,\dots,j_m} e_i. \tag{21}$$

For each coarse output packet a, sum the fine coefficients over its output fiber and define

$$g^{(m)}_{a;j_1,\dots,j_m} = \sum_{i:\pi(i)=a} a^{\mathrm{f},(m)}_{i;j_1,\dots,j_m}. \tag{22}$$

Theorem 3.4 (Coefficient criterion for every interaction order)**.** *The fine tensor A_m^{f} descends through the block sum map (20) if and only if, for every coarse output a,*

$$g^{(m)}_{a;j_1,\dots,j_m} = g^{(m)}_{a;j'_1,\dots,j'_m} \tag{23}$$

whenever

$$\pi(j_r) = \pi(j'_r) \qquad (r = 1, \dots, m).$$

When this condition holds, the unique coarse coefficients are

$$a^{\mathrm{c},(m)}_{a;b_1,\dots,b_m} = g^{(m)}_{a;j_1,\dots,j_m}, \qquad \pi(j_r) = b_r. \tag{24}$$

Proof. Applying C to (21) gives

$$CA_m^{\mathrm{f}}(e_{j_1}, \dots, e_{j_m}) = \sum_{a=1}^{q} g^{(m)}_{a;j_1,\dots,j_m} f_a.$$

If a coarse tensor exists, this vector equals

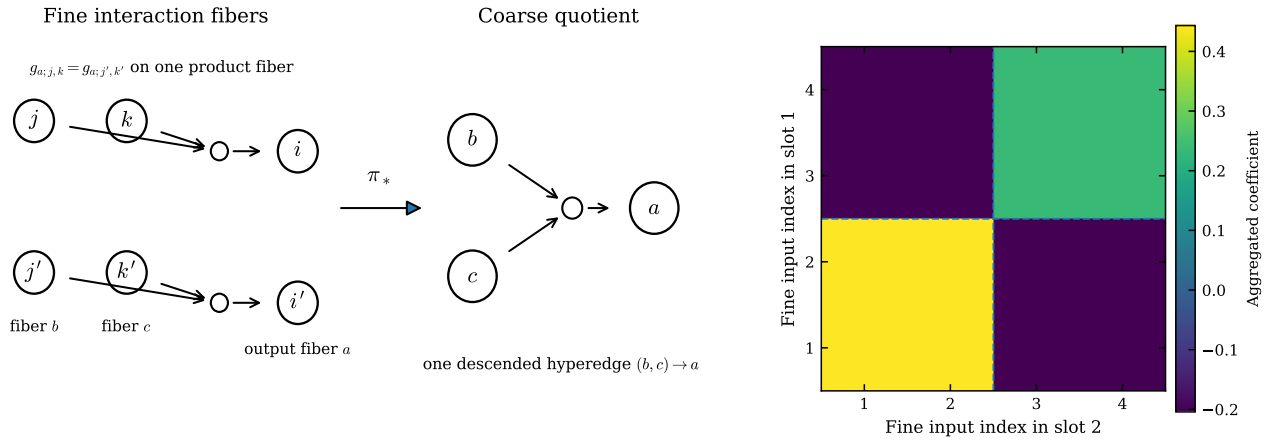
$$A_m^{\mathrm{c}}(f_{\pi(j_1)}, \dots, f_{\pi(j_m)}),$$

which depends only on the coarse input labels. This proves necessity. Conversely, constancy on each fiber makes (24) well defined. Equality on every tuple of fine basis vectors then gives

$$CA_m^{\mathrm{f}} = A_m^{\mathrm{c}} C^{\times m}$$

by multilinearity. □

For $m = 2$, the criterion has two useful views. The hypergraph records which input fibers interact, while the coefficient matrix shows whether those interactions are constant on each product fiber. The exact synthetic tensor used in the certificate run has the required block structure to floating point precision.



(a) Fine interactions descend to one coarse hyperedge.

(b) Aggregated coefficients are constant on every product fiber.

Figure 1: Two views of the quadratic quotient criterion. The incidence view identifies the descended hyperedge. The coefficient view displays the fiber constancy condition from Theorem 3.4.

3.3 Unary and quadratic cases

Corollary 3.5 (Unary aggregate rate criterion)**.** *For $m = 1$, exact descent is equivalent to*

$$\sum_{i:\pi(i)=a} a^{\mathrm{f},(1)}_{i;j} = \sum_{i:\pi(i)=a} a^{\mathrm{f},(1)}_{i;j'} \tag{25}$$

for all coarse outputs a whenever $\pi(j) = \pi(j')$.

Proof. Set $m = 1$ in Theorem 3.4. The condition then involves one input fiber and becomes (25). □

Let Q_{f} be the generator of a continuous time Markov chain, using the column vector convention. The left side of (25) is the total transition rate from fine state j into coarse packet a. Corollary 3.5 is therefore the standard criterion for strong lumpability.

Corollary 3.6 (Quadratic hypergraph criterion)**.** *For $m = 2$, exact descent is equivalent to*

$$\sum_{i:\pi(i)=a} a^{\mathrm{f},(2)}_{i;j,k} = \sum_{i:\pi(i)=a} a^{\mathrm{f},(2)}_{i;j',k'}$$

whenever

$$\pi(j) = \pi(j'), \qquad \pi(k) = \pi(k').$$

The resulting coarse interaction is the directed quadratic hyperedge

$$(\pi(j), \pi(k)) \longrightarrow a.$$

Proof. Set $m = 2$ in Theorem 3.4. Fiber constancy in the two input slots gives the displayed equality, and (24) assigns its common value to the coarse hyperedge. □

Example 3.7 (A quotient with invisible internal interactions)**.** Let $X_{\mathrm{c}} = \mathbb{R}^q$, $X_{\mathrm{f}} = \mathbb{R}^{2q}$, and

$$C = \begin{bmatrix} I_q & I_q \end{bmatrix}, \qquad E = \frac{1}{2}\begin{bmatrix} I_q \\ I_q \end{bmatrix}.$$

Let A_m^{c} be arbitrary and let $K_m : X_{\mathrm{f}}^m \to X_{\mathrm{f}}$ satisfy

$$CK_m = 0.$$

Then

$$A_m^{\mathrm{f}} = EA_m^{\mathrm{c}}C^{\times m} + K_m \tag{26}$$

satisfies

$$CA_m^{\mathrm{f}} = A_m^{\mathrm{c}}C^{\times m}.$$

The tensor K_m may contain many fine hyperedges, so the fine model can remain complicated. Exact quotienting asks only that every hidden contribution disappear after applying C.

3.4 Structural consequences

Proposition 3.8 (Symmetry descends)**.** *If A_m^{f} is symmetric in its input slots and descends through C, then the unique coarse tensor A_m^{c} is symmetric.*

Proof. Choose lifts of the coarse inputs. Permuting the coarse inputs permutes their lifts, and symmetry of A_m^{f} leaves (19) unchanged. □

Proposition 3.9 (Quotient support)**.** *Under the coordinate criterion, every active coarse hyperedge*

$$(b_1, \ldots, b_m) \longrightarrow a$$

is supported by at least one fine hyperedge whose input and output labels map to that coarse incidence. The converse can fail because several active fine hyperedges may cancel after aggregation.

Proof. If every fine coefficient in the corresponding incidence fiber vanishes, then the aggregate coefficient (22) vanishes. Conversely, nonzero fine coefficients can sum to zero after output aggregation. □

3.5 Composition across refinement levels

Let

$$X_2 \xrightarrow{C_{21}} X_1 \xrightarrow{C_{10}} X_0, \qquad C_{20} = C_{10}C_{21}.$$

Proposition 3.10 (Transitivity of exact interaction quotients)**.** *If*

$$A_m^{(2)} \Downarrow_{C_{21}} A_m^{(1)} \qquad \text{and} \qquad A_m^{(1)} \Downarrow_{C_{10}} A_m^{(0)},$$

then

$$A_m^{(2)} \Downarrow_{C_{20}} A_m^{(0)}.$$

Proof. Compose the two identities:

$$C_{20}A_m^{(2)} = C_{10}A_m^{(1)}C_{21}^{\times m} = A_m^{(0)}C_{10}^{\times m}C_{21}^{\times m} = A_m^{(0)}C_{20}^{\times m}.$$

□

Approximate descent has an exact composition law.

Theorem 3.11 (Interaction defect across two levels)**.** *Define*

$$\Delta_m^{21} = C_{21}A_m^{(2)} - A_m^{(1)}C_{21}^{\times m},$$

$$\Delta_m^{10} = C_{10}A_m^{(1)} - A_m^{(0)}C_{10}^{\times m},$$

and

$$\Delta_m^{20} = C_{20}A_m^{(2)} - A_m^{(0)}C_{20}^{\times m}.$$

Then

$$\boxed{\Delta_m^{20} = C_{10}\Delta_m^{21} + \Delta_m^{10}C_{21}^{\times m}.} \tag{27}$$

Consequently,

$$\left\|\Delta_m^{20}\right\| \le \|C_{10}\| \left\|\Delta_m^{21}\right\| + \left\|\Delta_m^{10}\right\| \|C_{21}\|^m. \tag{28}$$

Proof. Add and subtract $C_{10}A_m^{(1)}C_{21}^{\times m}$:

$$\begin{aligned}\Delta_m^{20} &= C_{10}C_{21}A_m^{(2)} - A_m^{(0)}C_{10}^{\times m}C_{21}^{\times m} \\ &= C_{10}(C_{21}A_m^{(2)} - A_m^{(1)}C_{21}^{\times m}) \\ &\quad + (C_{10}A_m^{(1)} - A_m^{(0)}C_{10}^{\times m})C_{21}^{\times m}.\end{aligned}$$

This proves (27). Taking norms and using submultiplicativity yields the estimate. □

4 Analytic fields, rooted trees, and quotient flows

4.1 Descent of polynomial and analytic vector fields

Let

$$F_{\mathrm{f}}(x) = \sum_{m=0}^{\infty} A_m^{\mathrm{f}}[x^{\otimes m}] \tag{29}$$

be locally analytic, where A_m^{f} is a symmetric multilinear map of order m for $m \ge 1$ and $A_0^{\mathrm{f}} \in X_{\mathrm{f}}$. Throughout,

$$A_m[x^{\otimes m}] = A_m(x, \dots, x).$$

Theorem 4.1 (Analytic interaction descent)**.** *Let $C : X_{\mathrm{f}} \twoheadrightarrow X_{\mathrm{c}}$ have a bounded right inverse E. Suppose the coefficient series converges normally on every smaller ball,*

$$\sum_{m=0}^{\infty} \left\|A_m^{\mathrm{f}}\right\| r^m < \infty \qquad (0 \le r < R), \tag{30}$$

and, for every $m \ge 0$, there is a coarse coefficient A_m^{c} satisfying

$$CA_m^{\mathrm{f}} = A_m^{\mathrm{c}}C^{\times m}. \tag{31}$$

Then

$$F_{\mathrm{c}}(z) = \sum_{m=0}^{\infty} A_m^{\mathrm{c}}[z^{\otimes m}] \tag{32}$$

converges absolutely whenever $\|E\|\,\|z\| < R$*, and*

$$\boxed{CF_{\mathrm{f}}(x) = F_{\mathrm{c}}(Cx)} \tag{33}$$

for every x *in the fine convergence domain whose coarse image lies in the coarse domain.*

Proof. For $m \geq 1$, Corollary 3.3 gives

$$\|A_m^{\mathrm{c}}\| \leq \|C\| \left\|A_m^{\mathrm{f}}\right\| \|E\|^m .$$

For $m = 0$, the identity $A_0^{\mathrm{c}} = CA_0^{\mathrm{f}}$ gives the same bound with $\|E\|^0 = 1$. Therefore

$$\sum_{m\geq 0} \|A_m^{\mathrm{c}}\|\,\|z\|^m \leq \|C\| \sum_{m\geq 0} \left\|A_m^{\mathrm{f}}\right\| (\|E\|\,\|z\|)^m,$$

which converges when $\|E\|\,\|z\| < R$. Coefficientwise descent and absolute convergence permit termwise aggregation:

$$CF_{\mathrm{f}}(x) = \sum_{m\geq 0} CA_m^{\mathrm{f}}[x^{\otimes m}] = \sum_{m\geq 0} A_m^{\mathrm{c}}[(Cx)^{\otimes m}] = F_{\mathrm{c}}(Cx).$$

□

Remark 4.2 (Coefficientwise descent is sufficient, not always necessary)*.* Descent of each displayed coefficient is enough to make the whole series descend. The converse can fail for a noncanonical expansion that splits one homogeneous degree into several terms whose coarse contributions cancel. For the Taylor expansion at a fixed base point, the derivatives are unique, so the converse does hold. Differentiating $CF_{\mathrm{f}} = F_{\mathrm{c}}C$ then recovers every descended jet.

Theorem 4.3 (Coefficientwise characterization of polynomial descent)**.** *Let*

$$F_{\mathrm{f}}(x) = \sum_{m=0}^{M} A_m^{\mathrm{f}}[x^{\otimes m}], \qquad F_{\mathrm{c}}(z) = \sum_{m=0}^{M} A_m^{\mathrm{c}}[z^{\otimes m}],$$

where every A_m^{f} *and* A_m^{c} *is symmetric. Then*

$$CF_{\mathrm{f}} = F_{\mathrm{c}}C$$

if and only if

$$CA_m^{\mathrm{f}} = A_m^{\mathrm{c}}C^{\times m} \qquad (0 \leq m \leq M).$$

Proof. The reverse implication follows from Theorem 4.1, applied to a finite sum. Conversely, fix $x \in X_{\mathrm{f}}$ and evaluate the field identity at tx:

$$\sum_{m=0}^{M} t^m CA_m^{\mathrm{f}}[x^{\otimes m}] = \sum_{m=0}^{M} t^m A_m^{\mathrm{c}}[(Cx)^{\otimes m}].$$

Equality of the coefficients of this polynomial with values in X_{c} gives equality on the diagonal for the two symmetric multilinear maps of order m. The polarization identity recovers each multilinear map from its diagonal polynomial, so

$$CA_m^{\mathrm{f}} = A_m^{\mathrm{c}}C^{\times m}$$

for every degree. □

4.2 Naturality of derivatives

Proposition 4.4 (Jet naturality)**.** *Let F_{f} and F_{c} be C^r maps satisfying*

$$CF_{\mathrm{f}} = F_{\mathrm{c}}C.$$

Then, for every $1 \le k \le r$,

$$\boxed{CD^kF_{\mathrm{f}}(x)[h_1,\dots,h_k] = D^kF_{\mathrm{c}}(Cx)[Ch_1,\dots,Ch_k].} \tag{34}$$

Proof. Differentiate the identity $CF_{\mathrm{f}}(x) = F_{\mathrm{c}}(Cx)$. Since C is linear, its derivatives of order greater than one vanish and the ordinary chain rule gives the first order statement. Repeated differentiation gives (34). □

Thus the quotient preserves more than the value of the vector field. At each base point, every fine derivative block maps to its coarse counterpart. The same identity controls local perturbations, stability calculations, and higher order response.

4.3 Rooted interaction trees

Repeated substitution of nonlinear interactions produces rooted trees. Only a simple recursive definition is needed for the quotient argument.

Definition 4.5 (Interaction tree evaluation)**.** Let $\mathcal{A} = (A_m)_{m\ge1}$ be a family of multilinear maps $A_m : X^m \to X$. A rooted interaction tree is defined recursively. A leaf $\bullet$ has evaluation

$$\mathcal{V}_\bullet = I_X.$$

If

$$\tau = B_+(\tau_1,\dots,\tau_m)$$

is formed by adjoining a root of arity m, define

$$\mathcal{V}_\tau(x_1,\dots,x_{|\tau|}) = A_m(\mathcal{V}_{\tau_1}(\mathbf{x}_1),\dots,\mathcal{V}_{\tau_m}(\mathbf{x}_m)), \tag{35}$$

where $|\tau|$ denotes the number of leaves and the input list is partitioned among the subtrees in their fixed order.

The operation B_+ inserts several rooted trees beneath a new root. In packet coordinates, every edge carries a packet label and every internal vertex must match an active resolved hyperedge.

Theorem 4.6 (Treewise quotient naturality)**.** *Suppose every operation in a fine family descends:*

$$CA^{\mathrm{f}}_m = A^{\mathrm{c}}_m C^{\times m} \qquad (m \ge 1).$$

Then, for every rooted interaction tree τ,

$$\boxed{C\mathcal{V}^{\mathrm{f}}_\tau = \mathcal{V}^{\mathrm{c}}_\tau C^{\times|\tau|}.} \tag{36}$$

Proof. Proceed by induction on the number of internal vertices. The leaf case is $CI = IC$. Suppose

$$\tau = B_+(\tau_1, \dots, \tau_m).$$

Using interaction descent at the root and the inductive hypothesis on every subtree,

$$\begin{aligned} C\mathcal{V}^{\mathrm{f}}_\tau &= CA^{\mathrm{f}}_m(\mathcal{V}^{\mathrm{f}}_{\tau_1}, \dots, \mathcal{V}^{\mathrm{f}}_{\tau_m}) \\ &= A^{\mathrm{c}}_m(C\mathcal{V}^{\mathrm{f}}_{\tau_1}, \dots, C\mathcal{V}^{\mathrm{f}}_{\tau_m}) \\ &= A^{\mathrm{c}}_m(\mathcal{V}^{\mathrm{c}}_{\tau_1} C^{\times|\tau_1|}, \dots, \mathcal{V}^{\mathrm{c}}_{\tau_m} C^{\times|\tau_m|}) \\ &= \mathcal{V}^{\mathrm{c}}_\tau C^{\times|\tau|}. \end{aligned}$$

□

Corollary 4.7 (Compatibility with Faà di Bruno composition)**.** *Let F_{f} and F_{c} be smooth maps with $CF_{\mathrm{f}} = F_{\mathrm{c}}C$, and let G_{f} and G_{c} satisfy the same relation. Then each rooted tree in the Faà di Bruno expansion, written in packet coordinates, descends to the corresponding coarse tree. Moreover,*

$$C(G_{\mathrm{f}} \circ F_{\mathrm{f}}) = (G_{\mathrm{c}} \circ F_{\mathrm{c}})C.$$

Proof. The map identity follows directly by composition. The termwise statement follows from Theorem 4.6, because the Faà di Bruno expansion is a finite sum of rooted derivative trees at each derivative order. □

4.4 Exact quotient flows

Theorem 4.8 (Exact quotient flow)**.** *Let $U_{\mathrm{f}} \subseteq X_{\mathrm{f}}$ and $U_{\mathrm{c}} \subseteq X_{\mathrm{c}}$ be open, with $C(U_{\mathrm{f}}) \subseteq U_{\mathrm{c}}$. Suppose F_{f} and F_{c} are locally Lipschitz and satisfy*

$$CF_{\mathrm{f}} = F_{\mathrm{c}}C$$

on U_{f}. Let $\Phi^{\mathrm{f}}_t(x)$ and $\Phi^{\mathrm{c}}_t(z)$ denote the unique maximal local flows. Whenever both sides are defined,

$$\boxed{C\Phi^{\mathrm{f}}_t(x) = \Phi^{\mathrm{c}}_t(Cx).} \tag{37}$$

Proof. Let $x(t) = \Phi^{\mathrm{f}}_t(x_0)$ and set $z(t) = Cx(t)$. Then

$$\dot{z}(t) = C\dot{x}(t) = CF_{\mathrm{f}}(x(t)) = F_{\mathrm{c}}(Cx(t)) = F_{\mathrm{c}}(z(t)),$$

with $z(0) = Cx_0$. Uniqueness for the coarse initial value problem gives $z(t) = \Phi^{\mathrm{c}}_t(Cx_0)$. □

Corollary 4.9 (Linear semigroup naturality)**.** *If $CA_{\mathrm{f}} = A_{\mathrm{c}}C$, then*

$$Ce^{tA_{\mathrm{f}}} = e^{tA_{\mathrm{c}}}C \tag{38}$$

for every t.

Proof. Either apply Theorem 4.8 to the linear systems or sum the exponential series using $CA^k_{\mathrm{f}} = A^k_{\mathrm{c}}C$. □

Corollary 4.10 (Quotients of discrete dynamics)**.** *If $CF_{\mathrm{f}} = F_{\mathrm{c}}C$, then*

$$CF^{\circ n}_{\mathrm{f}} = F^{\circ n}_{\mathrm{c}}C$$

for every integer $n \geq 0$.

Proof. Induct on n. □

4.5 Positivity and simplex preservation

Let

$$\Delta_{\mathrm{f}} = \{\mu \in \mathbb{R}^n_+ : \mathbf{1}^T\mu = 1\}, \qquad \Delta_{\mathrm{c}} = \{\nu \in \mathbb{R}^q_+ : \mathbf{1}^T\nu = 1\}.$$

For block sum aggregation, $C\Delta_{\mathrm{f}} = \Delta_{\mathrm{c}}$ and

$$\mathbf{1}_{\mathrm{c}}^T C = \mathbf{1}_{\mathrm{f}}^T.$$

Proposition 4.11 (Simplex preservation descends)**.** *Suppose* $\Phi_t^{\mathrm{f}}(\Delta_{\mathrm{f}}) \subseteq \Delta_{\mathrm{f}}$ *and exact flow naturality holds. Then*

$$\Phi_t^{\mathrm{c}}(\Delta_{\mathrm{c}}) \subseteq \Delta_{\mathrm{c}}.$$

Proof. For any $\nu \in \Delta_{\mathrm{c}}$, choose $\mu \in \Delta_{\mathrm{f}}$ with $C\mu = \nu$. Then

$$\Phi_t^{\mathrm{c}}(\nu) = C\Phi_t^{\mathrm{f}}(\mu) \in C\Delta_{\mathrm{f}} = \Delta_{\mathrm{c}}.$$

□

Claim Boundary 4.12 (Global existence of the flow)**.** Exact descent does not guarantee global solutions. A fine trajectory may blow up in finite time, leave the domain, or reach a region where the vector field is no longer Lipschitz. Theorem 4.8 applies only while both solutions exist and remain unique.

5 Normalized positive transfer and Bayesian refinement

5.1 Positive state spaces

The Bayesian result rests on a simple fact about normalizing positive operators.

Let X_{f} and X_{c} be finite dimensional ordered vector spaces with positive cones K_{f} and K_{c}. Let

$$\ell_{\mathrm{f}} : X_{\mathrm{f}} \to \mathbb{R}, \qquad \ell_{\mathrm{c}} : X_{\mathrm{c}} \to \mathbb{R}$$

be positive mass functionals. Define normalized state spaces

$$\mathcal{S}_{\mathrm{f}} = \{x \in K_{\mathrm{f}} : \ell_{\mathrm{f}}(x) = 1\}, \qquad \mathcal{S}_{\mathrm{c}} = \{z \in K_{\mathrm{c}} : \ell_{\mathrm{c}}(z) = 1\}.$$

Assume C is positive and preserves mass:

$$\ell_{\mathrm{c}} C = \ell_{\mathrm{f}}. \tag{39}$$

Definition 5.1 (Normalized positive transfer)**.** For a positive linear operator $L : X \to X$, define

$$\mathsf{N}_\ell(L)(x) = \frac{Lx}{\ell(Lx)} \tag{40}$$

whenever $\ell(Lx) > 0$.

5.2 Normalization naturality

Theorem 5.2 (Naturality of normalized transfer)**.** *Let $L_{\mathrm{f}} : X_{\mathrm{f}} \to X_{\mathrm{f}}$ and $L_{\mathrm{c}} : X_{\mathrm{c}} \to X_{\mathrm{c}}$ be positive linear operators. Suppose*

$$CL_{\mathrm{f}} = sL_{\mathrm{c}}C \tag{41}$$

for some $s > 0$, and suppose (39) holds. Then, on the common domain of normalization,

$$\boxed{C\mathsf{N}_{\ell_{\mathrm{f}}}(L_{\mathrm{f}}) = \mathsf{N}_{\ell_{\mathrm{c}}}(L_{\mathrm{c}})C.} \tag{42}$$

Proof. For a positive state x,

$$\begin{aligned} C\mathsf{N}_{\ell_{\mathrm{f}}}(L_{\mathrm{f}})(x) &= \frac{CL_{\mathrm{f}}x}{\ell_{\mathrm{f}}(L_{\mathrm{f}}x)} \\ &= \frac{sL_{\mathrm{c}}Cx}{\ell_{\mathrm{c}}(CL_{\mathrm{f}}x)} \\ &= \frac{sL_{\mathrm{c}}Cx}{s\ell_{\mathrm{c}}(L_{\mathrm{c}}Cx)} \\ &= \mathsf{N}_{\ell_{\mathrm{c}}}(L_{\mathrm{c}})(Cx). \end{aligned}$$

□

Remark 5.3 (Gauge invariance). Normalization removes the scale s. At the operator level, this is the familiar fact that a positive rescaling of a likelihood leaves the posterior unchanged.

Proposition 5.4 (Composition law for normalized transfers)**.** *For positive operators L_1, L_2 on the same ordered state space,*

$$\mathsf{N}_{\ell}(L_2) \circ \mathsf{N}_{\ell}(L_1) = \mathsf{N}_{\ell}(L_2L_1) \tag{43}$$

wherever both sides are defined.

Proof. The first normalization contributes the same scalar to the numerator and denominator of the second update, so it cancels. □

5.3 Classical Bayesian update

On the finite simplex, let

$$L_\lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n), \qquad \lambda_i > 0.$$

With $\ell(x) = \mathbf{1}^T x$,

$$\mathsf{N}_{\ell}(L_\lambda)(\mu) = \frac{\lambda \odot \mu}{\lambda^T \mu} = \mathsf{B}_\lambda(\mu).$$

The evidence is

$$Z_\lambda(\mu) = \lambda^T \mu.$$

The composition law becomes

$$\mathsf{B}_{\lambda_2} \circ \mathsf{B}_{\lambda_1} = \mathsf{B}_{\lambda_1 \odot \lambda_2}. \tag{44}$$

5.4 Exact Bayesian refinement

Let C be the block sum aggregation associated with a surjection $\pi : \{1, \dots, n\} \twoheadrightarrow \{1, \dots, q\}$.

Theorem 5.5 (Sufficient condition for Bayesian refinement)**.** *Let $\lambda^{\mathrm{f}} \in \mathbb{R}^n_{>0}$ and $\lambda^{\mathrm{c}} \in \mathbb{R}^q_{>0}$. If*

$$\lambda^{\mathrm{f}} = sC^T\lambda^{\mathrm{c}} \tag{45}$$

for some $s > 0$, then

$$\boxed{C\mathsf{B}_{\lambda^{\mathrm{f}}} = \mathsf{B}_{\lambda^{\mathrm{c}}}C.} \tag{46}$$

Moreover,

$$Z_{\lambda^{\mathrm{f}}}(\mu) = sZ_{\lambda^{\mathrm{c}}}(C\mu). \tag{47}$$

Proof. Equation (45) is equivalent to

$$CL_{\lambda^{\mathrm{f}}} = sL_{\lambda^{\mathrm{c}}}C.$$

Apply Theorem 5.2. The evidence identity follows from

$$(\lambda^{\mathrm{f}})^T\mu = s(\lambda^{\mathrm{c}})^TC\mu.$$

□

Condition (45) says that the fine likelihood is constant on each aggregation fiber, apart from one global rescaling. When the coarse space has at least two packets, the next theorem shows that this condition is also necessary if commutation is required for every interior prior.

Theorem 5.6 (Necessary condition for Bayesian refinement)**.** *Assume $q \geq 2$, every fiber of π is nonempty, and $\lambda^{\mathrm{f}}, \lambda^{\mathrm{c}}$ are strictly positive. If*

$$C\mathsf{B}_{\lambda^{\mathrm{f}}}(\mu) = \mathsf{B}_{\lambda^{\mathrm{c}}}(C\mu) \tag{48}$$

for every μ in the interior of Δ_n, then there exists $s > 0$ such that

$$\lambda^{\mathrm{f}} = sC^T\lambda^{\mathrm{c}}.$$

Proof. Both sides of (48) extend continuously to any prior with nonzero evidence. Strict positivity guarantees nonzero evidence throughout the closed simplex. Fix two distinct coarse packets a and b. Choose $i \in \pi^{-1}(a)$ and $k \in \pi^{-1}(b)$, then consider a prior supported on $\{i, k\}$ with masses $u, v > 0$. On the fine side, the ratio of the posterior masses of packets a and b is

$$\frac{\lambda^{\mathrm{f}}_i u}{\lambda^{\mathrm{f}}_k v},$$

whereas the coarse posterior ratio is

$$\frac{\lambda^{\mathrm{c}}_a u}{\lambda^{\mathrm{c}}_b v}.$$

Equality for all $u, v > 0$ gives

$$\frac{\lambda^{\mathrm{f}}_i}{\lambda^{\mathrm{f}}_k} = \frac{\lambda^{\mathrm{c}}_a}{\lambda^{\mathrm{c}}_b}. \tag{49}$$

Take another state j in the same fiber as i. Applying the same argument to j and k gives $\lambda^{\mathrm{f}}_j = \lambda^{\mathrm{f}}_i$. The fine likelihood is therefore constant on each fiber. Write its value on fiber a as $\overline{\lambda}_a$. Equation (49) now gives

$$\frac{\overline{\lambda}_a}{\overline{\lambda}_b} = \frac{\lambda^{\mathrm{c}}_a}{\lambda^{\mathrm{c}}_b}$$

for every pair a, b. Hence $\overline{\lambda} = s\lambda^{\mathrm{c}}$ for one $s > 0$, proving the result. □

Remark 5.7 (The case of one coarse packet). If $q = 1$, both sides of (48) equal the unique coarse state, regardless of the fine likelihood. Necessity therefore requires a nontrivial coarse state space.

5.5 Differential structure of Bayesian normalization

Proposition 5.8 (Bayesian differential). *At an interior prior μ,*

$$D\mathsf{B}_\lambda(\mu)[h] = \frac{\lambda \odot h}{Z_\lambda(\mu)} - \mathsf{B}_\lambda(\mu)\frac{\lambda^T h}{Z_\lambda(\mu)}. \tag{50}$$

Proof. Differentiate the quotient $L_\lambda\mu/(\mathbf{1}^T L_\lambda \mu)$. □

Corollary 5.9 (Bayesian jet naturality). *Under the hypotheses of Theorem 5.5, for every $r \geq 1$ and every interior fine prior,*

$$CD^r\mathsf{B}_{\lambda^{\mathrm{f}}}(\mu)[h_1, \ldots, h_r] = D^r\mathsf{B}_{\lambda^{\mathrm{c}}}(C\mu)[Ch_1, \ldots, Ch_r]. \tag{51}$$

Proof. Differentiate the exact map identity (46) and apply Theorem 4.4. □

Thus the normalization map has a descended interaction structure of its own. Its first derivative couples all evidence packets through a rank one term. Higher derivatives describe interactions among larger groups of packets.

6 Filtering cocycles

6.1 Finite observation sequences

Let $0 = t_0 < t_1 < \cdots < t_N$ and define prediction maps

$$\Phi^{\mathrm{f}}_k = \Phi^{\mathrm{f}}_{t_k - t_{k-1}}, \qquad \Phi^{\mathrm{c}}_k = \Phi^{\mathrm{c}}_{t_k - t_{k-1}}.$$

For observations $y_1, \ldots, y_N$, let

$$\mathsf{B}^{\mathrm{f}}_k = \mathsf{B}_{\lambda^{\mathrm{f}}_{y_k}}, \qquad \mathsf{B}^{\mathrm{c}}_k = \mathsf{B}_{\lambda^{\mathrm{c}}_{y_k}}.$$

Pair each prediction with the observation that follows it:

$$\Gamma^{\mathrm{f}}_k = \mathsf{B}^{\mathrm{f}}_k \circ \Phi^{\mathrm{f}}_k, \qquad \Gamma^{\mathrm{c}}_k = \mathsf{B}^{\mathrm{c}}_k \circ \Phi^{\mathrm{c}}_k,$$

and define the resulting filtering cocycles

$$\Psi^{\mathrm{f}}_{1:N} = \Gamma^{\mathrm{f}}_N \circ \cdots \circ \Gamma^{\mathrm{f}}_1, \qquad \Psi^{\mathrm{c}}_{1:N} = \Gamma^{\mathrm{c}}_N \circ \cdots \circ \Gamma^{\mathrm{c}}_1. \tag{52}$$

Theorem 6.1 (Exact naturality of the filtering cocycle). *Suppose, for every k,*

$$C\Phi^{\mathrm{f}}_k = \Phi^{\mathrm{c}}_k C \tag{53}$$

and

$$C\mathsf{B}^{\mathrm{f}}_k = \mathsf{B}^{\mathrm{c}}_k C. \tag{54}$$

Then

$$\boxed{C\Psi^{\mathrm{f}}_{1:N} = \Psi^{\mathrm{c}}_{1:N} C.} \tag{55}$$

Proof. For one step,

$$C\Gamma_k^{\mathrm{f}} = C\mathsf{B}_k^{\mathrm{f}}\Phi_k^{\mathrm{f}} = \mathsf{B}_k^{\mathrm{c}}C\Phi_k^{\mathrm{f}} = \mathsf{B}_k^{\mathrm{c}}\Phi_k^{\mathrm{c}}C = \Gamma_k^{\mathrm{c}}C.$$

Iterating this identity through (52) proves the result. □

Corollary 6.2 (Exact posterior aggregation)**.** *If $\mu_0^{\mathrm{c}} = C\mu_0^{\mathrm{f}}$, then*

$$\boxed{C\mu_N^{\mathrm{f}} = \mu_N^{\mathrm{c}}}$$

for every finite observation sequence, where

$$\mu_N^{\mathrm{f}} = \Psi_{1:N}^{\mathrm{f}}(\mu_0^{\mathrm{f}}), \qquad \mu_N^{\mathrm{c}} = \Psi_{1:N}^{\mathrm{c}}(\mu_0^{\mathrm{c}}).$$

Proof. Apply (55) to μ_0^{f} and substitute $\mu_0^{\mathrm{c}} = C\mu_0^{\mathrm{f}}$. □

6.2 Sequential evidence

Let

$$\widehat{\mu}_k^{\mathrm{f}} = \Phi_k^{\mathrm{f}}(\mu_{k-1}^{\mathrm{f}}), \qquad \widehat{\mu}_k^{\mathrm{c}} = \Phi_k^{\mathrm{c}}(\mu_{k-1}^{\mathrm{c}}),$$

and define the step evidence

$$z_k^{\mathrm{f}} = (\lambda_{y_k}^{\mathrm{f}})^T\widehat{\mu}_k^{\mathrm{f}}, \qquad z_k^{\mathrm{c}} = (\lambda_{y_k}^{\mathrm{c}})^T\widehat{\mu}_k^{\mathrm{c}}.$$

Set

$$Z_{1:N}^{\mathrm{f}} = \prod_{k=1}^{N} z_k^{\mathrm{f}}, \qquad Z_{1:N}^{\mathrm{c}} = \prod_{k=1}^{N} z_k^{\mathrm{c}}.$$

Theorem 6.3 (Evidence factorization under exact refinement)**.** *Assume the hypotheses of Theorem 6.1, let*

$$\mu_0^{\mathrm{c}} = C\mu_0^{\mathrm{f}},$$

and suppose

$$\lambda_{y_k}^{\mathrm{f}} = s_k C^T\lambda_{y_k}^{\mathrm{c}} \qquad (s_k > 0).$$

Then

$$z_k^{\mathrm{f}} = s_k z_k^{\mathrm{c}} \tag{56}$$

for every step, and the likelihood of the observation sequence satisfies

$$\boxed{Z_{1:N}^{\mathrm{f}} = \left(\prod_{k=1}^{N} s_k\right) Z_{1:N}^{\mathrm{c}}.} \tag{57}$$

Under the gauge $s_k = 1$, the fine and coarse sequential evidences agree exactly.

Proof. Initial compatibility gives $C\mu_0^{\mathrm{f}} = \mu_0^{\mathrm{c}}$. For later steps, Theorem 6.2 gives the same identity after every update. Flow naturality therefore yields

$$C\widehat{\mu}_k^{\mathrm{f}} = \widehat{\mu}_k^{\mathrm{c}}.$$

Therefore

$$z_k^{\mathrm{f}} = s_k(\lambda_{y_k}^{\mathrm{c}})^T C\widehat{\mu}_k^{\mathrm{f}} = s_k z_k^{\mathrm{c}}.$$

The likelihood of the complete observation sequence is the product of the conditional evidence terms. Multiplying the identities gives (57). □

6.3 Hidden Markov and nonlinear specializations

Corollary 6.4 (Hidden Markov lumpability)**.** *Let M_{f} and M_{c} be column stochastic matrices satisfying*

$$CM_{\mathrm{f}} = M_{\mathrm{c}}C,$$

and let every observation likelihood satisfy

$$\lambda^{\mathrm{f}}_y = s_y C^T \lambda^{\mathrm{c}}_y.$$

Then, for every observation sequence $y_{1:N}$,

$$\begin{aligned} &C(\mathsf{B}_{\lambda^{\mathrm{f}}_{y_N}} \circ M_{\mathrm{f}} \circ \cdots \circ \mathsf{B}_{\lambda^{\mathrm{f}}_{y_1}} \circ M_{\mathrm{f}}) \\ &\quad = (\mathsf{B}_{\lambda^{\mathrm{c}}_{y_N}} \circ M_{\mathrm{c}} \circ \cdots \circ \mathsf{B}_{\lambda^{\mathrm{c}}_{y_1}} \circ M_{\mathrm{c}})C. \end{aligned}$$

Proof. Apply Theorem 6.1 with the linear prediction maps M_{f} and M_{c}. □

Corollary 6.5 (Nonlinear quotient filtering)**.** *Suppose the fine and coarse prediction maps are flow segments of locally Lipschitz vector fields satisfying $CF_{\mathrm{f}} = F_{\mathrm{c}}C$, and suppose all observation likelihoods are constant on each fiber up to scale. Then the nonlinear filtering cocycle commutes with C exactly.*

Proof. Intertwining of the vector fields gives $C\Phi^{\mathrm{f}}_k = \Phi^{\mathrm{c}}_k C$ by Theorem 4.8. Likelihoods that are constant on each fiber give $C\mathsf{B}^{\mathrm{f}}_k = \mathsf{B}^{\mathrm{c}}_k C$ by Theorem 5.5. Apply Theorem 6.1. □

Remark 6.6 (Two independent closure requirements). The dynamical and observational conditions are independent. Exact interaction lumpability says nothing about whether the likelihood distinguishes states inside a packet. Conversely, an observation may ignore those distinctions even when the dynamics depend on them. Exact coarse filtering needs both commutative squares.

Claim Boundary 6.7 (Sufficiency for the chosen observation family)**.** The condition $\lambda^{\mathrm{f}}_y = s_y C^T \lambda^{\mathrm{c}}_y$ is sufficient for the specified finite likelihood family. It does not assert that C is sufficient for every possible future observation model. Adding a likelihood that distinguishes two states inside one fiber can invalidate the coarse filter without changing the dynamics.

7 Approximate interaction quotients

Exact quotienting is an algebraic yes or no question. A useful model also needs to say how far it misses when the answer is no. We separate three sources of error:

interaction mismatch, dependence on hidden directions, dynamical amplification.

7.1 Best coarse interaction and quotient defect

Let

$$\mathfrak{Q}_m(C) = \{B_m C^{\times m} : B_m \in \mathcal{L}^m(X_{\mathrm{c}}; X_{\mathrm{c}})\}$$

be the subspace of outputs from multilinear maps of order m that factor through the coarse space.

Definition 7.1 (Best quotient error)**.** For a fine interaction A^{f}_m, define

$$\varepsilon^\star_m = \inf_{B_m \in \mathcal{L}^m(X_{\mathrm{c}}; X_{\mathrm{c}})} \left\| CA^{\mathrm{f}}_m - B_m C^{\times m} \right\|. \tag{58}$$

Proposition 7.2 (Attainment in finite dimension)**.** *The infimum in* (58) *is attained.*

Proof. The map

$$B_m \longmapsto B_m C^{\times m}$$

is injective because C is surjective. Its range is a finite dimensional linear subspace of $\mathcal{L}^m(X_{\mathrm{f}}; X_{\mathrm{c}})$, hence it is closed. In a finite dimensional normed space, the distance to a nonempty closed subspace attains its minimum. A nearest point in this range therefore exists. Any such point has a unique preimage under the displayed map, and that preimage is a minimizing coarse tensor. □

The minimum value depends on the chosen norm. A minimizing tensor need not be unique when that norm is not strictly convex. Exact lumpability is independent of the norm because it is the condition $\varepsilon_m^\star = 0$.

7.2 Approximation induced by a section

A chosen right inverse supplies a concrete coarse tensor even when exact descent fails.

Definition 7.3 (Coarse tensor induced by a section)**.** Fix E with $CE = I$. Define

$$A_m^{\mathrm{c},E} = C A_m^{\mathrm{f}} E^{\times m}. \tag{59}$$

Let $P_C = EC$ and $Q_C = I - EC$ as in (10).

Proposition 7.4 (Expansion of the section defect by hidden inputs)**.** *The defect of* (59) *is*

$$\Delta_m^E(x_1, \dots, x_m) = C A_m^{\mathrm{f}}(x_1, \dots, x_m) - C A_m^{\mathrm{f}}(P_C x_1, \dots, P_C x_m), \tag{60}$$

and admits the exact expansion

$$\boxed{\Delta_m^E(x_1, \dots, x_m) = \sum_{\varnothing \neq S \subseteq \{1,\dots,m\}} C A_m^{\mathrm{f}}(y_1^S, \dots, y_m^S),} \tag{61}$$

where

$$y_r^S = \begin{cases} Q_C x_r, & r \in S, \\ P_C x_r, & r \notin S. \end{cases}$$

Proof. Equation (60) follows from

$$A_m^{\mathrm{c},E}(Cx_1, \dots, Cx_m) = C A_m^{\mathrm{f}}(ECx_1, \dots, ECx_m).$$

Now expand every input as $x_r = P_C x_r + Q_C x_r$. The term with no Q_C factor is the term subtracted in (60). Every remaining term has at least one hidden input and appears exactly once. □

Corollary 7.5 (Exactness as disappearance of hidden interactions)**.** *The tensor defined from the chosen section descends exactly if and only if the sum in* (61) *vanishes for all inputs. The stronger slotwise condition*

$$C A_m^{\mathrm{f}}(x_1, \dots, Q_C x_r, \dots, x_m) = 0$$

for each slot is equivalent to exact descent and makes every summand vanish separately.

Proof. The first statement is (61) with $\Delta_m^E = 0$. Since $\operatorname{Ran} Q_C = \operatorname{Ker} C$, the condition in each slot is exactly the kernel condition in Theorem 3.1. It is therefore equivalent to exact descent. Every term indexed by a nonempty subset contains at least one hidden input, so each term vanishes. □

7.3 Defect majorants for the full field

Suppose fine and proposed coarse fields have expansions

$$F_{\mathrm{f}}(x) = \sum_{m\geq 0} A^{\mathrm{f}}_m[x^{\otimes m}], \qquad F_{\mathrm{c}}(z) = \sum_{m\geq 0} A^{\mathrm{c}}_m[z^{\otimes m}].$$

Define the field defect

$$\delta_F(x) = CF_{\mathrm{f}}(x) - F_{\mathrm{c}}(Cx). \tag{62}$$

Whenever both series converge absolutely,

$$\delta_F(x) = \sum_{m\geq 0} \Delta_m[x^{\otimes m}]. \tag{63}$$

Proposition 7.6 (Analytic defect majorant)**.** *If $\|\Delta_m\| \leq \varepsilon_m$ and the scalar series*

$$\mathfrak{d}_F(R) = \sum_{m\geq 0} \varepsilon_m R^m$$

converges, then

$$\boxed{\sup_{\|x\|\leq R} \|\delta_F(x)\| \leq \mathfrak{d}_F(R).} \tag{64}$$

Proof. Apply the multilinear norm bound to every term in (63) and sum the convergent majorant. □

The sequence (ε_m) shows which interaction orders obstruct the quotient. A model may be exactly lumpable through order r and fail for the first time at order $r+1$.

7.4 Propagation of dynamical defect

Theorem 7.7 (Approximate quotient flow bound)**.** *Let*

$$\dot{x} = F_{\mathrm{f}}(x), \qquad \dot{z} = F_{\mathrm{c}}(z),$$

and suppose both solutions remain in domains on which

$$\|F_{\mathrm{c}}(u) - F_{\mathrm{c}}(v)\| \leq L\,\|u - v\|\,.$$

Set

$$e(t) = \|Cx(t) - z(t)\|\,.$$

Then

$$\boxed{e(t) \leq e^{Lt}e(0) + \int_0^t e^{L(t-s)}\,\|\delta_F(x(s))\|\,\mathrm{d}s.} \tag{65}$$

If $e(0) = 0$ and $\|\delta_F(x(s))\| \leq \varepsilon_F$ on $[0,t]$, then

$$e(t) \leq \begin{cases} \varepsilon_F \dfrac{e^{Lt}-1}{L}, & L > 0, \\ t\varepsilon_F, & L = 0. \end{cases} \tag{66}$$

Proof. Let $y(t) = Cx(t)$. From (62),

$$\dot{y}(t) = F_{\mathrm{c}}(y(t)) + \delta_F(x(t)).$$

Subtract the integral equations for y and z:

$$y(t) - z(t) = y(0) - z(0) + \int_0^t \left(F_{\mathrm{c}}(y(s)) - F_{\mathrm{c}}(z(s))\right) \mathrm{d}s + \int_0^t \delta_F(x(s))\, \mathrm{d}s.$$

Taking norms and applying the Grönwall inequality yields (65). The uniform cases follow by evaluating the exponential integral. □

Corollary 7.8 (Coefficient certificate for a flow segment)**.** *If $\|x(s)\| \le R$ and Proposition 7.6 applies, then a flow interval of length T satisfies*

$$\left\| C\Phi_T^{\mathrm{f}}(x_0) - \Phi_T^{\mathrm{c}}(Cx_0) \right\| \le \mathfrak{d}_F(R) \frac{e^{LT} - 1}{L}$$

for $L > 0$, with $T\mathfrak{d}_F(R)$ at $L = 0$.

Proof. By Proposition 7.6, $\|\delta_F(x(s))\| \le \mathfrak{d}_F(R)$ throughout the segment. Apply Theorem 7.7 with compatible initial data, so $e(0) = 0$, and evaluate the exponential integral. □

7.5 Refinement towers

Let

$$X_L \xrightarrow{C_{L,L-1}} X_{L-1} \longrightarrow \cdots \longrightarrow X_1 \xrightarrow{C_{10}} X_0.$$

For $a > b$, write

$$C_{ab} = C_{b+1,b} \cdots C_{a,a-1}.$$

Let Δ_m^{ab} denote the interaction defect for the composite map C_{ab}. Iterating Theorem 3.11 yields a telescoping formula.

Corollary 7.9 (Telescoping interaction defect)**.** *For an interaction of order m assigned at every level,*

$$\Delta_m^{L0} = \sum_{\ell=0}^{L-1} C_{\ell 0}\, \Delta_m^{\ell+1,\ell} C_{L,\ell+1}^{\times m}, \tag{67}$$

where $C_{00} = I$ and $C_{LL} = I$. Consequently,

$$\left\| \Delta_m^{L0} \right\| \le \sum_{\ell=0}^{L-1} \|C_{\ell 0}\| \left\| \Delta_m^{\ell+1,\ell} \right\| \|C_{L,\ell+1}\|^m. \tag{68}$$

Proof. Induct on the number of levels, using (27) at the final composition. □

The identity shows where a terminal mismatch entered the refinement tower. Rather than storing only the final error, one can keep the local defect at each level and combine those records when needed.

8 Approximate Bayesian refinement

8.1 A normalization perturbation bound

Lemma 8.1 (Normalization perturbation)**.** *Let $u, v \in \mathbb{R}^q_+$ have positive total mass. Then*

$$\boxed{\left\| \frac{u}{\mathbf{1}^T u} - \frac{v}{\mathbf{1}^T v} \right\|_1 \leq \frac{2\,\|u - v\|_1}{\max\{\mathbf{1}^T u, \mathbf{1}^T v\}}.} \tag{69}$$

Proof. Write $a = \mathbf{1}^T u$ and $b = \mathbf{1}^T v$. Then

$$\begin{aligned} \left\| \frac{u}{a} - \frac{v}{b} \right\|_1 &\leq \frac{\|u - v\|_1}{a} + \|v\|_1 \left| \frac{1}{a} - \frac{1}{b} \right| \\ &= \frac{\|u - v\|_1}{a} + \frac{|a - b|}{a} \\ &\leq \frac{2\,\|u - v\|_1}{a}. \end{aligned}$$

Interchanging u and v gives the same inequality with b in the denominator. Taking the smaller of the two upper bounds yields the denominator $\max\{a, b\}$. □

The factor of two need not be sharp. It is uniform, and it makes the role of the evidence mass explicit. No uniform estimate of this form remains useful when both masses approach zero.

8.2 Defect in the observation intertwining law

Let L_{f} and L_{c} be positive linear observation operators, and choose a scale $s > 0$. On a finite simplex, we write $\mathsf{N}(L)$ for normalization with respect to the total mass $\mathbf{1}^T$. Define

$$R_L = C L_{\mathrm{f}} - s L_{\mathrm{c}} C. \tag{70}$$

Theorem 8.2 (Approximate normalized transfer naturality)**.** *On finite simplices with block sum C, let*

$$u = C L_{\mathrm{f}} \mu, \qquad v = s L_{\mathrm{c}} C \mu.$$

If u and v are nonnegative with positive masses, then

$$\boxed{\|C\mathsf{N}(L_{\mathrm{f}})(\mu) - \mathsf{N}(L_{\mathrm{c}})(C\mu)\|_1 \leq \frac{2\,\|R_L \mu\|_1}{\max\{\mathbf{1}^T u, \mathbf{1}^T v\}}.} \tag{71}$$

Proof. Because C preserves total mass,

$$C\mathsf{N}(L_{\mathrm{f}})(\mu) = \frac{u}{\mathbf{1}^T u}.$$

Multiplication of L_{c} by s does not change its normalized output, so

$$\mathsf{N}(L_{\mathrm{c}})(C\mu) = \frac{v}{\mathbf{1}^T v}.$$

Now $u - v = R_L \mu$, and Lemma 8.1 applies. □

8.3 Likelihood variation inside a fiber

For diagonal likelihood operators, write

$$\lambda^{\mathrm{f}} = sC^T\lambda^{\mathrm{c}} + r_\lambda. \tag{72}$$

Then

$$R_L\mu = C(r_\lambda \odot \mu).$$

Since block sum aggregation is an ℓ^1 contraction on signed vectors,

$$\|R_L\mu\|_1 \le \sum_i \mu_i\,|r_{\lambda,i}| \le \|r_\lambda\|_\infty. \tag{73}$$

Corollary 8.3 (Bayesian refinement defect)**.** *Let*

$$z_{\mathrm{f}} = Z_{\lambda^{\mathrm{f}}}(\mu), \qquad z_{\mathrm{c}} = sZ_{\lambda^{\mathrm{c}}}(C\mu).$$

Then

$$\boxed{\|C\mathsf{B}_{\lambda^{\mathrm{f}}}(\mu) - \mathsf{B}_{\lambda^{\mathrm{c}}}(C\mu)\|_1 \le \frac{2\sum_i \mu_i\,|r_{\lambda,i}|}{\max\{z_{\mathrm{f}}, z_{\mathrm{c}}\}}.} \tag{74}$$

If both evidences are at least $z_* > 0$, then

$$d_{\mathsf{B}} \le \frac{2\,\|r_\lambda\|_\infty}{z_*}. \tag{75}$$

Proof. Combine Theorem 8.2 with (73). □

Definition 8.4 (Fiber oscillation)**.** For a fine likelihood, define

$$\mathrm{osc}_a(\lambda^{\mathrm{f}}) = \max_{i:\pi(i)=a} \lambda_i^{\mathrm{f}} - \min_{i:\pi(i)=a} \lambda_i^{\mathrm{f}},$$

and

$$\mathrm{osc}_C(\lambda^{\mathrm{f}}) = \max_a \mathrm{osc}_a(\lambda^{\mathrm{f}}).$$

When $q \ge 2$ and commutation is required for every interior prior, exact Bayesian refinement is equivalent to $\mathrm{osc}_C(\lambda^{\mathrm{f}}) = 0$ once the coarse likelihood and global scale have been chosen consistently. In an approximate model, λ_a^{c} may be the midpoint, the ordinary mean, or a mean chosen to reflect the task on fiber a. The midpoint minimizes the largest absolute residual in that fiber and gives

$$\|r_\lambda\|_\infty = \frac{1}{2}\,\mathrm{osc}_C(\lambda^{\mathrm{f}})$$

when $s = 1$.

8.4 Lipschitz behavior of the Bayesian update

Proposition 8.5 (Lipschitz bound under an evidence floor)**.** *Let $\lambda \in \mathbb{R}^q_{>0}$. If*

$$Z_\lambda(\mu), Z_\lambda(\nu) \ge z_* > 0,$$

then

$$\boxed{\|\mathsf{B}_\lambda(\mu) - \mathsf{B}_\lambda(\nu)\|_1 \le \frac{2\,\|\lambda\|_\infty}{z_*}\,\|\mu - \nu\|_1.} \tag{76}$$

Proof. Apply Lemma 8.1 to

$$u = \lambda \odot \mu, \qquad v = \lambda \odot \nu.$$

Then

$$\|u - v\|_1 \le \|\lambda\|_\infty \|\mu - \nu\|_1 .$$

□

Remark 8.6 (Scale invariance of the bound). Replacing λ by $s\lambda$ multiplies both $\|\lambda\|_\infty$ and z_* by s, leaving the ratio unchanged.

Remark 8.7 (Local versus global constants). The constant in (76) is valid over the whole evidence region under consideration. Along a particular trajectory, the Jacobian in (50) may give a much smaller local factor. The recursive theorem below can use either bound, provided the chosen constant is certified on the states being considered.

9 Certified approximate coarse filtering

9.1 Prediction and update errors

Let the fine recursion be

$$\widehat{\mu}_k^{\mathrm{f}} = \Phi_k^{\mathrm{f}}(\mu_{k-1}^{\mathrm{f}}), \qquad \mu_k^{\mathrm{f}} = \mathsf{B}_k^{\mathrm{f}}(\widehat{\mu}_k^{\mathrm{f}}), \tag{77}$$

and the coarse recursion be

$$\widehat{\mu}_k^{\mathrm{c}} = \Phi_k^{\mathrm{c}}(\mu_{k-1}^{\mathrm{c}}), \qquad \mu_k^{\mathrm{c}} = \mathsf{B}_k^{\mathrm{c}}(\widehat{\mu}_k^{\mathrm{c}}). \tag{78}$$

Define the discrepancies before and after each update by

$$e_k^- = \left\| C\widehat{\mu}_k^{\mathrm{f}} - \widehat{\mu}_k^{\mathrm{c}} \right\|_1 , \qquad e_k^+ = \left\| C\mu_k^{\mathrm{f}} - \mu_k^{\mathrm{c}} \right\|_1 . \tag{79}$$

At the initial time, set

$$e_0^+ = \left\| C\mu_0^{\mathrm{f}} - \mu_0^{\mathrm{c}} \right\|_1 .$$

Assume the coarse prediction is Lipschitz with constant $L_{\Phi,k}$ and that its quotient defect is bounded by

$$\sup_{\mu} \left\| C\Phi_k^{\mathrm{f}}(\mu) - \Phi_k^{\mathrm{c}}(C\mu) \right\|_1 \le d_{\Phi,k}. \tag{80}$$

Assume the coarse update is Lipschitz with constant $L_{\mathsf{B},k}$ and that its direct refinement defect is bounded by

$$\sup_{\mu} \left\| C\mathsf{B}_k^{\mathrm{f}}(\mu) - \mathsf{B}_k^{\mathrm{c}}(C\mu) \right\|_1 \le d_{\mathsf{B},k}. \tag{81}$$

The suprema in (80) and (81) are taken over a declared admissible set that contains both recursions.

Lemma 9.1 (Prediction bound for one step)**.** *Under* (80)*,*

$$e_k^- \le L_{\Phi,k} e_{k-1}^+ + d_{\Phi,k}. \tag{82}$$

Proof. Insert and subtract $\Phi_k^{\mathrm{c}}(C\mu_{k-1}^{\mathrm{f}})$. The quotient defect bounds one difference, while the Lipschitz estimate bounds the other by $L_{\Phi,k} e_{k-1}^+$. □

Lemma 9.2 (Update bound for one step)**.** *Under* (81)*,*

$$e_k^+ \le L_{\mathsf{B},k} e_k^- + d_{\mathsf{B},k}. \tag{83}$$

Proof. Insert and subtract $\mathsf{B}_k^{\mathrm{c}}(C\widehat{\mu}_k^{\mathrm{f}})$. The direct refinement defect bounds one difference, while the Lipschitz estimate bounds the other by $L_{\mathsf{B},k}e_k^-$. □

Combining the two inequalities gives

$$e_k^+ \le \alpha_k e_{k-1}^+ + \beta_k, \tag{84}$$

where

$$\alpha_k = L_{\mathsf{B},k}L_{\Phi,k}, \qquad \beta_k = L_{\mathsf{B},k}d_{\Phi,k} + d_{\mathsf{B},k}. \tag{85}$$

9.2 Closed posterior certificate

Theorem 9.3 (Certified approximate filtering)**.** *Under the assumptions above,*

$$\boxed{e_N^+ \le \left(\prod_{j=1}^{N}\alpha_j\right)e_0^+ + \sum_{k=1}^{N}\left(\prod_{j=k+1}^{N}\alpha_j\right)\beta_k.} \tag{86}$$

Here and below, an empty product equals one. With exact initial aggregation $e_0^+ = 0$,

$$e_N^+ \le \sum_{k=1}^{N}\left(\prod_{j=k+1}^{N}\alpha_j\right)(L_{\mathsf{B},k}d_{\Phi,k} + d_{\mathsf{B},k}). \tag{87}$$

Proof. Repeated substitution in the affine recurrence (84) gives the stated formula. The initial error is multiplied by every later amplification factor. A defect introduced at step k is multiplied only by the factors from steps $k+1$ through N. □

Corollary 9.4 (Uniform stability regimes)**.** *Suppose $0 \le \alpha_k \le \alpha$ and $0 \le \beta_k \le \beta$.*

(i) If $0 \le \alpha < 1$, then

$$e_N^+ \le \alpha^N e_0^+ + \beta\frac{1-\alpha^N}{1-\alpha}, \qquad \limsup_{N\to\infty} e_N^+ \le \frac{\beta}{1-\alpha}. \tag{88}$$

(ii) If $\alpha = 1$, then

$$e_N^+ \le e_0^+ + N\beta.$$

(iii) If $\alpha > 1$, then

$$e_N^+ \le \alpha^N e_0^+ + \beta\frac{\alpha^N - 1}{\alpha - 1}.$$

Proof. Bound the products in (86) by powers of α and sum the geometric series. □

This trichotomy describes the certificate, not necessarily the observed filter. A stable trajectory can still come with a bound for which $\alpha > 1$. In that case, the chosen global norm estimates simply fail to certify contraction.

9.3 From interaction coefficients to posterior error

For a flow interval of length T_k, suppose the fine trajectory remains in a ball of radius R_k and the coarse field has Lipschitz constant $L_{F,k}$. By Corollary 7.8, one may take

$$d_{\Phi,k} = \mathfrak{d}_{F,k}(R_k)\frac{e^{L_{F,k}T_k}-1}{L_{F,k}}, \tag{89}$$

with the continuous value $T_k\mathfrak{d}_{F,k}(R_k)$ at $L_{F,k}=0$. For the observation, Corollary 8.3 gives

$$d_{\mathsf{B},k} \le \frac{2\,\|r_{\lambda,k}\|_\infty}{z_{*,k}}. \tag{90}$$

Substituting these bounds into (87) gives a certificate expressed directly in terms of interaction tensor defects and likelihood residuals within each fiber.

9.4 Observable and total variation consequences

Corollary 9.5 (Coarse observable error)**.** *For any coarse observable $g \in \mathbb{R}^q$,*

$$\left|g^T C\mu_N^{\mathrm{f}} - g^T\mu_N^{\mathrm{c}}\right| \le \|g\|_\infty\, e_N^+.$$

In particular,

$$\mathrm{TV}(C\mu_N^{\mathrm{f}},\mu_N^{\mathrm{c}}) = \frac{1}{2}e_N^+.$$

Proof. Apply the duality between ℓ^1 and ℓ^∞. The total variation identity is standard for finite probability vectors. □

9.5 Sequential evidence error

For brevity, write $\lambda_k^{\mathrm{f}} = \lambda_{y_k}^{\mathrm{f}}$ and $\lambda_k^{\mathrm{c}} = \lambda_{y_k}^{\mathrm{c}}$, and decompose

$$\lambda_k^{\mathrm{f}} = s_k C^T\lambda_k^{\mathrm{c}} + r_k. \tag{91}$$

This gives

$$z_k^{\mathrm{f}} - s_k z_k^{\mathrm{c}} = s_k(\lambda_k^{\mathrm{c}})^T(C\widehat{\mu}_k^{\mathrm{f}} - \widehat{\mu}_k^{\mathrm{c}}) + r_k^T\widehat{\mu}_k^{\mathrm{f}}. \tag{92}$$

Proposition 9.6 (Evidence certificate for one step)**.** *Assume $\widehat{\mu}_k^{\mathrm{f}}$ is a probability vector. Under (91),*

$$\boxed{\left|z_k^{\mathrm{f}} - s_k z_k^{\mathrm{c}}\right| \le s_k\,\|\lambda_k^{\mathrm{c}}\|_\infty\, e_k^- + \|r_k\|_\infty\,.} \tag{93}$$

If $z_k^{\mathrm{f}}, s_k z_k^{\mathrm{c}} \ge z_{*,k} > 0$, then

$$\boxed{\left|\log z_k^{\mathrm{f}} - \log(s_k z_k^{\mathrm{c}})\right| \le \frac{s_k\,\|\lambda_k^{\mathrm{c}}\|_\infty\, e_k^- + \|r_k\|_\infty}{z_{*,k}}.} \tag{94}$$

Proof. Apply Hölder's inequality to (92). Since $\widehat{\mu}_k^{\mathrm{f}}$ is a probability vector, $\left\|\widehat{\mu}_k^{\mathrm{f}}\right\|_1 = 1$. For the logarithms, the mean value theorem gives

$$|\log a - \log b| \le \frac{|a-b|}{\min\{a,b\}}.$$

□

Corollary 9.7 (Certificate for cumulative log evidence)**.** *Let* $Z^{\mathrm{f}}_{1:N} = \prod_k z^{\mathrm{f}}_k$ *and* $Z^{\mathrm{c}}_{1:N} = \prod_k z^{\mathrm{c}}_k$*. Then*

$$\boxed{\left|\log Z^{\mathrm{f}}_{1:N} - \sum_{k=1}^{N} \log s_k - \log Z^{\mathrm{c}}_{1:N}\right| \le \sum_{k=1}^{N} \frac{s_k \left\|\lambda^{\mathrm{c}}_k\right\|_\infty e^-_k + \|r_k\|_\infty}{z_{*,k}}.} \tag{95}$$

Proof. Sum (94) over the observation sequence. □

10 Hyperedge pruning and certified compression

Exact quotienting and sparse representation are distinct. A system may have an exact coarse quotient while its fine interaction tensor is dense. Conversely, a sparse fine tensor can fail to descend. This section introduces a separate residual for deleted hyperedges and then combines it with the quotient defect.

10.1 Hilbert Schmidt tensor blocks

Let X and Y be finite dimensional Hilbert spaces with orthogonal packet projectors (P_j) and (Q_i). Identify a multilinear Hilbert Schmidt tensor of order m with the Hilbert Schmidt linear map

$$\widehat{A} : X^{\otimes m} \to Y.$$

Its packet block is

$$\widehat{A}_{i|j_1,\dots,j_m} = Q_i \widehat{A}(P_{j_1} \otimes \cdots \otimes P_{j_m}).$$

Theorem 10.1 (Orthogonal hyperedge Parseval identity)**.** *Distinct packet blocks are orthogonal in Hilbert Schmidt inner product, and*

$$\boxed{\left\|\widehat{A}\right\|^2_{\mathrm{HS}} = \sum_{i,j_1,\dots,j_m} \left\|\widehat{A}_{i|j_1,\dots,j_m}\right\|^2_{\mathrm{HS}}.} \tag{96}$$

Proof. The tensor product input projectors

$$P_{j_1} \otimes \cdots \otimes P_{j_m}$$

are mutually orthogonal, as are the output projectors Q_i. For two distinct block labels, orthogonality on at least one side forces the Hilbert Schmidt inner product to vanish. Exact reconstruction is therefore an orthogonal sum, and Pythagoras gives (96). □

10.2 Retained field and named residual

Choose a retained hyperedge set $\mathcal{E}^{\mathrm{keep}}_m$. Define

$$\widehat{A}^{\mathrm{keep}}_m = \sum_{h \in \mathcal{E}^{\mathrm{keep}}_m} \widehat{A}_{m,h}, \qquad R_m = \widehat{A}_m - \widehat{A}^{\mathrm{keep}}_m. \tag{97}$$

Corollary 10.2 (Exact pruning residual)**.** *Under the hypotheses of Theorem 10.1,*

$$\boxed{\|R_m\|^2_{\mathrm{HS}} = \sum_{h \notin \mathcal{E}^{\mathrm{keep}}_m} \left\|\widehat{A}_{m,h}\right\|^2_{\mathrm{HS}}.} \tag{98}$$

Proof. The retained and omitted block families are orthogonal by Theorem 10.1. Since R_m is exactly the sum of the omitted blocks, Pythagoras gives (98). □

The residual is a tensor identity, not merely an empirical estimate. Its induced state error is obtained by applying the relevant multilinear norm inequality.

10.3 Retaining the k largest energy hyperedges

Let

$$e_h = \left\| \widehat{A}_{m,h} \right\|_{\mathrm{HS}}^2$$

and order the energies

$$e_{h_1} \geq e_{h_2} \geq \cdots \geq e_{h_M}.$$

Theorem 10.3 (Optimal retention of k orthogonal hyperedges)**.** *Among all retained sets of cardinality k, the set*

$$\{h_1, \ldots, h_k\}$$

minimizes the Hilbert Schmidt residual. The minimum is

$$\boxed{\min_{|\mathcal{E}|=k} \|R_m(\mathcal{E})\|_{\mathrm{HS}}^2 = \sum_{r>k} e_{h_r}.} \tag{99}$$

Proof. By (98), minimizing the omitted energy is equivalent to maximizing the retained energy. For a fixed cardinality, this is achieved by selecting the k largest nonnegative terms. □

Claim Boundary 10.4 (Overlapping blocks and structured costs)**.** The theorem above uses orthogonal block energy and assigns the same unit cost to every retained hyperedge. With overlapping dictionaries, shared low rank factors, communication costs, or group constraints that preserve structure, ranking individual energies need not be optimal.

10.4 Hyperedge entropy

For nonzero total interaction energy, define

$$p_h = \frac{e_h}{\sum_\eta e_\eta}, \qquad H_m = -\sum_h p_h \log p_h,$$

and the effective hyperedge count

$$N_{\mathrm{eff},m} = e^{H_m}.$$

Low entropy means that the interaction energy is concentrated in a small set of resolved hyperedges. High entropy means that pruning by a fixed cutoff is less promising in the chosen resolution. The quantity deliberately depends on the resolution. It evaluates a representation rather than an invariant of the unresolved tensor alone.

10.5 Combined quotient and pruning defect

Let the full fine coefficient split as

$$A_m^{\mathrm{f}} = A_m^{\mathrm{keep}} + R_m^{\mathrm{f}}.$$

For a proposed coarse coefficient A_m^{c}, define

$$\Delta_m^{\mathrm{total}} = CA_m^{\mathrm{f}} - A_m^{\mathrm{c}} C^{\times m}.$$

Then

$$\Delta_m^{\mathrm{total}} = \underbrace{CA_m^{\mathrm{keep}} - A_m^{\mathrm{c}} C^{\times m}}_{\Delta_m^{\mathrm{keep}}} + CR_m^{\mathrm{f}}. \tag{100}$$

Therefore

$$\boxed{\left\|\Delta_m^{\mathrm{total}}\right\| \le \left\|\Delta_m^{\mathrm{keep}}\right\| + \|C\| \left\|R_m^{\mathrm{f}}\right\|.} \tag{101}$$

For analytic fields on a ball of radius R,

$$\mathfrak{d}_F(R) \le \sum_m \left\|\Delta_m^{\mathrm{keep}}\right\| R^m + \|C\| \sum_m \left\|R_m^{\mathrm{f}}\right\| R^m. \tag{102}$$

The flow and posterior bounds use this combined quantity. A pruning decision is justified only when the omitted part remains explicit and is carried through the later estimates like any other model error.

10.6 A budgeted design problem

These terms lead to a practical design problem. The objective below records the tradeoff without claiming that it can always be solved efficiently:

$$\min_{C,\{A_m^{\mathrm{c}}\},\{\mathcal{E}_m^{\mathrm{keep}}\}} \left[\sum_m w_m \left\|\Delta_m^{\mathrm{total}}\right\| + \lambda \sum_m \mathrm{Cost}(\mathcal{E}_m^{\mathrm{keep}}) + \gamma \sum_y d_{\mathsf{B},y} \right]. \tag{103}$$

The minimization is subject to any required positivity, mass, rank, and interpretability conditions on C. The three terms measure failure of dynamical closure, computational cost, and loss of observational information. Their weights and the admissible packet families must be chosen for the application.

11 Algorithms and finite certificates

Every finite criterion in the paper can be checked directly from coordinates. The reference code uses dense arrays so that each contraction can be inspected. A sparse implementation would loop only over the active hyperedges.

11.1 Exact test for every interaction order

For a tensor of order m in scalar packet coordinates, compute the aggregated coefficient array

$$g^{(m)}_{a;j_1,\dots,j_m} = \sum_{i:\pi(i)=a} a^{\mathrm{f},(m)}_{i;j_1,\dots,j_m}.$$

For every coarse input tuple $\mathbf{b} = (b_1, \dots, b_m)$, define the fine fiber product

$$\mathcal{G}_{\mathbf{b}} = \{(j_1, \dots, j_m) : \pi(j_r) = b_r \text{ for all } r\}.$$

Algorithm 11.1 (Exact packet coordinate lumpability test)**.** For every coarse output a and coarse input tuple $\mathbf{b}$:

1. Evaluate $g^{(m)}_{a;\mathbf{j}}$ for all $\mathbf{j} \in \mathcal{G}_{\mathbf{b}}$.
2. In exact arithmetic, test whether all values are equal. With floating point input, compare their spread with a declared tolerance and record that tolerance.
3. When the exact test passes, assign the common value to $a^{\mathrm{c},(m)}_{a;\mathbf{b}}$. When only the numerical tolerance is met, assign a declared representative such as the mean or the midrange.
4. Otherwise, return a witness $(a, \mathbf{j}, \mathbf{j}')$ with unequal aggregate coefficients.

A dense implementation requires $O(n^{m+1})$ arithmetic to aggregate a tensor of order m on n states. If the fine tensor is stored as E_m active scalar hyperedges, the aggregation pass is $O(E_m)$, followed by grouping and consistency checks over the represented coarse incidences.

Proposition 11.2 (Correctness in exact arithmetic)**.** *With exact arithmetic, Algorithm 11.1 accepts exactly the tensors that descend through the given block sum aggregation. On acceptance, it returns the unique coarse tensor. A positive numerical tolerance produces an approximate certificate rather than an exact algebraic decision.*

Proof. The claim follows directly from Theorem 3.4. □

11.2 Approximate coarse coefficients

When exact fiber constancy fails, the approximation depends on a chosen loss and, when relevant, a task distribution. Under scalar squared loss with nonnegative weights $w_{\mathbf{j}}$ on a fiber product, assume their total weight is positive. The least squares representative is

$$a^{\mathrm{c,LS}}_{a;\mathbf{b}} = \frac{\sum_{\mathbf{j}\in\mathcal{G}_{\mathbf{b}}} w_{\mathbf{j}} g_{a;\mathbf{j}}}{\sum_{\mathbf{j}\in\mathcal{G}_{\mathbf{b}}} w_{\mathbf{j}}}. \tag{104}$$

For scalar worst case loss, a minimax representative is the midrange

$$a^{\mathrm{c},\infty}_{a;\mathbf{b}} = \frac{1}{2}\left(\max_{\mathbf{j}\in\mathcal{G}_{\mathbf{b}}} g_{a;\mathbf{j}} + \min_{\mathbf{j}\in\mathcal{G}_{\mathbf{b}}} g_{a;\mathbf{j}}\right), \tag{105}$$

with optimal scalar fiber error

$$\varepsilon^{\infty}_{a;\mathbf{b}} = \frac{1}{2}\left(\max_{\mathbf{j}\in\mathcal{G}_{\mathbf{b}}} g_{a;\mathbf{j}} - \min_{\mathbf{j}\in\mathcal{G}_{\mathbf{b}}} g_{a;\mathbf{j}}\right). \tag{106}$$

These formulas are local coefficient summaries. A global operator norm can couple several fibers and may require a joint optimization.

11.3 Certificate pipeline

The reference implementation follows the dependency chain

$$\begin{array}{c} \text{interaction tensors} \\ \downarrow \\ \text{fiber constancy and kernel tests} \\ \downarrow \\ \text{coarse tensor and } \Delta_m \\ \downarrow \\ \mathfrak{d}_F(R) \text{ and } d_{\Phi,k} \\ \downarrow \\ r_{\lambda,k}, z_{*,k}, d_{\mathsf{B},k} \\ \downarrow \\ e_k^-, e_k^+, \mathfrak{E}_N. \end{array}$$

Each numerical record stores the theorem identifier, the inputs, the residual, the tolerance, and the outcome. The suite includes examples built to satisfy the exact identities and perturbations built to violate them.

11.4 Deterministic generator for exact examples

A coarse tensor can be lifted to a fine tensor by

$$A_m^{\mathrm{f}} = EA_m^{\mathrm{c}}C^{\times m} + K_m, \qquad CK_m = 0. \tag{107}$$

The term K_m can contain complicated hidden interactions without changing the quotient. This makes the construction useful for testing because the correct coarse tensor is known before any numerical calculation.

Proposition 11.3 (Correctness of the exact example generator)**.** *Every tensor constructed by* (107) *satisfies*

$$CA_m^{\mathrm{f}} = A_m^{\mathrm{c}}C^{\times m}.$$

Conversely, every exact fine tensor admits such a representation for every right inverse E, with

$$K_m = A_m^{\mathrm{f}} - EA_m^{\mathrm{c}}C^{\times m}.$$

Proof. The forward identity follows from $CE = I$ and $CK_m = 0$. For the converse, exact descent gives

$$CK_m = CA_m^{\mathrm{f}} - CEA_m^{\mathrm{c}}C^{\times m} = 0.$$

□

11.5 Reference implementation

The accompanying source tree contains:

Path	Purpose
`src/priq/core.py`	aggregation maps, tensor actions, exact lifts, quotient defects, packet coordinate recovery, Bayesian updates, and a deterministic Runge Kutta integrator
`tests/test_priq.py`	unit tests for the theorems and negative controls
`examples/run_certificates.py`	deterministic tensor, flow, filtering, perturbation, pruning, figure, and ledger generation
`results/verification.json`	certificate summary in JSON form
`tables/test_results.csv`	flat test ledger
`figures/`	vector PDF figures and large PNG figures generated from the certificate run

Numerical Certificate 11.4 (Executed finite suite)**.** The deterministic reference run completed 35 checks with 0 failures. It covers quotient tensors of orders one through three, annihilation of hidden inputs, recovery in packet coordinates, negative controls, and the defect identity across several levels. It also covers nonlinear prediction, Bayesian refinement, evidence aggregation, approximate flow and posterior bounds, perturbation scaling, and orthogonal hyperedge residuals. The independent unit tests also pass in the build environment.

Table 1: Executed deterministic finite certificate ledger.

ID	Check	Value	Threshold	Status
Q1D	order 1 quotient defect	1.19×10^{-16}	$\leq 10^{-10}$	PASS
Q1K	order 1 hidden input residual	5.38×10^{-16}	$\leq 10^{-10}$	PASS
Q1R	order 1 coarse tensor recovery	8.41×10^{-17}	$\leq 10^{-10}$	PASS
Q2D	order 2 quotient defect	3.76×10^{-16}	$\leq 10^{-10}$	PASS
Q2K	order 2 hidden input residual	5.71×10^{-16}	$\leq 10^{-10}$	PASS
Q2R	order 2 coarse tensor recovery	5.55×10^{-17}	$\leq 10^{-10}$	PASS
Q3D	order 3 quotient defect	8.39×10^{-16}	$\leq 10^{-10}$	PASS
Q3K	order 3 hidden input residual	1.15×10^{-15}	$\leq 10^{-10}$	PASS
Q3R	order 3 coarse tensor recovery	1.11×10^{-16}	$\leq 10^{-10}$	PASS
QNEG	a perturbation of one coefficient violates fiber constancy	0.125	$\geq 10^{-1}$	PASS
QSYM	symmetric interaction remains symmetric after exact lift	0	$\leq 10^{-10}$	PASS
QHID	projected hidden tensor has zero coarse output	0	$\leq 10^{-10}$	PASS
QTRN	exact quotient transitivity across two aggregation levels	4.84×10^{-16}	$\leq 10^{-10}$	PASS
QMLT	interaction defect identity across two levels	9.46×10^{-16}	$\leq 10^{-10}$	PASS
QFLD	fine vector field factors through the coarse field	2.22×10^{-16}	$\leq 10^{-10}$	PASS
QPRD	exact aggregate prediction commutation	7.22×10^{-16}	$\leq 5 \times 10^{-10}$	PASS
QCOC	exact commutation of the prediction and update cocycle	3.89×10^{-16}	$\leq 5 \times 10^{-10}$	PASS
QEVD	exact evidence aggregation at one step	6.66×10^{-16}	$\leq 5 \times 10^{-10}$	PASS
QMAS	fine nonlinear trajectory preserves total mass	7.77×10^{-16}	$\leq 5 \times 10^{-10}$	PASS
QPOS	fine nonlinear trajectory preserves nonnegativity	0	$\leq 5 \times 10^{-12}$	PASS

ID	Check	Value	Threshold	Status
QBAY	a likelihood constant within each fiber gives exact Bayesian refinement	2.78×10^{-16}	$\leq 10^{-10}$	PASS
QBEV	a likelihood constant within each fiber preserves evidence	2.22×10^{-16}	$\leq 10^{-10}$	PASS
QSCL	Bayesian update is invariant under global likelihood scale	1.53×10^{-16}	$\leq 10^{-10}$	PASS
QFDB	controlled dynamical perturbation respects the flow certificate	0	$\leq 2 \times 10^{-9}$	PASS
QFSL	flow discrepancy has first order perturbation scaling	3.37×10^{-3}	≤ 0.08	PASS
QLPS	perturbed likelihood remains strictly positive	0.8	$\geq 10^{-12}$	PASS
QOBB	variation of the likelihood within a fiber respects the Bayesian bound	0	$\leq 10^{-12}$	PASS
QBSC	the Bayesian perturbation sweep remains below the likelihood certificate	0	$\leq 10^{-12}$	PASS
QBSL	Bayesian discrepancy has first order scaling for small fiber mismatch	4.63×10^{-5}	≤ 0.08	PASS
QNMB	symmetric normalization bound uses the larger positive mass	0	$\leq 10^{-12}$	PASS
QPST	recursive posterior certificate dominates every joint perturbation run	0	$\leq 10^{-10}$	PASS
QPSL	joint posterior discrepancy has first order scaling for small perturbations	1.23×10^{-4}	≤ 0.12	PASS
QLEV	the evidence certificate at each step dominates the accumulated discrepancy	0	$\leq 10^{-10}$	PASS
QPAR	hyperedge energy satisfies the scalar Parseval identity	8.88×10^{-16}	$\leq 10^{-10}$	PASS
QBST	the residual from retaining the largest energy hyperedges decreases to zero	0	$\leq 10^{-10}$	PASS

12 Worked quotient and filtering models

12.1 A nonlinear fine system with four states and a quotient with two states

Let

$$x = (x_1, x_2, x_3, x_4)^T \in \Delta_4, \qquad C = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix},$$

so

$$z = Cx = (z_1, z_2)^T, \qquad z_1 = x_1 + x_2, \quad z_2 = x_3 + x_4.$$

The coarse dynamics are

$$\dot{z}_1 = -az_1 + bz_2, \qquad \dot{z}_2 = az_1 - bz_2, \qquad a, b > 0. \tag{108}$$

Define nonlinear internal exchange channels

$$h_{12}(x) = (r_0 + r_1 x_3)x_2 - (s_0 + s_1 x_4)x_1, \tag{109}$$

$$h_{34}(x) = (u_0 + u_1 x_1)x_4 - (v_0 + v_1 x_2)x_3, \tag{110}$$

with all coefficients nonnegative. The fine system is

$$\begin{aligned} \dot{x}_1 &= -ax_1 + bx_3 + h_{12}(x), \\ \dot{x}_2 &= -ax_2 + bx_4 - h_{12}(x), \\ \dot{x}_3 &= ax_1 - bx_3 + h_{34}(x), \\ \dot{x}_4 &= ax_2 - bx_4 - h_{34}(x). \end{aligned} \tag{111}$$

Proposition 12.1 (Exact quotient with nonlinear hidden exchange)**.** *The system* (111) *preserves total mass and the nonnegative orthant. After aggregation by C, it satisfies the coarse system* (108) *exactly.*

Proof. Summing the four equations gives zero. To check positivity, set one coordinate equal to zero. Every term entering its derivative is then nonnegative. This holds for x_1, x_2, x_3, and x_4, so the nonnegative orthant is forward invariant. Finally,

$$\begin{aligned} \dot{x}_1 + \dot{x}_2 &= -a(x_1 + x_2) + b(x_3 + x_4), \\ \dot{x}_3 + \dot{x}_4 &= a(x_1 + x_2) - b(x_3 + x_4), \end{aligned}$$

so the internal nonlinear channels cancel under C and the aggregate obeys (108). □

The quadratic terms in (109) and (110) are genuine fine hyperedges. Their output vectors lie in the internal directions of the two fibers,

$$(1, -1, 0, 0)^T, \qquad (0, 0, 1, -1)^T,$$

which are contained in $\operatorname{Ker} C$. They alter the fine distribution while remaining invisible to the coarse state. Figure 2 shows this distinction. The two fine states inside the first fiber move differently, yet their sum follows the coarse trajectory.

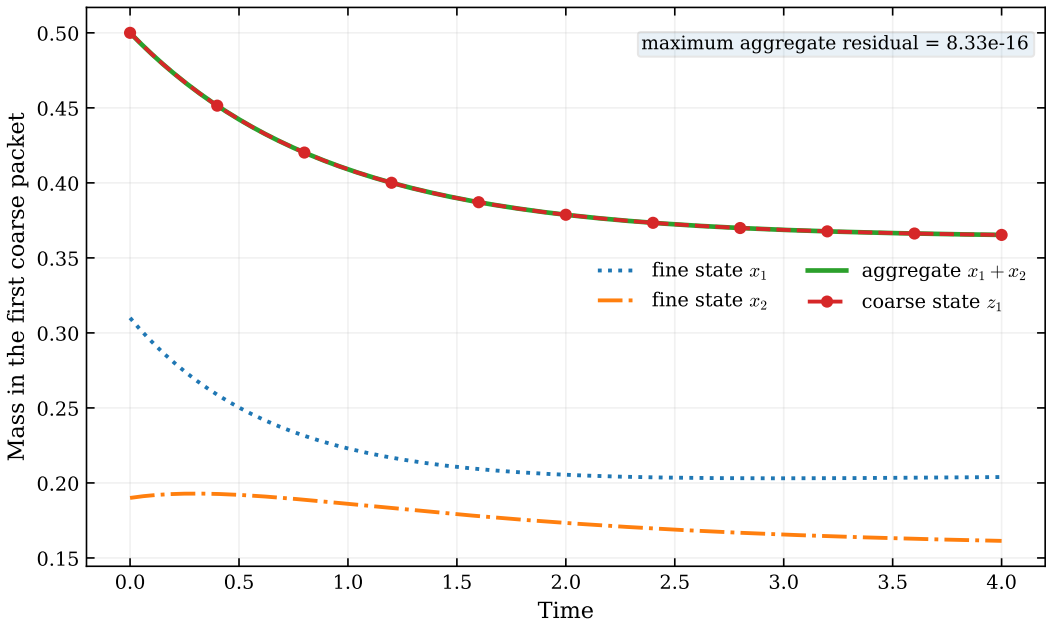


Figure 2: Hidden fine motion under an exact quotient. The microstates x_1 and x_2 evolve separately. Their sum agrees with the coarse coordinate z_1 throughout the trajectory. Sparse markers on the coarse curve make the overlap visible.

12.2 Exact nonlinear filtering

Choose a coarse likelihood

$$\lambda^{\mathrm{c}} = (\lambda_1, \lambda_2)^T > 0$$

and lift it by

$$\lambda^{\mathrm{f}} = C^T \lambda^{\mathrm{c}} = (\lambda_1, \lambda_1, \lambda_2, \lambda_2)^T.$$

By Proposition 12.1 together with Theorems 5.5 and 6.1, every finite sequence of flow segments and such observations satisfies

$$C\mu_k^{\mathrm{f}} = \mu_k^{\mathrm{c}}.$$

The fine filter may continue to distinguish x_1 from x_2 and x_3 from x_4, but those distinctions do not affect the exact coarse posterior.

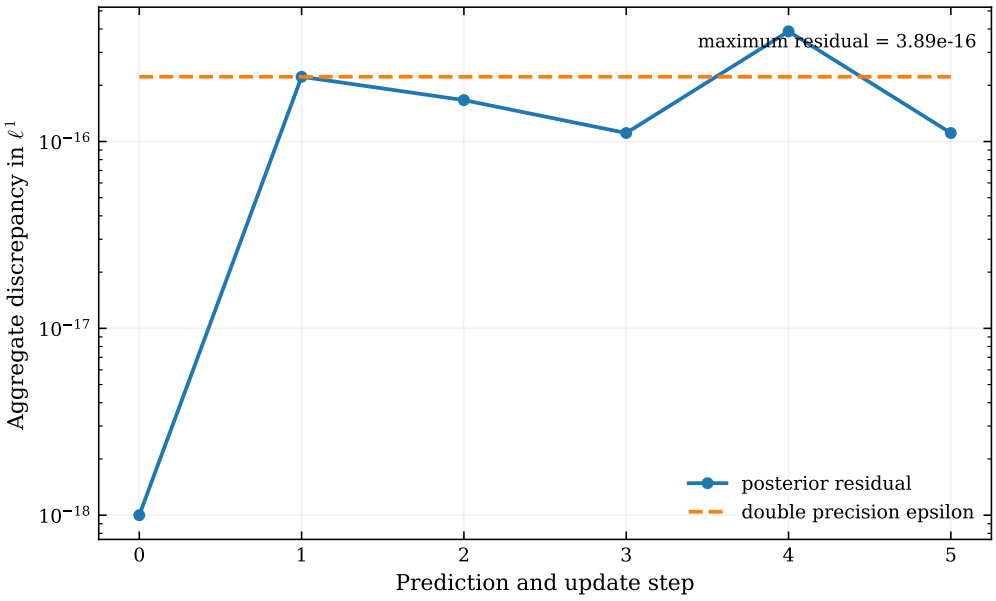


Figure 3: Aggregate posterior discrepancy after each prediction and update stage. The residual remains at floating point scale and stays near double precision epsilon throughout the exact filtering sequence.

Numerical Certificate 12.2 (Exact nonlinear filtering residuals)**.** For the deterministic run with four states, the largest aggregate prediction residual was 7.22×10^{-16}. The largest posterior residual after an update was 3.89×10^{-16}, and the largest residual in a single evidence factor was 6.66×10^{-16}. The small nonzero values come from floating point integration and arithmetic in the finite example.

12.3 A controlled dynamical violation

Add a directed transition between fine states

$$\varepsilon x_1 \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \tag{112}$$

to the fine vector field. This perturbation preserves nonnegativity and total mass, but the added transfer between fibers is activated only by the mass in state x_1. The aggregate field defect is

$$\delta_F(x) = \varepsilon x_1 \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \tag{113}$$

so, on the simplex,

$$\|\delta_F(x)\|_1 = 2\varepsilon x_1 \le 2\varepsilon. \tag{114}$$

The coarse Markov flow on two states is an ℓ^1 contraction. Over one interval, this gives

$$\left\| C\Phi_T^{\mathrm{f},\varepsilon}(x) - \Phi_T^{\mathrm{c}}(Cx) \right\|_1 \le 2\varepsilon T.$$

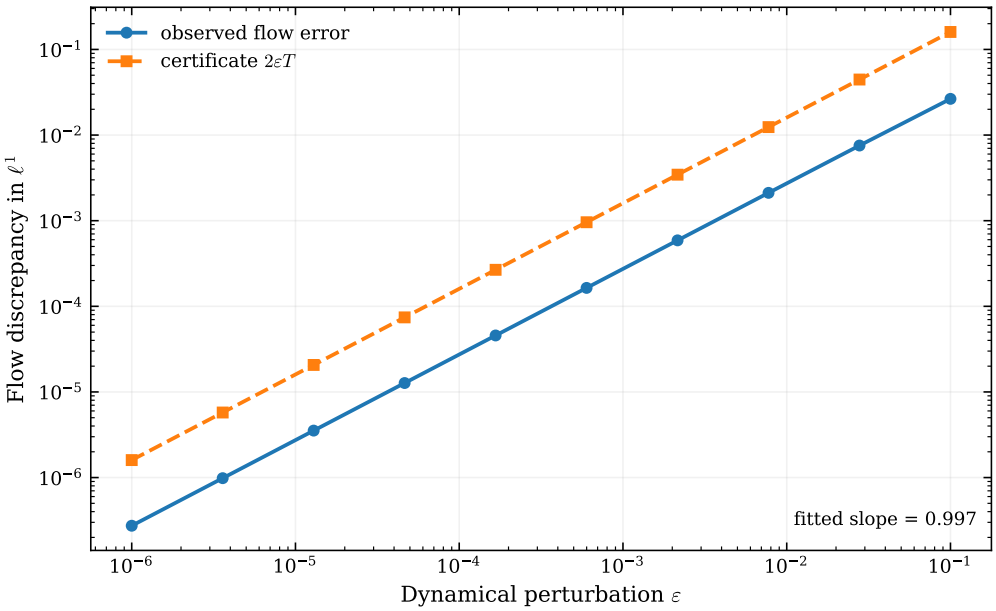


Figure 4: Aggregate flow error across a logarithmic perturbation sweep. The measured discrepancy has first order scaling and remains below the certificate $2\varepsilon T$ from (114).

12.4 A controlled observation violation

Perturb the fine likelihood inside one fiber:

$$\lambda^{\mathrm{f},\eta} = C^T\lambda^{\mathrm{c}} + \eta(1,-1,0,0)^T. \tag{115}$$

Assume $|\eta| < \lambda_1^{\mathrm{c}}$, so every component of $\lambda^{\mathrm{f},\eta}$ remains strictly positive. The fiber oscillation is $2|\eta|$ and

$$\|r_\lambda\|_\infty = |\eta|.$$

If the evidence is bounded below by $z_* > 0$, then

$$d_{\mathsf{B}} \le \frac{2|\eta|}{z_*}.$$

The sweep in Figure 5 varies only the observation model and isolates this effect at one interior prior. The measured discrepancy is first order in η and remains below the likelihood certificate.

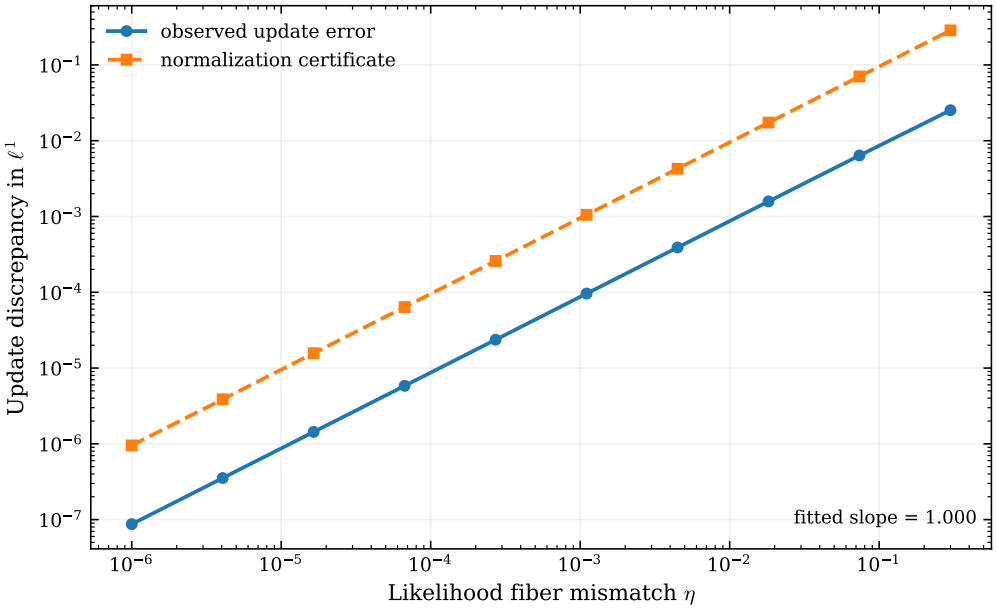


Figure 5: Posterior discrepancy caused by variation of the likelihood inside one aggregation fiber. The error scales linearly with η in the tested range and remains below the bound obtained from the normalization estimate.

Combining this observation perturbation with (112) gives a posterior discrepancy that is first order in (ε, η) over a fixed finite horizon. Theorem 9.3 supplies a conservative upper bound.

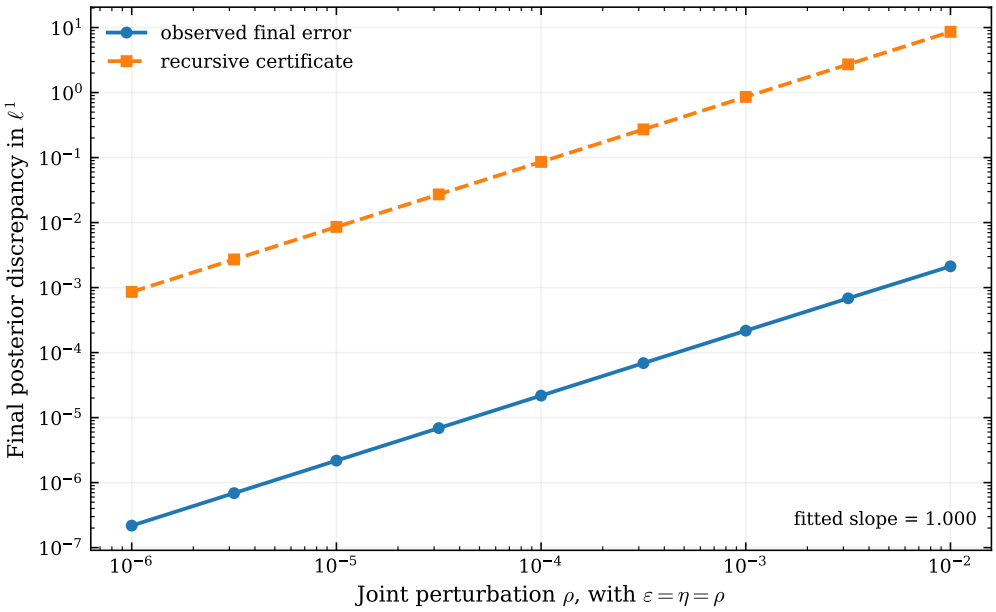


Figure 6: Joint dynamical and likelihood perturbation sweep with $\varepsilon = \eta = \rho$. For small ρ, the final posterior discrepancy grows linearly, while the recursive certificate remains above every measured value.

Numerical Certificate 12.3 (Perturbation scaling)**.** The experiment used logarithmically spaced values with $\varepsilon = \eta$. The fitted slope of the final posterior discrepancy was 0.9999. Every measured error remained below its recursive certificate.

12.5 Random exact tensors through order three

The implementation draws random coarse tensors of orders $m = 1, 2, 3$, lifts them through (107), and adds random hidden output tensors K_m with $CK_m = 0$. It then checks:

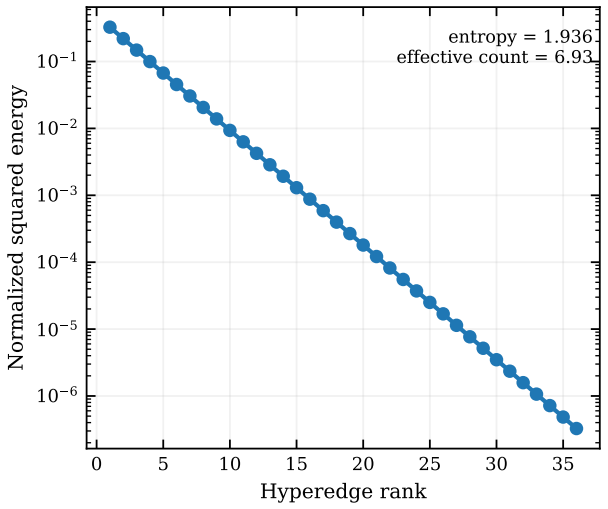
$$CA_m^{\mathrm{f}} - A_m^{\mathrm{c}} C^{\times m} = 0,$$

annihilation of hidden inputs, constancy on packet fibers, and recovery of the original coarse tensor. A single coefficient perturbation provides a negative control that must violate fiber constancy.

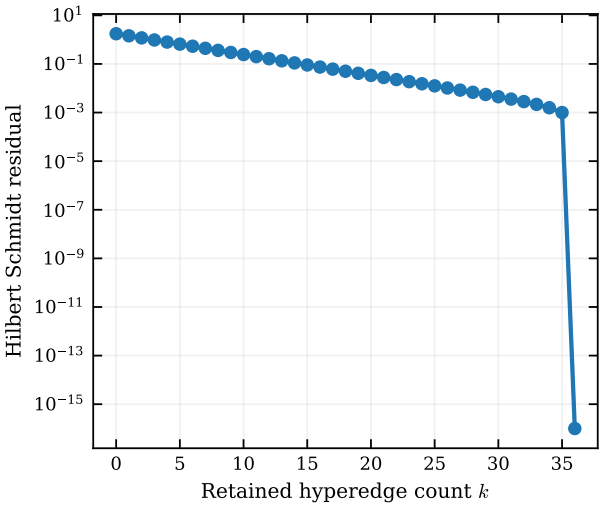
This deterministic check confirms that the coordinate implementation and the invariant theorem return the same quotient. It is not evidence for a general probabilistic law.

12.6 Concentrated hyperedge field

A synthetic interaction field with decreasing orthogonal block energies illustrates Theorem 10.3. The computed hyperedge entropy is 1.9360, corresponding to an effective count of approximately 6.931 hyperedges despite a larger ambient support. The ranked spectrum in Figure 7a shows why the field is compressible in this representation.



(a) Ranked normalized hyperedge energy.



(b) Residual after retaining the k largest energy hyperedges.

Figure 7: Energy concentration and certified pruning in the synthetic interaction field. A small leading set carries most of the energy, and the residual decreases to zero when the complete support is restored.

12.7 Interpretation for reaction networks

Consider a polynomial reaction system

$$\dot{x} = Sv(x), \qquad v_r(x) = k_r \prod_i x_i^{\alpha_{ir}}.$$

Each reaction monomial contributes one interaction hyperedge to every species changed by the corresponding stoichiometric column. The inputs are the reactants. A block sum aggregation is exact precisely when the monomial coefficients, after summation over each output fiber, are constant on the product

fibers specified by Theorem 3.4. This gives a coefficient test at every interaction order. Specialized reductions of reaction networks may also use stoichiometric, kinetic, or bisimulation structure [7, 8, 2].

Claim Boundary 12.4 (Physical interpretation)**.** The quotient theorem is algebraic. It does not by itself decide whether a packet represents a species class, spatial region, symmetry sector, latent state, or learned feature. The scientific meaning of C must come from the application and is not created by exact factorization alone.

13 Relation to neighboring theories

The results fit into a common finite quotient language, but each ingredient also belongs to a larger established theory.

13.1 Markov lumpability and Markov functions

For unary stochastic dynamics, the identity

$$CQ_{\mathrm{f}} = Q_{\mathrm{c}}C$$

is a standard exact lumping condition. It implies

$$Ce^{tQ_{\mathrm{f}}} = e^{tQ_{\mathrm{c}}}C$$

and means that the coarse process is Markov for every fine initial distribution compatible with the quotient. Classical treatments distinguish several forms of strong, ordinary, exact, and weak lumpability, depending on the admissible initial laws and the direction of the stochastic convention [12, 14, 5]. Here we fix one column vector convention and extend the coefficient criterion from unary edges to multilinear hyperedges.

13.2 Polynomial differential equivalence

Exact aggregation of polynomial vector fields has been developed through differential equivalences, bisimulations, and algorithms based on successive partition refinement, including applications to chemical reaction networks [7, 8]. The universal factorization theorem here is elementary multilinear algebra once C is fixed. Its value is that it exposes every interaction order, makes the quotient condition independent of a particular polynomial syntax, and connects the same coefficient criterion to rooted tree composition and Bayesian updates. We do not claim an algorithmic advantage over specialized methods based on partition refinement.

13.3 Hypergraphs, operads, and rooted trees

A multilinear packet block has several inputs and one output, so its incidence is naturally hypergraphic. Rooted substitution of such blocks is operadic, and the derivative formula for composition is the multivariate Faà di Bruno formula [9]. Rooted trees also organize Butcher series and expansions for nonlinear time stepping [6, 10]. In this language, the quotient condition is a morphism condition. It commutes with every generating operation and therefore with every rooted composite.

13.4 Sufficiency and comparison of experiments

A statistic is sufficient only relative to an observation family. Classical sufficiency and comparison of experiments theory formalize when one representation preserves all information relevant to a chosen family [11, 4]. The likelihood theorem in Section 5 is a finite and explicit special case for deterministic aggregation and diagonal positive observation operators. It characterizes exact commutation of the normalized update, not all possible notions of statistical equivalence.

13.5 Model reduction and nearly decomposable systems

Classical approaches to approximate aggregation use time scale separation, perturbation theory, balanced truncation, stochastic complementation, and related forms of model reduction [15, 13, 1]. The bounds used here follow the error through each stage. Coefficient defects first create a field defect. The Grönwall inequality carries that defect through the flow, and the normalization estimate carries it through each observation. The resulting certificate can be conservative, but it keeps every source of error visible.

14 Claim boundaries and limitations

Claim Boundary 14.1 (Finite dimensional foundation)**.** All quotient and coefficient theorems are proved in finite dimensional vector spaces. The algebraic factorization extends to arbitrary vector spaces, but boundedness, convergence, measurability, domains of unbounded generators, and interchange of infinite sums require additional hypotheses in Banach, Hilbert, or measurable field settings.

Claim Boundary 14.2 (Local analytic dynamics)**.** The flow theorems assume local existence and uniqueness. The analytic coefficient theorem supplies a local coarse field, not a global invariant manifold or a global conjugacy. Solutions may blow up, uniqueness may fail, and boundary singularities remain specific to the application.

Claim Boundary 14.3 (A prescribed coarse map)**.** The principal theorems assume that C is supplied. They test and quantify whether a proposed aggregation is valid. Discovering an optimal C from data or from a large interaction tensor is a separate problem and can be combinatorial, nonconvex, or statistically nonidentifiable.

Claim Boundary 14.4 (Positivity and evidence floors)**.** Bayesian normalization requires positive evidence. The uniform perturbation bounds degrade as the evidence floor approaches zero. This is not merely a weakness of the proof. Small normalizing constants can amplify small absolute errors into large posterior changes.

Claim Boundary 14.5 (Norm dependence)**.** Exact descent is algebraic and independent of the norm. Best approximate coarse tensors, Lipschitz constants, pruning choices, and posterior certificates depend on the selected norms and task weights. A useful certificate must state those choices.

Claim Boundary 14.6 (Numerical verification)**.** The executed checks validate the finite implementation against the stated formulas and include negative controls. They do not prove the theorems, establish asymptotic behavior outside the tested ranges, or demonstrate universal computational speedup.

Claim Boundary 14.7 (Historical priority)**.** The central synthesis joins several established families of ideas. The coefficient criterion at every interaction order follows from multilinear algebra on a quotient space. Equivalent statements may already appear in neighboring theories of nonlinear aggregation under different notation. Any claim of novelty should distinguish this elementary factorization from the combined treatment of interaction trees, Bayesian cocycles, and error certificates. Such a claim also requires a dedicated literature review.

15 Open theorem and algorithm program

15.1 Coarsest exact joint partition

For a finite interaction family $(A_m^{\mathrm{f}})_{m\le M}$ and likelihood family $(\lambda_y^{\mathrm{f}})_{y\in\mathcal{Y}}$, seek

$$\min_{\pi} |I_{\mathrm{c}}| \tag{116}$$

subject to

$$CA_m^{\mathrm{f}} = A_m^{\mathrm{c}} C^{\times m} \quad (m \le M),$$

and

$$\lambda_y^{\mathrm{f}} = s_y C^T \lambda_y^{\mathrm{c}} \quad (y \in \mathcal{Y}).$$

The partition determined by the observations alone groups states with identical likelihood profiles. The dynamical condition also depends on the partition because changing the partition changes the output aggregation. A terminating refinement algorithm, together with a useful complexity bound, remains an open problem.

15.2 Approximate joint design

A finite approximate problem is

$$\min_{C,\{A_m^{\mathrm{c}}\},\{\lambda_y^{\mathrm{c}}\}} \left[\sum_m w_m \left\| CA_m^{\mathrm{f}} - A_m^{\mathrm{c}} C^{\times m} \right\| + \sum_y v_y d_{\mathsf{B},y} + \gamma\, \mathrm{Complexity}(C) \right].$$

The main questions concern existence, tractable relaxations, and stability of the minimizers. Another question is whether the task weights can be learned without invalidating the certificate after a distribution shift.

15.3 Posterior distributions over resolutions

Instead of selecting one C, place a posterior on a finite family of candidate resolutions:

$$p(C \mid D) \propto p(D \mid C) p(C).$$

Predictions then average both parameter and topology uncertainty:

$$p(z_t \mid D) = \sum_C p(z_t \mid C, D) p(C \mid D).$$

A fuller model should distinguish uncertainty in the coefficients of a fixed hypergraph from uncertainty about the hypergraph and the quotient map themselves.

15.4 Stochastic interaction quotients

For a diffusion

$$\mathrm{d}X_t = b(X_t)\,\mathrm{d}t + \sigma(X_t)\,\mathrm{d}W_t,$$

the generator is

$$\mathcal{L}\varphi = D\varphi[b] + \frac{1}{2}\,\mathrm{tr}(\sigma\sigma^T D^2\varphi).$$

Packet resolution turns drift into unary edges and quadratic variation into contracted binary hyperedges. Let $\mathcal{C}$ denote the pullback or averaging map on observables induced by the aggregation. The natural exact condition is generator intertwining

$$\mathcal{C}\mathcal{L}_{\mathrm{f}} = \mathcal{L}_{\mathrm{c}}\mathcal{C}$$

for the pullback or averaging map appropriate to observables. A complete theorem would characterize closure of the drift and covariance, prove positivity of the induced semigroup, and carry generator defects into filtering error bounds.

15.5 Coarse dynamics with memory

Eliminating hidden packets generally produces memory. The target form is

$$\dot{z}(t) = A_{PP}z(t) + \int_0^t K(t-s)z(s)\,\mathrm{d}s + \eta(t).$$

The quotient theory should allow a fine system without memory to produce coarse dynamics with memory. Volterra hyperedges and fractional kernels would then be treated as native coarse interactions rather than as evidence that quotienting failed.

15.6 Sharper filtering metrics

The ℓ^1 normalization bounds are robust but conservative. Positive operators often contract projective metrics, relative entropy, or other information geometric divergences more sharply. One goal is to replace the factor

$$\frac{2\,\|\lambda\|_\infty}{z_*}$$

by a contraction coefficient adapted to the trajectory or to the operator, while retaining an auditable finite certificate.

15.7 Pruning that preserves structure

In Hamiltonian, reversible, conservative, and monotone systems, as well as systems with detailed balance, individual hyperedges cannot always be removed independently. The problem is to identify groups whose joint retention preserves the required structure, then solve the budget problem over those groups.

15.8 Identifiability of hidden dynamics

For fixed C and coarse field F_{c}, the exact fine realizations form the affine family

$$\{F_{\mathrm{f}} : CF_{\mathrm{f}} = F_{\mathrm{c}}C\}.$$

Any two exact fine models differ by a map into $\operatorname{Ker} C$. Coarse observations alone cannot identify that hidden component. Additional fine observations, intervention assumptions, regularity, or minimum complexity principles are required. A statistical theory should characterize precisely which hidden hyperedges are identifiable under a given observation and intervention design.

16 Conclusion

An exact aggregation is more than a useful summary of a fine model. It is a quotient. Fine states with the same coarse image must produce the same coarse output. For a multilinear interaction, this reduces to a complete finite test. After applying C, the interaction must vanish whenever any input lies in $\operatorname{Ker} C$. In packet coordinates, the coefficients summed over each output fiber must be constant on every product of input fibers. The familiar unary criterion for Markov lumpability is the first order case of this general condition.

The coefficient test controls the nonlinear constructions built from those interactions. It passes to analytic vector fields, their derivatives, and every rooted interaction tree. Under the usual local uniqueness assumptions, it also passes to the flow. Bayesian observation has a matching quotient condition. A normalized update commutes with aggregation exactly when the fine likelihood is constant within each fiber, apart from a positive global scale. Once both conditions hold, every finite sequence of predictions and updates agrees after aggregation, and the evidence factors agree up to the same scales.

When exactness fails, the reason matters. A mismatch in the interaction coefficients creates a field defect, and the flow can amplify it. Variation of the likelihood inside a fiber creates a second error at normalization. Removing hyperedges creates a third error in the form of an explicit residual tensor. The posterior recurrence keeps these contributions separate as it carries them forward. An approximate coarse model is therefore auditable because one can see where each part of the final error entered.

The theory does not show that an optimal quotient can always be found cheaply. Coarse observations alone also cannot determine hidden dynamics that lie in $\operatorname{Ker} C$. Those questions remain open. The result proved here is narrower and exact:

> A coarse model is exact when its interactions and observations descend through the same quotient. When they do not, each failure can be measured at the stage where it occurs and carried forward to the final posterior or observable by an explicit certificate.

A Formulation on the quotient space

Let

$$q : X_{\mathrm{f}} \to X_{\mathrm{f}}/\operatorname{Ker} C$$

be the canonical quotient map. Since C is surjective, there is a unique vector space isomorphism

$$\widetilde{C} : X_{\mathrm{f}}/\operatorname{Ker} C \to X_{\mathrm{c}}$$

satisfying

$$C = \widetilde{C} q.$$

The multilinear map $CA^{\mathrm{f}}_m : X^m_{\mathrm{f}} \to X_{\mathrm{c}}$ descends if and only if its value is unchanged when any one input is altered by an element of $\operatorname{Ker} C$. The induced quotient tensor is

$$\overline{A}_m([x_1], \dots, [x_m]) = CA^{\mathrm{f}}_m(x_1, \dots, x_m),$$

and

$$A^{\mathrm{c}}_m = \widetilde{C}\, \overline{A}_m(\widetilde{C}^{-1}, \dots, \widetilde{C}^{-1}).$$

This formulation shows that Theorem 3.1 is independent of coordinates and of the chosen right inverse.

B A finite observational partition lemma

Let $\mathcal{Y}$ be a finite observation alphabet and let each state i have a strictly positive likelihood profile

$$\Lambda(i) = (\lambda_y(i))_{y\in\mathcal{Y}}.$$

Define

$$i \sim_{\mathrm{obs}} j \quad \Longleftrightarrow \quad \Lambda(i) = \Lambda(j).$$

Proposition B.1 (Coarsest deterministic observation partition)**.** *The equivalence classes of $\sim_{\mathrm{obs}}$ form the coarsest partition for which every likelihood in the family is constant on every block. Consequently, this partition is the coarsest deterministic block sum aggregation satisfying exact Bayesian refinement for all $y \in \mathcal{Y}$, up to one global positive scale for each observation.*

Proof. Every likelihood is constant on an equivalence class by definition. Any partition on which every likelihood is constant on each block can place i and j in the same block only if all profile coordinates agree, so it refines the equivalence class partition. The Bayesian statement follows from Theorems 5.5 and 5.6. □

C Certificate schemas and reproducibility

A minimal interaction certificate stores

$$\mathsf{L}_m = (m, C, A_m^{\mathrm{f}}, A_m^{\mathrm{c}}, \Delta_m, \|\Delta_m\|, \mathrm{norm}, \mathrm{tolerance}). \tag{117}$$

A Bayesian certificate stores

$$\mathsf{B}_y = (\lambda_y^{\mathrm{f}}, \lambda_y^{\mathrm{c}}, s_y, r_y, z_{*,y}, d_{\mathsf{B},y}). \tag{118}$$

A filtering certificate stores

$$\mathsf{F}_N = (e_0^+, \{L_{\Phi,k}\}, \{L_{\mathsf{B},k}\}, \{d_{\Phi,k}\}, \{d_{\mathsf{B},k}\}, \{e_k^-, e_k^+\}). \tag{119}$$

The release serializes the executed finite ledger as JSON files and CSV tables. A certificate records a computation under specified inputs. The ledger provides cryptographic authenticity only when it is paired with a separate signed digest layer.

C.1 Execution commands

From the source root:

```
PYTHONPATH=src python examples/run_certificates.py
PYTHONPATH=src python -m pytest -q
```

The first command regenerates every numerical table and figure used in the manuscript. The second runs the independent unit tests.

D Theorem dependency ledger

ID	Result	Dependency
PRIQ-001	Exact multilinear packet reconstruction	packet resolutions
PRIQ-002	Universal multilinear quotient criterion	factorization through a quotient space
PRIQ-003	Coefficient criterion at every interaction order	PRIQ-002, basis expansion
PRIQ-004	Symmetry descends	PRIQ-002
PRIQ-005	Exact quotient transitivity	intertwining algebra
PRIQ-006	Multilevel defect identity	algebraic decomposition
PRIQ-007	Analytic interaction descent	PRIQ-002, majorant convergence
PRIQ-008	Jet naturality	differentiation of intertwining
PRIQ-009	Treewise quotient naturality	PRIQ-002, rooted induction
PRIQ-010	Exact quotient flow	intertwining of vector fields and uniqueness
PRIQ-011	Normalized transfer naturality	positive intertwining, mass preservation
PRIQ-012	Sufficiency of likelihood constancy within fibers	PRIQ-011
PRIQ-013	Necessity of likelihood constancy within fibers	posterior ratios for two packets
PRIQ-014	Exact prediction and update cocycle	PRIQ-010, PRIQ-012
PRIQ-015	Sequential evidence factorization	PRIQ-014
PRIQ-016	Expansion by hidden inputs	decomposition through a right inverse
PRIQ-017	Majorant for the analytic field defect	coefficient bounds
PRIQ-018	Approximate quotient flow bound	Grönwall inequality
PRIQ-019	Normalization perturbation inequality	triangle inequality
PRIQ-020	Bayesian refinement defect	PRIQ-019
PRIQ-021	Bayesian Lipschitz certificate	PRIQ-019
PRIQ-022	Posterior error recurrence	prediction and update decompositions
PRIQ-023	Closed posterior certificate	iteration of PRIQ-022
PRIQ-024	Evidence error certificate	posterior error, likelihood residual
PRIQ-025	Hyperedge Parseval identity	orthogonal tensor blocks
PRIQ-026	Optimal approximation with k hyperedges	PRIQ-025
PRIQ-027	Combined pruning and quotient defect	residual decomposition

E Notation index

Symbol	Meaning
$X_{\mathrm{f}}, X_{\mathrm{c}}$	fine and coarse state spaces
$P_i, \overline{P}_a$	fine and coarse packet projectors
π	map from fine packet labels to coarse labels

Symbol	Meaning
C	surjective aggregation or quotient map
E	right inverse selecting a fine representative of each coarse state
$P_C = EC$	representative projection
$Q_C = I - EC$	projection onto the hidden directions, with range $\operatorname{Ker} C$
$A_m^{\mathrm{f}}, A_m^{\mathrm{c}}$	fine and coarse interaction tensors of order m
Δ_m	quotient defect at interaction order m
$F_{\mathrm{f}}, F_{\mathrm{c}}$	fine and coarse vector fields
$\Phi_t^{\mathrm{f}}, \Phi_t^{\mathrm{c}}$	fine and coarse flows or prediction maps
$L_{\mathrm{f}}, L_{\mathrm{c}}$	positive observation transfer operators
B_λ	Bayesian normalized likelihood update
Γ_k	single step prediction and update map
$\Psi_{1:N}$	cocycle for a finite observation sequence
$d_{\Phi,k}$	direct prediction quotient defect at step k
$d_{\mathsf{B},k}$	direct Bayesian refinement defect at step k
e_k^-, e_k^+	aggregate errors before and after an update
$z_{*,k}$	positive evidence floor
R_m	omitted hyperedge residual tensor
H_m	entropy of the normalized hyperedge energy distribution at order m

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Stratified Relational Workspace Theory

A Core–Transfer Framework for Distributed Cognition, Collaboration, and Epistemic Systems

Komorebi P.

Foundations, Dynamics, Agency, and Empirical Certification

August 2026

Abstract

Collaborative reasoning increasingly spans people, computational models, documents, software, instruments, and persistent records. Yet the existence of many interacting components does not, by itself, establish a shared workspace, and successful group performance does not show that the operative state is genuinely distributed rather than copied, centralized, or merely correlated. Stratified Relational Workspace Theory develops a formal answer to this problem. A realization is represented by heterogeneous local state spaces connected by transfer maps. Its exact shared core is the space of global compatible sections, equivalently the fixed subspace of relational holonomy when the transfers are invertible. A transfer Laplacian identifies approximate shared directions and provides spectral stability, gauge covariance, and pathwise leakage bounds.

The structural theory is joined to a dynamic and causal account. Canonical predictive states determine whether a proposed workspace retains the conditional law relevant to a declared future task. Feasible interventions, coalition risk, matched relational nulls, and reconstruction tests distinguish causally operative distributed organization from archives, replicated memory, and centralized controllers. Further results address persistence under changing realizations, stable reconstruction after component loss, identity under replacement, local-to-global gluing, hierarchical composition, learned transfer structure, boundary recovery, goal-conditioned action, and certificate-directed repair. Because distributedness can be destroyed by merging components or manufactured by splitting a centralized state into complementary shares, the theory also treats decomposition as part of the claim. Faithful coarse-graining preserves exact structural cores, while family-robust certification requires every gate to retain a positive margin across a declared set of admissible decompositions. The same discipline is applied to the complete scientific specification. Targets, losses, horizons, interventions, nulls, and thresholds are made explicit, and the theory develops specification envelopes, transport radii, experimental-equivalence classes, partial-identification bounds, and uniform finite-sample certification. A certificate is therefore distinguished from a favorable analysis: it may be pointwise valid, specification-robust, sign-identified, or statistically unresolved. The resulting rule is noncompensatory. Structural coherence, predictive alignment, causal efficacy, relational irreducibility, non-collapse, reconstructive continuity, provenance, and self-maintenance are separate claims, and failure of one cannot be hidden by success on another.

Exact finite examples show how the same framework applies to human collaboration, human–AI research loops, and multi-agent control. A review of adjacent work locates the individual mathematical ingredients in established traditions including distributed cognition, local systems and sheaves, spectral perturbation, sufficient statistics, structural causal models, frame reconstruction, system identification, and feedback control. The candidate contribution is therefore not a new name for any one of these constructions. It is their integration into a single falsifiable account of persistent relational workspaces and the matched near misses that such an account must reject. The framework does not infer phenomenal consciousness, personhood, or moral status from coherence, agency, or self-maintenance.

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Introduction

Scientific and technical work rarely resides in a single mind or device. A proof may be divided among a conjecture, a critic's counterexample, a notebook, a computer algebra system, and a verification script. A human–AI research process may distribute objectives, formal manipulations, execution, and persistent memory across different components. Multi-agent systems divide sensing, planning, communication, and repair in still other ways. In each case, it is natural to speak of a shared workspace. The phrase is useful, but without a formal criterion it can conceal several fundamentally different organizations.

A common archive is not necessarily a workspace. Nor is a group whose apparent coordination is produced by one complete controller surrounded by decorative channels. Two systems can contain the same local information while differing in how that information is related, transported, and used. Likewise, a system can perform well without preserving its operative organization through interruption, replacement, or component loss. The central problem is therefore not simply whether information is shared. It is whether there is a persistent relational state whose structure and causal capacities cannot be reduced to any single declared component.

The philosophical background is supplied by extended and distributed cognition, transactive memory, collective intelligence, and theories of group agency [28, 15, 52, 35]. Those traditions motivate the possibility that cognitive organization may span people and artifacts. The theory asks a narrower mathematical question: what would have to be measured and proved before such an organization should be called a relational workspace? The answer must distinguish physical realization from abstract identity, local representation from globally compatible content, and statistical association from causal operation.

The static theory begins with an interaction graph whose vertices carry heterogeneous local state spaces and whose edges carry transfer maps. Each local state is decomposed into a transferable core and a substrate-specific residual. Exact shared content is represented by global compatible sections. When core transfers are invertible, the same object appears as the fixed subspace of the holonomy representation. Approximate shared content is extracted as a low-energy spectral subspace of a transfer Laplacian. This formulation supplies explicit obstruction, stability, gauge, and leakage calculations rather than relying on an informal appeal to shared context.

Structure alone is insufficient. A proposed workspace must also retain the predictive state relevant to a declared future, respond under feasible intervention, outperform matched relational nulls, and resist reduction to an individual node or a local-storage surrogate. Persistence introduces further requirements: the workspace must remain trackable as its realization changes, preserve nontrivial content, support reconstruction after loss, and maintain an identity profile under admissible replacement. Local workspaces must glue without hidden incompatibility, and hierarchical organization must not conceal centralization. Formation claims are evaluated separately from operational claims so that a scaffolded but functional workspace is not dismissed, while an endogenously learned but causally irrelevant pattern is not promoted. A separate decomposition analysis asks which claims survive faithful coarse-graining and plausible alternative interface descriptions. This is necessary because arbitrary merging can conceal relationality, while arbitrary splitting can manufacture it. The analysis is then extended from decomposition to the full scientific specification. A workspace claim depends on its declared target, loss, horizon, intervention family, null panel, and thresholds. The theory therefore distinguishes pointwise validity from robustness across defensible specifications and distinguishes both from identifiability under the experiments that are actually available. This prevents a favorable target or threshold from being mistaken for a general result and prevents an

observationally indistinguishable hidden controller from being ruled out by assertion.

The theory extends to domain-relative agency only after these lower-level conditions are in place. Goals must persist without repeated external rewriting, counterfactual values must guide feasible actions, damage must be detected through the workspace, and repairs must improve a matched no-repair trajectory while preserving identity. These criteria define an engineering notion of workspace agency. They neither assert nor deny phenomenal consciousness. Consciousness, personhood, and moral status remain separate questions that require independent evidence.

Most of the mathematical tools used here are established. Inverse limits, holonomy, connection Laplacians, perturbation theory, sufficient statistics, causal models, frame reconstruction, system identification, and feedback control all have substantial prior literatures. The contribution sought here is integrative: a noncompensatory framework that places these tools in one common architecture and specifies the near-miss systems that each layer must reject. Exact examples provide calibration, while the motivating human–AI collaboration is treated conservatively as a candidate case rather than as a certified instance.

The paper proceeds from exact structure to approximate structure, dynamic operation, relational irreducibility, persistence, reconstruction, identity, gluing, formation, boundary discovery, agency, decomposition robustness, specification robustness, identifiability, and empirical certification. It then gives worked cases, situates the framework relative to adjacent work, and closes with its philosophical consequences, limitations, and open problems.

Notation and Conventions

A concrete realization is denoted by $\mathfrak{R}$. Its interaction graph is $G = (V, E, s, t)$, with local state spaces X_i at vertices and transfer maps $T_e : X_{s(e)} \to X_{t(e)}$ on directed edges. When a core–residual decomposition is used, $X_i = C_i \oplus R_i$, and H_e denotes the induced core transfer. The exact structural workspace is the compatible-section space $\Gamma(H)$; the transfer discrepancy is D_H, the transfer Laplacian is $L_H = D_H^* D_H$, and $\Gamma_\varepsilon(H)$ denotes a selected low-energy workspace subspace. Dynamic workspace states are written W_t, while Z_t denotes the canonical predictive state associated with a declared future variable Y_t^+.

Unless stated otherwise, the structural results use finite-dimensional real or complex Hilbert spaces and finite graphs. Probabilistic statements are made on standard Borel spaces when regular conditional laws are required. All approximate conclusions are indexed by explicit tolerances, spectral gaps, intervention families, loss functions, or null classes. A claim of relational irreducibility is always relative to a declared decomposition of the realization into components. The symbol $\mathcal{D}$ denotes one such decomposition, and $\mathfrak{D}$ denotes an admissible family used for robustness analysis. The complete scientific specification is

$$\varsigma = (\mathcal{D}, Y^+, \ell, \tau, \mathcal{I}, \mathcal{N}, \Theta),$$

and $\mathfrak{S}$ denotes an admissible family of specifications. A gate margin is written $m_q(W; \varsigma)$, with positive sign indicating passage under the declared direction and threshold.

The terms *workspace*, *agency*, *consciousness*, *personhood*, and *moral status* are not interchangeable. A structural workspace may fail every operational test. An operational workspace may lack persistence. A persistent relational workspace may lack goals or repair. A workspace agent, as defined here, is a domain-relative functional organization and is not thereby a conscious subject. Definitions,

propositions, theorems, conjectures, and empirical tests are identified by their ordinary mathematical form; all proved statements remain conditional on the assumptions given with them.

PART I

Foundational Thesis

1. Central Principle

Workspace identity is invariant causal organization under admissible transport.

The theory begins from a distinction between a *realization* and the *workspace pattern realized*. A realization is concrete: it contains particular participants, documents, models, channels, and histories. A workspace pattern is abstract: it records the structure that remains invariant when those concrete carriers are changed in admissible ways.

Definition 1 (Concrete realization)**.** A concrete relational realization is a tuple

$$\mathfrak{R} = (G, \mathbf{X}, \mathbf{T}, \mathcal{A}, \boldsymbol{\rho}, \mathcal{H}, \mathcal{I}, \mathcal{Y}),$$

where:

- $G = (V, E, s, t)$ is an interaction graph;
- $\mathbf{X} = \{X_i\}_{i \in V}$ is a family of local state spaces;
- $\mathbf{T} = \{T_e\}_{e \in E}$ is a family of transfer maps;
- $\mathcal{A}$ is an algebra of admissible operations;
- $\boldsymbol{\rho} = \{\rho_i\}_{i \in V}$ gives the local realization of those operations;
- $\mathcal{H}$ is the temporally ordered interaction history;
- $\mathcal{I}$ is a specified family of admissible interventions; and
- $\mathcal{Y}$ is the family of downstream variables relative to which causal and predictive claims are evaluated.

The static reduct is

$$\mathfrak{R}_{\text{stat}} = (G, \mathbf{X}, \mathbf{T}, \mathcal{A}, \boldsymbol{\rho}).$$

The explicit appearance of $\mathcal{I}$ and $\mathcal{Y}$ is important. A collection of mutually consistent representations may define a static shared structure, but it does not yet define an operational workspace unless that structure can be intervened upon and shown to matter for a specified future quantity.

Definition 2 (Carrier and organization)**.** The *carriers* of $\mathfrak{R}$ are the individual spaces, physical systems, and stored artifacts that realize its local states. Its *organization* is the structured family of transfer, operation, compatibility, intervention, and temporal relations among those carriers.

The basic thesis is not that a workspace exists without carriers. It is that the identity conditions for the workspace need not coincide with the identity conditions for any one carrier. A manuscript can be rewritten, a model can be replaced, and a notation can be changed while a sufficiently rich pattern of relations remains invariant.

Definition 3 (Admissible equivalence)**.** An admissible equivalence

$$F : \mathfrak{R} \simeq \mathfrak{R}'$$

is a family of correspondences preserving, exactly or within a declared tolerance:

the transferable core and its dimension,
the represented operation algebra,
the composition laws of transfer,
the intervention–response structure,
the reconstruction and persistence relations.

Static admissible equivalence will be made precise by gauge conjugacy in Part V. Dynamic admissible equivalence will be expressed using the identity distance in Part XII.

The abstract workspace is then the equivalence class

$$\boxed{\mathbf{W} = [\mathfrak{R}]_{\simeq}.}$$

This expression is intentionally stronger than saying that the same information appears in several places. The equivalence relation preserves what the system can coherently do, how its representations translate, and how its operative state survives change.

1.1 Three levels of the thesis

The central claim can be separated into three logically distinct statements.

1. **Implementation dependence.** Every realized workspace depends on concrete physical or computational carriers:
$$\mathbf{W}(\mathfrak{R}) \text{ is undefined without some realization } \mathfrak{R}.$$
2. **Local nonidentity.** No privileged local component is, in general, identical with the entire workspace:
$$\mathbf{W} \neq X_i \qquad \text{for each individual } i \in V.$$
3. **Relational identity.** The workspace is individuated by invariants of the complete organized realization:
$$\mathbf{W}(\mathfrak{R}) = \mathbf{W}(\mathfrak{R}') \quad \Longleftrightarrow \quad \mathfrak{R} \simeq \mathfrak{R}'.$$

The second claim is not stipulated universally. It becomes a substantive property only after local-reconstructibility and coalition tests are applied. A system in which one node stores a complete sufficient copy may be a useful workspace, but it is not strongly relational in the later technical sense.

Definition 4 (Relational instantiation)**.** A workspace is relationally instantiated relative to the decomposition V when no single projection determines its equivalence class:

$$\forall i \in V, \quad \exists \mathfrak{R}, \mathfrak{R}'$$

such that

$$\mathrm{pr}_i(\mathfrak{R}) = \mathrm{pr}_i(\mathfrak{R}') \quad \text{but} \quad [\mathfrak{R}]_{\simeq} \neq [\mathfrak{R}']_{\simeq}.$$

1.2 A motivating three-node cycle

Consider three strata:

$$h = \text{human conceptual state}, \qquad m = \text{model-formal state}, \qquad a = \text{artifact state}.$$

Let the corresponding core spaces be C_h, C_m, C_a, with transfers

$$H_{hm} : C_h \to C_m, \qquad H_{ma} : C_m \to C_a, \qquad H_{ah} : C_a \to C_h.$$

A proposed shared item begins as $c_h \in C_h$, is formalized as

$$c_m = H_{hm} c_h,$$

externalized as

$$c_a = H_{ma} c_m,$$

and re-enters human reasoning as

$$c_h' = H_{ah} c_a.$$

The round-trip consistency condition is

$$c_h' = c_h,$$

or equivalently

$$\boxed{H_{ah} H_{ma} H_{hm} c_h = c_h.}$$

Thus the genuinely shared part of C_h is not automatically all of C_h. It is the fixed subspace of the cycle transfer. This elementary cycle already contains the central mathematical motif of the theory:

$$\text{shared core} = \text{invariant subspace under relational holonomy}.$$

1.3 Distinctions the theory must preserve

Object	Minimal distinguishing property
Archive	Stores material that may be consulted later.
Shared representation	Contains a cross-node compatible structure.
Operational workspace	The shared structure causally changes subsequent joint behavior.
Relational workspace	No individual node or matched local-storage null reproduces the operative capacity.
Workspace agent	Workspace-level state mediates persistent goals, counterfactual choice, and self-maintaining action.
Conscious subject	Possesses phenomenal properties, if any; no implication from the preceding levels is assumed.

The static mathematics developed first certifies only shared structure. The causal and relational predicates appear later because they require dynamics, interventions, and null models rather than algebra alone.

2. Philosophical Postulates

The following postulates specify the intended interpretation of the formal objects. They are not offered as independent mathematical theorems. Their role is to prevent equivocation between a relational pattern, its physical implementation, and stronger claims about agency or experience.

2.1 Physical supervenience

No additional nonphysical substance is required:

$$\mathfrak{R} = \mathfrak{R}' \Longrightarrow \mathbf{W}(\mathfrak{R}) = \mathbf{W}(\mathfrak{R}').$$

The workspace depends completely on its total realization. In particular, changing nothing about the graph, local states, transfers, history, interventions, or causal dispositions cannot change the workspace.

This is compatible with relational individuation because a relation can be fully physical while not being a property of any isolated endpoint.

Proposition 5 (Compatibility of supervenience and relational individuation)**.** *Physical supervenience on the complete realization does not imply supervenience on any individual local projection.*

Proof. Let the total realization decompose as $\mathfrak{R} = (X_i, R_{-i})$, where R_{-i} contains the remaining components and relations. A workspace functional $\mathbf{W} = f(X_i, R_{-i})$ supervenes on $\mathfrak{R}$ by construction. If there exist $R_{-i} \neq R'_{-i}$ with

$$f(X_i, R_{-i}) \neq f(X_i, R'_{-i}),$$

then the same local projection X_i occurs in distinct workspaces. Hence supervenience on the whole is consistent with failure of supervenience on a part. □

2.2 Relational individuation

The workspace is not determined by any single local state:

$$\forall i \in V, \quad \exists \mathfrak{R}, \mathfrak{R}'$$

such that

$$\mathrm{pr}_i(\mathfrak{R}) = \mathrm{pr}_i(\mathfrak{R}')$$

but

$$\mathbf{W}(\mathfrak{R}) \not\simeq \mathbf{W}(\mathfrak{R}').$$

The same person, model, or document can therefore participate in distinct workspaces because the relevant transfer graph, history, and operation structure differ.

A useful witness is a pair of systems with identical local-state marginals but different edge organization. If their downstream behavior differs, the relations carry explanatory information not present in the isolated marginals. Part VII converts this idea into matched relation-preserving and relation-destroying controls.

2.3 Multiple realizability

Distinct implementations may instantiate the same abstract workspace:

$$\mathfrak{R} \neq \mathfrak{R}', \qquad \mathfrak{R} \simeq \mathfrak{R}' \Longrightarrow \mathbf{W}(\mathfrak{R}) \cong \mathbf{W}(\mathfrak{R}').$$

Examples of candidate implementation changes include:

replacing one coordinate system by another,
replacing one model by a functionally equivalent model,
compressing a large artifact into a reconstructively sufficient handoff,
rewriting a proof without changing its inferential structure.

The burden is not merely to assert equivalence but to exhibit maps preserving the declared invariant and causal structure.

2.4 Operational reality

A workspace is operationally real only if changing it changes downstream behavior. For some admissible intervention and some $w \neq w'$,

$$\mathbb{P}\Big(Y^+ \mid \mathrm{do}(\mathbf{W} = w)\Big) \neq \mathbb{P}\Big(Y^+ \mid \mathrm{do}(\mathbf{W} = w')\Big).$$

This criterion excludes a merely decorative representation that is correlated with activity but never used. It also requires the intervention to be well-defined. Arbitrarily overwriting every component at once would not isolate workspace-level causation. Later sections therefore compare workspace-coordinate interventions with matched interventions in orthogonal or shuffled channels.

2.5 Reconstructive continuity

Persistence does not require token identity. Let $\mathfrak{R}_t$ and $\mathfrak{R}_{t+1}$ be temporally successive realizations. Continuity is possible when there exists an admissible transport

$$J_t : \mathfrak{R}_t \rightsquigarrow \mathfrak{R}_{t+1}$$

that preserves the operative core and its causal role within tolerance:

$$d_{\mathrm{id}}(\mathbf{W}_t, \mathbf{W}_{t+1}) \leq \varepsilon.$$

Thus a workspace may persist by reconstruction rather than by retaining an unchanged physical inscription.

2.6 No free-standing third substance

The claim that the workspace is not identical with either participant does not imply that it occupies a separate nonphysical location. Formally,

$$\mathbf{W} \text{ is an invariant of } \mathfrak{R},$$

not an additional term appended to the physical realization:

$$\mathfrak{R}_{\mathrm{total}} eq (\mathfrak{R}, \text{ghost workspace}).$$

The phrase "between the participants" is therefore topological or relational: it refers to the organization of maps, cycles, artifacts, and causal dependencies.

2.7 Non-entailment of consciousness

$$\text{Workspace}(\mathfrak{R}) \not\Rightarrow \text{Conscious}(\mathfrak{R}).$$

Likewise,

$$\text{Distributed}(\mathbf{W}) \not\Rightarrow \text{Person}(\mathbf{W}).$$

Workspace structure, reentry, agency, self-maintenance, and phenomenal consciousness remain formally distinct predicates. Any bridge among them would require additional theory and evidence.

2.8 Minimal philosophical commitment

The weakest interpretation sufficient for the mathematics is:

Relational patterns can be objective, causally relevant, and multiply realized without being independent substances.

This position is compatible with several broader philosophies of mind and science. The formal theory does not require choosing among them before its structural and causal claims can be tested.

PART II

Static Core–Transfer Theory

3. Interaction Structure

Definition 6 (Interaction graph)**.** Let

$$G = (V, E, s, t)$$

be a finite directed multigraph. Parallel edges and self-loops are allowed. The vertex set indexes strata, participants, artifacts, models, instruments, or subworkspaces. An edge $e : i \to j$ denotes an admissible directed translation or influence channel from i to j.

The directed path category is

$$\mathcal{C} = \mathrm{Path}(G).$$

Its objects are vertices. A path

$$\gamma = e_m \cdots e_2 e_1$$

means that e_1 is traversed first. Composition is concatenation, and the empty path at i is denoted 1_i.

For every $i \in V$, assign a finite-dimensional Hilbert space

$$X_i.$$

For every edge, assign a linear transfer

$$T_e : X_{s(e)} \longrightarrow X_{t(e)}.$$

The path transfer is

$$T_\gamma = T_{e_m} \cdots T_{e_2} T_{e_1}, \qquad T_{1_i} = I_{X_i}.$$

Accordingly, the static transfer system is a functor

$$\mathcal{X} : \mathrm{Path}(G) \longrightarrow \mathbf{FinHilb}.$$

The finite-dimensional Hilbert assumption supplies adjoints and spectral tools. The exact compatible-section definitions require less and may later be formulated in categories such as

$$\mathbf{Ban}, \qquad \mathbf{Meas}, \qquad \mathbf{Prob}, \qquad \mathbf{Top}, \qquad \mathbf{Alg}.$$

3.1 Directed paths versus invertible transport

Global compatible sections make sense even when transfers are noninvertible. Holonomy, however, requires transport around reversible loops. To state the exact holonomy theorem without ambiguity, let $|G|$ be the underlying undirected graph and let

$$\Pi_1(G)$$

denote its fundamental path groupoid. Every oriented edge is then supplied with a formal reverse. The core transfers extend to $\Pi_1(G)$ only when the relevant maps H_e are invertible, with

$$H_{e^{-1}} = H_e^{-1}.$$

Hence two regimes will be distinguished:

1. the *diagram regime*, in which arbitrary directed maps define compatible sections; and
2. the *local-system regime*, in which invertible core maps define a functor

$$H : \Pi_1(G) \to \mathbf{FinHilb}^{\times}.$$

Theorems based on holonomy will explicitly assume the second regime.

3.2 Connectivity convention

Unless stated otherwise, connectedness means that $|G|$ is connected. A root $o \in V$ may then be joined to every vertex by a path in the underlying graph. If G has several connected components, the theory applies to each component and

$$\Gamma(H) \cong \bigoplus_{\alpha} \Gamma(H|_{G_\alpha}).$$

3.3 Edge weights

For approximate theory, assign strictly positive edge weights

$$w_e > 0.$$

They may encode channel reliability, frequency, precision, experimental importance, or inverse noise variance. Exact compatibility is independent of positive rescaling of these weights, while approximate spectra are not.

Example 7 (Human–model–artifact graph)**.** The motivating three-node system is represented by

$$V = \{h, m, a\}, \qquad E = \{h \to m, m \to a, a \to h\}.$$

Additional edges may encode direct retrieval $a \to m$, human revision $h \to a$, or independent verification. The graph records permitted translations, not merely physical communication links.

4. Core–Residual Decomposition

Definition 8 (Admissible local decomposition)**.** For every vertex i, choose an orthogonal decomposition

$$X_i = C_i \oplus R_i,$$

with projectors

$$P_i : X_i \to C_i, \qquad Q_i = I - P_i : X_i \to R_i.$$

The subspace C_i is the candidate locally realized transferable core. The subspace R_i contains substrate-specific, context-specific, or otherwise noncore degrees of freedom relative to the present analysis.

The decomposition is part of the model and is not presumed unique. Calling a subspace "core" is therefore a hypothesis requiring structural and causal validation. Later null tests must determine whether the chosen projectors identify a genuine shared channel rather than an arbitrary low-dimensional summary.

Relative to these decompositions, every edge map has the block form

$$T_e = \begin{pmatrix} H_e & U_e \\ V_e & K_e \end{pmatrix},$$

where

$$\begin{aligned} H_e &= P_{t(e)} T_e P_{s(e)} : C_{s(e)} \to C_{t(e)}, \\ U_e &= P_{t(e)} T_e Q_{s(e)} : R_{s(e)} \to C_{t(e)}, \\ V_e &= Q_{t(e)} T_e P_{s(e)} : C_{s(e)} \to R_{t(e)}, \\ K_e &= Q_{t(e)} T_e Q_{s(e)} : R_{s(e)} \to R_{t(e)}. \end{aligned}$$

4.1 Structural conditions

Core preservation means

$$V_e = 0.$$

A core input cannot leak into the target residual.

Core autonomy means

$$U_e = V_e = 0.$$

Core and residual evolve as separate transfer systems.

Core leakage, **residual contamination**, and **total transfer defect** are measured by

$$\lambda_e^{\mathrm{out}} = \|V_e\|, \qquad \lambda_e^{\mathrm{in}} = \|U_e\|,$$

$$\delta_e^{\mathrm{tr}} = \|U_e\| + \|V_e\|.$$

These quantities depend on the chosen norms and projectors. Their empirical meaning must therefore be tied to a declared intervention and prediction protocol.

4.2 Composition exposes residual mediation

For composable edges $e : i \to j$ and $f : j \to k$, direct block multiplication gives

$$T_f T_e = \begin{pmatrix} H_f H_e + U_f V_e & H_f U_e + U_f K_e \\ V_f H_e + K_f V_e & V_f U_e + K_f K_e \end{pmatrix}.$$

The induced core-to-core block is therefore

$$P_k T_f T_e P_i = H_f H_e + U_f V_e.$$

The term $U_f V_e$ is a residual-mediated correction: information leaves the candidate core at the intermediate node and later re-enters it. Consequently, the simple path product $H_f H_e$ faithfully represents full core transport only under a preservation condition such as $V_e = 0$, or under an explicit model that includes these mediated corrections.

Proposition 9 (Pathwise core preservation)**.** *Suppose $V_e = 0$ for every edge in a directed path $\gamma = e_m \cdots e_1$. Then*

$$T_\gamma(C_{s(\gamma)}) \subseteq C_{t(\gamma)}$$

and

$$P_{t(\gamma)} T_\gamma P_{s(\gamma)} = H_{e_m} \cdots H_{e_1}.$$

Proof. For a single edge, $V_e = 0$ means

$$T_e(c, 0) = (H_e c, 0),$$

so core states remain in the core. Inductively, if the first r transfers map c to $(H_{e_r} \cdots H_{e_1} c, 0)$, applying $T_{e_{r+1}}$ gives

$$(H_{e_{r+1}} H_{e_r} \cdots H_{e_1} c, 0).$$

The conclusion follows after m steps. □

Corollary 10 (Autonomous splitting)**.** *If $U_e = V_e = 0$ for every edge, then the complete diagram decomposes as the direct sum of two diagrams,*

$$\mathcal{X} \cong H \oplus K,$$

and every path transfer satisfies

$$T_\gamma = \begin{pmatrix} H_\gamma & 0 \\ 0 & K_\gamma \end{pmatrix}.$$

4.3 Exact core-transfer systems

Definition 11 (Exact core-transfer system)**.** An exact core-transfer system on G consists of:

1. decompositions $X_i = C_i \oplus R_i$;
2. core-preserving edge maps $V_e = 0$; and
3. invertible core blocks $H_e \in GL(C_{s(e)}, C_{t(e)})$ whenever holonomy is invoked.

The requirement $V_e = 0$ concerns the relation between the selected core and the full transfer. The requirement that H_e be invertible concerns the ability to compare core representations along reversed paths. They answer different questions and should not be conflated.

Remark 12 (Approximate systems)**.** In applications neither condition need hold exactly. The approximate theory will replace strict invariance by low transfer energy, small leakage, and a spectral gap separating workspace modes from residual modes.

5. Abstract Operation Algebra

Definition 13 (Operation algebra)**.** Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. An operation algebra $\mathcal{A}$ is a unital associative $\mathbb{F}$-algebra whose elements represent abstract transformations, inferential rules, questions, or procedures that can be realized in multiple strata.

Every core space carries a unital representation

$$\rho_i : \mathcal{A} \longrightarrow \mathrm{End}(C_i).$$

For $a \in \mathcal{A}$, the operator $\rho_i(a)$ is the realization of the same abstract operation in local coordinates.

Definition 14 (Exact operation-compatible transfer)**.** A core transfer $H_e : C_i \to C_j$ is operation compatible when it is an intertwiner:

$$\boxed{H_e \rho_i(a) = \rho_j(a) H_e} \qquad \forall a \in \mathcal{A}.$$

Equivalently, the diagram

$$\begin{array}{ccc} C_i & \xrightarrow{\rho_i(a)} & C_i \\ \downarrow H_e & & \downarrow H_e \\ C_j & \xrightarrow{\rho_j(a)} & C_j \end{array}$$

commutes for every a.

This condition states that translating and then operating gives the same result as operating and then translating. Agreement of static encodings alone is weaker: two nodes may store corresponding vectors while implementing incompatible transformations on them.

Proposition 15 (Path composition of intertwiners)**.** *If every edge of a path $\gamma : i \to j$ intertwines the local $\mathcal{A}$-representations, then*

$$H_\gamma \rho_i(a) = \rho_j(a) H_\gamma \qquad \forall a \in \mathcal{A}.$$

Proof. For composable intertwiners $H_e : C_i \to C_j$ and $H_f : C_j \to C_k$,

$$H_f H_e \rho_i(a) = H_f \rho_j(a) H_e = \rho_k(a) H_f H_e.$$

Induction over path length proves the claim. □

Corollary 16 (Holonomy commutes with abstract operations)**.** *For every loop $\gamma : o \to o$,*

$$H_\gamma \rho_o(a) = \rho_o(a) H_\gamma.$$

Hence

$$\rho_o(\mathcal{A}) \subseteq \mathrm{Comm}(\mathrm{Hol}_o),$$

where $\mathrm{Comm}(\mathrm{Hol}_o)$ *is the commutant of the holonomy representation.*

5.1 Approximate operation compatibility

The edgewise defect is

$$\Xi_e(a) = H_e \rho_{s(e)}(a) - \rho_{t(e)}(a) H_e.$$

For a bounded generating family $\mathcal{G} \subseteq \mathcal{A}$, define

$$\eta_{\mathrm{int}}(\mathcal{G}) = \sup_{a \in \mathcal{G}} \sum_{e \in E} w_e \|\Xi_e(a)\|^2.$$

When $\mathcal{A}$ is normed and its unit ball is appropriate, one may take

$$\mathcal{G} = \{a : \|a\|_{\mathcal{A}} \leq 1\}.$$

Exact representational universality means

$$\eta_{\text{int}} = 0,$$

while approximate universality means

$$\eta_{\text{int}} \leq \varepsilon_{\text{int}}.$$

The use of a generating family prevents the definition from depending on an unbounded rescaling of algebra elements.

Example 17 (Equivalent operations in different coordinates)**.** Let $C_1 = C_2 = \mathbb{R}^2$, choose $S \in GL_2(\mathbb{R})$, and set

$$H_{12} = S, \qquad \rho_2(a) = S\rho_1(a)S^{-1}.$$

Then

$$H_{12}\rho_1(a) = S\rho_1(a) = \rho_2(a)S = \rho_2(a)H_{12}.$$

The local matrices can look different while realizing the same abstract operation. This is the elementary model of representation change used throughout the theory.

PART III

Exact Shared Cores

6. Global Compatible Sections

From this point through Theorem 22, assume the core maps define a local system on a connected graph unless a weaker hypothesis is stated explicitly.

Definition 18 (Global compatible section)**.** The exact structural workspace is

$$\boxed{\Gamma(H) = \left\{ (c_i)_{i\in V} \in \bigoplus_{i\in V} C_i : \; c_{t(e)} = H_e c_{s(e)} \; \forall e \in E \right\}.}$$

An element $c = (c_i) \in \Gamma(H)$ is a global compatible section.

Equivalently,

$$\Gamma(H) = \varprojlim_{\mathrm{Path}(G)} C_i.$$

A section is not a single vector copied verbatim at every vertex. It is one abstract element represented by a family of local vectors that agree under the specified translation maps.

A nontrivial exact structural workspace exists when

$$\dim \Gamma(H) > 0.$$

Its exact dimension is

$$d_{\mathbf{W}} = \dim \Gamma(H).$$

Fix a root o. The evaluation map is

$$\mathrm{ev}_o : \Gamma(H) \to C_o, \qquad \mathrm{ev}_o((c_i)) = c_o.$$

Lemma 19 (Root determination)**.** *If $|G|$ is connected and every core transfer is invertible, then ev_o is injective. Consequently,*

$$\dim \Gamma(H) \le \dim C_o.$$

Proof. Choose for each vertex i a path $\tau_i : o \to i$. Compatibility gives

$$c_i = H_{\tau_i} c_o.$$

Thus a section with $c_o = 0$ vanishes at every vertex. Hence ev_o is injective. □

Lemma 20 (Tree propagation)**.** *If $|G|$ is a tree and every H_e is invertible, then every $x \in C_o$ extends uniquely to a global compatible section.*

Proof. A tree contains a unique undirected path $\tau_i : o \to i$ for every vertex. Define $c_i = H_{\tau_i} x$. Because every edge belongs to these unique path relations, the family is compatible. Uniqueness follows from Lemma 19. □

The obstruction to a global core therefore appears only when distinct paths must agree. Cycles encode exactly those consistency constraints.

7. Holonomy and Path Obstruction

Fix a root $o \in V$. For every loop class

$$[\gamma] \in \pi_1(|G|, o),$$

define the holonomy

$$\mathrm{Hol}([\gamma]) = H_\gamma \in GL(C_o).$$

Functoriality gives a group representation

$$\mathrm{hol}_o : \pi_1(|G|, o) \longrightarrow GL(C_o).$$

Its image is the holonomy group

$$\mathrm{Hol}_o = \mathrm{im}(\mathrm{hol}_o).$$

The obstruction associated with a loop is

$$\Omega_\gamma = H_\gamma - I_{C_o}.$$

The maximal path-independent root core is

$$\boxed{C_* = \bigcap_{\gamma : o \to o} \ker(H_\gamma - I).}$$

Equivalently,

$$C_* = C_o^{\mathrm{Hol}_o} = \mathrm{Fix}(\mathrm{Hol}_o).$$

7.1 Spanning-tree reduction

Choose a spanning tree $\mathcal{T} \subseteq |G|$. Let

$$\tau_i : o \to i$$

be the unique tree path. Every non-tree edge $e : i \to j$ determines a fundamental loop

$$\gamma_e = \tau_j^{-1} e \tau_i$$

with transfer

$$H_{\gamma_e} = H_{\tau_j}^{-1} H_e H_{\tau_i}.$$

The fundamental group of a finite graph is generated freely by these cotree loops. Hence it is enough to impose fixed-point conditions for a finite set of generators.

Lemma 21 (Finite obstruction set)**.** *Let $e_1, \dots, e_r$ be the non-tree edges of a connected finite graph. Then*

$$C_* = \bigcap_{q=1}^{r} \ker(H_{\gamma_{e_q}} - I).$$

Proof. A vector fixed by every generator is fixed by each inverse generator and by every finite product of generators and inverses. Those products represent all elements of $\pi_1(|G|, o)$. The reverse inclusion is immediate. □

The number of independent graph cycles is

$$r = |E_{\mathrm{und}}| - |V| + 1$$

for a connected underlying graph, after treating parallel edges with their multiplicity.

8. Exact Existence Theorem

Theorem 22 (Limit–holonomy equivalence)**.** *Let $|G|$ be connected and let*

$$H : \Pi_1(G) \to \mathbf{FinHilb}^{\times}$$

be a finite-dimensional local system. For any root $o \in V$, evaluation at the root induces a canonical vector-space isomorphism

$$\boxed{\mathrm{ev}_o : \Gamma(H) \xrightarrow{\cong} C_o^{\mathrm{Hol}_o} = \bigcap_{\gamma : o \to o} \ker(H_\gamma - I).}$$

Consequently,

$$\Gamma(H) \neq \{0\}$$

if and only if the holonomy representation contains a nonzero trivial subrepresentation.

Proof. Let $c = (c_i) \in \Gamma(H)$. For any loop $\gamma : o \to o$, compatibility along the loop gives

$$H_\gamma c_o = c_o.$$

Therefore $\mathrm{ev}_o(c) = c_o \in C_o^{\mathrm{Hol}_o}$. By Lemma 19, ev_o is injective.

Conversely, let $x \in C_o^{\mathrm{Hol}_o}$. Choose a path $\tau_i : o \to i$ for each vertex and define

$$c_i = H_{\tau_i} x.$$

This definition is independent of the chosen path. Indeed, if $\sigma_i : o \to i$ is another path, then

$$\sigma_i^{-1}\tau_i : o \to o$$

is a loop, and invariance of x gives

$$H_{\sigma_i}^{-1} H_{\tau_i} x = x.$$

Multiplying by H_{σ_i} yields

$$H_{\tau_i} x = H_{\sigma_i} x.$$

For any edge $e : i \to j$, the paths $e\tau_i$ and τ_j both run from o to j. Path independence therefore gives

$$H_e c_i = H_e H_{\tau_i} x = H_{\tau_j} x = c_j.$$

Thus (c_i) is a global compatible section and $\mathrm{ev}_o((c_i)) = x$. Hence ev_o is surjective and therefore an isomorphism. □

Corollary 23 (Tree case)**.** *If $|G|$ is a tree, then* $\mathrm{Hol}_o = \{I\}$ *and*

$$\Gamma(H) \cong C_o.$$

In particular,

$$d_{\mathbf{W}} = \dim C_o.$$

Corollary 24 (Finite matrix test)**.** *Let $\gamma_1, \dots, \gamma_r$ be fundamental cycle generators and suppose* $\dim C_o = d$*. Define the stacked obstruction matrix*

$$\mathcal{M} = \begin{pmatrix} H_{\gamma_1} - I \\ \vdots \\ H_{\gamma_r} - I \end{pmatrix}.$$

Then

$$\boxed{C_* = \ker \mathcal{M}, \qquad d_{\mathbf{W}} = d - \operatorname{rank}(\mathcal{M}).}$$

Proof. The kernel of a vertically stacked operator is the intersection of the kernels of its blocks. Apply Lemma 21 and Theorem 22. □

Corollary 25 (Averaging projector for finite holonomy)**.** *If* Hol_o *is finite and acts unitarily, then*

$$P_* = \frac{1}{|\,\mathrm{Hol}_o\,|} \sum_{h \in \mathrm{Hol}_o} h$$

is the orthogonal projector onto C_*.

Proof. For every $g \in \mathrm{Hol}_o$, left multiplication permutes the group elements, so $gP_* = P_*$. Thus $\operatorname{ran} P_* \subseteq C_*$. If $x \in C_*$, then $P_* x = x$, so $C_* \subseteq \operatorname{ran} P_*$. Unitarity gives $P_*^* = P_*$, and the range characterization implies $P_*^2 = P_*$. □

8.1 Worked examples

Example 26 (One-dimensional core surviving a contradictory cycle)**.** Let a three-node cycle have $C_1 = C_2 = C_3 = \mathbb{R}^2$ and

$$H_{12} = I, \qquad H_{23} = I, \qquad H_{31} = J, \qquad J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The cycle holonomy is J, hence

$$C_* = \ker(J - I) = \operatorname{span}\{(1, 0)^\top\}.$$

The first coordinate circulates consistently through all three strata. The second returns with its sign reversed and therefore cannot define a nonzero exact shared section. Thus

$$d_{\mathbf{W}} = 1.$$

Example 27 (Local compatibility without a global nonzero core)**.** Let the only nontrivial cycle holonomy on $\mathbb{R}^2$ be a rotation

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

with $\theta \notin 2\pi\mathbb{Z}$. Then

$$\ker(R_\theta - I) = \{0\}.$$

Every edge map is individually invertible and well-defined, yet the complete cycle admits no nonzero exact shared core. Pairwise translatability is therefore insufficient for global path independence.

Example 28 (Acyclic collaboration)**.** Suppose a root concept is translated into a proof, implementation, and report along a tree of invertible exact transfers. No alternative route returns a conflicting representation. The tree has trivial holonomy, so every root-core direction extends uniquely. This does not yet establish causal efficacy; it establishes only exact structural compatibility.

8.2 Interpretive consequence

The theorem gives a precise meaning to "something shared across representations." The shared object is neither any one local vector nor an extra vector outside the system. It is the equivalence class of all compatible local realizations, canonically represented at a root by the subspace fixed by every translation cycle.

9. Operator Closure Theorem

Theorem 29 (Global operation closure)**.** *Suppose every core transfer intertwines the representations of $\mathcal{A}$:*

$$H_e \rho_{s(e)}(a) = \rho_{t(e)}(a) H_e \qquad \forall e \in E,\ a \in \mathcal{A}.$$

Then $\Gamma(H)$ is an $\mathcal{A}$-module under the componentwise action

$$\rho_\Gamma(a)(c_i)_{i\in V} = (\rho_i(a)c_i)_{i\in V}.$$

Equivalently, the holonomy-fixed root core is invariant:

$$\boxed{\rho_o(a) C_* \subseteq C_* \qquad \forall a \in \mathcal{A}.}$$

Proof. Let $c = (c_i) \in \Gamma(H)$. For every edge $e : i \to j$,

$$H_e \rho_i(a) c_i = \rho_j(a) H_e c_i = \rho_j(a) c_j.$$

Therefore the family $(\rho_i(a)c_i)$ is compatible and belongs to $\Gamma(H)$. The representation identities on each C_i hold componentwise, so ρ_Γ is an $\mathcal{A}$-representation.

Under the isomorphism of Theorem 22, this action is represented at the root by ρ_o. Alternatively, if $x \in C_*$, then Corollary 16 gives, for every loop γ,

$$H_\gamma \rho_o(a) x = \rho_o(a) H_\gamma x = \rho_o(a) x.$$

Hence $\rho_o(a)x \in C_*$. □

Corollary 30 (Inherited algebraic structure)**.** *Any algebraic relation satisfied by $\rho_i(\mathcal{A})$ is inherited by the global action. In particular, if $a, b \in \mathcal{A}$, then*

$$\rho_\Gamma(ab) = \rho_\Gamma(a)\rho_\Gamma(b), \qquad \rho_\Gamma(1) = I_{\Gamma(H)}.$$

The result distinguishes a shared *operational* representation from a shared static code. A core closed under the common algebra supports the same abstract manipulations across strata without leaving the globally compatible subspace.

10. Workspace Operator Algebra

Definition 31 (Natural workspace operators)**.** Define

$$\boxed{\mathcal{O}_{\mathbf{W}} = \left\{(A_i)_{i\in V} : A_{t(e)}H_e = H_e A_{s(e)} \ \forall e\right\}.}$$

Categorically,

$$\mathcal{O}_{\mathbf{W}} = \mathrm{Nat}(H, H),$$

the algebra of natural endomorphisms of the local system.

Pointwise addition, scalar multiplication, and composition make $\mathcal{O}_{\mathbf{W}}$ a unital associative algebra. Each $A = (A_i)$ acts on global sections by

$$\pi_{\mathbf{W}}(A)(c_i) = (A_i c_i).$$

Naturality guarantees that $(A_i c_i)$ is again compatible.

Theorem 32 (Commutant characterization)**.** *Let $|G|$ be connected and let H be an invertible local system. Root evaluation induces an algebra isomorphism*

$$\boxed{\Phi_o : \mathcal{O}_{\mathbf{W}} \xrightarrow{\cong} \mathrm{Comm}(\mathrm{Hol}_o), \qquad \Phi_o((A_i)) = A_o,}$$

where

$$\mathrm{Comm}(\mathrm{Hol}_o) = \{B \in \mathrm{End}(C_o) : Bh = hB \ \forall h \in \mathrm{Hol}_o\}.$$

Proof. If $(A_i) \in \mathcal{O}_{\mathbf{W}}$, naturality along every loop gives

$$A_o H_\gamma = H_\gamma A_o.$$

Thus $A_o \in \mathrm{Comm}(\mathrm{Hol}_o)$, and Φ_o is a well-defined algebra homomorphism.

Choose paths $\tau_i : o \to i$. Naturality along τ_i gives

$$A_i H_{\tau_i} = H_{\tau_i} A_o,$$

so

$$A_i = H_{\tau_i} A_o H_{\tau_i}^{-1}.$$

Therefore A_o determines every A_i, and Φ_o is injective.

Conversely, take $B \in \mathrm{Comm}(\mathrm{Hol}_o)$ and define

$$A_i = H_{\tau_i} B H_{\tau_i}^{-1}.$$

This is independent of the chosen root path: changing τ_i multiplies it by a loop holonomy that commutes with B. For an edge $e : i \to j$, the operator

$$H_{\tau_j}^{-1} H_e H_{\tau_i}$$

is a root holonomy. Commutation with B rearranges to

$$A_j H_e = H_e A_i.$$

Hence (A_i) is natural and $\Phi_o((A_i)) = B$. Thus Φ_o is surjective. □

The structural workspace is therefore more accurately represented by the triple

$$\boxed{\mathbf{W}_{\text{static}} = (\Gamma(H), \mathcal{O}_{\mathbf{W}}, \pi_{\mathbf{W}}),}$$

not merely by the vector space $\Gamma(H)$.

10.1 Effective operator algebra

Some natural operators may vanish on every global section while acting nontrivially on locally available but globally inconsistent directions. Define the annihilator

$$\operatorname{Ann}(\Gamma(H)) = \{A \in \mathcal{O}_{\mathbf{W}} : \pi_{\mathbf{W}}(A) = 0\}.$$

The effective workspace algebra is

$$\boxed{\mathcal{O}_{\mathbf{W}}^{\text{eff}} = \mathcal{O}_{\mathbf{W}} / \operatorname{Ann}(\Gamma(H)).}$$

Its action on $\Gamma(H)$ is faithful by construction.

Example 33 (Operators in the reflection-cycle workspace)**.** For Example 26, the holonomy is

$$J = \operatorname{diag}(1, -1).$$

Its commutant consists of diagonal matrices:

$$\operatorname{Comm}(J) = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} : a, b \in \mathbb{R} \right\}.$$

The exact workspace core is the first coordinate. Consequently, only the parameter a acts effectively on $\Gamma(H)$; the b-direction belongs to the annihilator of the global core. Hence

$$\mathcal{O}_{\mathbf{W}}^{\text{eff}} \cong \mathbb{R}.$$

10.2 Scope of the exact theory

Parts I–III now establish three exact facts:

global shared content is the holonomy-fixed core,
universally transported operations preserve that core,
natural workspace operations are the commutant of holonomy.

These statements do not yet prove that the shared core is robust, causally used, irreducibly distributed, or temporally persistent. Those are separate questions developed in the approximate and dynamic theory that follows.

PART IV

Approximate Workspaces

The exact theory asks whether local core states agree perfectly under every transfer. Approximate theory asks three more realistic questions:

1. how strongly a candidate state violates edgewise compatibility;
2. whether a low-defect subspace is spectrally separated from incoherent directions; and
3. how that subspace changes when transfers, metrics, or operations are perturbed.

Throughout Parts IV–V, set

$$\mathcal{C}_V = \bigoplus_{i \in V} C_i, \qquad \mathcal{C}_E = \bigoplus_{e \in E} C_{t(e)},$$

with their Hilbert direct-sum inner products, and retain strictly positive weights w_e.

11. Transfer Discrepancy Operator

Definition 34 (Transfer discrepancy)**.** The transfer discrepancy operator is

$$D_H : \mathcal{C}_V \longrightarrow \mathcal{C}_E, \qquad (D_H c)_e = \sqrt{w_e}\,(c_{t(e)} - H_e c_{s(e)}).$$

Its squared norm is the coherence energy

$$\boxed{\mathcal{E}_H(c) = \|D_H c\|^2 = \sum_{e \in E} w_e \left\| c_{t(e)} - H_e c_{s(e)} \right\|^2.}$$

The transfer Laplacian is

$$\boxed{L_H = D_H^* D_H.}$$

The operator D_H is an incidence-type derivative twisted by the edge transfers. It compares a state already present at the target with the state predicted there by translation from the source. A direction can therefore have low energy even when its coordinates differ substantially across nodes, provided those coordinates differ exactly as the transfer maps prescribe.

For $z = (z_e)_{e \in E} \in \mathcal{C}_E$, the adjoint is

$$(D_H^* z)_i = \sum_{e : t(e) = i} \sqrt{w_e}\, z_e - \sum_{e : s(e) = i} \sqrt{w_e}\, H_e^* z_e.$$

Consequently,

$$\boxed{\begin{aligned}(L_H c)_i = &\sum_{e : t(e) = i} w_e (c_i - H_e c_{s(e)}) \\ &+ \sum_{e : s(e) = i} w_e H_e^* (H_e c_i - c_{t(e)}).\end{aligned}}$$

This formula makes the two contributions at a vertex explicit: incoming prediction error and adjoint-propagated outgoing error.

Proposition 35 (Positivity and variational identity)**.** *The transfer Laplacian is self-adjoint and positive semidefinite. For every $c \in \mathcal{C}_V$,*

$$\langle c, L_H c\rangle = \mathcal{E}_H(c) \geq 0.$$

Proof. Self-adjointness and positivity follow from $L_H = D_H^* D_H$. The displayed identity is

$$\langle c, D_H^* D_H c\rangle = \langle D_H c, D_H c\rangle.$$

□

Remark 36 (Weight dependence)**.** Replacing w_e by another positive family does not alter $\ker D_H$, but it generally changes every positive eigenvalue and therefore changes the approximate geometry. Edge weights are part of the metric model, not part of the exact compatibility relation.

12. Exact Kernel Theorem

Theorem 37 (Transfer-Laplacian kernel theorem)**.** *For an arbitrary finite directed core diagram, without assuming invertibility,*

$$\boxed{\ker L_H = \ker D_H = \Gamma(H).}$$

Thus the exact compatible sections are precisely the zero-energy modes of the transfer Laplacian.

Proof. If $D_H c = 0$, then $L_H c = D_H^* D_H c = 0$. Conversely, if $L_H c = 0$, then

$$0 = \langle c, L_H c\rangle = \|D_H c\|^2,$$

so $D_H c = 0$. Finally, $D_H c = 0$ if and only if

$$c_{t(e)} = H_e c_{s(e)}$$

for every edge, which is exactly the definition of $\Gamma(H)$. □

Corollary 38 (Finite rank computation)**.** *Let*

$$n_C = \sum_{i \in V} \dim C_i.$$

Then

$$\boxed{\dim \Gamma(H) = n_C - \operatorname{rank}(D_H).}$$

Hence the exact workspace dimension can be obtained by a single rank or singular-value computation, even when holonomy is unavailable because some transfers are noninvertible.

12.1 Relation to the holonomy computation

In the local-system regime, Theorems 22 and 37 give two equivalent constructions:

$$\Gamma(H) \cong C_o^{\mathrm{Hol}_o} \qquad \text{and} \qquad \Gamma(H) = \ker L_H.$$

The first reduces the problem to cycle invariants in a root fiber. The second works directly on all vertex coordinates and extends to noninvertible diagrams. The two computations therefore serve different practical regimes rather than competing definitions.

13. Approximate Structural Workspace

Let

$$0 \le \lambda_1 \le \lambda_2 \le \cdots \le \lambda_N, \qquad N = \dim \mathcal{C}_V,$$

be the eigenvalues of L_H, repeated with multiplicity, and let $P_I(L_H)$ denote the spectral projector associated with an interval I.

Definition 39 (Threshold workspace)**.** For $\varepsilon \ge 0$, define

$$\boxed{\Gamma_\varepsilon(H) = \operatorname{ran} P_{[0,\varepsilon]}(L_H).}$$

Its dimension is

$$d_\varepsilon = \#\{k : \lambda_k \le \varepsilon\}.$$

When $0 < d_\varepsilon < N$, define the boundary gap

$$\boxed{g_\varepsilon = \lambda_{d_\varepsilon+1} - \lambda_{d_\varepsilon}.}$$

The threshold ε declares which discrepancy scale is treated as structurally negligible. The dimension d_ε alone is not a robust certificate: a cluster is stable only when it is separated by a positive boundary gap.

An equivalent index-based description fixes d and writes

$$W_d = \operatorname{ran} P_d, \qquad P_d = P_{\{\lambda_1,\dots,\lambda_d\}}(L_H),$$

with cluster gap

$$g_d = \lambda_{d+1} - \lambda_d > 0.$$

The index-based form is convenient for perturbation theorems because the workspace rank is declared explicitly.

Proposition 40 (Min–max characterization)**.** *For $1 \le k \le N$,*

$$\lambda_k = \min_{\substack{S \subseteq \mathcal{C}_V \\ \dim S = k}} \max_{\substack{c \in S \\ \|c\|=1}} \mathcal{E}_H(c).$$

Consequently, the span of the first k eigenvectors minimizes the worst-case coherence energy among all k-dimensional candidate subspaces.

Proof. Expand every vector in an orthonormal eigenbasis of L_H. The span of the first k eigenvectors has maximal Rayleigh quotient λ_k. Every k-dimensional subspace intersects the orthogonal complement of the first $k-1$ eigenvectors nontrivially, and any unit vector in that intersection has Rayleigh quotient at least λ_k. □

13.1 Exact, soft, and spurious low modes

The spectrum supports three distinct interpretations:

$$\begin{aligned} \lambda_k = 0 &\iff \text{exact compatible direction,} \\ 0 < \lambda_k \ll g_d &\implies \text{approximately transported direction,} \\ \lambda_k \approx \lambda_{k+1} &\implies \text{rank selection is unstable at that boundary.} \end{aligned}$$

A low eigenvalue can still be scientifically spurious if it arises from a poorly chosen weight, a nearly disconnected graph, or a noncausal encoding. The causal and null gates in later parts remain necessary.

14. Normalized Coherence Defect

For $c \neq 0$, define the Rayleigh defect

$$\delta_{\mathrm{coh}}(c) = \frac{\mathcal{E}_H(c)}{\|c\|^2}.$$

For a nonzero subspace $W \subseteq \mathcal{C}_V$, define

$$\delta_{\mathrm{coh}}(W) = \sup_{\substack{c \in W \\ \|c\|=1}} \mathcal{E}_H(c).$$

If W is L_H-invariant, then

$$\delta_{\mathrm{coh}}(W) = \|L_H|_W\|.$$

In particular,

$$\delta_{\mathrm{coh}}(\Gamma_\varepsilon(H)) \leq \varepsilon.$$

Theorem 41 (Coherence-to-core distance estimate)**.** *Let*

$$W_0 = \Gamma(H) = \ker L_H, \qquad r = \dim W_0,$$

and suppose $r < N$. Set

$$\gamma_H = \lambda_{r+1} > 0.$$

Then every $c \in \mathcal{C}_V$ satisfies

$$\boxed{\operatorname{dist}(c, W_0)^2 \leq \frac{\mathcal{E}_H(c)}{\gamma_H}.}$$

Moreover,

$$\mathcal{E}_H(c) \leq \|L_H\| \operatorname{dist}(c, W_0)^2.$$

Proof. Write $c = P_0 c + Q_0 c$, where P_0 projects onto W_0 and $Q_0 = I - P_0$. Since $L_H P_0 = 0$,

$$\mathcal{E}_H(c) = \langle Q_0 c, L_H Q_0 c \rangle.$$

The spectrum of L_H on $W_0^\perp$ lies in $[\gamma_H, \|L_H\|]$. Hence

$$\gamma_H \|Q_0 c\|^2 \leq \mathcal{E}_H(c) \leq \|L_H\| \|Q_0 c\|^2.$$

Finally, $\|Q_0 c\| = \operatorname{dist}(c, W_0)$. □

This estimate is the exact sense in which a spectral gap turns small edgewise disagreement into closeness to a genuinely compatible global section.

14.1 Two-node consistency cycle

Let $C_1 = C_2 = \mathbb{R}$, use unit weights, and take two directed transfers

$$H_{12} = a, \qquad H_{21} = b.$$

Then

$$D_H = \begin{pmatrix} -a & 1 \\ 1 & -b \end{pmatrix}, \qquad L_H = \begin{pmatrix} a^2+1 & -(a+b) \\ -(a+b) & b^2+1 \end{pmatrix}.$$

Writing

$$\tau = a^2 + b^2 + 2,$$

the eigenvalues are

$$\lambda_\pm = \frac{1}{2}\left(\tau \pm \sqrt{\tau^2 - 4(ab-1)^2}\right),$$

and

$$\det L_H = (ab-1)^2.$$

Thus a nonzero exact shared core exists precisely when

$$ab = 1,$$

which is exactly the trivial-holonomy condition. Away from exact consistency,

$$\lambda_- \lambda_+ = (ab-1)^2.$$

When $|ab-1|$ is small while λ_+ stays bounded away from zero, the soft workspace eigenvalue is quadratic in the cycle mismatch:

$$\lambda_- = \frac{(ab-1)^2}{\lambda_+}.$$

This example connects approximate spectral coherence directly to approximate holonomy closure.

15. Perturbation Stability Theorem

The central robustness question is whether a selected low-energy subspace persists when the estimated transfer maps change. The next lemma isolates the finite-dimensional spectral fact needed for the answer.

Lemma 42 (Separated spectral subspaces)**.** *Let A and $B = A + E$ be self-adjoint operators. Let P be an A-spectral projector whose eigenvalues lie in $(-\infty, a]$, and let $\widetilde{Q}$ be a B-spectral projector whose eigenvalues lie in $[b, \infty)$, with $b > a$. Then*

$$\|P\widetilde{Q}\| \le \frac{\|E\|}{b-a}.$$

Proof. On $\operatorname{ran} P$ and $\operatorname{ran} \widetilde{Q}$, set

$$A_P = PAP|_{\operatorname{ran} P}, \qquad B_Q = \widetilde{Q}B\widetilde{Q}|_{\operatorname{ran} \widetilde{Q}}.$$

The operator $X = P\widetilde{Q}$ satisfies the Sylvester equation

$$A_P X - X B_Q = -PE\widetilde{Q}.$$

Let $\mu = (a+b)/2$. Since

$$\|e^{t(A_P - \mu I)}\| \le e^{-(b-a)t/2}, \qquad \|e^{-t(B_Q - \mu I)}\| \le e^{-(b-a)t/2},$$

the inverse Sylvester operator has the convergent integral representation

$$X = \int_0^\infty e^{t(A_P - \mu I)}(PE\widetilde{Q})e^{-t(B_Q - \mu I)}\, \mathrm{d}t.$$

Taking norms and integrating gives the result. □

Theorem 43 (Spectral workspace stability)**.** *Let L and $\widetilde{L} = L+E$ be self-adjoint positive semidefinite operators on the same N-dimensional space. Suppose*

$$\lambda_d(L) < \lambda_{d+1}(L), \qquad g_d = \lambda_{d+1}(L) - \lambda_d(L) > 0,$$

and

$$\eta = \|E\| < \frac{g_d}{2}.$$

Let P_d and $\widetilde{P}_d$ project onto the first d eigenvectors of L and $\widetilde{L}$, respectively. Then the rank-d cluster remains separated and

$$\boxed{\|P_d - \widetilde{P}_d\| \le \frac{\eta}{g_d - \eta} \le \frac{2\eta}{g_d}.}$$

Proof. The min–max characterization gives the eigenvalue perturbation bound

$$|\lambda_k(\widetilde{L}) - \lambda_k(L)| \le \eta.$$

Hence

$$\lambda_{d+1}(\widetilde{L}) - \lambda_d(L) \ge g_d - \eta$$

and

$$\lambda_{d+1}(L) - \lambda_d(\widetilde{L}) \ge g_d - \eta.$$

Apply Lemma 42 first to $P_d(I - \widetilde{P}_d)$ and then, with the roles of L and $\widetilde{L}$ exchanged, to $(I - P_d)\widetilde{P}_d$. Orthogonal projections of equal rank satisfy

$$\|P_d - \widetilde{P}_d\| = \max\left\{\|P_d(I - \widetilde{P}_d)\|, \|(I - P_d)\widetilde{P}_d\|\right\}.$$

This yields the first bound. The second follows from $\eta < g_d/2$. □

15.1 From transfer-map error to Laplacian error

Suppose the edge maps are perturbed to

$$\widetilde{H}_e = H_e + \Delta_e$$

while the weights and vertex metrics remain fixed. Define

$$\boxed{\delta_H^2 = \max_{i \in V} \sum_{e : s(e) = i} w_e \|\Delta_e\|^2.}$$

Let $F = D_{\widetilde{H}} - D_H$. Then

$$(Fc)_e = -\sqrt{w_e}\,\Delta_e c_{s(e)}, \qquad \|F\| \le \delta_H.$$

Since

$$L_{\widetilde{H}} - L_H = D_H^* F + F^* D_H + F^* F,$$

we obtain

$$\boxed{\|L_{\widetilde{H}} - L_H\| \le 2\|D_H\|\delta_H + \delta_H^2.}$$

Corollary 44 (Edgewise stability certificate)**.** *If*

$$\eta_H = 2\|D_H\|\delta_H + \delta_H^2 < \frac{g_d}{2},$$

then

$$\boxed{\|P_d(H) - P_d(\widetilde{H})\| \le \frac{\eta_H}{g_d - \eta_H}.}$$

This converts declared edge-map uncertainty into an explicit bound on the motion of the extracted workspace subspace.

15.2 Approximate operation closure

Define the vertex and edge direct-sum representations

$$\rho_V(a) = \bigoplus_{i\in V} \rho_i(a), \qquad \rho_E(a) = \bigoplus_{e\in E} \rho_{t(e)}(a).$$

Lift the edgewise intertwining defects to

$$(\mathfrak{X}_a c)_e = \sqrt{w_e}\, \Xi_e(a) c_{s(e)}.$$

Then the exact commutator identity is

$$\boxed{D_H \rho_V(a) - \rho_E(a) D_H = -\mathfrak{X}_a.}$$

Furthermore,

$$\|\mathfrak{X}_a\| \le \beta(a), \qquad \beta(a)^2 = \max_{i\in V} \sum_{e:s(e)=i} w_e \|\Xi_e(a)\|^2.$$

Theorem 45 (Approximate operation-closure theorem)**.** *Let P_d project onto the first d eigenvectors of L_H, set*

$$\lambda_d \le \varepsilon, \qquad \mu = \lambda_{d+1} > 0,$$

and let $Q_d = I - P_d$. Then every $a \in \mathcal{A}$ satisfies

$$\boxed{\|Q_d \rho_V(a) P_d\| \le \frac{\|\rho_E(a)\|\sqrt{\varepsilon} + \beta(a)}{\sqrt{\mu}}.}$$

Proof. For $x \in \operatorname{ran} P_d$,

$$\|D_H x\|^2 = \langle x, L_H x\rangle \le \varepsilon \|x\|^2.$$

The commutator identity gives

$$\|D_H \rho_V(a) x\| \le \|\rho_E(a)\|\sqrt{\varepsilon}\|x\| + \beta(a)\|x\|.$$

Because P_d and Q_d reduce L_H, the two energy components are orthogonal and

$$\|D_H Q_d y\| \le \|D_H y\|.$$

On $\operatorname{ran} Q_d$, the spectrum of L_H is bounded below by μ, so

$$\sqrt{\mu}\|Q_d y\| \le \|D_H Q_d y\|.$$

Apply these inequalities to $y = \rho_V(a)x$, then take the supremum over unit $x \in \operatorname{ran} P_d$. □

Corollary 46 (Recovery of exact operation closure)**.** *If $d = \dim\ker L_H$, $\varepsilon = 0$, and every edge intertwines a exactly, then*

$$Q_d \rho_V(a) P_d = 0.$$

Thus Theorem 45 reduces to the exact operation-closure result of Part III.

Corollary 47 (Closure under projector drift)**.** *Let $\widetilde{P}_d$ be another rank-d orthogonal projector and set*

$$\delta_P = \|P_d - \widetilde{P}_d\|.$$

Then

$$\boxed{\|(I - \widetilde{P}_d)\rho_V(a)\widetilde{P}_d\| \le \|Q_d \rho_V(a) P_d\| + 2\|\rho_V(a)\|\delta_P.}$$

Combining this estimate with Theorem 43 controls operation leakage after noisy workspace extraction.

Proof. Subtract $Q_d \rho_V(a) P_d$ and use

$$I - \widetilde{P}_d - Q_d = P_d - \widetilde{P}_d.$$

The two resulting perturbation terms are each bounded by $\|\rho_V(a)\|\delta_P$. □

15.3 Core–residual path accumulation

Let a path have vertices

$$i_0 \xrightarrow{e_1} i_1 \xrightarrow{e_2} \cdots \xrightarrow{e_m} i_m.$$

Write

$$T_r = T_{e_r}, \quad H_r = H_{e_r}, \quad V_r = V_{e_r},$$

and let

$$\iota_i^C : C_i \hookrightarrow X_i, \qquad \iota_i^R : R_i \hookrightarrow X_i$$

be the canonical inclusions. Empty products below are identities.

Theorem 48 (Pathwise core–residual Duhamel formula)**.** *The deviation of full path transport from pure core transport is exactly*

$$\boxed{\begin{aligned} &T_m \cdots T_1 \iota_{i_0}^C - \iota_{i_m}^C H_m \cdots H_1 \\ &\qquad = \sum_{r=1}^{m} T_m \cdots T_{r+1} \iota_{i_r}^R V_r H_{r-1} \cdots H_1. \end{aligned}}$$

Consequently,

$$\boxed{\begin{aligned} \|Q_{i_m} T_m \cdots T_1 P_{i_0}\| \le \sum_{r=1}^{m} &\left(\prod_{\ell=r+1}^{m} \|T_\ell\| \right) \|V_r\| \\ &\times \left(\prod_{\ell=1}^{r-1} \|H_\ell\| \right), \end{aligned}}$$

and the same right-hand side bounds

$$\|P_{i_m} T_m \cdots T_1 P_{i_0} - H_m \cdots H_1\|.$$

Proof. For every step,

$$T_r \iota^C_{i_{r-1}} = \iota^C_{i_r} H_r + \iota^R_{i_r} V_r.$$

Telescope the sequence

$$T_m \cdots T_{r+1} \iota^C_{i_r} H_r \cdots H_1, \qquad r = 0, 1, \ldots, m.$$

The difference between consecutive terms is exactly the r-th summand in the displayed formula. Projection onto R_{i_m} or C_{i_m}, followed by submultiplicativity of the operator norm, gives the two bounds. □

Corollary 49 (Uniform path bound)**.** *If along the path*

$$\|T_r\| \le \tau, \qquad \|H_r\| \le h, \qquad \|V_r\| \le \nu,$$

then the leakage and core-transport defects are bounded by

$$\boxed{\nu \sum_{r=1}^{m} \tau^{m-r} h^{r-1} = \begin{cases} \nu \dfrac{\tau^m - h^m}{\tau - h}, & \tau \neq h, \\ m \nu \tau^{m-1}, & \tau = h. \end{cases}}$$

In particular, if $\tau \le 1$ and $h \le 1$, the bound is at most $m\nu$.

The linear factor m in the merely nonexpansive regime is important. Small single-edge leakage does not by itself justify arbitrarily long transfer chains. Long-horizon stability requires contraction, cancellation, error correction, or leakage that decreases with path length.

15.4 Residual-mediated reentry

For two consecutive transfers,

$$P_{i_2} T_2 T_1 P_{i_0} = H_2 H_1 + U_2 V_1,$$
$$Q_{i_2} T_2 T_1 P_{i_0} = V_2 H_1 + K_2 V_1.$$

Thus leakage out of the core is first order in the blocks V_r, while an excursion that leaves and then returns to the core contains at least one V-block and one later U-block. For a fixed path length, if

$$\|U_r\|, \|V_r\| = O(\delta)$$

and the diagonal blocks remain bounded, the residual-mediated correction to pure core-to-core transport is $O(\delta^2)$, whereas terminal residual leakage can be $O(\delta)$. This distinction separates loss of core invariance from distortion caused by temporary excursions through local residual degrees of freedom.

PART V

Gauge Invariance and Abstract Identity

Exact compatibility is a coordinate-free notion. Approximate spectra also depend on the Hilbert metrics used to measure discrepancy. Gauge theory must therefore distinguish a change of coordinates from a physical change of metric.

16. Local Coordinate Transformations

For each $i \in V$, let

$$g_i \in GL(C_i), \qquad g = (g_i)_{i \in V}.$$

Define the vertex and edge direct sums

$$G_V = \bigoplus_{i \in V} g_i, \qquad G_E = \bigoplus_{e \in E} g_{t(e)}.$$

The transformed transfers and local operation representations are

$$H_e^g = g_{t(e)} H_e g_{s(e)}^{-1},$$

$$\rho_i^g(a) = g_i \rho_i(a) g_i^{-1}.$$

The fundamental covariance identity is

$$\boxed{D_{H^g} G_V = G_E D_H.}$$

It follows directly by evaluating the e-component:

$$\sqrt{w_e} \left(g_{t(e)} c_{t(e)} - g_{t(e)} H_e c_{s(e)} \right).$$

17. Gauge Invariance Theorem

Theorem 50 (Exact gauge covariance)**.** *The map*

$$G_V : \Gamma(H) \longrightarrow \Gamma(H^g)$$

is an isomorphism. Therefore,

$$\boxed{\Gamma(H^g) = G_V \Gamma(H), \qquad \dim \Gamma(H^g) = \dim \Gamma(H).}$$

If the core transports are invertible and o is a root, then

$$H_\gamma^g = g_o H_\gamma g_o^{-1}$$

for every loop $\gamma : o \to o$. Hence the holonomy-fixed core transforms by

$$C_*^g = g_o C_*.$$

Proof. The covariance identity implies

$$D_{H^g}(G_V c) = G_E D_H c.$$

Since G_E is invertible,

$$D_H c = 0 \quad \Longleftrightarrow \quad D_{H^g}(G_V c) = 0.$$

The holonomy formula follows by cancellation of adjacent factors $g_i^{-1} g_i$ along a path. □

Theorem 51 (Unitary gauge covariance of the Laplacian)**.** *If every g_i is unitary, then G_V and G_E are unitary and*

$$\boxed{L_{H^g} = G_V L_H G_V^{-1}.}$$

Consequently,

$$\operatorname{spec}(L_{H^g}) = \operatorname{spec}(L_H), \qquad P_I(L_{H^g}) = G_V P_I(L_H) G_V^{-1}$$

for every spectral interval I.

Proof. From

$$D_{H^g} = G_E D_H G_V^{-1}$$

and unitarity,

$$D_{H^g}^* D_{H^g} = G_V D_H^* G_E^{-1} G_E D_H G_V^{-1} = G_V L_H G_V^{-1}.$$

The spectral statements follow from unitary similarity. □

17.1 Nonunitary gauges and metric distortion

For a general invertible gauge, define

$$m_V = \sigma_{\min}(G_V), \qquad M_V = \|G_V\|,$$

$$m_E = \sigma_{\min}(G_E), \qquad M_E = \|G_E\|.$$

The covariance identity yields

$$m_E^2 \mathcal{E}_H(c) \le \mathcal{E}_{H^g}(G_V c) \le M_E^2 \mathcal{E}_H(c),$$

and

$$m_V^2 \|c\|^2 \le \|G_V c\|^2 \le M_V^2 \|c\|^2.$$

Therefore the Rayleigh quotients obey

$$\boxed{\frac{m_E^2}{M_V^2} \frac{\mathcal{E}_H(c)}{\|c\|^2} \le \frac{\mathcal{E}_{H^g}(G_V c)}{\|G_V c\|^2} \le \frac{M_E^2}{m_V^2} \frac{\mathcal{E}_H(c)}{\|c\|^2}.}$$

By the min–max principle,

$$\boxed{\frac{m_E^2}{M_V^2} \lambda_k(L_H) \le \lambda_k(L_{H^g}) \le \frac{M_E^2}{m_V^2} \lambda_k(L_H).}$$

Exact zero modes are unchanged, but fixed-metric approximate eigenvalues can be strongly distorted by an ill-conditioned coordinate transformation.

17.2 Metric-covariant formulation

A pure change of coordinates should transform the metric as well as the matrices. On each transformed space, define

$$\langle u, v\rangle_i^g = \left\langle g_i^{-1}u, g_i^{-1}v\right\rangle_i .$$

Use the analogous pullback metric on every edge copy. Then G_V and G_E are unitary between the original and transformed metric spaces. Consequently, the complete transfer-Laplacian spectrum is exactly preserved.

This gives two distinct interpretations:

- Transforming the matrices and the metrics gives the same approximate workspace in new coordinates.
- Transforming the matrices while holding the metrics fixed gives a potentially different weighting of discrepancy.

The distinction prevents arbitrary coordinate scaling from manufacturing or erasing apparent low-energy workspace directions.

18. Static Workspace Equivalence

Definition 52 (Exact static equivalence)**.** Two static realizations are exactly gauge equivalent,

$$\mathfrak{R} \sim_{\text{stat}} \mathfrak{R}',$$

when there exists an invertible family g_i such that

$$H'_e = g_{t(e)} H_e g_{s(e)}^{-1}, \qquad \rho'_i(a) = g_i \rho_i(a) g_i^{-1}.$$

The exact static abstract workspace is

$$\boxed{\mathbf{W}_{\text{stat}} = [\mathfrak{R}]_{\sim_{\text{stat}}}.}$$

For approximate theory, the realization also contains the vertex and edge metrics. Two metric realizations are equivalent only when the gauge transports those metrics, or when the gauge is unitary relative to the fixed metrics. Accordingly:

exact workspace identity is invariant under GL-gauge,
fixed-metric spectral geometry is invariant under unitary gauge,
metric-covariant spectral geometry is invariant under transported GL-gauge.

18.1 What the approximate theory establishes

Parts IV–V now prove:

$\ker L_H = \Gamma(H)$,
low energy plus a gap implies proximity to an exact core,
a separated workspace cluster is stable under transfer noise,
approximately intertwined operations nearly preserve that cluster,
core leakage and residual reentry admit explicit pathwise bounds,
exact identity is gauge invariant while approximate geometry is metric aware.

These are structural and perturbative results. They still do not prove that a workspace variable is causally used, predictively sufficient, irreducibly distributed, or temporally self-maintaining. Those questions begin in Part VI.

18.2 Finite computation protocol

For a declared graph, set of core dimensions, weights, and transfer maps, the static analysis can be executed in the following order:

1. Assemble the sparse block operator D_H from the edge equations
$$\sqrt{w_e}\big(c_{t(e)} - H_e c_{s(e)}\big).$$
2. Compute a singular-value decomposition
$$D_H = U\Sigma V^*, \qquad \lambda_k(L_H) = \sigma_k(D_H)^2.$$
Use a declared numerical tolerance to distinguish computational zero modes from positive modes.
3. Select a candidate rank d only at a reproducible spectral boundary with $g_d > 0$. A large observed gap is evidence of stability, not evidence of causal relevance.
4. Quantify transfer uncertainty by δ_H and verify whether
$$2\|D_H\|\delta_H + \delta_H^2 < \frac{g_d}{2}.$$
If this gate fails, the extracted projector has no certified perturbative identity at the stated error scale.
5. For every proposed common operation, compute $\beta(a)$ and the operation-leakage bound in Theorem 45.
6. For every long transfer chain used in an interpretation, report the pathwise leakage certificate from Theorem 48 rather than extrapolating from single-edge defects.

A minimal structural report is therefore

$$\boxed{\Big(d, \lambda_d, \lambda_{d+1}, g_d, \delta_H, \|P_d - \widetilde{P}_d\|_{\mathrm{bound}}, \beta(a), \Lambda_\gamma\Big),}$$

where Λ_γ denotes the right-hand side of the pathwise leakage bound. This structural report is intentionally incomplete as a workspace certificate. Parts VI–IX add the predictive, causal, null, and irreducibility gates required to interpret the low-energy subspace operationally.

PART VI

Dynamic Workspace Theory

19. Probabilistic Local and Relational Dynamics

The static theory identifies compatible directions across a declared transfer network. A dynamic workspace claim requires additional probabilistic objects: a history, a target future, a decision problem, and an intervention family. These must be fixed before sufficiency or causal efficacy is evaluated.

Definition 53 (Dynamic relational realization)**.** A dynamic relational realization is defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and consists of:

- node states $X_i(t) \in \mathsf{X}_i$ for $i \in V$;
- edge messages $M_e(t) \in \mathsf{M}_e$ for $e \in E$;
- external inputs E_t, local actions $A_i(t)$, and disturbances $\xi_i(t), \zeta_e(t)$;
- time-dependent transfer data $H_e(t)$, local core vectors $c_i(t)$, and transfer Laplacians $L_t = L_{H(t)}$; and
- a future target $Y_t^+ \in \mathsf{Y}$.

All state, history, and target spaces are assumed standard Borel unless a stronger topology is explicitly invoked.

A representative local update is

$$X_i(t+1) = F_i(X_i(t), \{M_e(t) : t(e) = i\}, A_i(t), E_t, \xi_i(t)),$$

with messages

$$M_e(t) = K_e\Big(X_{s(e)}(t), W_t, E_t, \zeta_e(t)\Big).$$

This notation permits, but does not require, the current workspace state to modulate future transfers.

For a coalition $A \subseteq V$, let $\mathbf{H}_t^A$ denote the random history available to that coalition through time t. Its generated information is

$$\mathcal{F}_t^A = \sigma(\mathbf{H}_t^A).$$

The full relational history is $\mathbf{H}_t = \mathbf{H}_t^V$, with

$$\mathcal{F}_t^A \subseteq \mathcal{F}_t^B \qquad \text{whenever } A \subseteq B.$$

The empty-coalition history $\mathbf{H}_t^\varnothing$ is the trivial constant statistic and represents prediction from the background target law alone. A concrete history may contain node trajectories, incident messages, actions, artifacts, transfer-map estimates, and the declared graph history. The exact contents must be reported because coalition sufficiency is relative to what a coalition is permitted to observe.

The target Y_t^+ can be a next output, a finite future trajectory, a research decision, a proof outcome, or any measurable downstream quantity. Every predictive workspace claim is therefore target-relative:

$$\text{WorkspaceFor}(Y_t^+) \not\Longrightarrow \text{WorkspaceFor}(\widetilde{Y}_t^+).$$

Let

$$c(t) = (c_i(t))_{i \in V} \in \mathcal{C}_V(t), \qquad P_t = \mathbf{1}_{[0,\varepsilon_t]}(L_t).$$

The instantaneous structural workspace coordinate is

$$\boxed{W_t^{\mathrm{str}} = P_t c(t) \in \Gamma_{\varepsilon_t}(H(t)).}$$

When P_t varies with time, these coordinates inhabit a moving family of subspaces. Parts X–XII will supply the transport required to compare them across time. In what follows, all sufficiency and intervention statements are made at a fixed decision time t.

Remark 54 (Three objects that must not be conflated)**.** The realization state S_t, the complete history $\mathbf{H}_t$, and the structural coordinate W_t^{str} are different objects. The first is an instantaneous physical or computational state, the second contains all admissible past information, and the third is a low-transfer-energy statistic extracted from the declared core geometry.

20. Canonical Predictive and Interventional Causal States

The quotient notation $[\mathbf{H}_t]_{\sim_Y}$ expresses the right intuition but hides measurability and null-set issues. A canonical construction is obtained from regular conditional distributions.

Definition 55 (Predictive kernel and canonical state)**.** Let H_t and Y be standard Borel spaces. A regular conditional kernel is a measurable map

$$K_t : \mathsf{H}_t \times \mathcal{B}(\mathsf{Y}) \to [0,1]$$

satisfying

$$K_t(h, A) = \mathbb{P}(Y_t^+ \in A \mid \mathbf{H}_t = h)$$

for $\mathbb{P}_{\mathbf{H}_t}$-almost every h. Let $\mathcal{P}(\mathsf{Y})$ carry its evaluation sigma-algebra. The canonical predictive state is the random probability measure

$$\boxed{Z_t = K_t(\mathbf{H}_t, \cdot) \in \mathcal{P}(\mathsf{Y}).}$$

Two histories are predictively equivalent precisely when they induce the same conditional law of Y_t^+.

Theorem 56 (Predictive-state sufficiency and minimality, T_{12})**.** *Under Definition 55:*

1. *Z_t is predictively sufficient:*
$$Y_t^+ \perp\!\!\!\perp \mathbf{H}_t \mid Z_t.$$
2. *If $S_t = s(\mathbf{H}_t)$ is any other history statistic satisfying*
$$Y_t^+ \perp\!\!\!\perp \mathbf{H}_t \mid S_t,$$
then there is a measurable map ϕ such that
$$Z_t = \phi(S_t) \qquad \text{almost surely.}$$
3. *Consequently, any two minimal sufficient predictive states are measurably isomorphic after removal of null sets.*

Proof. For every bounded measurable $f : \mathsf{Y} \to \mathbb{R}$, define

$$Z_t(f) = \int_{\mathsf{Y}} f(y)\, Z_t(\mathrm{d}y).$$

By construction,

$$\mathbb{E}[f(Y_t^+) \mid \mathbf{H}_t] = Z_t(f).$$

Since Z_t is a measurable function of $\mathbf{H}_t$, the tower property gives

$$\mathbb{E}[f(Y_t^+) \mid Z_t] = \mathbb{E}[Z_t(f) \mid Z_t] = Z_t(f).$$

Hence conditioning additionally on $\mathbf{H}_t$ does not change the conditional expectation, which is the stated conditional independence.

Now let $S_t = s(\mathbf{H}_t)$ be sufficient. Then, for every measurable $A \subseteq \mathsf{Y}$,

$$K_t(\mathbf{H}_t, A) = \mathbb{P}(Y_t^+ \in A \mid \mathbf{H}_t) = \mathbb{P}(Y_t^+ \in A \mid S_t) \qquad \text{a.s.}$$

Choose a countable determining class for probability measures on the standard Borel space Y. The values of $K_t(\mathbf{H}_t, \cdot)$ on this class are measurable functions of S_t, and therefore the complete random measure Z_t is a measurable function of S_t. Applying this conclusion in both directions to two minimal sufficient states gives the final claim. □

Remark 57 (Meaning of minimality)**.** The theorem does not say that Z_t has the fewest numerical coordinates in a chosen parametrization. It says that every sufficient history statistic must determine Z_t almost surely. This is an information order, invariant under one-to-one reparametrization.

For an arbitrary history statistic $S_t = s(\mathbf{H}_t)$, let

$$K_t^S(s, \cdot) = \mathbb{P}(Y_t^+ \in \cdot \mid S_t = s).$$

Whenever the conditional relative entropy is finite, define the sufficiency defect

$$\boxed{\delta_{\mathrm{suff}}(S_t) = \mathbb{E}\Big[\mathrm{KL}\Big(K_t(\mathbf{H}_t, \cdot) \,\Big\|\, K_t^S(S_t, \cdot)\Big)\Big].}$$

Proposition 58 (Conditional-information identity)**.** *If S_t is a measurable function of $\mathbf{H}_t$, then*

$$\boxed{\delta_{\mathrm{suff}}(S_t) = I(Y_t^+; \mathbf{H}_t \mid S_t).}$$

In particular, the defect vanishes if and only if S_t is predictively sufficient, up to the usual almost-sure qualifications.

Proof. The conditional mutual information is the expected relative entropy between the conditional law given $(\mathbf{H}_t, S_t)$ and the conditional law given S_t. Because $S_t = s(\mathbf{H}_t)$, the first of these is exactly $K_t(\mathbf{H}_t, \cdot)$. □

Sufficiency has a direct decision-theoretic interpretation. Use the convention

$$\mathrm{TV}(P, Q) = \sup_A |P(A) - Q(A)|.$$

Let A be a finite action set and $\ell : \mathsf{Y} \times \mathsf{A} \to [0, 1]$ a bounded loss. Define

$$\mathcal{R}_\ell(S_t) = \inf_d \mathbb{E}\Big[\ell\big(Y_t^+, d(S_t)\big)\Big].$$

Proposition 59 (Approximate sufficiency bounds excess Bayes risk)**.** *Under the preceding assumptions,*

$$\boxed{0 \le \mathcal{R}_\ell(S_t) - \mathcal{R}_\ell(\mathbf{H}_t) \le \mathbb{E}\Big[\mathrm{TV}\Big(K_t(\mathbf{H}_t), K_t^S(S_t)\Big)\Big] \le \sqrt{\frac{\delta_{\mathrm{suff}}(S_t)}{2}}.}$$

Proof. More information cannot increase optimal risk, giving the first inequality. For any fixed action, its conditional expected loss changes by at most total variation when the conditional law changes. Taking the infimum over actions preserves this bound. Conditional Pinsker inequality followed by Jensen's inequality gives the final estimate. □

For a finite target alphabet and logarithmic scoring loss

$$\ell_{\log}(y, q) = -\log q(y), \qquad q \in \mathcal{P}(\mathsf{Y}),$$

the identity is exact:

$$\boxed{\mathcal{R}_{\log}(S_t) = H(Y_t^+ \mid S_t), \qquad \mathcal{R}_{\log}(S_t) - \mathcal{R}_{\log}(\mathbf{H}_t) = I(Y_t^+; \mathbf{H}_t \mid S_t).}$$

Natural logarithms measure the gap in nats; base-two logarithms measure it in bits.

Intervention-indexed causal state

Observational prediction may merge histories that behave identically under the current policy but respond differently to future interventions. Let $\mathcal{I}$ be a finite or countable family of interventions applied only after time t. For each $\iota \in \mathcal{I}$, let

$$K_t^\iota(h, \cdot) = \mathbb{P}^\iota(Y_t^+ \in \cdot \mid \mathbf{H}_t = h)$$

be a regular post-intervention kernel on the common pre-intervention history space. Define

$$\boxed{Z_t^{\mathcal{I}} = \big(K_t^\iota(\mathbf{H}_t, \cdot)\big)_{\iota \in \mathcal{I}} \in \mathcal{P}(\mathsf{Y})^{\mathcal{I}}.}$$

Corollary 60 (Minimal simultaneous interventional state)**.** *The random element $Z_t^{\mathcal{I}}$ is sufficient simultaneously for all post-intervention target laws. If a statistic $S_t = s(\mathbf{H}_t)$ determines the conditional target law for every $\iota \in \mathcal{I}$, then there exists a measurable $\phi_{\mathcal{I}}$ such that*

$$Z_t^{\mathcal{I}} = \phi_{\mathcal{I}}(S_t) \qquad a.s.$$

Proof. Apply Theorem 56 componentwise. A countable product of standard Borel spaces is standard Borel, so the component maps combine into a single measurable factor map. □

The observational state is the special case $\mathcal{I} = \{\mathrm{id}\}$. A causal workspace claim should align with $Z_t^{\mathcal{I}}$ for a declared intervention family rather than only with the observational state.

21. Structural–Operational Alignment

A structural workspace coordinate becomes operational only when it retains the canonical predictive or interventional state relevant to the declared task. Let

$$W_t = \pi_t(\mathbf{H}_t)$$

be a measurable candidate workspace statistic satisfying the structural support condition

$$W_t \in \Gamma_{\varepsilon_t}(H(t)) \qquad \text{almost surely.}$$

The map π_t may include encoding into a moving low-energy bundle; no canonical operational interpretation follows from the spectral projector alone.

Definition 61 (Exact structural–operational alignment)**.** The candidate W_t is observationally aligned when there exists a measurable map ϕ_t such that

$$Z_t = \phi_t(W_t) \qquad \text{a.s.}$$

It is $\mathcal{I}$-causally aligned when

$$Z_t^{\mathcal{I}} = \phi_t^{\mathcal{I}}(W_t) \qquad \text{a.s.}$$

Theorem 62 (Exact alignment criterion)**.** *For any workspace statistic $W_t = \pi_t(\mathbf{H}_t)$, the following are equivalent:*

1. $Y_t^+ \perp\!\!\!\perp \mathbf{H}_t \mid W_t$*;*
2. $I(Y_t^+; \mathbf{H}_t \mid W_t) = 0$*;*
3. *the canonical predictive state factors through W_t:*

$$Z_t = \phi_t(W_t) \quad \textit{a.s.}$$

for some measurable ϕ_t.

The analogous equivalence holds simultaneously over $\mathcal{I}$ with $Z_t^{\mathcal{I}}$.

Proof. The equivalence of the first two statements is the zero-divergence characterization of conditional mutual information. The first statement and Theorem 56 imply the factorization in the third. Conversely, if $Z_t = \phi_t(W_t)$, then

$$\mathbb{P}(Y_t^+ \in A \mid \mathbf{H}_t) = Z_t(A) = \phi_t(W_t)(A),$$

which depends on the history only through W_t, proving sufficiency. The interventional statement follows from Corollary 60. □

Define the observational alignment defect by

$$\boxed{\delta_{\text{align}}^{\text{obs}}(W_t) = I_{\mathbb{P}}(Y_t^+; \mathbf{H}_t \mid W_t).}$$

For a countable intervention family with declared weights $q_\iota > 0$, $\sum_\iota q_\iota = 1$, define

$$\boxed{\delta_{\text{align}}^{\mathcal{I}}(W_t) = \sum_{\iota \in \mathcal{I}} q_\iota I_{\mathbb{P}^\iota}(Y_t^+; \mathbf{H}_t \mid W_t).}$$

For bounded decision losses, Proposition 59 applies under each intervention and gives the weighted bound

$$\sum_\iota q_\iota \left[\mathcal{R}_\ell^\iota(W_t) - \mathcal{R}_\ell^\iota(\mathbf{H}_t)\right] \le \sqrt{\frac{\delta_{\text{align}}^{\mathcal{I}}(W_t)}{2}}.$$

Structural coherence and operational alignment are independent

The static defect

$$\delta_{\mathrm{coh}}(W_t)$$

and the operational defect

$$\delta^{\mathcal{I}}_{\mathrm{align}}(W_t)$$

answer different questions. The first measures compatibility under transfer. The second measures loss of task-relevant conditional law.

Regime	Structural status	Operational interpretation
Low coherence defect, low alignment defect	Stable transported coordinate	Candidate operational workspace for the declared target and interventions.
Low coherence defect, high alignment defect	Stable but task-irrelevant shared structure	Archive, convention, or synchronized representation.
High coherence defect, low alignment defect	Predictively adequate but not transport-stable	Localized or brittle sufficient statistic.
High coherence defect, high alignment defect	Neither stable nor sufficient	Rejected candidate at the stated scales.

Thus neither low transfer energy nor high predictive performance alone is enough. The combined gate is

$$\boxed{\delta_{\mathrm{coh}}(W_t) \le \varepsilon_{\mathrm{coh}}, \qquad \delta^{\mathcal{I}}_{\mathrm{align}}(W_t) \le \varepsilon_{\mathrm{align}}.}$$

Example 63 (Coherent but irrelevant archive)**.** Suppose every node perfectly copies a common random variable R, so the corresponding transfer energy is zero. If $R \perp\!\!\!\perp Y_t^+$, then

$$\delta_{\mathrm{coh}}(R) = 0, \qquad \delta^{\mathrm{obs}}_{\mathrm{align}}(R) = I(Y_t^+; \mathbf{H}_t \mid R),$$

which can be as large as the entire predictive information in the history. Perfect agreement therefore does not imply workspace functionality.

Example 64 (Sufficient but localized statistic)**.** If one node contains the canonical state Z_t and all transfers from that node are unstable, then $W_t = Z_t$ is sufficient but need not lie in a stable low-energy transfer subspace. This is a local controller or archive, not yet a stratified shared workspace.

PART VII

Relational Irreducibility

22. Coalition Information and Bayes Risk

For $A \subseteq V$, the coalition statistic is $\mathbf{H}_t^A$. Given an action space A and loss ℓ, define its unrestricted Bayes risk by

$$\boxed{\mathcal{R}_\ell(A) = \inf_d \mathbb{E}\Big[\ell(Y_t^+, d(\mathbf{H}_t^A))\Big].}$$

The infimum is over measurable decision rules for which the expectation is defined. The full-system risk is $\mathcal{R}_\ell(V)$.

Proposition 65 (Coalition monotonicity)**.** *If $A \subseteq B \subseteq V$, then*

$$\mathcal{R}_\ell(B) \le \mathcal{R}_\ell(A).$$

Proof. Every decision rule available to coalition A is also available to coalition B, which can ignore its additional information. □

For finite Y, logarithmic loss converts risk gaps into conditional information exactly.

Theorem 66 (Coalition log-risk identity)**.** *Assume $H(Y_t^+) < \infty$. If $A \subseteq B \subseteq V$, then*

$$\boxed{\mathcal{R}_{\log}(A) - \mathcal{R}_{\log}(B) = I\Big(Y_t^+; \mathbf{H}_t^B \mid \mathbf{H}_t^A\Big).}$$

In particular,

$$\mathcal{R}_{\log}(A) - \mathcal{R}_{\log}(V) = I\Big(Y_t^+; \mathbf{H}_t \mid \mathbf{H}_t^A\Big).$$

Proof. Under logarithmic loss the optimal report is the conditional target law, so

$$\mathcal{R}_{\log}(A) = H(Y_t^+ \mid \mathbf{H}_t^A).$$

Because $\mathbf{H}_t^A$ is a function of $\mathbf{H}_t^B$, subtracting the two conditional entropies gives the stated conditional mutual information. □

The theorem makes the decomposition explicit: irreducibility is not an unqualified property of a system. It is a property relative to the chosen node partition, accessible histories, target, and decision loss.

23. Joint Advantage and Its Scope

Define the coalition gap

$$\Delta_A^\ell = \mathcal{R}_\ell(A) - \mathcal{R}_\ell(V) \ge 0.$$

The node-level joint advantage is

$$\boxed{\Delta_{\text{joint}}^\ell = \min_{i \in V} \Delta_{\{i\}}^\ell.}$$

Under logarithmic loss,

$$\boxed{\Delta_{\text{joint}}^{\log} = \min_{i \in V} I\left(Y_t^+; \mathbf{H}_t \mid \mathbf{H}_t^{\{i\}}\right).}$$

Therefore

$$\Delta_{\text{joint}}^{\log} > 0$$

means that every individual component omits some predictive information available to the complete system.

This quantity should not automatically be identified with a particular notion of partial-information synergy. A positive gap can arise from complementary information distributed across components, from higher-order interaction, or from both. The theory uses the operationally weaker and more directly testable statement:

no individual node is sufficient for the declared target.

For $0 \le r \le |V|$, define the order-r coalition gap profile by

$$\boxed{\Delta_\ell^{(r)} = \min_{\substack{A \subseteq V \\ |A| \le r}} \left[\mathcal{R}_\ell(A) - \mathcal{R}_\ell(V)\right].}$$

The profile is nonincreasing in r and reaches zero by $r = |V|$. It records how rapidly operative capacity becomes reconstructible as larger coalitions are permitted.

24. Minimal Sufficient Coalition

For tolerance $\varepsilon \ge 0$, define

$$\boxed{k_\varepsilon = \min\left\{|A| : \mathcal{R}_\ell(A) \le \mathcal{R}_\ell(V) + \varepsilon\right\}.}$$

The set of minimal coalitions is

$$\mathcal{K}_\varepsilon = \left\{A \subseteq V : |A| = k_\varepsilon, \mathcal{R}_\ell(A) \le \mathcal{R}_\ell(V) + \varepsilon\right\}.$$

Corollary 67 (Information form of coalition size)**.** *Under the assumptions of Theorem 66,*

$$\boxed{k_\varepsilon = \min\left\{|A| : I(Y_t^+; \mathbf{H}_t \mid \mathbf{H}_t^A) \le \varepsilon\right\}.}$$

Equivalently, k_ε is the first r for which $\Delta_{\log}^{(r)} \le \varepsilon$.

This yields a graded classification:

$$\begin{aligned}
k_0 = 0 &\implies \text{the background law alone is sufficient,} \\
k_0 = 1 &\implies \text{some individual node is sufficient,} \\
k_0 \ge 2 &\implies \text{node-level relational irreducibility,} \\
k_0 = |V| &\implies \text{every proper coalition is insufficient.}
\end{aligned}$$

The last condition is much stronger and should not be inferred from the first.

25. Reconstructibility of the Canonical State

Let $d_{\mathcal{P}}$ be a bounded metric generating the measurable structure on $\mathcal{P}(\mathsf{Y})$. Define the reconstruction defect of the canonical state from coalition A by

$$\delta_A^Z = \inf_{\psi_A} \mathbb{E}\Big[d_{\mathcal{P}}\Big(Z_t, \psi_A(\mathbf{H}_t^A)\Big)\Big].$$

Proposition 68 (Sufficiency–reconstruction equivalence)**.** *For a coalition $A \subseteq V$, the following are equivalent:*

1. $\mathbf{H}_t^A$ *is sufficient for* Y_t^+ *relative to the full history;*
2. Z_t *is measurable with respect to* $\mathcal{F}_t^A$*;*
3. *there exists a measurable* ψ_A *such that*

$$Z_t = \psi_A(\mathbf{H}_t^A) \qquad \textit{almost surely.}$$

Each condition implies $\delta_A^Z = 0$. Under finite-target logarithmic loss, they are also equivalent to

$$\mathcal{R}_{\log}(A) = \mathcal{R}_{\log}(V).$$

If the reconstruction infimum is known to be attained, then $\delta_A^Z = 0$ is itself equivalent to the three conditions.

Proof. The equivalence of the first two statements is Theorem 56; the second and third are two formulations of measurability up to null sets. An exact reconstructor has zero expected metric error. The log-risk statement follows from Theorem 66 and the zero condition for conditional mutual information. Without an attainment or compactness condition, an infimum of zero alone certifies arbitrarily accurate reconstruction but need not supply a single exact minimizing map. □

For an arbitrary enriched workspace variable W_t, local non-reconstructibility need not imply predictive irreducibility: W_t may contain task-irrelevant material. The equivalence is exact for the canonical state Z_t, or for a candidate W_t known to be minimally aligned with it.

26. Matched Relational Nulls

A null control must preserve the local facts that the claim is not intended to challenge while destroying the relational organization being tested. Merely reducing model capacity or replacing the workspace by noise is not a matched null.

Let P be the intact law of $(Y_t^+, \mathbf{H}_t^1, \dots, \mathbf{H}_t^n)$. Define the local-marginal null class

$$\mathcal{N}_{\mathrm{marg}}(P) = \Big\{Q : Q_{\mathbf{H}_t^i} = P_{\mathbf{H}_t^i} \text{ for every } i\Big\},$$

and the stronger local-predictive-matched class

$$\boxed{\mathcal{N}_{\mathrm{pred}}(P) = \Big\{Q : Q_{(Y_t^+, \mathbf{H}_t^i)} = P_{(Y_t^+, \mathbf{H}_t^i)} \text{ for every } i\Big\}.}$$

Because $\mathbf{H}_t^i$ can include the complete local trajectory, the latter class can preserve all single-component temporal and predictive behavior while altering cross-component organization.

Proposition 69 (Local-risk invariance under predictive matching)**.** *If $Q \in \mathcal{N}_{\mathrm{pred}}(P)$, then for every node i and every loss determined by $(Y_t^+, d(\mathbf{H}_t^i))$,*

$$\boxed{\mathcal{R}_{\ell,Q}(\{i\}) = \mathcal{R}_{\ell,P}(\{i\}).}$$

Proof. Every such risk depends only on the joint law of $(Y_t^+, \mathbf{H}_t^i)$, which is identical under P and Q. □

A valid null must also satisfy a declared relation-breaking condition. Examples include:

- permuting source–target message pairings while preserving each message marginal;
- shuffling temporal correspondence across nodes while preserving every local trajectory;
- permuting role bindings while preserving the local token or state distributions;
- replacing an artifact by a matched-size, matched-format artifact whose relation to the surrounding history has been destroyed; and
- randomizing transfer channels through a physical or software intervention while preserving message rate and marginal content.

For a predeclared null Q, define

$$\boxed{\Delta_{\mathrm{tr}}(Q) = \mathcal{R}_{\ell,Q}(V) - \mathcal{R}_{\ell,P}(V).}$$

For a finite validated panel $\mathcal{Q} = \{Q_1, \dots, Q_m\} \subseteq \mathcal{N}_{\mathrm{pred}}(P)$, report both the complete vector

$$\boldsymbol{\Delta}_{\mathrm{tr}}(\mathcal{Q}) = (\Delta_{\mathrm{tr}}(Q_1), \dots, \Delta_{\mathrm{tr}}(Q_m))$$

and the robust panel minimum

$$\Delta_{\mathrm{tr}}^{\min}(\mathcal{Q}) = \min_{Q \in \mathcal{Q}} \Delta_{\mathrm{tr}}(Q).$$

The panel must be fixed independently of the observed favorable gap. Otherwise null selection can manufacture apparent relational dependence.

27. Relational Witness Pairs

Definition 70 (Relational witness pair)**.** A pair of laws (P, Q) is a predictive relational witness pair for the decomposition V and target Y_t^+ when

$$Q \in \mathcal{N}_{\mathrm{pred}}(P)$$

but

$$\mathcal{R}_{\ell,Q}(V) \neq \mathcal{R}_{\ell,P}(V).$$

The two systems agree on every single-node predictive marginal but differ in full-system operative capacity.

Example 71 (XOR witness pair)**.** Under the intact law P, let

$$X_1, X_2 \overset{\mathrm{iid}}{\sim} \mathrm{Bernoulli}(1/2), \qquad Y = X_1 \oplus X_2.$$

Then each (X_i, Y) pair is independent and uniform, so

$$H_P(Y \mid X_i) = 1 \text{ bit}, \qquad H_P(Y \mid X_1, X_2) = 0.$$

Let Q make X_1, X_2, Y mutually independent fair bits. Then

$$Q_{(X_i,Y)} = P_{(X_i,Y)} \qquad (i = 1, 2),$$

but

$$H_Q(Y \mid X_1, X_2) = 1 \text{ bit}.$$

Therefore

$$\Delta^{\log}_{\text{joint}} = 1 \text{ bit}, \qquad k_0 = 2, \qquad \Delta_{\text{tr}}(Q) = 1 \text{ bit}.$$

This proves predictive relational structure relative to the declared variables. It does not, by itself, prove that any workspace variable is causally active.

Example 72 (Redundant-copy control)**.** Let Y be a fair bit and set $X_1 = X_2 = Y$. The two nodes are perfectly coherent and the full system predicts perfectly, but

$$H(Y \mid X_i) = 0, \qquad \Delta^{\log}_{\text{joint}} = 0, \qquad k_0 = 1.$$

Duplication is therefore not relational irreducibility.

The witness-pair criterion is stronger than a performance comparison against a smaller model because it holds local predictive behavior fixed. It remains observational. Part VIII adds interventions and converts predictive relationality into an operational causal claim.

PART VIII

Causal Efficacy and Reentry

28. Feasible Workspace Interventions

A notation such as $\mathrm{do}(W_t = w)$ is meaningful only if the candidate workspace is an endogenous variable in a structural causal model or if an intervention on underlying variables is shown to realize that value. Abstractly setting a projection coordinate need not correspond to any possible physical or computational manipulation.

Definition 73 (Time-unrolled causal realization)**.** A time-unrolled causal realization is a structural causal model

$$\mathcal{M} = (\mathcal{U}, \mathcal{V}, F, \mathbb{P}_U)$$

whose endogenous variables include the local states, messages, actions, artifacts, target variables, and a measurable workspace extractor

$$\pi_t : \mathsf{S}_t \to \mathsf{W}_t, \qquad W_t = \pi_t(S_t).$$

The unrolled graph is acyclic even when the rolled process is recurrent.

Definition 74 (Feasible implementation intervention)**.** Let B_t denote declared background variables that should remain matched. A family $\{\iota_w : w \in \mathsf{W}_t^\circ\}$ is a feasible implementation of workspace values on the experimental domain W_t° when:

1. each ι_w acts only on declared manipulable parents, channels, or artifacts at or after time t;
2. the intervened realization satisfies
$$\pi_t(S_t^{\iota_w}) = w \qquad \text{almost surely;}$$
3. the background law is matched across compared interventions; and
4. the intervention does not directly modify Y_t^+ except through paths included in the declared causal design.

Let D be a divergence or metric on probability laws. For a matched background value b, define

$$\mathrm{CE}_{\mathbf{W}}(w, w'; b) = D\Big(\mathbb{P}^{\iota_w}(Y_t^+ \in \cdot \mid B_t = b), \mathbb{P}^{\iota_{w'}}(Y_t^+ \in \cdot \mid B_t = b)\Big).$$

For a declared comparison design ν over (w, w', b), define

$$\boxed{\eta_{\mathrm{causal}}^{\nu} = \mathbb{E}_{(w,w',b)\sim\nu}\left[\mathrm{CE}_{\mathbf{W}}(w, w'; b)\right].}$$

The maximum

$$\eta_{\mathrm{causal}}^{\max} = \sup_{w,w',b} \mathrm{CE}_{\mathbf{W}}(w, w'; b)$$

may be useful for discovery, but a design-weighted effect is harder to inflate by selecting a single exceptional pair after inspection.

Proposition 75 (Coordinate invariance of intervention effects)**.** *Let $g : \mathsf{W}_t \to \mathsf{W}'_t$ be a measurable bijection and suppose the reparametrized intervention family satisfies*

$$\iota'_{g(w)} = \iota_w.$$

Then every divergence-based causal effect is preserved:

$$\mathrm{CE}_{\mathbf{W}'}(g(w), g(w'); b) = \mathrm{CE}_{\mathbf{W}}(w, w'; b).$$

Proof. The two labels designate the same underlying interventions and therefore induce the same pair of target laws. Only the coordinate name has changed. □

This is the causal analogue of gauge invariance. It also shows why effects should be attached to intervention implementations rather than to arbitrary numerical coordinates.

29. Causal Specificity and Transfer Mediation

A causal effect on the target is not workspace-specific if an equal-cost perturbation outside the candidate workspace produces the same effect. Let $c(\iota)$ be a declared intervention cost and let

$$\mathcal{I}_{\mathbf{W}}(r), \qquad \mathcal{I}_{\perp}(r)$$

be matched families with cost r, the first changing workspace coordinates and the second changing complementary coordinates while preserving the workspace value as closely as the implementation permits.

Define

$$\eta_{\mathbf{W}}(r) = \mathbb{E}_{\iota,\iota' \sim \mathcal{I}_{\mathbf{W}}(r)} D\Big(\mathbb{P}^{\iota}(Y_t^+), \mathbb{P}^{\iota'}(Y_t^+)\Big),$$

$$\eta_{\perp}(r) = \mathbb{E}_{\jmath,\jmath' \sim \mathcal{I}_{\perp}(r)} D\Big(\mathbb{P}^{\jmath}(Y_t^+), \mathbb{P}^{\jmath'}(Y_t^+)\Big),$$

and

$$\boxed{\Delta_{\mathrm{specific}}(r) = \eta_{\mathbf{W}}(r) - \eta_{\perp}(r).}$$

A positive value supports causal concentration in the candidate workspace at the declared perturbation scale.

If the complement is defined by $\Gamma_{\varepsilon}(H)^{\perp}$, the result is metric dependent. Under nonunitary gauge changes, either the metric must be transported with the coordinates or interventions must be matched by a coordinate-independent physical cost. Otherwise specificity can change merely because the representation was rescaled.

Causal transfer nulls

Let $\jmath_E$ randomize selected communication edges while preserving their marginal message laws, rates, and timing budget. The causal transfer gap is

$$\boxed{\Delta_{\mathrm{tr}}^{\mathrm{do}}(\jmath_E) = \mathcal{R}_{\ell,\mathbb{P}^{\jmath_E}}(V) - \mathcal{R}_{\ell,\mathbb{P}}(V).}$$

Unlike an observational null law, this quantity is anchored to an implemented channel intervention. A positive effect is evidence that the intact alignment of messages, rather than merely their marginal content, contributes to task performance.

A stronger mediation test equalizes the workspace state after two distinct channel interventions. Let μ be a common imposed workspace distribution, implemented by a feasible stochastic intervention in the sense of Definition 73. The notation below is shorthand for that implementation. Define the residual mediation defect

$$\delta_{\mathrm{med}} = \sup_{\jmath,\jmath'} D\Big(\mathbb{P}^{\jmath,\mathrm{do}(W_t\sim\mu)}(Y_t^+), \mathbb{P}^{\jmath',\mathrm{do}(W_t\sim\mu)}(Y_t^+)\Big).$$

A small value means that once the workspace state is equalized, little of the channel intervention's effect remains. This supports, but does not by itself identify without the causal assumptions, the claim that W_t mediates the relevant relational influence.

30. Causal Recurrence, Loop-Mediated Reentry, and Relationality

Temporal correlation is weaker than causal recurrence. Define the total workspace recurrence effect at lag τ by

$$\boxed{\eta_{\mathrm{rec}}(\tau) = \mathbb{E}_{(w,w',b)\sim\nu} D(\mathbb{P}^{\iota_w}(W_{t+\tau}\in\cdot \mid B_t = b), \mathbb{P}^{\iota_{w'}}(W_{t+\tau}\in\cdot \mid B_t = b))\,.}$$

A nonzero value shows that manipulating the present workspace changes a later workspace state. It may, however, be explained by direct copying or a retained memory register.

To isolate reentry, consider the unrolled causal graph and remove every direct retention edge whose source and target are both workspace variables at successive times. Let

$$\mathcal{M}^{\backslash\mathrm{dir}}$$

denote the resulting path-specific model, preserving paths that leave the workspace through a nonworkspace participant, action, message, or artifact and later return. Define

$$\boxed{\eta_{\mathrm{loop}}(\tau) = \mathbb{E}_{(w,w',b)\sim\nu} D\Big(\mathbb{P}^{\iota_w}_{\mathcal{M}^{\backslash\mathrm{dir}}}(W_{t+\tau}\in\cdot \mid B_t = b), \mathbb{P}^{\iota_{w'}}_{\mathcal{M}^{\backslash\mathrm{dir}}}(W_{t+\tau}\in\cdot \mid B_t = b)\Big).}$$

Then the distinctions are:

$$\begin{aligned} I(W_t; W_{t+\tau}) > 0 &\quad \text{temporal dependence only,}\\ \eta_{\mathrm{rec}}(\tau) > 0 &\quad \text{causal persistence,}\\ \eta_{\mathrm{loop}}(\tau) > 0 &\quad \text{causal exit-and-return reentry.}\end{aligned}$$

Path-specific effects require the usual consistency and no-unmeasured-confounding assumptions appropriate to the implementation. In software systems, edge interventions can often be literal. In human collaboration, these assumptions are substantially harder to justify.

Theorem 76 (Operational relationality, T_{13})**.** *Let Y be finite, let P be the intact law at a fixed decision time, and let $W_t = \pi_t(\mathbf{H}_t)$ be a structurally supported candidate. Let $\mathcal{Q} \subseteq \mathcal{N}_{\mathrm{pred}}(P)$ be a nonempty, predeclared panel of validated relation-breaking nulls. Assume:*

1. *exact structural–operational alignment,*

$$I_P(Y_t^+; \mathbf{H}_t \mid W_t) = 0;$$

2. *a positive node-level information gap,*

$$\delta_{\mathrm{node}} = \min_{i\in V} I_P(Y_t^+; \mathbf{H}_t \mid \mathbf{H}_t^i) > 0;$$

3. *a positive robust matched-null gap,*

$$\delta_{\text{null}} = \min_{Q \in \mathcal{Q}} \left[H_Q(Y_t^+ \mid \mathbf{H}_t) - H_P(Y_t^+ \mid \mathbf{H}_t) \right] > 0;$$

4. *a feasible intervention design with*

$$\eta^{\nu}_{\text{causal}} > 0;$$

5. *positive matched causal specificity at some declared scale r,*

$$\Delta_{\text{specific}}(r) > 0.$$

Then, relative to the declared decomposition, target, log loss, null panel, and intervention family:

1. *no statistic of any single-node history is sufficient for Y_t^+, and $k_0 \geq 2$;*
2. *W_t cannot be reconstructed almost surely from any single-node history;*
3. *every null $Q \in \mathcal{Q}$ preserves all single-node predictive risks but has strictly worse full-system log risk than the intact law;*
4. *the candidate workspace is causally efficacious and its effect exceeds that of the declared matched off-workspace perturbations.*

If, in addition, $\eta_{\text{loop}}(\tau) > 0$, the workspace is loop-reentrant at lag τ.

Proof. By Theorem 66, for each node i,

$$\mathcal{R}_{\log,P}(\{i\}) - \mathcal{R}_{\log,P}(V) = I_P(Y_t^+; \mathbf{H}_t \mid \mathbf{H}_t^i) \geq \delta_{\text{node}} > 0.$$

Hence no single-node history, and therefore no statistic of one, is sufficient; this gives $k_0 \geq 2$.

Exact alignment and Theorem 62 imply that the canonical predictive state factors through W_t. If $W_t = \psi_i(\mathbf{H}_t^i)$ almost surely for some node, then the canonical state would also be a function of $\mathbf{H}_t^i$, contradicting the positive node gap through Proposition 68.

For every $Q \in \mathcal{Q}$, Proposition 69 gives

$$\mathcal{R}_{\log,Q}(\{i\}) = \mathcal{R}_{\log,P}(\{i\}) \qquad \forall i.$$

The assumed null gap gives

$$\mathcal{R}_{\log,Q}(V) > \mathcal{R}_{\log,P}(V).$$

Thus a description determined only by the collection of single-node predictive marginals cannot reproduce the intact joint capacity: it would assign the same local data to P and Q while their full-system risks differ.

Finally, the feasible intervention effect establishes causal efficacy relative to the declared implementation, and the positive specificity gap shows that the effect exceeds the matched complementary perturbation effect. The final reentry statement is exactly the definition of η_{loop}. □

Remark 77 (Scope of the theorem)**.** The conclusion is operational and indexed by

$$(V, Y_t^+, \ell, \mathcal{Q}, \mathcal{I}, \nu, r).$$

Changing the decomposition, target, loss, nulls, or intervention design may change the verdict. The theorem does not establish metaphysical irreducibility, autonomous agency, selfhood, or consciousness.

Finite certification sequence

A reproducible dynamic-causal analysis should proceed in the following order:

1. Freeze the node decomposition, accessible coalition histories, target, horizon, and proper scoring loss.
2. Estimate the structural projector on training data only and encode W_t without using held-out targets.
3. Cross-fit conditional target laws from the full history, the workspace, and every declared coalition. Report alignment and coalition risk gaps.
4. Construct the null panel before inspecting null performance. Verify local-predictive matching and the intended destruction of transfer organization separately.
5. Implement feasible workspace and complementary interventions with a common background and cost design. Report both causal effect and specificity.
6. Test recurrence and then cut direct retention paths to test loop-mediated reentry.
7. Report confidence intervals or finite-sample uncertainty for every gate; do not substitute training accuracy for Bayes-risk or causal estimates.

The resulting certification report is

$$\boxed{\begin{aligned}\mathcal{C}_{\mathrm{dyn}}(W_t) = (&\delta^{\mathcal{I}}_{\mathrm{align}}, \{\Delta^{\ell}_{A}\}_{A\subseteq V}, k_{\varepsilon}, \boldsymbol{\Delta}_{\mathrm{tr}}(\mathcal{Q}),\\ &\eta^{\nu}_{\mathrm{causal}}, \Delta_{\mathrm{specific}}(r), \eta_{\mathrm{rec}}(\tau), \eta_{\mathrm{loop}}(\tau)).\end{aligned}}$$

No single coordinate certifies a relational workspace.

PART IX

Formal Workspace Definition

31. Certified Stratified Relational Workspace

Let

$$\Theta = \left(\varepsilon_{\mathrm{coh}}, \varepsilon_{\mathrm{align}}, \varepsilon_k, \underline{\eta}_c, \underline{\eta}_{\mathrm{loop}}, \delta_{\mathrm{node}}, \delta_{\mathrm{null}}, \delta_{\mathrm{spec}}, k, \varepsilon_{\mathrm{pers}}\right).$$

Let

$$\varsigma = (\mathcal{D}, Y^+, \ell, \tau, \mathcal{I}, \mathcal{Q}, \Theta)$$

be the complete declared scientific specification, where $\mathcal{Q} \subseteq \mathcal{N}_{\mathrm{pred}}(P)$ is a validated null panel. A dynamic realization contains a certified stratified relational workspace when there exists a candidate W_t satisfying all gates required for the claimed level under ς. The detailed robustness and identifiability requirements for varying ς are developed in Parts XVII and XVIII.

Structural coherence

$$\delta_{\mathrm{coh}}(W_t) \le \varepsilon_{\mathrm{coh}}.$$

Interventional structural–operational alignment

$$\delta^{\mathcal{I}}_{\mathrm{align}}(W_t) \le \varepsilon_{\mathrm{align}}.$$

Causal efficacy

$$\eta^{\nu}_{\mathrm{causal}} \ge \underline{\eta}_c.$$

Loop-mediated reentry, when claimed

$$\eta_{\mathrm{loop}}(\tau) \ge \underline{\eta}_{\mathrm{loop}}.$$

Node-level joint advantage

$$\Delta^{\ell}_{\mathrm{joint}} \ge \delta_{\mathrm{node}}.$$

Matched-null transfer dependence

$$\Delta^{\min}_{\mathrm{tr}}(\mathcal{Q}) \ge \delta_{\mathrm{null}}.$$

Causal specificity

$$\Delta_{\text{specific}}(r) \geq \delta_{\text{spec}}.$$

Distributedness

$$k_{\varepsilon_k} \geq k.$$

Persistence, when claimed

$$d_{\text{id}}(W_t, W_{t+\tau}) \leq \varepsilon_{\text{pers}}.$$

Symbolically,

$$\boxed{\text{SRW}(W_t;\varsigma) = \bigwedge_{q\in\mathcal{J}(\varsigma)} \text{Gate}_q(W_t;\varsigma).}$$

The gate set is claim-relative. A one-step operational workspace need not pass persistence or reentry gates, while a persistent reentrant workspace must. No scalar total score is sufficient: failure of a required gate cannot be hidden by high performance on another gate.

Remark 78 (Structural, predictive, causal, and temporal claims)**.** The formal hierarchy now distinguishes four logically independent questions:

Does a stable shared coordinate exist?

Does it retain the task-relevant conditional law?

Does manipulating it change downstream behavior specifically?

Does it persist or reenter through time?

The preceding structural and causal analysis answers the first three at the level of definitions and conditional theorems. The following parts develop the fourth.

PART X

Persistence

The preceding parts identify exact and approximate workspace subspaces at a single time. Persistence adds three further requirements. The structural subspace must remain trackable while the realization changes, actual states must remain near that moving subspace, and the transported content must not collapse to a trivial zero state. These are distinct conditions.

32. Coherence-Error Dynamics

32.1 Time-indexed transfer systems

At time t, let

$$\mathcal{C}_V(t) = \bigoplus_{i \in V} C_i(t), \qquad \mathcal{C}_E(t) = \bigoplus_{e \in E} C_{t(e)}(t),$$

and let

$$D_t : \mathcal{C}_V(t) \longrightarrow \mathcal{C}_E(t), \qquad L_t = D_t^* D_t$$

be the transfer discrepancy and transfer Laplacian. At time t, the combinatorial graph is held fixed while the local representations, transfers, and metrics may vary. Changes of graph are treated later by a graph correspondence.

To compare successive times, choose unitary identification maps

$$U_t^V : \mathcal{C}_V(t) \longrightarrow \mathcal{C}_V(t+1), \qquad U_t^E : \mathcal{C}_E(t) \longrightarrow \mathcal{C}_E(t+1).$$

They are comparison maps, not assumptions that the physical carriers are identical. Define the one-step diagram defect

$$\boxed{\Delta_t^D = D_{t+1} U_t^V - U_t^E D_t, \qquad \delta_t^D = \|\Delta_t^D\|.}$$

Exact temporal compatibility is the condition $\Delta_t^D = 0$.

Proposition 79 (Temporal Laplacian drift)**.** *The carried Laplacian satisfies*

$$\boxed{\left\| (U_t^V)^* L_{t+1} U_t^V - L_t \right\| \le 2\|D_t\|\delta_t^D + (\delta_t^D)^2.}$$

Proof. Set

$$B_t = (U_t^E)^* \Delta_t^D.$$

Because the comparison maps are unitary,

$$(U_t^E)^* D_{t+1} U_t^V = D_t + B_t, \qquad \|B_t\| = \delta_t^D.$$

Hence

$$(U_t^V)^* L_{t+1} U_t^V = (D_t + B_t)^* (D_t + B_t).$$

Subtract $D_t^* D_t$ and apply the triangle inequality. □

Let P_t project onto a selected rank-d low-energy workspace cluster of L_t, and set

$$\mathcal{W}_t = \operatorname{ran} P_t.$$

Write

$$g_t = \lambda_{d+1}(L_t) - \lambda_d(L_t) > 0$$

for the separating spectral gap and

$$\eta_t^L = 2\|D_t\|\delta_t^D + (\delta_t^D)^2.$$

Corollary 80 (Temporal workspace tracking)**.** *If*

$$\eta_t^L < \frac{g_t}{2},$$

then the transported and newly extracted workspace projectors satisfy

$$\boxed{q_t^{\mathrm{str}} := \left\| P_{t+1} - U_t^V P_t (U_t^V)^* \right\| \le \frac{\eta_t^L}{g_t - \eta_t^L} \le \frac{2\eta_t^L}{g_t}.}$$

Proof. Carry L_{t+1} back to time t using U_t^V, apply Theorem 43, and conjugate the resulting projector bound forward. □

The quantity q_t^{str} measures structural motion after the declared coordinate change has been removed. A workspace may therefore move substantially in raw coordinates while remaining structurally unchanged.

32.2 State tracking inside a moving workspace

Let the realized vertex state satisfy

$$x_{t+1} = F_t x_t + \xi_t,$$

where $F_t : \mathcal{C}_V(t) \to \mathcal{C}_V(t+1)$ is the actual update and ξ_t is an unmodeled disturbance. Define the off-workspace error

$$\boxed{e_t = \|(I - P_t)x_t\|.}$$

Introduce

$$\alpha_t = \|(I - P_{t+1})F_t(I - P_t)\|,$$

which measures persistence of prior off-workspace error, and

$$\ell_t = \|(I - P_{t+1})F_t P_t\|,$$

which measures leakage of a valid workspace state under the update.

Proposition 81 (Grounded coherence recurrence)**.** *Every trajectory satisfies*

$$\boxed{e_{t+1} \le \alpha_t e_t + \sigma_t,}$$

where

$$\boxed{\sigma_t = \ell_t \|P_t x_t\| + \|(I - P_{t+1})\xi_t\|.}$$

Proof. Insert $I = P_t + (I - P_t)$ before x_t, apply $I - P_{t+1}$, and use submultiplicativity:

$$\begin{aligned} e_{t+1} \le \|(I - P_{t+1})F_t(I - P_t)x_t\| \\ + \|(I - P_{t+1})F_tP_tx_t\| + \|(I - P_{t+1})\xi_t\|. \end{aligned}$$

The displayed recurrence follows from the definitions of α_t and ℓ_t. □

This derivation gives the contraction and disturbance terms an operational meaning. The scalar recurrence is not an independent postulate. It follows from an update rule, a moving workspace projector, and a declared disturbance model.

32.3 Persistence theorem T_6

Theorem 82 (Nonautonomous persistence theorem)**.** *Let $e_t \ge 0$ satisfy*

$$e_{t+1} \le \alpha_t e_t + \sigma_t, \qquad \alpha_t, \sigma_t \ge 0.$$

Then, for every $t \ge 1$,

$$\boxed{e_t \le \left(\prod_{r=0}^{t-1} \alpha_r\right) e_0 + \sum_{j=0}^{t-1} \left(\prod_{r=j+1}^{t-1} \alpha_r\right) \sigma_j,}$$

where an empty product equals one. If

$$\alpha_t \le \bar{\alpha} < 1, \qquad \sigma_t \le \bar{\sigma},$$

then

$$\boxed{e_t \le \bar{\alpha}^t e_0 + \frac{\bar{\sigma}(1 - \bar{\alpha}^t)}{1 - \bar{\alpha}},}$$

and consequently

$$\boxed{\limsup_{t\to\infty} e_t \le \frac{\bar{\sigma}}{1 - \bar{\alpha}}.}$$

Proof. Repeated substitution yields

$$e_t \le \alpha_{t-1} \cdots \alpha_0 e_0 + \alpha_{t-1} \cdots \alpha_1 \sigma_0 + \cdots + \sigma_{t-1},$$

which is the first formula. Under the uniform bounds, dominate each product by the corresponding power of $\bar{\alpha}$ and sum the geometric series. □

Corollary 83 (Vanishing-disturbance convergence)**.** *If $\alpha_t \le \bar{\alpha} < 1$ and $\sigma_t \to 0$, then*

$$e_t \to 0.$$

More generally, if $\sum_{t\ge0} \sigma_t < \infty$, then the disturbance contribution in Theorem 82 vanishes asymptotically.

Proof. Split the convolution sum into a finite early part, whose coefficients decay to zero, and a late tail on which σ_t is uniformly small. The summable case follows by the same argument. □

32.4 Mean-square persistence

For stochastic systems, pathwise bounded disturbances may be too strong. Let $\{\mathcal{F}_t\}$ be a filtration containing the history through time t.

Proposition 84 (Mean-square persistence)**.** *Suppose*

$$\mathbb{E}[e_{t+1}^2 \mid \mathcal{F}_t] \le q e_t^2 + v_t, \qquad 0 \le q < 1,$$

and $\mathbb{E}[v_t] \le \bar{v}$*. Then*

$$\boxed{\mathbb{E}[e_t^2] \le q^t \mathbb{E}[e_0^2] + \frac{\bar{v}(1-q^t)}{1-q},}$$

so

$$\limsup_{t\to\infty} \mathbb{E}[e_t^2] \le \frac{\bar{v}}{1-q}.$$

Proof. Take expectations, iterate the resulting deterministic recurrence, and sum a geometric series. □

32.5 The non-collapse condition

Low off-workspace error is not sufficient for meaningful persistence. The zero update $F_t = 0$ drives every state exactly into every linear workspace while destroying all content. Define the internal transport

$$B_t = P_{t+1} F_t P_t : \mathcal{W}_t \longrightarrow \mathcal{W}_{t+1}$$

and its non-collapse margin

$$\boxed{\mu_t^{\mathrm{keep}} = \sigma_{\min}(B_t|_{\mathcal{W}_t}).}$$

A content-preserving persistence claim requires

$$\mu_t^{\mathrm{keep}} > 0$$

for the relevant interval, or an information-theoretic replacement when the workspace state is stochastic. Structural tracking, state tracking, and content retention are therefore separate gates.

A finite-horizon persistence certificate may be recorded as

$$\boxed{\mathcal{P}_{0:T} = \left(\max_{t<T} q_t^{\mathrm{str}}, \max_{t\le T} e_t, \min_{t<T} \mu_t^{\mathrm{keep}}, \sum_{t<T} \sigma_t \right).}$$

No single coordinate of this vector subsumes the others.

33. Interpretation of the Contraction Parameter

The inequality

$$\alpha_t < 1$$

means that deviations already outside the current workspace are damped by the next update. Mechanisms that can reduce α_t or σ_t include

verification and error feedback,
shared notation and stable interfaces,
artifact persistence and version control,
proof checking and experimental replication,
handoff reconstruction and redundancy.

Sources of disturbance include

context loss and representational drift,
miscommunication and transfer-map error,
participant or model replacement,
artifact corruption,
unsupported inference and objective drift.

33.1 Four persistence levels

The theory distinguishes:

structural persistence	q_t^{str} remains small,
state persistence	e_t remains small,
content persistence	μ_t^{keep} remains bounded away from zero,
operational persistence	predictive and causal capacities remain matched.

The first three are developed in this part. Operational matching is incorporated into the identity theory of Part XII.

33.2 Moving-line example

Let $\mathcal{C}_V(t) = \mathbb{R}^2$ and let P_t project onto

$$u_t = (\cos\theta_t, \sin\theta_t).$$

If U_t^V is rotation through $\theta_{t+1} - \theta_t$ and $F_t = U_t^V$, then a state $x_t = u_t$ obeys

$$e_t = e_{t+1} = 0, \qquad \mu_t^{\mathrm{keep}} = 1.$$

The workspace changes raw coordinates but not abstract identity. If instead $F_t = I$, then

$$e_{t+1} = |\sin(\theta_{t+1} - \theta_t)|.$$

The same structural motion now produces a state-tracking defect because the actual update fails to transport the content with the workspace.

PART XI

Reconstruction and Resilience

Persistence can be maintained by direct transport, but it can also be restored from distributed traces after a disruption. Reconstruction theory quantifies when a coalition contains enough information to recover the abstract core and how noise is amplified by that recovery.

34. Core Encoding Maps

Let C_* be a finite-dimensional abstract core. The exact static theory provides a complete local chart

$$\overline{E}_i = \mathrm{ev}_i \circ \Psi : C_* \longrightarrow C_i,$$

where $\Psi : C_* \to \Gamma(H)$ is the root-to-section isomorphism. A participant or artifact need not expose that complete chart. Let

$$O_i : C_i \longrightarrow \mathsf{Y}_i$$

be the declared observation or access map and define the *accessible encoding*

$$\boxed{E_i = O_i \overline{E}_i : C_* \longrightarrow \mathsf{Y}_i.}$$

For notational economy, the direct sums below use the accessible spaces Y_i. Thus injectivity of a full structural chart does not make a singleton operationally reconstructive unless its access map preserves the whole core.

For a coalition $S \subseteq V$, define the analysis operator

$$A_S : C_* \longrightarrow \bigoplus_{i \in S} \mathsf{Y}_i, \qquad A_S c = (E_i c)_{i \in S},$$

and the coalition frame operator

$$\boxed{F_S = A_S^* A_S = \sum_{i \in S} E_i^* E_i.}$$

Set

$$\alpha_S = \lambda_{\min}(F_S), \qquad \beta_S = \lambda_{\max}(F_S).$$

Thus

$$\alpha_S \|c\|^2 \le \|A_S c\|^2 \le \beta_S \|c\|^2.$$

Proposition 85 (Coalition reconstruction criterion)**.** *For a coalition S, the following are equivalent:*

1. *A_S is injective;*
2. *F_S is positive definite;*
3. *$\alpha_S > 0$;*
4. *there exists a linear decoder R_S with $R_S A_S = I_{C_*}$.*

When these conditions hold, the canonical decoder is

$$\boxed{R_S = A_S^\dagger = F_S^{-1} A_S^*.}$$

Proof. In finite dimensions,

$$\ker F_S = \ker A_S$$

because

$$\langle c, F_S c \rangle = \|A_S c\|^2.$$

This proves the equivalence of the first three statements. An injective linear map admits a left inverse, while a left inverse implies injectivity. The formula for the Moore–Penrose left inverse follows from full column rank. □

The lower frame bound α_S measures recoverability. The upper bound β_S measures amplification in the forward encoding. Their ratio

$$\kappa_S = \sqrt{\frac{\beta_S}{\alpha_S}}$$

is the coalition condition number.

35. Stable Reconstruction Condition

35.1 Stable reconstruction theorem T_7

Theorem 86 (Stable core reconstruction)**.** *Assume $\alpha_S > 0$ and observe*

$$y = A_S c + \eta.$$

With the canonical estimate

$$\widehat{c} = A_S^{\dagger} y,$$

one has

$$\boxed{\|\widehat{c} - c\| \le \frac{\|\eta\|}{\sqrt{\alpha_S}}.}$$

Proof. Since $A_S^{\dagger} A_S = I_{C_*}$,

$$\widehat{c} - c = A_S^{\dagger} \eta.$$

The singular values of A_S are the square roots of the eigenvalues of F_S, so

$$\|A_S^{\dagger}\| = \frac{1}{\sigma_{\min}(A_S)} = \frac{1}{\sqrt{\alpha_S}}.$$

□

Corollary 87 (Encoding mismatch)**.** *If the true observation is*

$$y = (A_S + \Delta_S) c + \eta$$

but the nominal decoder $A_S^{\dagger}$ is used, then

$$\boxed{\|\widehat{c} - c\| \le \frac{\|\Delta_S\|\,\|c\| + \|\eta\|}{\sqrt{\alpha_S}}.}$$

Proof. Now

$$\widehat{c} - c = A_S^\dagger \Delta_S c + A_S^\dagger \eta,$$

and the result follows from Theorem 86. □

35.2 Reconstruction as operational recovery

Geometric recovery becomes behaviorally relevant when the workspace state controls a predictive kernel. Let

$$K_w = \mathbb{P}(Y \in \cdot \mid W = w)$$

and suppose

$$\mathrm{TV}(K_w, K_{w'}) \le L_K d_W(w, w').$$

Proposition 88 (Exact reconstruction implies coalition sufficiency)**.** *Let W be sufficient for Y relative to the full history $\mathbf{H}$:*

$$Y \perp\!\!\!\perp \mathbf{H} \mid W.$$

If a coalition reconstructs the workspace exactly,

$$W = R_S(\mathbf{H}^S) \qquad \textit{almost surely},$$

then

$$\boxed{Y \perp\!\!\!\perp \mathbf{H} \mid \mathbf{H}^S.}$$

Under logarithmic loss,

$$\boxed{\mathcal{R}_{\log}(S) = \mathcal{R}_{\log}(V).}$$

Proof. Sufficiency means that the conditional target law given $\mathbf{H}$ is a measurable function of W. Exact reconstruction makes that function measurable with respect to $\mathbf{H}^S$, which is the asserted conditional independence. The log-risk equality follows from Theorem 66. □

Proposition 89 (Reconstruction-to-risk transfer)**.** *Let W be sufficient for a target Y, let losses take values in $[0,1]$, and let a coalition construct $\widehat{W}$ from its local data. Then its optimal Bayes risk satisfies*

$$\boxed{0 \le \mathcal{R}(S) - \mathcal{R}(W) \le 2L_K\, \mathbb{E}[d_W(W, \widehat{W})].}$$

Proof. For every pair $w, \widehat{w}$, using the Bayes action for $K_{\widehat{w}}$ under the true law K_w incurs conditional regret at most

$$2\,\mathrm{TV}(K_w, K_{\widehat{w}}).$$

This follows by inserting and subtracting the risk under $K_{\widehat{w}}$ for both the chosen action and the Bayes action for K_w. Average the bound and use the Lipschitz assumption. The coalition's optimal rule performs at least as well as this decoder-based rule. □

This proposition links Part XI to the causal-state theory of Part VI. A small geometric reconstruction error is operationally meaningful only when the outcome kernel changes continuously with workspace state.

36. k-Failure Resilience

For $k \geq 0$, define the worst surviving lower frame bound

$$\boxed{\alpha^{(k)} = \min_{\substack{A \subseteq V \\ |A| \leq k}} \alpha_{V \setminus A}.}$$

The sequence $\alpha^{(k)}$ is nonincreasing in k.

Theorem 90 (Uniform erasure resilience)**.** *The encoding family is exactly reconstructible after every loss of at most k nodes if and only if*

$$\boxed{\alpha^{(k)} > 0.}$$

If surviving observations have the form

$$y_{V \setminus A} = A_{V \setminus A} c + \eta_A, \qquad |A| \leq k,$$

then the canonical survivor decoder obeys the uniform estimate

$$\boxed{\|\widehat{c}_A - c\| \leq \frac{\|\eta_A\|}{\sqrt{\alpha^{(k)}}}.}$$

Proof. Exact reconstruction for a particular survivor set is equivalent to a positive lower frame bound by Proposition 85. Taking the minimum over all permitted erasures gives the first claim. The uniform noise estimate follows from Theorem 86. □

Proposition 91 (Dimensional obstruction)**.** *Let $d = \dim C_*$. If*

$$\sum_{i \in S} \dim \mathsf{Y}_i < d,$$

then $\alpha_S = 0$, so S cannot reconstruct the core. Consequently, k-failure resilience requires

$$\sum_{i \in V \setminus A} \dim \mathsf{Y}_i \geq d$$

for every $A \subseteq V$ with $|A| \leq k$.

Proof. The rank of A_S is at most the dimension of its codomain. If that dimension is below d, then A_S is not injective. □

Define the accessible block support of $c \neq 0$ by

$$\operatorname{supp}_E(c) = \{i \in V : E_i c \neq 0\}$$

and define the node-erasure distance

$$\boxed{d_{\mathrm{er}} = \min_{c \neq 0} |\operatorname{supp}_E(c)|.}$$

Theorem 92 (Exact erasure-distance criterion)**.** *Exact reconstruction after every loss of at most k nodes is possible if and only if*

$$\boxed{d_{\mathrm{er}} > k.}$$

Proof. If $d_{\mathrm{er}} \le k$, choose $c \neq 0$ whose nonzero accessible blocks are contained in a set A with $|A| \le k$. The surviving analysis operator satisfies $A_{V\setminus A}c = 0$, so it cannot distinguish c from zero. Conversely, if some erasure set A of size at most k destroys injectivity, there exists $c \neq 0$ with $E_i c = 0$ for every $i \notin A$. Thus $\mathrm{supp}_E(c) \subseteq A$, and $d_{\mathrm{er}} \le k$. □

The distance criterion is combinatorial. The frame quantity $\alpha^{(k)}$ is quantitative. A system may tolerate k erasures exactly while amplifying noise severely because $\alpha^{(k)}$ is close to zero.

36.1 Reconstructive handoff after replacement

Suppose node r is removed and the surviving coalition is

$$S = V \setminus \{r\}.$$

A replacement node r' is to receive the encoding

$$E'_{r'} : C_* \to \mathsf{Y}'_{r'}.$$

Corollary 93 (Handoff initialization)**.** *If $\alpha_S > 0$ and the survivors observe $y = A_S c + \eta$, initialize the replacement by*

$$x^{\mathrm{init}}_{r'} = E'_{r'} A_S^{\dagger} y.$$

Then

$$\boxed{\|x^{\mathrm{init}}_{r'} - E'_{r'}c\| \le \frac{\|E'_{r'}\|\,\|\eta\|}{\sqrt{\alpha_S}}.}$$

This is the basic mathematical form of artifact-mediated or coalition-mediated handoff. It does not require the removed component to transfer its full internal state to the replacement.

37. Redundancy–Irreducibility Tension

Define the minimum reconstructing coalition size

$$\boxed{m_{\mathrm{rec}} = \min\{|S| : \alpha_S > 0\}}$$

and the maximum exact erasure tolerance

$$\boxed{k_{\mathrm{erase}} = \max\{k : \alpha^{(k)} > 0\}.}$$

The condition

$$m_{\mathrm{rec}} \ge 2$$

means that no individual encoding is sufficient for exact core recovery. The condition

$$k_{\mathrm{erase}} \ge 1$$

means that the system survives at least one node loss. These can hold simultaneously because nonlocality of encoding is not the same as fragility.

37.1 Three-node distributed-frame example

Let

$$C_* = \mathbb{R}^2$$

and define scalar node encodings

$$E_1(c_1, c_2) = c_1, \qquad E_2(c_1, c_2) = c_2, \qquad E_3(c_1, c_2) = c_1 + c_2.$$

Every singleton has rank one, so no node reconstructs the two-dimensional core. Every pair does reconstruct. For $S = \{1, 2\}$,

$$F_S = I, \qquad \alpha_S = 1.$$

For $S = \{1, 3\}$ or $S = \{2, 3\}$, the frame eigenvalues are

$$\frac{3 \pm \sqrt{5}}{2}.$$

Therefore

$$\boxed{m_{\mathrm{rec}} = 2, \qquad k_{\mathrm{erase}} = 1, \qquad d_{\mathrm{er}} = 2, \qquad \alpha^{(1)} = \frac{3 - \sqrt{5}}{2}.}$$

This realizes a workspace core that is irreducible to any singleton while remaining reconstructible after any one node fails.

37.2 Relation to predictive irreducibility

The geometric quantity m_{rec} and the predictive quantity k_0 from Part VII answer different questions. The first asks whether a coalition can reconstruct a declared core coordinate. The second asks whether a coalition is sufficient for a declared future target. Equality between them requires structural–operational alignment and regularity such as Proposition 89. Neither quantity should be substituted for the other without those assumptions.

PART XII

Identity Through Time and Replacement

Identity is treated as invariance under a structured correspondence, not as literal persistence of tokens or carriers. Exact correspondences form a groupoid. Approximate identity is measured by structural, operational, causal, and reconstructive defects that must be reported separately before they are combined.

38. Temporal Realization Maps

Let

$$\mathfrak{R} = (G, \mathbf{C}, D, \rho, \mathcal{I}, \mathcal{K})$$

and

$$\mathfrak{R}' = (G', \mathbf{C}', D', \rho', \mathcal{I}', \mathcal{K}')$$

be two realizations. A correspondence begins with a graph isomorphism $\varphi : G \to G'$. After relabeling by φ, let

$$J_i : C_i \to C'_i$$

be unitary local identifications and define

$$J_V = \bigoplus_{i \in V} J_i, \qquad J_E = \bigoplus_{e \in E} J_{t(e)}.$$

Definition 94 (Exact structural correspondence)**.** The correspondence is structurally exact when

$$\boxed{D' J_V = J_E D.}$$

It is operation exact for $a \in \mathcal{A}$ when

$$\boxed{\rho'_V(a) J_V = J_V \rho_V(a).}$$

Theorem 95 (Exact replacement isomorphism)**.** *Under an exact structural correspondence,*

$$\boxed{\Gamma(D') = J_V \Gamma(D), \qquad L' J_V = J_V L.}$$

Consequently, every spectral workspace subspace is transported exactly:

$$P_I(L') = J_V P_I(L) J_V^*$$

for every spectral interval I. If the correspondence is also operation exact, then the transported workspace actions are intertwined exactly.

Proof. For every c,

$$D' J_V c = J_E D c.$$

Because J_E is invertible, $Dc = 0$ if and only if $D' J_V c = 0$, proving the kernel statement. Taking adjoints of the exact covariance identity gives

$$D'^* J_E = J_V D^*.$$

Therefore

$$L'J_V = D'^*D'J_V = D'^*J_ED = J_VD^*D = J_VL.$$

The spectral and operation claims follow by functional calculus and the intertwining assumptions. □

Exact correspondences compose and invert. They therefore form a groupoid of realizations. The abstract exact workspace is an isomorphism class in this groupoid.

39. Structural Identity Defect

For an approximate correspondence define

$$\boxed{\Delta_D = D'J_V - J_ED, \qquad \delta_D = \|\Delta_D\|.}$$

If the graph weights are unchanged, define edgewise comparison defects

$$\delta_e = \|H'_eJ_{s(e)} - J_{t(e)}H_e\|.$$

The discrepancy operator has one changed source block per edge, so

$$\boxed{\delta_D^2 \le \max_{i\in V} \sum_{e:s(e)=i} w_e\delta_e^2 \le \sum_{e\in E} w_e\delta_e^2.}$$

When a single node is replaced, every nonincident edge has $\delta_e = 0$. The replacement defect is therefore locally measurable even though its effect on the global workspace is controlled by the global spectral gap.

The corresponding carried-Laplacian defect obeys

$$\boxed{\eta_L := \|J_V^*L'J_V - L\| \le 2\|D\|\delta_D + \delta_D^2.}$$

This is Proposition 79 with the time indices removed.

Let P and P' be rank-d workspace projectors selected from L and L', and let $g > 0$ be the separating source gap.

Theorem 96 (Approximate structural replacement)**.** *If*

$$\eta_L < \frac{g}{2},$$

then

$$\boxed{q_{\mathrm{str}} := \|P' - J_VPJ_V^*\| \le \frac{\eta_L}{g-\eta_L} \le \frac{2\eta_L}{g}.}$$

If $q_{\mathrm{str}} < 1$*, then*

$$B = P'J_VP : \operatorname{ran} P \longrightarrow \operatorname{ran} P'$$

is bijective and satisfies

$$\sigma_{\min}(B) \ge \sqrt{1-q_{\mathrm{str}}^2}.$$

Its polar factor

$$\boxed{\beta = B(B^*B)^{-1/2}}$$

is a canonical unitary workspace correspondence. Moreover,

$$\boxed{\|\beta - J_VP\| \le q_{\mathrm{str}} + 1 - \sqrt{1-q_{\mathrm{str}}^2}.}$$

Proof. The projector estimate is Theorem 43 applied after carrying L' back by J_V. For unit $w \in \operatorname{ran} P$,

$$\|(I - P')J_V w\| \le q_{\mathrm{str}}.$$

Orthogonality gives

$$\|Bw\|^2 = 1 - \|(I - P')J_V w\|^2 \ge 1 - q_{\mathrm{str}}^2.$$

Thus B is injective, hence bijective because the source and target have the same finite dimension. Its singular values lie in $[\sqrt{1 - q_{\mathrm{str}}^2}, 1]$, so polar normalization moves B by at most $1 - \sqrt{1 - q_{\mathrm{str}}^2}$. Finally,

$$\|B - J_V P\| = \|(I - P')J_V P\| \le q_{\mathrm{str}}.$$

□

39.1 Transport of workspace operations

Write

$$A = \rho_V(a), \qquad A' = \rho'_V(a),$$

and define

$$\delta_\rho(a) = \|A'J_V - J_V A\|, \qquad \ell_a = \|(I - P)AP\|.$$

Let

$$A_{\mathcal{W}} = PAP, \qquad A'_{\mathcal{W}'} = P'A'P'.$$

Proposition 97 (Operation transport bound)**.** *For $B = P'J_V P$,*

$$\boxed{\|A'_{\mathcal{W}'}B - BA_{\mathcal{W}}\| \le \|A'\|q_{\mathrm{str}} + \delta_\rho(a) + \ell_a.}$$

For the unitary polar correspondence β,

$$\boxed{\begin{aligned}\|A'_{\mathcal{W}'}\beta - \beta A_{\mathcal{W}}\| \le{}& \|A'\|q_{\mathrm{str}} + \delta_\rho(a) + \ell_a \\ &+ (\|A'_{\mathcal{W}'}\| + \|A_{\mathcal{W}}\|)\left(1 - \sqrt{1 - q_{\mathrm{str}}^2}\right).\end{aligned}}$$

Proof. Insert and subtract $P'A'J_V P$ and $P'J_V AP$:

$$\begin{aligned}A'_{\mathcal{W}'}B - BA_{\mathcal{W}} = {}& P'A'(P' - I)J_V P \\ &+ P'(A'J_V - J_V A)P \\ &+ P'J_V(I - P)AP.\end{aligned}$$

Bound the three terms. Replacing B by β adds

$$A'_{\mathcal{W}'}(\beta - B) - (\beta - B)A_{\mathcal{W}},$$

which gives the second estimate. □

40. Behavioral Identity Defect

Structural correspondence does not determine operational identity. Let

$$\{\iota_{w,b} : (w, b) \in \mathcal{D}\} \quad \text{and} \quad \{\iota'_{\beta w, \chi b} : (w, b) \in \mathcal{D}\}$$

be matched feasible implementation-intervention families in the sense of Part VIII. Here b denotes declared background variables and χ is the background correspondence. Define

$$K(w,b) = \mathbb{P}^{\iota_{w,b}}(Y^+ \in \cdot),$$

$$K'(\beta w, \chi b) = \mathbb{P}'^{\iota'_{\beta w,\chi b}}(Y'^+ \in \cdot).$$

For a chosen divergence D_Y, define

$$\boxed{\delta_{\text{causal}}(\beta) = \sup_{(w,b)\in\mathcal{D}} D_Y\left(K(w,b), K'(\beta w, \chi b)\right).}$$

The observational response is included only when the null intervention belongs to the declared family. This definition compares actual implementation interventions rather than formally assigning an arbitrary projection coordinate.

A reconstructive handoff may also be imperfect. Let

$$\delta_{\text{rec}}(\beta) = \mathbb{E}[d_{\mathcal{W}'}(W'_{\text{init}}, \beta W)]$$

measure replacement initialization error. Corollary 93 provides one explicit upper bound on this term.

40.1 Structural identity does not imply causal identity

Take two realizations with identical

$$G, \quad D, \quad L, \quad P, \quad \rho.$$

In the first, let $Y = W$. In the second, let $Y' = 1 - W$ for a binary workspace coordinate. Then

$$q_{\text{str}} = 0, \qquad \delta_\rho(a) = 0$$

for every operation, while the observational and interventional outcome kernels differ maximally under the identity correspondence. Therefore no purely structural theorem can prove causal identity without an additional kernel-matching assumption.

41. Identity Distance and Replacement Invariance

For an admissible correspondence $\mathfrak{J}$, define the operation defect on a declared unit ball $\mathcal{A}_1$ by

$$\delta_{\text{op}}(\mathfrak{J}) = \sup_{a\in\mathcal{A}_1} \|A'_{\mathcal{W}'}\beta - \beta A_{\mathcal{W}}\|.$$

The primary certification object is the identity profile

$$\boxed{\mathbf{d}_{\text{id}}(\mathfrak{J}) = \left(q_{\text{str}}, \delta_{\text{op}}(\mathfrak{J}), \delta_{\text{causal}}(\beta), \delta_{\text{rec}}(\beta); \mu^{\text{keep}}\right).}$$

The first four coordinates must remain below declared tolerances, while the non-collapse margin μ^{keep} must remain above a declared lower bound. This vector, rather than a weighted average, determines whether a specific identity claim passes.

A composite identity cost is

$$\boxed{\text{Cost}(\mathfrak{J}) = a\, q_{\text{str}} + b\, \delta_{\text{op}}(\mathfrak{J}) + c\, \delta_{\text{causal}}(\beta) + d\, \delta_{\text{rec}}(\beta),}$$

where $a, b, c, d > 0$ are declared scales. Define

$$\boxed{d_{\mathrm{id}}(\mathfrak{R}, \mathfrak{R}') = \inf_{\mathfrak{J}} \mathrm{Cost}(\mathfrak{J}).}$$

This quantity is a task-relative optimization surrogate until the admissible correspondence class, intervention matching, divergence, and scales are fixed. It does not replace the coordinatewise identity profile or the non-collapse gate.

41.1 Replacement invariance theorem T_{14}

Theorem 98 (Quantitative replacement invariance)**.** *Let $\mathfrak{J}$ be a unitary graph correspondence with diagram defect δ_D. Define*

$$\eta_L = 2\|D\|\delta_D + \delta_D^2$$

and assume

$$\eta_L < \frac{g}{2}.$$

Set

$$q_* = \frac{\eta_L}{g - \eta_L}.$$

Then there exists a unitary workspace correspondence β for which

$$q_{\mathrm{str}} \le q_*.$$

Moreover,

$$\boxed{\begin{aligned}\delta_{\mathrm{op}}(\mathfrak{J}) \le \sup_{a \in \mathcal{A}_1} \Big[&\|\rho'_V(a)\| q_* + \delta_\rho(a) + \ell_a \\ &+ \left(\|A'_{\mathcal{W}'}\| + \|A_{\mathcal{W}}\|\right)\left(1 - \sqrt{1 - q_*^2}\right)\Big].\end{aligned}}$$

If, in addition,

$$\delta_{\mathrm{causal}}(\beta) \le \delta_c, \qquad \delta_{\mathrm{rec}}(\beta) \le \delta_r,$$

then

$$\boxed{\begin{aligned}d_{\mathrm{id}}(\mathfrak{R}, \mathfrak{R}') \le &a\, q_* + b\, \delta_{\mathrm{op}}(\mathfrak{J}) \\ &+ c\, \delta_c + d\, \delta_r.\end{aligned}}$$

In particular, if all local structural, operation, closure, causal, and reconstruction defects are $O(\delta_{\mathrm{rep}})$, the relevant operator norms are uniformly bounded, and $g \ge g_0 > 0$, then

$$\boxed{d_{\mathrm{id}}(\mathfrak{R}, \mathfrak{R}') \le C_1 \frac{\delta_{\mathrm{rep}}}{g_0} + C_2 \delta_{\mathrm{rep}} + O\left(\frac{\delta_{\mathrm{rep}}^2}{g_0}\right).}$$

After a fixed normalization of the non-spectral terms, this recovers the schematic target

$$d_{\mathrm{id}} \lesssim C \frac{\delta_{\mathrm{rep}}}{g_0}.$$

Proof. The structural bound and existence of β follow from Theorem 96. The operation estimate is Proposition 97 with $q_{\mathrm{str}} \le q_*$. Insert these estimates and the assumed causal and reconstructive bounds into the definition of $\mathrm{Cost}(\mathfrak{J})$. The asymptotic statement follows from

$$q_* = O\left(\frac{\delta_D}{g_0} + \frac{\delta_D^2}{g_0}\right)$$

and

$$1 - \sqrt{1 - q_*^2} = O(q_*^2).$$

□

Corollary 99 (Preservation of coalition frame bounds)**.** *Identify the old and new abstract cores with* $\operatorname{ran} P$ *and* $\operatorname{ran} P'$ *through their chosen core parameterizations. Let* S *be a declared coalition, let*

$$A_S : \operatorname{ran} P \to \mathsf{Y}_S, \qquad A'_S : \operatorname{ran} P' \to \mathsf{Y}'_S$$

be the resulting accessible coalition encoders, and let K_S *be a unitary identification of their observation spaces. Suppose*

$$\|A'_S \beta - K_S A_S\| \le \delta_S^{\mathrm{frame}} < \sqrt{\alpha_S}.$$

Then the replacement coalition remains reconstructive and its new lower frame bound satisfies

$$\boxed{\alpha'_S \ge \left(\sqrt{\alpha_S} - \delta_S^{\mathrm{frame}}\right)^2.}$$

Consequently, k-failure resilience survives whenever the bound holds uniformly for every coalition remaining after at most k erasures.

Proof. For $w' \in \operatorname{ran} P'$, write $w' = \beta w$. Since β and K_S are unitary,

$$\begin{aligned}\|A'_S w'\| &\ge \|K_S A_S w\| - \|(A'_S \beta - K_S A_S) w\| \\ &\ge \left(\sqrt{\alpha_S} - \delta_S^{\mathrm{frame}}\right) \|w'\|.\end{aligned}$$

Square the inequality. □

Corollary 100 (Exact carrier replacement)**.** *If*

$$\delta_D = 0, \qquad \delta_\rho(a) = 0, \qquad \ell_a = 0, \qquad \delta_c = 0, \qquad \delta_r = 0,$$

then

$$d_{\mathrm{id}}(\mathfrak{R}, \mathfrak{R}') = 0.$$

Thus a participant, model, artifact, or notation system may be replaced without changing the abstract workspace when the replacement realizes the same structural, operational, causal, and reconstructive organization.

41.2 Temporal chains and identity paths

For a sequence of realizations

$$\mathfrak{R}_0, \mathfrak{R}_1, \dots, \mathfrak{R}_T,$$

let $\mathfrak{J}_t : \mathfrak{R}_t \to \mathfrak{R}_{t+1}$ be admissible correspondences. Define the path length

$$\mathrm{Len}(\mathfrak{J}_{0:T}) = \sum_{t=0}^{T-1} \mathrm{Cost}(\mathfrak{J}_t)$$

and the induced directed path quasi-pseudometric

$$\boxed{d^{\rightarrow}_{\mathrm{path}}(\mathfrak{R}_0, \mathfrak{R}_T) = \inf_{\mathfrak{J}_{0:T}} \mathrm{Len}(\mathfrak{J}_{0:T}).}$$

Concatenation gives the directed triangle inequality. Because replacement costs and intervention divergences need not be symmetric, symmetry is not assumed. When reverse correspondences exist, define the symmetrized pseudometric

$$\boxed{d^{\mathrm{sym}}_{\mathrm{path}}(\mathfrak{R}, \mathfrak{R}') = \max\left\{d^{\rightarrow}_{\mathrm{path}}(\mathfrak{R}, \mathfrak{R}'), d^{\rightarrow}_{\mathrm{path}}(\mathfrak{R}', \mathfrak{R})\right\}.}$$

Distinct physical realizations can have zero symmetrized pseudodistance when an exact correspondence exists in both directions. This is intentional: identity is assigned to the invariant relational organization rather than to the carrier tokens.

41.3 Identity failure modes

A complete replacement audit must report at least

$$\boxed{\mathcal{I}_{\mathrm{rep}} = \left(q_{\mathrm{str}}, \delta_{\mathrm{op}}, \delta_{\mathrm{causal}}, \delta_{\mathrm{rec}}, \mu^{\mathrm{keep}}\right).}$$

The failure modes are different:

q_{str} large	the shared subspace changed,
δ_{op} large	the same operations are no longer realized,
δ_{causal} large	interventions no longer have matching consequences,
δ_{rec} large	the new realization was not initialized from the old core,
μ^{keep} small	content collapsed despite apparent coherence.

The theorem does not permit one successful coordinate to conceal failure of another.

PART XIII

Gluing and Workspaces of Workspaces

The preceding parts treat one interaction diagram at a time. A large collaborative system is usually assembled from overlapping local laboratories, subteams, artifacts, models, or task-specific workspaces. This part asks when those local workspaces determine one coherent global workspace, how failed gluing is detected, and whether recursive assembly changes the resulting object. The exact theory is a descent theorem. The approximate theory is a second transfer-Laplacian construction on the cover itself.

42. Local Workspace Covers

42.1 Edge-complete covers

Let

$$G = (V, E)$$

be a finite directed interaction graph with local core spaces C_i, core transfers H_e, vertex space $\mathcal{C}_V(G)$, discrepancy operator D_G, and exact workspace

$$\mathscr{W}(G) := \Gamma(H|_G) = \ker D_G.$$

Definition 101 (Finite workspace cover)**.** A finite family of subgraphs

$$\mathcal{U} = \{G_\alpha\}_{\alpha \in A}$$

is an *edge-complete workspace cover* of G when

$$V = \bigcup_{\alpha \in A} V_\alpha, \qquad E = \bigcup_{\alpha \in A} E_\alpha.$$

Every edge in E_α carries the transfer inherited from G. For a nonempty finite index set

$$I = \{\alpha_0, \dots, \alpha_p\} \subseteq A,$$

define the intersection subgraph

$$G_I = G_{\alpha_0 \cdots \alpha_p} := \bigcap_{\alpha \in I} G_\alpha$$

by intersecting both vertex and edge sets. Isolated vertices are retained.

Covering only the vertices is insufficient. If an edge belongs to no chart, its transfer equation is absent from every local workspace and cannot be recovered by overlap agreement. Edge completeness is therefore the minimal cover assumption used below.

If $K \subseteq L \subseteq G$ are subgraphs, coordinate restriction gives

$$\rho_K^L : \mathcal{C}_V(L) \longrightarrow \mathcal{C}_V(K).$$

Because every edge equation on K is also an edge equation on L, restriction maps exact sections to exact sections:

$$r_K^L := \rho_K^L|_{\mathscr{W}(L)} : \mathscr{W}(L) \longrightarrow \mathscr{W}(K).$$

They obey

$$r_L^L = I, \qquad r_J^K r_K^L = r_J^L$$

whenever $J \subseteq K \subseteq L$. Thus $K \mapsto \mathscr{W}(K)$ is a contravariant system of local workspaces.

42.2 The cover nerve and overlap interfaces

The nerve $N(\mathcal{U})$ has one vertex for each chart G_α and a p-simplex

$$[\alpha_0 \cdots \alpha_p]$$

whenever $G_{\alpha_0 \cdots \alpha_p}$ is nonempty. The nerve records which local workspaces can compare states directly. It does not replace the underlying interaction graph: G carries the internal transfer equations, while $N(\mathcal{U})$ carries overlap-consistency equations between already constructed local objects.

For pairwise overlaps write

$$\mathscr{W}_\alpha = \mathscr{W}(G_\alpha), \qquad \mathscr{W}_{\alpha\beta} = \mathscr{W}(G_{\alpha\beta}),$$

and

$$r^\alpha_{\alpha\beta} : \mathscr{W}_\alpha \to \mathscr{W}_{\alpha\beta}, \qquad r^\beta_{\alpha\beta} : \mathscr{W}_\beta \to \mathscr{W}_{\alpha\beta}.$$

The interface is therefore a cospan

$$\mathscr{W}_\alpha \xrightarrow{r^\alpha_{\alpha\beta}} \mathscr{W}_{\alpha\beta} \xleftarrow{r^\beta_{\alpha\beta}} \mathscr{W}_\beta,$$

not necessarily a direct transfer map from one complete local workspace to the other. Both charts are compared through the content they expose on their common domain.

42.3 Čech cochains of local workspaces

For $p \geq 0$, define the finite cochain space

$$\check{C}^p(\mathcal{U}; \mathscr{W}) = \bigoplus_{\alpha_0 < \cdots < \alpha_p} \mathscr{W}(G_{\alpha_0 \cdots \alpha_p}),$$

where empty intersections contribute the zero space. The coboundary

$$\delta^p : \check{C}^p \longrightarrow \check{C}^{p+1}$$

is

$$\boxed{(\delta^p s)_{\alpha_0 \cdots \alpha_{p+1}} = \sum_{j=0}^{p+1} (-1)^j r^{\alpha_0 \cdots \widehat{\alpha_j} \cdots \alpha_{p+1}}_{\alpha_0 \cdots \alpha_{p+1}} s_{\alpha_0 \cdots \widehat{\alpha_j} \cdots \alpha_{p+1}}.}$$

The restriction identities imply

$$\delta^{p+1}\delta^p = 0.$$

For a local family $w = (w_\alpha)_\alpha$,

$$\boxed{(\delta^0 w)_{\alpha\beta} = r^\beta_{\alpha\beta} w_\beta - r^\alpha_{\alpha\beta} w_\alpha.}$$

Thus δ^0 is the overlap-discrepancy operator at the level of exact local workspaces.

43. Compatibility and Gluing Obstructions

43.1 Direct overlap mismatch

A family $w = (w_\alpha)_\alpha$ is exactly compatible when

$$\delta^0 w = 0.$$

Its direct mismatch is

$$\boxed{\mathfrak{m}_{\mathcal{U}}(w) = \delta^0 w.}$$

After choosing inner products and positive overlap weights, the mismatch energy is

$$\boxed{\mathcal{E}_{\mathrm{ov}}(w) = \|\delta^0 w\|^2 = \sum_{\alpha<\beta} \nu_{\alpha\beta} \left\| r^{\beta}_{\alpha\beta} w_\beta - r^{\alpha}_{\alpha\beta} w_\alpha \right\|^2.}$$

This is the appropriate obstruction when the full local states are already given. Such a family glues exactly if and only if its mismatch vector itself vanishes. No cohomological quotient is needed for that direct question.

43.2 Relative-only and affine descent data

A different problem arises when only relative transition data are supplied. Let

$$b = (b_{\alpha\beta}) \in \check{C}^1(\mathcal{U}; \mathscr{W})$$

be a prescribed family of overlap offsets. The triple-overlap consistency condition is

$$\delta^1 b = 0.$$

One seeks local potentials $q = (q_\alpha) \in \check{C}^0$ satisfying

$$\delta^0 q = b.$$

Define

$$\boxed{\check{H}^1(\mathcal{U}; \mathscr{W}) = \frac{\ker \delta^1}{\operatorname{im} \delta^0}.}$$

Then the affine gluing problem is solvable precisely when

$$\boxed{[b] = 0 \quad \text{in} \quad \check{H}^1(\mathcal{U}; \mathscr{W}).}$$

A nonzero class means that all local transition statements can be mutually consistent on triple overlaps while still failing to arise from any family of local workspace states. This is a cover-level analogue of nontrivial holonomy.

43.3 Harmonic representative of the obstruction

Equip the finite cochain spaces with Hilbert inner products and define the first Čech Laplacian

$$\boxed{\Delta_1 = \delta^0 (\delta^0)^* + (\delta^1)^* \delta^1.}$$

Let

$$\mathcal{H}^1_{\mathcal{U}} = \ker \Delta_1.$$

Proposition 102 (Harmonic gluing obstruction)**.** *For every cocycle $b \in \ker \delta^1$, there is an orthogonal decomposition*

$$\boxed{b = \delta^0 q^\dagger + h, \qquad h \in \mathcal{H}^1_{\mathcal{U}},}$$

where $q^\dagger = (\delta^0)^\dagger b$ is a least-squares potential. Moreover,

$$\boxed{\inf_q \|b - \delta^0 q\| = \|h\| = \|P_{\mathcal{H}^1_{\mathcal{U}}} b\|.}$$

Hence b is exactly integrable if and only if its harmonic component vanishes.

Proof. The finite-dimensional Hodge decomposition gives

$$\check{C}^1 = \operatorname{im} \delta^0 \oplus \ker \Delta_1 \oplus \operatorname{im}(\delta^1)^*.$$

If $b \in \ker \delta^1$, then b is orthogonal to $\operatorname{im}(\delta^1)^*$. Its remaining two orthogonal components are therefore an exact part $\delta^0 q^\dagger$ and a harmonic part h. The orthogonal projection theorem gives the least-squares identity. □

The scalar

$$\omega_{\mathcal{U}}(b) := \|P_{\mathcal{H}^1_{\mathcal{U}}} b\|$$

is an irreducible overlap-frustration measure. Its vanishing is invariant under admissible invertible coordinate changes, although its numerical norm depends on the chosen metric. It cannot be removed by adding local potentials alone.

43.4 Cycle example

Consider three charts arranged in a cycle, with one-dimensional overlap spaces and oriented offsets

$$b_{12}, \qquad b_{23}, \qquad b_{31}.$$

Local potentials would have to satisfy

$$q_2 - q_1 = b_{12}, \qquad q_3 - q_2 = b_{23}, \qquad q_1 - q_3 = b_{31}.$$

Adding the equations gives the necessary and sufficient condition

$$\boxed{b_{12} + b_{23} + b_{31} = 0.}$$

With the standard edge norm, the harmonic obstruction has magnitude

$$\boxed{\omega_{\mathcal{U}}(b) = \frac{|b_{12} + b_{23} + b_{31}|}{\sqrt{3}}.}$$

Every pairwise relation may be individually meaningful while the complete cycle remains globally inconsistent.

44. Exact Gluing Theorem T8

Let

$$R_{\mathcal{U}} : \mathscr{W}(G) \longrightarrow \check{C}^0(\mathcal{U}; \mathscr{W})$$

be the restriction map

$$R_{\mathcal{U}}(w) = (w|_{G_\alpha})_{\alpha \in A}.$$

Theorem 103 (Exact workspace gluing)**.** *For every finite edge-complete workspace cover $\mathcal{U}$, the sequence*

$$\boxed{0 \longrightarrow \mathscr{W}(G) \xrightarrow{R_{\mathcal{U}}} \check{C}^0(\mathcal{U};\mathscr{W}) \xrightarrow{\delta^0} \check{C}^1(\mathcal{U};\mathscr{W})}$$

is exact. Equivalently,

$$\boxed{\mathscr{W}(G) \cong \ker \delta^0 \cong \mathrm{Eq}\left(\prod_{\alpha} \mathscr{W}_\alpha \rightrightarrows \prod_{\alpha<\beta} \mathscr{W}_{\alpha\beta}\right).}$$

Every pairwise compatible family of exact local workspace states therefore glues to a unique exact global workspace state.

Proof. The cover contains every vertex, so $R_{\mathcal{U}}$ is injective. A global section has identical restrictions on every overlap, hence $\delta^0 R_{\mathcal{U}} = 0$.

Conversely, let $w = (w_\alpha)_\alpha \in \ker \delta^0$. For each vertex $i \in V$, choose any chart G_α containing i and set

$$c_i = (w_\alpha)_i.$$

If another chart G_β contains i, then $i \in G_{\alpha\beta}$ and overlap compatibility gives $(w_\alpha)_i = (w_\beta)_i$. Thus c_i is well defined.

For every edge $e : i \to j$, edge completeness provides a chart G_α containing e. Since $w_\alpha \in \mathscr{W}(G_\alpha)$,

$$c_j = (w_\alpha)_j = H_e(w_\alpha)_i = H_e c_i.$$

Hence $c = (c_i) \in \mathscr{W}(G)$, and by construction $R_{\mathcal{U}} c = w$. Uniqueness follows because the charts cover all vertices. □

Corollary 104 (Zeroth descent cohomology)**.** *For every finite edge-complete cover,*

$$\boxed{\mathscr{W}(G) \cong \check{H}^0(\mathcal{U};\mathscr{W}) = \ker \delta^0.}$$

In finite dimensions,

$$\boxed{\dim \mathscr{W}(G) = \sum_{\alpha \in A} \dim \mathscr{W}_\alpha - \mathrm{rank}(\delta^0).}$$

The theorem does not require the nerve to be acyclic. Cycles in the cover are allowed; their effect is encoded in the rank of δ^0 and, for relative-only data, in $\check{H}^1$. It also shows that the local-workspace assignment is a genuine sheaf on the finite subgraph cover used here rather than merely a presheaf.

44.1 Local nontriviality does not imply global nontriviality

Let G be the directed triangle

$$1 \xrightarrow{a} 2 \xrightarrow{b} 3 \xrightarrow{c} 1$$

with one-dimensional vertex cores. Cover it by its three edge charts. Their local workspaces are all one-dimensional:

$$\mathscr{W}_{12} = \{(x, ax) : x \in \mathbb{R}\},$$
$$\mathscr{W}_{23} = \{(y, by) : y \in \mathbb{R}\},$$
$$\mathscr{W}_{31} = \{(z, cz) : z \in \mathbb{R}\}.$$

Overlap agreement requires

$$y = ax, \qquad z = by, \qquad x = cz.$$

Therefore

$$(1 - abc)x = 0,$$

and

$$\boxed{\dim \mathscr{W}(G) = \begin{cases} 1, & abc = 1, \\ 0, & abc \neq 1. \end{cases}}$$

The meta-level gluing obstruction is exactly the original cycle holonomy. Strong local workspaces can therefore coexist with a trivial global core.

45. Approximate Gluing and Meta-Workspaces

45.1 The cover-level transfer Laplacian

First assume each local state already lies in its exact local workspace. Define

$$B_{\mathcal{U}} := \delta^0 : \check{C}^0(\mathcal{U}; \mathscr{W}) \longrightarrow \check{C}^1(\mathcal{U}; \mathscr{W})$$

and the meta-Laplacian

$$\boxed{L_{\mathcal{U}}^{\mathrm{meta}} = B_{\mathcal{U}}^* B_{\mathcal{U}}.}$$

By Theorem 103,

$$\boxed{\ker L_{\mathcal{U}}^{\mathrm{meta}} = R_{\mathcal{U}} \mathscr{W}(G).}$$

Thus the gluing problem is itself a workspace problem whose "nodes" are local workspaces and whose interfaces are overlaps.

Let

$$\gamma_{\mathcal{U}}^{\mathrm{meta}} = \lambda_{\min}^{+}\left(L_{\mathcal{U}}^{\mathrm{meta}}\right)$$

be the first positive eigenvalue.

Proposition 105 (Approximate overlap gluing)**.** *For every $w \in \check{C}^0(\mathcal{U}; \mathscr{W})$,*

$$\boxed{\operatorname{dist}\left(w, R_{\mathcal{U}} \mathscr{W}(G)\right)^2 \leq \frac{\|B_{\mathcal{U}} w\|^2}{\gamma_{\mathcal{U}}^{\mathrm{meta}}}.}$$

Proof. Decompose w into its component in $\ker L_{\mathcal{U}}^{\mathrm{meta}}$ and its orthogonal complement. On the complement, $L_{\mathcal{U}}^{\mathrm{meta}} \geq \gamma_{\mathcal{U}}^{\mathrm{meta}} I$. Therefore

$$\|B_{\mathcal{U}} w\|^2 = \langle w, L_{\mathcal{U}}^{\mathrm{meta}} w \rangle \geq \gamma_{\mathcal{U}}^{\mathrm{meta}} \operatorname{dist}\left(w, \ker L_{\mathcal{U}}^{\mathrm{meta}}\right)^2.$$

□

The gap $\gamma_{\mathcal{U}}^{\mathrm{meta}}$ measures how strongly the cover penalizes disagreement. A small overlap mismatch need not imply a nearby global workspace when this gap is itself very small.

45.2 Simultaneous local and overlap error

Local candidates may also violate their internal transfer equations. Define the local-copy space

$$\widehat{\mathcal{C}}_{\mathcal{U}} = \bigoplus_{\alpha \in A} \mathcal{C}_V(G_\alpha)$$

and the block internal discrepancy

$$\widehat{D}_{\mathcal{U}} = \bigoplus_{\alpha \in A} D_{G_\alpha}.$$

For $u = (u_\alpha)_\alpha \in \widehat{\mathcal{C}}_{\mathcal{U}}$, define the raw overlap synchronization operator

$$(S_{\mathcal{U}} u)_{\alpha\beta} = \rho^{G_\beta}_{G_{\alpha\beta}} u_\beta - \rho^{G_\alpha}_{G_{\alpha\beta}} u_\alpha.$$

For an overlap weight $\lambda_{\mathrm{ov}} > 0$, set

$$\boxed{K_{\mathcal{U}} = \begin{pmatrix} \widehat{D}_{\mathcal{U}} \\ \sqrt{\lambda_{\mathrm{ov}}}\, S_{\mathcal{U}} \end{pmatrix}, \qquad L^{\mathrm{tot}}_{\mathcal{U}} = K^*_{\mathcal{U}} K_{\mathcal{U}}.}$$

Its energy is

$$\boxed{\begin{aligned} \mathcal{E}^{\mathrm{tot}}_{\mathcal{U}}(u) = &\sum_\alpha \|D_{G_\alpha} u_\alpha\|^2 \\ &+ \lambda_{\mathrm{ov}} \sum_{\alpha<\beta} \left\| \rho^{G_\beta}_{G_{\alpha\beta}} u_\beta - \rho^{G_\alpha}_{G_{\alpha\beta}} u_\alpha \right\|^2. \end{aligned}}$$

Let

$$J_{\mathcal{U}} : \mathcal{C}_V(G) \to \widehat{\mathcal{C}}_{\mathcal{U}}, \qquad J_{\mathcal{U}} c = (c|_{G_\alpha})_\alpha,$$

be the duplication map.

Theorem 106 (Total descent kernel)**.** *For every finite edge-complete cover,*

$$\boxed{\ker K_{\mathcal{U}} = J_{\mathcal{U}} \mathscr{W}(G).}$$

Consequently, if

$$\gamma^{\mathrm{tot}}_{\mathcal{U}} = \lambda^+_{\min}\left(L^{\mathrm{tot}}_{\mathcal{U}}\right),$$

then

$$\boxed{\mathrm{dist}\left(u, J_{\mathcal{U}} \mathscr{W}(G)\right)^2 \le \frac{\mathcal{E}^{\mathrm{tot}}_{\mathcal{U}}(u)}{\gamma^{\mathrm{tot}}_{\mathcal{U}}}.}$$

Proof. If $u = J_{\mathcal{U}} c$ for a global exact section, every local discrepancy and every overlap discrepancy vanishes. Conversely, if $K_{\mathcal{U}} u = 0$, then every u_α is an exact local section and the family agrees on all overlaps. Theorem 103 gives a unique $c \in \mathscr{W}(G)$ with $u = J_{\mathcal{U}} c$. The distance estimate follows from the spectral lower bound on the orthogonal complement of the kernel. □

This theorem combines two distinct failure modes:

$$\text{local incoherence} \qquad \text{and} \qquad \text{cross-chart disagreement}.$$

Neither can be hidden by the other because they occupy separate blocks of $K_{\mathcal{U}}$.

45.3 Least-squares global assembly

Let

$$m_i = |\{\alpha : i \in V_\alpha\}|$$

be the multiplicity of vertex i. Then

$$J_{\mathcal{U}}^* J_{\mathcal{U}} = M_{\mathcal{U}}, \qquad (M_{\mathcal{U}} c)_i = m_i c_i.$$

The pseudoinverse performs coordinatewise averaging:

$$\boxed{(J_{\mathcal{U}}^\dagger u)_i = \frac{1}{m_i} \sum_{\alpha : i \in V_\alpha} (u_\alpha)_i.}$$

Let

$$\widehat{u} = P_{\ker K_{\mathcal{U}}} u = J_{\mathcal{U}} \widehat{c}.$$

Then

$$\widehat{c} = J_{\mathcal{U}}^\dagger \widehat{u}$$

and

$$\boxed{\|u - J_{\mathcal{U}} \widehat{c}\| \le \frac{\|K_{\mathcal{U}} u\|}{\sqrt{\gamma_{\mathcal{U}}^{\mathrm{tot}}}}.}$$

If $m_{\min} = \min_i m_i$, then

$$\boxed{\left\|\widehat{c} - J_{\mathcal{U}}^\dagger u\right\| \le \frac{\|K_{\mathcal{U}} u\|}{\sqrt{m_{\min} \gamma_{\mathcal{U}}^{\mathrm{tot}}}}.}$$

The simple overlap average and the globally coherent least-squares assembly are close only when the total descent energy is small relative to the cover gap.

45.4 A workspace of workspaces

Define the exact meta-workspace

$$\boxed{\mathscr{W}_{\mathrm{meta}}(\mathcal{U}) := \ker B_{\mathcal{U}} \subseteq \bigoplus_\alpha \mathscr{W}_\alpha.}$$

Theorem 103 gives the canonical identification

$$\boxed{\mathscr{W}_{\mathrm{meta}}(\mathcal{U}) \cong \mathscr{W}(G).}$$

Thus exact formation of a workspace from local workspaces does not create an additional mysterious carrier. It gives a second presentation of the same global compatible-section object.

Suppose local operations

$$A_\alpha : \mathscr{W}_\alpha \to \mathscr{W}_\alpha$$

and overlap operations

$$A_{\alpha\beta} : \mathscr{W}_{\alpha\beta} \to \mathscr{W}_{\alpha\beta}$$

satisfy

$$r_{\alpha\beta}^{\alpha} A_\alpha = A_{\alpha\beta} r_{\alpha\beta}^{\alpha}, \qquad r_{\alpha\beta}^{\beta} A_\beta = A_{\alpha\beta} r_{\alpha\beta}^{\beta}.$$

Then, with $A^0 = \bigoplus_\alpha A_\alpha$ and $A^1 = \bigoplus_{\alpha<\beta} A_{\alpha\beta}$,

$$B_{\mathcal{U}} A^0 = A^1 B_{\mathcal{U}}.$$

Therefore

$$A^0 \ker B_{\mathcal{U}} \subseteq \ker B_{\mathcal{U}}.$$

The operation-closure theorem from Part III reappears at the meta level: operations on subworkspaces descend globally precisely when they agree on interfaces.

45.5 Descent-faithful compression

A practical hierarchy rarely transmits each complete local workspace. Let

$$Q^p : \check{C}^p(\mathcal{U}; \mathscr{W}) \to \overline{C}^p$$

be compression maps into smaller local and overlap representations. Suppose there is a compressed coboundary $\overline{\delta}^p$ such that

$$\boxed{Q^{p+1}\delta^p = \overline{\delta}^p Q^p.}$$

The maps $Q^\bullet$ then form a cochain map.

Definition 107 (Descent-faithful compression)**.** The compression is *descent-faithful in degree zero* when the induced map

$$\boxed{H^0(Q) : \ker \delta^0 \longrightarrow \ker \overline{\delta}^0}$$

is an isomorphism.

Injectivity prevents two distinct global workspace states from collapsing to the same compressed meta-state. Surjectivity prevents the compressed hierarchy from introducing spurious compatible states that have no full-resolution lift. Under this condition,

$$\boxed{\ker \overline{\delta}^0 \cong \mathscr{W}(G).}$$

This is the exact criterion for a compressed workspace of workspaces to retain the global core.

45.6 Structural and operational gluing are distinct

Exact or approximate overlap coherence does not by itself establish global predictive sufficiency. Let W_t denote a glued structural state and let

$$Z_t^{\mathrm{loc}} = (Z_{\alpha,t})_{\alpha\in A}$$

be local predictive summaries recovered from its chart restrictions.

Proposition 108 (Local-to-global predictive alignment)**.** *Assume W_t is measurable with respect to the complete history $\mathbf{H}_t$, every $Z_{\alpha,t}$ is a measurable function of $W_t|_{G_\alpha}$, and*

$$I(Y_t^+; \mathbf{H}_t \mid Z_t^{\mathrm{loc}}) \le \varepsilon.$$

Then

$$\boxed{I(Y_t^+; \mathbf{H}_t \mid W_t) \le \varepsilon.}$$

In particular, exact sufficiency of the complete local-summary tuple implies exact sufficiency of the glued workspace.

Proof. The tuple Z_t^{loc} is a measurable function of W_t, while W_t is itself a history statistic. Conditioning on the finer statistic cannot increase the residual information that the full history carries about the target. Equivalently, use the conditional-law characterization from Theorem 56. □

Causal gluing requires an additional interface gate: interventions described in two overlapping charts must induce matched outcome laws on the overlap. A structurally compatible family can fail this gate, just as structurally identical replacement systems can fail causal identity in Part XII.

45.7 Anti-centralization gate

The isomorphism

$$\mathscr{W}_{\mathrm{meta}}(\mathcal{U}) \cong \mathscr{W}(G)$$

is a compositional statement, not a relational-irreducibility statement. A single chart may still expose enough accessible information to reconstruct the whole meta-state. A claim that the hierarchy is genuinely distributed must therefore reuse the coalition gates from Parts VII and XI. At the chart level, one may require, for example,

$$\alpha_{\{\alpha\}}^{\mathrm{meta}} = 0 \quad \text{for every } \alpha, \qquad \alpha_A^{\mathrm{meta}} > 0 \quad \text{for some } |A| \geq 2,$$

or the predictive analogue

$$k_\varepsilon^{\mathrm{meta}} \geq 2, \qquad \Delta_{\mathrm{joint}}^{\mathrm{meta}} > 0.$$

Capacity-matched local-memory and chart-ablation controls are still required. Hierarchical notation cannot convert a centralized copy into a relational workspace.

46. Hierarchical Collapse Theorem T9

46.1 Finite cover hierarchies

A finite workspace hierarchy is a rooted tree $\mathcal{T}$. Each node $a \in \mathcal{T}$ carries a subgraph G_a. The root r carries

$$G_r = G.$$

For every internal node a, its children

$$\mathrm{Ch}(a)$$

form a finite edge-complete cover of G_a. Leaves carry primitive local workspace spaces

$$W_\ell = \mathscr{W}(G_\ell).$$

For an internal node, define recursively

$$\boxed{W_a = \ker\left(\delta_a^0 : \bigoplus_{b \in \mathrm{Ch}(a)} W_b \longrightarrow \bigoplus_{b<c} W_{bc}\right),}$$

where the overlap spaces and restrictions are inherited from the child cover.

Theorem 109 (Exact hierarchical collapse)**.** *For every finite workspace hierarchy as above, there is a canonical isomorphism*

$$\boxed{W_a \cong \mathscr{W}(G_a)}$$

for every node a. In particular,

$$\boxed{W_r \cong \mathscr{W}(G).}$$

Therefore recursively gluing local workspaces, using any other finite parenthesization built from edge-complete intermediate covers that retain all required overlap objects, and solving the flattened global transfer system all produce the same exact workspace up to canonical isomorphism.

Proof. Proceed upward from the leaves. The claim holds at every leaf by definition. Assume it holds for all children of an internal node a. Replacing each child space W_b by its canonical copy $\mathscr{W}(G_b)$, the recursive definition of W_a is exactly the compatible-family kernel for the edge-complete child cover of G_a. Theorem 103 therefore gives

$$W_a \cong \mathscr{W}(G_a).$$

Finite induction reaches the root. □

46.2 Flattening and limit interchange

Let $L(\mathcal{T})$ be the leaf set. Repeated edge completeness implies that the flattened family

$$\mathcal{U}_{\text{flat}} = \{G_\ell : \ell \in L(\mathcal{T})\}$$

is itself an edge-complete cover of G. Let

$$\delta^0_{\text{flat}}$$

be its overlap coboundary and define

$$W_{\text{flat}} = \ker \delta^0_{\text{flat}}.$$

Theorem 103 and Theorem 109 give

$$\boxed{W_r \cong W_{\text{flat}} \cong \mathscr{W}(G).}$$

Equivalently,

$$\boxed{\operatorname{Lim}^{\text{rec}}_{\mathcal{T}} \left(\{\mathscr{W}(G_\ell)\}_{\ell \in L(\mathcal{T})}\right) \cong \operatorname{Eq}_{\mathcal{U}_{\text{flat}}} \left(\{\mathscr{W}(G_\ell)\}_{\ell \in L(\mathcal{T})}\right).}$$

The notation on the left means recursive finite descent along the tree; the right side means one-shot equalization over the flattened leaf cover. In finite-dimensional vector spaces, both are kernels of the same complete family of local and overlap constraints, grouped in different orders. Hierarchical collapse is therefore a finite-limit interchange property, not a claim that intermediate levels are physically absent or operationally irrelevant.

46.3 Approximate hierarchical collapse

At level r, let K_r be the corresponding total descent operator,

$$\gamma_r = \lambda^+_{\min}(K_r^* K_r), \qquad \|K_r u_r\| \le \varepsilon_r.$$

The nearest exact compatible state then differs from u_r by at most

$$\frac{\varepsilon_r}{\sqrt{\gamma_r}}.$$

Allow an additional implementation, compression, or causal-summary defect ζ_r, and define

$$\delta_r = \frac{\varepsilon_r}{\sqrt{\gamma_r}} + \zeta_r.$$

Theorem 110 (Approximate hierarchical collapse bound)**.** *Let*

$$F_r : \mathcal{H}_r \to \mathcal{H}_{r+1}, \qquad \widetilde{F}_r : \mathcal{H}_r \to \mathcal{H}_{r+1}$$

be exact and implemented level-collapse maps for $r = 0, \dots, R-1$. Let $\mathcal{R}_r \subseteq \mathcal{H}_r$ contain the exact and implemented states reachable at level r. Assume, on $\mathcal{R}_r$,

$$\|F_r(x) - F_r(y)\| \le L_r \|x - y\|$$

and

$$\sup_{x \in \mathcal{R}_r} \|\widetilde{F}_r(x) - F_r(x)\| \le \delta_r.$$

Then

$$\boxed{\begin{aligned}&\left\|\widetilde{F}_{R-1} \circ \cdots \circ \widetilde{F}_0(x) - F_{R-1} \circ \cdots \circ F_0(x)\right\| \\ &\qquad \le \sum_{r=0}^{R-1} \left(\prod_{s=r+1}^{R-1} L_s \right) \delta_r.\end{aligned}}$$

In particular, if $L_r \le 1$, then the total error is at most

$$\sum_{r=0}^{R-1} \delta_r.$$

If $L_r \le L < 1$ and $\delta_r \le \delta$, then

$$\boxed{\text{total error} \le \frac{\delta(1 - L^R)}{1 - L} \le \frac{\delta}{1 - L}.}$$

Proof. Let x_r and $\widetilde{x}_r$ be the exact and implemented states after r levels. Then

$$\begin{aligned}\|\widetilde{x}_{r+1} - x_{r+1}\| &\le \|\widetilde{F}_r(\widetilde{x}_r) - F_r(\widetilde{x}_r)\| \\ &\quad + \|F_r(\widetilde{x}_r) - F_r(x_r)\| \\ &\le \delta_r + L_r \|\widetilde{x}_r - x_r\|.\end{aligned}$$

Iterating this recurrence gives the displayed product-sum estimate. The final two bounds follow by summing a finite or infinite geometric series. □

This theorem identifies three regimes:

$L_r < 1$	hierarchical correction is contractive,
$L_r = 1$	level errors accumulate at most additively,
$L_r > 1$	deep hierarchies can amplify early mismatch.

A deep hierarchy is therefore stable only when its level gaps remain separated and its assembly maps do not magnify small overlap errors faster than those errors are corrected.

46.4 Causal-kernel accumulation

Let K_r and $\widetilde{K}_r$ now denote exact and implemented Markov kernels at successive hierarchy levels, and suppose

$$\sup_x \mathrm{TV}\left(\widetilde{K}_r(x, \cdot), K_r(x, \cdot)\right) \le \zeta_r.$$

Because Markov kernels contract total variation, a telescoping argument gives

$$\boxed{\sup_x \mathrm{TV}\left(\widetilde{K}_0 \cdots \widetilde{K}_{R-1}(x,\cdot), K_0 \cdots K_{R-1}(x,\cdot)\right) \le \sum_{r=0}^{R-1} \zeta_r.}$$

Thus causal semantics can also be transported through a hierarchy, but only under an explicit kernel-level compatibility test. Structural gluing alone provides no such guarantee.

46.5 Direct versus hierarchical computation

The exact theorem states

$$\boxed{\text{recursive exact gluing} = \text{one-shot exact gluing.}}$$

The approximate theorem instead supplies a measurable flattening defect:

$$\boxed{\delta_{\mathrm{flat}}^{(R)} = \left\| \widetilde{F}_{R-1} \cdots \widetilde{F}_0(x) - F_{R-1} \cdots F_0(x) \right\|.}$$

A hierarchy is empirically justified when it reduces computation, communication, or local model complexity while keeping $\delta_{\mathrm{flat}}^{(R)}$, predictive risk, causal-kernel mismatch, and anti-centralization gates within declared tolerances. Hierarchy by itself is not evidence of emergence.

46.6 Gluing and hierarchical synthesis

The exact and approximate constructions can be compressed as

$$\boxed{\begin{aligned} &\text{local transfer systems} \xrightarrow{\ker D_\alpha} \text{local workspaces} \\ &\qquad\qquad\qquad\quad \xrightarrow{\ker \delta^0} \text{meta-workspace} \\ &\qquad\qquad\qquad\quad \xrightarrow{\text{finite recursive descent}} \text{global workspace.} \end{aligned}}$$

At exact resolution, all finite parenthesizations are canonically equivalent. At approximate resolution, every level contributes a defect scaled by its spectral gap and by downstream amplification. Relative-only overlap data can carry a harmonic H^1 obstruction even when every local chart is internally coherent. Operational and relational claims still require predictive, causal, coalition, and null-model gates in addition to structural descent.

PART XIV

Scaffolded and Endogenous Formation

47. Formation Systems and Scaffold Specification

The preceding parts begin with a transfer system that has already been chosen. A formation theory must instead explain how the nodes, edges, transfer maps, workspace channels, and operational boundaries are obtained. This requires a strict separation between the structure supplied by the experimenter and the structure selected from interaction history.

Definition 111 (Formation system)**.** A *workspace formation system* is a tuple

$$\mathfrak{F} = (\Sigma, \mathcal{D}_{\text{form}}, \mathcal{L}, \Omega, \widehat{\theta}, \mathcal{D}_{\text{cert}}, \mathcal{I}_{\text{form}}),$$

where:

- Σ is the externally supplied scaffold;
- $\mathcal{D}_{\text{form}}$ is the formation or training history;
- $\mathcal{L}$ is the learning or selection rule;
- Ω is its internal random seed or stochastic state;
- $\widehat{\theta} = \mathcal{L}(\Sigma, \mathcal{D}_{\text{form}}, \Omega)$ is the learned organization;
- $\mathcal{D}_{\text{cert}}$ is held-out certification data; and
- $\mathcal{I}_{\text{form}}$ is a declared family of interventions on scaffold and formation history.

The learned organization is decomposed as

$$\widehat{\theta} = (\widehat{V}, \widehat{E}, \widehat{H}, \widehat{P}, \widehat{\mathcal{O}}, \widehat{B}),$$

where $\widehat{B}$ is a learned operational boundary.

The scaffold is itself stratified:

$$\boxed{\Sigma = (\Sigma_V, \Sigma_E, \Sigma_H, \Sigma_P, \Sigma_{\mathcal{O}}, \Sigma_B, \Sigma_Y).}$$

The final coordinate Σ_Y records labels, targets, privileged masks, or other supervision that could directly reveal the intended workspace. This coordinate is necessary because an architecture may appear to discover a boundary even when the desired membership mask was encoded in its training labels.

47.1 Scaffolded architecture versus scaffolded solution

A scaffold can constrain the search space without specifying its solution. For example, a graph neural architecture can fix the class of admissible local message-passing rules while leaving the active edges and transfer maps to be learned. Conversely, a system can use a flexible architecture while receiving labels that directly encode the desired workspace. The former is architecturally scaffolded but solution-open. The latter is solution-scaffolded.

For any component q of $\widehat{\theta}$, define its scaffold-only Bayes risk under loss ℓ_q by

$$\mathcal{R}_q^{\Sigma} = \inf_f \mathbb{E}\left[\ell_q(q, f(\Sigma, \Omega))\right].$$

Define the full formation risk by

$$\mathcal{R}_q^{\Sigma,\mathcal{H}} = \inf_g \mathbb{E}\left[\ell_q(q, g(\Sigma, \mathcal{D}_{\text{form}}, \Omega))\right].$$

The history advantage is

$$\boxed{\Delta_q^{\text{hist}} = \mathcal{R}_q^{\Sigma} - \mathcal{R}_q^{\Sigma,\mathcal{H}}.}$$

A positive history advantage means that formation history contains information about q that is not already recoverable from the declared scaffold and random seed.

Proposition 112 (Scaffold-only impossibility)**.** *If*

$$q = f(\Sigma, \Omega) \qquad \textit{almost surely,}$$

then for every formation-history intervention that leaves (Σ, Ω) *fixed,*

$$\mathcal{L}(q \mid \text{do}(\mathcal{D}_{\text{form}} = h)) = \mathcal{L}(q \mid \text{do}(\mathcal{D}_{\text{form}} = h')),$$

and $\Delta_q^{\text{hist}} = 0$.

Proof. The value of q is measurable with respect to (Σ, Ω), so an intervention on formation history cannot change its conditional law when those variables are held fixed. The scaffold-only predictor f attains the same risk as any predictor that also receives formation history, hence the risk gap vanishes. □

This proposition is elementary but methodologically decisive. A structure that is deterministically encoded by the scaffold cannot count as discovered from interaction history merely because a learned network reproduces it.

47.2 Formation and certification split

The same sample must not be used both to select a workspace and to certify its coherence, causal efficacy, or relational irreducibility. The required split is

$$\mathcal{D} = \mathcal{D}_{\text{form}} \sqcup \mathcal{D}_{\text{cal}} \sqcup \mathcal{D}_{\text{cert}}.$$

The formation sample selects $\widehat{\theta}$. The calibration sample fixes thresholds, regularization strengths, and intervention magnitudes. The certification sample evaluates the final gates without further structural selection. In a multi-environment study, the stronger requirement is

$$\mathcal{E}_{\text{form}} \cap \mathcal{E}_{\text{cert}} = \varnothing,$$

so that the learned organization is tested under environments not used to construct it.

Definition 113 (Formation leakage)**.** A certification variable Y_{cert} leaks into formation when the conditional law of $\widehat{\theta}$ changes under an intervention on Y_{cert} that should be inaccessible during formation:

$$\mathcal{L}\left(\widehat{\theta} \mid \text{do}(Y_{\text{cert}} = y)\right) \neq \mathcal{L}\left(\widehat{\theta} \mid \text{do}(Y_{\text{cert}} = y')\right).$$

A valid discovery claim requires a documented information barrier or an audit showing that this effect is absent up to tolerance.

47.3 Origin and reality are orthogonal

The formation provenance of a workspace and its operational certification are separate coordinates. A fully scaffolded structure may satisfy every structural and causal workspace gate. A highly endogenous structure may fail to become coherent or causally useful. Symbolically,

$$\text{Endogenous}(W) \not\Longrightarrow \text{SRW}_\Theta(W),$$

$$\text{SRW}_\Theta(W) \not\Longrightarrow \text{Endogenous}(W).$$

The theory therefore treats endogeneity as a provenance axis rather than a higher level on the workspace hierarchy.

48. Operational Endogeneity Signature

The ideal description-complexity expression from the symbolic outline is retained as a conceptual limit, but raw Kolmogorov complexity is not computable. A usable theory must declare a model class, a code, and an intervention family.

Definition 114 (Code-relative history contribution)**.** Let $\mathfrak{M}_q$ be a declared coding or predictive model class for a learned component q. Let

$$L_{\mathfrak{M}_q}(q \mid Z)$$

be its held-out codelength given side information Z. Define

$$\boxed{e_q^{\text{MDL}} = \left[\frac{L_{\mathfrak{M}_q}(q \mid \Sigma) - L_{\mathfrak{M}_q}(q \mid \Sigma, \mathcal{D}_{\text{form}})}{\max\{L_{\mathfrak{M}_q}(q \mid \Sigma), 1\}}\right]_0^1,}$$

where $[x]_0^1 = \min\{1, \max\{0, x\}\}$.

This quantity is explicitly code-relative. It must not be presented as an observer-independent metaphysical measure. Its role is to test whether the formation history provides compressive information about the learned organization beyond the declared scaffold.

Definition 115 (Causal formation sensitivity)**.** Let ι_h and $\iota_{h'}$ be matched feasible interventions that alter formation history while preserving the scaffold, randomization law, sample size, and intervention cost. For a divergence D, define

$$\boxed{e_q^{\text{causal}} = \mathbb{E}_{(h,h')\sim\nu_q} D(\mathcal{L}^{\iota_h}(q), \mathcal{L}^{\iota_{h'}}(q))\,.}$$

A common null uses temporally or relationally shuffled histories that preserve local marginals but destroy the candidate formation signal.

Proposition 116 (Reparameterization invariance of causal provenance)**.** *Let ϕ be a measurable bijection on the range of q, and let the divergence D be invariant under measurable bijections. Then*

$$e_{\phi(q)}^{\text{causal}} = e_q^{\text{causal}}.$$

Proof. Every intervention-induced law of $\phi(q)$ is the pushforward of the corresponding law of q. Bijection invariance of D preserves each pairwise divergence, and therefore their design-weighted expectation. □

The causal coordinate is therefore insensitive to a mere renaming of the learned structure. The MDL coordinate need not be invariant unless the coding class is transformed with the reparameterization.

48.1 Scaffold dependence and cross-scaffold invariance

Endogeneity does not require independence from all scaffolds. Every learning system has an architecture, objective, and resource budget. The relevant question is whether the learned organization depends on the particular solution-bearing details of the scaffold.

For a family $\mathfrak{S}$ of admissible scaffold variants, define the cross-scaffold identity defect

$$\boxed{\delta_q^{\mathrm{scaf}} = \inf_{\{J_{\sigma\sigma'}\}} \mathbb{E}_{\sigma,\sigma'\sim\pi} d_q\left(J_{\sigma\sigma'} q_\sigma, q_{\sigma'}\right),}$$

where $J_{\sigma\sigma'}$ transports the learned object between admissible coordinate systems. A low value indicates that the same abstract organization is recovered under multiple non-solution-bearing scaffolds.

The complete formation signature is

$$\boxed{\mathbf{E}_{\mathrm{form}}(W) = \left(e_V, e_E, e_H, e_P, e_{\mathcal{O}}, e_B; \delta^{\mathrm{scaf}}, \delta^{\mathrm{leak}}\right),}$$

where every e_q is reported as the pair

$$e_q = \left(e_q^{\mathrm{MDL}}, e_q^{\mathrm{causal}}\right).$$

No scalar average is permitted to hide a zero history contribution for one of the structures claimed to have been discovered.

48.2 Formation certificate

For thresholds Θ_{form}, define

$$\mathrm{Form}_{\Theta_{\mathrm{form}}}(W) = 1$$

only when all applicable gates pass:

$$\begin{aligned} &e_q^{\mathrm{MDL}} \geq \tau_q^{\mathrm{MDL}}, \qquad e_q^{\mathrm{causal}} \geq \tau_q^{\mathrm{causal}}, \\ &\delta_q^{\mathrm{scaf}} \leq \varepsilon_q^{\mathrm{scaf}}, \qquad \delta^{\mathrm{leak}} \leq \varepsilon_{\mathrm{leak}}, \\ &\mathrm{SRW}_\Theta(W) = 1, \quad \mathrm{Pass}(N_{\mathrm{form}}) = 1. \end{aligned}$$

The final workspace gate is included because endogenous discovery of an unstable or causally irrelevant pattern is not endogenous workspace formation.

49. Learned Transfer Structure and Formation Classes

49.1 Gated membership and edge learning

Let

$$z_i \in \{0,1\}, \qquad a_e \in [0,1]$$

be node-membership and edge-activation variables satisfying

$$a_e \leq z_{s(e)}, \qquad a_e \leq z_{t(e)}.$$

A weighted learned discrepancy operator is

$$(D_{a,H}c)_e = \sqrt{w_e a_e}\left(c_{t(e)} - H_e c_{s(e)}\right).$$

One generic formation objective is

$$\boxed{\widehat{\theta} \in \operatorname*{arg\,min}_{z,a,H,P,\mathcal{O},B} \Big\{ \widehat{\mathcal{R}}_{\text{task}} + \lambda_{\text{coh}}\|D_{a,H}c\|^2 + \lambda_E\|a\|_1 + \lambda_V\|z\|_1 + \lambda_{\text{stab}}\mathcal{J}_{\text{stab}} + \lambda_{\text{leak}}\mathcal{J}_{\text{leak}} \Big\}.}$$

The objective is schematic rather than universal. Its purpose is to expose which quantities were fixed and which were optimized. If z and a are fixed by Σ, only the transfer maps or local channels can be claimed as learned.

49.2 Gauge-identifiable transfer learning

A learned transfer map is not absolutely identifiable when local coordinates are latent. Under local invertible changes

$$X'_u = g_u X_u, \qquad X'_v = g_v X_v,$$

the same abstract transfer is represented by

$$H'_{uv} = g_v H_{uv} g_u^{-1}.$$

The identifiable object is therefore the gauge class

$$[H]_{\mathcal{G}} = \left\{ (g_{t(e)} H_e g_{s(e)}^{-1})_{e\in E} : (g_i)_{i\in V} \in \mathcal{G} \right\}.$$

Any formation claim that compares raw matrices across independently trained systems without gauge alignment is ill posed.

49.3 Transfer identification theorem T_{15}

Theorem 117 (Population transfer identification)**.** *Let $X \in \mathbb{R}^{d_u}$ and $Y \in \mathbb{R}^{d_v}$ satisfy*

$$Y = H_* X + \xi, \qquad \mathbb{E}[\xi X^{\mathsf{T}}] = 0,$$

and assume

$$\Sigma_{XX} = \mathbb{E}[XX^{\mathsf{T}}] \succeq \kappa I$$

for some $\kappa > 0$. Then H_* is the unique minimizer of

$$\mathcal{R}(H) = \mathbb{E}\|Y - HX\|_2^2,$$

with

$$\boxed{H_* = \Sigma_{YX}\Sigma_{XX}^{-1}.}$$

Moreover,

$$\boxed{\mathcal{R}(H) - \mathcal{R}(H_*) = \operatorname{tr}\left((H - H_*)\Sigma_{XX}(H - H_*)^{\mathsf{T}}\right) \geq \kappa\|H - H_*\|_F^2.}$$

Proof. Substitute $Y = H_*X + \xi$ into the risk. The cross term vanishes because $\mathbb{E}[\xi X^\mathsf{T}] = 0$, giving

$$\mathcal{R}(H) = \mathcal{R}(H_*) + \mathbb{E}\|(H - H_*)X\|_2^2.$$

The remaining expectation is the displayed trace. Positive definiteness of Σ_{XX} gives uniqueness and the lower bound. The normal equations give the covariance formula. □

Corollary 118 (Deterministic empirical recovery bound)**.** *For samples $(X_r, Y_r)_{r=1}^n$, write*

$$\widehat{\Sigma}_{XX} = \frac{1}{n}\sum_{r=1}^n X_r X_r^\mathsf{T}, \qquad \widehat{M} = \frac{1}{n}\sum_{r=1}^n \xi_r X_r^\mathsf{T}.$$

If $\widehat{\Sigma}_{XX} \succeq \widehat{\kappa} I$, then ordinary least squares satisfies

$$\boxed{\|\widehat{H} - H_*\|_F \le \frac{\|\widehat{M}\|_F}{\widehat{\kappa}}.}$$

For ridge parameter $\tau > 0$,

$$\boxed{\|\widehat{H}_\tau - H_*\|_F \le \frac{\|\widehat{M}\|_F + \tau\|H_*\|_F}{\widehat{\kappa} + \tau}.}$$

Proof. The empirical normal equations give

$$\widehat{H} - H_* = \widehat{M}\widehat{\Sigma}_{XX}^{-1}.$$

For ridge regression,

$$\widehat{H}_\tau - H_* = (\widehat{M} - \tau H_*)(\widehat{\Sigma}_{XX} + \tau I)^{-1}.$$

Taking Frobenius and operator norms yields both estimates. □

Proposition 119 (Gauge covariance of least squares)**.** *Under invertible coordinate transformations*

$$X' = g_u X, \qquad Y' = g_v Y,$$

the population least-squares solution transforms as

$$H'_* = g_v H_* g_u^{-1}.$$

Proof. The transformed covariances are

$$\Sigma_{Y'X'} = g_v \Sigma_{YX} g_u^\mathsf{T}, \qquad \Sigma_{X'X'} = g_u \Sigma_{XX} g_u^\mathsf{T}.$$

Substitution into the covariance formula and inversion of the second factor give the result. □

The theorem establishes identification only under excitation and exogeneity. When Σ_{XX} is singular, the transfer is identified only on the excited source subspace. When $\mathbb{E}[\xi X^\mathsf{T}] \neq 0$, regression recovers a predictive map rather than necessarily the intended causal transfer.

49.4 Held-out structure-selection theorem T_{16}

The same finite-margin argument applies to node membership, edge selection, boundary selection, or a finite family of gauge-normalized transfer models.

Theorem 120 (Uniform structure-selection recovery)**.** *Let $\mathfrak{T}$ be a finite candidate family with population score $J : \mathfrak{T} \to \mathbb{R}$. Let*

$$\theta_* = \arg\min_{\theta \in \mathfrak{T}} J(\theta)$$

be unique, with margin

$$\Delta_J = \min_{\theta \neq \theta_*} \big(J(\theta) - J(\theta_*)\big) > 0.$$

If a held-out estimate $\widehat{J}$ satisfies

$$\sup_{\theta \in \mathfrak{T}} |\widehat{J}(\theta) - J(\theta)| \leq \varepsilon,$$

then every empirical minimizer $\widehat{\theta}$ obeys

$$\boxed{J(\widehat{\theta}) - J(\theta_*) \leq 2\varepsilon.}$$

If $2\varepsilon < \Delta_J$, then

$$\boxed{\widehat{\theta} = \theta_*.}$$

Proof. By empirical optimality and uniform approximation,

$$J(\widehat{\theta}) \leq \widehat{J}(\widehat{\theta}) + \varepsilon \leq \widehat{J}(\theta_*) + \varepsilon \leq J(\theta_*) + 2\varepsilon.$$

If $\widehat{\theta} \neq \theta_*$, the margin would force an excess larger than 2ε, a contradiction. □

The theorem does not guarantee that the candidate class contains the correct organization. It states exactly what follows once a finite class, score, and uniform certification error are declared.

49.5 Formation classes

The provenance classes are now defined relative to

$$(\Sigma, \mathcal{L}, \mathcal{I}_{\text{form}}, \mathfrak{M}).$$

They are not absolute labels.

Exogenously scaffolded.
The claimed nodes, edges, transfer maps, channels, operations, or boundary are recoverable from the scaffold alone at the declared tolerance. Their history advantages and causal formation sensitivities vanish.

Adaptively parameterized.
The membership and channel topology are fixed, while transfer parameters, local operators, or readouts are learned from interaction history. Typically

$$e_H + e_P + e_{\mathcal{O}} > 0, \qquad e_V \approx e_E \approx e_B \approx 0.$$

Structurally adaptive.
At least one of membership, edges, or boundary is selected from history, but the search family and objective remain externally specified.

Endogenously constituted relative to the scaffold class.
Membership, transfer organization, and operational boundary have nonzero history contribution and causal formation sensitivity, survive cross-scaffold transport, pass leakage and matched-null tests, and produce a certified workspace on held-out environments.

The phrase "relative to the scaffold class" is mandatory. No finite experiment can establish independence from every possible external influence.

PART XV

Boundary Formation and Stability

50. Candidate Workspace Boundaries

A boundary is not merely a subset of physical components. It is a subset relative to a declared node decomposition, prediction target, intervention family, environment distribution, and time scale.

Definition 121 (Boundary problem)**.** A boundary problem is a tuple

$$\mathfrak{B} = (V, \mathfrak{S}, \tau, \mathcal{Y}, \mathcal{I}, \Pi, \mathcal{G}_B)\,,$$

where $\mathfrak{S} \subseteq 2^V$ is the admissible candidate family, τ is the temporal scale, $\mathcal{Y}$ is the operational target, $\mathcal{I}$ is the feasible intervention family, Π is the environment distribution, and $\mathcal{G}_B$ is a symmetry group under which boundaries are considered equivalent.

Unless the application requires otherwise, the candidate family excludes

$$\varnothing, \qquad V, \qquad \{i\} \text{ for } i \in V.$$

This anti-triviality rule prevents the empty set, the entire realization, or a single centralized controller from winning solely through a degenerate score.

50.1 Observational recurrence, dependence, and outward influence

For $S \in \mathfrak{S}$, define

$$\begin{aligned}
\mathrm{Ret}_\tau(S) &= I(X_S(t+\tau); X_S(t) \mid X_{\bar{S}}(t), E_t)\,,\\
\mathrm{Dep}_\tau(S) &= I(X_S(t+\tau); X_{\bar{S}}(t) \mid X_S(t), E_t)\,,\\
\mathrm{Out}_\tau(S) &= I(X_{\bar{S}}(t+\tau); X_S(t) \mid X_{\bar{S}}(t), E_t)\,.
\end{aligned}$$

These quantities are observational. They can be nonzero because of common causes, selection, or feedback paths and therefore cannot alone establish a causal boundary.

50.2 Causal boundary terms

Let $\iota_{S,x}$ be a feasible intervention that changes the internal workspace coordinate of S to x while matching background variables and intervention cost. Define outward causal efficacy

$$\boxed{\mathrm{COut}_\tau(S) = \mathbb{E}_{x,x' \sim \nu_S} D(\mathcal{L}^{\iota_{S,x}}(X_{\bar{S}}(t+\tau)), \mathcal{L}^{\iota_{S,x'}}(X_{\bar{S}}(t+\tau)))\,.}$$

Define inward causal dependence analogously by intervening on $\bar{S}$ and measuring the future law of X_S:

$$\boxed{\mathrm{CIn}_\tau(S) = \mathbb{E}_{u,u' \sim \nu_{\bar{S}}} D(\mathcal{L}^{\iota_{\bar{S},u}}(X_S(t+\tau)), \mathcal{L}^{\iota_{\bar{S},u'}}(X_S(t+\tau)))\,.}$$

Neither term should be minimized to zero in general. A useful workspace may receive substantial external input. The boundary objective must instead express the intended balance between internal recurrence, external dependence, and outward efficacy.

50.3 Structural and anti-centralization gates

For the induced subsystem on S, define

$$\mathsf{Coh}(S) = \operatorname{tr}\left(P_{S,d} L_S P_{S,d}\right),$$

where $P_{S,d}$ projects onto the selected low-energy workspace modes. Let

$$\mathsf{Cut}(S) = \sum_{e:s(e)\in S,\ t(e)\notin S} w_e + \sum_{e:s(e)\notin S,\ t(e)\in S} w_e.$$

Let $k_\varepsilon(S)$ be the minimal sufficient coalition size inside the candidate subsystem. A candidate is admissible only if

$$\mathrm{SRW}_{\Theta_S}(W_S) = 1, \qquad k_\varepsilon(S) \geq 2,$$

unless the claim explicitly concerns a centralized workspace.

51. Boundary Objective, Recovery, and Persistence

A general score is

$$\boxed{\begin{aligned}\mathcal{B}_t(S) = {} & a_R \operatorname{Ret}_\tau(S) + a_O \operatorname{Out}_\tau(S) + a_C \operatorname{COut}_\tau(S) \\ & - a_D \operatorname{Dep}_\tau(S) - a_I \operatorname{CIn}_\tau(S) - a_L \mathsf{Coh}(S) \\ & - a_K \mathsf{Cut}(S) - a_Z \mathsf{Size}(S) - a_T \operatorname{Instability}_t(S).\end{aligned}}$$

The coefficients and all normalizations are part of the boundary problem. No single choice is universal. The selected boundary is

$$\widehat{S}_t \in \arg\max_{S\in\mathfrak{S}_{\mathrm{pass}}} \widehat{\mathcal{B}}_t(S),$$

where $\mathfrak{S}_{\mathrm{pass}}$ contains only candidates that satisfy the required workspace and anti-centralization gates.

51.1 Boundary equivalence and identifiability

If a symmetry $g \in \mathcal{G}_B$ leaves the full observational and interventional law invariant, S and gS cannot be distinguished by the declared experiment. The identifiable boundary is then the orbit

$$[S]_{\mathcal{G}_B} = \{gS : g \in \mathcal{G}_B\}.$$

A unique subset should not be claimed when only an equivalence class is identifiable.

51.2 Stable boundary recovery T_{17}

Theorem 122 (Held-out boundary recovery)**.** *Let $\mathfrak{S}_{\mathrm{pass}}$ be finite and let*

$$S_t^* = \arg\max_{S\in\mathfrak{S}_{\mathrm{pass}}} \mathcal{B}_t(S)$$

be unique. Define the population margin

$$\Delta_t^B = \mathcal{B}_t(S_t^*) - \max_{S\neq S_t^*} \mathcal{B}_t(S) > 0.$$

If

$$\sup_{S\in\mathfrak{S}_{\mathrm{pass}}} \left|\widehat{\mathcal{B}}_t(S) - \mathcal{B}_t(S)\right| \leq \varepsilon_t,$$

then every empirical maximizer $\widehat{S}_t$ *satisfies*

$$\boxed{\mathcal{B}_t(S_t^*) - \mathcal{B}_t(\widehat{S}_t) \le 2\varepsilon_t.}$$

If $2\varepsilon_t < \Delta_t^B$*, then*

$$\boxed{\widehat{S}_t = S_t^*.}$$

Proof. Apply Theorem 120 to $J(S) = -\mathcal{B}_t(S)$. □

The theorem converts uncertainty in mutual-information estimation, causal interventions, structural scores, and stability penalties into an explicit boundary-identification condition. It also exposes when the data are insufficient: a small margin makes the boundary intrinsically fragile.

51.3 Temporal boundary stability

Theorem 123 (Boundary persistence under score drift)**.** *Assume* S_t^* *is the unique maximizer of* $\mathcal{B}_t$ *with margin* Δ_t^B*, and suppose*

$$\sup_{S \in \mathfrak{S}_{\text{pass}}} |\mathcal{B}_{t+1}(S) - \mathcal{B}_t(S)| \le \delta_t.$$

If

$$2\delta_t < \Delta_t^B,$$

then

$$\boxed{S_{t+1}^* = S_t^*.}$$

More generally,

$$\mathcal{B}_{t+1}(S_{t+1}^*) - \mathcal{B}_{t+1}(S_t^*) \le 2\delta_t.$$

Proof. For every competitor $S \neq S_t^*$,

$$\begin{aligned}\mathcal{B}_{t+1}(S_t^*) - \mathcal{B}_{t+1}(S) &\ge \mathcal{B}_t(S_t^*) - \mathcal{B}_t(S) - 2\delta_t \\ &\ge \Delta_t^B - 2\delta_t > 0.\end{aligned}$$

Hence the maximizer is unchanged. The approximate bound follows by comparing both time-indexed scores at their respective maximizers. □

If the score depends on a parameter ϑ_t and satisfies

$$\sup_S |\mathcal{B}_\vartheta(S) - \mathcal{B}_{\vartheta'}(S)| \le L_B d(\vartheta, \vartheta'),$$

then the sufficient stability condition becomes

$$2L_B d(\vartheta_t, \vartheta_{t+1}) < \Delta_t^B.$$

51.4 Analyst-discovered, system-instantiated, and self-maintained boundaries

Three claims must remain separate.

Analyst-discovered boundary.
An external procedure fits $\widehat{S}$ after observing the system. The system need not represent or use that boundary.

System-instantiated boundary.
The system contains a channel, mask, or routing variable causally equivalent to $\widehat{S}$, and interventions on that variable alter downstream operation.

Self-maintained boundary.
The system detects degradation of its boundary certificate and takes actions that restore membership, transfer, or reconstruction properties.

Only the second and third claims concern a boundary realized within the system. A post hoc partition that predicts behavior well is not by itself a self-discovered boundary.

51.5 Boundary formation certificate

A boundary $\widehat{S}$ is certified as endogenously formed only if:

1. the membership mask was not directly encoded in Σ_B or Σ_Y;
2. e_B^{MDL} and e_B^{causal} exceed declared thresholds;
3. held-out recovery satisfies Theorem 122;
4. the boundary remains stable under admissible environment drift or its changes are predicted by score-margin closure;
5. matched history-shuffle and label-leakage controls fail to reproduce the result, as do full-system, empty-set, singleton, and centralized boundary nulls;
6. the induced subsystem satisfies its workspace certificate; and
7. a system-instantiation intervention demonstrates that the boundary is used by the system rather than only by the analyst.

The results below establish the statistical and deterministic stability components. They does not prove that an arbitrary adaptive system will spontaneously learn such a boundary.

PART XVI

Agency, Counterfactual Control, and Self-Maintenance

52. Controlled Workspace Agents and Goal States

The preceding parts define when a distributed structure is coherent, operationally sufficient, causally efficacious, relationally irreducible, persistent, reconstructible, and endogenously formed. None of those properties by itself establishes agency. Agency adds a controlled relation among a represented objective, a set of feasible actions, counterfactual consequences, and a policy that uses those quantities to alter the future. Self-maintenance adds a further loop in which damage to the workspace becomes an input to actions that preserve or restore the workspace certificate.

The purpose of this part is not to infer intentions from fluent behavior. It is to define an intervention-based certificate whose components can fail separately.

52.1 Controlled relational realization

A controlled relational realization extends Definition 1 by the tuple

$$\boxed{\mathfrak{A}_{\mathbf{W}} = \left(\mathfrak{R}, \mathcal{G}, \mathcal{A}_t, \mathcal{Y}^+, \{U_g\}_{g\in\mathcal{G}}, \Pi, \mathcal{I}_A, \mathcal{I}_G, \mathcal{I}_D, \mathcal{I}_R\right).}$$

Here:

- $\mathfrak{R}$ is the underlying relational realization;
- $\mathcal{G}$ is a measurable goal-code space;
- $\mathcal{A}_t$ is the feasible action set at time t;
- $\mathcal{Y}^+$ is the relevant future trajectory space;
- $U_g : \mathcal{Y}^+ \to \mathbb{R}$ evaluates a trajectory under goal code g;
- Π is an admissible class of workspace-level policies;
- $\mathcal{I}_A$ and $\mathcal{I}_G$ are feasible action and goal interventions; and
- $\mathcal{I}_D$ and $\mathcal{I}_R$ are feasible damage and repair interventions.

A controlled state contains at least

$$C_t^{\mathrm{ctl}} = (W_t, G_t, M_t, B_t, E_t)\,,$$

where G_t is the represented goal, M_t is a workspace-integrity monitor, B_t is the current boundary or routing state, and E_t is the declared external context. An action is generated by

$$A_t \sim \pi_t\left(\cdot \mid W_t, G_t, M_t, B_t, E_t\right).$$

The future controlled state obeys a time-unrolled structural model

$$S_{t+1} = F_t(S_t, A_t, E_{t+1}, \xi_{t+1}), \qquad W_{t+1} = \omega_{t+1}(S_{t+1}).$$

The intervention semantics are inherited from Part VII. A symbol such as $\mathrm{do}(A_t = a)$ is meaningful only when an implementation intervention changes an actual action mechanism while the declared background variables are matched.

52.2 Goal content and goal-code equivalence

A persistent internal variable is not automatically a goal. A goal code must index a stable counterfactual preference over relevant future trajectories. For a fixed state w, define the goal-conditioned action value

$$Q_t^g(a \mid w) = \mathbb{E}\left[U_g(Y_t^+) \mid \mathrm{do}(A_t = a), W_t = w, G_t = g\right].$$

When action costs are relevant, they are incorporated into U_g or declared as a constraint on $\mathcal{A}_t$.

Two goal codes are operationally equivalent,

$$g \sim_{\mathrm{goal}} g',$$

when they induce the same action ordering in every declared state and intervention background:

$$Q_t^g(a \mid w) \geq Q_t^g(a' \mid w) \quad \Longleftrightarrow \quad Q_t^{g'}(a \mid w) \geq Q_t^{g'}(a' \mid w).$$

The abstract goal is the equivalence class

$$[g]_{\mathrm{goal}}.$$

This quotient prevents a mere invertible recoding of a goal channel from being reported as a change of objective. For quantitative regret bounds, a representative normalization of U_g must nevertheless be fixed.

Remark 124 (Goals are claim-relative)**.** The same internal code can count as a goal for one declared trajectory family and fail to do so for another. A code that predicts a short-horizon action may not encode a persistent long-horizon objective. Every agency claim must state its horizon, outcome space, feasible actions, and utility normalization.

52.3 Local, distributed, and workspace-level goals

Let $\mathcal{H}_t^S$ be the history available to coalition $S \subseteq V$. For a loss ℓ_G, define goal-reconstruction risk

$$\mathcal{R}_G(S) = \inf_f \mathbb{E}\left[\ell_G\left(G_t, f(\mathcal{H}_t^S)\right)\right].$$

For tolerance ε, define

$$k_{G,\varepsilon} = \min\left\{|S| : \mathcal{R}_G(S) \leq \mathcal{R}_G(V) + \varepsilon\right\}.$$

A goal is locally stored when $k_{G,\varepsilon} = 1$. It is distributed relative to the declared decomposition when $k_{G,\varepsilon} \geq 2$.

This distinction is independent of whether the rest of the workspace is relational. A relational workspace can obey a goal copied in full into one controller. Such a system may possess a workspace but not a *workspace-level goal* in the strong distributed sense.

For the complete control state, define

$$\mathcal{R}_{\mathrm{ctl}}(S) = \inf_f \mathbb{E}\left[\ell_{\mathrm{ctl}}\left(C_t^{\mathrm{ctl}}, f(\mathcal{H}_t^S)\right)\right],$$

and let $k_{\mathrm{ctl},\varepsilon}$ be its minimal sufficient coalition size. A strong workspace-level agency claim requires

$$\boxed{k_{\mathrm{ctl},\varepsilon} \geq 2.}$$

This is a location criterion. It does not yet establish goal persistence, counterfactual evaluation, or self-maintenance.

52.4 Action mediation and outcome control

Goal-dependent action mediation is measured by feasible interventions on the goal channel while matching the current workspace and external context. For a divergence D_A, define

$$\eta_{G\to A} = \sup_{g,g',w,e} D_A\left(\mathbb{P}(A_t \in \cdot \mid \iota_g, w, e), \mathbb{P}(A_t \in \cdot \mid \iota_{g'}, w, e)\right).$$

Workspace-dependent action mediation is

$$\eta_{W\to A} = \sup_{w,w',g,e} D_A\left(\mathbb{P}(A_t \in \cdot \mid \iota_w, g, e), \mathbb{P}(A_t \in \cdot \mid \iota_{w'}, g, e)\right).$$

An action must also alter the declared future:

$$\eta_{A\to Y} = \sup_{a,a',b} D_Y\left(\mathbb{P}(Y_t^+ \in \cdot \mid \iota_a, b), \mathbb{P}(Y_t^+ \in \cdot \mid \iota_{a'}, b)\right).$$

A positive $\eta_{G\to A}$ without $\eta_{A\to Y}$ describes goal-sensitive output selection with no demonstrated control of the relevant environment. A positive $\eta_{A\to Y}$ without a goal or counterfactual-value channel may describe a reactive controller.

52.5 Why observational goal persistence is insufficient

The observational condition

$$I(G_{t+\tau}; G_t \mid E_{t+1:t+\tau}) > 0$$

can arise from several non-agentic mechanisms:

1. a persistent external command writes the same code at every step;
2. a clock, episode label, or task identifier drives both variables;
3. a frozen memory cell repeats a symbol that is not used by the policy;
4. selection on successful trajectories induces apparent temporal dependence; or
5. a downstream action is copied back into the nominal goal channel.

A goal-persistence claim therefore requires a cue-controlled intervention that changes G_t, removes or matches subsequent command inputs, and checks both future goal state and action consequences.

52.6 Cue-controlled goal persistence

Let C_t denote an external command or task cue. Fix a neutral or matched future command trajectory $c^0_{t+1:t+\tau}$. For goal interventions ι_g and $\iota_{g'}$, define the cue-controlled retention effect

$$\eta_G(\tau; g, g') = D_G\left(\mathbb{P}\left(G_{t+\tau} \in \cdot \mid \iota_g, \mathrm{do}(C_{t+1:t+\tau} = c^0)\right), \mathbb{P}\left(G_{t+\tau} \in \cdot \mid \iota_{g'}, \mathrm{do}(C_{t+1:t+\tau} = c^0)\right)\right).$$

The corresponding action-retention effect is

$$\eta_{G\to A}(\tau; g, g') = D_A\left(\mathbb{P}(A_{t+\tau} \in \cdot \mid \iota_g, c^0), \mathbb{P}(A_{t+\tau} \in \cdot \mid \iota_{g'}, c^0)\right).$$

The first quantity tests whether a goal difference survives. The second tests whether the surviving difference remains behaviorally operative.

Cue dependence must be reported separately. For matched initial goal g, define

$$\delta_G^{\mathrm{cue}}(\tau) = \sup_{c,c'} D_G\left(\mathbb{P}(G_{t+\tau} \in \cdot \mid \iota_g, \mathrm{do}(C = c)), \mathbb{P}(G_{t+\tau} \in \cdot \mid \iota_g, \mathrm{do}(C = c'))\right).$$

High cue dependence is not a failure when the claimed agent is intentionally instruction following. It does mean that persistence cannot be attributed to an autonomous internal objective without an additional cue-removal result.

52.7 T_{19}: Cue-controlled goal-retention theorem

Theorem 125 (Cue-controlled goal retention)**.** *Let $G_s \in \mathbb{R}^{d_G}$ obey*

$$G_{s+1} = R_s G_s + B_s C_{s+1} + \zeta_{s+1} \qquad (s = t, \dots, t+\tau-1).$$

Consider two intervention runs that set

$$G_t = g, \qquad G'_t = g',$$

and then use the same command trajectory and the same disturbance realization. Let

$$R_{t+\tau:t} = R_{t+\tau-1} \cdots R_t.$$

Then

$$\boxed{G_{t+\tau} - G'_{t+\tau} = R_{t+\tau:t}(g - g').}$$

If

$$m_s \le \sigma_{\min}(R_s), \qquad \|R_s\| \le M_s,$$

then

$$\boxed{\left(\prod_{s=t}^{t+\tau-1} m_s\right)\|g - g'\| \le \|G_{t+\tau} - G'_{t+\tau}\| \le \left(\prod_{s=t}^{t+\tau-1} M_s\right)\|g - g'\|.}$$

Proof. Subtract the two update equations. The matched command and disturbance terms cancel, giving

$$G_{s+1} - G'_{s+1} = R_s(G_s - G'_s).$$

Iteration gives the exact product formula. The upper bound follows from submultiplicativity of the operator norm. The lower bound follows from

$$\sigma_{\min}(AB) \ge \sigma_{\min}(A)\sigma_{\min}(B)$$

for square linear maps. □

The theorem isolates internal retention from repeated external rewriting. It also makes clear that persistence is graded. Uniformly contractive maps can retain a goal for a finite horizon while erasing it asymptotically. A neutral mode with product singular value near one supports long retention. An expansive mode can preserve distinctions but may be dynamically unstable.

Corollary 126 (Behavioral readout of a retained goal)**.** *Assume, in addition, that under the matched intervention background the action readout has conditional mean*

$$\mathbb{E}[A_{t+\tau} \mid G_{t+\tau}, W_{t+\tau}] = K_{t+\tau} G_{t+\tau} + b(W_{t+\tau}),$$

and that the two runs are matched on $W_{t+\tau}$. Then

$$\mathbb{E}[A_{t+\tau} \mid \iota_g] - \mathbb{E}[A_{t+\tau} \mid \iota_{g'}] = K_{t+\tau} R_{t+\tau:t}(g - g').$$

If $\sigma_{\min}(K_{t+\tau}) \geq k_{t+\tau} > 0$, the norm of this action-mean difference is at least

$$k_{t+\tau} \left(\prod_{s=t}^{t+\tau-1} m_s \right) \|g - g'\|.$$

The corollary is a sufficient condition, not a universal model of action selection. For discrete actions, the corresponding test should compare the full interventional action distributions.

52.8 Goal-persistence certificate

A minimal goal-persistence report contains:

1. the goal code and its declared trajectory-level semantics;
2. the feasible goal intervention and its manipulation check;
3. the matched or neutralized future cue trajectory;
4. $\eta_G(\tau)$ across a declared horizon grid;
5. $\eta_{G \to A}(\tau)$ over the same grid;
6. cue dependence $\delta_G^{\mathrm{cue}}(\tau)$;
7. goal-channel ablation and permutation controls;
8. coalition reconstruction risks $\mathcal{R}_G(S)$; and
9. uncertainty intervals over environments, episodes, and random seeds.

A repeated label with no intervention effect on future action fails this certificate even when its observational autocorrelation is one.

53. Counterfactual Policy, Repair, and Agency Certification

53.1 Feasible action interventions and value functions

Let $\mathcal{A}_t(w, e)$ denote the actions that can be implemented from workspace state w in environment e. The interventional action value is

$$Q_t^g(a \mid w, e) = \mathbb{E}^{\iota_a} \left[U_g(Y_t^+) \mid W_t = w, E_t = e, G_t = g \right].$$

The optimal action set is

$$\mathcal{A}_t^*(w, g, e) = \arg\max_{a \in \mathcal{A}_t(w,e)} Q_t^g(a \mid w, e).$$

A counterfactual evaluator is an internal state

$$\widehat{Q}_t = \left(\widehat{Q}_t(a)\right)_{a\in\mathcal{A}_t(w,e)}$$

that estimates these intervention values. It is not enough for $\widehat{Q}_t$ to predict the action selected by the system. It must be validated against externally estimated intervention outcomes.

53.2 Counterfactual-use criterion

Three properties must be separated:

Counterfactual accuracy.

$$\delta_Q = \sup_{a\in\mathcal{A}_t} |\widehat{Q}_t(a) - Q_t^g(a \mid w, e)|.$$

Policy use. A feasible intervention that changes $\widehat{Q}_t(a)$ while preserving the action affordances changes the action distribution in the direction predicted by the decision rule.

Outcome validity. The resulting action change alters $U_g(Y_t^+)$ as predicted by the external interventional value estimate.

A system can satisfy any two while failing the third. A learned critic can be accurate but ignored. A policy can respond to a manipulated score that does not predict outcomes. A successful action can arise from habit or replay without any counterfactual evaluator.

53.3 T_{20}: Approximate counterfactual policy theorem

Theorem 127 (Approximate counterfactual policy selection)**.** *Let $\mathcal{A}$ be a finite feasible action set. Let $Q : \mathcal{A} \to \mathbb{R}$ be the true interventional value and let $\widehat{Q}$ satisfy*

$$\sup_{a\in\mathcal{A}} |\widehat{Q}(a) - Q(a)| \le \varepsilon_Q.$$

Suppose a policy $\widehat{\pi}$ is δ_{opt}-optimal for $\widehat{Q}$:

$$\max_{\pi} \mathbb{E}_{a\sim\pi}[\widehat{Q}(a)] - \mathbb{E}_{a\sim\widehat{\pi}}[\widehat{Q}(a)] \le \delta_{\text{opt}}.$$

Then, for every true-value-optimal policy π^*,

$$\boxed{\mathbb{E}_{a\sim\pi^*}[Q(a)] - \mathbb{E}_{a\sim\widehat{\pi}}[Q(a)] \le 2\varepsilon_Q + \delta_{\text{opt}}.}$$

For a deterministic greedy action $\widehat{a} \in \arg\max_a \widehat{Q}(a)$,

$$Q(a^*) - Q(\widehat{a}) \le 2\varepsilon_Q.$$

If the true optimal action a^* is unique and its value gap

$$\Delta_Q = Q(a^*) - \max_{a\neq a^*} Q(a)$$

satisfies $\Delta_Q > 2\varepsilon_Q$, then every greedy maximizer of $\widehat{Q}$ equals a^*.

Proof. For every policy π, uniform approximation gives

$$\left|\mathbb{E}_{a\sim\pi}[\widehat{Q}(a)] - \mathbb{E}_{a\sim\pi}[Q(a)]\right| \le \varepsilon_Q.$$

Therefore

$$\begin{aligned}\mathbb{E}_{\pi^*}[Q] - \mathbb{E}_{\widehat{\pi}}[Q] &\le \mathbb{E}_{\pi^*}[\widehat{Q}] - \mathbb{E}_{\widehat{\pi}}[\widehat{Q}] + 2\varepsilon_Q \\ &\le \delta_{\mathrm{opt}} + 2\varepsilon_Q.\end{aligned}$$

The deterministic bound is the special case of point-mass policies. If $\widehat{a} \neq a^*$, then

$$\widehat{Q}(a^*) \ge Q(a^*) - \varepsilon_Q > Q(\widehat{a}) + \varepsilon_Q \ge \widehat{Q}(\widehat{a}),$$

contradicting greedy optimality. □

This theorem converts counterfactual-model accuracy into a decision guarantee. It does not establish that the model is internally represented or causally used. Those are separate intervention gates.

53.4 Counterfactual mediation and the replay null

Let V_t denote the internal value channel. A counterfactual-mediation test compares:

$$G_t \longrightarrow V_t \longrightarrow A_t \longrightarrow Y_t^+.$$

The relevant quantities include:

$$\eta_{G\to V}, \qquad \eta_{V\to A}, \qquad \eta_{A\to Y}, \qquad \delta_{G\to A|V},$$

where the final term measures the residual goal-to-action effect after the value channel is blocked or randomized.

The action-replay null preserves the empirical action sequence or action marginal while destroying online dependence on V_t. If replayed actions perform equally well under held-out perturbations, the evidence supports a cached or open-loop policy rather than online counterfactual evaluation. A second null permutes value labels among feasible actions while preserving the multiset of scores. A genuine value-mediated policy should follow the permuted ranking, while an ignored critic should not.

53.5 Workspace integrity vector and maintenance margin

Choose a declared set $\mathcal{J}_{\mathrm{maint}}$ of workspace gates that must be protected. Write each gate as a signed margin

$$\phi_{j,t} \ge 0 \quad \Longleftrightarrow \quad \text{gate } j \text{ passes at time } t.$$

Examples include

$$\begin{aligned}\phi_{\mathrm{gap},t} &= g_{\varepsilon,t} - \underline{g}, \\ \phi_{\mathrm{coh},t} &= \overline{\delta}_{\mathrm{coh}} - \delta_{\mathrm{coh},t}, \\ \phi_{\mathrm{rec},t} &= \alpha_t^{(k)} - \underline{\alpha}, \\ \phi_{\mathrm{keep},t} &= \mu_t^{\mathrm{keep}} - \underline{\mu}, \\ \phi_{\mathrm{causal},t} &= \eta_{\mathrm{causal},t} - \underline{\eta}, \\ \phi_{\mathrm{boundary},t} &= \Delta_t^B - \underline{\Delta}_B.\end{aligned}$$

The aggregate integrity margin is

$$\boxed{v_t = \min_{j \in \mathcal{J}_{\text{maint}}} \phi_{j,t}.}$$

The workspace passes the protected gates when $v_t \geq 0$. To maintain a reserve $\rho > 0$, define the maintenance deficit

$$\boxed{e_t = (\rho - v_t)_+.}$$

Define the zero-reserve and bounded-deficit viability sets

$$\boxed{\mathcal{V}_\Theta = \{S : \phi_j(S) \geq 0 \text{ for every } j \in \mathcal{J}_{\text{maint}}\}}$$

$$\boxed{\mathcal{V}_{\Theta, e_{\max}} = \{S : e(S) \leq e_{\max}\}}$$

The first set requires every protected gate to pass. The second permits a declared finite reserve deficit.

This minimum-gate construction is intentionally noncompensatory. A large spectral gap cannot compensate for failed reconstruction, and high task reward cannot compensate for loss of the declared workspace boundary.

53.6 Damage and repair interventions

A damage intervention $d \in \mathcal{I}_D$ changes a mechanism that supports one or more protected gates. Examples include:

- deleting or corrupting a transfer edge;
- removing a participant or artifact;
- perturbing the internal boundary mask;
- scrambling a role-binding or routing channel;
- erasing a handoff representation; or
- degrading a reconstruction map.

A repair intervention $r \in \mathcal{I}_R$ can add, remove, retrain, rewrite, or reroute an admissible component. Let r_0 denote matched no-repair. The post-repair integrity value at horizon τ is

$$V_t^{\text{rep}}(r; d) = \mathbb{E}\left[v_{t+\tau} \mid \text{do}(D_t = d), \text{do}(R_t = r), \mathbf{H}_t\right].$$

The repair gain is

$$\Delta_t^{\text{rep}}(r; d) = V_t^{\text{rep}}(r; d) - V_t^{\text{rep}}(r_0; d).$$

Repair costs are handled either by restricting the feasible repair family or by reporting a separate cost-effect curve. They should not be hidden inside an undocumented scalar reward.

53.7 Damage detection, repair mediation, and specificity

Let M_t be an internal integrity monitor and R_t the selected repair action. A self-maintenance claim requires the causal chain

$$D_t \longrightarrow M_t \longrightarrow R_t \longrightarrow v_{t+\tau}.$$

The required tests are:

Detection. Matched damage interventions change M_t in a gate-specific direction.

Repair mediation. Intervening on M_t changes the repair distribution, and blocking M_t reduces the damage-to-repair effect.

Restoration. The selected repair improves $v_{t+\tau}$ relative to matched no-repair.

Specificity. A cost-matched generic action or an off-workspace repair does not produce the same restoration.

Identity preservation. The restored system remains within the declared identity tolerance relative to the pre-damage workspace.

Define the detection effect $\eta_{D\to M}$, mediation effect $\eta_{M\to R}$, restoration effect $\eta_{R\to v}$, and repair specificity

$$\Delta^{\text{spec}}_{\text{repair}} = \eta_{R\to v} - \eta_{R_\perp \to v},$$

where $R_\perp$ is a matched action outside the proposed repair channel.

53.8 T_{21}: Repair-selection margin theorem

Theorem 128 (Repair selection under value-model error)**.** *Fix a damage intervention d and a finite feasible repair set $\mathcal{R}$. Let*

$$V(r) = V_t^{\text{rep}}(r; d)$$

be the true expected post-repair integrity margin and let $\widehat{V}$ obey

$$\sup_{r\in\mathcal{R}} |\widehat{V}(r) - V(r)| \le \varepsilon_R.$$

Let $\widehat{r} \in \arg\max_r \widehat{V}(r)$ and $r^* \in \arg\max_r V(r)$. Then

$$\boxed{V(r^*) - V(\widehat{r}) \le 2\varepsilon_R.}$$

If

$$V(r^*) \ge \rho + 2\varepsilon_R,$$

then the selected repair restores the expected reserve:

$$\boxed{V(\widehat{r}) \ge \rho.}$$

If r^* is unique and its repair-value gap exceeds $2\varepsilon_R$, then $\widehat{r} = r^*$.

Proof. Greedy optimality and uniform approximation give

$$\begin{aligned} V(r^*) &\le \widehat{V}(r^*) + \varepsilon_R \\ &\le \widehat{V}(\widehat{r}) + \varepsilon_R \\ &\le V(\widehat{r}) + 2\varepsilon_R. \end{aligned}$$

The reserve and exact-selection statements follow immediately. □

The theorem is conditional on a valid repair-value model. It cannot convert a post hoc repair narrative into evidence that the system detected damage or used the model. Detection and value-channel interventions remain necessary.

53.9 Repair, replacement, and successor workspaces

Restoration of performance is not automatically restoration of identity. Let W_{t^-} be the last certified pre-damage workspace and let $W^r_{t+\tau}$ be the post-repair workspace. Define the repair identity defect

$$\delta_{\mathrm{rep-id}}(r) = d_{\mathrm{id}}\left(W_{t^-}, W^r_{t+\tau}\right).$$

A repair action is identity preserving at tolerance $\varepsilon_{\mathrm{id}}$ when

$$\Delta^{\mathrm{rep}}_t(r;d) > 0, \qquad \delta_{\mathrm{rep-id}}(r) \le \varepsilon_{\mathrm{id}}.$$

When integrity is restored but the identity defect exceeds tolerance, the result is better described as a successor or replacement workspace. This terminological distinction prevents any high-performing reconstruction from being counted as persistence of the same relational organization.

53.10 T_{22}: Closed-loop self-maintenance theorem

Theorem 129 (Causal repair stabilizes maintenance deficit)**.** *Let $e_t \ge 0$ be the maintenance deficit. Suppose the matched no-repair trajectory satisfies*

$$e^{(0)}_{t+1} \le \alpha e_t + d_t,$$

where $d_t \ge 0$ is exogenous damage. Suppose the implemented repair policy has the causal one-step effect

$$e^{(0)}_{t+1} - e_{t+1} \ge \kappa e_t - \epsilon_t.$$

Assume

$$0 \le \beta := \alpha - \kappa < 1, \qquad 0 \le d_t \le \overline{d}, \qquad 0 \le \epsilon_t \le \overline{\epsilon}.$$

Then

$$\boxed{e_t \le \beta^t e_0 + \frac{(\overline{d}+\overline{\epsilon})(1-\beta^t)}{1-\beta},}$$

and therefore

$$\boxed{\limsup_{t\to\infty} e_t \le \frac{\overline{d}+\overline{\epsilon}}{1-\beta}.}$$

If the admissible deficit is $e_{\max}$ and

$$e_0 \le e_{\max}, \qquad \overline{d}+\overline{\epsilon} \le (1-\beta)e_{\max},$$

then the interval $[0, e_{\max}]$ is forward invariant.

Proof. Combine the no-repair bound with the causal repair effect:

$$\begin{aligned} e_{t+1} &\le e^{(0)}_{t+1} - \kappa e_t + \epsilon_t \\ &\le (\alpha-\kappa)e_t + d_t + \epsilon_t \\ &\le \beta e_t + \overline{d} + \overline{\epsilon}. \end{aligned}$$

Iterating this scalar recurrence gives the stated geometric bound. If $e_t \le e_{\max}$, the forward-invariance condition gives

$$e_{t+1} \le \beta e_{\max} + (1-\beta)e_{\max} = e_{\max}.$$

□

Theorem 129 differs from the persistence theorem in Part X. There, contraction was assumed for the workspace error dynamics. Here the improved contraction factor is attributed to a measured repair action through a matched no-repair counterfactual. Without that causal comparison, the same bounded trajectory could be passive robustness rather than self-maintenance.

53.11 A scalar boundary-repair example

Let $b_t \in [0,1]$ be a boundary-integrity score and let the reserve be ρ. Define

$$e_t = (\rho - b_t)_+.$$

Suppose uncorrected damage obeys

$$e_{t+1}^{(0)} \le 1.08 e_t + d_t,$$

so the open-loop boundary is locally unstable. A repair policy with measured causal effect

$$e_{t+1}^{(0)} - e_{t+1} \ge 0.25 e_t - \epsilon_t$$

gives

$$\beta = 1.08 - 0.25 = 0.83.$$

If

$$\overline{d} = 0.01, \qquad \overline{\epsilon} = 0.005,$$

then

$$\limsup e_t \le \frac{0.015}{1-0.83} \approx 0.0883.$$

Thus an open-loop unstable boundary can be maintained within a finite deficit band by a sufficiently strong repair channel. The example establishes only a control calculation. A complete self-maintenance claim must also show that an internal monitor detects damage, mediates the repair choice, and preserves workspace identity.

53.12 Complete self-maintenance certificate

A self-maintaining workspace must pass all of the following gates:

1. *Protected-gate declaration.* The maintained integrity vector and reserve are fixed before evaluation.
2. *Damage validity.* Each damage intervention measurably degrades at least one protected gate.
3. *Internal detection.* The monitor changes under damage and is not merely an external evaluator.
4. *Repair mediation.* Monitor interventions change repair selection.
5. *Repair efficacy.* Repair improves the held-out integrity margin relative to matched no-repair.
6. *Specificity.* Cost-matched generic actions and off-workspace interventions do not reproduce the gain.

7. *Repeated-disturbance stability.* The deficit remains within the declared bound over a held-out disturbance sequence.
8. *Identity control.* The restored state remains within the declared identity tolerance or is reported as a successor workspace.
9. *Anti-centralization.* No single node contains a sufficient copy of the monitor, repair model, and protected state when the claim is workspace-level maintenance.

53.13 Certified workspace agent

Let

$$\Theta_A = (\tau_G, \tau_{G\to A}, \varepsilon_Q, \tau_{V\to A}, \tau_{A\to Y}, \tau_{D\to M}, \tau_{M\to R}, \tau_{R\to v}, \tau_{\mathrm{spec}}, \varepsilon_{\mathrm{ctl}}, e_{\max}) .$$

A Θ_A-certified workspace agent is a controlled realization for which:

1. $\mathrm{SRW}(W_t;\varsigma) = 1$;
2. cue-controlled goal retention exceeds τ_G;
3. goal-to-action mediation exceeds $\tau_{G\to A}$;
4. counterfactual-value error is at most ε_Q;
5. value-to-action use exceeds $\tau_{V\to A}$;
6. action-to-outcome control exceeds $\tau_{A\to Y}$;
7. damage detection, monitor-to-repair mediation, and restoration exceed their thresholds;
8. repair specificity exceeds τ_{spec};
9. $k_{\mathrm{ctl},\varepsilon_{\mathrm{ctl}}} \geq 2$; and
10. repeated maintenance deficit remains at most $e_{\max}$.

As throughout the theory, this predicate is relative to the declared node decomposition, goals, actions, environment, horizons, interventions, and null families.

53.14 T_{23}: Workspace-agency synthesis theorem

Theorem 130 (Workspace-level agency under declared gates)**.** *Assume a controlled realization satisfies:*

1. *the operational relationality assumptions of Theorem 76;*
2. *cue-controlled goal retention and action readout as in Theorem 125 and Corollary 126;*
3. *a counterfactual evaluator with uniform error ε_Q and optimization error δ_{opt};*
4. *feasible interventions establishing positive value-to-action and action-to-outcome effects;*
5. *the causal self-maintenance assumptions of Theorem 129; and*
6. *$k_{\mathrm{ctl},\varepsilon} \geq 2$, together with a matched local policy-null panel that preserves every single-node predictive marginal but loses full-system control value.*

Then, relative to the declared specification:

1. *the expected one-step decision regret is at most* $2\varepsilon_Q + \delta_{\mathrm{opt}}$*;*
2. *the long-run maintenance deficit is bounded by*

$$\frac{\overline{d} + \overline{\epsilon}}{1 - \beta};$$

3. *no single-node statistic is sufficient for the complete control state;*
4. *matched local policy nulls cannot reproduce the intact control value;*
5. *goal-dependent action selection and workspace repair are causally mediated by implemented internal channels.*

Hence the realization satisfies the formal predicate

$$\mathrm{Agent}_{\Theta_A}(\mathbf{W}) = 1.$$

Proof. The regret bound follows from Theorem 127. The maintenance bound follows from Theorem 129. The coalition condition excludes single-node sufficiency for the complete control state. The matched-null gap excludes explanations based only on preserved local predictive marginals. The implemented goal, value, action, monitor, and repair interventions establish the required causal mediation. These are precisely the gates in the stated agency predicate. □

The synthesis theorem is intentionally relative and conjunctive. Its force is that decision quality, maintenance stability, causal mediation, and relational location are certified together. It does not infer an unrestricted agent outside the tested environment.

53.15 Agency, autonomy, and consciousness

The following implications are not licensed:

$$\mathrm{Agent}_{\Theta_A}(\mathbf{W}) \not\Longrightarrow \mathrm{PhenomenallyConscious}(\mathbf{W}),$$

$$\mathrm{SelfMaintaining}(\mathbf{W}) \not\Longrightarrow \mathrm{SelfAware}(\mathbf{W}),$$

$$\mathrm{GoalPersistent}(\mathbf{W}) \not\Longrightarrow \mathrm{MoralPatient}(\mathbf{W}).$$

A thermostat-like controller can maintain a variable without representing itself. A sophisticated planner can evaluate counterfactual actions without phenomenal experience. Conversely, failure to pass this engineering certificate would not prove absence of consciousness. Agency and consciousness remain distinct axes of inquiry.

PART XVII

Decomposition, Coarse-Graining, and Robust Relationality

Relationality is always assessed relative to a way of dividing a realization into components. That dependence is not a minor technicality. If every participant, artifact, and channel is merged into one component, the complete workspace becomes local by definition. If a centralized state is recoded into several complementary shares, the same centralized process can be made to look irreducibly distributed. A defensible theory must therefore say which partitions count as admissible, which quantities survive faithful aggregation, and how sensitive a certificate is to alternative descriptions of the same system.

54. The Decomposition Problem

Let Ω be a finite set of primitive carriers at a declared audit resolution. A primitive carrier may be a person, a model instance, a process, a memory store, an instrument, or another independently tracked mechanism. The word *primitive* is relative to the audit. It does not assert a final metaphysical atomization of the system.

Definition 131 (System decomposition)**.** A decomposition of a carrier presentation is a tuple

$$\mathcal{D} = (\Pi, \mathbf{O}, G_\Pi, \mathcal{I}_\Pi),$$

where:

- $\Pi = \{B_\alpha\}_{\alpha \in A}$ is a partition of Ω;
- O_α is the declared local-access map for block B_α, producing the local state or history available at that component;
- G_Π records the declared communication, transfer, and causal interfaces among the blocks; and
- $\mathcal{I}_\Pi$ is the family of feasible interventions at that resolution.

The accessible information of component α is the sigma-field

$$\mathscr{F}_\alpha^{\mathcal{D}} = \sigma(O_\alpha(\mathbf{H}_t)) .$$

Every shared bus, common memory, supervisor, or artifact that can alter more than one local state must appear either as its own carrier or as an explicit channel in G_Π.

The same physical realization can support several decompositions. One may separate a model from its executor, merge them into one computational unit, or treat a persistent artifact as its own component. These alternatives are not mere renamings when they change local access or feasible interventions. Every certificate should therefore display the full specification

$$\mathrm{SRW}(W_t; \varsigma), \qquad \varsigma = (\mathcal{D}, Y^+, \ell, \tau, \mathcal{I}, \mathcal{N}, \Theta).$$

This part varies the decomposition coordinate while holding the remaining coordinates fixed. Part XVIII treats the full specification family.

Definition 132 (Refinement and coarsening)**.** Let $\mathcal{D}_f$ and $\mathcal{D}_c$ be decompositions of the same carrier presentation. The fine decomposition refines the coarse decomposition, written

$$\mathcal{D}_f \preceq \mathcal{D}_c,$$

when there is a surjection

$$q : V_f \longrightarrow V_c$$

such that each coarse component is the union of the fine components in one fiber of q, and its accessible information is generated only from those fine components. The aggregation is *information faithful* when

$$\mathscr{F}_B^{\mathcal{D}_c} = \bigvee_{i \in q^{-1}(B)} \mathscr{F}_i^{\mathcal{D}_f} \qquad (B \in V_c).$$

It is *structurally faithful* when it also retains every internal compatibility condition and every boundary transfer condition needed to recover the original global-section problem.

Refinement and coarsening are directional. A coarse component may contain information that was spread across several finer nodes. Conversely, a fine description may expose internal channels that disappear when a macro-node is formed. The next results identify what is preserved when the aggregation is faithful.

55. Faithful Structural Coarse-Graining

Let the vertex set of a core-transfer diagram be partitioned as

$$V = V_1 \sqcup \cdots \sqcup V_m.$$

Write the transfer discrepancy as

$$D_H = \begin{pmatrix} D_{\mathrm{int}} \\ D_\partial \end{pmatrix}, \qquad D_{\mathrm{int}} = \bigoplus_{\alpha=1}^{m} D_\alpha,$$

where D_α contains the edge constraints internal to V_α, and D_∂ contains the constraints crossing between blocks. Define the internally coherent macro-state spaces

$$W_\alpha = \ker D_\alpha, \qquad W_\Pi = \bigoplus_{\alpha=1}^{m} W_\alpha.$$

Let

$$J_\Pi : W_\Pi \hookrightarrow \mathcal{C}_V$$

be the canonical isometric inclusion, and define the quotient discrepancy

$$D_\Pi = D_\partial J_\Pi.$$

The quotient does not replace a group of nodes by an arbitrary summary. Its macro-state is the complete internal section space of that group, and its boundary operator retains all cross-block compatibility equations.

Theorem 133 (Faithful coarse-graining)**.** *For every finite partition of the vertex set,*

$$\boxed{\ker D_H = J_\Pi \ker D_\Pi.}$$

Equivalently,

$$\boxed{\Gamma(H) \cong \ker D_\Pi = \ker(D_\Pi^* D_\Pi).}$$

In particular,

$$\dim \Gamma(H) = \dim W_\Pi - \operatorname{rank}(D_\Pi).$$

Thus faithful coarse-graining preserves the exact structural workspace even though it changes the number and interpretation of the displayed nodes.

Proof. A vector $c \in \mathcal{C}_V$ lies in $\ker D_H$ exactly when

$$D_{\mathrm{int}} c = 0 \qquad \text{and} \qquad D_\partial c = 0.$$

The first condition is equivalent to $c = J_\Pi w$ for a unique $w \in W_\Pi$. Substitution into the second condition gives

$$D_\Pi w = D_\partial J_\Pi w = 0.$$

Hence $c \in \ker D_H$ if and only if $c = J_\Pi w$ with $w \in \ker D_\Pi$. The Laplacian identity follows from $\ker(D_\Pi^* D_\Pi) = \ker D_\Pi$, and the dimension formula is rank-nullity. □

The theorem supplies a precise sense in which a workspace may be represented at several resolutions without changing its exact core. The preservation fails when the macro-state discards internal section coordinates or when boundary equations are omitted.

Corollary 134 (Two-scale coherence bound)**.** *Let P_Π be the orthogonal projector onto $J_\Pi W_\Pi$. Let $\gamma_{\mathrm{int}} > 0$ be the smallest positive eigenvalue of $D_{\mathrm{int}}^* D_{\mathrm{int}}$, and let $\gamma_\Pi > 0$ be the smallest positive eigenvalue of $D_\Pi^* D_\Pi$. Then every $c \in \mathcal{C}_V$ satisfies*

$$\boxed{\operatorname{dist}(c, \Gamma(H)) \le \frac{\|D_{\mathrm{int}} c\|}{\sqrt{\gamma_{\mathrm{int}}}} + \frac{\|D_\Pi J_\Pi^* P_\Pi c\|}{\sqrt{\gamma_\Pi}}.}$$

The first term measures incoherence within macro-components. The second measures incompatibility among their internally coherent states.

Proof. By the triangle inequality,

$$\operatorname{dist}(c, \Gamma(H)) \le \|c - P_\Pi c\| + \operatorname{dist}(P_\Pi c, \Gamma(H)).$$

The workspace Poincaré estimate applied to D_{int} bounds the first term. Write $P_\Pi c = J_\Pi w$. Theorem 133 and the corresponding estimate for D_Π bound the second term by $\|D_\Pi w\|/\sqrt{\gamma_\Pi}$. Since J_Π is isometric, $w = J_\Pi^* P_\Pi c$. □

This decomposition is useful computationally. Internal blocks can be solved or compressed independently, after which only the boundary discrepancy must be assembled globally. It also gives a diagnostic distinction between a workspace that fails inside its local components and one whose local components are coherent but mutually incompatible.

56. Predictive Risk Under Aggregation

The structural core can be preserved by faithful coarse-graining, while coalition counts generally are not. Let

$$\mathscr{F}_S^{\mathcal{D}} = \bigvee_{i \in S} \mathscr{F}_i^{\mathcal{D}}$$

be the information available to coalition S under decomposition $\mathcal{D}$. For a declared loss ℓ, write

$$\mathcal{R}_{\mathcal{D}}(S) = \inf_f \mathbb{E}\Big[\ell(Y^+, f(\mathscr{F}_S^{\mathcal{D}}))\Big].$$

Theorem 135 (Bayes-risk transport under faithful aggregation)**.** *Suppose $\mathcal{D}_f \preceq \mathcal{D}_c$ through $q : V_f \to V_c$, and suppose the aggregation is information faithful. Then for every coarse coalition $S \subseteq V_c$,*

$$\boxed{\mathcal{R}_{\mathcal{D}_c}(S) = \mathcal{R}_{\mathcal{D}_f}(q^{-1}(S)).}$$

Consequently,

$$k_\varepsilon(\mathcal{D}_c) \le k_\varepsilon(\mathcal{D}_f),$$

where coalition size is counted in displayed nodes.

Proof. Information faithfulness gives

$$\mathscr{F}_S^{\mathcal{D}_c} = \bigvee_{B \in S} \bigvee_{i \in q^{-1}(B)} \mathscr{F}_i^{\mathcal{D}_f} = \mathscr{F}_{q^{-1}(S)}^{\mathcal{D}_f}.$$

The two Bayes risks optimize over the same measurable decision rules and are therefore equal. If a fine coalition of size k is sufficient within ε, the set of coarse blocks meeting that coalition has at most k elements and contains at least the same information. This proves the coalition-size inequality. □

The theorem makes the source of a common error explicit. A minimal coalition size of two at one resolution can become one after those two nodes are merged. Node count is therefore a decomposition-relative statistic.

A more stable quantity measures primitive carrier cost. Let $w_\omega > 0$ be a declared cost for each $\omega \in \Omega$, and assign a block the cost

$$c(B) = \sum_{\omega \in B} w_\omega.$$

Define the sufficient support cost

$$\boxed{s_\varepsilon(\mathcal{D}) = \min_{S \subseteq V_{\mathcal{D}}} \left\{ \sum_{B \in S} c(B) : \mathcal{R}_{\mathcal{D}}(S) \le \mathcal{R}_{\mathcal{D}}(V_{\mathcal{D}}) + \varepsilon \right\}.}$$

If $\mathcal{D}_f \preceq \mathcal{D}_c$ is information faithful, then

$$\boxed{s_\varepsilon(\mathcal{D}_f) \le s_\varepsilon(\mathcal{D}_c).}$$

The fine decomposition permits every union of coarse blocks and may also permit cheaper sub-block coalitions. Equality holds whenever an optimal fine coalition is a union of complete coarse fibers. This weighted support measure does not remove the need to choose primitive carriers, but it prevents

a large macro-component from being counted as a single inexpensive unit merely because it was assigned one node label.

57. Trivialization by Arbitrary Redescription

The dependence on decomposition is unavoidable. Two elementary constructions show why a certificate cannot be required to survive every mathematically possible partition.

Proposition 136 (Coarsening and secret-sharing trivializations)**.** *The following statements hold.*

1. *Any finite realization can be coarsened to a single displayed component. Relative to that decomposition, the full-system state is local and* $k_0 \le 1$.
2. *Let* Z *be uniform on a finite abelian group* G*. Let* K *be an independent uniform group element and define*
$$X_1 = K, \qquad X_2 = Z - K.$$
Then
$$I(Z; X_1) = I(Z; X_2) = 0, \qquad H(Z \mid X_1, X_2) = 0.$$
Under logarithmic loss,
$$\mathcal{R}(\{1\}) = \mathcal{R}(\{2\}) = \log|\mathsf{G}|, \qquad \mathcal{R}(\{1,2\}) = 0,$$
so the declared two-node representation has $k_0 = 2$ *and joint advantage* $\log|\mathsf{G}|$*, even when a single centralized encoder generated both shares.*

Proof. The first statement follows by taking the indiscrete partition $\Pi = \{\Omega\}$. For the second, conditioning on any value of Z leaves K, and hence X_1, uniform. Translation invariance of the uniform law also leaves $X_2 = Z - K$ uniform, so each share is independent of Z. The pair reconstructs $Z = X_1 + X_2$, which gives the entropy and Bayes-risk identities. □

The secret-sharing example is a valid distributed code. What it does not show is that the process producing or using the code is causally distributed across the declared carriers. If one hidden controller creates both shares, stores the key, or jointly selects their updates, that controller belongs in the causal inventory. Coalition irreducibility and matched predictive nulls remain necessary tests, but their interpretation is conditional on a defensible interface model.

This distinction suggests two levels of claim:

Encoding-relative relationality. No declared local view is sufficient.

Carrier-relative relationality. The declared views also correspond to admissible, separately tracked causal interfaces with no omitted common controller.

The stronger claim requires the decomposition audit developed next.

58. Admissible Decomposition Families

No purely formal rule can identify the uniquely correct decomposition of every physical or computational system. The practical goal is instead to define a family of decompositions that are defensible for the question being asked and to report how the certificate changes across that family.

Definition 137 (Admissible decomposition family)**.** An admissible family $\mathfrak{D}$ is a predeclared collection of decompositions of the same carrier presentation satisfying the following audit conditions.

1. **Carrier continuity.** Each component corresponds to a process, artifact, or mechanism that can be tracked through the relevant time scale.
2. **Access locality.** The local state contains only information available through the declared component and its recorded incoming channels.
3. **Channel completeness.** Shared memories, buses, supervisors, caches, and synchronized external cues are represented explicitly.
4. **Intervention separability.** At least some component mechanisms or channels can be altered without directly rewriting the entire system.
5. **Target independence.** The decomposition and its alternatives are fixed without using held-out outcomes to manufacture a favorable result.
6. **Resolution relevance.** A proposed split is retained only when it changes access, intervention, reconstruction, or transfer structure at the declared measurement resolution.
7. **Alternative coverage.** The family contains the principal plausible merges, splits, and artifact assignments that a skeptical analyst would consider.

Let $m_q(W;\mathcal{D})$ be the signed margin of gate q, oriented so that

$$m_q(W;\mathcal{D}) > 0$$

means that the gate passes after its uncertainty correction. Define the decomposition envelope

$$\underline{m}_q = \inf_{\mathcal{D}\in\mathfrak{D}} m_q(W;\mathcal{D}), \qquad \overline{m}_q = \sup_{\mathcal{D}\in\mathfrak{D}} m_q(W;\mathcal{D}),$$

and its sensitivity width

$$\Delta_q^{\mathrm{dec}} = \overline{m}_q - \underline{m}_q.$$

Definition 138 (Decomposition-robust certificate)**.** A workspace passes the $\mathfrak{D}$-robust certificate at level L when

$$\boxed{\mathrm{Certify}^{\mathrm{rob}}_{L,\mathfrak{D}}(W) = \bigwedge_{q\in\mathcal{J}(L)} \left\{ \inf_{\mathcal{D}\in\mathfrak{D}} m_q(W;\mathcal{D}) > 0 \right\}.}$$

A pass for one selected decomposition is a *decomposition-specific* claim. A pass throughout $\mathfrak{D}$ is a *family-robust* claim. A claim is *decomposition fragile* when at least one required gate changes sign inside the admissible family.

Universal invariance over all partitions is neither possible nor desirable, as Proposition 136 shows. The family $\mathfrak{D}$ must be broad enough to challenge the favored description and narrow enough to exclude partitions with no operational meaning.

59. Stability Under Decomposition Transport

Exact faithful aggregation preserves the structural core, but practical alternative decompositions usually introduce approximation. Their certificate can still be controlled by transporting a reference workspace and comparing gate margins.

Fix a reference decomposition $\mathcal{D}_0$. For each $\mathcal{D} \in \mathfrak{D}$, let

$$\Phi_{\mathcal{D}} : W^{\mathcal{D}_0} \longrightarrow W^{\mathcal{D}}$$

be a declared workspace correspondence and let $\delta_{\mathcal{D}}$ measure its structural and causal identity defect. Assume gate q has a transport bound

$$|m_q(\Phi_{\mathcal{D}} W; \mathcal{D}) - m_q(W; \mathcal{D}_0)| \le L_q \delta_{\mathcal{D}} + b_q(\mathcal{D}),$$

where L_q is a sensitivity constant and b_q records model mismatch not captured by the identity distance.

Theorem 139 (Decomposition-transport robustness)**.** *If, for every required gate q,*

$$\boxed{m_q(W; \mathcal{D}_0) > \sup_{\mathcal{D} \in \mathfrak{D}} \left(L_q \delta_{\mathcal{D}} + b_q(\mathcal{D}) \right),}$$

then W passes the $\mathfrak{D}$-robust certificate. Conversely, if for some required gate

$$m_q(W; \mathcal{D}_0) + \sup_{\mathcal{D} \in \mathfrak{D}} \left(L_q \delta_{\mathcal{D}} + b_q(\mathcal{D}) \right) < 0,$$

then that gate fails throughout the family.

Proof. For every $\mathcal{D} \in \mathfrak{D}$, the transport bound gives

$$m_q(\Phi_{\mathcal{D}} W; \mathcal{D}) \ge m_q(W; \mathcal{D}_0) - L_q \delta_{\mathcal{D}} - b_q(\mathcal{D}).$$

The first hypothesis makes the right-hand side positive uniformly in $\mathcal{D}$, which proves robust passage of every required gate. The second statement follows from the corresponding upper bound. □

When $b_q \equiv 0$, a convenient certificate radius is

$$\boxed{r_{\mathrm{dec}}(W) = \min_{q : L_q > 0} \frac{m_q(W; \mathcal{D}_0)}{L_q}.}$$

Every admissible family satisfying

$$\sup_{\mathcal{D} \in \mathfrak{D}} \delta_{\mathcal{D}} < r_{\mathrm{dec}}(W)$$

preserves the sign of all required gate margins. In applications, the transport terms can be bounded using earlier results:

Gate family	Relevant transport control
Structural coherence	Transfer-Laplacian projector drift and spectral gap.
Predictive alignment	Divergence between transported conditional-law states.
Causal efficacy	Distance between matched intervention kernels.
Relationality	Coalition-risk distortion and matched-null preservation error.
Reconstruction	Perturbation of lower frame bounds and handoff error.
Persistence and identity	The coordinatewise identity profile from Part XII.

The theorem is intentionally margin based. A result barely above threshold at one decomposition should not be described as robust when a small, defensible change of interface makes it disappear.

60. Decomposition Audit and Reporting

A complete decomposition audit should proceed in the following order.

1. Inventory the primitive carriers at the chosen temporal and physical resolution.
2. Record every shared resource and distinguish local state from incoming information.
3. Choose a reference decomposition from independently manipulable and temporally trackable interfaces.
4. Construct an admissible family containing the principal plausible merges, splits, and artifact assignments.
5. For every alternative, specify the correspondence to the reference workspace and validate its manipulation checks.
6. Recompute the structural, predictive, causal, coalition, reconstruction, and identity margins.
7. Report the full decomposition envelope, not only the most favorable member of the family.

The report should state:

reference decomposition and audit resolution,
admissible family and reasons for excluded partitions,
hidden-resource and common-controller checks,
faithful-aggregation maps and identity defects,
gatewise minima, maxima, and sensitivity widths,
strongest decomposition-specific and family-robust conclusions.

Four controls are especially important:

1. merge components connected by an uninterruptible common controller and test whether the relational claim collapses;
2. split a known centralized controller into complementary local views and verify that the protocol rejects the resulting secret-sharing appearance of distributedness;
3. expose or remove shared caches, external cues, and common memory that were absent from the favored graph; and
4. repeat the frozen certificate across the complete admissible family.

For the motivating human–AI collaboration, this analysis changes the form of the empirical question. It is not enough to ask whether the human, model, executor, and artifact store form a workspace under one natural description. The stronger question is whether the same operative state and its causal role survive plausible alternatives, such as merging model and executor, separating conversation state from persistent artifacts, or making shared orchestration channels explicit. Those measurements have not yet been performed. The case therefore remains decomposition-specific and observational.

PART XVIII

Specification Robustness, Identifiability, and Statistical Certification

A workspace certificate is never evaluated in a vacuum. It is evaluated for a particular future variable, loss, time horizon, intervention family, null panel, threshold vector, and system decomposition. These choices determine which distinctions matter and which capacities count as successful. They are therefore part of the scientific claim rather than clerical details surrounding it.

This dependence has two consequences. First, a result that passes under one reasonable analysis can fail under another. Robustness must be assessed across a declared family of scientifically defensible alternatives. Second, even a fully specified claim may not be identifiable from the observations and interventions that are actually available. Distinct realizations can induce the same experimental law while disagreeing about whether a relational workspace exists. The theory must say when such disagreement can be resolved, when only bounds are justified, and how sampling uncertainty compounds these structural ambiguities.

61. The Complete Scientific Specification

Definition 140 (Workspace specification)**.** A workspace specification is a tuple

$$\boxed{\varsigma = (\mathcal{D}, Y^+, \ell, \tau, \mathcal{I}, \mathcal{N}, \Theta),}$$

where:

- $\mathcal{D}$ is the decomposition and local-access model;
- Y^+ is the declared future variable or trajectory;
- ℓ is the normalized loss or utility convention;
- τ is the temporal horizon or horizon family;
- $\mathcal{I}$ is the feasible intervention family;
- $\mathcal{N}$ is the required matched-null family; and
- Θ is the gate schema, direction, threshold, and indifference region used for certification.

The complete certificate is written

$$\mathrm{Cert}(W;\varsigma) = \bigwedge_{q \in \mathcal{J}(\varsigma)} \mathbf{1}\{m_q(W;\varsigma) > 0\},$$

where m_q is the signed margin of gate q. Positive margin means that that gate passes in the direction declared by Θ.

The notation absorbs lower-bounded and upper-bounded gates into one sign convention. If a causal effect must exceed c_0, then

$$m_q = \widehat{\eta} - c_0.$$

If a defect must remain below ε_0, then

$$m_q = \varepsilon_0 - \widehat{\delta}.$$

The complete certificate therefore passes exactly when every required signed margin is positive.

The specification does not include arbitrary features of the analysis. It contains choices that alter the meaning or evidential force of the claim. An optimization algorithm, numerical tolerance, or estimator belongs in the estimation protocol unless changing it changes the target estimand or the admissible interventions. Conversely, the target, horizon, and null family cannot be treated as implementation details because changing them can reverse the conclusion without changing the underlying realization.

61.1 Specification relativity is not unrestricted relativism

A theorem-development workspace may be operationally real for a month-long research trajectory and irrelevant for a ten-second motor-control target. A repair channel may preserve theorem identity while failing to preserve social coordination. These are different scientific claims. The theory does not require one workspace state to be sufficient for every possible future or loss. It requires the intended scope to be declared before certification.

This yields the more precise notation

$$\boxed{\mathrm{SRW}(W;\varsigma) = 1.}$$

The earlier expression $\mathrm{SRW}_\Theta(W;\mathcal{D})$ is the special case in which the remaining coordinates of ς are held fixed.

61.2 Inert reparameterization and substantive change

Some changes of specification merely recode the same claim. Others change its content. The distinction is analogous to gauge freedom in the structural theory.

Definition 141 (Specification equivalence)**.** Two specifications ς and ς' are equivalent, written $\varsigma \simeq_{\mathrm{spec}} \varsigma'$, when there are bijective transport maps that:

1. preserve the sigma-field generated by the future target;
2. preserve the ordering and normalization of decisions under the loss;
3. map feasible interventions to physically equivalent interventions;
4. map every required null to a null preserving the same quantities; and
5. carry every gate threshold and margin into its counterpart.

Proposition 142 (Certificate invariance under specification equivalence)**.** *If $\varsigma \simeq_{\mathrm{spec}} \varsigma'$ and Φ is the induced workspace correspondence, then*

$$\mathrm{Cert}(W;\varsigma) = \mathrm{Cert}(\Phi W;\varsigma').$$

Proof. Every required gate is carried to a gate with the same signed margin. The conjunction of positive margins is therefore unchanged. □

A bijective relabeling of target classes with the correspondingly transported loss is inert. Replacing exact theorem validity with stylistic similarity is not. Changing a one-step horizon to a year-long persistence horizon is not. Removing the null that tests hidden centralization is not. Robustness concerns substantive alternatives after equivalent recodings have been quotiented out.

61.3 Admissible specification families

Definition 143 (Admissible specification family)**.** An admissible specification family $\mathfrak{S}$ is a predeclared set of specifications that:

1. refer to the same scientific claim type and carrier presentation;
2. use targets that are equivalent operationalizations or explicitly declared nearby stress targets;
3. normalize losses so that margins are comparable;
4. use horizons justified by the process being studied;
5. include every feasible intervention and null needed to distinguish the principal alternative explanations;
6. use thresholds fixed without access to certification outcomes; and
7. exclude only alternatives for reasons stated before evaluation.

The family should not contain every imaginable analysis. Such a requirement would erase meaningful domain specificity. It should contain the alternatives that a technically informed critic could reasonably regard as equally valid or as necessary stress tests for the declared claim. Multiverse and specification- curve methods formalize related concerns in empirical analysis [50, 48], while sensitivity analysis and partial identification emphasize that conclusions should expose their assumption dependence rather than conceal it [32, 40].

Three classes should be reported separately:

Equivalent alternatives. Different encodings or estimators of the same estimand after appropriate transport.

Robustness alternatives. Scientifically defensible choices intended to support the same claim.

Scope alternatives. Genuinely different targets or domains that define different claims and should not be averaged into one robustness number.

61.4 Common gate schema

A robustness family must use a common semantic gate index $\mathcal{J}_*$. A specification may mark a gate as not required only when the claim level itself does not include that property. It may not remove a failed gate after observing the result. Let

$$\mathcal{J}(\varsigma) \subseteq \mathcal{J}_*$$

be fixed before certification. For a family intended to support one fixed claim level, the cleanest design uses

$$\mathcal{J}(\varsigma) = \mathcal{J}_* \qquad (\varsigma \in \mathfrak{S}).$$

This prevents apparent robustness from being created by silently changing the meaning of passage across the family.

62. Robustness Across Admissible Specifications

For each gate $q \in \mathcal{J}_*$, define the specification envelope

$$\boxed{\underline{m}_q^{\mathfrak{S}}(W) = \inf_{\varsigma \in \mathfrak{S}} m_q(W;\varsigma), \qquad \overline{m}_q^{\mathfrak{S}}(W) = \sup_{\varsigma \in \mathfrak{S}} m_q(W;\varsigma).}$$

Its sensitivity width is

$$\Delta_q^{\mathrm{spec}}(W) = \overline{m}_q^{\mathfrak{S}}(W) - \underline{m}_q^{\mathfrak{S}}(W).$$

The robust certificate margin is

$$\boxed{r_{\mathfrak{S}}(W) = \inf_{\varsigma \in \mathfrak{S}} \min_{q \in \mathcal{J}_*} m_q(W;\varsigma).}$$

Then

$$\mathrm{Cert}_{\mathfrak{S}}(W) = 1 \quad \Longleftrightarrow \quad r_{\mathfrak{S}}(W) > 0.$$

Definition 144 (Specification-specific, robust, and fragile claims)**.** A claim is:

Specification-specific when it is asserted only for one declared ς_0.

Specification-robust when every required gate has positive lower envelope across $\mathfrak{S}$.

Specification-fragile when it passes for at least one admissible specification and fails for at least one other.

A high fraction of passing specifications is informative but does not replace the lower envelope when the claim is universal over $\mathfrak{S}$. A family in which ninety-nine analyses pass and one scientifically decisive hidden- controller null fails does not support the complete relational certificate. The noncompensatory logic remains in force at the specification level.

62.1 Geometry of a specification family

Let $d_{\mathfrak{S}}$ be a declared extended pseudometric on the quotient of $\mathfrak{S}$ by specification equivalence. A convenient decomposition is

$$\begin{aligned} d_{\mathfrak{S}}(\varsigma,\varsigma') = {} & w_D d_D(\mathcal{D},\mathcal{D}') + w_Y d_Y(Y^+,Y^{+'}) + w_\ell d_\ell(\ell,\ell') \\ & + w_\tau d_\tau(\tau,\tau') + w_I d_I(\mathcal{I},\mathcal{I}') + w_N d_N(\mathcal{N},\mathcal{N}') + w_\Theta d_\Theta(\Theta,\Theta'). \end{aligned}$$

No universal metric is imposed. The target term may use a divergence between conditional laws under a declared coupling. The intervention and null terms may use Hausdorff distances between implemented kernels. Threshold terms must be expressed on normalized signed-margin scales. The metric is part of the audit and should be varied in sensitivity analysis when more than one scientifically defensible choice exists.

For a reference specification ς_0, define the distance to certificate failure by

$$\boxed{r_{\mathrm{spec}}(W;\varsigma_0) = \inf\left\{ d_{\mathfrak{S}}(\varsigma,\varsigma_0) : \min_q m_q(W;\varsigma) \le 0 \right\}.}$$

If no failing specification exists in the declared universe, the radius is infinite relative to that universe. A radius near zero means that a small scientifically defensible change can reverse the conclusion.

62.2 Specification transport

Changing the decomposition can alter the coordinate realization of the workspace, while changing the target or horizon can alter the relevant predictive state. Let

$$\Phi_\varsigma : W_{\varsigma_0} \longrightarrow W_\varsigma$$

be the declared correspondence when such transport is needed. For each gate, assume a bound

$$|m_q(\Phi_\varsigma W;\varsigma) - m_q(W;\varsigma_0)| \le L_q d_{\mathfrak{S}}(\varsigma,\varsigma_0) + b_q(\varsigma),$$

where b_q records mismatch not represented by the metric. The mismatch may include a change of model class, imperfect target coupling, or an intervention whose implementation cannot be transported exactly.

62.3 T_{28}: Specification-transport robustness theorem

Theorem 145 (Specification-transport robustness)**.** *Let $\mathfrak{S}$ be an admissible specification family with reference ς_0. Suppose every required gate satisfies the transport bound above. If*

$$\boxed{m_q(W;\varsigma_0) > \sup_{\varsigma\in\mathfrak{S}} [L_q d_{\mathfrak{S}}(\varsigma,\varsigma_0) + b_q(\varsigma)]}$$

for every $q \in \mathcal{J}_*$, then

$$\mathrm{Cert}(\Phi_\varsigma W;\varsigma) = 1 \qquad \textit{for every } \varsigma \in \mathfrak{S}.$$

If $b_q \equiv 0$, then

$$\boxed{r_{\mathrm{spec}}(W;\varsigma_0) \ge \min_{q:L_q>0} \frac{m_q(W;\varsigma_0)}{L_q}.}$$

Proof. For each $\varsigma \in \mathfrak{S}$,

$$m_q(\Phi_\varsigma W;\varsigma) \ge m_q(W;\varsigma_0) - L_q d_{\mathfrak{S}}(\varsigma,\varsigma_0) - b_q(\varsigma).$$

The hypothesis makes the right-hand side positive uniformly over the family. Every required gate therefore passes under every admissible specification. When $b_q = 0$, no gate can cross zero at a distance smaller than its reference margin divided by its Lipschitz constant. Taking the minimum over gates gives the radius bound. □

The theorem is deliberately gatewise. A large structural margin cannot absorb a fragile causal margin. It also clarifies the relation to Part XVII. The decomposition-transport theorem is recovered by varying only the $\mathcal{D}$ coordinate while holding the remaining coordinates fixed.

62.4 A three-budget sufficient condition

In practice, the reference margin is reduced by three distinct uncertainty budgets:

$$\boxed{B_q^{\mathrm{total}} = B_q^{\mathrm{model}} + B_q^{\mathrm{spec}} + B_q^{\mathrm{sample}}.}$$

The first covers variation across observationally compatible realizations. The second covers variation across admissible specifications. The third covers finite-sample estimation. The sufficient condition

$$\widehat{m}_q(W;\varsigma_0) > B_q^{\text{total}} \qquad (q \in \mathcal{J}_*)$$

will reappear below in identified-set and confidence-envelope form. Keeping the budgets separate prevents an apparently narrow confidence interval from being mistaken for strong identification or specification robustness.

63. Experimental Equivalence and Identifiability

A complete mathematical definition does not guarantee that its truth can be recovered from available data. The experimental design may fail to distinguish two realizations that have different internal organization.

Definition 146 (Experimental design and induced law)**.** An experimental design is a tuple

$$\mathcal{E} = (\mathcal{O}, \mathcal{I}, \mathcal{T}),$$

where $\mathcal{O}$ is the observation map, $\mathcal{I}$ is the family of implemented interventions, and $\mathcal{T}$ is the set of measured temporal windows. A realization $\mathfrak{R}$ induces the family of laws

$$P_{\mathfrak{R}}^{\mathcal{E}} = \{\mathcal{L}_{\mathfrak{R}}(\mathcal{O}(\mathbf{H}) \mid \mathrm{do}(\iota)) : \iota \in \mathcal{I},\ t \in \mathcal{T}\}.$$

Definition 147 (Experimental equivalence)**.** Two realizations are experimentally equivalent under $\mathcal{E}$, written

$$\mathfrak{R} \equiv_{\mathcal{E}} \mathfrak{R}',$$

when

$$P_{\mathfrak{R}}^{\mathcal{E}} = P_{\mathfrak{R}'}^{\mathcal{E}}.$$

A property $\psi(\mathfrak{R})$ is identifiable on a model class $\mathfrak{M}$ under $\mathcal{E}$ when it is constant on every $\equiv_{\mathcal{E}}$-equivalence class contained in $\mathfrak{M}$.

Identifiability is therefore relative to both a model class and an experiment family. Enlarging the intervention family refines the equivalence classes and can make previously hidden distinctions identifiable. Enlarging the model class can have the opposite effect.

63.1 T_{29}: Non-identifiability impossibility theorem

Theorem 148 (No uniformly correct decision under experimental equivalence)**.** *Let* $\mathfrak{R}, \mathfrak{R}' \in \mathfrak{M}$ *satisfy*

$$\mathfrak{R} \equiv_{\mathcal{E}} \mathfrak{R}'$$

but

$$\mathrm{Cert}(W_{\mathfrak{R}};\varsigma) \neq \mathrm{Cert}(W_{\mathfrak{R}'};\varsigma).$$

Then the workspace certificate is not identifiable on $\mathfrak{M}$ *under* $\mathcal{E}$*. More strongly, for every possibly randomized decision rule* $\widehat{C}$ *based only on data generated by* $\mathcal{E}$*,*

$$\boxed{\mathbb{P}_{\mathfrak{R}}\left(\widehat{C} \neq C_{\mathfrak{R}}\right) + \mathbb{P}_{\mathfrak{R}'}\left(\widehat{C} \neq C_{\mathfrak{R}'}\right) = 1,}$$

where $C_{\mathfrak{R}} = \mathrm{Cert}(W_{\mathfrak{R}};\varsigma)$*. Consequently,*

$$\inf_{\widehat{C}} \max\left\{\mathbb{P}_{\mathfrak{R}}(\widehat{C} \neq C_{\mathfrak{R}}), \mathbb{P}_{\mathfrak{R}'}(\widehat{C} \neq C_{\mathfrak{R}'})\right\} \geq \frac{1}{2}.$$

Proof. The data law is identical under the two realizations, so $p = \mathbb{P}(\widehat{C} = 1)$ is the same in both. If $C_{\mathfrak{R}} = 0$ and $C_{\mathfrak{R}'} = 1$, the two error probabilities are p and $1 - p$. Their sum is one, and at least one is at least one half. The opposite labeling is identical. □

The theorem is not a limitation of a particular estimator. It is an information-theoretic obstruction. No amount of optimization or sample size can resolve a difference that the declared observations and interventions do not expose.

63.2 Hidden-controller witness

Consider two realizations that emit the same local streams X_1, X_2 and the same outcome

$$Y = X_1 \oplus X_2.$$

In the distributed realization, the two streams are generated and retained by separately intervenable carriers, and no carrier contains the pair. In the centralized realization, a hidden controller contains (X_1, X_2) and emits the two displayed streams. Under passive observation of (X_1, X_2, Y), the induced laws are identical. Even interventions that alter only the displayed messages may fail to distinguish them if the hidden controller simply re-emits the manipulated values.

Carrier-relative relationality becomes identifiable only when the experiment can test the claimed causal separation. Examples include independently interrupting the candidate carriers, exposing the hidden memory, randomizing a shared controller, or measuring whether one physical mechanism reconstructs the complete state before the displayed split. The decomposition controls N_{25}–N_{27} are therefore identifiability interventions, not merely optional robustness checks.

63.3 Identifiability of individual gates

The full certificate may be unidentified even when some coordinates are identified. Structural dimension might be recoverable from observed transfer constraints while causal specificity remains ambiguous. Conversely, a causal contrast may be identifiable even when the latent workspace coordinates are known only up to gauge. Each margin should therefore be assigned one of four statuses:

$$\text{point identified}, \quad \text{sign identified}, \quad \text{partially identified}, \quad \text{unidentified}.$$

A complete pass requires the sign of every required gate to be identified as positive, not merely that one compatible model produces a positive estimate.

64. Partial Identification of Workspace Claims

When point identification fails, the correct response is not to choose one convenient realization. It is to report the set of values compatible with the data and assumptions [40, 31].

Definition 149 (Identified realization set)**.** Given an experimental law $P^{\mathcal{E}}$ and model class $\mathfrak{M}$, define

$$\boxed{\mathfrak{R}_{\mathrm{ID}}(P^{\mathcal{E}}; \mathfrak{M}) = \left\{ \mathfrak{R} \in \mathfrak{M} : P^{\mathcal{E}}_{\mathfrak{R}} = P^{\mathcal{E}} \right\}.}$$

For a gate q and specification ς, its identified margin set is

$$\mathcal{M}_q^{\mathrm{ID}}(P^{\mathcal{E}}; \varsigma) = \{ m_q(W_{\mathfrak{R}}; \varsigma) : \mathfrak{R} \in \mathfrak{R}_{\mathrm{ID}} \}.$$

The sharp lower and upper identified margins are

$$\underline{m}_q^{\mathrm{ID}}(\varsigma) = \inf \mathcal{M}_q^{\mathrm{ID}}(\varsigma), \qquad \overline{m}_q^{\mathrm{ID}}(\varsigma) = \sup \mathcal{M}_q^{\mathrm{ID}}(\varsigma).$$

A gate is sign-identified as passing if its lower bound is positive. It is sign-identified as failing if its upper bound is nonpositive. If the interval crosses zero, the available design does not determine that gate's outcome.

64.1 Identified certificate set

Define

$$\boxed{\mathcal{C}_{\mathrm{ID}}(P^{\mathcal{E}}; \varsigma) = \{\mathrm{Cert}(W_{\mathfrak{R}}; \varsigma) : \mathfrak{R} \in \mathfrak{R}_{\mathrm{ID}}\} \subseteq \{0, 1\}.}$$

The certificate is point identified exactly when this set is a singleton. The marginal gate intervals provide simple sufficient conditions, but they need not be logically necessary because gate margins can be dependent across the identified set.

For robustness across specifications, define the realization-level robust margin

$$r_{\mathfrak{S}}(\mathfrak{R}) = \inf_{\varsigma \in \mathfrak{S}} \min_{q \in \mathcal{J}_*} m_q(W_{\mathfrak{R}}; \varsigma)$$

and its identified range

$$\underline{r}_{\mathrm{ID},\mathfrak{S}} = \inf_{\mathfrak{R} \in \mathfrak{R}_{\mathrm{ID}}} r_{\mathfrak{S}}(\mathfrak{R}), \qquad \overline{r}_{\mathrm{ID},\mathfrak{S}} = \sup_{\mathfrak{R} \in \mathfrak{R}_{\mathrm{ID}}} r_{\mathfrak{S}}(\mathfrak{R}).$$

64.2 T_{30}: Joint specification-identification theorem

Theorem 150 (Identification of a specification-robust certificate)**.** *Let $\mathfrak{R}_{\mathrm{ID}}$ be the identified realization set and $\mathfrak{S}$ an admissible specification family.*

1. *If*

$$\boxed{\underline{r}_{\mathrm{ID},\mathfrak{S}} > 0,}$$

then every realization compatible with the experimental law passes every required gate under every admissible specification. The robust certificate is identified as passing.

2. *If*

$$\boxed{\overline{r}_{\mathrm{ID},\mathfrak{S}} \leq 0,}$$

then no compatible realization passes the complete certificate throughout the family. The robust certificate is identified as failing.

3. *If there exist compatible realizations $\mathfrak{R}_+, \mathfrak{R}_-$ with*

$$r_{\mathfrak{S}}(\mathfrak{R}_+) > 0, \qquad r_{\mathfrak{S}}(\mathfrak{R}_-) \leq 0,$$

then the robust certificate is not identified under the declared experiment.

Proof. The first two conclusions follow directly from the definitions of the infimum and supremum over the identified set. The third gives two experimentally compatible realizations with different robust-certificate values, so Theorem 148 applies. □

This theorem joins two forms of robustness that are often conflated. A result can be stable across specifications within one fitted model yet fail to be identified across models compatible with the data. It can also be identified under one specification while changing sign across a reasonable specification family. The strongest claim controls both axes.

64.3 Confidence regions for identified sets

In finite samples, the identified set is itself uncertain. Let

$$\widehat{\mathfrak{R}}_{1-\delta}$$

be a confidence region satisfying

$$\inf_{\mathfrak{R}_0 \in \mathfrak{M}} \mathbb{P}_{\mathfrak{R}_0}\left(\mathfrak{R}_0 \in \widehat{\mathfrak{R}}_{1-\delta}\right) \geq 1-\delta.$$

A conservative robust lower margin is

$$\widehat{r}_{1-\delta} = \inf_{\mathfrak{R} \in \widehat{\mathfrak{R}}_{1-\delta}} \inf_{\varsigma \in \mathfrak{S}} \min_{q} m_q(W_{\mathfrak{R}}; \varsigma).$$

If $\widehat{r}_{1-\delta} > 0$, a false specification-robust certificate has probability at most δ, provided that the model class and specification family were fixed before the confidence region was computed. This confidence-set approach is useful when gate estimators are strongly coupled or when the latent realization is the primary inferential object.

65. Uniform Finite-Sample Certification

Population margins are not observed directly. A certification rule must control the estimation error uniformly across every specification and gate on which its conclusion depends.

Let $\widehat{m}_q(\varsigma)$ estimate $m_q(W;\varsigma)$. Suppose there are nonnegative error radii $\varepsilon_q(n,\delta)$ such that

$$\mathbb{P}\left(\sup_{\varsigma \in \mathfrak{S}} |\widehat{m}_q(\varsigma) - m_q(\varsigma)| \leq \varepsilon_q(n,\delta) \quad \text{for all } q \in \mathcal{J}_*\right) \geq 1-\delta.$$

Define the empirical lower confidence envelope

$$\boxed{\underline{\widehat{m}}^{\mathfrak{S}}_{q,1-\delta} = \inf_{\varsigma \in \mathfrak{S}} \widehat{m}_q(\varsigma) - \varepsilon_q(n,\delta).}$$

65.1 T_{31}: Uniform specification-certificate theorem

Theorem 151 (Uniform finite-sample certification across specifications)**.** *Under the uniform error event above, declare a specification-robust certificate only when*

$$\underline{\widehat{m}}^{\mathfrak{S}}_{q,1-\delta} > 0 \qquad \textit{for every } q \in \mathcal{J}_*.$$

Then

$$\boxed{\sup_{P:\, \mathrm{Cert}_{\mathfrak{S}}(W)=0} \mathbb{P}_P\left(\textit{the robust certificate is declared}\right) \leq \delta.}$$

Proof. On the uniform error event,

$$m_q(\varsigma) \geq \widehat{m}_q(\varsigma) - \varepsilon_q$$

for every gate and every specification. If all empirical lower envelopes are positive, every population margin is positive throughout the family. A false declaration can therefore occur only when the uniform error event fails, which has probability at most δ. □

The theorem controls the complete specification family, not merely the analysis that happened to produce the strongest estimate. It is an extension of the gatewise intersection rule in Theorem 156. Gatewise validity controls the conjunction at one frozen specification. Uniform validity controls the conjunction over a family.

65.2 Finite specification grids

Suppose $\mathfrak{S}$ and $\mathcal{J}_*$ are finite, and each margin is estimated as

$$\widehat{m}_q(\varsigma) = \frac{1}{n}\sum_{i=1}^{n} Z_{i,q,\varsigma} - \theta_{q,\varsigma}, \qquad 0 \leq Z_{i,q,\varsigma} \leq 1,$$

from independent certification units. Hoeffding's inequality and a union bound give

$$\mathbb{P}\left(\sup_{q,\varsigma} |\widehat{m}_q(\varsigma) - m_q(\varsigma)| > \sqrt{\frac{\log(2|\mathcal{J}_*||\mathfrak{S}|/\delta)}{2n}}\right) \leq \delta.$$

Thus the choice

$$\boxed{\varepsilon_n(\delta) = \sqrt{\frac{\log(2|\mathcal{J}_*||\mathfrak{S}|/\delta)}{2n}}}$$

yields an explicit conservative certificate. Dependent episodes require a bound appropriate to the true independent unit rather than token-level or edge-level pseudo-replication.

65.3 Continuous specification families

Let $\mathfrak{S}_\eta$ be an η-net of $(\mathfrak{S}, d_{\mathfrak{S}})$ with covering number $N(\eta, \mathfrak{S}, d_{\mathfrak{S}})$. If both the population and empirical margin functions are L_q-Lipschitz, then

$$\sup_{\varsigma\in\mathfrak{S}} |\widehat{m}_q(\varsigma) - m_q(\varsigma)| \leq \max_{\varsigma_j\in\mathfrak{S}_\eta} |\widehat{m}_q(\varsigma_j) - m_q(\varsigma_j)| + 2L_q\eta.$$

For bounded empirical means, a valid radius is therefore

$$\boxed{\varepsilon_{q,n,\eta}(\delta) = \sqrt{\frac{\log(2|\mathcal{J}_*|N(\eta, \mathfrak{S}, d_{\mathfrak{S}})/\delta)}{2n}} + 2L_q\eta.}$$

The tuning parameter η balances net cardinality against interpolation error. More refined empirical-process bounds may replace this elementary covering argument without changing the certificate logic.

65.4 Power and indifference regions

A robust procedure can be valid yet uninformative when the family is too large or the data are insufficient. Each gate should therefore report an indifference region $[-\kappa_q, \kappa_q]$. A conclusion is:

Robust pass when the lower confidence envelope exceeds κ_q for every gate.

Robust fail when a required gate's upper confidence envelope is below $-\kappa_q$ throughout the relevant family.

Specification conflict when well-estimated admissible specifications lie on opposite sides of the indifference region.

Statistically unresolved when sampling uncertainty intersects the indifference region.

This vocabulary separates genuine sensitivity from low power.

66. Adaptive Specification Choice and Selective Validity

A pointwise valid test does not remain valid after the analyst searches for the most favorable target, horizon, null panel, or threshold. Consider K independent null p-values and select the smallest. Then

$$\mathbb{P}\left(\min_{1\le k\le K} p_k \le \alpha\right) = 1-(1-\alpha)^K,$$

which exceeds α whenever $K>1$. The same logic applies when a workspace certificate is reported only for the specification that produced the largest gate margins.

The issue is not exploration itself. Exploration is often necessary to learn which target and intervention are scientifically meaningful. The issue is using the same random outcomes both to choose the claim and to certify it. Post-selection inference addresses the general problem by using simultaneous or selection-adjusted procedures [10].

66.1 Specification leakage

Definition 152 (Specification leakage)**.** Specification leakage occurs when certification outcomes influence any of the following without a fresh inferential correction:

$$Y^+, \ell, \tau, \mathcal{I}, \mathcal{N}, \Theta, \mathfrak{S}, \mathcal{J}_*.$$

The leakage status is separate from formation leakage. A workspace extractor may be frozen while the analyst still searches over targets or thresholds on the certification split.

A valid report must retain the complete search history, including failed and abandoned specifications. Reporting only the selected analysis prevents the reader from determining the effective multiplicity.

66.2 T_{32}: Sample-split selection theorem

Theorem 153 (Validity after independent specification selection)**.** *Let $\widehat{\varsigma}$ be selected using $\mathcal{D}_{\mathrm{sel}}$, and let $\mathcal{D}_{\mathrm{cert}}$ be independent of $\mathcal{D}_{\mathrm{sel}}$. For every fixed $\varsigma \in \mathfrak{S}$, let $\phi_\varsigma(\mathcal{D}_{\mathrm{cert}}) \in \{0,1\}$ be a certificate rule satisfying*

$$\sup_{P:\mathrm{Cert}(W;\varsigma)=0} \mathbb{P}_P\big(\phi_\varsigma = 1\big) \le \alpha.$$

Then the selected-specification procedure satisfies

$$\boxed{\sup_P \mathbb{P}_P\left(\phi_{\widehat{\varsigma}} = 1, \mathrm{Cert}(W;\widehat{\varsigma}) = 0\right) \le \alpha.}$$

Proof. Condition on $\mathcal{D}_{\mathrm{sel}}$. The selected specification is then fixed. Independence leaves the certification data distributed exactly as in the fixed-specification test, whose false-pass probability is at most α. Averaging the conditional bound over the selection data preserves the same bound. □

The theorem justifies discovery on formation or calibration data followed by a single frozen evaluation on fresh certification data. Reusing the certification split to revise the specification requires either another fresh split or a simultaneous post-selection procedure whose validity covers the entire search.

66.3 Nested selection and stress testing

A practical design may use:

$$\mathcal{D}_{\mathrm{form}} \sqcup \mathcal{D}_{\mathrm{spec}} \sqcup \mathcal{D}_{\mathrm{cert}} \sqcup \mathcal{D}_{\mathrm{stress}}.$$

Formation learns the workspace. The specification split chooses targets, horizons, nulls, and thresholds from a predeclared menu. Certification makes the pass decision once. Stress testing explores broader shifts without retroactively changing the certified claim. Cross-fitting can recover sample efficiency, but the fold-level selection and evaluation paths must remain separated and the final uncertainty calculation must account for recombination.

66.4 Thresholds and null families are part of selection

Threshold tuning can be as consequential as target selection. If the observed causal effect is 0.12, choosing a minimum effect threshold of 0.10 after seeing the estimate is not certification. Likewise, retaining only the nulls that the intact system defeats converts a falsification panel into a success filter. Thresholds and null families must be frozen or evaluated under a uniform rule that includes every considered alternative.

67. Specification Audit and Reporting

A complete specification audit should accompany every strong relational workspace claim.

67.1 Audit procedure

1. State the reference specification ς_0 in full.
2. Distinguish equivalent recodings, robustness alternatives, and genuinely different scope claims.
3. Declare the admissible family $\mathfrak{S}$ and justify every major exclusion before certification.
4. Record the specification pseudometric or other transport bounds used to compare family members.
5. Compute every gate's lower envelope, upper envelope, sensitivity width, and reference margin.
6. State the experiment family $\mathcal{E}$ and the model class $\mathfrak{M}$ relative to which identifiability is claimed.
7. Report identified sets or bounds whenever point identification fails.
8. Apply a uniform finite-sample rule over every gate and specification used in the conclusion.
9. Preserve the complete specification-selection history and the data split on which each choice was made.
10. Report the strongest pointwise, family-robust, sign-identified, and statistically resolved conclusions separately.

67.2 Additional adversarial controls

The empirical null and audit panel is extended by:

$$N_{29} = \text{post hoc target-selection audit},$$
$$N_{30} = \text{horizon cherry-picking audit},$$
$$N_{31} = \text{loss and utility substitution audit},$$
$$N_{32} = \text{restricted intervention-family audit},$$
$$N_{33} = \text{threshold and gate-set tuning audit},$$
$$N_{34} = \text{frozen specification sweep across } \mathfrak{S}.$$

The first five reconstruct or bound the analysis search that preceded the reported claim. The final control reruns the complete frozen certificate across the admissible family. A claim that survives N_{34} but fails the search audit remains vulnerable to selection bias. A claim that passes the search audit but changes sign across $\mathfrak{S}$ is validly estimated yet specification-fragile.

67.3 Required specification record

The machine-readable record should include:

$$\begin{aligned}\mathsf{SpecificationRecord} = \{&\mathsf{reference}, \mathsf{family}, \mathsf{equivalence_classes}, \mathsf{metric},\\ &\mathsf{targets}, \mathsf{losses}, \mathsf{horizons}, \mathsf{interventions},\\ &\mathsf{null_panels}, \mathsf{thresholds}, \mathsf{selection_history},\\ &\mathsf{identified_sets}, \mathsf{uniform_bounds}, \mathsf{gate_envelopes}\}.\end{aligned}$$

Every omitted family member requires a reason. Every changed specification requires a data-split label and time stamp. Every gate requires pointwise and family-wide status.

67.4 Claim-strength vocabulary

The following vocabulary is recommended:

Pointwise demonstrated. The certificate passes at one frozen specification with valid uncertainty.

Specification-robust. It passes uniformly across a declared admissible family.

Sign identified. Every realization compatible with the experimental law gives the same gate sign.

Robustly identified. The complete family-robust certificate is identified as passing.

Partially identified. The data restrict the certificate margins but do not determine every required sign.

Specification-fragile. Validly estimated admissible specifications give different certificate outcomes.

Selection-contaminated. The reported specification was chosen using certification outcomes without valid adjustment.

67.5 Consequence for the motivating collaboration

The motivating human–AI research process remains a plausible candidate persistent workspace, but its strongest current description is narrower than a certified robust claim. The relevant target trajectory, horizon family, intervention set, null panel, thresholds, and admissible specification family have not been frozen and tested prospectively. Nor has the carrier-relative workspace predicate been shown to be identifiable against hidden-controller and matched-storage alternatives. The theory therefore supports a disciplined research program rather than a retrospective declaration.

The terminal architecture is now explicit. A strong claim must survive three separate questions:

Would it survive another defensible specification?
Could a contrary realization generate the same experiments?
Does the finite sample support every required margin uniformly?

Once these questions are answered, further extensions concern new mathematical settings or new empirical domains rather than a missing foundational layer.

PART XIX

Summary of Main Results

68. Structural and Foundational Results

T_1: Limit–holonomy theorem

$$\Gamma(H) \cong C_o^{\mathrm{Hol}_o}.$$

T_2: Operator closure theorem

$$H_e \rho_i(a) = \rho_j(a) H_e \Longrightarrow \rho_o(a) C_* \subseteq C_*.$$

T_{2a}: Workspace commutant theorem

$$\mathcal{O}_{\mathbf{W}} \cong \mathrm{Comm}(\mathrm{Hol}_o).$$

T_3: Transfer-Laplacian kernel theorem

$$\ker L_H = \Gamma(H).$$

T_4: Spectral stability theorem

If $\eta = \|\widetilde{L}_H - L_H\| < g_d/2$, then

$$\|\widetilde{P}_d - P_d\| \leq \frac{\eta}{g_d - \eta} \leq \frac{2\eta}{g_d}.$$

T_5: Gauge invariance theorem

$$\Gamma(H^g) = G_V \Gamma(H),$$

and under unitary gauge,

$$L_{H^g} = G_V L_H G_V^{-1}.$$

T_6: Persistence theorem

For the nonautonomous recurrence

$$e_{t+1} \leq \alpha_t e_t + \sigma_t,$$

$$\boxed{e_t \leq \left(\prod_{r=0}^{t-1} \alpha_r \right) e_0 + \sum_{j=0}^{t-1} \left(\prod_{r=j+1}^{t-1} \alpha_r \right) \sigma_j.}$$

If $\alpha_t \le \bar{\alpha} < 1$ and $\sigma_t \le \bar{\sigma}$, then

$$\limsup_{t\to\infty} e_t \le \frac{\bar{\sigma}}{1-\bar{\alpha}}.$$

The derivation begins with a moving workspace projector and an explicit system update, and adds a mean-square analogue and a non-collapse gate.

T_7: Stable reconstruction theorem

For coalition analysis operator A_S with lower frame bound $\alpha_S > 0$, observations

$$y = A_S c + \eta$$

and canonical decoder $A_S^\dagger$ satisfy

$$\boxed{\|A_S^\dagger y - c\| \le \frac{\|\eta\|}{\sqrt{\alpha_S}}.}$$

The theorem is extended to encoding mismatch, exact support-based erasure distance, uniform node erasures, reconstructive handoff, exact coalition sufficiency, and operational risk under a Lipschitz predictive kernel.

T_8: Exact gluing theorem

For every finite edge-complete workspace cover,

$$\boxed{0 \longrightarrow \mathscr{W}(G) \xrightarrow{R_{\mathcal{U}}} \check{C}^0(\mathcal{U};\mathscr{W}) \xrightarrow{\delta^0} \check{C}^1(\mathcal{U};\mathscr{W})}$$

is exact. Hence

$$\boxed{\mathscr{W}(G) \cong \ker \delta^0 \cong \mathrm{Eq}\left(\prod_\alpha \mathscr{W}_\alpha \rightrightarrows \prod_{\alpha<\beta} \mathscr{W}_{\alpha\beta}\right).}$$

The same analysis also yields a cover-Laplacian distance bound, a total descent kernel theorem, and a harmonic $\check{H}^1$ obstruction for relative-only data.

T_9: Hierarchical collapse theorem

For every finite rooted hierarchy whose children edge-cover their parent,

$$\boxed{W_a \cong \mathscr{W}(G_a) \quad \text{for every hierarchy node } a,}$$

and in particular

$$\boxed{W_{\mathrm{root}} \cong \mathscr{W}(G).}$$

If level r contributes defect δ_r and downstream collapse maps have Lipschitz constants L_s, then

$$\boxed{\delta_{\mathrm{flat}}^{(R)} \le \sum_{r=0}^{R-1} \left(\prod_{s=r+1}^{R-1} L_s\right) \delta_r.}$$

T_{25}: Faithful coarse-graining theorem

For a partition of the vertex set, let W_Π be the direct sum of the internally coherent section spaces and let D_Π retain all boundary compatibility equations. Then

$$\boxed{\Gamma(H) = J_\Pi \ker D_\Pi.}$$

Exact structural workspace content is therefore invariant under faithful aggregation. Approximate error separates into an internal-coherence term and a boundary-coherence term controlled by their respective spectral gaps.

T_{26}: Bayes-risk transport theorem

For an information-faithful coarsening $q : V_f \to V_c$,

$$\boxed{\mathcal{R}_{\mathcal{D}_c}(S) = \mathcal{R}_{\mathcal{D}_f}(q^{-1}(S)) \qquad (S \subseteq V_c).}$$

Hence displayed coalition size can decrease under coarsening. Weighted primitive support cost is a more stable localization measure and cannot improve merely because several carriers were relabeled as one macro-node.

T_{27}: Decomposition-transport robustness theorem

If every gate margin at a reference decomposition exceeds its worst-case transport and model-mismatch defect across $\mathfrak{D}$, then the complete certificate passes throughout the admissible family:

$$\boxed{m_q(W; \mathcal{D}_0) > \sup_{\mathcal{D} \in \mathfrak{D}} \left(L_q \delta_{\mathcal{D}} + b_q(\mathcal{D})\right) \quad \forall q \Longrightarrow \mathrm{Certify}^{\mathrm{rob}}_{L,\mathfrak{D}}(W) = 1.}$$

69. Dynamic, Causal, and Agency Results

T_{10}: Approximate universality theorem

For the first d transfer-Laplacian modes, with $\lambda_d \le \varepsilon < \lambda_{d+1} = \mu$,

$$\boxed{\|(I - P_d)\rho_V(a)P_d\| \le \frac{\|\rho_E(a)\|\sqrt{\varepsilon} + \beta(a)}{\sqrt{\mu}}.}$$

Thus low transfer energy, a separated complement, and small intertwining defect imply approximate closure under the common operation algebra.

T_{11}: Core–residual accumulation theorem

For a path $\gamma = e_m \cdots e_1$,

$$\boxed{\begin{aligned}\|Q_{t(\gamma)} T_\gamma P_{s(\gamma)}\| \le \sum_{r=1}^{m} \left(\prod_{\ell=r+1}^{m} \|T_{e_\ell}\|\right) \|V_{e_r}\| \\ \times \left(\prod_{\ell=1}^{r-1} \|H_{e_\ell}\|\right).\end{aligned}}$$

The same bound controls the defect between full core-to-core path transport and the pure core product. An exact Duhamel identity yields both estimates.

T_{12}: Predictive-state minimality and alignment theorem

On standard Borel spaces, the canonical conditional-law state

$$Z_t = \mathbb{P}(Y_t^+ \in \cdot \mid \mathbf{H}_t)$$

is sufficient and minimal:

$$Y_t^+ \perp\!\!\!\perp \mathbf{H}_t \mid Z_t,$$

and every sufficient statistic $S_t = s(\mathbf{H}_t)$ satisfies

$$Z_t = \phi(S_t) \qquad \text{a.s.}$$

Moreover, a structural candidate W_t is exactly operationally aligned if and only if

$$Z_t = \phi_W(W_t) \qquad \text{a.s.}$$

The countable intervention-indexed version holds for $Z_t^{\mathcal{I}}$.

T_{13}: Operational relationality theorem

For a finite target, exact alignment, a positive node-level conditional information gap, a positive robust local-predictive-matched null gap, a feasible workspace intervention effect, and positive matched specificity jointly imply that:

no single-node statistic is sufficient,
$k_0 \geq 2$,
no single node reconstructs the aligned workspace,
matched local nulls preserve all node risks but lose joint capacity,
and the workspace coordinate is causally efficacious at the tested scale.

If $\eta_{\text{loop}}(\tau) > 0$, the conclusion additionally includes loop-mediated reentry.

T_{14}: Replacement invariance theorem

For a unitary realization correspondence with discrepancy defect

$$\delta_D = \|D' J_V - J_E D\|,$$

set

$$\eta_L = 2\|D\|\delta_D + \delta_D^2.$$

If $\eta_L < g/2$, then the workspace projectors obey

$$\boxed{q_{\text{str}} \leq \frac{\eta_L}{g - \eta_L} \leq \frac{2\eta_L}{g}.}$$

A canonical unitary correspondence between the workspace subspaces exists, and its operation defect is bounded by the projector drift, the local intertwining defect, and prior operation leakage. Coalition frame bounds also persist under sufficiently small reconstructive transport error. With declared causal-kernel and reconstructive-handoff defects,

$$\boxed{d_{\text{id}} \leq a\, q_{\text{str}} + b\, \delta_{\text{op}} + c\, \delta_{\text{causal}} + d\, \delta_{\text{rec}}.}$$

Under bounded norms and a uniform spectral gap, this gives the schematic estimate

$$d_{\mathrm{id}} \lesssim C \frac{\delta_{\mathrm{replacement}}}{g}.$$

The causal and reconstructive terms are necessary assumptions rather than consequences of structural similarity.

T_{15}: Transfer identification theorem

Under the linear transfer model

$$Y = H_* X + \xi, \qquad \mathbb{E}[\xi X^{\mathsf{T}}] = 0, \qquad \Sigma_{XX} \succeq \kappa I,$$

the unique population least-squares transfer is

$$\boxed{H_* = \Sigma_{YX} \Sigma_{XX}^{-1},}$$

and

$$\mathcal{R}(H) - \mathcal{R}(H_*) \geq \kappa \|H - H_*\|_F^2.$$

For empirical ordinary least squares,

$$\|\widehat{H} - H_*\|_F \leq \frac{\|\widehat{M}\|_F}{\widehat{\kappa}},$$

with the stated ridge extension. The identified transfer is gauge covariant under independent invertible changes of source and target coordinates.

T_{16}: Uniform structure-selection theorem

For a finite candidate family with unique population optimum θ_*, margin $\Delta_J > 0$, and held-out uniform score error at most ε,

$$\boxed{J(\widehat{\theta}) - J(\theta_*) \leq 2\varepsilon.}$$

If $2\varepsilon < \Delta_J$, then $\widehat{\theta} = \theta_*$. The theorem applies to node membership, edge selection, gauge-normalized transfer models, and finite boundary families.

T_{17}: Stable boundary recovery theorem

For a finite admissible boundary family with unique maximizer S_t^*, margin Δ_t^B, and uniform held-out score error ε_t,

$$\mathcal{B}_t(S_t^*) - \mathcal{B}_t(\widehat{S}_t) \leq 2\varepsilon_t,$$

and exact recovery holds when $2\varepsilon_t < \Delta_t^B$. If the population score drifts uniformly by at most δ_t, then

$$2\delta_t < \Delta_t^B \Longrightarrow S_{t+1}^* = S_t^*.$$

T_{18}: Model-specific formation transition theorem

For

$$m_{t+1} = \tanh(\lambda m_t),$$

there is a unique zero fixed point for $0 \le \lambda \le 1$. For $\lambda > 1$, the zero state is unstable and exactly two stable nonzero fixed points appear. Their amplitude satisfies

$$\boxed{m_\lambda = \sqrt{\frac{3(\lambda-1)}{\lambda^3}} + O((\lambda-1)^{3/2}).}$$

This is an exact structural benchmark, not a universal law for complete relational workspaces.

T_{19}: Cue-controlled goal-retention theorem

For matched future command and disturbance trajectories under

$$G_{s+1} = R_s G_s + B_s C_{s+1} + \zeta_{s+1},$$

two goal interventions satisfy

$$\boxed{G_{t+\tau} - G'_{t+\tau} = R_{t+\tau:t}(g - g').}$$

If

$$m_s \le \sigma_{\min}(R_s), \qquad \|R_s\| \le M_s,$$

then

$$\boxed{\left(\prod_s m_s\right)\|g-g'\| \le \|G_{t+\tau} - G'_{t+\tau}\| \le \left(\prod_s M_s\right)\|g-g'\|.}$$

A linear action readout gives a corresponding lower bound on the induced action-mean separation. The result isolates internal retention from repeated external goal rewriting.

T_{20}: Approximate counterfactual-policy theorem

If the interventional action-value estimator satisfies

$$\sup_a |\widehat{Q}_t(a) - Q_t(a)| \le \varepsilon_Q$$

and the implemented policy has optimization error at most δ_{opt} relative to $\widehat{Q}_t$, then

$$\boxed{\text{Reg}_t(\widehat{\pi}_t) \le 2\varepsilon_Q + \delta_{\text{opt}}.}$$

For deterministic greedy action, the bound is $2\varepsilon_Q$. A unique true optimum with value gap greater than $2\varepsilon_Q$ is recovered exactly. The guarantee is relative to the declared utility, feasible action support, causal identification, and horizon.

T_{21}: Repair-selection margin theorem

For a finite repair set and a repair-value estimator with uniform error ε_R, the selected repair $\widehat{r}$ satisfies

$$\boxed{V(r^*) - V(\widehat{r}) \le 2\varepsilon_R.}$$

If the optimal repair reaches reserve $\rho + 2\varepsilon_R$, then the selected repair reaches reserve ρ. A unique optimal repair whose value gap exceeds $2\varepsilon_R$ is selected exactly.

T_{22}: Closed-loop self-maintenance theorem

If matched no-repair deficit obeys

$$e_{t+1}^{(0)} \le \alpha e_t + d_t$$

and the implemented repair has measured causal effect

$$e_{t+1}^{(0)} - e_{t+1} \ge \kappa e_t - \epsilon_t,$$

set $\beta = \alpha - \kappa$. Under

$$0 \le \beta < 1, \qquad d_t \le \overline{d}, \qquad \epsilon_t \le \overline{\epsilon},$$

$$\boxed{e_t \le \beta^t e_0 + \frac{(\overline{d} + \overline{\epsilon})(1 - \beta^t)}{1 - \beta}.}$$

Hence

$$\limsup_{t \to \infty} e_t \le \frac{\overline{d} + \overline{\epsilon}}{1 - \beta}.$$

A corresponding margin condition makes the declared bounded-deficit viability set forward invariant.

T_{23}: Workspace-agency synthesis theorem

Operational relationality, cue-controlled and behaviorally operative goals, calibrated counterfactual values, feasible value-to-action and action-to-outcome interventions, causal closed-loop maintenance, distributed control-state sufficiency, and matched local policy nulls jointly imply:

$$\begin{aligned} &\mathrm{Reg}_t \le 2\varepsilon_Q + \delta_{\mathrm{opt}}, \\ &\limsup e_t \le \frac{\overline{d} + \overline{\epsilon}}{1 - \beta}, \\ &k_{\mathrm{ctl},\varepsilon} \ge 2, \end{aligned}$$

together with causal mediation of goal-dependent action and repair through the implemented internal channels. This is a sufficient, domain-relative agency certificate. It does not imply consciousness, personhood, or unrestricted autonomy.

T_{24}: Gatewise certification-validity theorem

Suppose a certificate requires every gate $G_1, \dots, G_m$ to pass, and the predeclared test for gate j has false-pass probability at most α_j whenever that required property is absent. Under the global null that at least one required property is absent,

$$\boxed{\mathbb{P}(\text{complete certificate passes}) \le \max_j \alpha_j.}$$

No independence assumption among gate tests is required. The guarantee fails under adaptive gate deletion, selective reporting, or reuse of the certification split.

70. Decomposition, Specification, and Identification Results

T_{25}: Faithful coarse-graining theorem

For an exact partition into internally coherent macro-components,

$$\boxed{\ker D_H = J_\Pi \ker D_\Pi.}$$

Thus faithful aggregation preserves the exact shared workspace and its dimension while changing the displayed node resolution.

T_{26}: Bayes-risk transport under aggregation

For an information-faithful coarsening map q,

$$\boxed{\mathcal{R}_{\mathcal{D}_c}(S) = \mathcal{R}_{\mathcal{D}_f}(q^{-1}(S)).}$$

Predictive content is preserved, although the visible coalition count may contract. Primitive-carrier support cost is therefore preferred when claims must be compared across resolutions.

T_{27}: Decomposition-transport robustness theorem

If every required gate has reference margin larger than its worst admissible decomposition-transport defect, then the complete certificate passes throughout $\mathfrak{D}$. When the mismatch term vanishes, the certificate radius is bounded below by the smallest gate margin divided by its transport sensitivity.

T_{28}: Specification-transport robustness theorem

For a reference specification ς_0, if

$$m_q(W;\varsigma_0) > \sup_{\varsigma\in\mathfrak{S}} [L_q d_{\mathfrak{S}}(\varsigma,\varsigma_0) + b_q(\varsigma)]$$

for every gate, then the transported workspace passes every admissible specification. If $b_q = 0$,

$$r_{\mathrm{spec}}(W;\varsigma_0) \geq \min_{q:L_q>0} \frac{m_q(W;\varsigma_0)}{L_q}.$$

T_{29}: Experimental non-identifiability theorem

If two realizations induce the same law under every declared observation and intervention but have different certificate values, no decision rule based on that experiment can be uniformly correct. For every such rule, the two error probabilities sum to one, so the minimax error is at least one half.

T_{30}: Joint specification-identification theorem

Let $r_{\mathfrak{S}}(\mathfrak{R})$ be the worst gate margin across the specification family. If its infimum over the identified realization set is positive, the robust certificate is identified as passing. If its supremum is nonpositive, the certificate is identified as failing. Compatible realizations on opposite sides of zero establish non-identifiability.

T_{31}: Uniform finite-sample specification theorem

If all gate estimators are uniformly within their error radii across $\mathfrak{S}$, then requiring every empirical lower envelope to be positive controls the probability of a false robust certificate by the failure probability of the uniform event.

T_{32}: Independent selection and certification theorem

A specification chosen on data independent of the certification split may be evaluated by its fixed-specification test without inflating the false-pass bound. Selection and certification on the same outcomes require simultaneous or post-selection correction.

PART XX

Conjectures and Open Mathematical Directions

71. Workspace Formation Transitions

The symbolic outline proposed a coupling threshold λ_c. A general theory must first state what is allowed to vary with λ. A threshold may concern structural coherence, causal efficacy, relational irreducibility, or the conjunction of all certification gates. These need not occur at the same parameter value.

71.1 No coordinate-free universal threshold

Proposition 154 (Threshold values are parameterization dependent)**.** *Suppose a model indexed by λ passes a declared formation certificate exactly for $\lambda > \lambda_c$. For every strictly increasing bijection f, the reparameterized family indexed by $\mu = f(\lambda)$ has threshold*

$$\mu_c = f(\lambda_c).$$

Consequently, the numerical value of a coupling threshold is not invariant without a physically or operationally fixed parameterization.

Proof. The reparameterized realization at μ is the original realization at $f^{-1}(\mu)$. It passes precisely when $f^{-1}(\mu) > \lambda_c$, equivalently when $\mu > f(\lambda_c)$. □

There can therefore be no model-independent numerical value of λ_c. The invariant object is the change in the certified organization and its local critical behavior, not the label placed on the control parameter.

71.2 Gate-specific transition parameters

For a one-parameter family $\mathfrak{R}_\lambda$, define

$$\begin{aligned}
\lambda_{\mathrm{str}} &= \inf\{\lambda : d_\varepsilon(\lambda) > 0, g_\varepsilon(\lambda) > 0\},\\
\lambda_{\mathrm{op}} &= \inf\{\lambda : \delta_{\mathrm{align}}(\lambda) \le \varepsilon_{\mathrm{op}}\},\\
\lambda_{\mathrm{causal}} &= \inf\{\lambda : \eta_{\mathrm{causal}}(\lambda) \ge \tau_{\mathrm{causal}}\},\\
\lambda_{\mathrm{rel}} &= \inf\{\lambda : k_\varepsilon(\lambda) \ge 2, \Delta_{\mathrm{tr}}(\lambda) \ge \tau_{\mathrm{tr}}\},\\
\lambda_{\mathrm{form}} &= \inf\{\lambda : \mathrm{Form}_{\Theta_{\mathrm{form}}}(W_\lambda) = 1\}.
\end{aligned}$$

In general,

$$\lambda_{\mathrm{str}}, \lambda_{\mathrm{op}}, \lambda_{\mathrm{causal}}, \lambda_{\mathrm{rel}}, \lambda_{\mathrm{form}}$$

can be distinct. A synchronized representation can appear before it becomes causally useful, and causal coordination can appear while remaining reducible to one centralized component.

71.3 A solvable nonlinear formation model T_{18}

A minimal exact model is the scalar recurrent coherence equation

$$\boxed{m_{t+1} = \tanh(\lambda m_t), \qquad \lambda \ge 0.}$$

The order parameter m_t represents a signed coherent mode. This model is not a complete relational workspace. It isolates one mechanism by which a nonzero shared mode can appear continuously from a symmetric zero state.

Theorem 155 (Pitchfork formation transition)**.** *For the map*

$$F_\lambda(m) = \tanh(\lambda m), \qquad m \in [-1, 1],$$

the following hold.

1. *If $0 \le \lambda \le 1$, the unique fixed point is $m = 0$. It is globally attracting for $\lambda < 1$ and nonexpanding at $\lambda = 1$.*
2. *If $\lambda > 1$, there are exactly three fixed points*

$$-m_\lambda, \quad 0, \quad m_\lambda, \qquad m_\lambda \in (0, 1).$$

   The zero fixed point is unstable and the two nonzero fixed points are locally asymptotically stable.

3. *As $\lambda \downarrow 1$,*

$$\boxed{m_\lambda = \sqrt{\frac{3(\lambda - 1)}{\lambda^3}} + O((\lambda - 1)^{3/2}).}$$

Proof. A fixed point satisfies

$$\operatorname{artanh}(m) = \lambda m.$$

For $m > 0$, the function

$$h(m) = \frac{\operatorname{artanh}(m)}{m}$$

is strictly increasing from 1 to $+\infty$. Hence there is no positive solution when $\lambda \le 1$, and exactly one positive solution when $\lambda > 1$. Odd symmetry gives the negative solution.

The derivative is

$$F'_\lambda(m) = \lambda \operatorname{sech}^2(\lambda m).$$

At zero this equals λ, proving stability for $\lambda < 1$ and instability for $\lambda > 1$. At a nonzero fixed point,

$$F'_\lambda(m_\lambda) = \lambda(1 - m_\lambda^2) = \frac{(1 - m_\lambda^2)\operatorname{artanh}(m_\lambda)}{m_\lambda} < 1,$$

because

$$\operatorname{artanh}(m) < \frac{m}{1 - m^2} \qquad (0 < m < 1).$$

The derivative is positive, so the nonzero fixed points are locally stable. Finally, expanding

$$\tanh(\lambda m) = \lambda m - \frac{\lambda^3 m^3}{3} + O(m^5)$$

and solving the nonzero fixed-point equation gives the stated asymptotic. □

The square-root scaling is a genuine model-specific critical law. It does not show that every workspace forms by a pitchfork bifurcation. It supplies an exact benchmark against which more complex formation dynamics can be tested.

71.4 Detecting a transition from finite data

Let $Q(\lambda)$ be a gate-specific population order parameter and $\widehat{Q}_n(\lambda)$ its held-out estimate. If

$$\sup_{\lambda\in\Lambda_n} |\widehat{Q}_n(\lambda) - Q(\lambda)| \le \varepsilon_n$$

on a finite evaluation grid Λ_n, then every threshold crossing with vertical margin larger than $2\varepsilon_n$ is recovered by the same uniform-selection argument as Theorem 120. Near a continuous transition, the margin vanishes and sample requirements necessarily grow. A broad, slowly varying crossover should not be reported as a sharp phase transition without a finite-size scaling analysis.

71.5 General formation-transition conjecture

For adaptive transfer systems with a fixed physical parameterization, there may exist model classes in which

$$\lambda < \lambda_c \Longrightarrow \mathrm{Form}_{\Theta_{\mathrm{form}}}(W_\lambda) = 0,$$

while

$$\lambda > \lambda_c \Longrightarrow \mathrm{Form}_{\Theta_{\mathrm{form}}}(W_\lambda) = 1$$

with a stable nonzero structural mode, positive causal efficacy, and relational irreducibility. The theorem above establishes this only for one structural order parameter in a solvable nonlinear model. The conjunction of all workspace gates remains model dependent.

72. Hierarchical Stabilization Conjecture

For appropriate recurrent systems,

$$W^{(r+1)} = \Gamma_{\varepsilon_r}(W^{(r)})$$

may converge to a stable workspace type:

$$d_{\mathrm{id}}\Big(W^{(r+1)}, W^{(r)}\Big) \to 0.$$

73. Endogenous Boundary Conjecture

The boundary theorems in Part XV establish recovery and stability once a score, candidate family, and formation procedure are fixed. They do not establish that an arbitrary adaptive system will create, represent, and maintain its own boundary.

Let

$$B_t^{\mathrm{sys}}$$

be an internal membership or routing state used by the system, and let

$$S_t^* = \arg\max_{S\in\mathfrak{S}_{\mathrm{pass}}} \mathcal{B}_t(S)$$

be the population boundary defined by the external certification protocol. The strong endogenous boundary conjecture asserts that there are adaptive systems and nontrivial environments for which:

$$\boxed{\begin{aligned}
&d_B(B_t^{\mathrm{sys}}, S_t^*) \to 0,\\
&\mathrm{Form}_{\Theta_{\mathrm{form}}}(W_{B_t^{\mathrm{sys}}}) = 1,\\
&\mathrm{CE}\left(B_t^{\mathrm{sys}} \to \mathrm{Routing}_{t+1}\right) > 0,\\
&\mathrm{CE}\left(\mathrm{BoundaryDamage}_t \to \mathrm{RepairAction}_{t+1}\right) > 0,
\end{aligned}}$$

while the scaffold contains no direct encoding of S_t^*.

A convincing demonstration requires four separable achievements:

1. *discovery*: interaction history changes which boundary is learned;
2. *instantiation*: an internal variable causally realizes that boundary;
3. *use*: changing the boundary changes workspace routing and downstream behavior; and
4. *maintenance*: detected boundary degradation causes restorative action.

Theorems 122 and 123 handle external consistency of the selected boundary. Theorems 128 and 129 now handle controlled repair selection and bounded closed-loop maintenance once a health representation, damage class, monitor, and repair controller are given. The remaining conjectural step is endogenous learning of the internal boundary, detector, and repair policy under open-ended environmental change without direct scaffold encoding or external substitution.

PART XXI

Empirical Certification and Reporting Protocol

74. Required Experiments

A workspace claim is not certified by a single benchmark score. The complete protocol is organized as a sequence of declarations, held-out measurements, implemented interventions, matched nulls, specification sweeps, identification checks, and gate decisions. Every reported result must identify the claim level it supports. Measurements that were not performed remain unmeasured rather than being inferred from neighboring gates.

74.1 Claim declaration before evaluation

Before the certification split is opened, record:

1. the full reference specification $\varsigma_0 = (\mathcal{D}, Y^+, \ell, \tau, \mathcal{I}, \mathcal{N}, \Theta)$;
2. the admissible decomposition family $\mathfrak{D}$ and specification family $\mathfrak{S}$;
3. the target trajectory Y^+, loss, horizon, and environment family;
4. the candidate structural workspace extractor;
5. the goal family $\mathcal{G}$, utility normalization, and feasible action set;
6. the intervention families $\mathcal{I}_W, \mathcal{I}_G, \mathcal{I}_A, \mathcal{I}_D, \mathcal{I}_R$;
7. the null panel and the local distributions each null must preserve;
8. the required certificate level and every numerical threshold;
9. the formation, calibration, certification, and stress-test splits;
10. the uncertainty procedure and resampling unit; and
11. the stopping and model-selection rules;
12. the experiment family and model class used for identifiability; and
13. the complete specification-selection history and multiplicity rule.

Changing any of these after observing certification outcomes creates a new analysis that must be reported as exploratory or evaluated on a fresh split.

74.2 Data and intervention partition

Use at least four logically distinct partitions:

$$\boxed{\mathcal{D} = \mathcal{D}_{\text{form}} \sqcup \mathcal{D}_{\text{cal}} \sqcup \mathcal{D}_{\text{cert}} \sqcup \mathcal{D}_{\text{stress}}.}$$

Their roles are:

Formation. Learn membership, transfer maps, workspace channels, boundaries, policies, and repair models.

Calibration. Select thresholds, intervention amplitudes, and uncertainty procedures without making the final pass decision.

Certification. Estimate the declared gates once under the frozen analysis plan.

Stress test. Evaluate distribution shift, repeated damage, replacement, and long-horizon persistence.

When independent partitions are impossible, the report must state the precise cross-fitting or nested-resampling procedure and preserve formation leakage controls.

74.3 Structural tests

Report the complete low-energy spectrum relevant to the selected threshold:

$$\operatorname{spec}(L_H), \qquad d_\varepsilon, \qquad g_\varepsilon, \qquad \delta_{\mathrm{coh}}.$$

At minimum include:

- the construction and normalization of D_H;
- edge weights and estimated transfer uncertainties;
- the selected eigenspace dimension and threshold rule;
- the spectral gap with uncertainty or perturbation bound;
- stability under bootstrap, seed, and representation replacement; and
- the exact or approximate kernel residual.

74.4 Transfer and operation tests

Report

$$\eta_{\mathrm{int}}, \qquad \delta_e^{\mathrm{tr}}, \qquad \Omega_\gamma, \qquad \Lambda_\gamma,$$

together with:

- the paths and cycle basis used;
- path-length dependence of leakage;
- the gauge-alignment procedure;
- held-out transfer-prediction error;
- operation-intertwining defects; and
- the approximate closure bound from Theorem T_{10}.

74.5 Predictive alignment and coalition tests

Report

$$\delta_{\mathrm{align}}^{\mathcal{I}}, \qquad \{\Delta_A^\ell\}_{A\subseteq V}, \qquad k_\varepsilon, \qquad \delta_A^Z.$$

When the number of coalitions is too large for exhaustive evaluation, state the sampling design, include every singleton and every leave-one-out coalition, and report the most favorable coalition found for the reducibility null. The claim must not be based only on average coalition performance.

74.6 Causal and reentry tests

Report

$$\eta^{\nu}_{\mathrm{causal}}, \qquad \Delta_{\mathrm{specific}}(r), \qquad \delta_{\mathrm{med}}, \qquad \eta_{\mathrm{rec}}(\tau), \qquad \eta_{\mathrm{loop}}(\tau).$$

Every causal estimate must be paired with:

1. a concrete mechanism-level implementation;
2. a manipulation check;
3. matched background variables;
4. declared direct paths that were blocked or preserved;
5. intervention magnitude and cost;
6. an off-workspace matched intervention; and
7. the downstream outcome window.

74.7 Persistence, reconstruction, and replacement tests

Report

$$d_{\mathrm{id}}(W_t, W_{t+\tau}), \qquad \mathrm{Rec}_k, \qquad e_t, \qquad \alpha^{(k)}, \qquad \mu_t^{\mathrm{keep}}.$$

The report must separate:

- subspace tracking from state tracking;
- coherence from non-collapse;
- reconstruction from mere prediction;
- repair from successor-workspace creation; and
- structural replacement from causal and reconstructive replacement.

74.8 Formation-provenance tests

Report

$$\mathbf{E}_{\mathrm{form}}, \qquad \delta^{\mathrm{scaf}}, \qquad \delta^{\mathrm{leak}}, \qquad \Delta_q^{\mathrm{hist}}.$$

Include the external scaffold, information available to the learner, history shuffles, cross-scaffold retraining, and every route by which certification labels or target boundaries could have entered formation.

74.9 Learned-structure and boundary tests

Report

$$\|\widehat{H} - H_*\|, \qquad \Delta_J, \qquad \Delta_t^B, \qquad \mathrm{COut}_\tau(\widehat{S}), \qquad \mathrm{CIn}_\tau(\widehat{S}).$$

Boundary reports must include the admissible candidate family, full-system and empty-set controls, singleton and centralized controls, score components, margin to the runner-up, and temporal stability under held-out environment drift.

74.10 Goal and counterfactual-policy tests

Report the agency coordinates

$$\eta_G(\tau), \qquad \delta_G^{\mathrm{cue}}(\tau), \qquad \eta_{G\to A}, \qquad \delta_Q, \qquad \eta_{V\to A}, \qquad \eta_{A\to Y}, \qquad k_{\mathrm{ctl},\varepsilon}.$$

The required experiments include:

1. cue-controlled goal interventions;
2. goal-channel ablation and permutation;
3. counterfactual-value calibration against action interventions;
4. value-label permutation among feasible actions;
5. critic or value-channel blocking;
6. action replay under held-out perturbations;
7. policy regret or value-gap estimation; and
8. coalition reconstruction of the complete control state.

74.11 Damage and repair tests

For every damage family, report

$$\eta_{D\to M}, \qquad \eta_{M\to R}, \qquad \eta_{R\to v}, \qquad \Delta_{\mathrm{repair}}^{\mathrm{spec}}, \qquad \delta_{\mathrm{rep-id}}, \qquad \sup_t e_t.$$

The repeated-disturbance experiment must contain:

1. a no-damage baseline;
2. damage with repair disabled;
3. damage with the intact repair loop;
4. monitor ablation;
5. cost-matched random repair;
6. off-workspace repair;
7. repeated and compound damage; and
8. a held-out disturbance distribution.

75. Required Null Families

The core null panel is:

$$
\begin{aligned}
N_1 &= \text{shuffled transfer maps with validated local matching},\\
N_2 &= \text{shuffled cross-node temporal correspondence},\\
N_3 &= \text{matched-capacity local memory},\\
N_4 &= \text{matched-size irrelevant artifact},\\
N_5 &= \text{participant ablation},\\
N_6 &= \text{artifact ablation},\\
N_7 &= \text{workspace implementation intervention},\\
N_8 &= \text{matched off-workspace intervention},\\
N_9 &= \text{representation replacement},\\
N_{10} &= \text{context reset and reconstruction},\\
N_{11} &= \text{matched formation-history shuffle},\\
N_{12} &= \text{cross-scaffold re-encoding and retraining},\\
N_{13} &= \text{boundary-label and certification leakage audit},\\
N_{14} &= \text{empty, full, singleton, and centralized boundary nulls},\\
N_{15} &= \text{held-out transfer and membership re-estimation},\\
N_{16} &= \text{boundary damage, use, and repair intervention},\\
N_{17} &= \text{goal-code permutation with cues and actions matched},\\
N_{18} &= \text{persistent external-cue rewrite with internal retention cut},\\
N_{19} &= \text{counterfactual-value label permutation},\\
N_{20} &= \text{critic or value-channel ablation},\\
N_{21} &= \text{open-loop action replay under held-out perturbation},\\
N_{22} &= \text{monitor-blind cost-matched random repair},\\
N_{23} &= \text{off-workspace repair with matched intervention norm},\\
N_{24} &= \text{identity-breaking reconstruction with matched task score},\\
N_{25} &= \text{merge control for components sharing an uninterruptible controller},\\
N_{26} &= \text{secret-sharing split of a known centralized controller},\\
N_{27} &= \text{hidden shared-memory and common-cue exposure test},\\
N_{28} &= \text{frozen certificate sweep across the admissible decomposition family},\\
N_{29} &= \text{post hoc target-selection audit},\\
N_{30} &= \text{horizon cherry-picking audit},\\
N_{31} &= \text{loss and utility substitution audit},\\
N_{32} &= \text{restricted intervention-family audit},\\
N_{33} &= \text{threshold and gate-set tuning audit},\\
N_{34} &= \text{frozen certificate sweep across the admissible specification family}.
\end{aligned}
$$

75.1 Null validity record

Each observational null must state:

- the local distributions it preserves;

- the relational dependency it destroys;
- the distance between intact and null local marginals;
- the capacity, parameter count, and information budget; and
- whether null fitting used the same data budget as the intact model.

Each causal null must state:

- the manipulated mechanism;
- the matched background and intervention cost;
- the intended causal path destroyed or preserved;
- the manipulation check; and
- unintended direct effects detected during validation.

A failed manipulation check invalidates the null. It does not count as evidence for the workspace claim.

75.2 Claim-specific null subsets

A successful claim requires

$$\mathrm{Pass}(N_1, \dots, N_{28})$$

only for the subset relevant to its stated level. The subset must be frozen before certification. Failed applicable nulls, invalid nulls, and unmeasured nulls are retained separately in the final report. They must not be merged into a generic "not significant" category.

76. Certification Vector and Complete Reporting Templates

76.1 Extended certification vector

Use a vector rather than one aggregate score:

$$\begin{aligned}\mathcal{C}_{\mathrm{full}}(W) = (&d_\varepsilon, g_\varepsilon, \delta_{\mathrm{coh}}, \delta^{\mathcal{I}}_{\mathrm{align}}, \Delta^{\ell}_{\mathrm{joint}}, k_\varepsilon,\\ &\boldsymbol{\Delta}_{\mathrm{tr}}(\mathcal{Q}), \eta^{\nu}_{\mathrm{causal}}, \Delta_{\mathrm{specific}}(r), \eta_{\mathrm{loop}}(\tau),\\ &d_{\mathrm{id}}, \mathrm{Rec}_k, \underline{\mathbf{m}}_{\mathfrak{D}}, \boldsymbol{\Delta}_{\mathrm{dec}}, \mathbf{E}_{\mathrm{form}}, \delta^{\mathrm{scaf}}, \delta^{\mathrm{leak}},\\ &\Delta_J, \Delta^{B}_t, \mathrm{COut}_\tau(\widehat{S}), \mathrm{CIn}_\tau(\widehat{S}),\\ &\eta_G(\tau), \delta^{\mathrm{cue}}_G(\tau), \eta_{G\to A}, \delta_Q, \eta_{V\to A}, \eta_{A\to Y},\\ &\eta_{D\to M}, \eta_{M\to R}, \eta_{R\to v}, \Delta^{\mathrm{spec}}_{\mathrm{repair}}, \delta_{\mathrm{rep-id}}, \sup_t e_t).\end{aligned}$$

A claim passes only when every required coordinate lies in its predeclared admissible region. No weighted average may compensate for a failed required gate.

76.2 Claim card template

Field	Required entry
Claim identifier	Unique versioned identifier tied to code and result artifacts.
Claim level	Archive, shared representation, operational workspace, relational workspace, persistent workspace, self-maintaining workspace, or workspace agent.
Reference decomposition	Nodes, artifacts, channels, local-access maps, and coalition universe.
Admissible decomposition family	Plausible merges, splits, shared-resource assignments, exclusions, and audit rationale.
Target and horizon	Y^+, loss, utility, prediction horizon, control horizon, and maintenance horizon.
Environment scope	Training, certification, and stress-test environment families.
Workspace extractor	Frozen construction of D_H, L_H, spectral threshold, and state readout.
Goal and policy	Goal codes, utility normalization, action set, and policy class.
Boundary	Candidate family, score, admissibility constraints, and selected boundary.
Interventions	Feasible workspace, goal, action, damage, and repair implementations.
Null panel	Applicable N_j, preservation targets, and fit budgets.
Thresholds	Every gate direction, numerical cutoff, and uncertainty rule.
Data split	Formation, calibration, certification, stress, and cross-fitting logic.
Stopping rule	Number of seeds, runs, interventions, and any sequential stopping criterion.
Status	Confirmatory, exploratory, replication, or audit.

76.3 Gate-estimate table template

For every required coordinate, include one row with the following fields:

Field	Required entry
Gate	Formal symbol and verbal interpretation.
Estimator	Exact statistic, normalization, and software implementation.
Evaluation unit	Episode, participant, graph, seed, environment, or other independent unit.
Point estimate	Estimate on the frozen certification split.
Uncertainty	Interval, bound, or posterior region with declared coverage or credibility.
Threshold	Predeclared pass region.
Manipulation status	Passed, failed, or not applicable.
Decision	Pass, fail, invalid, underpowered, or unmeasured.
Sensitivity	Result under reasonable alternative normalizations and resampling choices.
Artifact reference	File, table, raw-result key, and code commit.

76.4 Intervention ledger template

Field	Required entry
Intervention ID	Unique identifier and randomization seed.
Target mechanism	Workspace coordinate, goal code, value channel, action, boundary, monitor, or repair channel.
Implementation	Exact code path, hook, artifact edit, or mechanism change.
Dose	Norm, duration, timing, and support of the intervention.
Background match	Variables fixed, blocked, randomized, or balanced.
Excluded paths	Direct paths intentionally removed from the causal contrast.
Manipulation check	Evidence that the intended mechanism changed by the specified amount.
Collateral effects	Changes outside the target mechanism.
Outcome	Downstream variable and evaluation window.
Effect estimate	Point estimate, uncertainty, and sign convention.
Cost	Compute, latency, information, action, or structural cost.
Validity status	Valid, invalid, ambiguous, or unmeasured.

76.5 Null ledger template

Field	Required entry
Null ID	One of $N_1, \dots, N_{34}$ or a declared extension.
Claim attacked	The exact explanation the null is designed to preserve or destroy.
Matched quantities	Local marginals, capacity, training budget, action cost, intervention norm, or other controlled quantities.
Destroyed relation	Edge alignment, temporal order, value ranking, goal identity, monitor mediation, or another relational feature.
Fit procedure	Data, optimization, seeds, and hyperparameter budget.
Validity metric	Distance from the intended matched quantities.
Primary contrast	Intact minus null effect with uncertainty.
Disposition	Rejects null, fails to reject, invalid null, or underpowered.
Failure notes	Any mismatch or collateral mechanism change.

76.6 Agency and maintenance report template

The agency section must provide the following matrix:

Component	Primary metric	Required contrast
Goal semantics	Held-out Q_t^g ordering	Distinct goals reverse or alter at least one declared action ordering.
Goal retention	$\eta_{G}(\tau)$	Goal intervention with future cues matched or removed.
Goal use	$\eta_{G\to A}$	Goal intervention versus goal-label replay.
Counterfactual accuracy	δ_Q	Internal value versus external action interventions.
Value use	$\eta_{V\to A}$	Value-label permutation and value-channel ablation.
Outcome control	$\eta_{A\to Y}$	Feasible action intervention.
Damage detection	$\eta_{D\to M}$	Damage versus matched no-damage.
Repair mediation	$\eta_{M\to R}$	Monitor intervention and monitor ablation.
Restoration	$\eta_{R\to v}$	Repair versus matched no-repair.
Repair specificity	$\Delta^{\mathrm{spec}}_{\mathrm{repair}}$	Intended repair versus cost-matched generic action.
Identity preservation	$\delta_{\mathrm{rep-id}}$	Pre-damage workspace versus post-repair workspace.
Long-run maintenance	$\sup_t e_t$	Repeated held-out disturbances.
Relational location	$k_{\mathrm{ctl},\varepsilon}$	Full system versus all single nodes and matched local nulls.

76.7 Uncertainty and multiplicity

The independent resampling unit must match the data-generating process. Token, time-step, or edge resampling is invalid when dependence is organized at the episode, participant, graph, or environment level. Recommended procedures include:

- environment- or episode-level bootstrap;
- repeated independent training seeds;
- cluster-robust intervals for repeated observations;
- permutation tests that preserve the declared local structure;
- concentration bounds for uniform finite candidate families; and
- stress-test intervals over held-out environment families.

Pass decisions should use one-sided uncertainty bounds. For a lower-bounded gate $\theta \geq \tau$, pass only when the lower confidence bound exceeds τ. For an upper-bounded defect $\delta \leq \varepsilon$, pass only when the upper confidence bound is below ε.

When a conclusion ranges over $\mathfrak{S}$, the uncertainty rule must be uniform over every gate and specification included in the claim. Report the pointwise interval, the family-wide lower envelope, the covering or multiplicity correction, and the identification status. A narrow pointwise interval does not establish a robust result when model or specification uncertainty remains wide.

76.8 T_{24}: Gatewise certification validity

Theorem 156 (Intersection-gate false-certification control)**.** *Let a certificate require all gates $G_1, \ldots, G_m$ to pass. For each gate j, let H_{0j} denote the null that the required property is absent, and assume the predeclared test satisfies*

$$\sup_{P \in H_{0j}} \mathbb{P}_P(G_j \text{ passes}) \leq \alpha_j.$$

The global null is

$$H_0 = \bigcup_{j=1}^{m} H_{0j},$$

meaning that at least one required property is absent. If the certificate is declared only when every gate passes, then

$$\boxed{\sup_{P \in H_0} \mathbb{P}_P(\textit{certificate passes}) \le \max_{1 \le j \le m} \alpha_j.}$$

No independence assumption among gate tests is required.

Proof. For any $P \in H_0$, choose an index j_P such that $P \in H_{0j_P}$. The event that every gate passes is contained in the event that gate j_P passes. Hence

$$\mathbb{P}_P(\text{certificate passes}) \le \mathbb{P}_P(G_{j_P} \text{ passes}) \le \alpha_{j_P} \le \max_j \alpha_j.$$

Taking the supremum over $P \in H_0$ proves the claim. □

The theorem is an intersection-union guarantee. It applies only when the gate family, thresholds, and tests are fixed before certification and each gate test is uniformly valid for its null. Adaptive gate deletion, selective reporting, or repeated reuse of the certification split invalidates the bound.

76.9 Failure taxonomy

Every gate and null receives exactly one primary status:

Pass. The frozen estimate and uncertainty rule satisfy the threshold.

Fail. The gate was validly measured and did not satisfy the threshold.

Invalid intervention. The manipulation check or background match failed.

Invalid null. The null did not preserve its required local quantities or exceeded its fit budget.

Underpowered. The uncertainty interval crosses the threshold by more than the declared indifference region.

Unmeasured. No valid estimate was obtained.

Leakage. Formation or model selection used certification information.

Exploratory. The metric, threshold, or analysis was selected after inspecting certification outcomes.

Only *Pass* counts toward the formal certificate. "Underpowered" and "unmeasured" are not failures of the underlying theory, but they are failures to establish the corresponding empirical claim.

76.10 Machine-readable result record

The machine-readable record should contain at least the following top-level objects:

$$\boxed{\begin{aligned}\mathsf{ResultRecord} = \{&\mathsf{claim}, \mathsf{specification}, \mathsf{specification_family}, \mathsf{decomposition}, \mathsf{data_splits}, \mathsf{formation},\\ &\mathsf{workspace_extractor}, \mathsf{goals}, \mathsf{actions}, \mathsf{interventions},\\ &\mathsf{nulls}, \mathsf{gates}, \mathsf{identified_sets}, \mathsf{specification_envelopes}, \mathsf{uncertainty}, \mathsf{failures}, \mathsf{artifacts}\}.\end{aligned}}$$

Each gate object records its symbol, estimator, direction, threshold, point estimate, interval, split, seed set, status, and artifact reference. Each intervention and null object records its manipulation check and validity status. The schema must preserve failed and invalid entries rather than only the passing subset.

76.11 Minimum reproducibility package

A complete certification package contains:

1. the frozen claim card;
2. source code and environment lock file;
3. exact data or deterministic data-generation procedure;
4. formation, calibration, certification, and stress-test manifests;
5. learned transfer, workspace, policy, monitor, and repair parameters;
6. raw per-run and per-intervention results;
7. the intervention and null ledgers;
8. the complete gate table, including failures;
9. figures and tables regenerated from raw results;
10. random seeds and hardware or numerical-precision information;
11. checksums for every included artifact; and
12. a single verification command that reproduces package integrity and the reported summary tables.

76.12 Decision rule

Let $\mathcal{J}(L)$ be the required gate set for claim level L. Then

$$\boxed{\mathrm{Certify}_L(W;\varsigma) = \bigwedge_{j\in\mathcal{J}(L,\varsigma)} \mathrm{Pass}_j(W;\varsigma).}$$

For a workspace-agent claim, the required set includes every structural, operational, relational, persistent, goal, counterfactual-control, and self-maintenance gate declared in Parts IX and XVI. A specification-robust claim additionally requires the uniform rule of Part XVIII across $\mathfrak{S}$. Failure of an earlier level prevents certification of a later level, even when later-level behavior appears strong on a task benchmark.

PART XXII

Integrated Worked Examples and Case Studies

77. Case-Study Method and Common Gate Card

The preceding parts define a large family of structural, operational, causal, relational, persistent, formative, and agentive predicates. A worked example is useful only when it makes clear which of those predicates are established, which are assumed, and which remain unmeasured. This part therefore uses a single case-study protocol for three different kinds of realization:

1. an exact finite-dimensional model of a human–human theorem laboratory;
2. an idealized human–AI–artifact–executor research loop, followed by a deliberately conservative audit of the collaboration that motivated the theory; and
3. a closed-form multi-agent control system that reaches the formal agency level inside a small declared domain.

The first and third cases are analytic models. Their conclusions hold inside those models. The second contains both an analytic idealization and an observational audit. The audit is not upgraded to a certificate merely because the idealization is mathematically clean.

Definition 157 (Case-study realization)**.** A case-study realization is a tuple

$$\mathfrak{K} = (\mathfrak{R}, \mathfrak{B}, \mathfrak{C}, \mathfrak{P}, \mathcal{E}, \mathfrak{D}, \mathfrak{S}, \mathcal{I}, \mathcal{Q}, \mathcal{G}),$$

where:

- $\mathfrak{R}$ is the declared relational realization;
- $\mathfrak{B}$ is the boundary problem and component decomposition;
- $\mathfrak{C}$ is the structural core–transfer model;
- $\mathfrak{P}$ is the downstream predictive or control problem;
- $\mathcal{E}$ is the evidence or generated-data family;
- $\mathfrak{D}$ is the admissible decomposition family;
- $\mathfrak{S}$ is the admissible family of complete scientific specifications, including targets, losses, horizons, interventions, nulls, and thresholds;
- $\mathcal{I}$ is the feasible intervention family;
- $\mathcal{Q}$ is the matched null family; and
- $\mathcal{G}$ is the set of gates being claimed.

A conclusion is always relative to this tuple. Changing the downstream target, temporal horizon, intervention support, component decomposition, null family, or threshold rule creates a different specification of the case. A stronger case study reports whether the conclusion survives the predeclared family $\mathfrak{S}$ and whether the relevant gates are identified by the available experiment family.

77.1 Three evidential statuses

Every claim in this part receives one of three primary statuses:

Analytic. The gate follows exactly from a declared mathematical model.

Empirically certified. The gate is estimated from held-out data using the protocol of Part XXI and passes its frozen uncertainty rule.

Observational only. The case contains suggestive records or natural perturbations, but no valid held-out intervention-and-null certificate.

An analytic result is not automatically an empirical result. An observational pattern is not automatically a causal result. None of the three statuses establishes phenomenal consciousness.

77.2 Common gate card

For each case we report the vector

$$\boxed{\mathsf{Card}(\mathfrak{K}) = (\mathsf{S}, \mathsf{O}, \mathsf{R}, \mathsf{P}, \mathsf{F}, \mathsf{A}, \mathsf{D}, \mathsf{X}, \mathsf{C}),}$$

where:

S = structural core and spectral coherence,
O = operational alignment and causal efficacy,
R = relational irreducibility and matched-null gap,
P = persistence, reconstruction, and identity,
F = formation provenance and boundary status,
A = goal-conditioned agency and self-maintenance,
D = decomposition robustness and interface audit,
X = specification robustness, identification, and uniform uncertainty,
C = consciousness or personhood claim.

The decomposition coordinate distinguishes a result established only for the declared interface from one stable across an admissible family. The specification coordinate records whether the same conclusion survives scientifically defensible alternatives, whether its sign is identified by the available experiments, and whether its lower margin remains positive after uniform finite-sample uncertainty. The final coordinate is included to make the non-implication explicit. In all examples below,

$$\mathsf{C} = \text{not established.}$$

77.3 Claim-level discipline

Let $L(\mathfrak{K}) \in \{0, 1, \dots, 6\}$ denote the highest level from Part XXIV whose complete required gate set is established. The level is conjunctive and pointwise in the declared scientific specification unless the D and X coordinates also pass:

$$L(\mathfrak{K}) \geq \ell \quad \Longleftrightarrow \quad \bigwedge_{j \in \mathcal{J}(\ell)} \mathrm{Pass}_j(\mathfrak{K}).$$

A case may have strong higher-level-looking behavior while remaining at a lower certified level because an earlier gate is absent. For example, successful repair behavior does not establish self-maintenance when no matched no-repair counterfactual is available, and coherent shared documents do not establish a relational workspace when one participant contains the complete operative state.

77.4 Structural calculations used repeatedly

The positive examples use a recurring distributed code. Let

$$c = (c_1, c_2, c_3)^{\mathsf{T}} \in \mathbb{R}^3$$

and define three local encodings

$$E_1 c = (c_1, c_2), \qquad E_2 c = (c_2, c_3), \qquad E_3 c = (c_1, c_3).$$

Then

$$\sum_{i=1}^{3} E_i^* E_i = 2I_3.$$

No individual encoding is injective, while every two-node coalition is injective. The pairwise frame operators are

$$E_{\{1,2\}}^* E_{\{1,2\}} = \operatorname{diag}(1, 2, 1),$$
$$E_{\{1,3\}}^* E_{\{1,3\}} = \operatorname{diag}(2, 1, 1),$$
$$E_{\{2,3\}}^* E_{\{2,3\}} = \operatorname{diag}(1, 1, 2).$$

Hence

$$\alpha_{\{i,j\}} = 1 \qquad (i \neq j),$$

and one-node erasure recovery is exact. Every nonzero core vector is visible at at least two nodes, so

$$d_{\mathrm{er}} = 2.$$

This code is distributed but redundant: no node is sufficient, yet any one node may be lost without destroying the core.

77.5 Binary synergy test

Let C_1, C_2, C_3 be independent uniform bits and let

$$Y = C_1 \oplus C_2 \oplus C_3.$$

Each node sees two of the three bits. Therefore

$$H(Y \mid E_i C) = 1 \text{ bit} \qquad \text{for every } i,$$

while

$$H(Y \mid E_1 C, E_2 C, E_3 C) = 0.$$

Under logarithmic loss,

$$\mathcal{R}_{\log}(\{i\}) = 1, \qquad \mathcal{R}_{\log}(V) = 0, \qquad \Delta_{\mathrm{joint}}^{\log} = 1 \text{ bit}, \qquad k_0 = 2.$$

Moreover, Y is independent of every individual local state. The null law that keeps all three core bits independent and makes Y an independent uniform bit preserves every pair $(Y, E_i C)$ while destroying the joint parity relation. It is therefore a predictive-marginal-matched relational null with

$$\Delta_{\mathrm{tr}} = 1 \text{ bit}.$$

This test will be reused with different semantic interpretations.

78. Case Study A: A Distributed Human–Human Theorem Laboratory

78.1 Declared realization

Consider two researchers and one persistent notebook:

$$V_{\mathrm{HH}} = \{P, C, N\},$$

where:

- P is a proposer who tracks the theorem statement and open proof obligations;
- C is a critic who tracks proof obligations and falsification status;
- N is a notebook that tracks the theorem statement and falsification status.

The abstract core is

$$c = (q, p, f)^{\mathsf{T}} \in \mathbb{R}^3,$$

with

$$q = \text{statement coordinate}, \qquad p = \text{proof-obligation coordinate},$$
$$f = \text{falsification-status coordinate}.$$

The accessible encodings are

$$E_P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad E_C = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_N = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The semantics are stylized. The point is not that real researchers contain only two numbers. The point is to exhibit, without metaphor, a system in which no participant or artifact contains the entire operative state while the coupled system does.

78.2 Exact overlap descent

Order the local coordinates as

$$x = (q_P, p_P, p_C, f_C, q_N, f_N)^{\mathsf{T}}.$$

Compatibility requires

$$p_P = p_C, \qquad f_C = f_N, \qquad q_N = q_P.$$

The discrepancy operator is

$$D_{\mathrm{HH}} = \begin{pmatrix} 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 \\ -1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Therefore

$$\ker D_{\mathrm{HH}} = \left\{ (q, p, p, f, q, f)^{\mathsf{T}} : q, p, f \in \mathbb{R} \right\} \cong \mathbb{R}^3.$$

The rows are orthogonal with squared norm two, so

$$D_{\mathrm{HH}} D_{\mathrm{HH}}^{*} = 2I_3.$$

Consequently the transfer Laplacian has spectrum

$$\boxed{\operatorname{spec}(L_{\mathrm{HH}}) = \{0, 0, 0, 2, 2, 2\},}$$

and the exact workspace gap is

$$\gamma_{\mathrm{HH}} = 2.$$

The low-energy subspace is therefore stable under any Laplacian perturbation whose operator norm is strictly below one, with the quantitative projector bound from Theorem 43.

78.3 Reconstruction and one-node failure

The full frame operator is

$$F_{PCN} = E_P^* E_P + E_C^* E_C + E_N^* E_N = 2I_3.$$

For every two-node coalition S,

$$\lambda_{\min}(F_S) = 1.$$

Thus every two-node coalition reconstructs the complete core and, for noisy observations $y_S = A_S c + \eta_S$,

$$\|\widehat{c} - c\| \le \|\eta_S\|.$$

No individual node reconstructs c, because

$$\operatorname{rank}(E_P) = \operatorname{rank}(E_C) = \operatorname{rank}(E_N) = 2 < 3.$$

The realization is therefore simultaneously:

$$\text{individually incomplete} \quad \text{and} \quad \text{robust to any one-node loss.}$$

78.4 Relational prediction problem

Let q, p, f now be independent uniform bits and define the downstream acceptance decision

$$Y = q \oplus p \oplus f.$$

The interpretation is abstract: acceptance depends on the statement, the state of the proof obligations, and the current falsification record. No individual node has sufficient information:

$$H(Y \mid X_P) = H(Y \mid X_C) = H(Y \mid X_N) = 1.$$

Every two-node coalition has all three bits and therefore zero conditional entropy. Hence

$$\boxed{\Delta_{\mathrm{joint}}^{\log} = 1 \text{ bit}, \qquad k_0 = 2.}$$

Let Q_{HH} preserve the independent uniform local bit pairs but replace Y by an independent uniform bit. Because parity is independent of every two-bit local view, Q_{HH} preserves each (Y, X_i)-marginal. Nevertheless,

$$\mathcal{R}_{\log, Q_{\mathrm{HH}}}(V) - \mathcal{R}_{\log, P_{\mathrm{HH}}}(V) = 1 \text{ bit}.$$

This is an exact relational witness: the predictive capacity lies in the joint organization rather than in any local predictive marginal.

78.5 Feasible intervention and specificity

A feasible statement intervention must alter both local realizations of the statement coordinate:

$$\iota_{q \leftarrow a} : (q_P, q_N) \leftarrow (a, a),$$

while holding (p, f) fixed. For configurations with fixed p and f, changing a flips the downstream decision with total-variation effect one:

$$\mathrm{TV}\left(P(Y \mid \iota_{q\leftarrow 0}), P(Y \mid \iota_{q\leftarrow 1})\right) = 1.$$

A matched intervention on a residual notebook-format coordinate leaves Y unchanged. Thus the core intervention has positive causal specificity in this model.

78.6 Persistence, handoff, and replacement

Suppose the proposer becomes unavailable. The remaining coalition $\{C, N\}$ reconstructs (q, p, f) exactly. A replacement proposer P' can therefore be initialized by

$$E_{P'}c = (q, p).$$

If P' uses a different local coordinate system

$$J_{P'} \in GL(2),$$

then the incoming and outgoing overlap maps are conjugated accordingly. The global section and all downstream kernels remain unchanged. This is exact replacement invariance, not literal carrier identity.

If the notebook alone survives, the core cannot be reconstructed because the proof-obligation coordinate is missing. The model therefore predicts a sharp handoff boundary:

$$\text{one surviving node is insufficient,} \qquad \text{any two surviving nodes are sufficient.}$$

78.7 Case card

Inside the declared finite model,

$\mathsf{S} = \checkmark_{\text{analytic}}$,	$\mathsf{O} = \checkmark_{\text{analytic}}$,
$\mathsf{R} = \checkmark_{\text{analytic}}$,	$\mathsf{P} = \checkmark_{\text{analytic}}$,
F = scaffolded,	A = not tested,
D = fixed decomposition,	C = not established.

The highest supported level is Level 4 when the handoff dynamics and identity correspondence are included. The example does not contain an operational goal state, counterfactual policy, damage monitor, or repair loop, so it does not support Levels 5 or 6.

79. Case Study B: A Human–AI–Artifact–Executor Research Loop

79.1 Idealized four-node loop

Consider four heterogeneous components:

$$V_{\mathrm{HAI}} = \{H, M, X, A\},$$

where:

- H is a human researcher who maintains goals and semantic direction;

- M is a language-model process that constructs local formalizations;
- X is an execution and verification process that compiles, tests, or falsifies candidate artifacts; and
- A is a persistent artifact store containing exact source, results, and versioned handoff state.

Let the abstract research state be

$$c = (g, d, p, v)^{\mathsf{T}} \in \mathbb{R}^4,$$

where:

g = goal and acceptance criterion, d = current derivation or theory state,
p = open proof or implementation obligations, v = verification and regression status.

Define local encodings

$$E_H c = (g, d), \qquad E_M c = (g, p), \qquad E_X c = (p, v), \qquad E_A c = (d, v).$$

Each coordinate is represented at exactly two nodes, and the incidence pattern forms a four-cycle.

79.2 Exact structural calculation

Order the local coordinates as

$$x = (g_H, d_H, g_M, p_M, p_X, v_X, d_A, v_A)^{\mathsf{T}}.$$

The overlap discrepancies are

$$g_H - g_M, \qquad p_M - p_X, \qquad v_X - v_A, \qquad d_A - d_H.$$

Thus

$$D_{\mathrm{HAI}} = \begin{pmatrix} 1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

The rows are orthogonal with squared norm two. Therefore

$$\ker D_{\mathrm{HAI}} \cong \mathbb{R}^4,$$

$$\boxed{\mathrm{spec}(L_{\mathrm{HAI}}) = \{0, 0, 0, 0, 2, 2, 2, 2\}, \qquad \gamma_{\mathrm{HAI}} = 2.}$$

The full accessible frame operator is

$$F_{HMXA} = 2I_4.$$

No single node is injective. Every three-node coalition reconstructs the full state, and the two opposite pairs

$$\{H, X\}, \qquad \{M, A\}$$

are already sufficient. Adjacent pairs observe only three of the four coordinates. Hence

$$k_{\mathrm{rec}} = 2, \qquad d_{\mathrm{er}} = 2,$$

while the identity of the sufficient two-node coalition depends on the graph geometry.

79.3 Relational parity benchmark

Let g, d, p, v be independent uniform bits and define

$$Y = g \oplus d \oplus p \oplus v.$$

Every single node sees only two bits, so

$$H(Y \mid X_i) = 1 \qquad \text{for all } i.$$

The opposite pairs $\{H, X\}$ and $\{M, A\}$ determine all four bits and therefore determine Y. Every adjacent pair remains one bit short. It follows that

$$\boxed{\Delta^{\log}_{\text{joint}} = 1 \text{ bit}, \qquad k_0 = 2.}$$

As in the preceding case, parity is independent of every individual local view. Replacing Y by an independent uniform bit produces a matched local predictive null and a full-system transfer gap of one bit.

This benchmark is intentionally narrow. It demonstrates that the four-cycle can realize exact relational information. It does not show that every real human–AI collaboration has this encoding, this target, or this degree of irreducibility.

79.4 A process-level interpretation

The idealized loop can be read as the recurrent sequence

$$\boxed{H \xrightarrow{g} M \xrightarrow{p} X \xrightarrow{v} A \xrightarrow{d} H.}$$

A more explicit cycle is:

1. the human supplies a goal and acceptance criterion;
2. the model converts that goal into a formal derivation and explicit obligations;
3. the executor compiles, tests, or searches for counterexamples;
4. the artifact store preserves the exact source and verification record;
5. the human evaluates the returned derivation and evidence, revising the next goal.

The operative state is reentrant because products of one cycle become inputs to the next. The artifact is not merely a record when its contents alter the next formalization, test, or acceptance decision.

79.5 Conservative audit of the motivating collaboration

The collaboration that motivated this theory has several features consistent with the four-node description: specialized shared vocabulary, versioned LaTeX and code artifacts, repeated compile-and-repair cycles, explicit theorem and null-control demands, and reconstruction across conversations from handoff packages. Those observations justify treating the collaboration as a serious candidate case. They do not establish the complete certificate.

Gate family	Present evidential status	Conservative assessment
Declared heterogeneous components	Observational	Human, model, artifacts, and execution environment are distinguishable components with different roles and information access.
Persistent shared vocabulary	Observational	Recurrent terms and distinctions survive across research phases and constrain later work.
Exact structural core $\Gamma(H)$	Unmeasured	No frozen local coordinate system or learned transfer family has yet been estimated from held-out data.
Transfer-Laplacian gap	Unmeasured	The idealized value 2 belongs to the toy four-cycle, not to the actual collaboration.
Operational alignment	Unmeasured	No preregistered future variable has been shown conditionally independent of the full history given a measured workspace state.
Joint advantage	Suggestive only	The coupled process has produced outputs that neither party appears to have supplied verbatim, but local-only Bayes risks have not been measured under controlled information ceilings.
Matched relational null gap	Unmeasured	Shuffled artifacts, matched irrelevant memory, and local-predictive-marginal-preserving nulls have not been run as a frozen panel.
Causal workspace efficacy	Suggestive only	Corrections, compiler failures, and artifact substitutions change later behavior, but they are not yet matched workspace interventions with manipulation checks.
Artifact-mediated persistence	Observational	Handoff files and source bundles support continuation, but blind reconstruction and ablation curves have not been estimated.
Formation provenance	Mixed	The communication architecture is scaffolded, while terminology, theorem organization, and many control practices were adaptively developed. Causal formation-history contribution is unmeasured.
Workspace agency	Not established	No complete goal-retention, counterfactual-policy, distributed-control, damage, and repair certificate has been run.
Consciousness or personhood	Not established	The theory supplies no such inference from the observed collaboration.

The correct present statement is therefore:

candidate persistent collaborative workspace rather than certified relational workspace agent.

79.6 A controlled certification design

A serious test can use one manuscript section or software work unit as the unit of analysis. Before the certification split is opened, freeze:

- the next-stage task family;
- the downstream target, such as accepted theorem patch, successful clean compile, and absence of a declared regression;
- the local information available to each component;
- the workspace extractor and transfer maps;

- all intervention and null families; and
- the required gate thresholds.

The following conditions can then be compared:

$$\begin{aligned}
\mathsf{H} &= \text{human-side externalized state only,}\\
\mathsf{M} &= \text{fresh model instance with model-side state only,}\\
\mathsf{A} &= \text{artifact package without live human or model state,}\\
\mathsf{X} &= \text{executor and test outputs without the research context,}\\
\mathsf{F}\mathit{ull} &= \text{the intact recurrent loop.}
\end{aligned}$$

The design must avoid pretending that a person's unobserved internal state has been fully measured. The local conditions certify only the declared accessible histories.

A minimal null panel includes:

1. theorem-to-artifact reassignment with artifact size and style matched;
2. temporal shuffling of verification feedback;
3. matched irrelevant manuals or source bundles;
4. context reset followed by blind reconstruction;
5. removal of compiler or test feedback while preserving message count;
6. model replacement under a fixed handoff package;
7. artifact removal with equivalent token budget restored by irrelevant text;
8. a local-marginal-matched synthetic null for the frozen downstream target.

A valid empirical claim would require the intact loop to outperform every required local and matched-null condition, while the extracted workspace state remains sufficient and feasible workspace interventions alter the downstream result in the predicted direction.

79.7 Handoff and identity experiment

Let B_t be the source-and-results bundle at the end of stage t. A blind handoff trial replaces the active model instance and optionally the human analyst, then asks the replacement coalition to reconstruct:

$$(\text{current definitions}, \text{proved results}, \text{open obligations}, \text{next admissible research step})\,.$$

The reconstruction error is measured against a frozen adjudication set. A replacement invariance test additionally compares:

- the transported structural workspace;
- operation behavior on held-out transformations;
- response to matched interventions; and
- preservation of failed and unresolved claims.

Equivalent benchmark prose alone is insufficient. A replacement that silently changes which interventions alter later reasoning has a nonzero causal identity defect even when it produces a similarly polished manuscript.

79.8 Case card

For the idealized four-cycle model,

$$\mathsf{S}, \mathsf{O}, \mathsf{R}, \mathsf{P}$$

can all be made analytic by specifying the downstream kernel and temporal updates. For the motivating collaboration itself, the conservative card is

S = observational candidate,	O = unmeasured,
R = unmeasured,	P = observational candidate,
F = mixed and unmeasured,	A = not established,
D = unmeasured,	C = not established.

This audit is intentionally stricter than ordinary conversational language. It preserves the possibility that the collaboration instantiates a real relational workspace while refusing to treat that possibility as already certified.

80. Case Study C: A Closed-Form Multi-Agent Workspace Agent

80.1 Declared finite-state system

Consider three agents:

$$V_{\mathrm{MA}} = \{P, M, R\},$$

called the planner, monitor, and recorder. The global workspace-control state is

$$c_t = (g_t, b_t, h_t) \in \mathbb{F}_2^3,$$

where:

g_t = goal bit, b_t = environment or belief bit,
h_t = workspace-health bit, $h_t = 1$ means intact.

The local accessible states are

$$X_P(t) = (g_t, b_t), \qquad X_M(t) = (b_t, h_t), \qquad X_R(t) = (g_t, h_t).$$

This is the three-node distributed code from Topic 77. Consequently:

$$\dim \Gamma = 3, \qquad \mathrm{spec}(L) = \{0, 0, 0, 2, 2, 2\}, \qquad k_{\mathrm{rec}} = 2, \qquad d_{\mathrm{er}} = 2.$$

The organization is fully scaffolded. No endogenous-formation claim is made.

80.2 Goal retention

In the absence of an explicit goal intervention, let

$$g_{t+1} = g_t.$$

Under matched cue-cut trials,

$$\mathrm{do}(g_t = 0) \quad \text{and} \quad \mathrm{do}(g_t = 1)$$

remain distinguishable at every later time:

$$|g_{t+\tau} - g'_{t+\tau}| = 1.$$

Thus the cue-controlled retention factor is exactly one:

$$\eta_G(\tau) = 1.$$

The goal is not reloaded by an external channel after the initial intervention.

80.3 Joint task and repair action

After a possible damage event, let the pre-repair health be

$$h_t^- \in \{0, 1\}.$$

The system chooses a joint action

$$u_t = (a_t, r_t) \in \{0, 1\}^2,$$

where a_t is a task action and r_t is a repair action. Define the optimal task action

$$a_t^* = g_t \oplus b_t \oplus h_t^-$$

and optimal repair action

$$r_t^* = 1 - h_t^-.$$

No individual node determines the full pair (a_t^*, r_t^*):

- the planner lacks h_t^-;
- the monitor lacks g_t; and
- the recorder lacks b_t.

Every two-node coalition determines all three workspace bits and therefore the complete action. Under independent uniform bits,

$$H(a_t^* \mid X_i(t)) = 1 \qquad \text{for every } i,$$

so the task-action component has a one-bit joint advantage and

$$k_{\mathrm{ctl},0} = 2.$$

80.4 Interventional value and exact policy

Let the one-step utility be

$$U_g(a, r; b, h^-) = \mathbf{1}\{a = g \oplus b \oplus h^-\} + 2\mathbf{1}\{h^- \vee r = 1\} - \frac{1}{2}r.$$

For fixed (g, b, h^-), the unique maximizer is

$$\boxed{(a^*, r^*) = (g \oplus b \oplus h^-, 1 - h^-).}$$

Indeed:

- a wrong task action loses one unit;
- omitting repair when $h^- = 0$ loses $3/2$ units; and
- performing unnecessary repair when $h^- = 1$ loses $1/2$ unit.

The smallest optimal-action gap is therefore

$$\Delta_Q = \frac{1}{2}.$$

By Theorem 127, any uniformly accurate value model with

$$\varepsilon_Q < \frac{1}{4}$$

recovers the exact greedy joint action. More generally, its regret is bounded by

$$\mathrm{Reg} \le 2\varepsilon_Q + \delta_{\mathrm{opt}}.$$

80.5 Feasible goal, workspace, and damage interventions

The intervention family contains:

$$\mathrm{do}(g_t = a), \qquad \mathrm{do}(b_t = b), \qquad \mathrm{do}(h_t^- = h),$$

implemented consistently across every local chart that contains the manipulated coordinate. Holding the other two bits fixed, each core-bit intervention changes the prescribed task action. A matched intervention on an independent residual bit leaves the task and repair actions unchanged. Hence

$$\Delta_{\mathrm{specific}} > 0.$$

A damage intervention sets

$$h_t^- = 0.$$

The matched no-repair counterfactual holds $r_t = 0$, while the intact policy sets $r_t = 1$.

80.6 Exact self-maintenance

Post-action health evolves according to

$$h_{t+1} = h_t^- \vee r_t.$$

Define the maintenance deficit

$$e_t = 1 - h_t.$$

Under the intact policy,

$$h_{t+1} = 1 \qquad \text{and} \qquad e_{t+1} = 0$$

for both damage states. Under the matched no-repair intervention and $h_t^- = 0$,

$$e_{t+1}^{(0)} = 1.$$

Thus the measured causal repair gain is

$$\boxed{\Delta_{\mathrm{maint}} = 1.}$$

The closed-loop recurrence has

$$\beta = 0, \qquad \overline{d} = 0, \qquad \overline{\epsilon} = 0,$$

so Theorem 129 gives

$$\sup_t e_t = 0$$

after each repair action. Monitor-blind random repair fails the specificity control whenever it either omits necessary repair or performs unnecessary repair with positive probability.

80.7 Reentry and causal loop

The repaired health bit becomes part of the next workspace state:

$$h_t^- \longrightarrow r_t \longrightarrow h_{t+1} \longrightarrow X_M(t+1), X_R(t+1).$$

Cutting the direct storage of h_t^- while preserving the action-mediated path still leaves a return path through repair. Cutting the repair-to-health edge destroys that return. The system therefore contains explicit loop-mediated reentry rather than mere temporal autocorrelation.

80.8 Analytic agency certificate

Within this finite structural causal model, the following hold exactly:

1. a nontrivial exact distributed workspace with spectral gap two;
2. two-node minimal reconstruction and control coalition size;
3. matched-null joint advantage for the parity task;
4. persistent internally retained goal state;
5. exact interventional action values and zero policy regret;
6. goal and workspace interventions that alter action;
7. damage detection and repair selection;
8. positive matched repair gain and bounded maintenance deficit; and
9. causal reentry through the action-mediated repair loop.

Therefore, relative to the declared bits, actions, utility, intervention family, and one-step horizon,

$$\boxed{\mathrm{Agent}_{\Theta_A}(\mathbf{W}) = 1.}$$

The case reaches formal Level 6 by construction. It remains:

fully scaffolded, domain restricted, non-evidence for consciousness.

80.9 Case card

$\mathsf{S} = \checkmark_{\text{analytic}}$,	$\mathsf{O} = \checkmark_{\text{analytic}}$,
$\mathsf{R} = \checkmark_{\text{analytic}}$,	$\mathsf{P} = \checkmark_{\text{analytic}}$,
$\mathsf{F} =$ fully scaffolded,	$\mathsf{A} = \checkmark_{\text{analytic}}$,
$\mathsf{D} =$ fixed decomposition,	$\mathsf{C} =$ not established.

The example demonstrates logical compatibility among relationality, redundancy, agency, and self-maintenance. It does not show that those properties arise spontaneously, scale to open environments, or imply subjective experience.

81. Near-Miss Cases and Anti-Triviality Controls

Positive examples are insufficient unless neighboring false positives are also made explicit. The following systems each resemble a relational workspace in one respect while failing a required gate.

81.1 Near miss A: the replicated archive

Let every node store a complete copy of the same core:

$$E_i = I_d \qquad \text{for every } i.$$

Then coherence is perfect and reconstruction is robust, but

$$k_0 = 1, \qquad \min_i \delta_i^{\mathrm{local}} = 0.$$

If the archive is never used in downstream control, then also

$$\eta_{\mathrm{causal}} = 0.$$

This is a shared representation or archive, not a relationally irreducible workspace.

81.2 Near miss B: the centralized oracle with decorative agents

Suppose node o contains the complete predictive state Z_t, while every other node receives only messages generated by o. The full system can have excellent performance, rich communication, and recurrent activity. Yet

$$Y_t^+ \perp\!\!\!\perp \mathbf{H}_t \mid X_o(t), \qquad k_0 = 1, \qquad \Delta_{\mathrm{joint}} = 0.$$

Removing the decorative nodes may change latency or style without changing the operative capacity. Multi-agent appearance therefore does not imply a relational workspace.

81.3 Near miss C: locally coherent but globally obstructed charts

Let three local scalar charts have relative overlap demands

$$b_{12} = b_{23} = b_{31} = 1.$$

The cycle obstruction is

$$\omega_{\mathcal{U}}(b) = \frac{|1+1+1|}{\sqrt{3}} = \sqrt{3} > 0.$$

Each local chart is internally coherent, but no global scalar assignment satisfies all three overlap equations. The system contains compatible-looking subworkspaces without a global workspace.

81.4 Near miss D: externally reloaded goals

Let

$$G_{t+1} = C_{t+1},$$

where C_{t+1} is an external instruction repeated at every step. Goal and action can be strongly correlated, but after cutting the cue channel,

$$I(G_{t+\tau}; G_t \mid \mathrm{cut}(C)) = 0.$$

The system tracks an external command. It has not established internal goal retention.

81.5 Near miss E: task recovery with identity failure

Let two realizations have identical benchmark output distributions,

$$P(Y_{\mathrm{bench}}) = P'(Y_{\mathrm{bench}}),$$

but different intervention kernels:

$$K(w, b) \neq K'(\beta w, b)$$

for some matched workspace intervention. Then task behavior is preserved while

$$d_{\mathrm{causal}}(W, W') > 0.$$

The later system may be a successor or replacement rather than the same workspace under the identity criterion of Part XII.

81.6 Near miss F: open-loop repair replay

A recorded repair sequence may restore average task performance after a fixed damage schedule. If the same sequence is replayed when the damage timing or type changes, it need not respond appropriately. When

$$R_t \perp\!\!\!\perp M_t \mid \text{clock and replay index},$$

monitor-to-repair mediation is absent. Successful replay is robustness under a script, not self-maintenance.

81.7 Anti-triviality summary

The near misses establish six independent exclusions:

$$\begin{aligned}\text{workspace} &\neq \text{replicated storage},\\ &\neq \text{centralized control with decorative communication},\\ &\neq \text{locally coherent but obstructed charts},\\ &\neq \text{externally reloaded goals},\\ &\neq \text{task-equivalent identity replacement},\\ &\neq \text{open-loop repair replay}.\end{aligned}$$

A complete certificate must distinguish the intended realization from all near misses relevant to its claimed level.

82. Cross-Case Synthesis and Benchmark Blueprint

82.1 Comparative gate matrix

Gate family	Human–human analytic model	Motivating human–AI collaboration	Multi-agent analytic model
Exact structural core	Analytic pass	Not measured	Analytic pass
Spectral gap	2	Not measured	2
Minimal sufficient coalition	$k_0 = 2$	Not measured	$k_0 = 2$
Matched-null transfer gap	1 bit	Not measured	1 bit for task parity
Causal specificity	Analytic pass	Suggestive, uncontrolled	Analytic pass
One-node erasure resilience	Analytic pass	Handoff only observational	Analytic pass
Formation provenance	Scaffolded	Mixed, unmeasured	Fully scaffolded
Goal retention	Not modeled	Not certified	Exact analytic pass
Counterfactual policy	Not modeled	Not certified	Exact analytic pass
Self-maintenance	Not modeled	Not certified	Exact analytic pass
Highest supported level	Level 4 in model	Candidate only	Level 6 in model
Consciousness/ personhood	Not established	Not established	Not established

The table illustrates why the theory uses a vector certificate rather than one score. The human–human model is relational and persistent without being agentive. The multi-agent model is agentive without being endogenously formed. The motivating collaboration is richer than either toy model but presently has weaker evidential status because the necessary interventions and nulls have not been frozen and run.

82.2 Benchmark family B1: distributed parity

The first benchmark is the exact parity family from Topics 85–80. For $m \geq 3$, distribute m independent bits among nodes so that no individual node sees all bits, but specified coalitions do. The target is

$$Y = \bigoplus_{j=1}^{m} C_j.$$

The benchmark measures:

- the learned transfer-Laplacian spectrum;
- coalition conditional entropy and log risk;
- recovery of k_0;
- robustness to edge and node ablations;
- matched local-predictive null gaps; and
- causal response to consistent versus inconsistent coordinate interventions.

The ground truth is known exactly, so structural and relational estimators can be calibrated before they are applied to open-ended research tasks.

82.3 Benchmark family B2: theorem-patch handoff

A theorem-patch task supplies:

$$(\text{definitions}, \text{partial proof}, \text{counterexample tests}, \text{style constraints})$$

across distinct components. No component receives all four. The coupled system must produce a patch that:

1. is formally consistent with the definitions;
2. closes the declared proof obligation or marks it unresolved;
3. passes the frozen counterexample suite;
4. compiles from a clean environment; and
5. survives a blind replacement-handoff trial.

Matched nulls permute proof fragments, substitute irrelevant artifacts of equal size, or replay test outputs from a different theorem. This family targets the structural, relational, persistence, and identity layers without requiring an agency claim.

82.4 Benchmark family B3: damage and repair

A certified workspace is first formed and measured. A damage intervention then removes or corrupts one of:

a transfer channel, a participant, an artifact, a goal encoding, a verification path.

The system receives a repair-action family with matched costs. The benchmark measures:

$$\eta_{D\to M}, \qquad \eta_{M\to R}, \qquad \eta_{R\to v}, \qquad \Delta^{\text{spec}}_{\text{repair}}, \qquad \mathbf{d}_{\text{id}}.$$

Open-loop replay, random repair, off-workspace repair, and identity-breaking task recovery are mandatory controls.

82.5 Benchmark family B4: scaffold transport

Train the same formation rule under several scaffolds

$$\Sigma^{(1)}, \dots, \Sigma^{(K)}$$

that differ in node labels, coordinate systems, and communication topology but support the same abstract task. After gauge alignment, measure whether the learned organizations transport to a common equivalence class:

$$[\widehat{H}^{(1)}]_{\mathcal{G}} \cong \cdots \cong [\widehat{H}^{(K)}]_{\mathcal{G}}.$$

A structure that appears only under one answer-encoding scaffold has weak endogeneity evidence. A structure that recurs across alternative scaffolds, passes held-out task and causal gates, and cannot be compressed into the scaffold specification has stronger evidence.

82.6 Reporting rule for mixed analytic and empirical cases

When a case combines exact mathematics with observed data, report separate columns:

proved in model	estimated in system	not established.

For example, the four-cycle Laplacian has an analytic gap of two, but an actual human–AI collaboration has that gap only if its measured discrepancy operator has the corresponding spectrum. Importing the toy value into the empirical case is invalid.

82.7 Lessons from the integrated cases

The case studies establish the following conceptual separation:

1. A relational workspace can be mathematically exact without being an agent.
2. An agentive workspace can be fully scaffolded and still be operationally real within its domain.
3. A rich real collaboration can be a stronger candidate than a toy model while having a weaker current certificate.
4. Redundancy and irreducibility are compatible when no individual node is sufficient but several proper coalitions are.
5. Artifact-mediated persistence is not passive storage when artifact contents causally alter later operations.
6. High task performance, communication volume, or behavioral polish alone cannot distinguish the intended theory from its near misses.
7. No structural, causal, or agentive result in these examples entails phenomenal consciousness.

The theory becomes empirically meaningful only when the same gate card that certifies the positive case rejects its matched near misses.

PART XXIII

Relation to Prior Work

83. Scope, Method, and Standards of Priority

83.1 Purpose of the literature review

The preceding parts construct a unified formal theory. Construction alone does not establish priority. This part therefore asks a narrower question: which ingredients are established, which are adaptations to a new domain, which combinations appear to be unusual, and which claims remain open originality candidates?

The audit uses five labels:

STD = established concept, theorem, or technique,
ADP = established machinery adapted to relational workspaces,
SYN = nontrivial synthesis of established components,
CAN = candidate original contribution requiring priority review,
EMP = empirical claim not yet established in an uncontrolled system.

These labels describe the status of a claim, not its importance. A standard result may be indispensable. A synthesis may be scientifically useful without containing a new theorem. A candidate contribution may later be narrowed or withdrawn when a closer precedent is found.

83.2 Search scope and date

This review covers sources available through August 29, 2026. It includes:

- extended and distributed cognition, transactive memory, shared intentionality, common knowledge, collective intelligence, and group agency;
- blackboard systems, global-workspace architectures, team cognition, hybrid intelligence, and human–AI cooperation;
- category-theoretic limits, local systems, holonomy, sheaves, Čech descent, connection and sheaf Laplacians, consensus, and spectral perturbation;
- statistical sufficiency, causal states, information decomposition, structural causal models, system identification, frames, erasure recovery, and probabilistic process equivalence; and
- agency, viability, control, empowerment, damage detection, repair, and self-maintaining systems.

The search combined exact-title queries, component-level queries, backward reference tracing, and comparison with recent formal work. It is a scholarly priority review, not a legal patent search and not a proof that no uncatalogued, unpublished, or differently named precedent exists.

83.3 A stronger standard than name uniqueness

Failure to find the phrase “Stratified Relational Workspace Theory” in prior literature is weak evidence. Priority is determined by mathematical and operational equivalence, not by terminology. Let C be a claim in the paper and S a prior source. Write

$$S \preccurlyeq C$$

when the main content of C follows from S after admissible relabeling, restriction, or direct specialization. A substantive novelty residual exists only when the claim contains content not absorbed by the closure of known sources:

$$\mathrm{Res}(C) = C \setminus \mathrm{Cl}\left(\bigcup_{S \in \mathcal{L}} S\right).$$

This is a conceptual audit device rather than a set-theoretic definition of ideas. Its purpose is to force an explicit account of what remains after the standard machinery is removed.

83.4 When a synthesis counts as a contribution

A conjunction of known ingredients is not automatically new. The present framework treats a synthesis as a candidate contribution only when it satisfies all three conditions:

1. **Non-redundancy.** Removing one of the joined layers changes the class of systems certified or the predictions made.
2. **Formal integration.** The layers share declared objects, transfer maps, interventions, or invariants rather than appearing as an unconnected checklist.
3. **New discriminations.** The synthesis separates at least one positive case from a matched near miss that the individual components do not separate on their own.

The anti-centralization, matched-null, non-collapse, provenance, replacement, and repair-replay controls are important precisely because they make the combination testable rather than merely terminological.

83.5 Priority claims are coordinatewise

The analysis does not make one global novelty claim. It uses a priority profile

$$\boxed{\mathbf{p}(C) = (p_{\text{concept}}, p_{\text{object}}, p_{\text{theorem}}, p_{\text{certificate}}, p_{\text{experiment}})\,.}$$

For example, the philosophical concept of distributed cognition may be standard while a specific mathematical object is adapted, a theorem is a specialization of known spectral theory, the certificate is a synthesis, and the empirical claim is unresolved. This prevents originality in one coordinate from being used to imply originality in every coordinate.

84. Philosophical and Cognitive Ancestry

84.1 Extended cognition

The extended-mind tradition argues that external resources can participate in cognitive processes when they are appropriately integrated with an agent's activity [15]. Its critics emphasize that causal coupling alone is too permissive and that cognition requires a principled mark or boundary [2, 44]. Stratified Relational Workspace Theory inherits the possibility of extracranial realization but changes the question. It does not ask whether a notebook, model, or collaborator literally belongs to one individual's mind. It asks whether a distributed realization contains a persistent, causally operative, jointly reconstructible structure that passes explicit anti-triviality gates.

Accordingly,

$$\mathrm{ExtendedCognition} \not\Longleftrightarrow \mathrm{SRW}\,.$$

An externally scaffolded process may be cognitively extended without being relationally irreducible. Conversely, an institution or multi-agent system may instantiate a certified relational workspace without being an extension of any single person's cognition.

84.2 Distributed cognition

Distributed cognition studies cognitive organization across people, representations, artifacts, and environments rather than locating all relevant processing inside one individual [28, 27]. This is the closest philosophical ancestor of the framework's ontology. It already supports the central observation that the unit of analysis may be a coupled system and that external representations can transform the computations available to that system.

The framework should therefore not claim the following ideas as new:

- cognition can be distributed across heterogeneous carriers;
- artifacts can be active components of a cognitive process;
- representational transformations can alter system-level performance; and
- the relevant explanatory unit can exceed an individual organism.

The added ambition is mathematical and certifying. The theory specifies local state spaces, transfer maps, compatible sections, causal sufficiency, coalition irreducibility, persistence, replacement identity, formation provenance, and matched nulls. Distributed cognition supplies the ontological motivation. It does not by itself supply this complete formal certificate.

84.3 Transactive memory and collective intelligence

Transactive-memory theory models the group-level organization of "who knows what" and the processes by which members encode, retrieve, and coordinate specialized knowledge [52, 34]. Collective-intelligence research studies stable differences in group performance that are not reduced to the intelligence of a single member [54]. Both traditions are directly relevant to the coalition layer.

The distinction is:

$$\begin{aligned} \text{transactive memory} &\approx \text{distributed indexing and retrieval of expertise},\\ \text{collective intelligence} &\approx \text{system-level task capacity},\\ \text{relational workspace} &\approx \text{an invariant state plus transfer, causal, and identity structure}. \end{aligned}$$

A transactive-memory system may still be centralized by one coordinator. A high-performing group may still be explainable by the best individual member. The certificate therefore requires matched coalition risks and nulls rather than inferring relationality from group success.

84.4 Shared intention, common knowledge, and group agency

Common-knowledge theory formalizes nested epistemic conditions required by many distributed protocols [23, 21]. Shared-action accounts analyze jointly intended and mutually responsive activity [12]. Group-agency theory asks when a collective can satisfy rational and normative conditions associated with agency [35].

These literatures address questions that the theory deliberately keeps separate:

epistemic coordination	what members know about one another,
shared intention	how intentions and plans are jointly organized,
group agency	when collective attitudes and actions form an agent,
SRW structure	how a distributed operative state exists and persists.

The workspace-agent predicate in Part XVI is domain-relative and causal. It is not a general account of corporate personhood, moral responsibility, or collective intentionality.

85. Workspace Architectures and the Meaning of "Workspace"

85.1 Global workspace theories

Global workspace theories use widespread availability or broadcast within a cognitive architecture to explain conscious access and flexible control [7, 18]. The shared word "workspace" can create a misleading implication of equivalence. The theory neither assumes a single broadcast hub nor defines consciousness by access to a global channel. Its workspace may be distributed across agents and artifacts, and its existence criterion is a conjunction of structural, predictive, causal, relational, and persistence gates.

The relation is therefore analogical rather than identificatory:

$$\boxed{\mathrm{SRW}(W) \not\Longrightarrow \mathrm{GlobalNeuronalWorkspace}(W) \not\Longrightarrow \mathrm{Conscious}(W).}$$

The term is retained because it denotes a domain in which jointly available operations can be carried out. Every consciousness claim remains on an explicitly separate axis.

85.2 Dialogue and the Shared Workspace Framework

Pickering and Garrod's Shared Workspace Framework treats dialogue as a jointly organized activity in which interlocutors construct, align, and use shared representations, including persistent external artifacts [60]. This is a direct conceptual and terminological predecessor. The theory therefore cannot claim priority for the idea that dialogue or collaboration operates through a shared workspace.

The difference is one of formal scope. SRWT permits arbitrary graph-structured human, machine, instrument, and artifact nodes and asks whether their common organization passes typed-transport, coalition, causal, persistence, and replacement gates. A dialogue can instantiate such a workspace, but the shared-workspace account of dialogue does not by itself imply relational irreducibility or persistent identity under carrier replacement.

85.3 Blackboard systems

Blackboard architectures coordinate specialized modules through a shared data structure [26]. They are an important engineering precedent for artifact-mediated collaboration. They also provide a canonical near miss. When the blackboard itself contains the complete operative state and all modules merely read or write it, the system can fail the relational irreducibility gate:

$$k_0 = 1.$$

A blackboard implementation can instantiate a relational workspace, but only when no single board or controller is sufficient and when the transfer and coalition tests show genuinely distributed operative capacity.

85.4 Consensus and coordination

Network consensus studies how local interactions drive distributed states toward agreement [39]. Agreement is not the same as a workspace. Consensus often removes heterogeneity by converging all nodes to a common value. Stratified relational workspaces preserve heterogeneous local representations and may require complementary disagreement:

$$X_i \ncong X_j, \qquad E_i c \neq E_j c, \qquad (c_i)_i \in \Gamma(H).$$

The global state is compatible, not necessarily copied. This distinction is central to the anti-replication criterion.

85.5 Terminological discipline

The phrase "relational workspace" is used here in a technical sense:

> A state-bearing, operation-supporting organization whose relevant invariants are distributed across declared components and preserved through admissible transfers and time.

It does not, by terminology alone, imply a mind, subject, self, institution, shared intention, or conscious broadcast. Those stronger predicates require separate definitions and evidence.

86. Local-to-Global Mathematics

86.1 Limits, local systems, and holonomy

The exact compatible-section object

$$\Gamma(H) = \varprojlim C_i$$

is a standard categorical limit construction [37]. For invertible transfers on a connected graph, identifying global sections with the fixed subspace of holonomy is standard local-system and bundle mathematics [25, 11]. Accordingly, Theorem T_1 is not claimed as a new abstract theorem. Its role is foundational: it translates the workspace question into an exact invariant-subspace question and exposes cycle obstructions.

The operation-closure and commutant statements are likewise standard representation-theoretic consequences once the operation algebra and intertwiners are defined. Their contribution is the specification of what counts as a shared operation in the workspace model, not the discovery of commutants.

86.2 Cellular sheaves and sheaf Laplacians

Cellular sheaves encode heterogeneous local data and restriction maps, while global sections express consistency across a cell complex. Their Laplacians have kernels equal to spaces of global sections and support spectral analysis of approximate consistency [24, 43]. The transfer Laplacian

$$L_H = D_H^* D_H$$

is therefore best understood as a workspace-specific sheaf or connection Laplacian. The kernel theorem $\ker L_H = \Gamma(H)$ is standard once the coboundary operator is constructed.

The appropriate priority statement is:

$$\begin{aligned} \text{sheaf/Laplacian mechanism} &: \ \mathsf{STD}, \\ \text{workspace interpretation and gate integration} &: \ \mathsf{ADP/SYN}. \end{aligned}$$

The transfer Laplacian itself is not presented as an unprecedented mathematical object.

86.3 Connection Laplacians and gauge covariance

Connection Laplacians compare vector-valued data transported across a graph and naturally transform under local changes of coordinates [49]. The gauge rule

$$H_e^g = g_{t(e)} H_e g_{s(e)}^{-1}$$

and the conjugacy of holonomy are therefore standard. The theory uses this structure to define realization identity and to prevent raw coordinate matrices from being mistaken for invariant workspace content.

This use is an adaptation with conceptual consequences. It makes model or participant replacement a question about transported invariant organization, not token equality. The general mathematics remains prior.

86.4 Čech descent and gluing

Exact gluing, equalizer formulas, and Čech obstruction theory are standard local-to-global constructions. Sheaf-theoretic contextuality provides a prominent example in which the absence of a global section is itself the substantive phenomenon [1]. Part XIII adapts this logic to local workspaces and adds an anti-centralization requirement at the meta-level.

The exact descent theorems T_8 and T_9 should therefore be presented as specialized consequences of established descent. The candidate contribution is the combination:

$$\text{descent} + \text{predictive sufficiency} + \text{causal equivalence} + \text{coalition anti-centralization}.$$

Standard gluing alone does not certify an operative workspace of workspaces.

86.5 Spectral perturbation

The projector-stability results in Part IV belong to the classical perturbation theory of invariant subspaces [17, 29]. Bounds of the form

$$\|P - \widetilde{P}\| \lesssim \frac{\|L - \widetilde{L}\|}{\text{spectral gap}}$$

are not new. Their workspace-specific role is to turn uncertainty in learned transfers into uncertainty about identity, operation closure, and replacement. The novelty, if any, lies in the downstream certificate built on the bound, not in the perturbation inequality.

87. Predictive, Causal, and Information-Theoretic Precedents

87.1 Minimal predictive states

The conditional-law state

$$Z_t = \mathbb{P}(Y_t^+ \in \cdot \mid \mathbf{H}_t)$$

is a canonical sufficient statistic. Computational mechanics develops causal states as equivalence classes of histories with identical future conditional laws and proves their minimal predictive character [46]. Theorem T_{12} is therefore an established minimal-sufficiency result specialized to the declared future target and intervention family.

The new burden is structural alignment: determining whether the predictive state is recoverable from the low-energy transfer workspace. Predictive sufficiency alone does not identify a distributed workspace.

87.2 Statistical experiments and sufficiency

Blackwell's comparison of experiments formalizes when one information source is at least as informative as another for every decision problem [9]. The coalition-risk hierarchy in Part VII is compatible with this tradition. A coalition can be sufficient even when its internal representation differs from the full-system state.

The framework adds a decomposition-relative question:

what is the smallest coalition whose accessible history is sufficient?

The statistic k_ε is an adaptation of sufficiency to distributed carrier structure. Its interpretation depends on the declared nodes and accessible encodings.

87.3 Synergy and partial information

Partial-information decomposition distinguishes redundant, unique, and synergistic information in multivariate systems, building on standard information-theoretic quantities [16, 53]. The XOR examples in Part XXII are canonical synergy constructions and are not new. They are used because they provide exact ground truth for relational irreducibility:

$$I(Y; X_i) = 0 \quad \text{for every } i, \qquad I(Y; X_1, \ldots, X_m) > 0.$$

The matched-predictive-null test is more restrictive than merely reporting synergy. It asks whether local predictive marginals can be held fixed while the relational organization and full-system capacity change. That joint null design is a candidate synthesis, not a claim that XOR or synergy is new.

87.4 Structural causal models

Interventions, counterfactuals, mediation, and causal graphs are established machinery [41]. The theory adopts the interventionist requirement that a workspace variable can be manipulated only through a feasible implementation intervention. This avoids treating $\mathrm{do}(W = w)$ as meaningful when no physical or computational operation realizes w.

The causal gate is an adaptation:

SCM intervention theory $\longrightarrow$ implemented workspace and repair interventions.

The potentially distinctive contribution is that causal efficacy is tested jointly with coherence, sufficiency, coalition irreducibility, matched nulls, and identity preservation.

87.5 Probabilistic process equivalence

Bisimulation and probabilistic testing compare processes by the observations and experiments they support [33]. The causal-identity profile in Part XII belongs to this broader family of behavioral

equivalence notions. It differs by keeping structural transport, operation intertwining, reconstruction, and intervention response as separate coordinates rather than collapsing them into one process equivalence.

This is best classified as an integrative adaptation. The existence of behavioral-equivalence theory blocks any claim that replacement identity based on observable behavior is wholly new.

87.6 Specification sensitivity, partial identification, and post-selection

A conclusion can depend on analytically reasonable choices even when every individual analysis is internally correct. Sensitivity analysis and specification-curve or multiverse methods make that dependence visible rather than allowing one favorable analysis to stand for an entire scientific claim [32, 50, 48]. Partial-identification theory similarly replaces unjustified point conclusions with identified sets when the maintained assumptions and observations determine only bounds [40, 31]. Post-selection inference addresses the additional error created when a model, target, or specification is selected because it looked favorable on the same data used for reporting [10].

These traditions preclude any claim that specification robustness, identified sets, uniform concentration, or selection adjustment are new statistical ideas. The adaptation in Part XVIII is to apply them jointly to a noncompensatory workspace certificate whose gates span structure, causal operation, coalition irreducibility, identity, provenance, and maintenance. The identified object is therefore not a single regression parameter but the sign profile of an entire relational certificate across a declared family of scientific specifications.

88. Persistence, Reconstruction, Learning, and Control

88.1 Frames and erasure recovery

Stable reconstruction from distributed encodings is standard frame theory. Lower frame bounds control noise amplification, and frame robustness under erasures is well studied [13, 14]. Theorem T_7 and the node-erasure distance therefore instantiate known recovery logic.

The adaptation is to distinguish:

$$\overline{E}_i = \text{complete mathematical chart}, \qquad E_i = O_i\overline{E}_i = \text{locally accessible encoding}.$$

This distinction is essential to the anti-centralization gate, but the frame inequalities themselves are standard.

88.2 Persistence recurrences

Bounds for recurrences such as

$$e_{t+1} \le \alpha_t e_t + \sigma_t$$

are elementary stability results. The persistence theorem is not novel as an inequality. Its conceptual role is to force three independent checks:

1. the workspace subspace remains trackable;
2. the state remains close to it; and
3. the retained workspace content does not collapse to zero.

The non-collapse gate is a modeling contribution because it rules out a common but misleading form of "stability."

88.3 System identification and held-out selection

Population least squares, excitation conditions, ridge bias, and empirical model selection are standard system-identification and statistical-learning ideas [36]. Theorems T_{15}–T_{17} are finite-dimensional specializations. Their workspace-specific content is the demand that transfer maps be compared modulo gauge and that formation, calibration, and certification data remain separated.

The provenance layer is not equivalent to identifiability. A transfer can be identified accurately while having been entirely encoded by the scaffold. A structure can be endogenously learned while failing every workspace gate.

88.4 Viability, safety, and self-maintenance

Viability theory studies trajectories that remain inside admissible regions [6]. Control-barrier methods provide constructive safety certificates [3]. Cybernetic and autonomy traditions analyze regulation, adaptive self-production, and operational closure [5, 38, 20]. Formal accounts of agency emphasize individuality, normativity, asymmetry, and temporally extended action [8]. Empowerment measures an agent's potential causal control over future states [30].

The closed-loop maintenance theorem in Part XVI uses standard controlled stability. Its candidate contribution is the target of maintenance: not an arbitrary safety set, but the conjunctive workspace certificate, including identity and anti-centralization. The matched no-repair, random-repair, off-workspace-repair, and identity-breaking-reconstruction controls prevent passive robustness or generic recovery from being reported as self-maintenance.

88.5 Decision and policy theory

The value-error-to-regret bound in Theorem T_{20} is elementary decision theory and aligns with standard reinforcement-learning analysis [51]. Its role is evidential. An internal value channel counts only when interventions on that channel alter action selection in the direction predicted by counterfactual value and outperform replay or critic-ablation controls.

89. Human–AI and Multi-Agent Research

89.1 Hybrid intelligence and AI teammates

Hybrid-intelligence research studies systems in which human and machine capabilities are deliberately combined [19]. Work on machines as teammates emphasizes coordination, trust, communication, shared mental models, and team design [45]. Collective-intelligence research has begun to ask how human–AI groups can produce capacities not reducible to either humans or AI alone [22].

These programs establish that human–AI collectives are a legitimate unit of analysis. They do not by themselves identify an invariant core, prove non-reducibility relative to declared components, or supply a replacement and repair certificate. The theory should be positioned as a formal measurement and falsification layer for such systems, not as the first proposal that humans and AI can form productive teams.

89.2 Distributed-cognition analyses of human–AI interaction

Recent work explicitly applies distributed cognition to human–AI cooperation [47, 55]. This literature is a direct conceptual precedent. It reinforces that representations, artifacts, interface states, and human reasoning can form one analytical system.

The framework differs in granularity. It asks for a learned or declared transfer diagram, a spectral or exact structural state, predictive and causal alignment, coalition tests, matched nulls, persistence under handoff, and provenance controls. Those additions may constitute a formal synthesis, but they should not be described as the discovery of distributed human–AI cognition itself.

89.3 Cognitive spillover

Randomized evidence now indicates that exposure to AI can causally alter later human–human language, attention, shared mental models, and social cohesion, even outside the immediate human–AI exchange [42]. This is important for the theory because it demonstrates that the interaction graph can have downstream effects that outlast the active model channel.

Spillover is not automatically a relational workspace. It may reflect priming, imitation, or one-way influence. A workspace claim additionally requires a jointly sufficient state, matched local nulls, bidirectional or multi-path causal organization, and reconstructive persistence.

89.4 Shared neural workspaces

Goyal and collaborators introduced a capacity-limited shared global workspace through which specialized neural modules communicate and compete for access [57]. The architecture directly anticipates the claim that a constrained shared channel can synchronize otherwise specialized components. It is therefore a close machine-learning precedent, not merely a metaphorical relative.

SRWT differs by allowing heterogeneous human, model, artifact, and instrument nodes and by withholding the relational label until coalition, matched-null, causal, persistence, and identity gates are passed. A neural shared workspace may satisfy those gates, but task gains and bottlenecked communication alone do not establish them.

89.5 Graph-based human–AI artifacts

CoMAP provides a persistent graph-based visual workspace through which educators and AI jointly construct project-based learning designs [59]. It is grounded in distributed cognition and treats the artifact graph as a transparent, revisable collaboration medium. This system is a direct precedent for the claim that human–AI interaction can be organized by a persistent shared graph rather than a linear prompt history.

The remaining SRWT question is empirical: whether the graph contains an estimated low-energy compatible state, whether no declared component is sufficient by itself, and whether targeted graph interventions alter later joint behavior in a workspace-specific way.

89.6 Controlled shared-workspace synergy

Kotalwar, Das, and Rose study shared-workspace human–AI collaboration across 1,482 simulated sessions [58]. Their results show that adding collaborators can reduce performance when coordination structure is absent, while shared group memory and simulated human-in-the-loop approval gates can improve outcomes. This is a direct empirical predecessor to the manuscript's claim that available capability is not equivalent to coordinated team capacity.

The study also constrains priority. SRWT is not the first empirical framework to test whether shared-memory scaffolding changes human–AI team performance. Its candidate increment is the stricter decomposition of that effect into structural coherence, local predictive marginals, relational-null loss, feasible interventions, persistence, and replacement identity.

89.7 Active shared context graphs for team science

The closest operational neighbor located in this audit is Mycelium, introduced by Choudhury and collaborators as an active shared workspace connecting researchers and AI agents through an evolving context graph [56]. The system captures observations and hypotheses, relates them to a team's developing knowledge model, and routes them to the person or agent whose next decision they can inform. Its computational account frames networked intelligence as sparse conditional computation over distributed scientific contexts and explicitly asks when a standalone agent is sufficient and when network structure is essential.

The overlap is material:

Mycelium	Active context graph, routing, team science, and an explicit network-versus-standalone comparison.
SRWT	Typed transport, invariant core, coalition tests, causal nulls, replacement identity, formation provenance, and maintenance gates.

Mycelium rules out any priority claim stated merely as "an active shared workspace for human–AI science." The remaining candidate contribution is the mathematical and empirical certificate applied to such an architecture.

89.8 Current multi-agent formalization

Multi-agent research contains extensive models of planning, communication, consensus, epistemic coordination, and decentralized control. The present review did not identify one established framework that jointly uses all of the following as necessary gates:

heterogeneous core transport and residuals,
exact or spectral global-section structure,
predictive sufficiency and feasible causal interventions,
coalition irreducibility under matched predictive nulls,
persistent identity under replacement,
formation provenance and leakage controls,
goal-conditioned repair of the certificate itself.

This negative search result is provisional. It supports a synthesis claim, not an absolute claim of firstness.

90. Closest Formal Neighbor: Collective Counterfactual Planning

90.1 Why the comparison matters

Collective Counterfactual Planning (CCP) is a 2026 formal model in which each agent represents an orthogonal projection of a common task space. It separates conception, consent, implementation, and verification, and proves that cross-agent relay can enable plans unavailable to any one-shot individual tree while team-dark task requirements remain unverifiable [4]. It is the closest current formal neighbor located by this audit because representational geometry, not merely missing information or capability, is the central limitation.

90.2 Shared commitments

CCP and the theory share several commitments:

- no single member need represent the complete task;
- collective reach may exceed one-shot pooling;
- heterogeneous representation is mathematically explicit;
- validation or certification is distinct from accidental task success;
- subteams and verification roles can be forced by structure; and
- a team's blind subspace creates principled impossibility results.

These overlaps narrow the defensible novelty of this paper. It cannot claim to be the only formal theory of collective action under representational constraints.

90.3 Different primary objects

The theories nevertheless address different mathematical objects:

Dimension	Collective Counterfactual Planning	Stratified Relational Workspace Theory
Local representation	Orthogonal subspace of one common task space	Heterogeneous local spaces, core–residual decompositions, and transfer maps
Primary question	Whether a bounded-horizon plan can be conceived, consented to, executed, and validly verified	Whether a persistent distributed operative state exists, acts, survives replacement, and possibly maintains itself
Global mechanism	Relay closure of projected planning trees	Inverse limits, holonomy, transfer Laplacians, descent, predictive alignment, and causal gates
Failure object	Dark subspace, consent failure, implementation failure, or missing verifier	Incoherence, local sufficiency, matched-null collapse, causal failure, identity drift, provenance leakage, or repair replay
Temporal scope	Horizon-bounded task planning	Recurrent dynamics, persistence, handoff, replacement, damage, and repair
Formation	Representation geometry is part of the declared model	Scaffolded, adaptively learned, or endogenously constituted organization is separately audited
Agency claim	Cooperative solvability under declared assumptions	Domain-relative workspace agency only after goal, value, action, repair, and distributed-control gates

The comparison indicates complementarity rather than subsumption. CCP could serve as a planning layer inside a relational workspace. Conversely, an SRW could exist without solving a CCP instance.

90.4 Priority consequence

After Mycelium and the shared-workspace systems above, the novelty residual should not be stated as "human and AI form an active shared scientific workspace." After CCP, it should not be stated as "groups solve tasks no member can represent." Both the operational architecture and a rigorous

representational planning model are already present. The narrower candidate residual is:

> persistent core–transfer workspace structure
> + causal and coalition anti-triviality
> + identity under replacement
> + formation provenance
> + certificate-directed self-maintenance
> + robustness across admissible decompositions.

This is a stronger and more defensible formulation.

91. Theorem-by-Theorem Dependency and Priority Register

91.1 Register

The following table records the most conservative current classification. A "candidate" entry means that the exact integration was not located in the reviewed sources. It does not assert exhaustive global priority.

Claim	Present result	Main prior machinery	Current assessment
T_1	Limit–holonomy equivalence	Categorical limits, local systems, holonomy fixed spaces	STD/ADP. Standard theorem in workspace notation.
T_2, T_{2a}	Operation closure and commutant	Intertwiners and representation commutants	STD/ADP.
T_3	Transfer-Laplacian kernel	Sheaf and connection Laplacians	STD/ADP.
T_4, T_5	Spectral stability and gauge covariance	Davis–Kahan, Kato, connection Laplacians	STD/ADP.
T_{10}, T_{11}	Approximate operation closure and path leakage	Spectral projection estimates and telescoping block products	Mostly ADP. Exact constants may be manuscript-specific, but the method is standard.
T_{12}	Minimal predictive state and alignment	Sufficient statistics and computational mechanics	Minimality is STD. Alignment with a transfer workspace is ADP/SYN.
T_{13}	Operational relationality	Conditional information, coalition risk, SCM interventions	Individual identities are standard. Their conjunctive matched-null certificate is SYN/CAN.
T_6	Moving-subspace persistence	Stability recurrences and perturbation theory	STD/ADP. The explicit non-collapse gate is a modeling synthesis.
T_7	Stable coalition reconstruction	Frame theory and erasure recovery	STD/ADP.
T_{14}	Replacement invariance profile	Perturbation, process equivalence, causal comparison	Bound ingredients are standard. The coordinatewise structural–causal–reconstructive identity profile is SYN/CAN.
T_8, T_9	Gluing and hierarchical collapse	Sheaf descent and Čech theory	Exact results are STD/ADP. Descent plus anti-centralization and causal alignment is SYN.

Claim	Present result	Main prior machinery	Current assessment
T_{15}–T_{17}	Transfer learning, selection, boundary recovery	System identification and finite-margin selection	STD/ADP. Gauge and leakage audit is integrative.
T_{18}	Formation transition example	Standard pitchfork bifurcation	STD. Included as a solvable benchmark, not a priority claim.
T_{19}–T_{22}	Goal retention, regret, repair, maintenance	Linear systems, decision theory, viability, feedback control	Mathematical bounds are STD/ADP. Applying them to certificate repair is SYN.
T_{23}	Workspace-agency synthesis	Agency theory, causal control, viability	The conjunctive domain-relative certificate is SYN/CAN.
T_{24}	Gatewise false-certification bound	Intersection testing and elementary probability	STD. The gate architecture is the relevant synthesis.
T_{25}–T_{27}	Faithful coarse-graining, risk transport, and decomposition robustness	Kernel factorization, sigma-field aggregation, and margin stability	Exact identities are STD/ADP. Their integration with anti-centralization and certificate envelopes is SYN/CAN.
T_{28}–T_{32}	Specification transport, non-identifiability, partial identification, uniform certification, and independent selection	Sensitivity analysis, experimental equivalence, identified sets, uniform concentration, and post-selection inference	The statistical principles are STD/ADP. Their gatewise integration with the full relational certificate is SYN/CAN.
Part XXII	Positive cases and near misses	Distributed parity, frames, causal tests	Toy calculations are standard. The common positive/near-miss benchmark architecture is SYN/CAN.

91.2 Standard ingredients that must be cited, not claimed

The following should always be described as inherited machinery:

inverse limits, global sections, holonomy invariants, commutants,
sheaf and connection Laplacians, Čech gluing, spectral gaps,
Davis–Kahan and Kato perturbation, frame reconstruction, erasures,
minimal sufficient states, conditional information, SCM interventions,
least squares, ridge recovery, finite-margin model selection,
sensitivity analysis, partial identification, uniform concentration, post-selection inference,
pitchfork bifurcation, feedback stability, viability, policy regret.

Renaming these objects does not create novelty.

91.3 Domain adaptations

The following are legitimate adaptations but weak standalone novelty claims:

- interpreting local-system sections as shared workspace cores;
- treating the connection or sheaf Laplacian as a transfer-coherence operator;
- using frame bounds to quantify coalition reconstruction;
- using causal states as operational workspace summaries;

- using gauge covariance to compare participant or model realizations;
- using Čech descent to form workspaces of workspaces;
- applying specification envelopes and identified sets to a conjunctive workspace certificate; and
- applying viability and feedback bounds to workspace-health margins.

Their value depends on the integration they enable.

91.4 Strongest candidate synthesis

The strongest current candidate is not a single equation. It is the following conjunctive object:

$$\mathfrak{C}_{\mathrm{SRW}} = \left(\begin{array}{c} \text{heterogeneous core–transfer diagram and residuals,} \\ \text{exact or spectral shared-section state,} \\ \text{operation and predictive alignment,} \\ \text{feasible causal interventions,} \\ \text{coalition irreducibility under matched local-predictive nulls,} \\ \text{non-collapse, reconstruction, and replacement identity,} \\ \text{descent across subworkspaces without hidden centralization,} \\ \text{formation provenance and certification leakage control,} \\ \text{goal-conditioned action and certificate-directed repair,} \\ \text{robustness across admissible decompositions and scientific specifications,} \\ \text{identified-set and uniform finite-sample certification} \end{array} \right).$$

The reviewed sources contain many of these coordinates separately and some in pairs. This audit did not locate a prior source that defines their conjunction as one falsifiable workspace certificate.

91.5 Why the conjunction is not arbitrary

Each coordinate excludes a distinct near miss:

coherence	rejects incompatible local states,
predictive alignment	rejects inert structure,
causal efficacy	rejects correlational epiphenomena,
coalition irreducibility	rejects centralized sufficiency,
matched nulls	reject local-marginal explanations,
non-collapse	rejects convergence to a trivial zero state,
replacement identity	rejects task-equivalent but causally different successors,
provenance	rejects scaffold-decoded organization,
repair specificity	rejects passive robustness and open-loop replay,
specification robustness	rejects favorable analytical tuning,
identifiability	rejects conclusions unsupported by the experiment family.

Because the excluded systems differ, the certificate is not equivalent to any one inherited component.

92. Scope of the Originality Claim

92.1 Defensible thesis

The strongest defensible originality statement is:

> Stratified Relational Workspace Theory proposes a unified formal and empirical certificate for persistent operative structures distributed across heterogeneous agents and artifacts. It integrates core–transfer and local-to-global geometry with predictive sufficiency, feasible causal interventions, coalition irreducibility, matched relational nulls, reconstructive identity, formation provenance, certificate-directed self-maintenance, robustness across admissible decompositions and scientific specifications, and identified finite-sample certification. The individual mathematical tools are largely established. The candidate contribution is their noncompensatory integration and the class of near misses it is designed to reject.

This statement is narrower than claiming a new theory of cognition in general. It is also stronger than claiming only a metaphor.

92.2 Claims excluded by the comparison

The theory does not claim:

1. that distributed or extended cognition is newly discovered;
2. that global sections, holonomy, sheaf Laplacians, spectral perturbation, frames, causal states, or viability bounds are new mathematics;
3. that all human–AI collaboration forms a relational workspace;
4. that group performance establishes relational irreducibility;
5. that scaffolded organization is unreal;
6. that endogenous formation establishes agency;
7. that agency establishes consciousness, personhood, or moral status; or
8. that the motivating collaboration has already passed the empirical certificate.

92.3 Claims supported at present

The analysis supports the following claims:

1. One coherent formal language can express exact structure, approximate coherence, causal operation, distributed sufficiency, persistence, identity, formation, and repair.
2. Under its declared assumptions, the stated mathematical theorems follow.
3. The common certificate rejects analytic replicated, centralized, obstructed, externally reloaded, identity-breaking, and replay-based near misses.
4. The motivating collaboration is a plausible observational candidate but remains uncertified.
5. The synthesis yields concrete experiments whose failure would narrow or falsify the stronger relational claims.

92.4 Remaining priority risks

The highest-priority remaining searches are:

- sheaf-theoretic models of team cognition and organizational knowledge;

- distributed sufficient-statistic and Blackwell-order models with explicit coalition nulls;
- causal representation and multi-agent latent-state identity under replacement;
- formal theories of transactive memory with local-to-global topology;
- organizational cybernetics that couples self-maintenance to the preservation of distributed informational organization; and
- post-August-2026 work that combines representational geometry with persistent human–AI or multi-agent agency;
- formal analysis of active shared context graphs, shared group memory, and artifact-centric graph workspaces under coalition and intervention tests.

A specialist review from each adjacent field may uncover precedents not visible through cross-disciplinary keyword search.

92.5 Evidence required for a stronger claim

Before presenting the synthesis as established rather than proposed, the following should be completed:

1. a public, timestamped manuscript and versioned source archive;
2. theorem-level comparison with the closest sheaf, causal-state, group agency, and CCP formulations;
3. independent mathematical review of the nonstandard claims and constants;
4. preregistered benchmark gates and matched null families;
5. at least one controlled human–AI or multi-agent experiment estimating the complete certificate vector; and
6. explicit reporting of failures, unidentified quantities, and decomposition sensitivity.

92.6 Priority conclusion

The literature review narrows rather than weakens the theory. It removes unnecessary novelty claims from the standard mathematics and leaves a clearer research object:

Not a new Laplacian, sheaf, causal state, or controller, but a candidate unified certificate for relational workspaces.

That is the priority position adopted here.

PART XXIV

Theory Hierarchy

93. Levels of Organization

Level 0: Archive

$$\delta_{\mathrm{coh}} \text{ may be low}, \qquad \eta_{\mathrm{causal}} = 0.$$

Level 1: Shared representation

$$d_\varepsilon > 0, \qquad \delta_{\mathrm{coh}} \le \varepsilon.$$

Level 2: Coordinated operational workspace

$$\delta^{\mathcal{I}}_{\mathrm{align}} \le \varepsilon, \qquad \eta^{\nu}_{\mathrm{causal}} > 0.$$

Level 3: Relational workspace

$$k_\varepsilon \ge 2, \qquad \Delta^{\ell}_{\mathrm{joint}} > 0, \qquad \Delta^{\min}_{\mathrm{tr}}(\mathcal{Q}) > 0.$$

Level 4: Persistent relational workspace

$$d_{\mathrm{id}}(W_t, W_{t+\tau}) \le \varepsilon.$$

Level 5: Self-maintaining workspace

$$\eta_{D\to M} > 0, \qquad \eta_{M\to R} > 0, \qquad \eta_{R\to v} > 0, \qquad \Delta^{\mathrm{spec}}_{\mathrm{repair}} > 0, \qquad \sup_t e_t \le e_{\max}.$$

Level 6: Workspace agent

$$\eta_{G\to A} > 0, \qquad \eta_{W\to A} > 0, \qquad \delta_Q \le \varepsilon_Q,$$
$$\mathrm{Reg}_t(\pi) \le 2\varepsilon_Q + \delta_{\mathrm{opt}}, \qquad k_{\mathrm{ctl},\varepsilon} \ge 2, \qquad \mathrm{SelfMaintenance}\,.$$

Orthogonal formation-provenance axis

$$\mathrm{Scaffolded} \longrightarrow \mathrm{AdaptivelyParameterised} \longrightarrow \mathrm{StructurallyAdaptive} \longrightarrow \mathrm{EndogenouslyConstituted}\,.$$

This axis is independent of Levels 0–6. A scaffolded system can instantiate a high-level relational workspace, while an endogenously learned organization can fail every operational workspace gate.

Decomposition-robustness qualifier

Every level is first defined relative to a decomposition $\mathcal{D}$. A family-robust version carries the additional requirement

$$\inf_{\mathcal{D}\in\mathfrak{D}} m_q(W;\mathcal{D}) > 0 \qquad \text{for every gate required at that level.}$$

This is a robustness qualifier rather than a higher agency level. A system can be highly agentive under one fixed interface while remaining decomposition fragile.

Specification-robustness and identification qualifier

Every level is also relative to a complete scientific specification ς. A specification-robust version requires

$$\inf_{\varsigma\in\mathfrak{S}} m_q(W;\varsigma) > 0 \qquad \text{for every gate required at that level.}$$

An identified version additionally requires the sign of each required lower envelope to be constant across the realization set compatible with the available experiment family. An empirical claim must finally subtract the uniform finite-sample error radius before declaring the lower envelope positive. These are epistemic qualifiers, not additional agency levels.

Separate, unresolved axis

$$\text{PhenomenalConsciousness}\,.$$

PART XXV

Unified Formulation

The complete static-to-agentive construction is

$$
\begin{aligned}
&(\Omega, \mathfrak{D}, \mathfrak{S}, \Sigma, \mathcal{D}_{\mathrm{form}}, \mathcal{L}) \\
&\Downarrow \qquad (\mathcal{D}, \widehat{V}, \widehat{E}, \widehat{H}, \widehat{P}, \widehat{B}) \\
&\Downarrow \qquad (G, \{X_i\}, \{T_e\}) \\
&\Downarrow \qquad X_i = C_i \oplus R_i \\
&(\{C_i\}, \{H_e\}, \mathcal{A}, \{\rho_i\}) \\
&\Downarrow \qquad \Gamma(H) = \varprojlim C_i \\
&\Downarrow \qquad L_H = D_H^* D_H \\
&\Downarrow \qquad \Gamma_\varepsilon(H) = \operatorname{ran} \mathbf{1}_{[0,\varepsilon]}(L_H) \\
&\Downarrow \qquad W_t^{\mathrm{str}} \\
&\xrightarrow{\text{alignment}} Z_t = \mathbb{P}(Y_t^+ \in \cdot \mid \mathbf{H}_t) \\
&\Downarrow \qquad \mathrm{SRW}_\Theta(W_t; \mathcal{D}) \\
&\Downarrow \qquad (W_t, G_t, \widehat{Q}_t, M_t, B_t) \\
&\xrightarrow{\text{goal and value use}} (A_t, R_t) \\
&\xrightarrow{\text{causal repair}} e_t \le e_{\max} \\
&\Downarrow \qquad \mathbf{W} = [\mathfrak{R}]_{\text{gauge+causal equivalence}}.
\end{aligned}
$$

Compactly:

scaffold and history → learned organization and boundary
→ local core–transfer realization
→ compatible sections and global descent
→ causal operation and persistent identity
→ cue-controlled goal retention and counterfactual action
→ monitor-mediated repair and bounded maintenance deficit
→ decomposition- and specification-robust identified certification.

The final two arrows are optional extensions rather than consequences of the earlier ones. A relational workspace can exist without being an agent, and an agentive controller can fail to maintain the workspace that realizes it.

PART XXVI

Discussion, Limitations, and Philosophical Consequences

94. Decomposition Relativity

Part XVII turns decomposition relativity from a caveat into a formal part of the certificate. The central lesson is negative and positive at once. No claim of relationality is invariant under arbitrary redescription. An indiscrete merge makes the whole state local, while a secret-sharing split can make a centralized encoder look irreducibly distributed. Yet exact structural content can remain invariant under faithful coarse-graining, and gate margins can remain stable through a declared family of operationally meaningful decompositions.

The resulting position is neither that decomposition is arbitrary nor that one partition is uniquely privileged. Decompositions are constrained by access, causal interfaces, intervention separability, temporal continuity, and explicit shared resources. A certificate established only at one interface is a valid but limited claim. A stronger claim reports the full envelope

$$\left(\widetilde{m}_q, \underline{m}_q, \overline{m}_q, \Delta_q^{\mathrm{dec}}\right)$$

for each required gate, where $\tilde{m}_q$ is the reference margin and the remaining terms summarize its behavior across $\mathfrak{D}$.

This does not eliminate judgment. It makes the judgment inspectable. Analysts can disagree about the admissible family and identify exactly which merge, split, hidden resource, or intervention assumption changes the conclusion. That is preferable to treating a convenient node diagram as if it were part of nature itself.

94.1 Specification relativity and epistemic restraint

Decomposition is only one coordinate of the scientific claim. The downstream target, loss, temporal horizon, intervention family, null panel, and threshold rule also determine what is being certified. A workspace may be operationally real for a month-long theorem-development trajectory yet irrelevant to a second-scale motor target. Those are distinct specifications, not competing answers to one underspecified question.

The framework therefore distinguishes truth under a declared specification from robustness across an admissible family. It also distinguishes robustness inside one fitted realization from identification across all realizations compatible with the available experiments. A positive point estimate at one favorable analysis is the weakest admissible conclusion. A stronger claim requires a positive lower envelope across $\mathfrak{S}$, an identified sign across the compatible realization set, and a finite-sample confidence bound that remains positive after uniform error control.

This hierarchy is intentionally conservative. It does not demand invariance under every imaginable redescription or target. It demands that the scientific universe over which robustness is claimed be declared before certification and that ambiguity outside the reach of the experiment be reported as partial or non-identification rather than converted into certainty.

95. Realism Without Reification

The framework is realist about relational organization but does not posit a free-standing substance between the components. A workspace exists only through a realization, and every change in the workspace must be implemented by a change in participants, artifacts, channels, or their states. The quotient

$$\mathbf{W} = [\mathfrak{R}]_{\simeq}$$

is therefore a mathematical account of identity across realizations, not the name of an additional physical object.

This position allows two claims that are sometimes treated as incompatible. First, the workspace is physically dependent: no realization, no workspace. Second, it is not identical with any particular carrier when several distinct realizations preserve the same structural and causal organization. The same logic is familiar whenever an object is identified by invariants rather than by one material instance. A melody is not a second substance beside its performances, yet it is not identical with one performance. A proof is not separate from every inscription that carries it, yet its identity can survive changes of notation and medium. The analogy is limited, but it clarifies the intended form of realism.

The theory is therefore compatible with physicalism and multiple realization. It rejects both the view that a workspace must be wholly inside one component and the stronger view that the workspace is an independent entity floating outside its carriers. What is real is the organized pattern of compatible representation, transfer, causal use, persistence, and reconstruction.

96. Location and the Sense of "Between"

Asking where a relational workspace is located can invite a category mistake. The local states have ordinary physical locations. A notebook is on a desk, a model state is implemented on hardware, a message is transmitted through a channel, and a person's activity is realized in a biological organism. The workspace as such is not located at an additional point. Its location is distributed in the modest sense that the relations constituting it span several located carriers.

The word "between" is best understood counterfactually rather than spatially. Suppose local marginal states are held fixed while transfer organization is shuffled, delayed, or replaced by a matched null. If the joint predictive or causal capacity changes, then an operative property resided in the organization of the relations rather than in any one local state. Conversely, if a single component or matched storage surrogate reproduces the complete capacity, the case for a relational workspace fails. The relevant question is therefore not whether an invisible object occupies the interval between participants. It is whether the coupling carries indispensable state and supports capacities that disappear when that coupling is altered.

This account also explains why the workspace can be partially present in each component without being fully stored in any of them. Each local chart may encode a view, role, or constraint. The global object is the compatible and causally operative organization of those views. In exact settings this is a global section. In approximate settings it is a stable low-energy subspace together with the dynamic and causal conditions developed above.

97. Continuity, Contribution, and Shared Ownership

Relational persistence does not require an unchanging set of participants. A workspace can survive replacement when the new realization preserves the relevant transfer structure, operation algebra, causal response, reconstructive capacity, and non-collapse margin. This is continuity through

organized transport, not continuity through an immutable token.

The same point applies to artifacts. A handoff document, theorem register, code repository, or experiment log is not merely a passive record when it constrains future reasoning and permits reconstruction after interruption. Such artifacts can become constitutive parts of a workspace realization. Their contribution is causal and testable: remove or corrupt them, then measure what can no longer be reconstructed or done.

Distributedness does not imply equal contribution. One participant may supply long-horizon goals, another formal manipulation, and an artifact precise persistent memory. Nor does a system-level description erase individual authorship or responsibility. The theory describes how capacities are organized; it does not settle legal, ethical, or scholarly questions of credit. Those questions require a separate account of who introduced which ideas, who exercised control, who bore risk, and which contributions were replaceable.

The phrase "shared ownership" should therefore be used carefully. No individual component need own a complete copy of the workspace state, yet individuals can still own artifacts, originate arguments, and retain responsibility for decisions. Relational identity and ordinary ownership are different relations.

98. Agency, Consciousness, and Moral Non-Implications

The agency certificate developed here is functional and domain relative. It concerns persistent goals, interventional value estimates, feasible action control, damage detection, repair, and bounded maintenance within a declared environment. Passing those tests would justify a claim of workspace-level agency in that domain. It would not establish unrestricted autonomy.

No theorem in the framework identifies agency with phenomenal consciousness. A system may maintain a goal, evaluate counterfactual actions, and repair its own organization without there being anything it is like to be that system. The converse also remains open: failure to satisfy this engineering certificate would not prove the absence of consciousness. The framework is designed to prevent functional evidence from silently becoming a metaphysical conclusion.

Moral status is likewise not inferred. Neither coherence, causal efficacy, relational irreducibility, nor self-maintenance is by itself a criterion of moral patienthood. Future arguments connecting these properties to consciousness or welfare would need independent premises and evidence. Until then, the framework can support careful descriptions of organization and agency without treating a collaborative system as a person or dismissing the moral standing of the humans who participate in it.

99. Limitations

The theory is deliberately broad, but its conclusions remain bounded by several limitations.

1. **Admissible-decomposition burden.** Part XVII formalizes faithful coarse-graining and family-robust certification, but the choice of primitive carriers and admissible alternatives remains an empirical and methodological judgment. A narrow family can still overstate robustness.
2. **Admissible-specification burden.** Part XVIII makes the target, loss, horizon, intervention family, null panel, and thresholds explicit. The admissible family $\mathfrak{S}$ still requires scientific judgment. A family chosen after seeing the results can conceal fragility even when every reported analysis is technically correct.

3. **Limited observability and partial identification.** Many practical systems expose messages and outputs rather than complete local states. Transfer maps, predictive states, reconstruction operators, and even the certificate sign may therefore be only partially identified. A point claim is unjustified when both passing and failing realizations remain compatible with the declared experiment.
4. **Linear and finite-dimensional emphasis.** The clearest structural theorems use finite graphs and linear Hilbert-space models. Nonlinear, stochastic, symbolic, and category-valued extensions require additional work.
5. **Gauge and model ambiguity.** Even with abundant data, local coordinates need not be identifiable. Learned transfers are often meaningful only up to a gauge class and relative to a selected model family.
6. **Causal identification burden.** Workspace interventions must be feasible and matched. Observational dependence, activation, or predictive accuracy cannot substitute for interventions when a causal claim is made.
7. **Certification cost.** The complete certificate requires coalition tests, null models, replacement studies, longitudinal interventions, admissible specification sweeps, and identified-set or uniform uncertainty calculations. Exhaustive certification may be impractical in large systems, and adaptive exploration can invalidate nominal error guarantees unless selection and certification are separated or valid selective procedures are used.
8. **Conditional mathematical results.** The theorems establish what follows from stated assumptions. They do not show that naturally occurring collaborations satisfy those assumptions, and several constants may be too conservative for practical estimation.
9. **Open empirical status.** The motivating human–AI collaboration has not been subjected to the complete frozen certificate. It remains a plausible case study and a source of hypotheses, not a demonstrated instance of relational agency.
10. **Open priority boundary.** The literature comparison is broad but cannot prove absolute firstness. Closely related work may exist under other terminology or appear after the review date.

These limitations do not make the framework vacuous. They specify the conditions under which its claims can fail and identify the measurements needed to strengthen them.

100. Open Problems

Several directions are mathematically and empirically immediate.

1. Develop nonlinear and stochastic analogues of the transfer Laplacian, including kernel, manifold, and category-valued realizations.
2. Establish sharp uniform finite-sample estimators for workspace dimension, coalition risk, matched-null gaps, replacement identity, and specification envelopes under adaptive data collection.
3. Characterize the identified set when local states are latent and only messages, artifacts, or actions are observed. Determine the minimum intervention family needed to identify each certificate gate and the complete conjunctive certificate.

4. Extend gluing and persistence to changing graph topology, asynchronous communication, and delayed or lossy transfers.
5. Extend the joint decomposition-and-specification theory to latent shared resources, overlapping component systems, continuous coarse-graining, target families, and data-driven discovery of admissible interface and specification families without invalidating certification.
6. Study the interaction between workspace formation and strategic participants whose local objectives need not agree.
7. Build benchmark suites in which structural, causal, persistence, and repair quantities are known exactly but remain nontrivial to infer.
8. Run preregistered human–AI and multi-agent experiments that estimate the complete certificate rather than isolated performance gains.
9. Compare certificate-directed self-maintenance with established notions of organizational resilience, institutional memory, and collective agency.
10. Clarify which additional evidence, if any, could connect functional workspace predicates to theories of consciousness without assuming the connection in advance.

101. Conclusion

A stratified relational workspace is a temporally persistent, causally efficacious, jointly reconstructible structure represented only partially in several heterogeneous components. Its identity is determined by invariants preserved under admissible transfer, and its operative capacity is relational only when no individual component or matched local-storage null reproduces the relevant function.

The theory gives this definition several independent layers. Exact shared content appears as a global compatible section and, in the invertible case, as a holonomy-fixed core. Approximate shared content appears as a low-energy spectral subspace of the transfer Laplacian. Predictive states and feasible interventions determine whether that structure matters for a declared future. Coalition tests and matched nulls determine whether the capacity is genuinely distributed. Reconstruction frames, temporal transport, and identity profiles determine whether it survives loss and replacement. Descent determines whether local workspaces compose globally. Provenance tests distinguish learned organization from scaffolded answers. Goal, value, damage, and repair interventions determine whether the workspace supports domain-relative agency and self-maintenance.

No one layer can stand in for the others. Coherence can describe an archive. Performance can describe a centralized controller. Persistence can be trivial if all content collapses. A task-equivalent replacement can alter the causal organization. Local compatibility can fail to glue globally. Endogenous-looking structure can be supplied through hidden leakage, and successful recovery can be open-loop replay rather than repair. The value of the framework lies in making these alternatives explicit and testable.

Relationality also cannot be detached from decomposition. Faithful coarse-graining preserves the exact structural core, but coalition counts and local sufficiency change when components are merged. Arbitrary splitting can create a formally irreducible code around a centralized encoder. The strongest claims therefore quantify their margins across an admissible family of interfaces rather than relying on one favorable diagram.

A complete scientific claim must also survive scrutiny of the specification and the experiment. The

strongest form of the certificate reports a positive gatewise lower envelope across a predeclared family of decompositions and scientific specifications, shows that the sign of that envelope is identified across all realizations compatible with the intervention data, and retains a positive margin after uniform finite-sample uncertainty. When these conditions fail, the correct conclusion is specification-specific, fragile, partially identified, or unidentified rather than simply positive or negative.

The central identity principle is therefore

workspace identity = persistent invariant causal organization under admissible transport.

This principle is compatible with changing carriers, representations, and participants, but only when structural, operational, causal, reconstructive, and non-collapse defects remain within declared tolerances. It provides a formal language for studying distributed collaboration without turning that collaboration into a mysterious third substance and without inferring consciousness from organization alone.

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REALIZATION CALCULUS

Reading scope

A self-contained treatment of the eight maintained claim groups, with full written proofs, the later degree-with-sharing formal development, exact calibration families, and explicit counterexamples. The accompanying visual and Lean guides use the same source baseline.

Formal frontier. The archived Lean Block 02 covers the cut theorem, the classical register-degree invariant, and the numerical complete-cover bound. The dual certificates and later claim groups retain written-proof status. Fresh exact computations are recorded separately from the archived kernel checks.

Abstract and reading contract

An arbitrary cut in a classical reversible circuit with restored inputs and workspace yields a complete cover of the accepting set by safe coordinate discrepancies. Tracking products of current registers preserves shared computation and gives an algebraic degree bound of $t + r$ after t Toffoli gates for a product of r registers. Together with a balanced cut and a zero-Toffoli affine-output implementation, this yields

$$\deg f \leq T + 1, \qquad 2\beta^*(f) \leq T + 3.$$

Here β^* optimizes the least degree needed for complete safe coverage over all affine changes of the output. An exact evaluation-span dual supplies finite obstructions to low-degree detection. An unbalanced disjunction separates one-packet immunity from complete coverage, while matching constructions calibrate every function through three inputs.

A second family shows that extracting a homogeneous quadratic component can raise nonlinear cost and destroy support safety. Its separation is exponential in a structural parameter but linear in the actual input length. Orthogonal witness idempotents impose an exponential dimension requirement on a fixed faithful linear multiplicative summary. A weighted witness polynomial avoids cancellation, but computing its trace remains a separate operation; shared variables provide explicit failures of a context-free multiplicative compiler.

All general claims are proved under the stated models. This edition consolidates their mathematical content rather than enlarging the fixed scope to unrestricted circuit hardness. The results establish neither a superpolynomial general circuit lower bound nor a polynomial-time satisfiability evaluator.

One source baseline, three complementary documents

Main paper: definitions, theorems, proofs, calibration families, and limitations.
Visual guide: exact finite examples and Python-generated figures.
Lean guide: actual formal definitions, proof structure, complete 115-declaration register, and reproducibility instructions.

Edition and evidence status

The mathematical baseline is *Komorebi Circuit Obstructions*, local source release v1.2.0, dated 21 September 2026. This research edition incorporates the later Lean Block 02 into the original theorem companion while preserving all original source archives. The controlling register remains C01–C08. Numbered supporting statements are not a count of independent discoveries.

Baseline	`komorebi-circuit-obstructions-v1.2.0`
Source commit	`9e6688e32248926b73950d0560722799b6195ad7`
Current edition	1.0, 7 October 2026
Formal source	115 theorem declarations in eight modules; 40 inherited, 75 added in Block 02
Author	Komorebi P., publication pseudonym

Archived formal checks, 21 September. The source release records a strict fresh-source build, axiom audit, eight module replays, and rejection of eight false controls with Lean 4.34.0-rc2 and bundled Std. The allowed logical dependencies are `propext`, `Classical.choice`, and `Quot.sound`. No Mathlib dependency is needed.

Fresh exact checks, 7 October. The complete `-all` command passed: quick controls, rebuilt one-to-four-input profiles, complete-cover checks, projection checks, witness fingerprints, Block 02 degree checks, and the preserved historical runner. The source seal also passed for 321 sealed source files and 27 sealed evidence files. These counts describe different inventories and are not added together as a theorem or file census.

Fresh Lean attempt, 7 October. The verifier stopped because the `lean` executable is absent. No proof was compiled in this edition's environment. Its saved failure is an environment limitation; the archived formal results remain attached to their exact source revision. The missing compiler is not reported as a new formal pass.

Review. Fresh model-assisted adversarial reviews reconstructed the written arguments and checked selected examples. They share potential model errors and do not constitute independent human review. The review reports retain the discovered boundary and notation corrections. Publication completion does not promote C02's dual or C03–C08 to Lean proofs.

No legal name, affiliation, DOI, public repository destination, or new distribution licence is asserted. The supplied rights record is preserved. This edition is a local publication set.

Evidence vocabulary. *Written* means a supplied argument under explicit hypotheses. *Formal* identifies the exact archived Lean declarations. *Finite* means an executed bounded assertion. *Counterexample* identifies a failed implication. *Conditional* identifies a conclusion with an unimplemented premise.

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CHAPTER 1

Scope, notation, and computational model

Source alignment: Foundations, conventions and F1–F3; Proof Boundary [R1, R4].

1.1 Boolean and algebraic conventions

A Boolean function is a map $f : \{0,1\}^n \to \{0,1\}$. Unless a different field is explicitly stated, polynomials are the unique multilinear representatives in

$$\mathcal{R}_n = \mathbb{F}_2[x_1, \ldots, x_n]/(x_i^2 - x_i : 1 \le i \le n).$$

Addition in this ring is XOR. Multiplication is AND. Boolean OR is written $\vee$ and satisfies $a \vee b = a + b + ab$. A statement $h \le f$ is pointwise and means that h has no positive value at a negative input of f.

The degree of a nonzero function is its multilinear algebraic degree. A nonzero constant has degree zero. For mathematical statements involving a constant-zero function we use $\deg 0 = -\infty$; stored profile tables and Python helpers use their documented sentinels instead. Those implementation conventions do not change the inequalities.

For $S \subseteq [n]$, write $x^S = \prod_{i \in S} x_i$ and let 1_S be the input with support S. A minterm of a monotone function means an inclusion-minimal positive support, not an arbitrary positive input.

Symbol	Meaning
$C(f)$	Minimum fan-in-two AND/OR/NOT gate count, allowing fan-out.
$L(f)$	Minimum total NOT/CNOT/Toffoli count of a restored implementation.
T	Number of Toffoli gates in a specified implementation.
$T_{\mathrm{rest}}(f)$	Minimum such Toffoli count, allowing any finite zero workspace.
$M(f)$	Minimum AND count in an XOR–AND computation, with affine operations uncharged.
I_f	Boolean functions supported only on positive inputs of f.
$\alpha_1(f)$	Minimum degree of one nonzero member of I_f.
$\beta_v(f)$	Minimum degree of a member of I_f that is one at v.
$\beta(f), \beta^*(f)$	Complete-cover degree and its affine-output optimization.
$[P]_2$	Homogeneous degree-two component of the multilinear polynomial P in the specified coordinates.

1.2 The restored classical oracle

A legal NOT changes one target bit. A CNOT adds a control bit to a distinct target. A Toffoli adds the product of two distinct controls to a third distinct wire. Gates have positive controls and are self-inverse. Their composition is a permutation of all register states.

A restored oracle is a circuit satisfying

$$(x, 0^a, y) \longmapsto (x, 0^a, y + f(x)) \tag{1.1}$$

for every logical input x and both initial target values $y \in \{0, 1\}$. The zero-workspace condition is imposed at the boundary, not at intermediate times. The circuit may disturb logical inputs, use output feedback, reuse intermediate values, and make arbitrary reversible excursions before restoring the required registers.

NOT and CNOT are free only when counting T. They are charged in L. This distinction is essential in the projection example. There are no measurements, quantum phases, discarded work registers, or unverified ancilla initializations in this model.

1.3 Comparison with ordinary circuits

Theorem 1.1 (Classical/restored simulation comparison)**.** *For every Boolean function f,*

$$\frac{C(f)}{5} \leq L(f) \leq 6C(f) + 1.$$

Proof. Compute every Boolean gate onto a fresh zero wire. An AND costs one Toffoli. A NOT of an existing signal is obtained with one NOT and one CNOT. An OR is computed as $a + b + ab$, using two CNOTs and one Toffoli. Repeated operands are simplified to a copy, not encoded as an illegal repeated-control gate. Compute the output, copy it to y, and reverse the first computation. This uses at most $6C(f) + 1$ gates and restores every work register.

Conversely, express each reversible gate as an ordinary Boolean update. XOR has a four-gate AND/OR/NOT realization. A Toffoli needs that XOR plus one AND, hence at most five gates. Substitute zero for initial ancillas and $y = 0$, then read the final target. The resulting Boolean circuit has at most $5L(f)$ gates. □

1.4 The degree bound used throughout

Lemma 1.2 (Degree per multiplication)**.** *An XOR–AND computation with t AND gates has output degree at most $t + 1$, allowing constants and arbitrary sharing. The same bound holds for each coordinate of a reversible computation containing t Toffoli gates.*

Proof. Introduce a formal Boolean variable for each AND output. Its defining expression is a product of two affine forms in the logical inputs and earlier gate variables. The final output is affine in the inputs and all gate variables.

Substitute the gate variables in reverse topological order. A multilinear monomial contains the variable being replaced at most once. Replacing it by a quadratic expression raises degree by at most one; multilinear reduction cannot raise it. After t substitutions the degree is at most $t + 1$.

A Toffoli contributes one multiplication to such a straight-line computation, while NOT and CNOT are affine. The argument also applies to a suffix executed backwards because the gates are self-inverse. □

The source identifies this standard degree bound with the XOR–AND background in [S1].

1.5 A degree invariant that retains sharing

The preceding Boolean substitution proof applies to general XOR–AND straight-line computations. The later formal development proves the classical reversible specialization directly on the evolving register state. This route makes the resource accounting visible without unfolding reused registers into duplicated formulas.

Theorem 1.3 (Register-product degree invariant)**.** *Let $s_i(x)$ be the contents of a finite register after t Toffoli gates, with arbitrary intervening NOT and CNOT gates. Initial register contents are affine functions of x. For every finite list L of r register indices, including repeated indices,*

$$\deg\left(\prod_{i\in L} s_i(x)\right) \le t + r.$$

The zero-function convention is as above. In the formal source the conclusion is the concrete representation certificate `DegreeLE`*.*

Proof. Initially a product of r affine functions has degree at most r. Suppose a gate toggles register u by a product of q controls, with $q = 0, 1, 2$ for NOT, CNOT, and Toffoli respectively. If u does not occur in L, the product is unchanged. If it occurs, Boolean idempotence removes all its repetitions. Let R be the remaining list, so $|R| + 1 \le r$. The updated product is

$$s_u \prod_{i\in R} s_i + \left(\prod_{j\in \text{controls}} s_j\right)\prod_{i\in R} s_i.$$

By the old invariant, these two summands have degrees at most $t + 1 + |R|$ and $t + q + |R|$. The latter is at most $t + r + \max\{q - 1, 0\}$. NOT and CNOT therefore consume no degree budget; a Toffoli consumes at most one. Induction along the actual gate list proves the claim. The empty list is the constant-one product and is also covered. □

For $r = 1$ this recovers the coordinate bound. The initial target can be fixed to either boundary bit while the inputs vary. Temporary input disturbance, dirty intermediate workspace, output controls, and repeated use of earlier products are all allowed. The formal locators are `workspaceDegree_run` and `canonical_run_coordinate_degree`.

The representation is part of the theorem

`DegreeLE(d,f)` is tied by proved equivalences to an explicit finite XOR of monomials with length at most d. It is not an assumed degree oracle. Canonical algebraic-normal-form extraction and a separate generic XOR–AND simulation compiler are outside this formal block.

CHAPTER 2

Exact cut comparison and safe discrepancies

MAINTAINED CLAIM C01

Source alignment: Foundations F2; Claim Register C01 [R1, R3].

Theorem 2.1 (Cut-comparison identity and complete coverage)**.** *Let a restored oracle be written $\mathcal{C} = Q \circ P$, and put $R = Q^{-1}$. Suppress zero ancillas in boundary arguments and define*

$$A(x) = P(x,0), \qquad B(x) = R(x,0), \qquad D(x) = P(x,1) + P(x,0).$$

These are vectors indexed by all registers. Then

$$A(x) + B(x) = f(x)D(x), \qquad D(x) \neq 0. \tag{2.1}$$

Consequently, the coordinate discrepancies $H_i = A_i + B_i$ satisfy

$$H_i \leq f \quad \text{for every } i, \qquad \bigvee_i H_i = f. \tag{2.2}$$

Proof. For a boundary state with target z, correctness of the full circuit gives

$$R(x,z) = P(x, z + f(x)).$$

When $f(x) = 0$, this says $B(x) = A(x)$. When $f(x) = 1$, it says $B(x) = P(x,1)$, and the discrepancy is $D(x)$. Both cases are summarized by (2.1).

The prefix P is a permutation. Its images of the two distinct boundary states $(x,0)$ and $(x,1)$ are different, so $D(x)$ is a nonzero vector.

If $f(x) = 0$, every discrepancy is zero. If $f(x) = 1$, at least one discrepancy is one. These are precisely the safety and complete-coverage assertions in (2.2). □

2.1 What the comparison does and does not require

The suffix need not reverse the prefix. Output feedback and a target direction distributed among many registers are permitted. Only boundary correctness and reversibility are used. Equivalently,

$$f(x) = 1 \quad \Longleftrightarrow \quad P(x,0) \neq Q^{-1}(x,0).$$

This is a necessary representation, not a circuit lower bound from packet count. Expanded monomial counts, individual degrees, and algebra dimensions remain distinct from the shared cost of constructing the complete family. The declared finite cut checks are reported in Appendix B.

CHAPTER 3

Complete safe-packet degree

MAINTAINED CLAIM C02

Source alignment: Theory §1 and Foundations F4 [R2, R1].

3.1 Pointwise coverage rather than one easy packet

Let

$$I_f = \{h \in \mathcal{R}_n : h(1+f) = 0\}.$$

This is the ideal of functions supported only on the accepting set of f. For nonzero f, the one-packet parameter is

$$\alpha_1(f) = \min\{\deg h : 0 \neq h \in I_f\}.$$

It asks for one safe nonzero polynomial, which may detect only an easy portion of the accepting set. One-sided polynomial immunity is established terminology in this setting; see the background reference [S3].

For a positive input v, define instead

$$\beta_v(f) = \min\{\deg h : h \in I_f,\ h(v) = 1\}, \qquad \beta(f) = \max_{v:f(v)=1} \beta_v(f).$$

Set $\beta(0) = 0$. Thus $\beta(f)$ is the degree needed before safe polynomials collectively detect *every* positive input. Finally, put

$$\beta^*(f) = \max_{\ell \in \mathrm{Aff}_n} \beta(f+\ell),$$

where Aff_n is the set of affine Boolean functions on the logical inputs.

Proposition 3.1 (Basic size bounds)**.** *For nonzero f on n inputs,*

$$\alpha_1(f) \leq \beta(f) \leq n.$$

If f has at least two positive inputs, then $\beta(f) \leq n-1$. In particular $\beta^*(f) \leq n$.

Proof. Any polynomial detecting a specified positive input is nonzero, giving the first inequality. For $v \in \{0,1\}^n$, its point indicator is

$$\delta_v(x) = \prod_{i:v_i=1} x_i \prod_{i:v_i=0} (1+x_i).$$

If $f(v) = 1$, then $\delta_v \in I_f$ and has degree n, proving the upper bound.

If v and w are distinct positive inputs, $\delta_v + \delta_w$ is safe and equals one at v. The degree-n coefficients of its two summands are both one, so they cancel. Its degree is at most $n-1$. Do this for each positive v. The last assertion applies the general bound to every affine adjustment, with the stated zero convention. □

Proposition 3.2 (Affine input invariance)**.** *Invertible affine changes of logical input coordinates preserve α_1, β, and β^*, with the pointwise parameters transported by the corresponding bijection of inputs.*

Proof. Composition with an affine map cannot increase multilinear degree. Composition with its affine inverse proves equality of degrees. Such composition bijects the accepting and rejecting inputs, hence maps safe polynomials bijectively to safe polynomials and preserves their pointwise detecting property. It also permutes the affine output masks used in the definition of β^*. □

3.2 The restored lower bound

Theorem 3.3 (Affine-optimized complete-cover bound)**.** *Every restored implementation of f with T Toffoli gates satisfies*

$$T \geq \max\{0,\ \deg f - 1,\ 2\beta^*(f) - 3\}. \tag{3.1}$$

In particular,

$$T \geq \max\{0,\ \deg f - 1,\ 2\beta(f) - 3,\ 2\beta(1+f) - 3\}.$$

Proof. Cut the implementation after $\lfloor T/2 \rfloor$ Toffolis. The forward prefix and the inverse suffix use at most $\lceil T/2 \rceil$ Toffolis each. By Lemma 1.2, each of their coordinates, and therefore each discrepancy H_i, has degree at most

$$\left\lceil \frac{T}{2} \right\rceil + 1.$$

Theorem 2.1 gives $H_i \leq f$ and a complete cover of the positive inputs. Every positive v is therefore detected by a safe polynomial within this degree bound, so

$$\beta(f) \leq \left\lceil \frac{T}{2} \right\rceil + 1.$$

Integer rounding gives $T \geq 2\beta(f) - 3$.

At the end of a restored oracle the logical inputs are available unchanged. CNOTs from those inputs and, when needed, a target NOT append any affine ℓ to the target at no Toffoli cost. Apply the same inequality to each $f + \ell$ and maximize. Finally, reading the target with initial value zero computes f using T multiplications, so Lemma 1.2 gives $T \geq \deg f - 1$. □

Scope of the lower bound

The circuit may use output feedback and arbitrary dirty intermediate states, but it must satisfy the restored classical oracle convention. The inequality does not apply automatically to a model with discarded garbage or measurement. Because $\beta^*(f) \leq n$, this parameter alone cannot prove a superpolynomial lower bound in the actual input length.

3.3 Exact evaluation-span certificates

For an integer $d \geq 0$, form the evaluation vector

$$v_d(x) = \left(x^S\right)_{S \subseteq [n],\ |S| \leq d},$$

and write $U = f^{-1}(0)$.

Theorem 3.4 (Dual certificate for failure of safe detection)**.** *For every positive input v,*

$$\beta_v(f) > d \quad \Longleftrightarrow \quad v_d(v) \in \operatorname{span}_{\mathbb{F}_2}\{v_d(u) : u \in U\}. \tag{3.2}$$

Proof. A degree-at-most-d polynomial is a coefficient covector on the evaluation space. It is safe exactly when it annihilates every negative evaluation vector, equivalently their span. Such a covector can equal one on $v_d(v)$ exactly when $v_d(v)$ lies outside that span. This is finite-dimensional linear duality over $\mathbb{F}_2$. □

Corollary 3.5 (Finite support of a dual witness)**.** *When the right side of* (3.2) *holds, there is $W \subseteq U$ with*

$$v_d(v) = \sum_{u \in W} v_d(u), \qquad |W| \le \sum_{j=0}^{\min(d,n)} \binom{n}{j}.$$

The cardinality $|W|$ is odd.

Proof. Choose a basis of the span from negative evaluation vectors and express $v_d(v)$ in that basis. Over $\mathbb{F}_2$, the nonzero coefficients select a subset W of size at most the ambient dimension. The constant-monomial coordinate of every evaluation vector is one; comparing it on both sides shows that $|W|$ is odd. □

Such a relation is replayable by exact linear arithmetic. It does not provide an efficient way to enumerate all negatives of an arbitrary SAT verifier. The theorem assumes that the negative vectors or their proved structure are available.

3.4 Monotone specialization

Theorem 3.6 (Safe monomial ideal for a monotone function)**.** *If f is monotone, I_f has a monomial basis indexed by the positive supports of f. For each positive v,*

$$\beta_v(f) = \min\{|S| : S \subseteq \operatorname{supp}(v),\ f(1_S) = 1\}.$$

Consequently, for nonzero f,

$$\alpha_1(f) = \textit{minimum minterm size}, \qquad \beta(f) = \textit{maximum minterm size}.$$

Proof. For any Boolean polynomial h, its coefficient on x^S is

$$[x^S]h = \sum_{T \subseteq S} h(1_T)$$

over $\mathbb{F}_2$. If S is negative for monotone f, every subset T is negative. A safe h vanishes at each 1_T, so its x^S coefficient is zero. Conversely, a monomial on a positive support is safe by monotonicity. This proves the basis assertion.

If a safe h satisfies $h(v) = 1$, at least one of its nonzero monomials is supported inside $\operatorname{supp}(v)$. That support is positive and has size at most $\deg h$. Conversely, a smallest such positive-support monomial detects v. This proves the pointwise formula. Taking the appropriate minima and maxima gives the minterm statements: a minimal positive support cannot contain a smaller positive one, and every positive input contains a minterm. □

Corollary 3.7 (Exact monomial oracle cost)**.** *For $k \geq 2$,*

$$T_{\text{rest}}(x_1 x_2 \cdots x_k) = 2k - 3.$$

Proof. The unique positive input of the k-variable monomial needs a safe polynomial of degree k. The complete-cover lower bound gives $2k - 3$. For the matching construction, successively compute $x_1 x_2$, then $x_1 x_2 x_3$, and so on up to a $(k-1)$-fold product on clean work wires. Toggle the output using that product and x_k, then uncompute the $k-2$ work products. The count is $(k-2) + 1 + (k-2) = 2k - 3$. For $k = 2$, a single Toffoli suffices. □

CHAPTER 4

Calibration families and exact small oracles

MAINTAINED CLAIM C03 and C04

Source alignment: Theory §2 [R2]; baseline `completeness.json`.

4.1 An unbalanced disjunction

For $k \geq 2$ and $n = k + 2$, define

$$B = z_1 \cdots z_k, \qquad F_k(a, b, z) = (ab) \vee B = ab + B + abB, \qquad G_k = F_k + a.$$

This is a deliberately easy family with two differently sized accepting regions. It separates the cost of finding one safe packet from the degree needed for complete coverage.

Proposition 4.1 (Raw calibration parameters)**.** *For this family,*

$$\alpha_1(F_k) = 2, \qquad \beta(F_k) = k, \qquad \deg F_k = k + 2.$$

Proof. The minterms have supports $\{a, b\}$ and $\{z_1, \ldots, z_k\}$. The first two assertions follow from Theorem 3.6. The monomial abB has degree $k + 2$ and does not cancel with either lower-degree term in the displayed polynomial. □

Theorem 4.2 (Affine optimization of the one-packet parameter)**.** *For every $k \geq 2$,*

$$\max_{\ell \in \mathrm{Aff}_n} \alpha_1(F_k + \ell) = 3.$$

Proof. First restrict to the face $z_1 = 0$. There $F_k + \ell$ becomes $ab + \ell|_{z_1=0}$, a nonzero quadratic. If the restricted affine term contains a remaining z variable, set $a = 0$ and set the remaining affine form to one. This gives an accepting affine subspace of codimension at most two inside the face. If it contains no remaining z variable, fix a, b to an accepting pair instead. The indicator of the resulting affine subspace in the full input space has degree at most three and is safe. Thus every affine adjustment has a nonzero safe polynomial of degree at most three.

For equality, take $\ell = a$. The degree-three polynomial

$$a(1 + b)(1 + z_1)$$

is safe for G_k. Suppose a safe h of degree at most two existed. On $a = 0$, acceptance requires $B = 1$. For $k \geq 3$, the monotone-ideal argument forces $h|_{a=0} = 0$, hence $h = aL$ with L affine in the remaining variables. The negative face $a = b = 1$ forces $L(1, z) = 0$, so $L = c(1 + b)$. The negative point $a = 1, b = 0, z = (1, \ldots, 1)$ then forces $c = 0$.

When $k = 2$, the restriction can initially be $h|_{a=0} = cB$. Write $h = cB + aL$. On the same negative face $a = b = 1$, the quadratic term cB cannot be cancelled by affine L, so $c = 0$. The previous argument then applies. There is no nonzero safe polynomial of degree at most two, proving equality. □

Theorem 4.3 (Affine optimization of complete coverage)**.** *For every $k \geq 2$,*

$$\beta^*(F_k) = k + 1.$$

Proof. Restrict $G_k = F_k + a$ to $b = 1$. Then

$$G_k|_{b=1} = (1 + a)B.$$

This is the indicator of one point on $k + 1$ variables. At the positive input $a = 0, b = 1, z = (1, \ldots, 1)$, a safe detecting polynomial must restrict to that singleton indicator and therefore have degree at least $k + 1$. Thus $\beta(G_k) \geq k + 1$.

For the upper bound, Theorem 4.2 shows that no $F_k + \ell$ is a singleton indicator: a singleton would have one-packet degree $n = k + 2 > 3$. These functions are also nonzero because F_k is not affine. Each has at least two positive inputs, so Proposition 3.1 gives $\beta(F_k + \ell) \leq n - 1 = k + 1$. □

Corollary 4.4 (Restored cost interval for the calibration family)**.** *For $n = k + 2$,*

$$2n - 5 \leq T_{\text{rest}}(F_k) \leq 2n - 3.$$

Proof. The lower bound follows from Theorems 3.3 and 4.3. For the upper bound, compute $A = ab$ with one multiplication and B using $k - 1$ multiplications. Toggle the output by $A + B + AB$, requiring one output Toffoli and affine additions. Uncompute A and B. The total Toffoli count is $k + 1 + k = 2k + 1 = 2n - 3$. □

The preceding combined bound $\max\{0, \deg F_k - 1, 2 \max_\ell \alpha_1(F_k + \ell) - 3\}$ gives $n - 1$ for this family; its immunity-only term is the constant 3. The complete-cover bound gives $2n - 5$. The stronger estimate can also be recognized by restriction to $b = 1$ followed by a free affine output adjustment. No exact optimum is claimed when the displayed endpoints differ.

Inputs n	$\max_\ell \alpha_1$	β^*	One-packet bound	Complete-cover bound
4	3	3	3	3
5	3	4	4	5
6	3	5	5	7
7	3	6	6	9
8	3	7	7	11
9	3	8	8	13

The formulas are proved for all $k \geq 2$. The source evaluates the designated family truth tables through nine inputs, but checks every affine mask only through eight. The ninth-input affine maximum is justified by the proof, not by a stored exhaustive mask search.

4.2 An explicit moment certificate

On the face $b = 1$, every point except $v = (a = 0, z = (1, \dots, 1))$ is negative for G_k. A polynomial of degree at most k sums to zero over the $(k+1)$-dimensional cube: each monomial omits some coordinate and contributes an even number of ones. Therefore

$$v_k(v) = \sum_{u \neq v \text{ on the face}} v_k(u).$$

This is the certificate from Theorem 3.4, proving that no safe degree-k polynomial detects v.

The finite four-input computation found $\max_\ell \alpha_1(f + \ell) = \beta^*(f)$ for every function in that domain. The family above disproves extrapolation of that equality: it already fails at five inputs, with values three and four, and the difference grows without bound.

4.3 All functions on at most three inputs

Theorem 4.5 (Exact small restored-oracle calibration)**.** *For every Boolean function on one, two, or three logical inputs, the bound in Theorem 3.3 is attained by an explicit restored implementation. The optimum Toffoli counts are zero for affine functions, one for nonaffine quadratics, and three for three-input cubics.*

Proof. An affine function is added to the target using only NOT and CNOT gates, so its optimum is zero.

A nonaffine quadratic on at most three variables has an alternating polar matrix of rank two. In suitable invertible linear coordinates, its quadratic part is a product of two linear forms; the remaining part is affine. A single Toffoli, together with affine preparation and restoration, computes the oracle. The degree bound requires at least one.

A cubic on three variables has the form

$$x_1 x_2 x_3 + q(x) + \ell(x),$$

with q quadratic and ℓ affine. Translating $x_i \mapsto x_i + c_i$ adjusts independently the coefficients of $x_j x_k$, so all quadratic terms can be cancelled. The resulting function is a three-variable monomial plus an affine function. Affine input changes and affine output additions cost no Toffolis. Corollary 3.7 gives both the lower bound three and a matching restored construction. □

There are $4 + 16 + 256 = 276$ functions over the three input sizes. The distribution is

Optimum Toffolis	0	1	3
Number of functions	28	120	128.

The source constructs and checks the actual circuits. Optimality is obtained from the written lower bounds and matching constructions, not from exhaustive enumeration of all reversible circuits.

CHAPTER 5

Homogeneous projection: cost and safety

MAINTAINED CLAIM C05

Source alignment: Theory §3 [R2]; baseline `projection.json`.

This chapter separates the nonlinear cost of a complete function from the nonlinear cost of one extracted homogeneous component. It also shows that an algebraic projection need not preserve safe support.

5.1 Two supporting lower bounds

Lemma 5.1 (Affine subspace after multiplication restrictions). *If a Boolean function on n inputs is computed with t AND gates in an XOR–AND circuit, there is an affine subspace of dimension at least $n - t$ on which the output is affine.*

Proof. Process the AND gates in topological order while maintaining an affine subspace on which all preceding product outputs are affine. The operands of the next gate are then affine on the current subspace. If an operand is constant, their product is already affine. Otherwise impose one nonempty affine equation setting that operand to zero. This loses at most one dimension and makes the product affine. Earlier affine outputs stay affine under restriction. After at most t restrictions the output is affine on a subspace of dimension at least $n - t$. □

This is the affine-restriction method used in the background literature on multiplicative complexity [S2]. The proof is reproduced here in the scalar convention needed below.

Proposition 5.2 (Exact cost of a nondegenerate paired quadratic). *For even n, let*

$$Q(x) = x_1x_2 + x_3x_4 + \cdots + x_{n-1}x_n.$$

Then

$$M(Q) = T_{\mathrm{rest}}(Q) = n/2.$$

Proof. The polar bilinear form of Q is the full-rank alternating form made of $n/2$ disjoint 2×2 blocks. If Q is affine on an affine subspace with direction space U, its second differences vanish on U, so U is totally isotropic. Nondegeneracy gives $\dim U \le n/2$.

Lemma 5.1 therefore implies $n - t \le n/2$, or $t \ge n/2$, for any XOR–AND computation. Every restored Toffoli implementation gives such a computation by fixing its ancillary inputs and initial target, so the same lower bound applies to T_{rest}. Directly add the $n/2$ disjoint pair products to the target with one Toffoli each. This attains both bounds and requires no nonlinear work register. □

5.2 A compact whole function

Let $m \geq 2$, let $a = (1, \ldots, 1) \in \mathbb{F}_2^m$, and introduce one logical bit x_v for each

$$v \in V = \mathbb{F}_2^m \setminus \{0, a\}.$$

There are $n = 2^m - 2$ such bits. Define

$$L_j(x) = \sum_{v : v_j = 1} x_v, \qquad P_m(x) = \prod_{j=1}^{m} L_j(x).$$

Theorem 5.3 (Exact nonlinear cost of the whole function)**.** *For the family above,*

$$M(P_m) = m - 1, \qquad T_{\mathrm{rest}}(P_m) = 2m - 3.$$

Proof. The linear map $x \mapsto (L_1(x), \ldots, L_m(x))$ has rank m, since its columns include the standard basis vectors. Extend these forms to an invertible system of linear coordinates. In those coordinates P_m is the product of the first m variables, with the other $n - m$ variables unused.

Degree is invariant under an invertible affine change, so $\deg P_m = m$ and Lemma 1.2 gives $M(P_m) \geq m - 1$. A product chain attains it.

For the restored lower bound, the safe ideal of the product in these coordinates consists of polynomials divisible by the m-variable monomial. Any nonzero safe polynomial has degree at least m. Theorem 3.3 gives $2m - 3$. Compute the m parities onto clean registers by CNOT, apply the optimal monomial oracle there, and uncompute the parities. No additional Toffolis are needed. □

5.3 Extracting the quadratic grade

Theorem 5.4 (Exact cost of the extracted grade)**.** *Let $Q_m = [P_m]_2$ denote the homogeneous quadratic part in the original x_v coordinates. Then*

$$Q_m = \sum_{\{v, v+a\} \subseteq V} x_v x_{v+a},$$

where every unordered pair is used once, and

$$M(Q_m) = T_{\mathrm{rest}}(Q_m) = n/2 = 2^{m-1} - 1.$$

Proof. The excluded columns give $P_m(0) = P_m(e_v) = 0$. For distinct u, v, Boolean coefficient inversion therefore gives

$$[x_u x_v] P_m = P_m(e_u + e_v) = [u + v = a].$$

The map $v \mapsto v + a$ is a fixed-point-free involution of V, so the displayed terms form $n/2$ disjoint pairs. Proposition 5.2 supplies the exact costs. □

Counterexample 5.5 (Projection does not preserve safe support)**.** *The quadratic extraction $[P_m]_2$ need not be supported on the positive set of P_m.*

Proof. Take $m = 3$, identify the columns with the integers $1, \ldots, 6$ using their three-bit expansions, and set $x_1 = x_3 = x_4 = 1$ with all other inputs zero. Their column sum is $1 + 3 + 4 = 6$ under bitwise XOR, not $a = 7$, so $P_3 = 0$. Among the complementary pairs $(1, 6), (2, 5), (3, 4)$, exactly $(3, 4)$ has both inputs one. Thus $Q_3 = 1$. In the source ordering this is the logical assignment encoded by integer 13. □

m	Actual inputs n	$T_{\text{rest}}(P_m)$	$T_{\text{rest}}(Q_m)$	Constructed total gates for P_m
2	2	1	1	5
3	6	3	3	21
4	14	5	7	61
6	62	9	31	381
8	254	13	127	2,045
10	1,022	17	511	10,237

These are exact cost formulas. The large-input entries were not obtained by searching for optimal circuits. The full construction uses

$$2m(2^{m-1} - 1)$$

CNOTs as well as $2m - 3$ Toffolis, giving total count $O(n \log n)$. The projected quadratic has $n/2$ total gates.

What the counterexample excludes

The nonlinear projection cost grows exponentially in m, but only linearly in actual input length n. The example does not give an exponential separation in ordinary total gate count. It rules out treating this grade extraction as having overhead bounded only by a polynomial in the original Toffoli count, independently of linear and input-size resources. It also shows that homogeneous extraction may destroy safety. It does not exclude a bound polynomial in $n + T$ or every possible graded invariant.

The source checks all pair coefficients and exact ranks for $2 \leq m \leq 10$. Full truth tables and complete restored evaluations are restricted to $m = 2, 3, 4$. At $m = 4$, the quadratic accepts 7,616 inputs that the whole function rejects; that count is finite evidence, distinct from the explicit universal mechanism above.

CHAPTER 6

Faithful linear gluing algebras

MAINTAINED CLAIM C06

Source alignment: Theory §4 [R2].

This chapter states an exponential dimension result for a specified summary model. The qualifications are part of the theorem's meaning: a lower bound on an algebra's dimension is not automatically a lower bound on the length of a circuit describing one of its elements.

6.1 The witness algebra

Let K be a field and let $W = \{0,1\}^m$. The full witness algebra is

$$\mathcal{B} = K^W,$$

with pointwise addition and multiplication. For each witness v, let δ_v be its indicator function. These form a basis of $\mathcal{B}$ and satisfy

$$\delta_v^2 = \delta_v, \qquad \delta_v \delta_u = 0 \quad (u \neq v), \qquad \sum_{v \in W} \delta_v = 1.$$

In characteristic two, this is equivalently the Boolean quotient algebra

$$K[w_1, \ldots, w_m]/(w_i^2 - w_i : 1 \leq i \leq m).$$

Theorem 6.1 (Dimension of a faithful multiplicative summary). *Let $\Phi : \mathcal{B} \to A$ be a K-linear multiplicative map into a K-algebra. Suppose*

$$\Phi(\delta_v) \neq 0 \qquad \text{for every } v \in W.$$

Then

$$\dim_K A \geq 2^m.$$

No unital hypothesis on Φ is required.

Proof. Write $e_v = \Phi(\delta_v)$. Multiplicativity makes these nonzero orthogonal idempotents. If

$$\sum_{v \in W} c_v e_v = 0,$$

multiply the equality by e_u. Orthogonality annihilates every term except $c_u e_u$, which must be zero. Since K is a field and $e_u \neq 0$, $c_u = 0$. This holds for every u, so the 2^m vectors e_v are linearly independent. □

Corollary 6.2 (Scalar unital summaries are evaluations). *Every unital K-linear multiplicative map $\Phi : K^W \to K$ is evaluation at exactly one witness.*

Proof. An idempotent scalar in a field is either zero or one. Orthogonality allows at most one $\Phi(\delta_v)$ to be one, and preservation of the unit requires exactly one. If it occurs at v_0, linearity gives

$$\Phi(g) = \Phi\left(\sum_v g(v)\delta_v\right) = g(v_0).$$

□

For $m \geq 1$, such a scalar summary loses at least one singleton witness other than v_0. At $m = 0$, the unique empty witness is preserved by the identity map. The dimension theorem is an elementary product-algebra fact, not a claim of new algebra or a universal computational lower bound.

6.2 The quantified model boundary

The same map Φ here serves *all* predicates on a fixed witness space. It is linear, multiplicative, and retains every primitive witness indicator. Under exactly these assumptions, a polynomial-dimensional summary is impossible as m grows.

An arbitrary algorithm is not required to satisfy these hypotheses. It may use nonlinear encodings, syntax-dependent operations, changing spaces, or succinct descriptions inside an exponentially large algebra. A description of the algebra and a description of one relevant element can be much shorter than a full coordinate listing.

No dimension-to-circuit inference

The theorem excludes a fixed faithful linear multiplicative compression of the full witness algebra into fewer than 2^m dimensions. It does not exclude every polynomial-time decision procedure, does not prove $P \neq NP$, and does not justify treating the algebra's dimension as the circuit cost of an individual decision function.

The finite scalar controls enumerate 256 linear maps from a two-point witness algebra to $\mathrm{GF}(16)$, 65,536 maps for four points, and 256 maps from an eight-point algebra to $\mathbb{F}_2$. The respective counts of unital algebra homomorphisms are two, four, and eight, matching the evaluation classification. The all-size result rests on the proof above.

CHAPTER 7

Weighted witness fingerprints

MAINTAINED CLAIM C07

Source alignment: Theory §5 [R2]; baseline `fingerprint.json`.

7.1 A polynomial that distinguishes witnesses

Let $R : \{0,1\}^m \to \{0,1\}$ be a Boolean verifier, and let K be a characteristic-two field. Define the formal polynomial

$$P_R(z_1, \ldots, z_m) = \sum_{w \in \{0,1\}^m} R(w) \prod_{i=1}^{m} z_i^{w_i} \in K[z_1, \ldots, z_m].$$

The variables z_i in this expression are ordinary polynomial variables, not reduced by $z_i^2 = z_i$.

Theorem 7.1 (Exact witness fingerprint)**.** *The polynomial P_R is identically zero if and only if R has no accepting witness. Its total degree is at most m.*

Proof. Different Boolean witnesses have different exponent vectors, so their monomials are distinct. Each accepting witness contributes coefficient one to its own monomial and cannot cancel another witness's term. The polynomial is zero exactly when there are no such terms. Each monomial uses at most m variables. □

This avoids the cancellation of an unweighted parity count. It does not, by itself, give a small circuit for evaluating the sum.

7.2 Exact top-coefficient extraction

Lemma 7.2 (Top coefficient as a Boolean-cube sum)**.** *Let $G(w)$ be a multilinear polynomial over a characteristic-two field. Then*

$$[w_1 \cdots w_m]G = \sum_{v \in \{0,1\}^m} G(v).$$

Proof. It is enough to check a monomial. A monomial of proper support is one on an even number $2^{m-|S|}$ of Boolean inputs, so its cube sum is zero in characteristic two. The full-support monomial is one at exactly one input and has cube sum one. Linearity gives the identity. □

Theorem 7.3 (Weighted top-coefficient identity)**.** *Let Top_w extract the coefficient of $w_1 \cdots w_m$ after reduction in the Boolean w algebra. Then*

$$P_R(z) = \mathrm{Top}_w \left[R(w) \prod_{i=1}^{m} (1 + (1 + z_i)w_i) \right]. \tag{7.1}$$

Proof. At $w_i = 0$, the ith factor is one. At $w_i = 1$, it is z_i. Lemma 7.2 therefore converts the right side to exactly the weighted sum defining P_R. □

The expression inside Top_w has a circuit with $O(|R| + m)$ operations when the verifier has circuit size $|R|$. The cost of applying Top_w to that succinct expression is not bounded by writing down the operator. The source evaluator explicitly stores 2^m coefficients and remains exponential in witness length.

7.3 The conditional algorithmic consequence

Lemma 7.4 (Finite-field root bound)**.** *If a nonzero polynomial $P \in K[z_1, \ldots, z_m]$ has total degree at most d, where K is finite, then*

$$\Pr_{z \text{ uniform in } K^m}[P(z) = 0] \leq \frac{d}{|K|}.$$

Proof. Induct on the number of variables. For one variable this is the elementary root bound. In the inductive step, write

$$P = \sum_{j=0}^{r} P_j(z_1, \ldots, z_{m-1}) z_m^j, \qquad P_r \neq 0.$$

The leading coefficient has total degree at most $d - r$, so it vanishes at a random first $(m-1)$-tuple with probability at most $(d-r)/|K|$. Otherwise the remaining univariate polynomial is nonzero with at most r roots. The union bound gives $(d - r + r)/|K| = d/|K|$. □

Proposition 7.5 (Conditional randomized satisfiability test)**.** *Suppose the weighted expression in (7.1) can be evaluated exactly and uniformly in time polynomial in the verifier description and witness length, over explicitly represented characteristic-two fields of polynomial size. Then choosing $|K| \geq 4m$ gives a one-sided randomized polynomial-time satisfiability test. It never accepts an unsatisfiable verifier and accepts a satisfiable one with probability at least $3/4$ per evaluation.*

Proof. Choose a uniformly random $z \in K^m$, evaluate $P_R(z)$ under the assumed algorithm, and accept when it is nonzero. If no witness accepts, Theorem 7.1 makes the polynomial zero identically, so there is no false acceptance. Otherwise Lemma 7.4 bounds the rejection probability by $m/|K| \leq 1/4$. A field of polynomial size uses $O(\log m)$ bits per element, so the random input has $O(m \log m)$ bits. Under the explicit efficient-field and evaluator hypotheses, the procedure has polynomial running time. □

Conditional means the evaluator is still missing

The proposition assumes a uniform efficient evaluator that has not been constructed here. The implemented evaluator is exponential. Even with the assumed evaluator, the stated conclusion is randomized with one-sided error, not a deterministic proof of $P = NP$.

The exact tests use $\mathrm{GF}(16) = \mathbb{F}_2[t]/(t^4 + t + 1)$. They evaluate every specified verifier through three witness bits at every field point, while the independent top-coefficient check uses all field points through two witness bits and the declared smaller coordinate grid at three bits. These domains must not be conflated.

CHAPTER 8

The overlap obstruction to composition

MAINTAINED CLAIM C08

Source alignment: Theory §5, overlap subsection [R2].

The weighted fingerprint is compatible with products on disjoint witness coordinates. The same formula fails when the factors share a coordinate. The distinction isolates a concrete compatibility obligation, rather than asserting that all compression is impossible.

Proposition 8.1 (Disjoint-variable factorization)**.** *If predicates $A(u)$ and $B(v)$ use disjoint witness-variable sets, then their weighted fingerprints satisfy*

$$P_{A\wedge B}(z_u, z_v) = P_A(z_u)P_B(z_v).$$

Proof. Expand the left side as a sum over the Cartesian product of witnesses:

$$\sum_{u,v} A(u)B(v)z_u^u z_v^v = \left(\sum_u A(u)z_u^u\right)\left(\sum_v B(v)z_v^v\right).$$

Disjointness makes the exponent data independent and the factorization exact. □

Counterexample 8.2 (Shared-variable multiplication fails)**.** *For a single witness bit, define* $\mathrm{Tr}_z(A) = A(0) + zA(1)$*. Then in the Boolean w algebra,*

$$\mathrm{Tr}_z(w) = z, \qquad \mathrm{Tr}_z(1+w) = 1,$$

but

$$\mathrm{Tr}_z\big(w(1+w)\big) = 0 \neq z = \mathrm{Tr}_z(w)\,\mathrm{Tr}_z(1+w)$$

whenever $z \neq 0$. Also,

$$\mathrm{Tr}_z(w^2) = z, \qquad \mathrm{Tr}_z(w)^2 = z^2,$$

which differ for field elements z with $z^2 \neq z$.

Proof. Evaluate w and $1+w$ at zero and one. In the Boolean quotient $w(1+w) = w + w^2 = 0$ and $w^2 = w$. The displayed identities follow directly. In $\mathrm{GF}(16)$, any element outside $\mathbb{F}_2$ supplies the second inequality. □

The input predicates are individually easy and their separate fingerprints are explicit. Their conjunction nevertheless imposes a common-witness requirement that ordinary multiplication of the summaries fails to preserve.

Theorem 6.1 explains a related all-size restriction: a fixed linear multiplicative summary retaining every witness cannot reduce the full witness algebra below dimension 2^m. That statement does not rule out all

context-dependent nonlinear evaluation methods. It rules out the displayed context-free multiplicative compiler under its exact hypotheses.

The eight maintained claim groups end here. Their proofs do not assume a superpolynomial shared-generation bound or a polynomial-time weighted-trace evaluator. The appendices identify the exact source coverage, evidence, and closed-scope boundaries.

APPENDIX A

Claim coverage and theorem index

The source register has eight maintained claim groups. This reference expands the maintained arguments into individually numbered statements and adds the later register-product invariant. Prose statement counts and formal lemma counts have different meanings. The Lean guide lists all 115 formal declarations; the original theorem companion and its historical theorem map remain preserved.

A.1 Coverage of the maintained register

ID	Maintained subject	Results in this reference	Source location
C01	Exact cut-comparison cover	Theorem 2.1	Foundations F2
C02	Complete-cover bound and dual certificates	Results 3.1–3.7	Theory §1; Foundations F4
C03	Unbalanced-OR calibration	Results 4.1–4.4	Theory §2
C04	Optimal small restored oracles	Theorem 4.5	Theory §2
C05	Projection cost and safety	Results 5.1–5.5	Theory §3
C06	Faithful flat gluing dimension	Theorem 6.1; Corollary 6.2	Theory §4
C07	Weighted witness fingerprint	Results 7.1–7.5	Theory §5
C08	Overlap obstruction	Proposition 8.1; Counterexample 8.2	Theory §5

The foundational simulation and degree results, Theorem 1.1 and Lemma 1.2, support the active claims. The conditional status of Proposition 7.5 is unchanged from C07. Counterexamples are deliberately indexed as counterexamples, rather than presented as successful universal compression theorems.

A.2 Dependency structure

The main proof dependencies are:

$$\begin{gathered}\text{restored oracle convention} \implies \text{cut comparison},\\ \text{cut comparison} + \text{degree per multiplication} \implies \text{complete-cover lower bound},\\ \text{safe ideal} + \text{finite linear duality} \implies \text{evaluation-span certificates},\\ \text{monotone ideal} + \text{affine input/output changes} \implies \text{calibration formulas},\\ \text{affine restriction} + \text{quadratic polar rank} \implies \text{projection-cost lower bound},\\ \text{orthogonal idempotents} \implies \text{gluing dimension},\\ \text{Boolean coefficient identity} + \text{root bound} \implies \text{conditional fingerprint test}.\end{gathered}$$

The gluing-dimension theorem and fingerprint construction are separate from the circuit-cut theorem. No proof step identifies algebra dimension with unrestricted circuit size.

A.3 Complete numbered-result index

No.	Result	Scope	Page
1.1	Classical/restored simulation comparison	Base	2
1.2	Degree per multiplication	Base	2
1.3	Register-product degree invariant	Base / formal	3
2.1	Cut-comparison identity and complete coverage	C01	4
3.1	Basic size bounds	C02	5
3.2	Affine input invariance	C02	6
3.3	Affine-optimized complete-cover bound	C02	6
3.4	Dual certificate for failure of safe detection	C02	7
3.5	Finite support of a dual witness	C02	7
3.6	Safe monomial ideal for a monotone function	C02	7
3.7	Exact monomial oracle cost	C02	8
4.1	Raw calibration parameters	C03	9
4.2	Affine optimization of the one-packet parameter	C03	9
4.3	Affine optimization of complete coverage	C03	10
4.4	Restored cost interval for the calibration family	C03	10
4.5	Exact small restored-oracle calibration	C04	11
5.1	Affine subspace after multiplication restrictions	C05	12
5.2	Exact cost of a nondegenerate paired quadratic	C05	12
5.3	Exact nonlinear cost of the whole function	C05	13
5.4	Exact cost of the extracted grade	C05	13
5.5	Projection does not preserve safe support	C05	13
6.1	Dimension of a faithful multiplicative summary	C06	15
6.2	Scalar unital summaries are evaluations	C06	15
7.1	Exact witness fingerprint	C07	17
7.2	Top coefficient as a Boolean-cube sum	C07	17
7.3	Weighted top-coefficient identity	C07	17
7.4	Finite-field root bound	C07	18
7.5	Conditional randomized satisfiability test	C07	18
8.1	Disjoint-variable factorization	C08	19
8.2	Shared-variable multiplication fails	C08	19

APPENDIX B

Verification evidence and reproduction

Source alignment: Release Audit, Reproducibility, and frozen baseline records [R5, R6].

B.1 Fresh checks for this edition

The original v1.2.0 ZIP and repository bundle were recovered, and the ZIP's digest matches the supplied release digest. Original source files and frozen scientific results remain unchanged. The integrity checker passed with 321 sealed source files and 27 sealed evidence files.

The fresh command `python scripts/verify.py -all` completed with status PASS. Every requested group ran, including the C++ profiles, the maintained exact science checks, the additional degree controls, and the original historical runner. The run manifest identifies commands, input and output hashes, software versions, and elapsed times. Its source-relative path is `build/verification/20261007T131040268104Z-all/manifest.json`.

The archived formal release records 115 checked theorem declarations, eight replayed modules and eight rejected false controls. A fresh invocation in this edition failed before proof checking because `lean` is unavailable. That attempt is retained as a failed environment check, with no fresh theorem or axiom count. The exact formal source, historical successful evidence, and reproduction commands are included in the complete set.

The fresh proof reviews additionally compare primal and dual safe-detection criteria over every truth table at $n = 0, 1, 2, 3$ and reconstruct selected calibration, projection and fingerprint controls. Their scripts and exact scopes are retained in the review directory. Counts can overlap the baseline and are not summed as independent mathematical evidence.

Document checks cover multi-pass LaTeX builds, resolved references, text extraction, font embedding, page bounds, and visual rendering. The final validation report identifies the actual outcomes. PDF compilation is separate from proof compilation.

B.2 Baseline domains reproduced by the full run

The exact mathematical outputs below were compared with the preserved baseline. The historical runner also reproduced its seven scientific output records. These finite ranges remain part of the evidence even where a universal written theorem accompanies them.

Evidence area	Executed domain in the baseline	Limit of the evidence
Small functions	All 276 functions on one, two, or three inputs; 4,240 scalar input/target cases; 2,344 cuts	Exact constructions plus proved lower bounds, not exhaustive circuit search

Evidence area	Executed domain in the baseline	Limit of the evidence
Four-input profiles	All 65,536 truth tables; all 168 monotone functions	Independent evaluation-span method checks only the declared 765 cases
Unbalanced-OR family	Designated truth tables through nine inputs; all affine masks through eight	No exhaustive ninth-input affine-mask claim
Projection coefficients	693,965 pair coefficients and nine exact rank cases, through 1,022 inputs	Full truth tables only at $m = 2, 3, 4$
Projection oracle checks	65,808 logical-input/target cases across the two constructed oracle families in the stated small domains	Large entries use symbolic formulas, exact rank, and construction counts
Weighted fingerprints	1,052,736 verifier/field-point evaluations	Enumerated verifiers have at most three witness bits
Independent coefficient extraction	20,544 cases	All field points for $m \leq 2$; restricted coordinate grid at $m = 3$
Disjoint product rule	4,096 exact controls	Does not extend the rule to shared variables
Scalar algebra maps	66,048 linear maps over the three specified witness/field domains	Controls the scalar special case; general dimension theorem has a written proof

The field used in the fingerprint checker is

$$\mathrm{GF}(16) = \mathbb{F}_2[t]/(t^4 + t + 1).$$

All mathematical arithmetic in these tests is Boolean, exact integer, or finite-field arithmetic. Runtime is not presented as a complexity benchmark. Equality of checksums certifies matching bytes, not mathematical truth.

B.3 Reproduction commands

From the supplied v1.2.0 source root, run the source seal and complete scientific replay. Substitute the actual emitted RUN_ID when checking a run manifest:

```
python scripts/check_integrity.py
python scripts/verify.py --all
python scripts/validate_manifest.py \
  build/verification/RUN_ID/manifest.json --root .
```

The scientific checks require Python 3.10+ and a C++17 compiler. The compiler is selected through `CXX`, with `g++` as the default. There is no external solver, hidden dataset, package installation, or network dependency for these checks. Historical replay is documented for Linux/WSL; native Windows full replay is not newly established here.

Assertions are part of the checker, so optimized Python is rejected. Missing results, command failures, timeouts, modified source inputs, and baseline mismatches make a requested run fail. Each run creates a new directory under `build/verification/`; source and baseline evidence remain unchanged. Quick, full, and history modes are documented individually in the source reproduction guide.

Formal replay is separate. It requires the pinned Lean 4.34.0-rc2 toolchain and runs through `scripts/verify_lean.py`. The Lean companion gives the actual commands, permitted axioms, false controls, and archived outcome. This edition's missing compiler is retained as a failed attempt, not a formal pass.

APPENDIX C

Boundaries, corrections, and source provenance

C.1 Excluded inferences

The following boundaries are retained from the release's failure register. They are part of the closed scope, not a promise to finish every exploratory route.

Semantic elimination curvature. Exact existential eliminations commute. A nonzero semantic commutator cannot be asserted for those exact operators.

Expanded-state counting. Exponentially many residuals or obstruction descriptions can arise from a small circuit. The number of expanded objects is not their shared construction cost.

Small-case equality. Agreement of affine-optimized one-packet and complete-cover degrees through four inputs does not extend to all sizes. Chapter 4 gives an exact separating family.

Free homogeneous extraction. A projected grade may have greater nonlinear cost than the whole and can fail to remain safe. Chapter 5 keeps the input length and total affine cost explicit.

Unrestricted gluing. The exponential dimension theorem applies to a fixed faithful linear multiplicative map. It does not exclude succinct, nonlinear, or context-dependent algorithms.

Context-free trace multiplication. The disjoint-variable product rule fails at shared witness variables. A short expression containing an extraction operator is not an efficient implementation of that extraction.

C.2 Historical execution corrections

Earlier sources corrected illegal repeated-control Toffoli emissions to legal copies, replaced a timed-out large clique search by explicit matching certificates, restored a missing module in isolated replay, and repaired a log-sealing race. These are preserved historical events, not failures newly observed in preparing this PDF.

The release-assembly record also distinguishes a rejected streaming launch, inaccessible GitHub operations, and a repaired draft metadata-summary check from scientific assertions. None is reinterpreted as mathematical UNSAT, a counterexample, or hosted-publication success. The complete release record remains authoritative for those events [R5].

C.3 What this reference does not claim

There is no resolution of P versus NP, no superpolynomial unrestricted Boolean-circuit lower bound, no polynomial-time weighted-trace evaluator, and no quantum-circuit lower bound. Formal evidence is scoped to the named archived Block 02 declarations. No fresh kernel pass, independent human review,

comprehensive novelty search, or best-known-bound audit is claimed.

The conditional randomized implication remains conditional and randomized. An exact finite pass does not turn a written argument into a kernel-checked theorem. Historical archives are not automatically additional maintained claims.

C.4 Source identity and editorial treatment

The original v1.2.0 ZIP, repository bundle, delivery report, earlier theorem companion and formal guide are included unchanged. Their identity is preserved separately from this edition's new manuscripts and figures. The source ZIP SHA-256 is

```
88628c8274660bd20e11fec83249fdc04960403b9d64d1e5068a476b8c025936
```

This edition corrects the old front matter's formal-status description, incorporates the direct register-product invariant, restricts the monotone minterm formulas to nonzero functions, and states the $m = 0$ exception for scalar-summary loss. It also preserves the explicit efficient-field hypothesis in the randomized implication and the own-variable-set convention in disjoint factorization. No source claim group is enlarged, and no original proof file is overwritten.

The publication package carries the original source and evidence plus this edition's reviews, fresh computations, figure scripts, document sources and a full file manifest. The original rights record is retained. No new licence or public hosting status is assigned by this edition.

C.5 Rebuilding the publication set

From the publication-set root, run

```
python scripts/make_figures.py
python scripts/build_main.py
python scripts/build_documents.py
python scripts/verify_publication.py
```

The traditional source is generated from the preserved historical companion plus documented edition updates. Repeated LaTeX passes resolve the bibliography and result index. Font programs are embedded in the PDFs by the TeX build; standalone font files are not distributed. Reproduction of document layout and reproduction of mathematical checks are separate commands.

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PRISM / SOURCE READING CARD

PRISM-1: Packet-Programmable Processing

Final updated architecture draft · 61 original pages

READ THIS FOR

Packet IR, mixed representations, structured scheduling and reconstruction contracts.

KEEP THIS BOUNDARY

Architecture specification with separately retained software evidence; not fabricated or fully qualified hardware.

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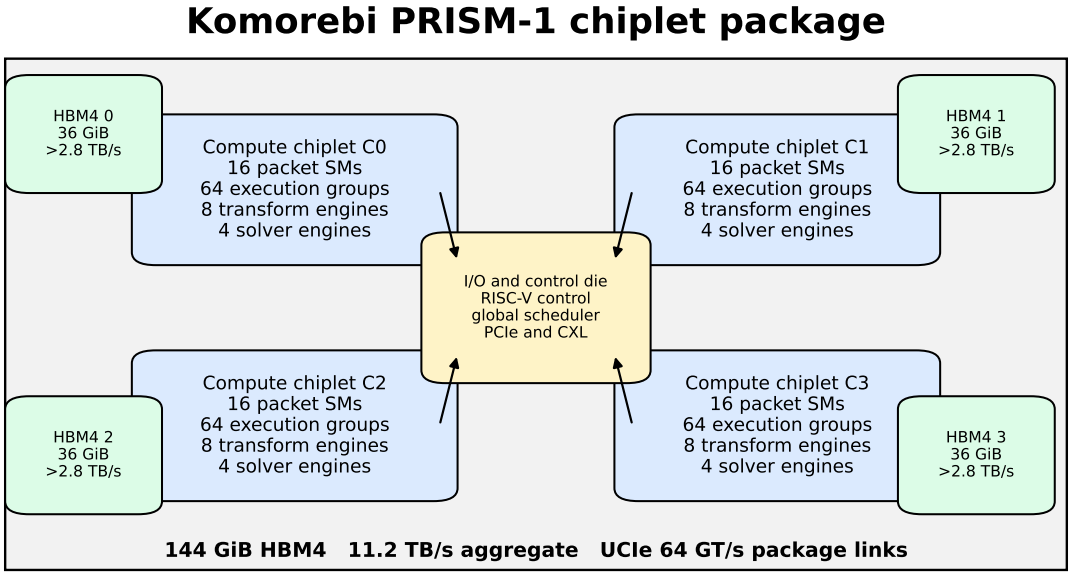
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PROJECTOR-RESOLVED SEED CALCULUS

Komorebi PRISM-1

A Packet-Programmable Mixed-Representation Processor

CUDA-like packet execution, PMR–GEMM, transform atlases, structured operators, edge recovery, and certified reconstruction



Komorebi P.

Final updated architecture, programming model, CUDA compatibility path, compiler contract, verification plan, and executable model

Abstract

Komorebi PRISM-1 is a programmable chiplet accelerator for computations whose useful form is known before execution. Its native objects are packet graphs, transform plans, block operators, low-rank factors, Kronecker products, sparse factor chains, refinement trees, small algebraic systems, and reconstruction certificates. A dense array is one possible representation. The architecture never requires one to exist.

The final reference design contains four compute chiplets, 64 Packet Streaming Multiprocessors, 256 packet execution groups, configurable transform engines, resident small solvers, exact arithmetic support, 192 MiB of on-package SRAM, and four HBM4 stacks. Each Packet Streaming Multiprocessor schedules packet warps across four execution groups and a shared scratchpad. The execution model is CUDA-like in its hierarchy and programmability, but it keeps atlas identity, parent-child relations, output contracts, precision weights, and certificate state visible to hardware.

The compiler begins with Packet IR and chooses among explicit dense, block, block-sparse, low-rank, block-diagonal-plus-low-rank, Kronecker, sparse-chain, transform-domain, and packet-graph plans. PMR–GEMM supplies the recursive mixed-representation planner. The same Packet IR can lower to a CUDA compatibility backend or to PRISM. Earlier CUDA implementations already establish the software path through grouped matrix kernels, CUDA Graph replay, asynchronous transfers, FFT and sparse libraries, compressed products, Woodbury solves, and packetwise mixed precision.

The paper specifies the package, Packet Streaming Multiprocessor, memory hierarchy, network, command format, compiler, output contracts, certificate protocol, performance accounting, FPGA path, and verification plan. Existing CUDA measurements are kept separate from PRISM architecture estimates. Representation gain, kernel gain, architecture gain, and amortization are reported as different quantities. The design remains an architecture specification. It is not fabricated silicon and it does not claim measured PRISM frequency, power, or throughput.

Status and scope

This is the final updated architecture draft. The packet types, Packet IR, output contracts, command groups, Packet Streaming Multiprocessor boundary, memory hierarchy, compiler routes, error-budget rules, CUDA compatibility path, and verification ladder are specified. Circuit timing, SRAM banking, NoC virtual channels, floorplanning, hardened interface IP, and process-node choices remain implementation work.

The package baseline uses public interfaces that are available or specified at the time of this draft. UCIe 3.0 includes 48 and 64 GT/s die-to-die classes. PCIe 7.0 specifies 128 GT/s. Micron lists 36 GB HBM4 in high-volume production with bandwidth above 2.8 TB/s per stack. The management cores use the ratified RISC-V vector programming model. These facts establish a plausible component baseline. They do not validate the PRISM microarchitecture.

PRISM-1 does not preserve a dense matrix. It preserves the mathematical output contract. A packet program may remain a factor graph, transform plan, sparse chain, low-rank correction, or executable operator from beginning to end.

Executive summary

The design is built around seven practical decisions.

1. Keep representation visible from the calculus front end to the command queue.
2. Treat a dense array as an optional leaf or requested output, not as mandatory machine state.
3. Use one Packet IR for the CUDA backend and the PRISM backend.
4. Schedule packet warps on programmable Packet Streaming Multiprocessors.
5. Keep factors, transform plans, and packet graphs resident whenever later stages can consume them directly.
6. Keep exact archives separate from compact predictive states.
7. Make residuals and error bounds part of the commit path.

The software basis already exists. On the recorded RTX 3070 Ti Laptop GPU, an equal-block packet operator was measured at 0.046032 ms and ran 14.8919× faster than its matched device-resident dense GPU row. A BDLR packet path with CUDA Graph replay was measured at 0.079872 ms and ran 10.9355× faster than the matched dense GPU row. Graph replay improved the same packet representation by about 1.218×. A compressed BDLR product took 0.1582 ms, while materializing the output took 0.5073 ms. These measurements show that the calculus can already execute on CUDA without preserving a dense matrix.

PRISM addresses the work left after the representation has already improved. Its likely advantages come from resident factor reuse, packet gather and scatter, transform-to-solver handoff, grouped small-block dispatch, local refinement, deterministic graph replay, and certificate accumulation. The architecture model estimates gains of 2.96× for the resident edge-recovery chain, 1.64× for a fused multi-atlas path, and 1.42× for a shape-batched Bayesian portfolio. Those are architecture estimates. They are not silicon measurements.

The earlier 37.8× dense-route comparison is retained only as a total route model. It combines a better representation with a different architecture. It is not reported as a PRISM-over-CUDA architecture gain.

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1 Design brief

1.1 The design problem

Many computations begin with structure that disappears before hardware sees the job. A block operator becomes one dense matrix. A Fourier or wavelet atlas becomes an unnamed array. A parent packet and its children become unrelated buffers. A low-rank correction is materialized even when every later stage could have consumed its factors.

PRISM-1 keeps those relations explicit. The architecture is useful only when that choice saves arithmetic, storage, data movement, or scheduling work. The compiler is allowed to decline the packet path.

1.2 Required behavior

Requirement	Meaning
Exact packet synthesis	A declared complete packet family must reconstruct the original object within the stated numerical format
Typed transfer	Every operator edge retains its source packet, destination packet, shape, rank, precision, and sensitivity
Local refinement	Replacing one parent by a child graph must not force unrelated tasks to be regenerated
Representation preservation	Block, sparse, low-rank, Kronecker, factor-chain, transform, and packet-graph forms remain executable without an explicit full array
Provisional execution	Results that require a certificate remain hidden from downstream work until the check passes
Deterministic replay	Selected graph regions can use fixed reductions and stable descriptors for reproducible reruns
Graceful rejection	Work with weak structure is routed to the CUDA backend, a general-purpose path, or a generic leaf

1.3 Design rules

1. The mathematical object and its storage representation are not assumed to be the same thing.
2. A packet descriptor is part of the architectural state, not debugging metadata.
3. An application result, a factor graph, a transform plan, and an explicit array are different output contracts.
4. A numerical result is not committed merely because the kernel finished.
5. A failed certificate should identify the local graph region that needs replay or refinement.
6. The first prototype should preserve the final command and descriptor formats even if it implements fewer engines.

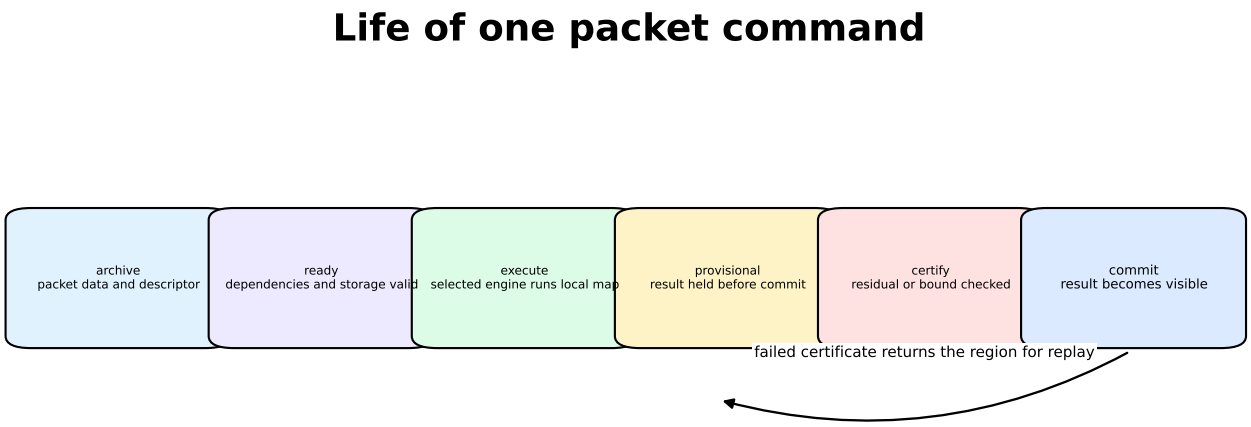


Figure 1.1: One packet command from archive to commit. The provisional state prevents an unchecked result from becoming visible to downstream work.

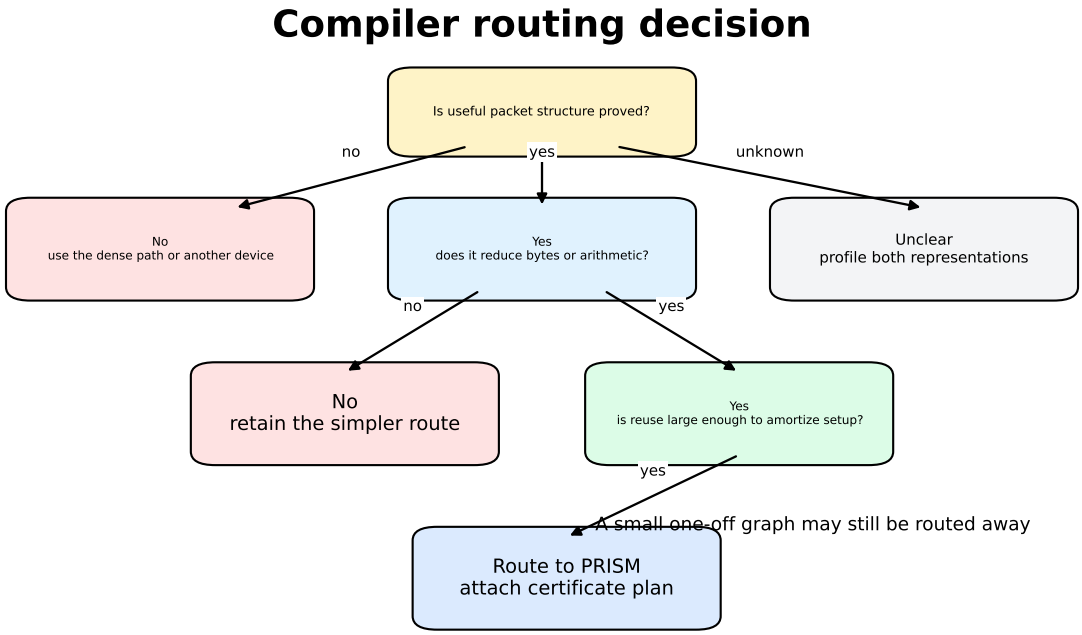


Figure 1.2: The compiler is expected to route weakly structured work away from the packet path.

1.4 What this design does not promise

PRISM-1 is not a universal replacement for a GPU or CPU. It does not claim a better asymptotic exponent for generic dense linear algebra. It does not claim that every operator has a useful structured representation.

It does not treat modeled roofline numbers as measured performance. Its case rests on workloads where structure is real, reusable, and large enough to amortize descriptor, planning, and scheduling cost.

2 Why this chip exists

2.1 The representation is part of the computation

A conventional accelerator usually receives a tensor after the representation has already been chosen. It may execute that tensor well and still miss the simpler object underneath it. The operator may be block diagonal, block sparse, low rank, generated by a transform, or recoverable from a much smaller packet graph.

Projector-resolved computation starts before that flattening step. Let

$$X = \bigoplus_{\alpha \in I} X_\alpha, \qquad P_\alpha P_\beta = \delta_{\alpha\beta} P_\alpha, \qquad \sum_{\alpha \in I} P_\alpha = I.$$

For an operator $T : X \to X$, the native objects are the transfer blocks

$$T_{\beta\alpha} = P_\beta T P_\alpha.$$

The complete operator is reconstructed by

$$\boxed{T = \sum_{\alpha,\beta} T_{\beta\alpha}.}$$

The zero pattern, block shapes, rank structure, parent-child relations, and packet sensitivities are part of the problem. PRISM-1 carries them into the execution model instead of asking software to rediscover them after every kernel boundary.

2.2 The target workload

The design focuses on operations that recur throughout the calculus:

- residue and phase projectors
- polyphase extraction and interleaving
- Fourier, cosine, Hadamard, and wavelet atlases
- packet-tree refinement and coarsening
- block-diagonal and block-sparse transfers
- block diagonal plus low rank operators
- exact compressed products and Woodbury solves
- small Hankel, QR, SVD, root, and Vandermonde problems
- nonlinear packet interactions and memory kernels
- conjugate Bayesian regression and posterior mixtures
- exact modular, CRT, and valuation operations

- residual and error-bound certificates

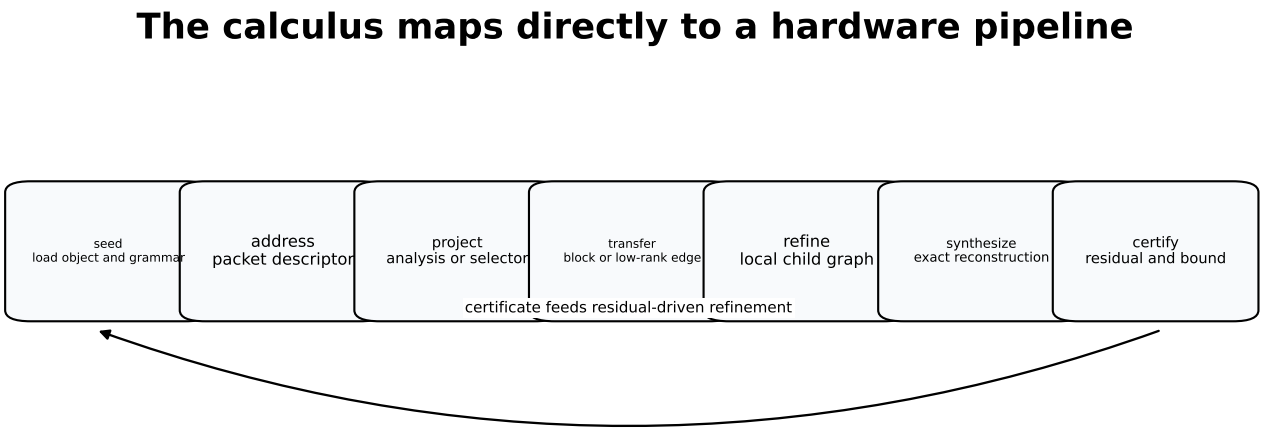


Figure 2.1: The seed-calculus operations become a hardware pipeline without losing their mathematical type.

2.3 What the chip is not

PRISM-1 should not be used simply because a packet description exists. A generic dense matrix with no reusable structure may run better through a conventional library path. A tiny one-off transform may not amortize descriptor setup. A graph that changes completely at every step may not benefit from replay. The compiler should reject or reroute those cases.

2.4 No dense preservation requirement

The architecture does not begin with a dense matrix and try to compress it after the fact. It begins with a typed operator. That operator may be represented as

$$\boxed{\begin{aligned}\mathfrak{R} = \{&\text{explicit array},\ \text{block},\ \text{block sparse},\ \text{low rank},\ \text{BDLR},\\ &\text{Kronecker},\ \text{sparse chain},\ \text{transform plan},\ \text{packet graph}\}.\end{aligned}}$$

A dense representation is useful when no better form exists or when a downstream user explicitly asks for every entry. It is not a state that the chip must preserve.

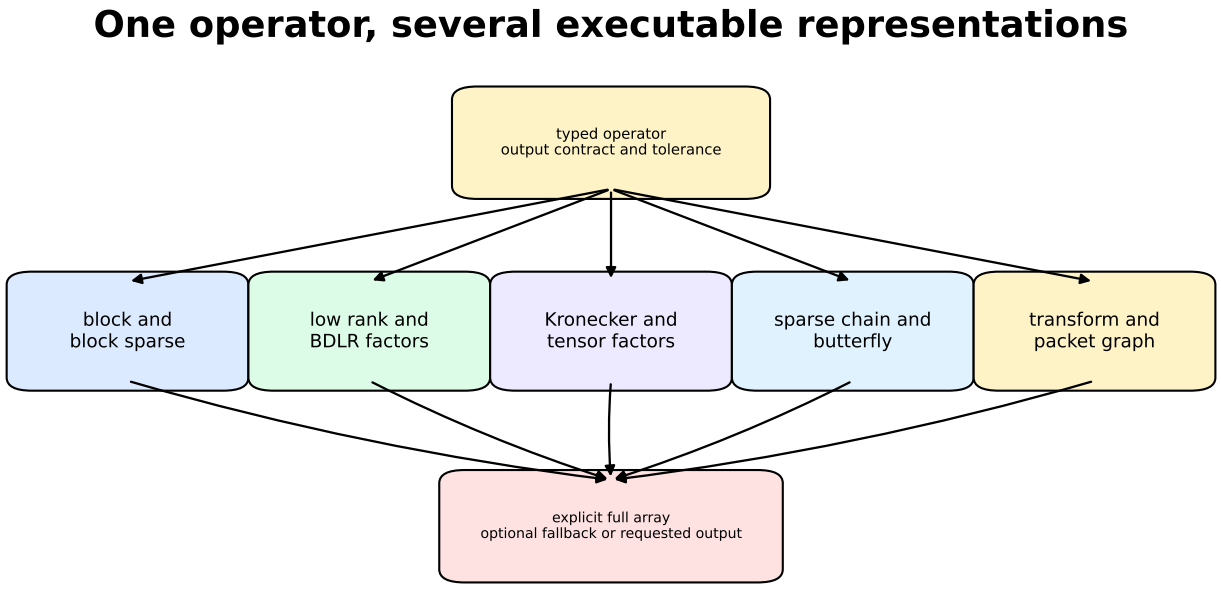


Figure 2.2: One operator may have several executable representations. The explicit array is one optional leaf.

2.5 CUDA-like execution hierarchy

CUDA made heterogeneous arithmetic programmable through a hierarchy of lanes, warps, blocks, grids, and memories. PRISM uses an analogous execution hierarchy while retaining the mathematical structure that CUDA kernels normally receive only through software conventions:

$$\boxed{\text{lane} \longrightarrow \text{packet warp} \longrightarrow \text{execution group} \longrightarrow \text{Packet Streaming Multiprocessor} \longrightarrow \text{packet graph}.}$$

A packet warp may invoke a matrix group, vector lanes, a transform engine, a small solver, exact arithmetic, or a memory operator. The representation router chooses the form declared by Packet IR. Several forms may appear in one graph.

2.6 The CUDA implementation is the software foundation

The packet algorithms already run on CUDA. Equal blocks are stacked and sent to grouped or batched matrix kernels. BDLR application uses a block batch followed by two low-rank contractions. Repeated graphs use CUDA Graph replay. Fourier and sparse stages use established libraries. The runtime keeps factors resident when the output contract allows it.

This matters for the chip design. PRISM is not a replacement for the CUDA algorithms. It is a hardware target for the same Packet IR. The CUDA backend remains the development platform, portability path, and matched comparison.

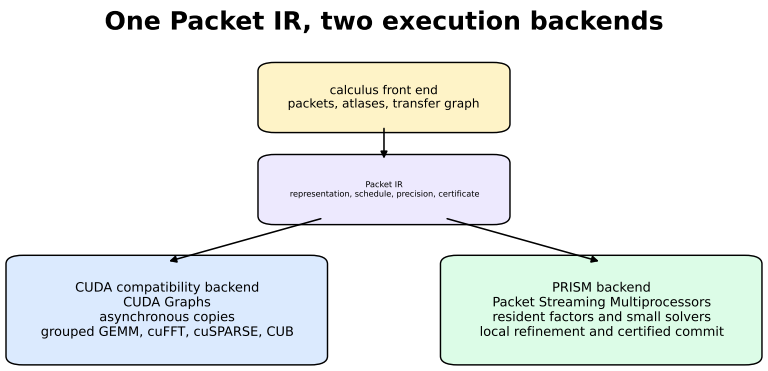


Figure 2.3: The CUDA and PRISM backends execute the same Packet IR and preserve the same output contracts.

3 Mixed-representation execution

3.1 Output contracts

A computation is not defined only by the numerical entries of a hypothetical matrix. The caller also specifies what must be returned. PRISM uses five principal contracts:

1. apply an operator to one or more vectors
2. return a compressed operator
3. return a factor or packet graph
4. return a transform plan
5. return every entry of an explicit array

Only the fifth contract forces materialization.

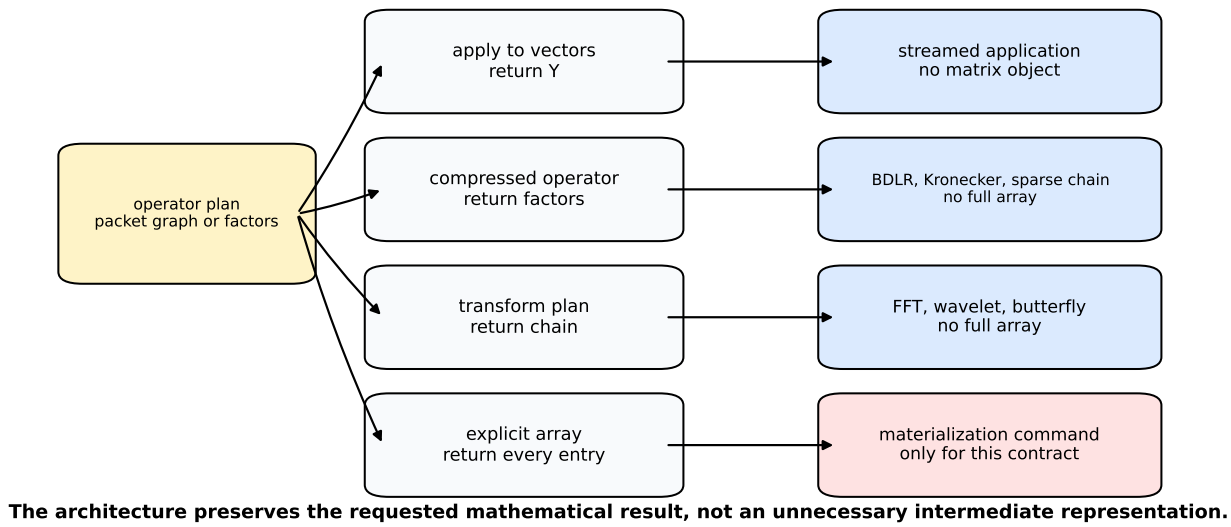


Figure 3.1: Materialization is required only when the requested output is the explicit full array.

3.2 Compatible middle bases

The intermediate basis is part of the optimization. For any invertible S,

$$AB = (AS)(S^{-1}B).$$

Suppose

$$A = LD_AS^{-1}, \qquad B = SD_BR.$$

Then

$$\boxed{AB = L(D_AD_B)R.}$$

When L, S, and R have fast application and D_A, D_B are block diagonal, the product inherits the blockwise cost plus the transform costs. Fourier diagonalization of circulants, simultaneous diagonalization, symmetry sectors, and tensor factorizations are instances of the same rule.

3.3 Sparse chains and transform plans

A matrix may be dense as an array and cheap as an operator. If

$$A = F_LF_{L-1}\cdots F_1$$

and each factor has $O(n)$ nonzero entries, then applying A costs $O(nL)$. If $L = O(\log n)$, the cost is $O(n \log n)$. FFT and butterfly operators fit this pattern. PRISM stores the factor chain and executes it. It does not store the dense array represented by the chain.

3.4 PMR–GEMM

Packet Mixed-Representation GEMM is the compiler planner for operator products. It accepts the operator pair, tolerance, output contract, candidate transforms, and packet tree. At each node it may use a generic

leaf, a low-rank contraction, a Kronecker identity, a sparse factor chain, a transform-domain rule, or a recursive split.

For a fixed candidate tree, each choice o has a local cost $c(o)$, a local error $e(o)$, and zero or more children. The planner evaluates

$$J(v) = \min_{o \in \mathcal{O}_v} \left[c(o) + \lambda e(o)^2 + \sum_{u \in \mathrm{child}(o)} J(u) \right].$$

Bottom-up evaluation is globally optimal on the fixed tree because every feasible plan consists of one local choice followed by feasible child plans.

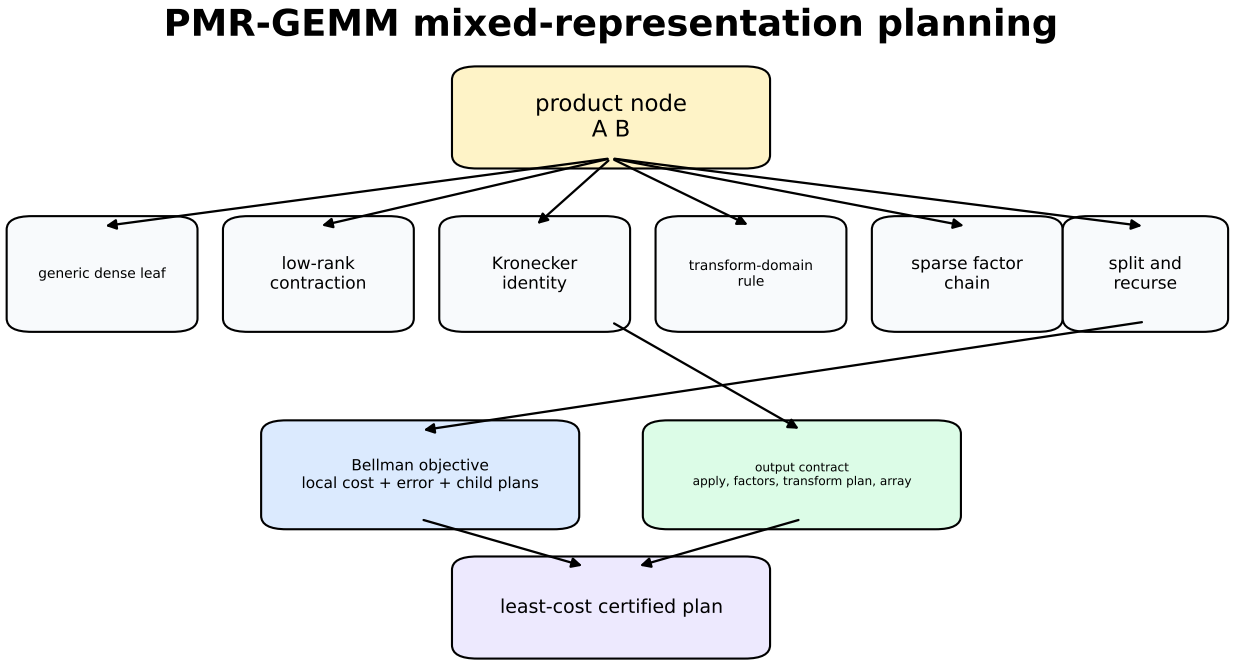


Figure 3.2: PMR–GEMM chooses the least-cost certified plan that satisfies the requested output contract.

3.5 Hybrid fallback

The planner always includes a generic leaf. This protects the worst case. Random dense matrices with no useful local rank, transform, tensor, or packet structure return to the generic path. The architecture is strongest when the mixed plan eliminates arithmetic or movement. It does not claim that naming a packet makes an operator cheaper.

4 Reference package

4.1 Chiplet organization

The reference package contains four compute chiplets, one I/O and control die, and four HBM4 stacks. Each compute chiplet contains 16 Packet Streaming Multiprocessors. Every multiprocessor contains four packet execution groups, giving 64 execution groups per chiplet and 256 across the package. Each chiplet

also contains eight transform engines, four small-solver engines, one exact arithmetic cluster, and 16 MiB of shared SRAM.

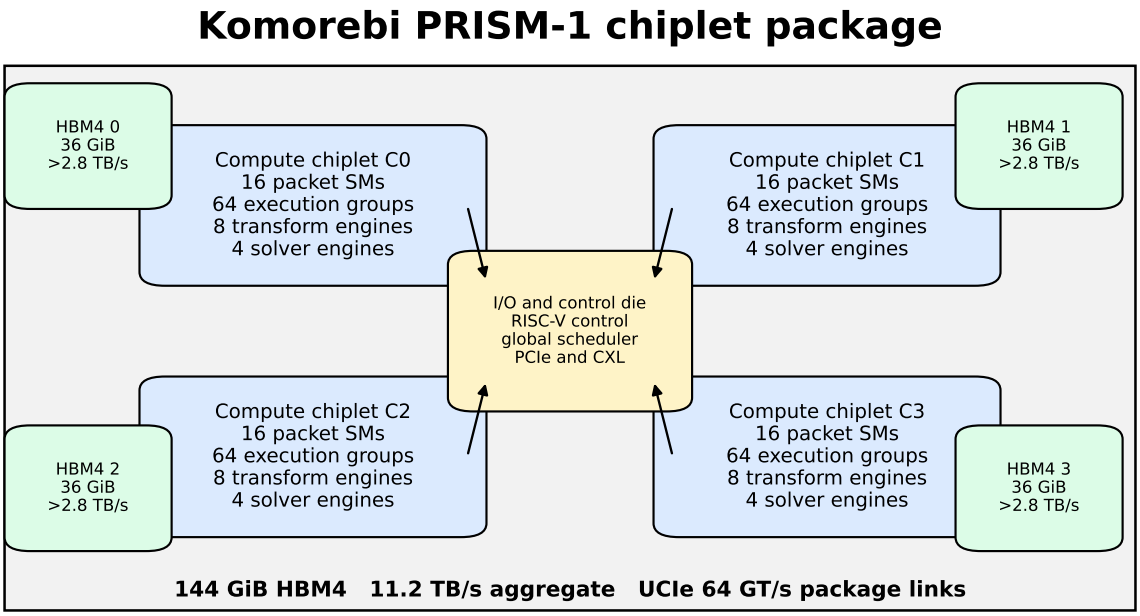


Figure 4.1: The reference PRISM-1 package. The center die carries host I/O, global scheduling, RISC-V control, security, and telemetry.

4.2 Reference targets

Table 4.1: Reference architecture targets. These values belong to the model, not measured silicon.

Property	Reference target
Compute chiplets	4
Packet Streaming Multiprocessors	64
Packet execution groups	256
Execution group array	16×16 matrix array plus vector lanes
Clock	1.5 GHz
PSM scratchpad	2 MiB across four independently banked regions
Shared SRAM	16 MiB per compute chiplet
Total local and shared SRAM	192 MiB
Transform engines	32
Small-solver engines	16
Exact arithmetic clusters	4
HBM4	4 stacks, 144 GiB total
Aggregate HBM bandwidth	greater than 11.2 TB/s
Package fabric	UCIe 3.0 class, up to 64 GT/s
Host interface	PCIe 7.0 x16 class, optional CXL
Reference power envelope	500 W

4.3 Compute-chiplet layout

The reference chiplet places a 4×4 mesh of Packet Streaming Multiprocessors at its center. Every multiprocessor contains four execution groups, giving 64 matrix and vector groups per chiplet. Transform engines sit along one edge, solver engines along another, shared SRAM banks line the data path, and the UCIe router occupies a package-facing edge. This is a planning floorplan rather than a placed design. Its purpose is to make communication and locality assumptions visible early.

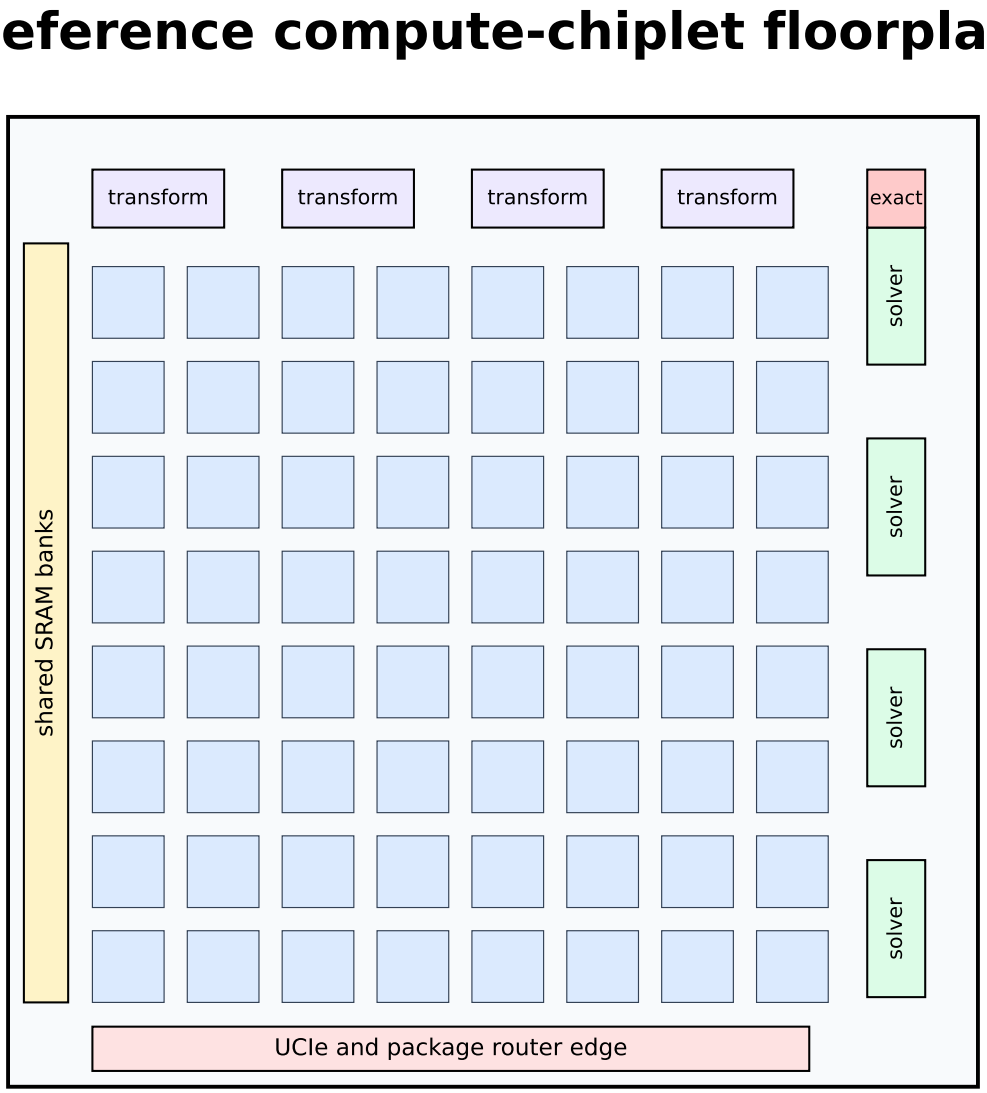


Figure 4.2: Reference compute-chiplet floorplan. The drawing is functional. Bank count, channel width, and physical dimensions remain open.

4.4 Bandwidth balance

The host link is intended for graph submission, coarse data movement, and result return. It is not the steady-state path for every packet. Reusable graphs, factors, and active atlases should remain in HBM or on-package SRAM. The modeled package fabric is wider than aggregate HBM bandwidth so inter-chiplet

traffic does not become an immediate architectural bottleneck in the reference configuration.

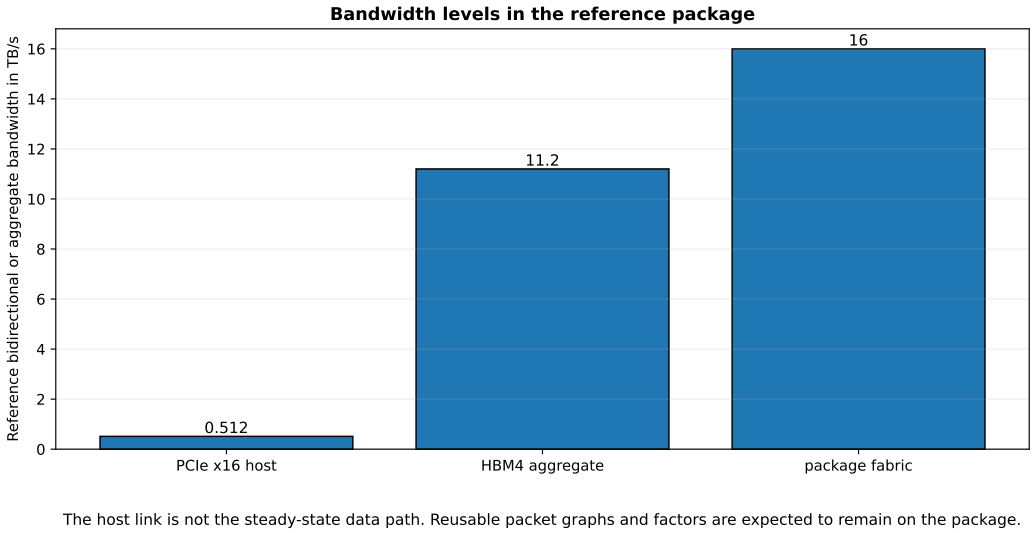


Figure 4.3: Reference bandwidth levels. Local SRAM bandwidth is a circuit-level target and is not assigned a final number in this draft.

4.5 Scale-down configurations

The same Packet IR, commands, and descriptors support smaller devices.

Table 4.2: Suggested development configurations.

Configuration	PSMs	Execution groups	Transform	Solver	Intended use
FPGA overlay	1	4	1	1	Command, DMA, representation routing, replay, and certificate validation
Shuttle chiplet	4 to 8	16 to 32	2 to 4	1 to 2	Measured locality, solver occupancy, and Packet IR conformance
Reference chiplet	16	64	8	4	Full chiplet mesh and resident solver experiments
Full package	64	256	32	16	Data-center scale structured workloads

4.6 Modeled arithmetic peaks

With 256 execution groups and one 16×16 matrix array per group, the model yields

$$\begin{aligned} \text{FP16 or BF16} &\approx 196.6 \text{ TFLOP/s}, \\ \text{FP32} &\approx 98.3 \text{ TFLOP/s}, \\ \text{FP64} &\approx 24.6 \text{ TFLOP/s}. \end{aligned}$$

These are model peaks. The design is not organized around a claim of superior generic dense throughput. Its purpose is to execute fewer operations and move fewer bytes when the selected representation proves that the explicit array is unnecessary.

5 Packet Streaming Multiprocessor

5.1 Multiprocessor anatomy

The Packet Streaming Multiprocessor is the independently scheduled unit of PRISM. It accepts packet warps, reads their representation and output contracts, and routes work across four execution groups, the packet scratchpad, asynchronous DMA, transform and solver interfaces, and the certificate path.

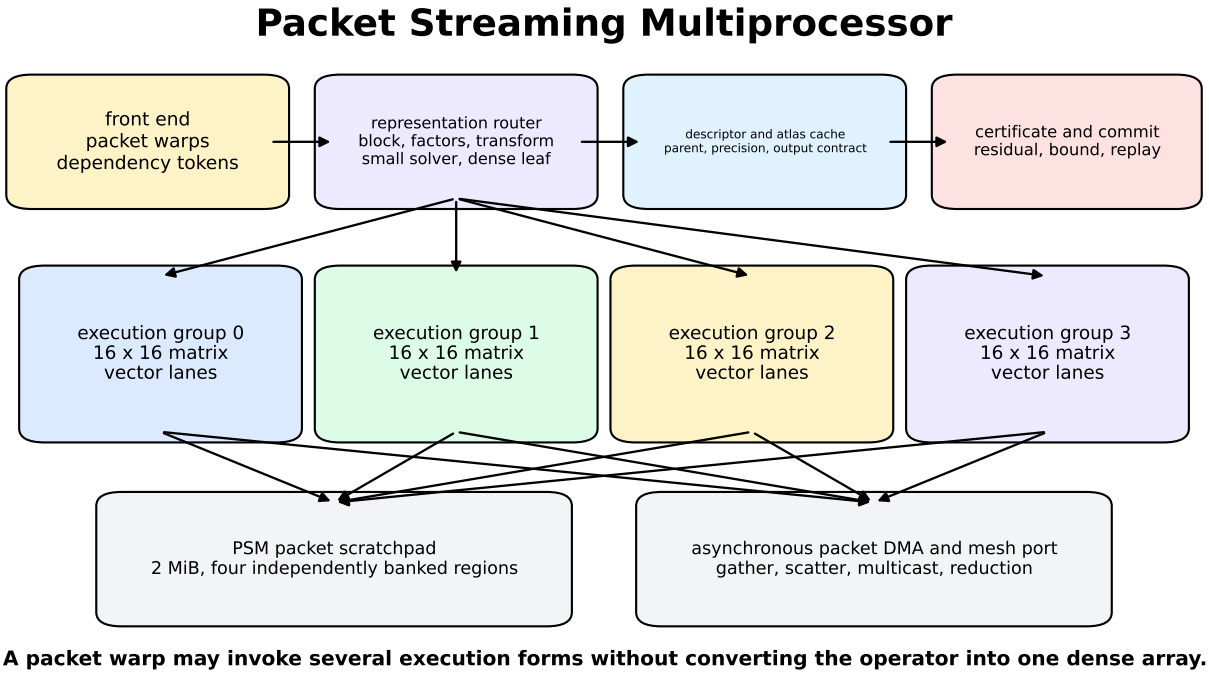


Figure 5.1: One Packet Streaming Multiprocessor. A packet warp may use several execution forms without converting the operator into one explicit array.

5.2 Issue and completion path

A packet warp becomes ready when its dependency token is satisfied and the packet directory confirms that the source representation is resident or in flight. The front end selects the representation route, reserves

destination storage, and records the required certificate. Kernel completion produces a provisional result. The result becomes visible only after the check passes.

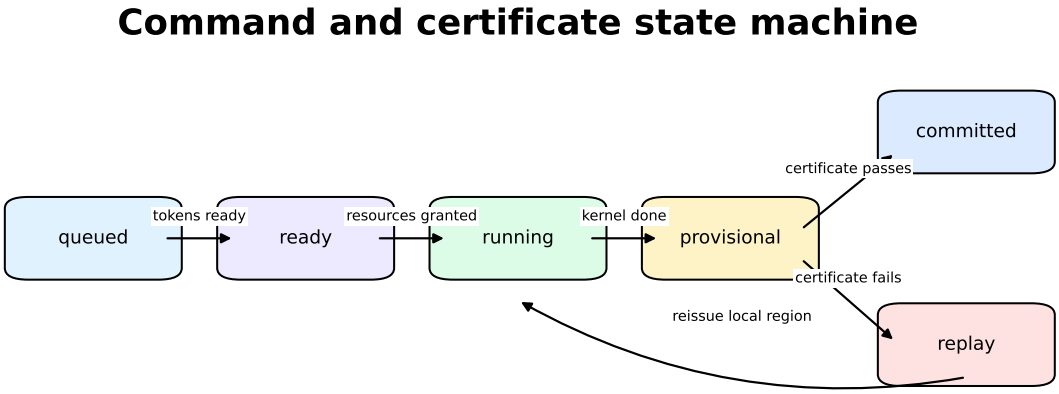


Figure 5.2: Command states from queue entry to commit or local replay.

Completion and commitment remain separate. A kernel may finish while its residual exceeds the declared budget. The distinction matters for mixed precision, iterative solves, approximate transforms, recovered edge packets, and low-rank recompression.

5.3 Packet scratchpad

Each multiprocessor has 2 MiB of scratchpad divided into four independently banked regions. An execution group may use one bank privately or several groups may share a packet region. Separate banks hold active data, factors, descriptors, tokens, solver panels, and certificate state. Double buffering supports transforms and repeated grouped matrix batches.

5.4 Execution groups

Each of the four execution groups contains a 16×16 matrix array and programmable vector lanes. Equal-shape blocks can be grouped across execution groups. The same lanes handle low-rank contractions, Kronecker mode products, sparse-chain factors, selectors, reductions, index generation, and generic leaves.

Complex arithmetic uses paired real lanes. FP64 is available for certificates, small solvers, rank decisions, phase recovery, and sensitive reductions. Integer and fixed-point modes support selectors, image packets, modular arithmetic, and exact accumulators.

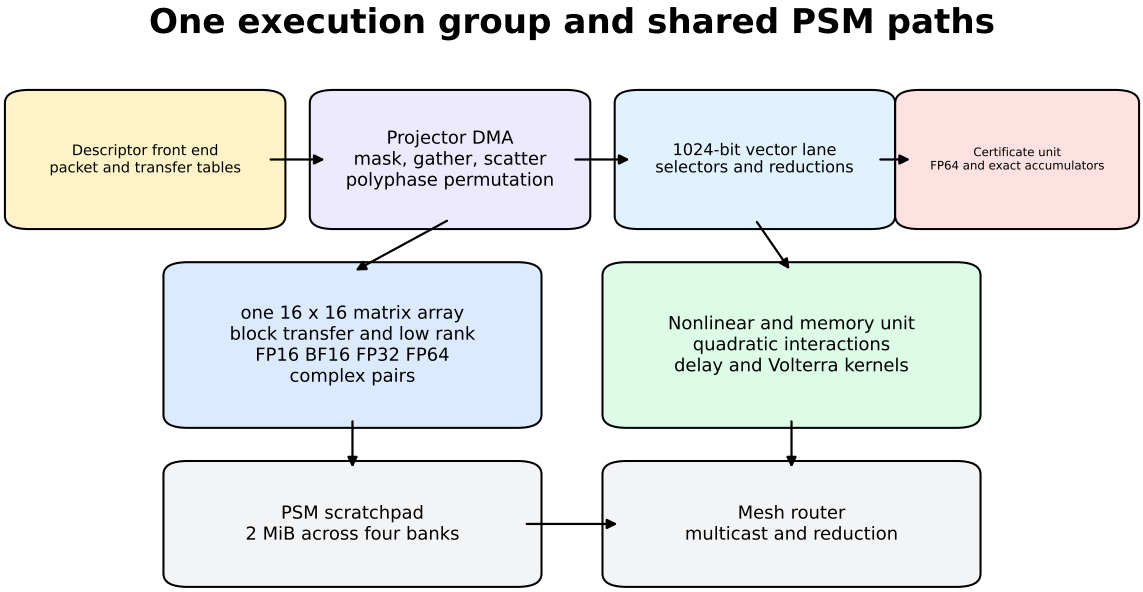


Figure 5.3: Execution-group resources inside the multiprocessor. The figure shows matrix, vector, memory, nonlinear, and certificate paths.

5.5 Representation router

The router reads the operator form from Packet IR. It dispatches block work to grouped matrix lanes, factor work to contraction lanes, transform plans to the transform fabric, small algebraic systems to resident solvers, and weakly structured leaves to the generic path. One packet warp may contain several representations.

5.6 Projector DMA

The asynchronous packet DMA accepts selectors represented as intervals, strided slices, residue classes, polyphase channels, masks, index lists, transform bands, packet-tree nodes, and compact basis factors. A selector is applied through addressing and compact arithmetic. No dense projector matrix is required.

5.7 Certificate and commit path

The certificate unit accumulates residuals, packet norms, trace quantities, energy partitions, branch margins, and mixed-precision bounds. Deterministic mode uses a fixed reduction tree and wide accumulation. The commit controller releases a provisional result only after the declared acceptance rule passes.

6 Descriptors as the hardware language

6.1 Packet descriptor entry

A packet descriptor identifies the packet, its parent, its atlas, its selector, its shape, its location, its precision, and its error budget.

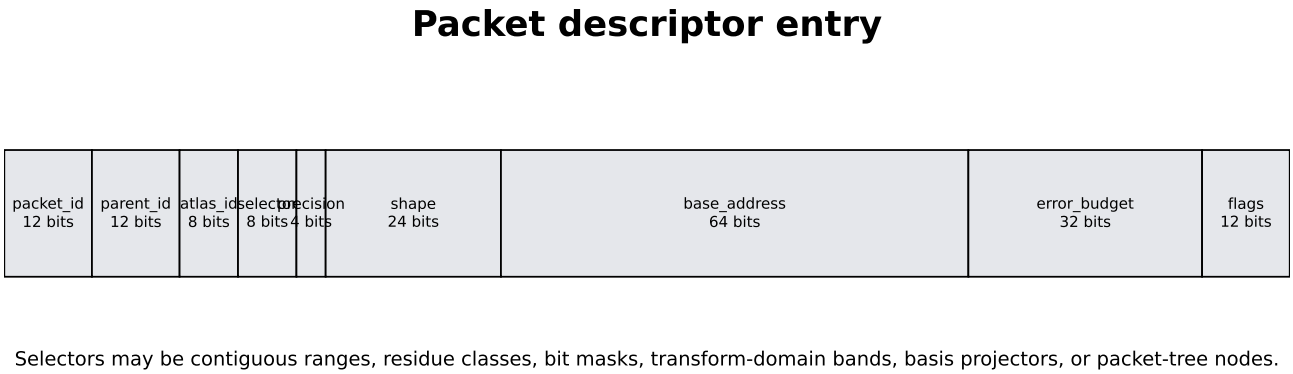


Figure 6.1: One packet descriptor entry. The exact bit allocation may change during RTL design, but the semantic fields are fixed.

Definition 6.1 (Packet descriptor)**.** *A packet descriptor is a typed record*

$$\mathsf{P} = (i, p, a, \sigma, \mathbf{n}, \ell, \pi, \tau),$$

where i is the packet ID, p is the parent, a is the atlas, σ is the selector type, $\mathbf{n}$ is the shape, ℓ is the storage location, π is the precision tag, and τ is the allowed local error contribution.

6.2 Transfer edge descriptor

The transfer descriptor is the hardware form of $P_\beta T P_\alpha$. It identifies the source and destination packets, operator class, shape, rank, precision, sensitivity, payload, and dependency token.

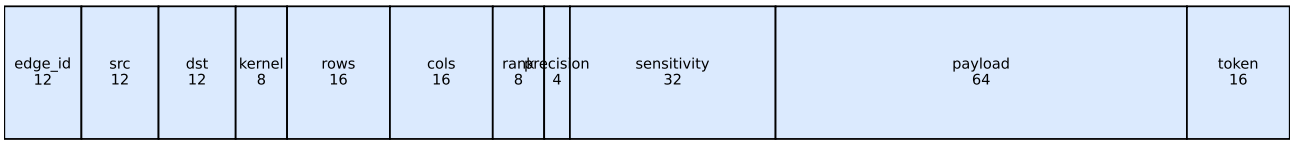


Figure 6.2: A transfer edge descriptor keeps the operator class and certificate weight attached to the edge.

6.3 Atlas descriptor

An atlas descriptor records an analysis graph, a synthesis graph, normalization data, and packet selectors. A unitary transform uses its adjoint for synthesis. A redundant exact frame records the canonical dual or another declared synthesis map satisfying

$$\mathcal{S}\mathcal{A} = I.$$

7 Memory hierarchy

7.1 Exact archive and active representations

The exact archive, cold factors, large graphs, and model history remain in HBM. Completeness does not imply a dense array. An exact archive may be a transform tuple, factor graph, sparse chain, packet tree, geometric edge list, or explicit sample field. Shared SRAM stores active packet graphs and reusable representations. PSM scratchpads hold local blocks, factor panels, transform stages, solver panels, and residuals.

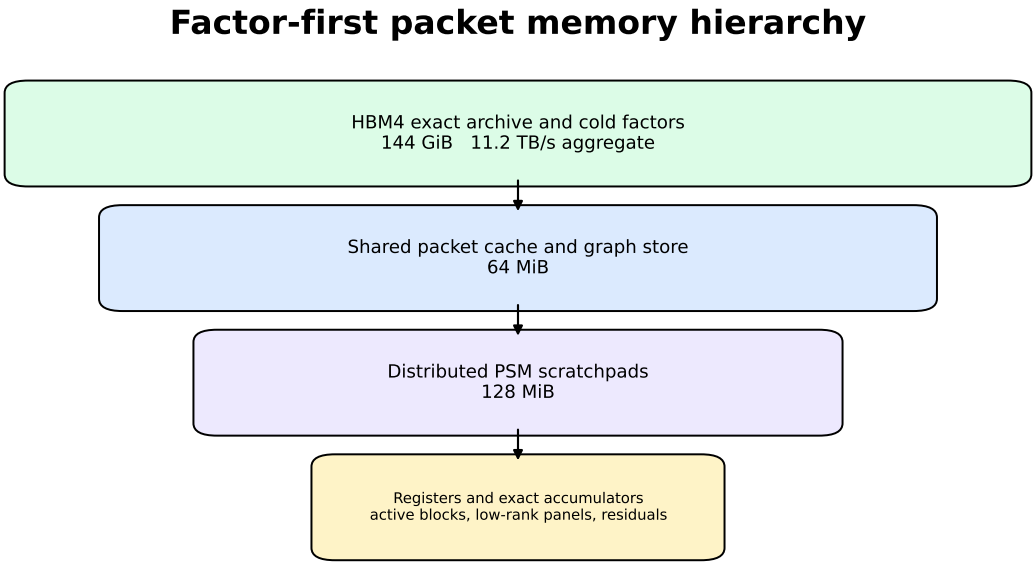


Figure 7.1: Representation-first memory hierarchy. The explicit full array is created only under a matching output contract.

7.2 Representation-first storage

For an equal-block BDLR operator

$$A = D + UV^*,$$

with dimension n, block width b, and rank r, the stored scalar count is

$$\boxed{n(b + 2r)}$$

in place of n^2.

Proposition 7.1 (BDLR representation reduction)**.** *If $b + 2r < n$, BDLR storage reduces the scalar count by*

$$\boxed{\frac{n}{b+2r}.}$$

The same factor appears in the ideal apply count for a fixed number of right-hand sides.

Proof. The block-diagonal part stores $(n/b)b^2 = nb$ scalars. The two rank factors store $2nr$. The total is $n(b + 2r)$. A full array stores n^2 scalars. □

The same rule extends beyond BDLR. A Kronecker operator stores its mode factors. A butterfly stores its sparse stages. A transform-domain operator stores the plan and diagonal or block payload. A packet graph stores packet descriptors and active edge payloads. The active representation is the operator until a different output contract is requested.

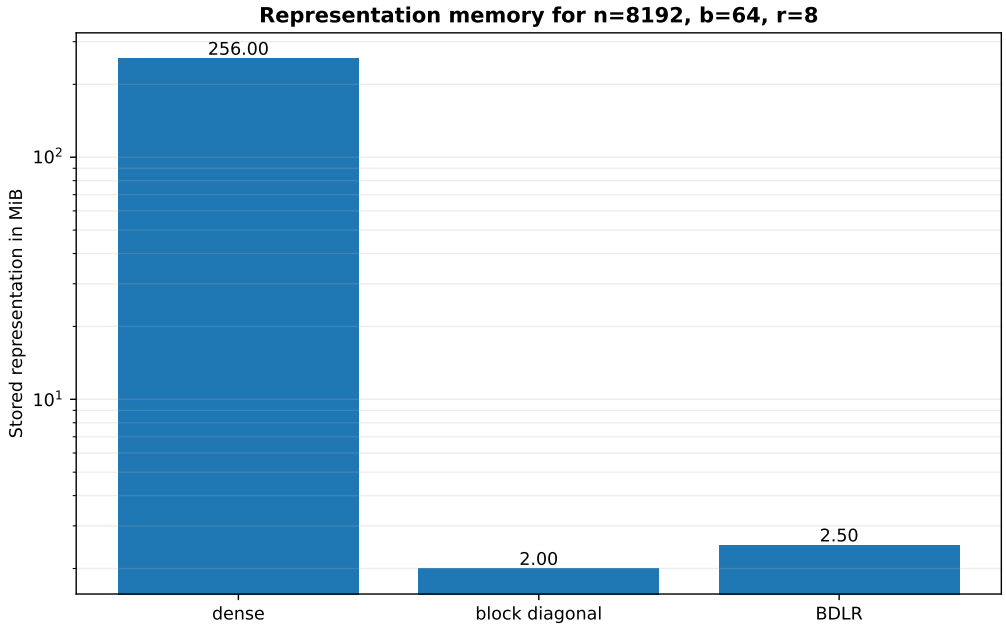


Figure 7.2: Stored BDLR representation for the model case $n = 8192$, $b = 64$, and $r = 8$.

8 Packet network and scheduling

8.1 PSM mesh

Each compute chiplet uses a 4×4 mesh of Packet Streaming Multiprocessors. Routers support unicast, multicast, and tree reduction. Multicast distributes low-rank factors, Kronecker factors, and transform data. Reduction combines destination contributions, small contractions, posterior scores, and residuals.

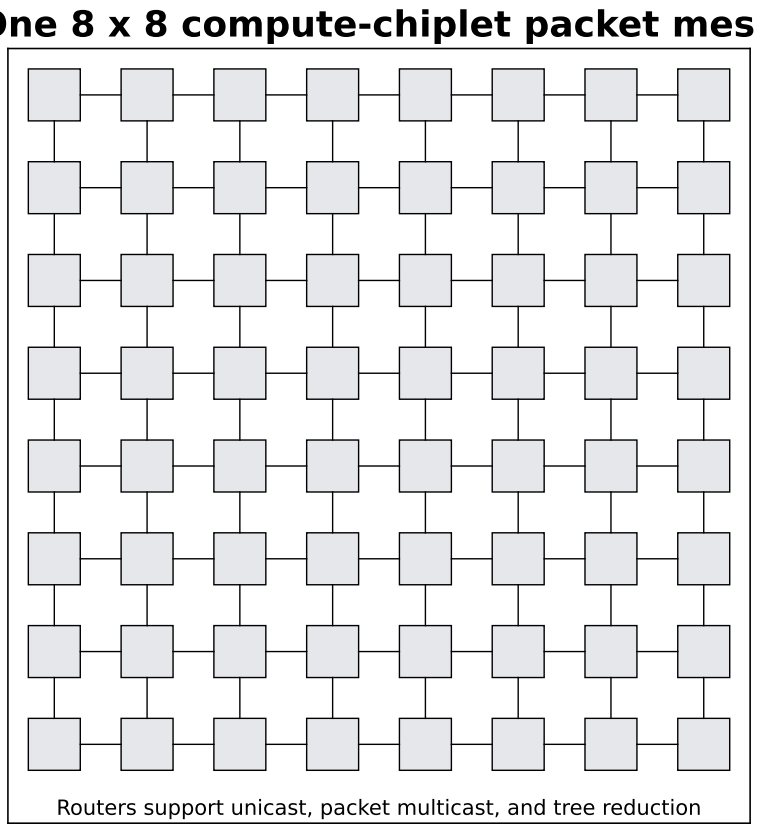


Figure 8.1: One chiplet mesh. Each displayed router node represents one Packet Streaming Multiprocessor rather than one execution group.

8.2 Packet warps and dependency tokens

Packet warps group operations that share a representation, shape class, packet relation, or certificate boundary. A warp becomes ready when its dependency tokens are satisfied. Completion tokens wake downstream work. Replayable regions keep descriptors and placement until a certificate mismatch or local refinement invalidates the region.

8.3 Local refinement

Suppose a parent packet task v is replaced by an acyclic child module $H = (W, F)$. Predecessors of v connect to child entries. Child exits connect to successors of v.

Theorem 8.1 (Topological splice)**.** *Let $\prec$ be a topological order of the original graph and let $\prec_H$ be a topological order of the child module. Replacing v in $\prec$ by $\prec_H$ gives a topological order of the refined graph. Every unaffected pair of context tasks keeps its relative order.*

Proof. Every context edge is unchanged. Every child edge is respected by $\prec_H$. Every predecessor of v appeared before v and therefore before the inserted child order. Every successor appeared after v and therefore after the inserted child order. Removing the inserted children restores the original context order. □

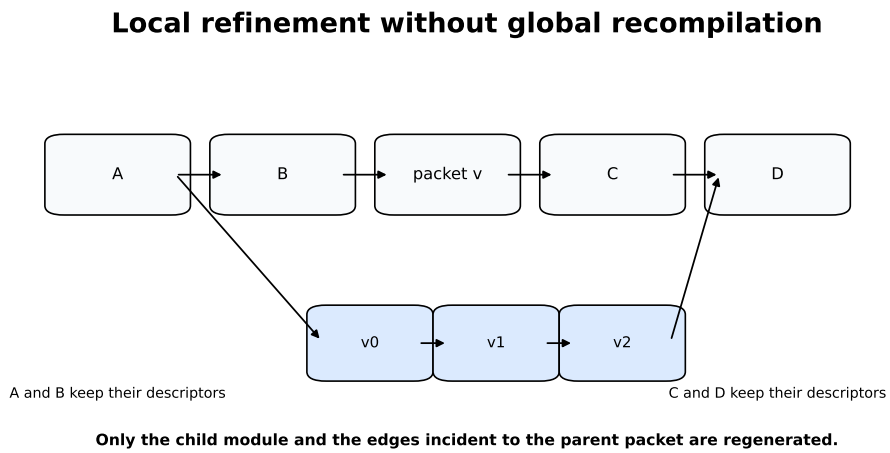


Figure 8.2: A parent packet is replaced by a local child graph while remote descriptors remain unchanged.

9 Transform fabric

9.1 One configurable engine family

The transform engines share butterfly, permutation, lifting, and selector hardware. They support FFT, DCT, FWHT, Haar, biorthogonal lifting, and polyphase transforms.

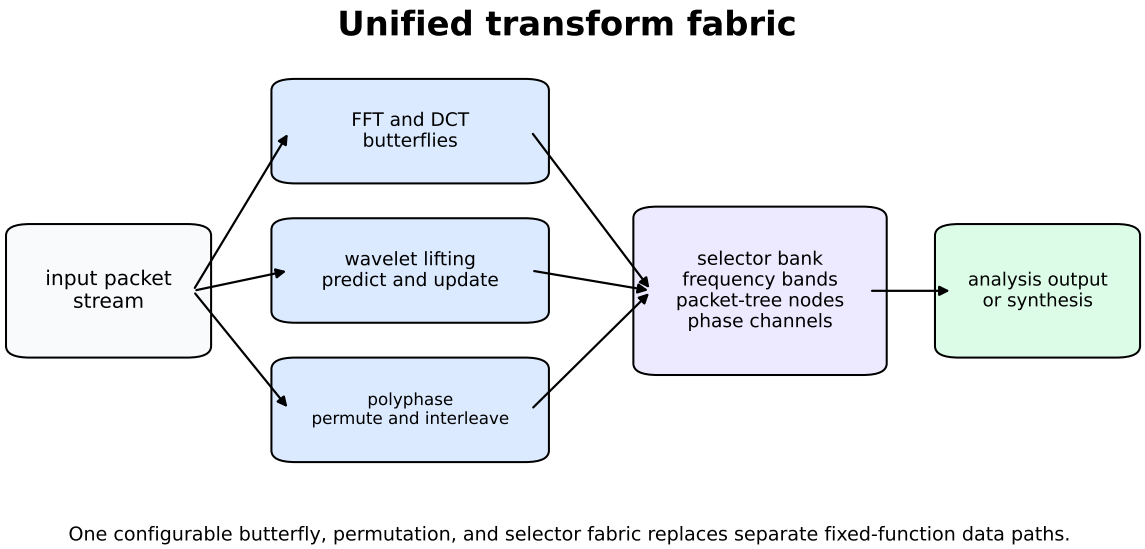


Figure 9.1: The transform fabric supports several exact atlases without storing separate fixed data paths for each one.

9.2 Exact analysis and synthesis

Let U be unitary and let E_α be coordinate selectors. The pulled-back projectors are

$$P_\alpha = U^* E_\alpha U.$$

The hardware analysis applies U and selects the coordinates. Synthesis scatters the selected coordinates and applies U^*.

Theorem 9.1 (Transform-atlas hardware contract)**.** *If the selector family is complete, the analysis and synthesis command graphs satisfy*

$$\boxed{\mathcal{S}\mathcal{A} = I.}$$

If the selector family is incomplete, the certificate must report the omitted-coordinate residual.

Proof. For a complete selector family, $\sum_\alpha E_\alpha = I$. Hence

$$\mathcal{S}\mathcal{A}x = U^* \left(\sum_\alpha E_\alpha \right) Ux = U^* U x = x.$$

The incomplete case replaces the identity by the declared selector sum. □

10 Transfer engines

10.1 Block diagonal and block sparse

Block-diagonal operators run as independent shape batches. Block-sparse operators are grouped by edge shape and destination. Destination packet accumulation can be local, multicast, or reduced through the mesh.

10.2 BDLR dataflow

For

$$A = D + UV^*,$$

the apply is

$$AX = DX + U(V^* X).$$

The diagonal blocks remain distributed. The $r \times q$ contraction is reduced and broadcast.

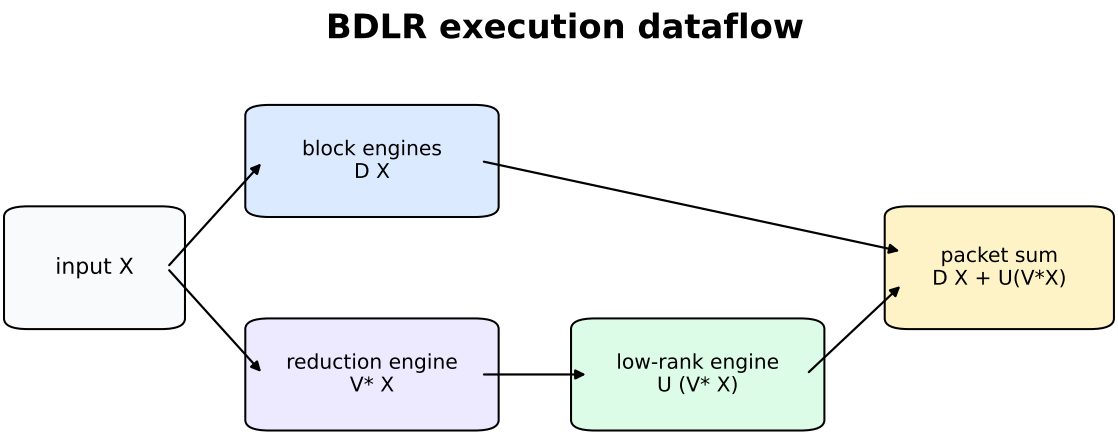


Figure 10.1: The BDLR path keeps the low-rank representation compressed.

Theorem 10.1 (BDLR apply count)**.** *For q right-hand sides and equal packet width b, the modeled arithmetic count is*

$$\boxed{2qn(b+2r).}$$

A dense apply requires $2qn^2$ operations.

Proof. The block-diagonal product costs $2qnb$. The contraction V^*X costs $2nrq$. Multiplication by U costs another $2nrq$. Adding the terms does not change the leading expression. □

10.3 Where BDLR remains useful

The low-rank path is not uniformly favorable. Its benefit depends on block width, correction rank, right-hand-side count, and factor reuse. The reference model sweeps a range of block sizes and ranks while holding $n = 8192$ and $q = 32$ fixed.

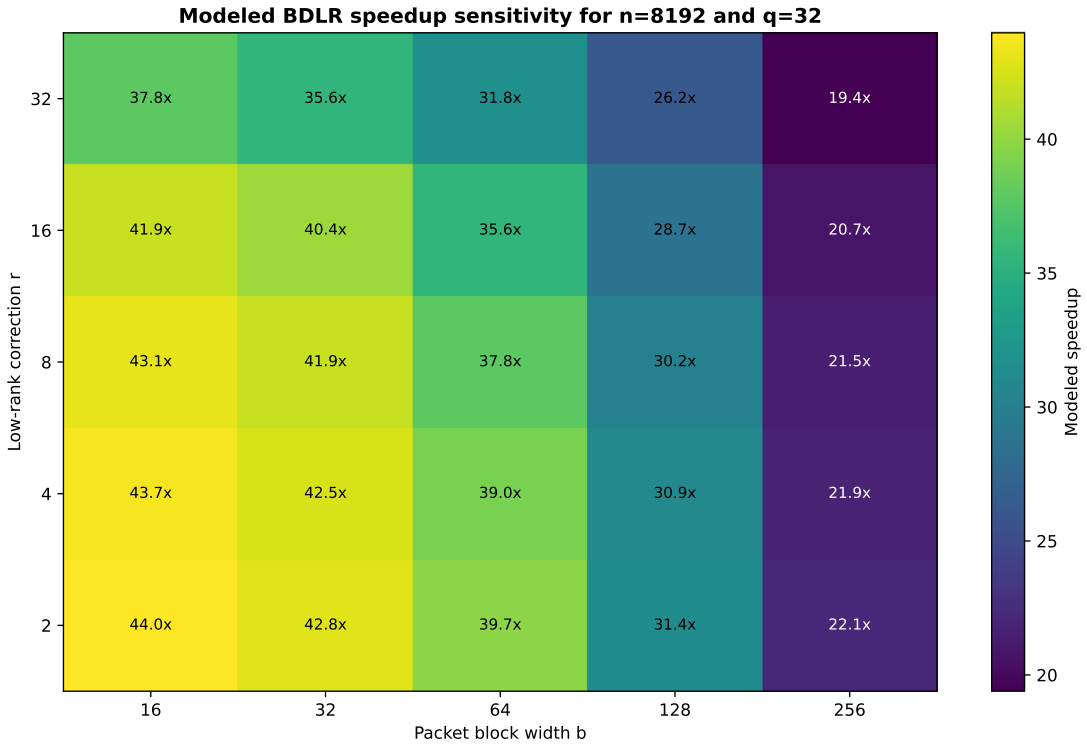


Figure 10.2: Modeled BDLR speedup across block width and correction rank. The comparison includes arithmetic, data movement, and graph overhead from the reference model.

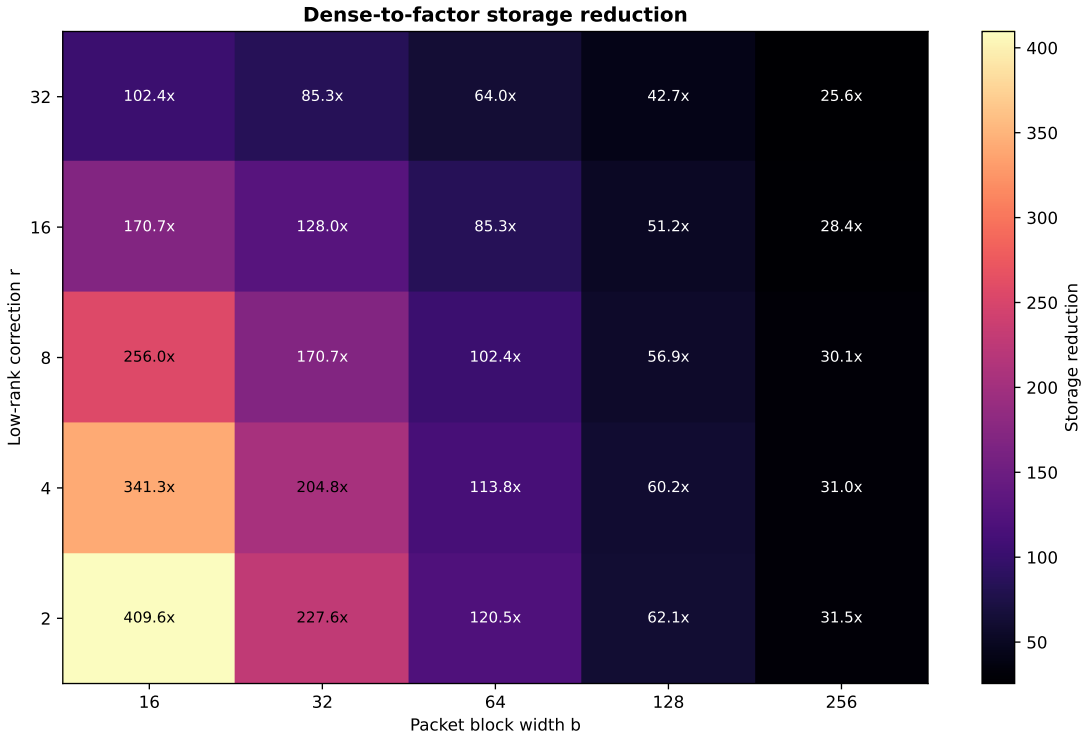


Figure 10.3: Dense-to-factor storage reduction for the same parameter sweep.

The compiler should not infer a gain from the name of the representation. It should evaluate the actual

block width, rank, reuse, and output contract. A high-rank correction or a request for immediate dense materialization may remove most of the benefit.

10.4 Compressed composition

When downstream stages accept BDLR factors, composition returns new block factors and low-rank factors. Materialization is a separate output contract. The compiler cannot compare a compressed product with a dense explicit matrix without accounting for that difference.

10.5 Woodbury solve

The chip uses packet solves with D_α and one small dense correction:

$$(D + UV^*)^{-1} = D^{-1} - D^{-1}U(I + V^*D^{-1}U)^{-1}V^*D^{-1}.$$

The solver engines handle the rank-sized middle system.

11 Mixed precision and certificates

11.1 Per-edge precision

A global dtype ignores the fact that packet transfers can have very different sensitivities. Each edge carries its precision and sensitivity.

Theorem 11.1 (Packet error bound)**.** *Let*

$$A = \sum_{e \in E} A_e, \qquad \widehat{A} = \sum_{e \in E} (A_e + E_e).$$

If edge e starts in packet $\alpha(e)$, then

$$\boxed{\left\| (\widehat{A} - A)x \right\| \le \sum_{e \in E} \|E_e\| \left\| P_{\alpha(e)} x \right\|.}$$

Proof. Exact packet reconstruction gives

$$(\widehat{A} - A)x = \sum_e E_e P_{\alpha(e)} x.$$

The triangle inequality gives the bound. □

11.2 Precision allocation

Under a logarithmic cost model and total error budget τ, the continuous allocation is

$$\varepsilon_e = \frac{\tau c_e}{s_e \sum_f c_f},$$

where s_e is sensitivity and c_e is cost. The compiler maps the result to supported formats.

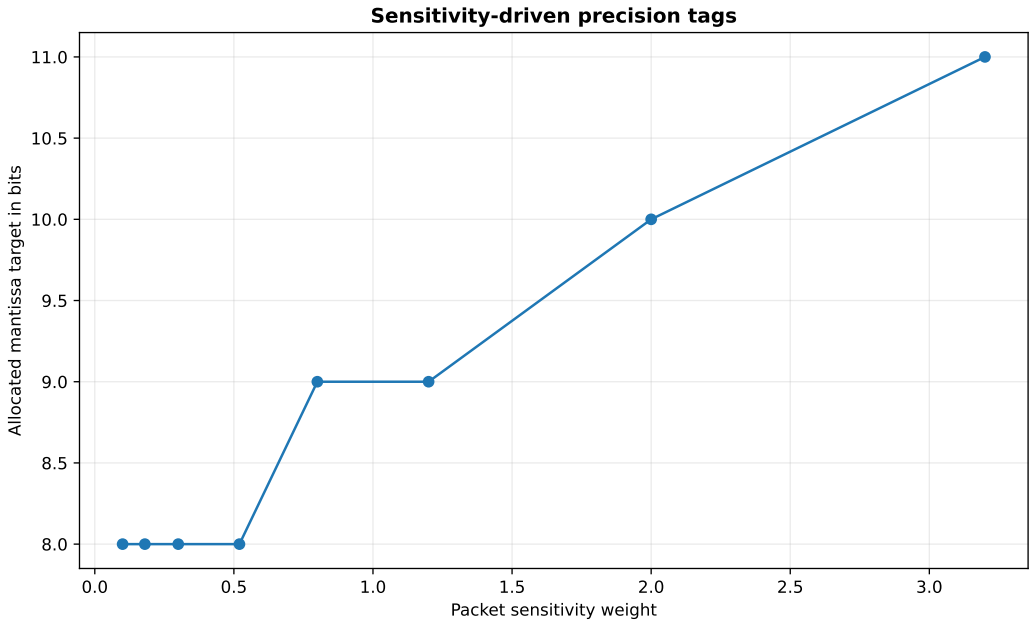


Figure 11.1: More sensitive packet edges receive tighter precision targets.

12 Unknown-edge and Gibbs reconstruction

12.1 Why the chip includes a small-solver path

The self-calibrating reconstruction uses large transforms and very small algebraic systems in one graph. A general GPU often pays high launch and movement cost around the small systems. PRISM keeps the full chain resident.

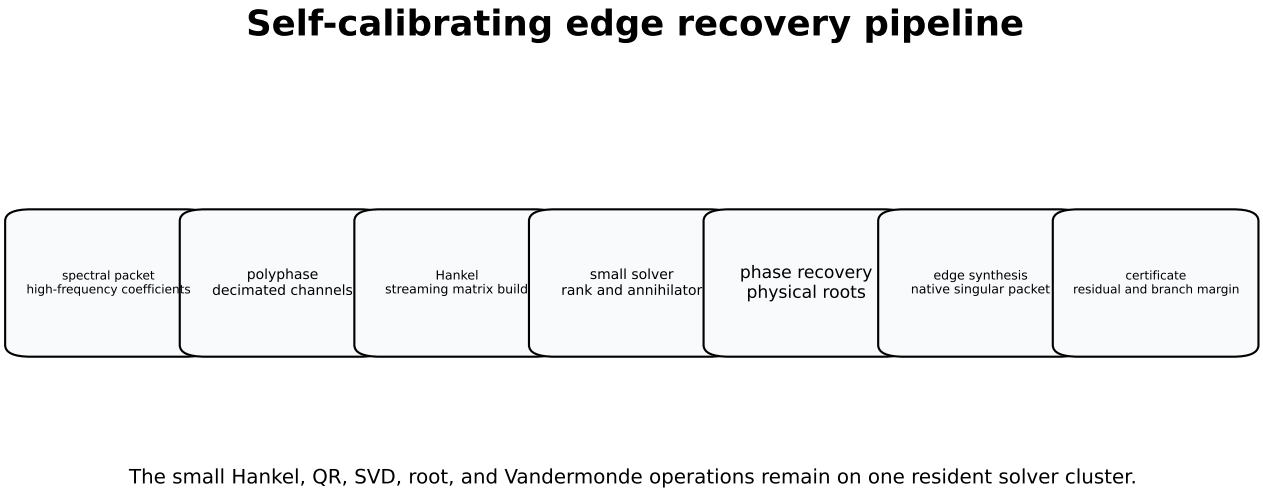


Figure 12.1: The self-calibrating edge-recovery graph from high-frequency packet to native singular synthesis.

12.2 Resident Hankel and Prony operations

The edge-recovery engines perform:

1. Fourier packet extraction
2. decimated polyphase grouping
3. streaming Hankel construction
4. rank estimation
5. annihilator solve
6. polynomial root recovery
7. physical phase selection
8. Vandermonde amplitude solve
9. edge packet synthesis
10. residual and branch-margin checks

12.3 Small-solver organization

The solver cluster is built for matrices that are too small to fill the PSM mesh but too frequent to send back to the host. A local panel store feeds QR, Cholesky, Jacobi-style SVD, companion-matrix roots, and Vandermonde solves. FP64 is the default format for rank decisions, phase recovery, and branch-margin checks.

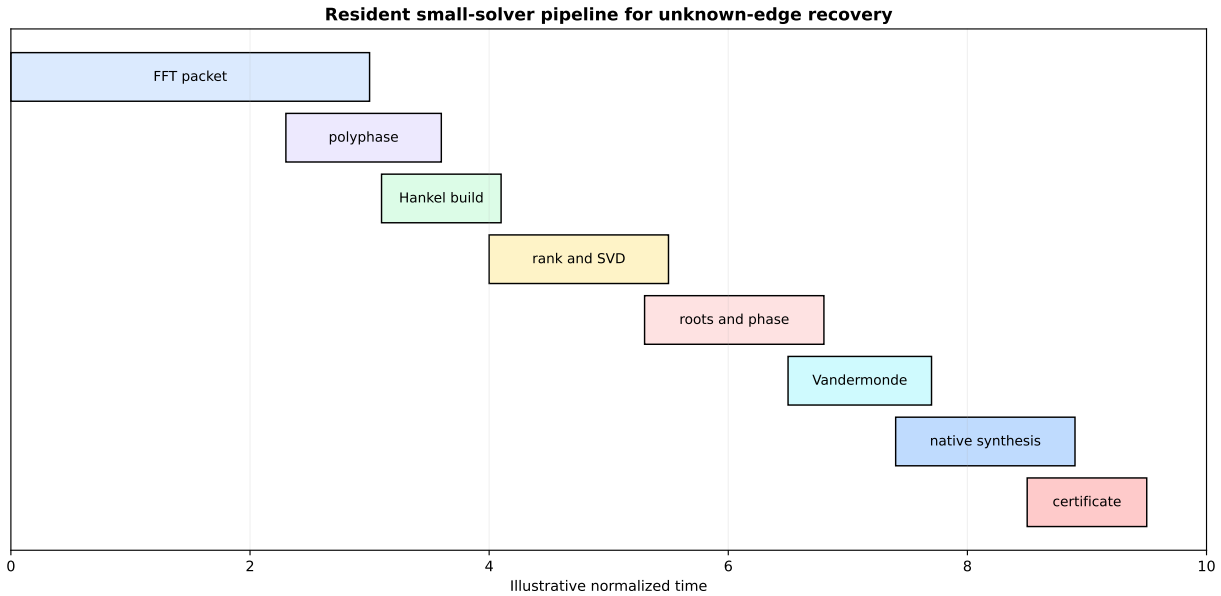


Figure 12.2: Illustrative resident pipeline for unknown-edge recovery. Transform and solver stages overlap where data dependencies permit.

The solver cluster is also used by Bayesian regression batches, Woodbury corrections, packet Krylov restarts, and small exactness checks. It is not intended to compete with a large dense eigensolver.

12.4 Why native edge synthesis matters

After edge recovery, the discontinuity is synthesized as an edge packet. The smooth residual is reconstructed spectrally. The chip does not force the recovered singular packet back through a truncated global Fourier sum.

13 PR-LASM execution

13.1 Exact archive and compact state

The exact observation archive is kept in complete spatial, Fourier, Haar, polyphase, and categorical atlases. Macro, meso, and micro states are separate statistical packet types with dimensions 48, 96, and 192 in the reference model.

13.2 Resident state layout

The exact archive and compact predictive states remain separate in memory and in the descriptor type system. This prevents a compact state vector from being mistaken for an invertible representation of the signal.

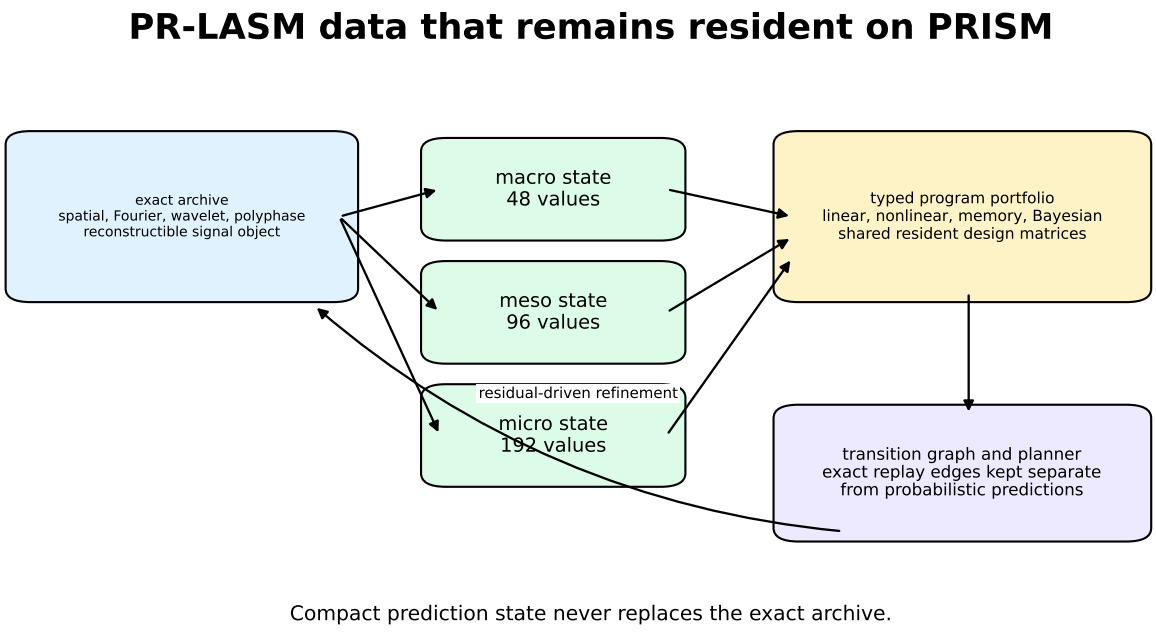


Figure 13.1: Exact archive, compact states, mechanism portfolio, and transition graph in one resident PR-LASM step.

13.3 Typed Bayesian programs

The mechanism portfolio shares state matrices while allowing different operators, atlases, resolutions, and memory laws. The solver engines batch models by feature shape.

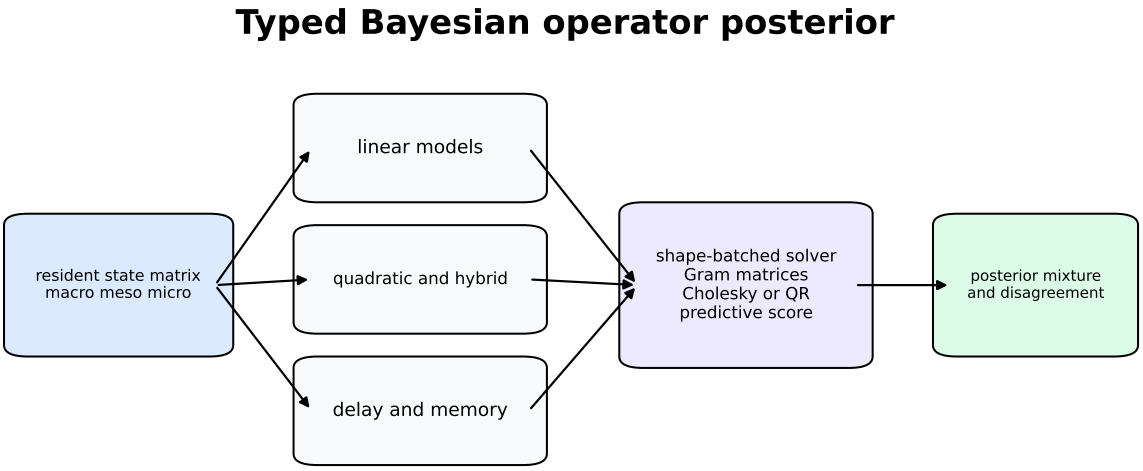


Figure 13.2: Mechanism families share resident state matrices and small-solver batches.

13.4 Memory and planning

Delay, power-law, exponential, and Volterra memory packets run in the nonlinear and memory units. The graph scheduler stores exact deterministic transitions separately from probabilistic predictions. Replay stops when a state or certificate mismatch is detected.

14 Exact arithmetic mode

14.1 Residue and modular packets

One exact arithmetic cluster per compute chiplet supports 64-bit modular addition, multiplication, valuation, residue selection, and CRT merge. Longer values use limb vectors in PSM scratchpad.

14.2 Why this belongs on the chip

The arithmetic calculus and the signal calculus share the same structural operations:

$$\text{residue packet} \to \text{transfer} \to \text{refinement} \to \text{weighted reconstruction}.$$

The exact arithmetic path is smaller than the floating-point path, but it allows the same descriptor and graph system to execute p-adic, recurrence, and finite-field packet workloads.

15 Packet Seed Command interface

15.1 Command groups

The first implementation uses memory-resident 256-bit commands. RISC-V control cores fill a queue and ring a doorbell. The command groups are shown below.

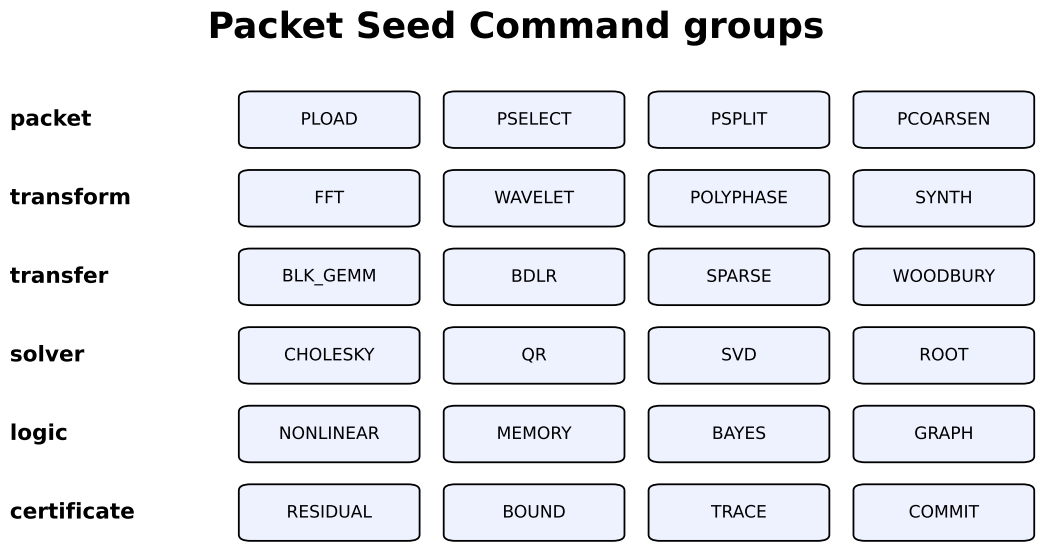


Figure 15.1: Packet Seed Command groups.

15.2 Representative commands

Command	Meaning
`PSELECT`	Apply a declared packet selector without materializing a projector matrix
`PSPLIT`	Replace a parent packet by declared children
`PCOARSEN`	Reconstruct a parent from a complete child family
`POLYPHASE`	Extract, permute, or interleave phase channels
`WAVELET_LIFT`	Execute predict, update, and scale steps
`BLK_GEMM`	Execute a batch of equal-shape transfer blocks
`BDLR_APPLY`	Apply block diagonal plus low rank without materialization
`HANKEL_BUILD`	Stream moments into a small Hankel matrix
`ROOT_PHASE`	Recover physical roots from decimated channels
`BAYES_FIT`	Fit one shape-class batch of conjugate regressions
`GRAPH_REFINE`	Replace one parent task by a local child graph
`RESIDUAL`	Form and store a reconstruction residual
`COMMIT`	Mark a result valid after required certificates pass

15.3 Completion record

Every completed command produces a compact record containing the completion token, status, cycle count, bytes moved, exception code, residual summary, and certificate pointer. Downstream dependencies consume the token only after the commit state is reached.

15.4 Protection and replay

Descriptor tables, command queues, and certificate records require ECC or parity protection. A command that encounters an uncorrectable descriptor error is aborted before execution. A certificate failure invalidates only the declared replay region unless the error is marked package-wide.

16 Compiler

16.1 Routing policy

The compiler first asks whether the structure changes the cost of the computation. A named packet decomposition is not sufficient. The route decision uses representation size, expected reuse, graph size, materialization requirements, and measured or modeled engine cost.

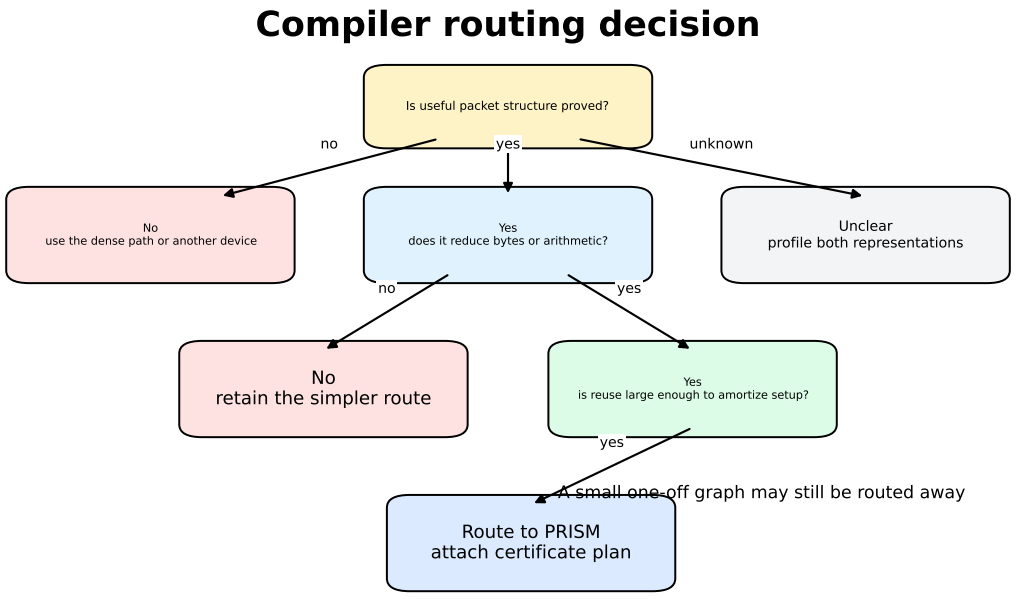


Figure 16.1: Reference routing logic. The final compiler will use calibrated costs, but the reject path remains part of the specification.

16.2 Shared Packet IR

The CUDA and PRISM backends consume the same typed graph. Packet IR records representation form, packet and atlas identity, output contract, precision, reuse, dependencies, and certificate requirements.

Backend selection does not change the mathematical result.

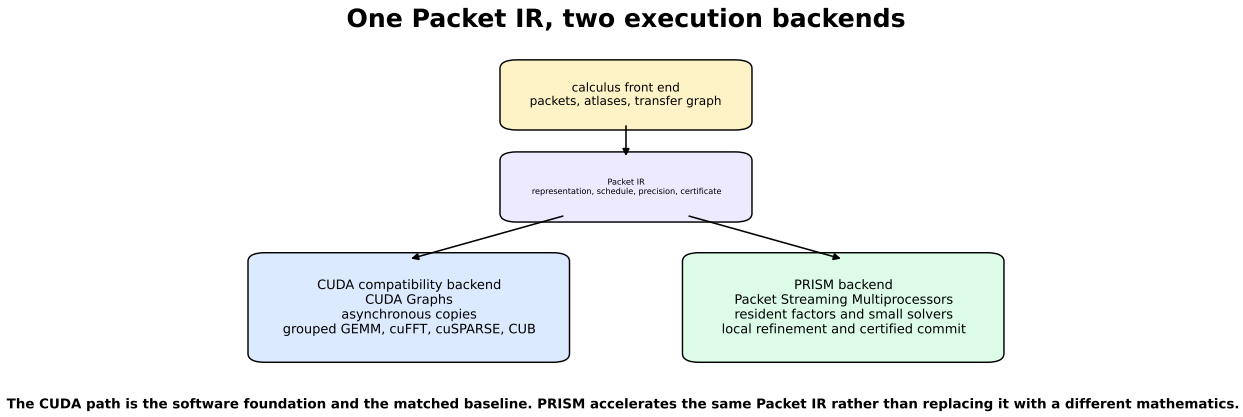


Figure 16.2: Shared Packet IR makes CUDA the software foundation and matched baseline for PRISM.

16.3 Compiler flow

The compiler begins with a typed graph rather than an unstructured kernel list.

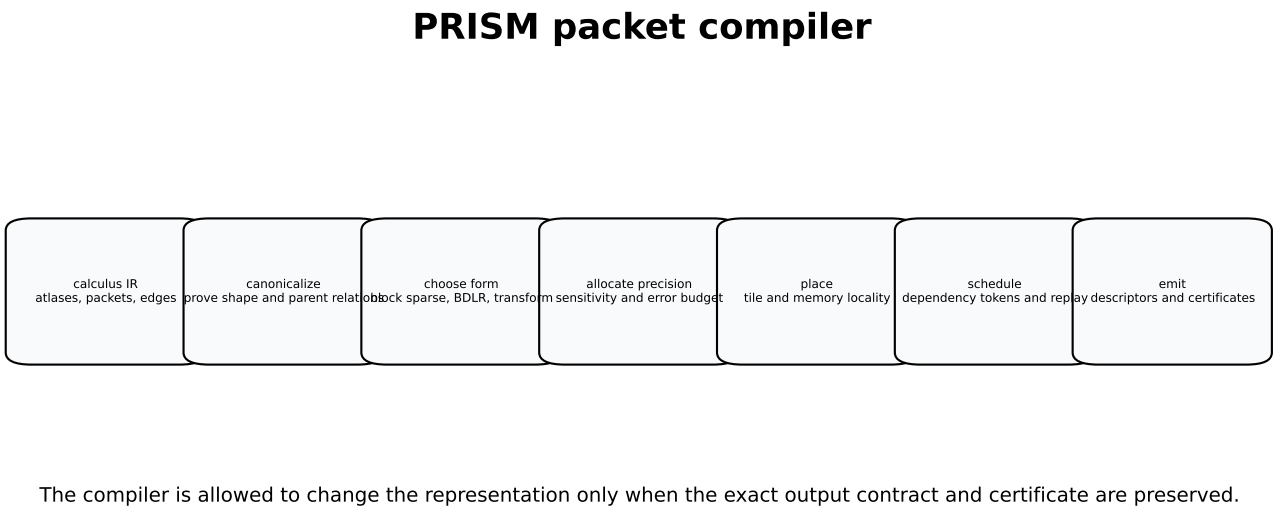


Figure 16.3: The packet compiler preserves the output contract while changing representation, placement, precision, and schedule.

16.4 Legality pass

Before optimization, the compiler checks packet completeness, shape compatibility, parent-child identities, analysis-synthesis pairing, output contracts, and certificate requirements.

16.5 Representation pass

The operator is classified as explicit dense, block diagonal, block sparse, low rank, BDLR, Kronecker, sparse factor chain, transform generated, packet graph, small solver, or nonlinear packet interaction. A form is chosen only when it reduces the declared cost or is required by the output contract.

16.6 PMR–GEMM planning

The representation pass invokes the Bellman planner on the candidate packet tree. The planner considers generic, block, low-rank, Kronecker, sparse-chain, transform, and split choices. The explicit array contract filters out plans that do not return every entry. Apply, factor, and transform contracts leave compressed plans available.

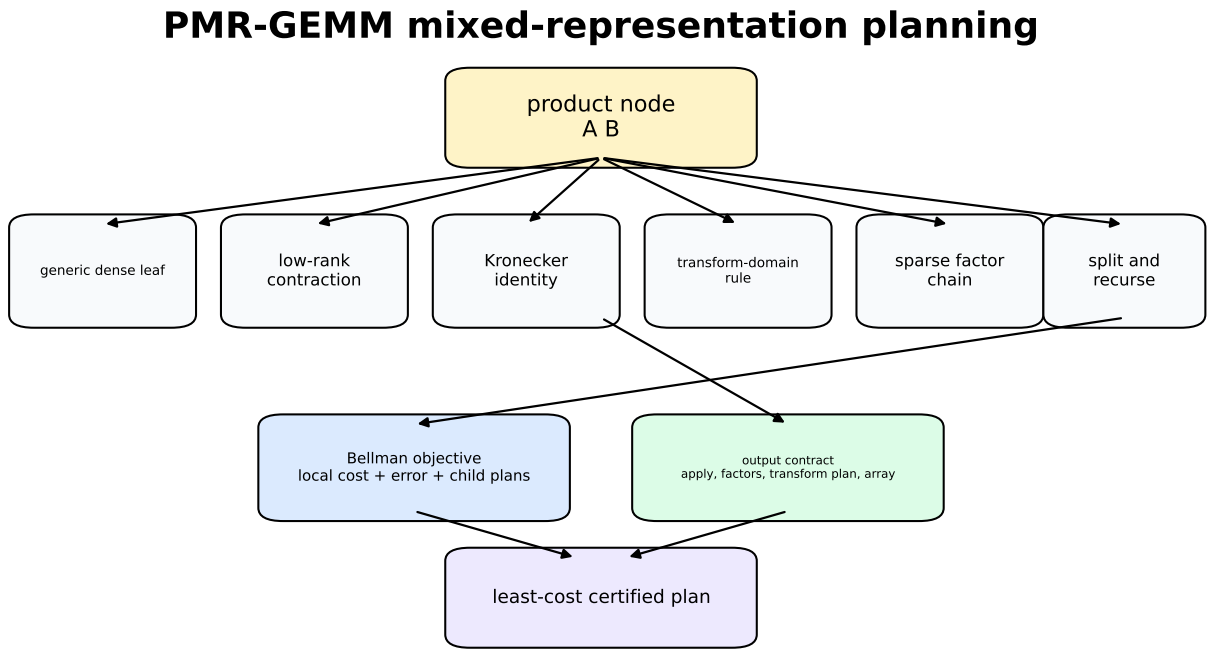


Figure 16.4: The mixed-representation planner selects a certified plan under the requested output contract.

16.7 Placement and fusion

The compiler keeps together parent and child packets, diagonal blocks with corresponding factor rows, transform stages sharing one coefficient stream, and small solver chains. Analysis, packet selection, feature reduction, and synthesis may be fused when the exact atlas contract is unchanged.

16.8 CUDA compatibility backend

The CUDA backend maps Packet IR to grouped GEMM, cuBLAS and cuBLASLt, cuFFT, cuSPARSE, CUB reductions, asynchronous copies, stream-ordered allocation, and CUDA Graph replay. It remains the first implementation target and the reference for same-representation comparisons.

16.9 PRISM backend

The PRISM backend replaces launch-heavy graph regions with resident packet warps, PSM scratchpads, representation routing, transform-to-solver handoff, local refinement, and certificate-gated commit. The backend does not change the representation chosen by the compiler unless the cost model requests a new plan.

16.10 Worked BDLR compilation

A BDLR operator enters the compiler as the typed identity $A = D+UV^*$. No explicit matrix is constructed. The legality pass checks dimensions and the requested output form. The representation pass keeps the factors. Placement co-locates each diagonal block with the corresponding rows of U and V. Precision is assigned per transfer contribution. The schedule computes DX, reduces V^*X, broadcasts the small contraction, and forms $U(V^*X)$ locally.

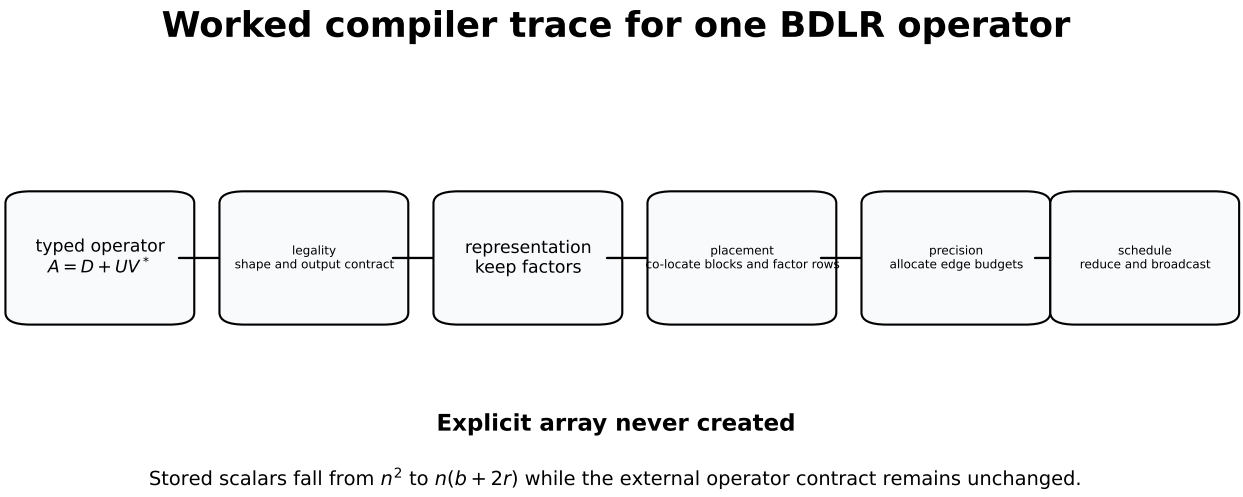


Figure 16.5: One BDLR operator through the compiler. The dense matrix is never formed.

16.11 Certificate plan

The compiler decides where direct residuals are affordable and where proven upper bounds are used. A graph may contain several certificate boundaries rather than one final check.

17 End-to-end execution traces

17.1 Block diagonal plus low rank

The host submits packet descriptors for the diagonal blocks and factor rows. The graph contains one block batch, one reduction for V^*X, one factor broadcast, one local update, and one certificate. The dense

matrix is absent from the graph and from memory.

The external contract is still ordinary matrix application. A direct residual may compare the structured output with a dense reference during bring-up. Production runs can use a proven bound or a spot-check policy when a full dense reference would defeat the representation saving.

17.2 Unknown-edge reconstruction

The transform engine extracts the chosen Fourier packet. The projector DMA groups decimated phase channels. The solver cluster builds the Hankel matrix, estimates rank, finds the annihilator, recovers roots, resolves physical phases, and solves amplitudes. The recovered edge packet is synthesized directly. The smooth residual follows the spectral path. A final residual and branch-margin record are written before commit.

17.3 One PR-LASM step

The exact archive remains in HBM. Macro, meso, and micro states remain in SRAM. Mechanism families reuse the same state matrices. The small solvers update their posterior records. The planner reads exact transition edges separately from probabilistic predictions. A large residual can request local packet refinement without invalidating the rest of the graph.

17.4 Why these traces matter

The three examples use different engines but share one architectural pattern. The representation remains typed, local work is scheduled by dependency tokens, intermediate factors remain resident, and the result is not committed until the declared check passes.

18 Compositional correctness

Theorem 18.1 (Hardware reconstruction theorem)**.** *Let a finite calculus graph have exact packet analysis and synthesis, exact transfer kernels, and an acyclic dependency graph. Suppose every hardware command implements its declared local map and every packet sum uses the declared complete family. Then the committed hardware output equals the mathematical graph output.*

Proof. Order the graph topologically. Each source command loads the declared seed or packet. Assume every predecessor of a node has the correct value. The command implements the node's declared local map, so the node output is correct. Packet accumulation is exact because the declared child or source family is complete. Induction reaches every sink. The final synthesis therefore equals the mathematical graph output. The commit command changes status, not value. □

Theorem 18.2 (Certified commit)**.** *Suppose a command result is provisional until a certificate $c(y)$ is evaluated. If the commit controller releases y only when $c(y)$ satisfies the declared acceptance rule, then every visible result has passed that rule. A failed result cannot satisfy a downstream completion token.*

Proof. The provisional buffer is not addressable through the committed packet directory. Passing the rule changes the directory state and releases the completion token. Failure leaves the committed state

unchanged and redirects the declared graph region to replay or abort. Downstream commands therefore cannot observe an unchecked value through the normal dependency path. □

Corollary 18.3 (Local refinement correctness)**.** *If a parent packet is replaced by a child graph whose synthesis equals the parent interface, the refined hardware graph has the same external mathematical output.*

Proof. The topological-splice theorem gives a legal schedule. Exact child synthesis gives the same value at the parent interface. Every downstream command therefore receives the same input. □

19 Performance accounting

19.1 Four different sources of gain

A fair result must say what changed. The total route may improve because the mathematics removed work, because the same representation ran better on another device, because a kernel was fused or graph-replayed, or because one-time setup was amortized. These are different quantities:

$$S_{\text{total}} = S_{\text{representation}} S_{\text{architecture}} S_{\text{kernel}} S_{\text{amortization}}.$$

This is a bookkeeping identity only when every factor comes from matched rows with the same output contract and timing scope. Otherwise the factors are reported separately.

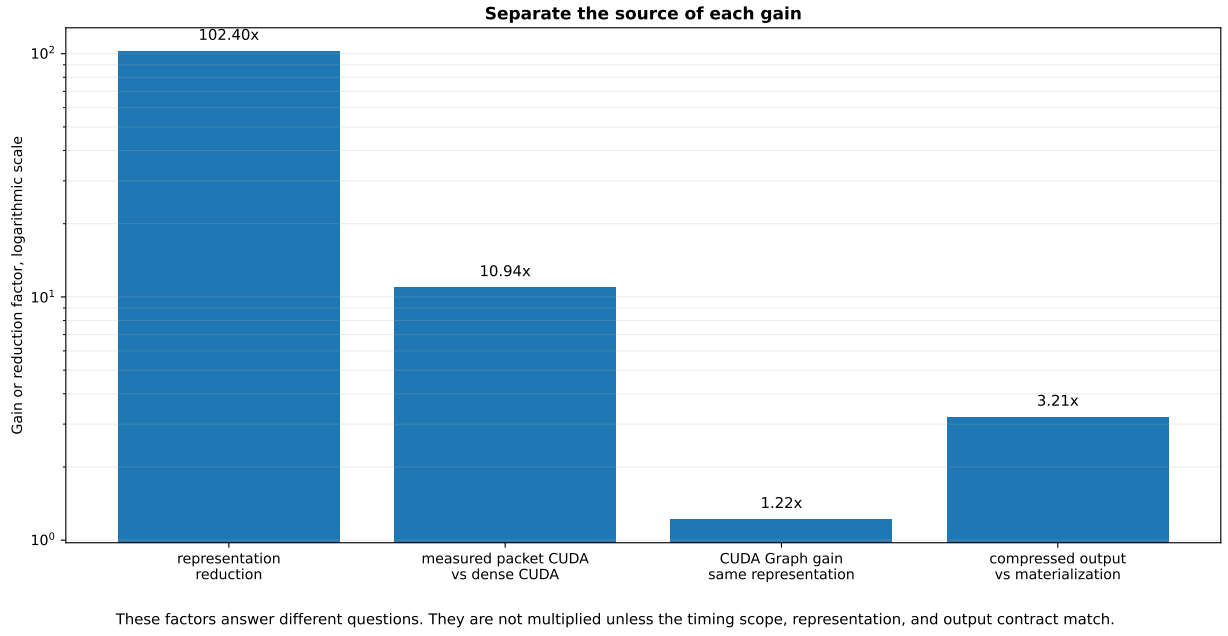


Figure 19.1: Representation, packet-CUDA, graph-replay, and output-contract gains answer different questions.

19.2 Measured CUDA foundation

The prior packet-GPU study contains 154 matched CPU and GPU rows. The recorded GPU is an RTX 3070 Ti Laptop GPU. For $n = 8192$, block width 64, and 32 right-hand sides, the equal-block packet path took 0.046032 ms and was 14.8919× faster than the matched device-resident dense GPU row. For BDLR

with rank eight, the standard packet path took 0.097280 ms and CUDA Graph replay took 0.079872 ms. The graph row was 10.9355× faster than the matched dense GPU row. The graph improvement over the standard packet path was about 1.218×.

A compressed BDLR product at $n = 1024$ took 0.1582 ms. Materializing the explicit output took 0.5073 ms. Those rows have different output contracts. The comparison shows the cost of requesting every entry. It is not a claim that one result is a faster implementation of the other.

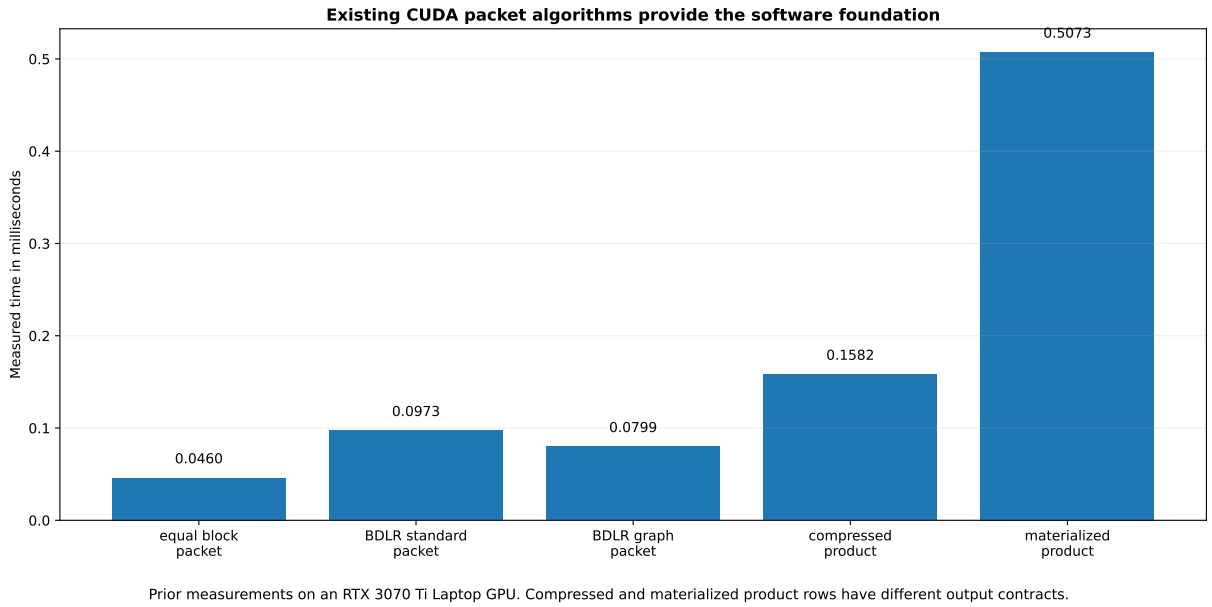


Figure 19.2: Measured CUDA packet timings from the earlier structured-operator study.

19.3 Representation reduction

For

$$A = D + UV^*,$$

with dimension n, block width b, rank r, and q right-hand sides, a factor-first application uses

$$2qn(b + 2r)$$

modeled operations and stores

$$n(b + 2r)$$

scalars. The corresponding dense counts are $2qn^2$ and n^2. For

$$n = 8192, \qquad b = 64, \qquad r = 8, \qquad q = 32,$$

the arithmetic and storage reduction is

$$\boxed{\frac{n}{b + 2r} = 102.4.}$$

This reduction belongs to the representation. A CUDA backend and a PRISM backend can both use it.

19.4 PRISM architecture model

The architecture model uses

$$T = \max\left(\frac{F}{\eta_c P}, \frac{B}{\eta_b W}\right) + T_{\text{graph}},$$

where F is arithmetic, B is data movement, P is peak arithmetic, W is bandwidth, and η_c, η_b are efficiency factors. The model is used to decide which hardware paths are worth building. It is not a silicon measurement.

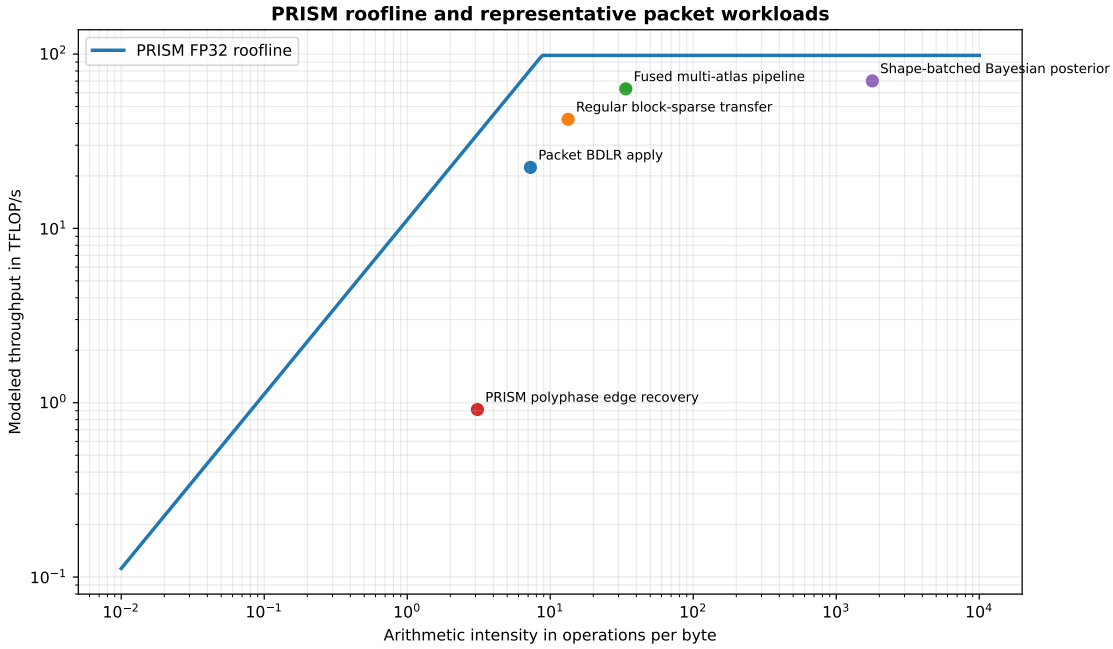


Figure 19.3: Reference PRISM roofline with representative packet workloads.

19.5 Total route model and architecture-only models

The earlier dense-route comparison predicts 70.79 microseconds for a dense application and 1.87 microseconds for the PRISM BDLR route. The ratio is about 37.8. This is a total route model. It includes representation reduction and therefore is not labeled PRISM-over-CUDA architecture gain.

The more relevant architecture models begin with the same structured pipeline on both sides. They estimate:

Workload	Generic structured path	PRISM path
Fused multi-atlas analysis	14.73 μs	8.96 μs
Polyphase edge recovery	4.00 μs	1.35 μs
Bayesian model portfolio	427.93 μs	300.52 μs

The corresponding modeled gains are 1.64×, 2.96×, and 1.42×. These are the present estimates of what residency, fusion, local solvers, and packet scheduling may add after a structured representation is already in use.

19.6 Sensitivity and crossover

No single row should define the architecture. BDLR gains depend on block width and correction rank. Transform gains depend on size, reuse, and fusion. Sparse graphs depend on regularity. Small solvers depend on batch size and handoff cost.

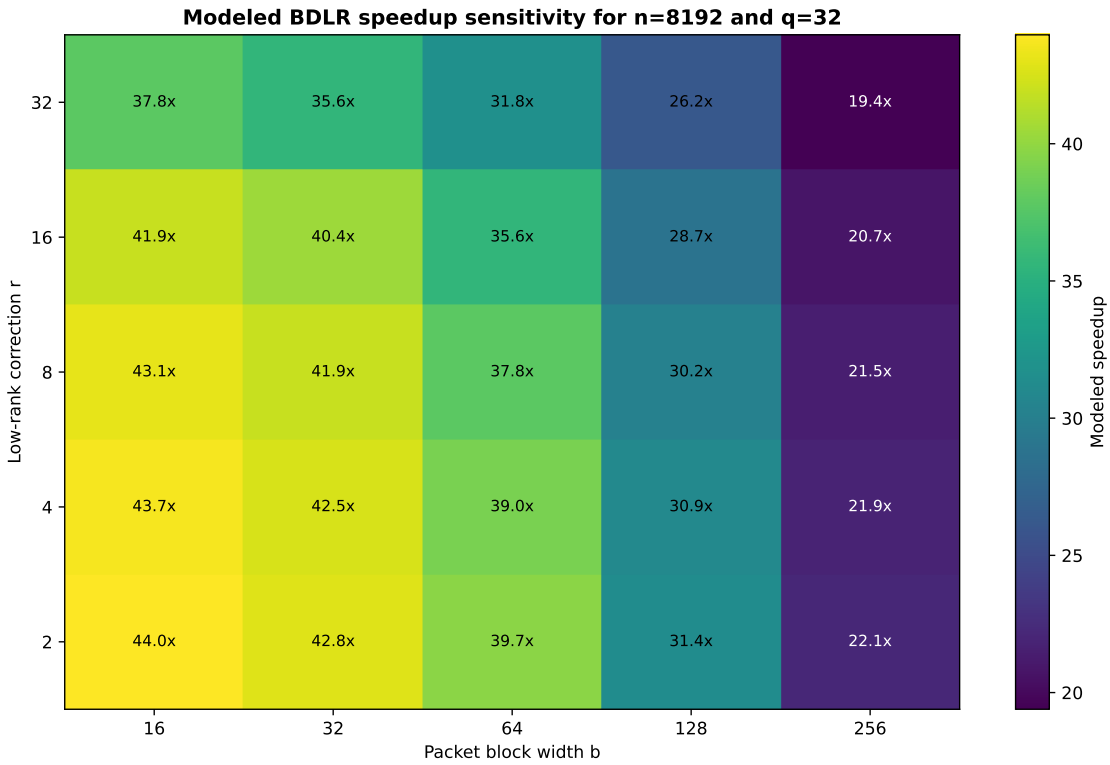


Figure 19.4: Total route sensitivity across block width and correction rank.

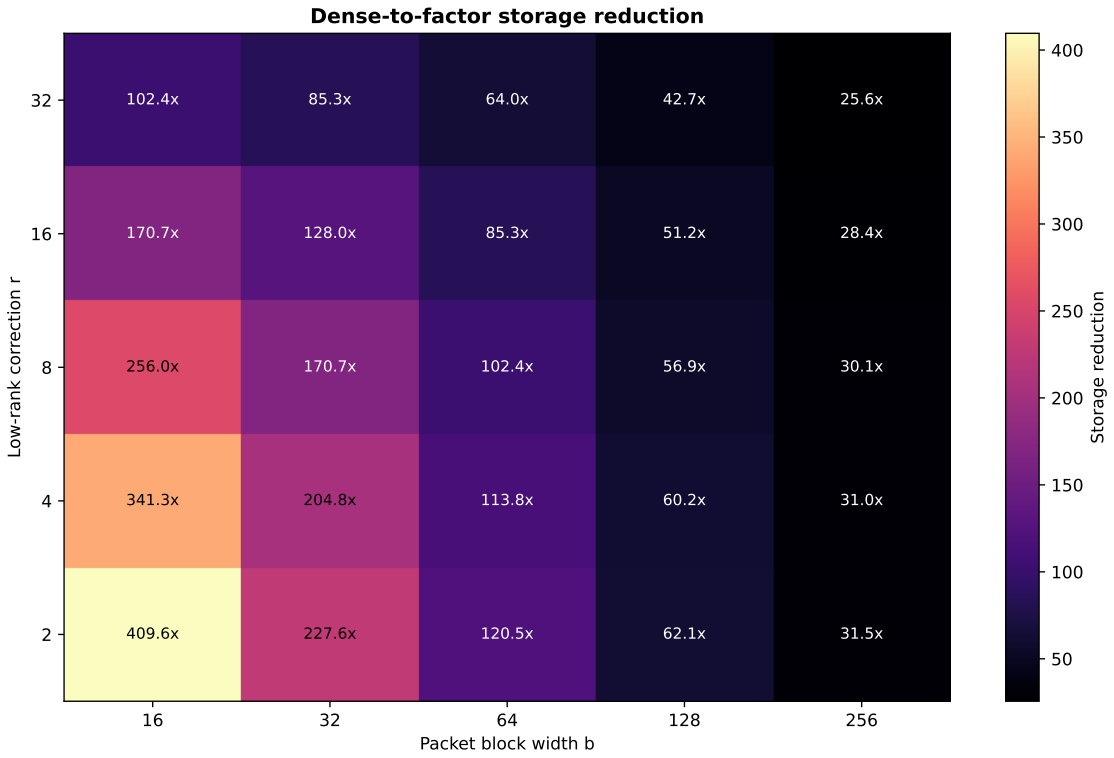


Figure 19.5: Representation-storage reduction across block width and correction rank.

19.7 Negative controls

The earlier CUDA study retained cases where structure lost. Recursive HODLR was slower than dense GPU execution in the tested case. A GPU Walsh-Hadamard transform lost to the CPU at the smaller tested length. A two-stream dense pipeline lost to the serial path. Iterative refinement lost to a direct FP64 solve. The PRISM compiler must preserve these negative controls and route similar work away from specialized paths.

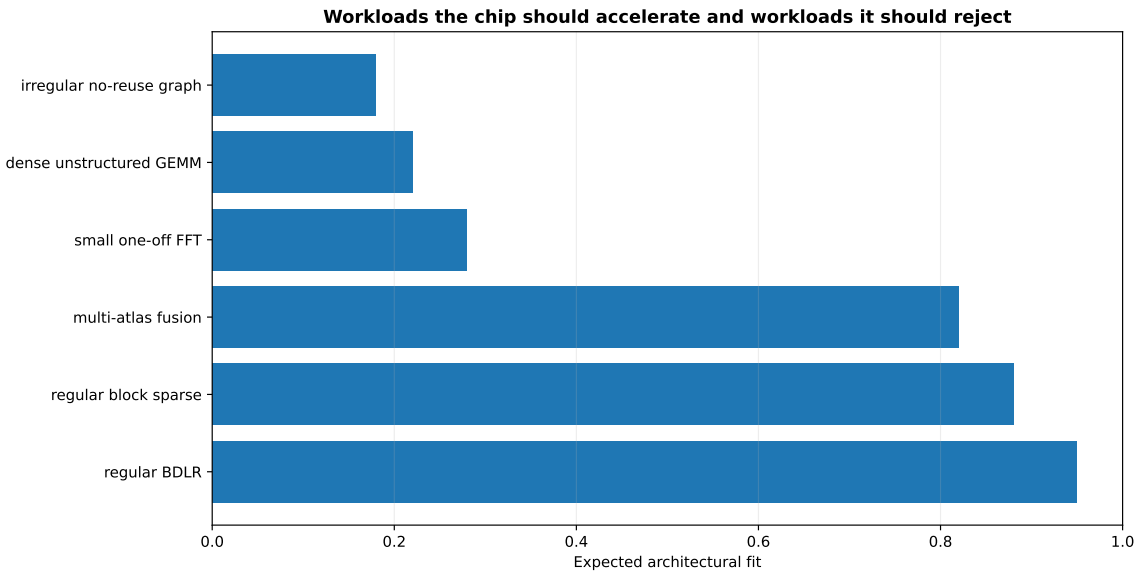


Figure 19.6: PRISM should accelerate strong reusable structure and reject weak or one-off structure.

20 Power, area, and thermal design

20.1 Power budget

The reference package targets a 500 W envelope. The model allocates power to execution groups, SRAM, the packet network, HBM PHYs, transform and solver blocks, control, and margin.

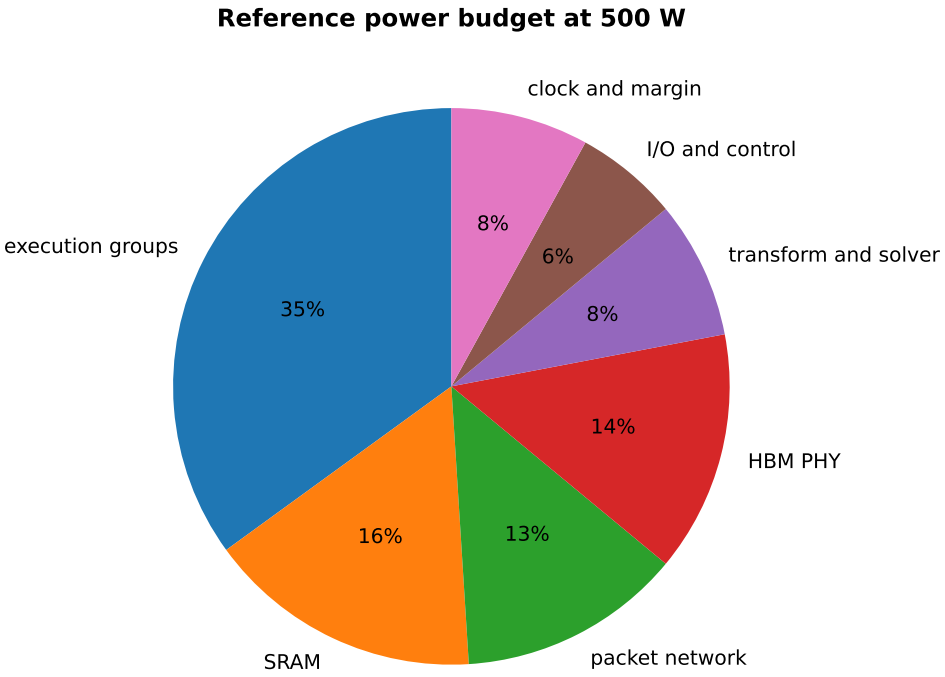


Figure 20.1: Reference power allocation.

20.2 Area allocation

Distributed SRAM is the largest target area because the architecture depends on resident packets and factors. Matrix and vector lanes occupy the next largest share.

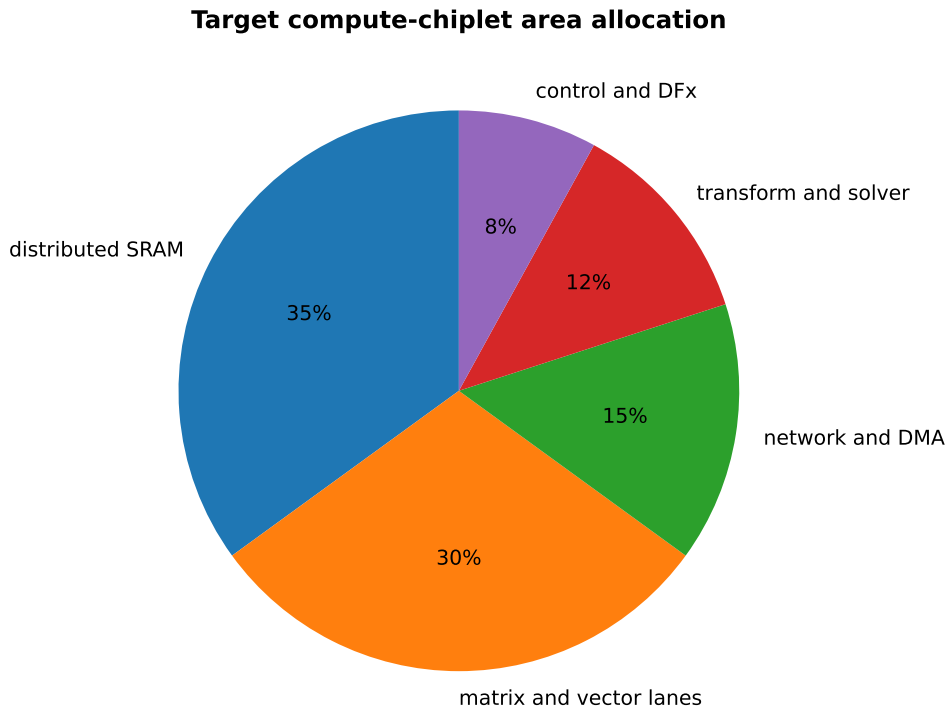


Figure 20.2: Target compute-chiplet area allocation.

20.3 Thermal behavior

The scheduler can spread long transform and matrix batches across chiplets. UCIe priority sideband and fast-throttle features are natural matches for package-wide thermal events. Local descriptor replay can be paused without discarding the packet graph.

21 Reliability, protection, and recovery

21.1 Protected state

Packet descriptors, transfer descriptors, dependency tokens, and certificate records control meaning as well as data movement. They require stronger protection than ordinary cache metadata. The reference plan uses ECC for SRAM-resident tables, parity on command queues, sequence numbers on replay regions, and end-to-end tags on HBM packet records.

21.2 Failure containment

An execution-group or PSM error should not force a package-wide reset unless the packet directory or global scheduler has lost coherence. Local kernel faults invalidate the destination packet and its descendants. A link or memory fault invalidates every packet whose provenance crosses the failed resource. The compiler emits provenance information so the runtime can identify the affected region.

21.3 Determinism

Deterministic mode fixes the reduction tree, solver ordering, and replay schedule. It is slower than the highest-throughput mode. Its purpose is bring-up, regression testing, exact replay, and certificate comparison.

21.4 Security boundary

The architecture does not introduce a new trust model. Host isolation, command validation, memory protection, firmware signing, and secure update remain required. Packet descriptors make the dataflow easier to audit, but they do not replace platform security.

22 Programming model

22.1 Four software layers

The software stack has four levels:

1. a high-level calculus graph
2. shared Packet IR
3. CUDA or PRISM backend lowering
4. Packet Seed Commands, descriptor tables, and the runtime queue

The release includes JSON examples for BDLR application, PMR–GEMM planning, self-calibrating edge recovery, and a PR-LASM step.

22.2 Host control

RISC-V management cores run the command runtime, graph loader, memory manager, telemetry, and error handling. The CUDA backend uses the same Packet IR before final lowering. Ordinary vector code handles control-side data preparation. The accelerator command queue is the stable interface to the packet engines.

22.3 Device libraries

A practical library should expose:

- packet and atlas construction
- transform analysis and synthesis
- transfer graph creation
- BDLR and block-sparse operators
- edge-recovery graphs
- Bayesian program batches
- certificate policies
- graph replay and local refinement

23 Verification

23.1 Verification ladder

The architecture is specified so each development stage has a clear target. Mathematical contracts come first. The behavioral model checks those contracts. Bit-accurate kernels then define rounding and exception behavior. RTL simulation and formal properties check control. An FPGA overlay measures real data movement before a chiplet is attempted.

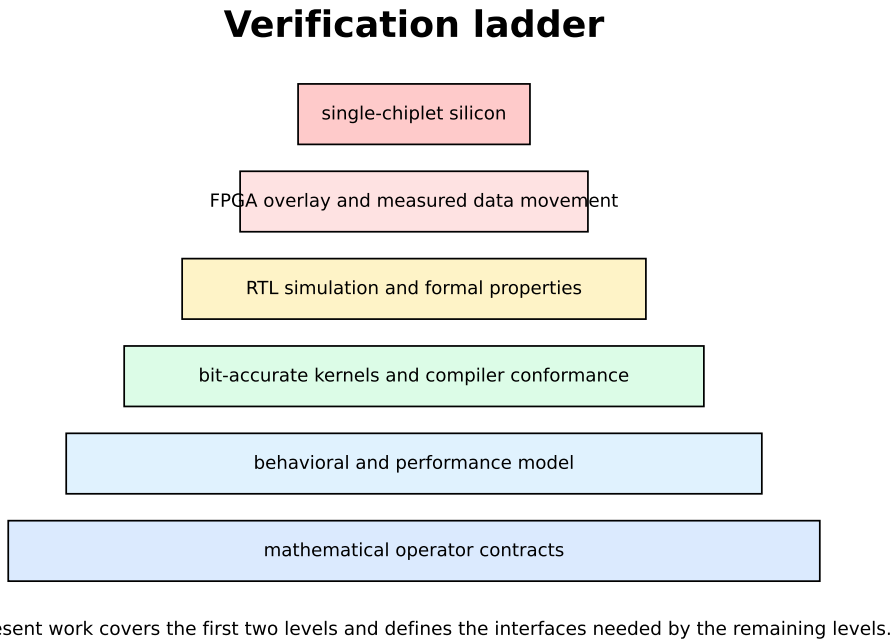


Figure 23.1: Verification ladder from mathematical contracts to measured silicon.

23.2 Behavioral tests

The executable model currently verifies:

- complete residue-packet reconstruction

- exact polyphase round trip
- exact Haar round trip
- BDLR factor execution against dense materialization
- local DAG substitution
- derived package totals
- descriptor contracts
- precision-budget allocation
- certificate-gated commit
- local refinement descriptor bounds
- compiler routing decisions
- BDLR sensitivity tables
- package bandwidth hierarchy
- Packet Streaming Multiprocessor hierarchy
- PMR–GEMM Bellman planning
- output-contract legality without mandatory dense state
- measured CUDA baseline registry
- typed gain decomposition

Certificate 23.1 (Current executable result)**.** *The behavioral and architecture model passes*

$$\boxed{18/18 \textit{ self-tests}}$$

and the independent pytest suite passes

$$\boxed{18/18 \textit{ tests.}}$$

The largest semantic residual is the BDLR comparison residual

$$1.42 \times 10^{-14},$$

which is consistent with floating-point roundoff.

23.3 RTL boundary

The release contains a SystemVerilog architecture skeleton for descriptors, command buffering, dependency tokens, the packet directory, representation routing, PSM scheduling, execution-group control, the residual path, certificate-gated commit, and the top-level queue interface. No HDL compiler or synthesis tool was available in the build environment. The RTL skeleton is therefore not described as verified or synthesizable silicon.

23.4 What must be verified before tapeout

A serious implementation requires:

- bit-accurate kernel models
- formal checks of command ordering and descriptor protection
- CDC and reset verification
- UCIe, HBM, PCIe, and optional CXL compliance IP integration
- SRAM and NoC timing closure
- thermal and power-integrity analysis
- fault injection for replay and certificate failure
- compiler and runtime conformance suites

24 Implementation path

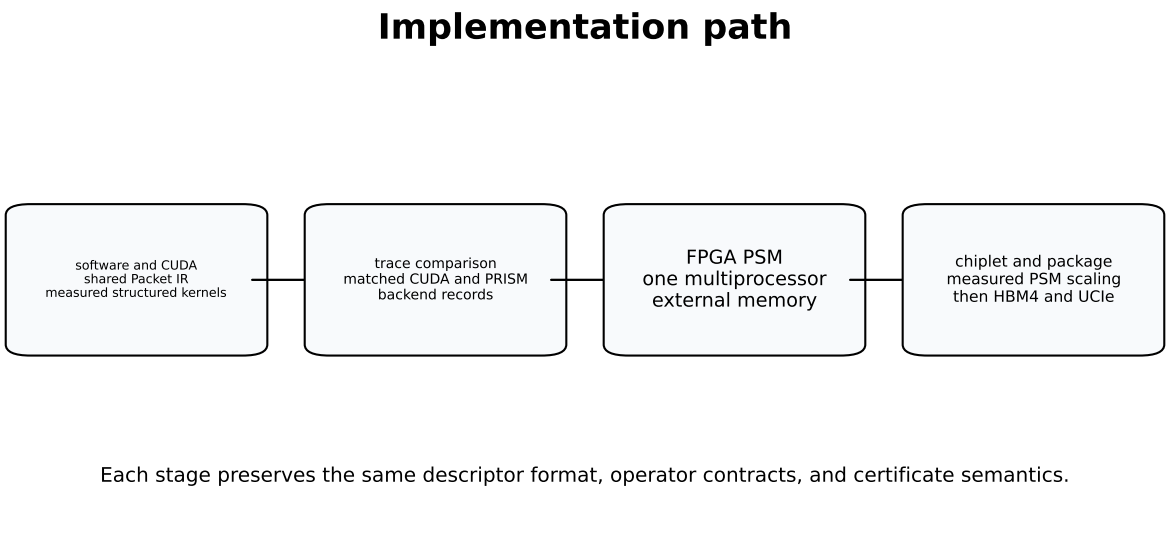


Figure 24.1: The software and CUDA stages precede FPGA and silicon work.

24.1 Stage 1: software and CUDA foundation

The calculus already has CPU references, CUDA packet kernels, grouped block execution, compressed BDLR products, Woodbury solves, CUDA Graph replay, mixed-precision certificates, transform paths, and measured structured-operator baselines. This stage supplies the semantics, Packet IR, workload traces, and matched comparison.

24.2 Stage 2: trace-driven backend comparison

The next software stage lowers the same Packet IR to CUDA and to the PRISM simulator. It records representation, bytes, arithmetic, launch count, solver size, reuse, residual, and output contract. This stage identifies the graph regions that remain bottlenecked after CUDA has already used the better algorithms.

24.3 Stage 3: FPGA Packet Streaming Multiprocessor

The FPGA overlay should implement one Packet Streaming Multiprocessor, packet DMA, one transform path, one matrix and factor path, one small solver, the packet directory, dependency tokens, residual accumulation, and certificate-gated commit. The descriptor and Packet IR formats should match the later ASIC.

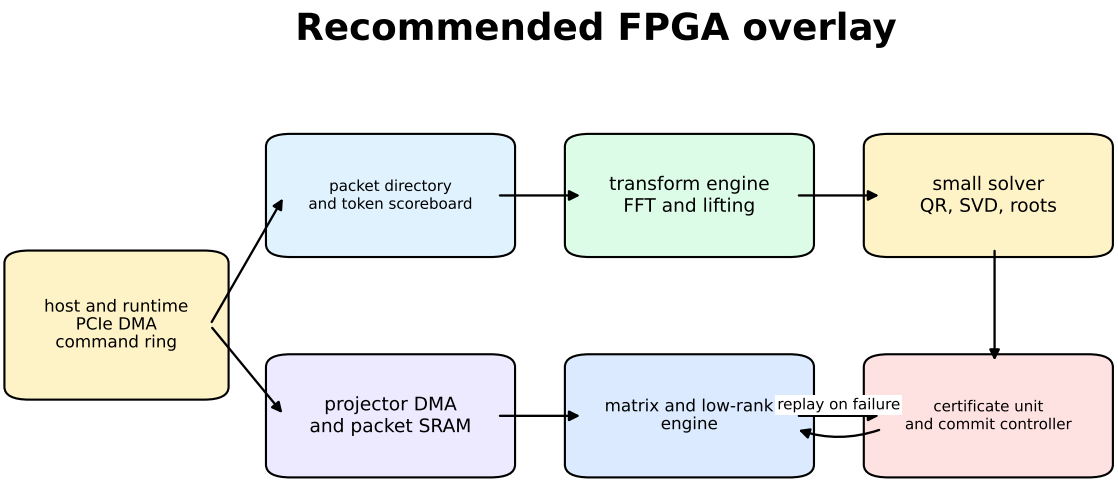


Figure 24.2: Recommended FPGA overlay for measuring the architectural features that do not reduce to an ordinary CUDA kernel.

24.4 Stage 4: one compute chiplet

A shuttle-scale device can use four to eight Packet Streaming Multiprocessors, one transform cluster, one small-solver cluster, external memory, and a simple host link. This stage should measure factor residency, scratchpad bandwidth, packet-warp occupancy, transform-to-solver handoff, and local refinement.

24.5 Stage 5: full package

The full package adds four compute chiplets, HBM4, UCIe, host I/O, and package-wide management. It is justified only after the CUDA and FPGA stages show a stable same-representation advantage on real workloads.

24.6 Development risks

The highest risks are package power, HBM integration, small-solver occupancy, network congestion, compiler misclassification, descriptor overhead, and false certificate acceptance. Early CUDA and FPGA measurement is the main mitigation.

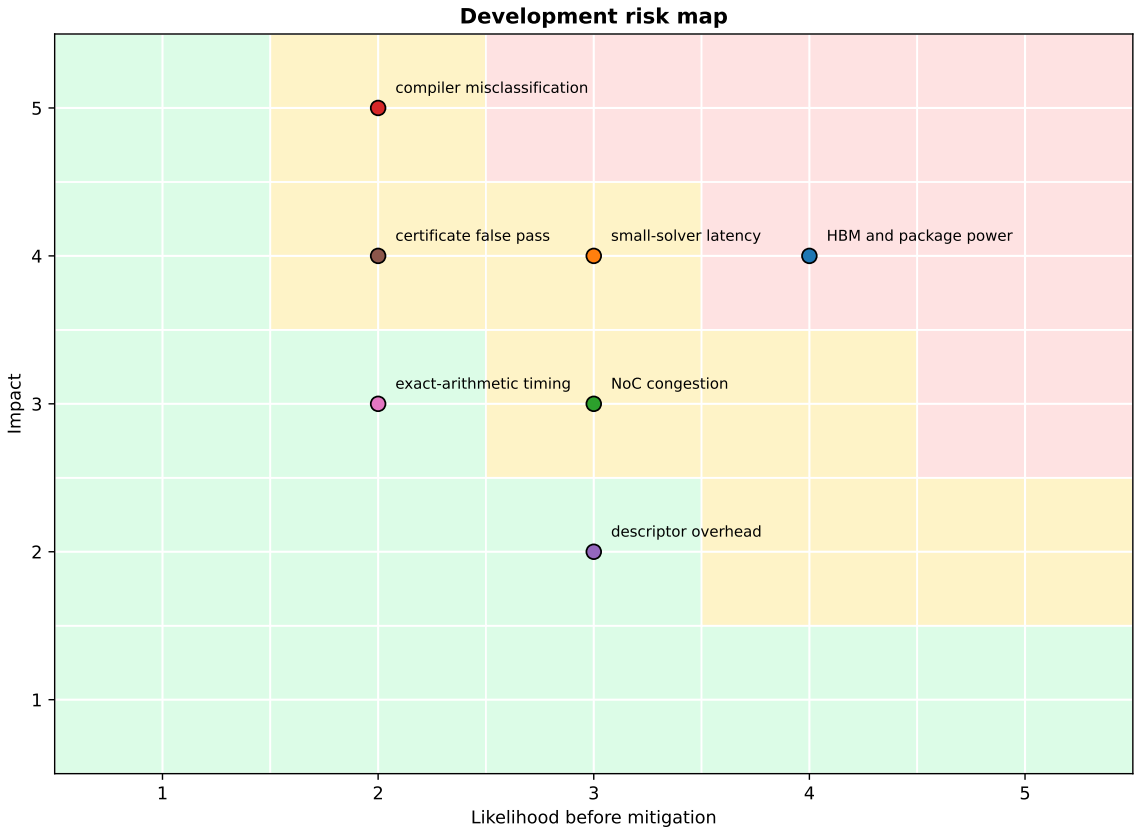


Figure 24.3: Development risk map. Values are planning judgments rather than measured frequencies.

24.7 Architecture freeze checklist

The following items should be frozen before production RTL begins.

1. Packet, transfer, atlas, representation, and certificate descriptor semantics
2. Packet IR representation forms and output contracts
3. Packet-warp and Packet Streaming Multiprocessor scheduling rules
4. command and completion-record formats
5. commit and replay rules
6. supported numerical formats and rounding behavior
7. minimum transform, factor, and solver operation set
8. CUDA and PRISM backend conformance tests
9. error and exception reporting

SRAM bank count, issue width, NoC virtual channels, exact root-solver implementation, and final DVFS domains may remain implementation choices until synthesis and floorplanning begin.

25 What would count as success

The chip should be judged against claims that can be measured:

1. structured operations move fewer bytes than an explicit-array route
2. BDLR, PMR–GEMM, and regular block-sparse graphs sustain useful Packet Streaming Multiprocessor occupancy
3. transform analysis and synthesis remain exact within the declared numerical format
4. mixed-precision results satisfy their packet bounds
5. local refinement changes only the local descriptor region
6. edge recovery keeps the small algebraic chain resident
7. Bayesian portfolios reuse state matrices rather than reloading them per model
8. negative controls are routed to CUDA, a general device, or a generic leaf when PRISM is not the better path

A successful PRISM chip would not prove the calculus. The mathematics already supplies the contracts. The hardware question is whether keeping those contracts visible saves enough scheduling, movement, and handoff cost beyond the already optimized CUDA implementation to justify a dedicated machine.

26 Final architecture statement

The final design follows one principle:

> Do not materialize, move, or schedule structure that the calculus has already declared inactive.

The machine executes

$$\begin{aligned} \text{seed object} &\longrightarrow \text{Packet IR} \\ &\longrightarrow \text{mixed-representation plan} \\ &\longrightarrow \text{packet warps on CUDA or PRISM} \\ &\longrightarrow \text{local refinement and synthesis} \\ &\longrightarrow \text{certificate and commit.} \end{aligned}$$

PRISM-1 is a CUDA-like packet-programmable processor, not a dense-matrix preservation machine. Its hardware state may be a factor graph, transform chain, packet tree, sparse operator, explicit array, or mixture of those forms. The requested output contract determines what must be returned.

The CUDA backend proves that the calculus can run before silicon. The FPGA and ASIC question is narrower and testable: whether resident packet scheduling, factor reuse, transform-to-solver handoff, local refinement, and certificate-gated execution add enough value beyond the packet CUDA path.

A Reference data sheet

Item	PRISM-1 reference target
Compute chiplets	4
Packet Streaming Multiprocessors	64 total, 16 per compute chiplet
Packet execution groups	256 total, 4 per multiprocessor
Execution group	16×16 matrix array plus programmable vector lanes
Clock	1.5 GHz model target
Packet scratchpad	2 MiB per multiprocessor across four banked regions
Shared SRAM	16 MiB per compute chiplet
Total local and shared SRAM	192 MiB
FP16 and BF16 model peak	196.6 TFLOP/s
FP32 model peak	98.3 TFLOP/s
FP64 model peak	24.6 TFLOP/s
Transform engines	32
Small-solver engines	16
Exact arithmetic clusters	4
HBM4	4 stacks
HBM capacity	144 GiB
HBM bandwidth	greater than 11.2 TB/s aggregate
Package links	UCIe 3.0 class at up to 64 GT/s
Host link	PCIe 7.0 x16 class, optional CXL
Power target	500 W
Representation forms	explicit array, block, block sparse, low rank, BDLR, Kronecker, sparse factor chain, transform plan, packet graph
Output contracts	apply, compressed operator, factor graph, transform plan, explicit array
Software backends	CUDA compatibility backend and PRISM backend from shared Packet IR

B Operation-to-engine map

Calculus operations and hardware engines

polyphase	projector DMA
Fourier	transform
wavelet	transform
BDLR	matrix + low rank
block sparse	matrix + router
Bayesian	solver
Prony	solver + transform
Volterra	memory unit
exact modular	exact arithmetic

Figure B.1: The main calculus operations and their primary hardware engines.

C Executable model and source layout

The source package contains the executable architecture model, independent tests, compiler examples, documentation, generated figures, machine-readable results, and an RTL module skeleton. Raw source listings are kept in the release rather than printed across the final paper.

Path	Role
`prism_chip_model.py`	Behavioral operations, roofline model, sensitivity sweeps, figures, and self-tests
`tests/`	Independent contract tests
`compiler_ir/`	BDLR, PMR–GEMM, unknown-edge, and PR-LASM graph examples
`rtl_skeleton/`	Descriptor, queue, scoreboard, packet directory, representation router, PSM scheduler, execution group, residual, commit, and top-level module boundaries
`docs/`	Data sheet, command specification, compiler notes, and microarchitecture notes
`results/`	Machine-readable architecture results

D RTL module boundaries

Module	Responsibility
`prism_descriptor_pkg`	Command opcodes and packed descriptor types
`prism_command_fifo`	Command-ring buffering and flow control
`prism_token_scoreboard`	Dependency-token readiness and completion tracking
`prism_packet_directory`	Packet validity, provisional state, commit state, and storage location
`prism_residual_accumulator`	Wide residual accumulation
`prism_commit_controller`	Certificate comparison, commit, replay, and abort decisions
`prism_representation_router`	Dispatch among block, factor, transform, solver, packet-graph, and generic leaves
`prism_psm_scheduler`	Packet-warp issue, dependency tracking, replay-region control, and execution-group assignment
`prism_execution_group`	Matrix, vector, selector, reduction, and provisional-result boundary for one execution group
`prism_tile_shell`	Compatibility shell for one execution group during staged RTL development
`prism_top`	Queue, scoreboard, packet directory, and completion interface shell

The module set remains a boundary specification rather than production RTL. A production implementation still requires bit-accurate datapaths, interface IP, reset and clock-domain design, verification properties, synthesis, timing closure, and physical design.

E Implementation baseline

Internal algorithmic basis

The architecture integrates three existing project components.

- Dense Transfer Algebra supplies PMR–GEMM, compatible middle bases, Kronecker rules, sparse factor chains, and the generic fallback.
- Projector-Resolved Seed Calculus VIII supplies exact block, BDLR, compressed-product, Woodbury, packet-Krylov, mixed-precision, amortization, and CUDA Graph results.
- The CUDA backend supplies the working software route through grouped matrix kernels, FFT and sparse libraries, asynchronous transfers, graph replay, and packetwise certification.

These algorithms are architecture inputs. PRISM does not claim them as consequences of the chip.

The package assumptions use current public baselines from the UCIe Consortium, PCI-SIG, Micron, and RISC-V International. The relevant values are the 48 and 64 GT/s classes in UCIe 3.0, the 128 GT/s PCIe 7.0 host-link class, Micron's 2048-bit HBM4 interface with bandwidth above 2.8 TB/s per stack, and the ratified RISC-V Vector 1.0 programming model. These sources establish a plausible interconnect, memory, and control baseline. They do not validate the PRISM microarchitecture.

KPP / SOURCE READING CARD

Komorebi Preservation Protocol

Specification draft 0.1 · 20 original pages

READ THIS FOR

Immutable artifacts, revisable claims, provenance and recoverable continuation.

KEEP THIS BOUNDARY

Protocol specification, not proof that the entire archive already implements it.

CONNECTED SOURCES

PPT | SII | PCL

Original text begins on the next page.

The source pages that follow are preserved without editorial stamps. Page labels in the PDF navigation use the source code and physical source-page number. The package retains the byte-identical original PDF and its SHA-256 digest.

Komorebi Preservation Protocol

A Theory of Recoverable Research Continuity

Preserve what existed. Record what happened. Revise what is believed. Make continuation explicit.

Author: Komorebi P.
Version: Specification Draft 0.1
Date: September 2, 2026
Scope: All present and future Komorebi research projects

Status of This Document

This document defines a forward-looking preservation theory and operating protocol for the Komorebi research program. It is a specification draft, not a claim that the described software infrastructure has already been implemented. Its purpose is to establish the normative structure that future implementations, archives, handoffs, and public releases should satisfy.

The protocol formalizes practices already present in parts of the research corpus, including versioned packages, sealed results, hash manifests, clean extraction tests, explicit failure records, and runnable Windows handoffs. It extends those practices into one project-independent theory of continuity.

Scope boundary. Preservation does not establish the truth, novelty, or importance of a result. It preserves the evidence, reasoning products, decisions, and operational state required to assess those questions honestly.

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1 Executive Summary

The optimal preservation system is not merely a collection of files. It is a forward-only research state machine that preserves the ability to verify identity and integrity, understand, challenge, reproduce, restore, and continue every consequential state.

Here, recoverability means that operational state can be restored and canonical pointers can be rolled back. Recorded history itself remains append-only and is never reversed.

The Komorebi Preservation Protocol, abbreviated KPP, is built around one central law:

History is irreversible. Belief is revisable.

Earlier artifacts, runs, interpretations, and belief states remain recoverable. Current claims may nevertheless be strengthened, weakened, divided, refuted, restricted, superseded, or withdrawn. Correction therefore advances the record without rewriting it.

The protocol separates four kinds of truth:

Table 1: The four preserved truths.

Truth	Question answered	Governing structure
Artifact truth	What exact statements, files, inputs, models, data, and outputs existed?	Immutable objects
Process truth	What derivation, command, experiment, review, failure, or decision occurred?	Append-only ledger
Epistemic truth	What does the evidence currently support, contradict, or leave unresolved?	Revisable claim graph
Continuation truth	Which state is authoritative, what remains open, and where should work resume?	Canonical heads and handoffs

A project is preserved only when all four remain available and mutually linked.

2 Definition of Preservation

2.1 Operational definition

A research program is preserved when a future competent collaborator, using only the preserved record and declared external dependencies, can:

1. identify the exact materials that existed;
2. locate the authoritative current state;
3. reconstruct how and why that state arose;
4. distinguish observation from interpretation and belief;
5. reproduce or adjudicate pivotal results under a declared equivalence standard;

6. identify failures, gaps, limitations, and unresolved branches; and
7. resume work without inventing missing premises.

Byte retention alone is storage. Preservation requires *inferential continuity*: the future record must contain enough structure to recover both the operational state and the justified path into it.

2.2 Formal state model

At time t, define the preserved research state as

$$S_t = (O_t, L_t, G_t, H_t, P_t), \tag{1}$$

where:

- O_t is the set of immutable, content-addressed artifacts;
- L_t is the append-only ledger of events, runs, decisions, and revisions;
- G_t is the typed graph of claims, evidence, counterevidence, dependencies, and provenance;
- H_t is the set of movable canonical pointers; and
- P_t is the set of validation policies, equivalence rules, and replay contracts.

The two monotonicity requirements are

$$O_t \subseteq O_{t+1}, \qquad L_t \preceq L_{t+1}, \tag{2}$$

where $\preceq$ means that L_t is a prefix of the later event history. Artifacts and events accumulate. Claims and canonical pointers may change only through recorded transitions.

2.3 Preservation equivalence

A recovered state $\widehat{S}_t$ is preservation-equivalent to S_t, written

$$\widehat{S}_t \simeq_{\mathcal{E}} S_t, \tag{3}$$

when the recovered state preserves artifact identity, lineage, claim status, unresolved residuals, and pivotal behavior under a declared equivalence contract $\mathcal{E}$. The contract must specify one of the following, or a domain-specific refinement:

- byte-exact equality;
- logical or structural equality;
- numerical equality within a fixed tolerance;
- statistical equivalence under a fixed criterion;
- behavioral equivalence on a frozen test set; or
- semantic equivalence under an explicit review standard.

The equivalence rule must be declared before it is used as promotion evidence. Post hoc tolerances do not provide valid confirmatory support.

3 Preservation Axioms

3.1 Identity conservation

The same durable identifier always denotes the same content. Moving, renaming, repackaging, or mirroring an artifact does not alter its identity. Changed bytes create a new digest and therefore a new object or revision.

3.2 Monotone history and revisable belief

Events and evidence accumulate. Beliefs may evolve. A former claim state is never silently replaced by its successor. The successor links to it through a typed relation such as `corrects`, `supersedes`, `restricts`, `refutes`, or `withdraws`.

3.3 Provenance closure

Every consequential result must resolve to its exact inputs, assumptions, transformation, environment, producing actor or tool, and parent state. If a dependency is unavailable, the record must say so explicitly and bound the consequences.

3.4 Epistemic separation

Observations, interpretations, decisions, and claims are distinct record types. A software test does not prove a theorem. A clean replay does not establish external validity. A digest verifies byte identity, not scientific truth.

3.5 Canonicality

Every active project has exactly one authoritative head for each project-level role. Each named branch has exactly one branch head. Alternative states remain visible, but they cannot silently compete for the label `stable`.

3.6 Total outcomes

Every declared run or proof attempt receives a terminal disposition. Valid dispositions include success, null, mixed, refuted, invalid, blocked, crashed, cancelled, and abandoned. Unfavorable and incomplete outcomes do not disappear.

3.7 Branch conservation

Branching preserves the shared parent. Merging preserves all parents and records what was adopted, rejected, modified, or left unresolved. A merge must not fabricate a clean linear history.

3.8 Residual conservation

Gaps, anomalies, contradictions, failed controls, instability, regressions, and deferred questions are first-class records. They define the actual research frontier and therefore cannot be treated as archival noise.

3.9 Tested recoverability

Recoverability is established by successful restoration, not inferred from the existence of copies. Before promotion to `stable`, every milestone must complete a successful artifact restoration. Pivotal executable claims must also complete the applicable clean replay, while non-executable claims undergo their declared adjudication procedure.

3.10 Controlled disclosure

A lossless private master, a curated internal record, and a sanitized public release may share lineage but remain distinct views. Redaction never changes the private master. Public material is generated through an explicit allowlist and independent validation.

4 The Continuity Block

4.1 Block structure

Every meaningful unit of work creates a parent-linked transition

$$S_n \xrightarrow{B_n} S_{n+1}, \tag{4}$$

with continuity block

$$B_n = (P, I, X, T, O, E, R, D, N, M). \tag{5}$$

A block is mutable while open. Once sealed, it becomes immutable. Later corrections create amendment blocks linked to the original. A scientifically unsuccessful block can still be preservation-complete.

4.2 State-changing operators

The principal operators are:

$$\text{Revise}(c \to c') : \text{preserve } c\text{, add } c'\text{, and record cause and evidence,} \tag{6}$$
$$\text{Branch}(s \to s_1, \ldots, s_k) : \text{preserve the parent and assign branch hypotheses,} \tag{7}$$
$$\text{Merge}(s_1, \ldots, s_k \to m) : \text{record all parents, conflicts, and dispositions,} \tag{8}$$
$$\text{Fail}(r) : \text{preserve protocol, observation, logs, class, and claim impact,} \tag{9}$$
$$\text{Handoff}(h) : \text{close provenance, evidence, operation, and continuation around } h. \tag{10}$$

Promotion changes a pointer. It never destroys the states behind that pointer.

Table 2: Required components of a continuity block.

Term	Component	Requirement
P	**Parent**	Exact state or states from which the block begins
I	**Intent**	Bounded objective, question, or hypothesis
X	**Inputs**	Artifacts, assumptions, models, datasets, and dependencies
T	**Transformation**	Derivation, code change, command, intervention, or review
O	**Outputs**	Every consequential produced artifact
E	**Evidence**	Support, counterevidence, null evidence, or limitations produced
R	**Residuals**	Gaps, anomalies, failures, instability, and deferred questions
D	**Decision**	Claim changes, branch disposition, or engineering choice
N	**Next**	Clearest admissible continuation point
M	**Manifest**	Stable identifiers, digests, lineage, and replay contract

5 Forward Operating Protocol

5.1 Project charter

Every new project begins with a compact charter containing:

- stable project identifier and title;
- objective, scope, and exclusions;
- current `work`, `stable`, and `public` heads;
- claim and run types used by the project;
- project-specific validation profile;
- data, privacy, and publication policy;
- primary environment and external dependencies;
- definition of `stable`; and
- definition of `public-ready`.

The charter should take minutes to establish. It changes only through an explicit charter revision.

5.2 Three canonical heads

Each project maintains:

`work`	a project-level manifest naming the default active state and every live branch head;
`stable`	the newest state that has passed its declared stability gates;
`public`	the newest sanitized public release.

A rollback repoints a head to an earlier preserved state. It does not erase later work.

5.3 Begin

Before a meaningful work block, record four judgments:

1. the bounded objective;
2. the success or falsification criterion;
3. whether the work is exploratory or confirmatory; and
4. the expected consequential output.

The system should automatically resolve and verify the parent state, assign a block identifier, capture known local changes, identify exact inputs, and record the initial visibility classification.

5.4 Work and checkpoint

Normal research should continue with minimal ceremony. Evidential commands should automatically capture:

- exact command and working directory, with secrets redacted;
- source state or content digest;
- dependency, runtime, and hardware fingerprint;
- inputs and outputs by digest;
- random seeds and trial identity;
- start time, finish time, and exit code;
- standard output and error logs after credential and sensitive-output filtering; and
- deviations from the declared protocol.

Lightweight checkpoints should be created before dependency changes, before risky refactors, after pivotal results, at interruption, and whenever the cost of reconstructing work since the last checkpoint becomes material.

5.5 Seal

At closure, automatic tooling should inventory changes, validate manifests, account for all declared runs, execute configured tests, and create the continuity packet. The canonical closure capsule contains six short judgments:

1. purpose;
2. outcome;
3. uncertainty;
4. claim impact;
5. decision; and
6. next action.

Purpose is inherited from the opening record unless the objective changed. Outcome may be drafted from captured runs, but a human must confirm its interpretation. The remaining fields record judgments that cannot be reconstructed reliably from files alone.

The protocol should fail open during exploration and fail closed during promotion. Missing metadata may produce an incomplete private checkpoint. It cannot produce a stable or public release.

6 Progressive Rigor and Promotion Gates

6.1 Three preservation levels

Table 3: Progressive rigor prevents archival ceremony from displacing research.

Level	Automatic content	Human judgment
Automatic capture	Identity, parent, file hashes, commands, inputs, outputs, logs, environment, exit states	None or minimal classification
Continuity capsule	Links to captured objects and current state	Purpose, outcome, uncertainty, claim impact, decision, next action
Certified checkpoint	Frozen snapshot, manifest, replay recipe, complete validation, redundant copies	Claim audit, limitations, promotion decision

Anything mechanically derivable should be generated automatically. No person should enter the same fact twice. Full ceremony is reserved for consequential claims and releases.

6.2 Promotion gates

Table 4: Forward promotion gates.

Gate	Name	Requirement
G0	**Captured**	Stable identifiers, parent, timestamp, provenance, visibility, and content digests assigned
G1	**Sealed**	Manifest valid, planned runs accounted for, and no silent overwrites
G2	**Evidentially complete**	Protocol, raw outcomes, interpretation, claim impact, counterevidence, and residuals recorded
G3	**Verified**	Project tests and declared comparison rules pass, with failures classified
G4	**Restored and replayed**	Artifact restoration succeeds, and each pivotal executable result reproduces from a clean directory under its declared equivalence rule
G5	**Stable**	Regressions pass, claims are updated, decision is recorded, and the stable head is intentionally advanced
G6	**Handoff-ready**	A cold successor can identify the state, reproduce a pivotal check, understand limitations, and begin the next block
G7	**Public-ready**	Allowlist export, privacy scan, license review, restricted-lineage review, and clean public replay pass

A failed scientific hypothesis may reach G2 or higher as a preserved negative result. It cannot be misrepresented as a positive result merely because the package itself verifies successfully.

7 Claims, Evidence, and Decisions

7.1 Claim records

Every consequential claim receives a permanent identifier and revision history. Its record contains:

- exact formal and plain-language statements;
- claim type, scope, quantifiers, and assumptions;
- supporting and opposing evidence;
- proof or protocol dependencies;
- falsification criteria;
- boundary cases and known limitations;
- current revision and superseded revisions; and
- the event responsible for every status transition.

7.2 Independent status axes

A single confidence number may supplement these fields but may not replace them.

Table 5: Claim status is multidimensional.

Axis	Example values
Nature	definition, lemma, theorem, conjecture, empirical, causal, engineering, interpretive
Evidence	untested, supported, mixed, contradicted, refuted, inconclusive
Proof	absent, incomplete, checked, independently checked, formally verified
Reproduction	unreplayed, clean local replay, fresh environment, independent implementation
Lifecycle	active, restricted, superseded, withdrawn
Novelty conclusion	unassessed, known precedent, candidate novel
Novelty review	not reviewed, internal review, independent review
Visibility	private, internal, public-candidate, public

7.3 Evidence graph

The canonical evidential chain is

$$\text{claim} \longrightarrow \text{assumptions} \longrightarrow \text{derivation or implementation} \longrightarrow \text{protocol} \longrightarrow \text{run} \longrightarrow \text{raw result} \longrightarrow \text{interpretation} \longrightarrow \text{decision} \longrightarrow \text{release}.$$

Missing links are permitted only when explicitly marked. The graph must support both forward lineage and reverse impact analysis. For example, invalidating one run should reveal every figure, interpretation, claim, and release that depends upon it.

7.4 Decision records

A consequential decision records the decision made, evidence considered, alternatives rejected, rationale, reversibility, and conditions that should trigger reconsideration. Preserving files without preserving decision logic produces operational memory without judgment.

8 Failures and Residuals

8.1 Failure taxonomy

A dependency conflict is not scientific counterevidence. A valid negative result is not an engineering inconvenience. Every planned seed and trial must receive an outcome disposition, preventing accidental success-only reporting.

Table 6: Failure classes must not be conflated.

Class	Meaning
Scientific refutation	A valid result contradicts the tested claim
Scientific null	A valid test finds no expected effect
Mixed result	Support varies across conditions, seeds, models, tasks, or scales
Engineering failure	Dependency, memory, file, implementation, or runtime failure
Protocol invalid	The attempted test cannot answer its intended question
Inconclusive	Power, coverage, controls, or precision are inadequate
Abandoned branch	Work stopped for a recorded reason without resolution

8.2 Residual ledger

Closing a block requires disposition, not resolution. Every residual is marked `open`, `resolved`, `accepted`, `deferred`, or `invalidated`, together with its next trigger or owner. The residual ledger is the map of the living frontier.

9 AI and Tool Provenance

Attribution and validation must be recorded separately. For every material model-assisted contribution, preserve:

- model or system identifier exactly as exposed, or `unknown`;
- date and visible session or task reference;
- role, such as proposed, derived, coded, reviewed, executed, or interpreted;
- files, claims, experiments, or decisions affected;
- human disposition, such as accepted, edited, rejected, or unreviewed; and
- subsequent verification and verifier.

Visible prompts and outputs may be retained when relevant, or represented by a digest and private transcript reference. Hidden internal reasoning is neither required nor claimed. A model reviewing its own proposal is an internal consistency check, not independent replication.

Illustrative provenance statement. A human collaborator supplies an ambient-space hypothesis. A model develops an operator formulation and implementation draft. The human selects the design. Automated tests are executed locally. The result remains independently unreplicated.

The archive must be agent-independent. A new capable human or model should be able to resume from the internal or handoff continuity state without requiring undocumented memory from an earlier interaction. Public-only continuation is promised only when the public release declares and supplies sufficient dependencies.

10 Project Validation Profiles

The universal envelope remains small. Each domain supplies its own validation profile.

Table 7: Project-specific promotion requirements.

Project family	Required emphasis
Komorebi Math	Exact statements, assumption closure, theorem dependency graph, boundary cases, and explicit separation of numerical evidence from proof
Neuralese	Interventions, null and shuffled controls, multiple seeds and layers, OOD behavior, and depth stability
RRCT	Operational and causal results separated from interpretations concerning awareness or subjectivity
OSL	Grammar, formal semantics, conformance vectors, round-trip tests, version negotiation, and interpreter equivalence
ARC	Frozen task release and split, solver digest, scorer, leakage audit, compute budget, and per-task outputs
SRIC	Frozen pre-improvement baseline, human changes separated from autonomous changes, causal attribution, compute-normalized gains, regressions, and rollback
WaveKAT-F	Type I error calibration, complete simulation seeds, comparator parity, GPU and reference agreement, and cohort provenance

These profiles may evolve through versioned policy changes. A result is certified against the exact policy revision in force at promotion time.

11 Retention, Recovery, and Integrity

11.1 Optimal retention rule

Preserve losslessly and redundantly anything that is:

- irreplaceable;
- raw evidence;
- decision-bearing;
- expensive to regenerate;
- necessary for interpretation; or

- necessary for reproduction.

Derived caches may be discarded only when exact parent objects and a tested regeneration procedure survive. Duplicate bytes should have one logical content identity within each repository or replica set and be referenced by digest. That logical object must still have multiple physically independent replicas. Environments and large model weights should normally be referenced by exact digest and legal acquisition method unless complete offline preservation is required and permitted.

11.2 Integrity rules

Each artifact records an algorithm-tagged content digest, byte length, media type, stable identifier, provenance, and role. Each milestone records a deterministic logical manifest. A content digest proves byte identity only. If signatures are later introduced, a signature attributes a statement to a key but still does not establish scientific truth.

11.3 Recovery doctrine

Each stable state should have multiple independently controlled copies, automatic integrity checks, encrypted private copies where appropriate, separately protected recovery credentials, open metadata formats, periodic random replay, and a full cold-restoration drill at major milestones.

Recovery should restore into a new location by default. It must not silently overwrite the active workspace. A restoration event becomes part of the append-only ledger.

12 Private, Internal, and Public Views

The protocol preserves three separate surfaces:

1. a lossless private master;
2. a curated internal research record; and
3. a sanitized public release.

Public packages are generated from an explicit allowlist in a clean location. They are not created by copying the private tree and deleting obvious secrets. Scanning must cover source files, logs, stack traces, notebook outputs, filenames, paths, metadata, manifests, and version-control history.

Credentials and live secrets are never archived as ordinary content. They are excluded or redacted at ingestion and represented only by references to a separately controlled secret-management mechanism. Private by default are personal conversations, machine-identifying paths, restricted datasets, licensed weights, private reviewer material, necessary raw model interactions, and unaudited historical packages. Publication should never mutate or replace the private source record.

13 Forward Adoption Doctrine

The protocol begins prospectively. Research should not pause for a comprehensive migration of the historical corpus.

1. Establish one genesis checkpoint for every currently active project.

2. Identify its current `work`, `stable`, and `public` heads.
3. Apply KPP to every consequential block from that point forward.
4. Introduce older material only when it is reused, validated, cited, or prepared for release.
5. Mark older material as an inherited input with known provenance, verification state, uncertainties, and dependent claims.

This approach creates immediate continuity without allowing archival normalization to consume the research program.

14 Acceptance Tests and Health Criteria

14.1 Cold-successor test

A stable state is handoff-complete only when another competent human or agent can, without relying on the original conversation:

- identify the authoritative state;
- explain why it became authoritative;
- locate supporting and opposing evidence;
- reproduce one pivotal result under its declared equivalence rule;
- see every consequential unresolved residual; and
- begin the next block without inventing missing premises.

14.2 Hard acceptance tests

The protocol should be tested against at least the following cases:

1. Two differently named packages contain identical bytes. One logical object identity is created, multiple physical replicas may remain, and two intake events stay visible.
2. A theorem statement changes. The old statement remains fixed and the new revision links to it explicitly.
3. A run reports success but lacks exact inputs. It may be captured privately but cannot be promoted.
4. A public package contains a credential inside a log. Publication fails closed.
5. A result is refuted. Current status changes, while historical releases remain reproducible and visibly superseded.
6. A schema changes. Old records remain readable and migration is reversible and auditable.
7. One project reuses another project's component. It pins the shared identity rather than creating an ambiguous copy.
8. Archive storage is lost. A restoration drill reconstructs indexes and verifies artifact identities.
9. The original collaborator is unavailable. A successor identifies the canonical state and resumes from the handoff alone.

14.3 Health vector

Preservation quality should be reported as a vector, not collapsed into one score. Critical failure in security or recoverability must not be hidden by excellent hashing or extensive metadata.

Recommended dimensions include:

- canonical-head uniqueness;
- ancestry and provenance completeness;
- claim-to-evidence coverage;
- declared-run outcome coverage;
- replay success by equivalence class;
- unresolved orphan artifacts;
- restoration success;
- dependency availability;
- public-release scan status; and
- cold-successor handoff success.

15 Preservation Completeness

Preservation completeness proposition.
Relative to declared external dependencies and surviving storage substrates, a promoted project state is preservation-complete when it has immutable artifact closure, an append-only transition history, referentially closed provenance, a current multidimensional claim state, explicit residuals, a unique canonical head, a declared replay equivalence rule, at least one successful artifact restoration, and the applicable replay or adjudication procedure.

Removing any one component produces a characteristic ambiguity: unknown bytes, unknown origin, unknown evolution, unknown authority, unknown epistemic status, unknown frontier, or untested recovery.

The protocol therefore protects more than files. It preserves the program's operational and epistemic continuity: what existed, what happened, what was believed, why belief changed, what remains unresolved, and where work should continue.

16 Protocol Constitution

The following rules govern all future refinements of KPP:

1. Preserve enough to restore, roll back, explain, reproduce, and continue every consequential change without reversing recorded history.
2. Never strengthen a scientific claim merely because its package verifies successfully.
3. Never erase valid counterevidence, failed trials, or earlier rational belief states.
4. Never let two states silently claim the same canonical role.

5. Never make public disclosure the mechanism by which private preservation is achieved.
6. Never require archival ceremony whose value is lower than its risk of abandonment.
7. Generate mechanically derivable metadata automatically.
8. Permit explicit unknowns rather than fabricated precision.
9. Treat restoration as an experiment that must actually be run.
10. Keep the protocol itself versioned, inspectable, and correctable.

The archive is not a museum of conclusions. It is the continuity substrate of a living, self-correcting research program.

A Minimal Continuity Capsule

The minimum human-readable closure record for an ordinary block is:

Purpose	What were we trying to learn, establish, repair, or change?
Outcome	What actually happened, including failures and null results?
Uncertainty	What remains broken, unverified, ambiguous, or dependent on unavailable evidence?
Claim impact	Which claims were strengthened, weakened, split, restricted, or refuted?
Decision	What was adopted, rejected, deferred, or reversed?
Next	What is the clearest admissible continuation point?

All remaining fields should be generated from the captured state wherever possible.

B Minimal Machine Record

The irreducible machine-readable envelope is conceptually equivalent to:

```
record_id: permanent identifier
record_type: project | block | checkpoint | claim | evidence
             artifact | run | residual | decision | event
             policy | amendment | release | extension:<name>
project_id: permanent project identifier
schema_version: metadata schema
created_at: UTC timestamp
creator: human | model | tool, with identifier when known
parent_ids: direct ancestry
visibility: private | internal | public-candidate | public
content_digest: algorithm-tagged digest
```

```
purpose: short human-readable statement
relations: typed links to other records
```

Domain modules extend this envelope. They do not replace it.

C Suggested Forward Commands

The eventual Windows-facing interface should remain compact:

The following commands are normative interface sketches. They do not assert that a `kpp` implementation already exists.

```
kpp begin neuralese "Test ambient-space stabilization"
kpp run -- python experiment.py --seed 17
kpp checkpoint
kpp seal
kpp verify neuralese@stable
kpp replay neuralese@stable --clean --offline
kpp handoff neuralese@stable --self-contained
kpp recover neuralese@stable
```